

FIREFLY SOFTWARE FOUNDATION



PyFly

by Example

Event-Driven Python Microservices
with the Firefly Framework



PyFly by Example

Event-Driven Python Microservices with the Firefly Framework

Firefly Software Foundation

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The original Lumen chapters target PyFly 26.6.x. Appendix E documents the native webapp and model-administration implementation in the accompanying 26.9.5 source checkout; run its catalog sample from that checkout.

Published by the Firefly Software Foundation.

For Jacinto Arias and the Taidy team —

for pushing us, again and again, to build a real framework for Python.

Because of you there is, at last, one way to do things.

One. Single. Way. (We give it a week.)

Preface

Enterprise Python has long meant stitching together a dozen independent libraries — one for dependency injection, another for routing, yet another for async database access — with no shared idiom to bind them. **PyFly** changes that. It brings the cohesive, convention-over-configuration experience that Spring Boot gave the Java world, rebuilt from the ground up for Python 3.12+ and `async / await`.

This book teaches PyFly **by doing**. You build one real application from an empty folder to a secured, observable, event-driven service — making every concept concrete before moving to the next. Crucially, the code in these pages is not illustrative pseudocode: it is taken from a **real project that compiles, boots, and passes its tests** against PyFly v26.6.110. Every listing was verified against the running sample, so what you read is what actually works.

Appendix E extends the book with the separate `samples/webapp` catalog, verified against the current PyFly checkout. It covers native HTML, forms, route/static helpers, custom errors, and administration of the same SQLAlchemy/Beanie models. Lumen chapters retain their historical version references.

Who This Book Is For

This book is for intermediate Python developers comfortable with `async / await`, type hints, and the basics of HTTP services. You need no prior framework expertise — if you have built anything with FastAPI, Flask, or SQLAlchemy, you are well prepared.

Spring Boot developers will feel especially at home. Wherever PyFly mirrors a Spring concept — beans, stereotypes, declarative transactions, application events — a **Spring parity** callout draws the parallel explicitly, so you map what you already know rather than learning from zero.

What You Will Build

Every chapter advances **Lumen**, a digital-wallet and ledger service. The journey follows a deliberate arc, one part at a time:

- **Quick Start — Build Lumen Step by Step.** Before the deep dive, a single guided walkthrough takes you from an empty folder to a running, tested wallet feature — open a wallet, deposit, read the balance over HTTP — so you see the whole shape of a PyFly application before zooming into any one part. Every later chapter then expands one slice of what you built here.
- **Part I — Foundations (Chapters 1–4).** You scaffold the first Lumen service with `pyfly new`, run it under an ASGI server, wire PyFly's dependency-injection container, bind typed configuration and profiles, and expose your first validated REST endpoints.
- **Part II — Modeling & Persisting (Chapters 5–7).** You introduce the repository pattern over a port, persist wallets with async SQLAlchemy (SQLite, no infrastructure required), model the domain with a `Money` value object and a `Wallet` aggregate, and split reads from writes with CQRS command and query handlers dispatched through a bus.
- **Part III — Event-Driven (Chapters 8–10).** The aggregate raises domain events; a listener projects them; an **event-sourced ledger** rebuilds every balance by replaying its event stream; and the same events flow out to Kafka or RabbitMQ for other services.
- **Part IV — Into Microservices (Chapters 11–13).** Lumen reaches beyond its own process: a typed HTTP client calls an external Payments service, an orchestrated **transfer saga** moves money across wallets and *compensates* when a step fails, and caching plus resilience patterns keep the system fast and fault-tolerant.
- **Part V — Secure · Observe · Ship (Chapters 14–18).** You secure the endpoints with JWT and `@secure`, make the service observable with metrics, tracing, health

checks, and the admin dashboard, test the whole stack, connect it to the outside world with scheduling, notifications, and webhooks, and finally extend and ship it to production.

By the last page you have a working, tested, observable, secured service — and the mental model to extend it.

How to Use This Book

Read sequentially. Each chapter builds on the one before, and the Lumen codebase grows incrementally; skipping ahead leaves gaps.

Type every listing yourself. Reading and typing code at the same time is how the patterns stick. Resist copy-pasting until you have written each listing at least once.

Run it. Lumen really runs — `uv run pyfly run` boots the service and `uv run --extra dev pytest` exercises it. Whenever a chapter adds a feature, start the app or the tests and watch it work. Seeing real JSON come back from a real endpoint is worth a hundred diagrams.

Each chapter closes with a **Recap** of what changed in the Lumen codebase and a set of **Exercises** that push one step further. The exercises are optional but recommended for anything you intend to apply immediately.

Conventions in Brief

Typographic and structural conventions — code-listing captions, callout types, and figure numbering — are demonstrated, with live examples, in the **Conventions** section that follows.

The Companion Code

The complete, runnable Lumen project lives in the framework's `samples/lumen` directory. It is a single, layered PyFly project — `interfaces`, `models`, `core`, `web` — that you grow chapter by chapter; the finished source there is the destination this book walks you to. Set it up once with `uv sync`, and use it to compare your work, catch up if you fall behind, or simply run the parts you are reading about.

Conventions

This page explains the typographic and structural conventions used throughout the book.

Code Listings

Every multi-line code example has a **file-name tab** in the top-left corner showing the file it belongs to, and a "**Listing N.N**" caption below the block identifying it by chapter and sequence number. For example:

wallet/domain/wallet.py

```
from pyfly.core import component
from dataclasses import dataclass, field
from decimal import Decimal

@Component
@dataclass
class Wallet:
    id: str
    balance: Decimal = field(default=Decimal("0.00"))

    def deposit(self, amount: Decimal) → None:
        if amount ≤ 0:
            raise ValueError("Deposit amount must be positive")
        self.balance += amount
```

Listing 5.1 — Wallet aggregate root

Inline code references within prose use `monospace` font, as in "the `@component` decorator registers the class with PyFly's container."

Callouts

Four callout styles appear in the margins and body:

NOTE

Notes provide supplementary context or clarify a subtlety in the main text — worth reading, but not blocking.

TIP

Tips share a shortcut, idiom, or best practice that will save you time in real projects.

WARNING

Warnings flag a common mistake or a sharp edge that can cause hard-to-debug problems if ignored.

SPRING PARITY

Spring parity callouts map a PyFly concept directly to its Spring Boot equivalent — ideal for developers migrating from the JVM ecosystem.

Figures

Diagrams are numbered **Figure N.N** and captioned below the image. They are embedded as inline SVG, so they render crisply at any zoom level in both the screen and print editions. You will meet the first one on the opening page of Chapter 1.

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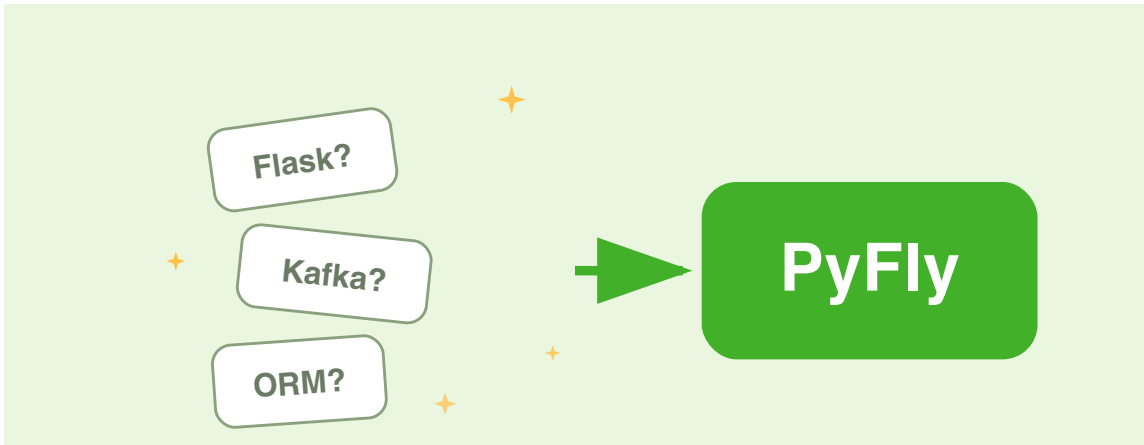
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Quick Start

QUICK START

Build Lumen Step by Step



Welcome. This is the very first thing you will build with PyFly, and we are going to take it slowly. By the end of this chapter you will have gone from an *empty folder* to a *running, tested* slice of a real digital-wallet service — opening a wallet, persisting it to a database, reading its balance back over HTTP, and reacting to a domain event. Every concept gets a small, complete piece of code and a "Run it" checkpoint so you can see it working before moving on.

This is a *tour*, not the deep dive. Each step previews a topic that Part I and Part II cover thoroughly later. The goal here is momentum: by the time you reach Chapter 1 you will have already met dependency injection, configuration, HTTP, persistence, CQRS, and events — in the small — and the rest of the book will fill in the *why*.

The application you build is called **Lumen**: a DDD-flavoured digital wallet. A wallet can be opened, deposited to, and withdrawn from, protecting one core rule — **the balance never goes negative** — and modelling money with exact integer arithmetic so there is never any floating-point drift. It is the same application the entire book builds, so nothing you learn here is throwaway.

 NOTE

This chapter is written against PyFly **v26.6.110**. Every listing is taken from the real, running `samples/lumen` project that ships with the book — the code compiles, boots, and passes its tests. You will recognise these exact files again, in more depth, in later chapters.

Step 1 — Prerequisites and install

PyFly is a Python framework, and the smoothest way to work with it is through **uv**, Astral's fast Python package and project manager. `uv` handles your Python version, your virtual environment, and your dependencies in one tool, and the `pyfly` command-line tool runs through it.

You need two things:

- **Python 3.12 or newer.** PyFly uses modern typing features (`StrEnum` , `X | None` unions, PEP 695 generics).
- **uv.** Install it once, system-wide.

Install `uv` with the official one-liner:

```
curl -Lsf https://astral.sh/uv/install.sh | sh # macOS / Linux
# Windows (PowerShell):
# powershell -c "irm https://astral.sh/uv/install.ps1 | iex"
```

Run it

Confirm both tools are available:

```
uv --version
uv python install 3.12 # ensures a 3.12+ interpreter is available
```

You should see a uv version printed, and uv will report that Python 3.12 is installed (or already present).

FOLLOWING ALONG WITH THE FINISHED SAMPLE

Everything in this chapter exists, finished, in the book's repository under `samples/lumen`. If at any point you want to compare your code with the real thing — or just run it — clone the repo and do:

```
cd samples/lumen
uv sync --extra dev # framework + pytest
uv run pyfly run --server uvicorn
```

You can build alongside it, copying one file at a time, or read the finished version when a step is unclear. Both work.

Step 2 — Scaffold the project

PyFly ships a project generator, `pyfly new`, the same way Spring has the Spring Initializr. It writes a conventional project layout so you do not start from a blank page. Because `pyfly` lives inside the framework package, the very first thing we do is create a project directory and add PyFly to it.

For this tour we will build the directories by hand as we go — it keeps every file visible and nothing hidden behind a generator — but the generator is there when you want a head start:

```
pyfly new lumen --archetype hexagonal --features web,data-relational
```

That command creates a `lumen/` folder, a `pyproject.toml`, a `pyfly.yaml`, and a layered source tree. Let's create the same shape ourselves so you see exactly what each piece is for. Start with the project and its dependencies:

```
mkdir lumen && cd lumen
uv init --package --name lumen
uv add "pyfly[cli,web,data-relational]" "pydantic≥2.5"
uv add --dev "pytest≥8" "pytest-asyncio≥0.24" "httpx≥0.27"
```

The three PyFly extras you just added map to the three things Lumen needs: `cli` brings the `pyfly` command itself, `web` brings the ASGI server, and `data-relational` brings SQLAlchemy 2 (async) plus `aiosqlite` so we can persist to a SQLite file with no external database to install.

The project layout

PyFly applications follow a layered structure that separates the public contract, the domain model, the application logic, and the web edge. Create these packages under `src/lumen`:

```

lumen/
├── pyproject.toml
├── pyfly.yaml           # framework configuration
├── src/lumen/
│   ├── interfaces/     # the public contract: DTOs + enums
│   │   ├── dtos/v1/
│   │   └── enums/v1/
│   ├── models/         # the domain model + persistence
│   │   ├── entities/v1/
│   │   └── repositories/
│   └── core/           # application logic: commands, queries,
handlers
├── services/
├── mappers/
├── web/                # the HTTP edge: controllers
│   └── controllers/
├── app.py              # the application class
└── main.py             # the ASGI entry point

```

Each layer has one job. `interfaces` is the boundary other code (and other services) talks to. `models` holds the rich domain objects and the rows they persist as. `core` holds the business operations. `web` exposes them over HTTP. We will fill these in one layer at a time.

🍃 SPRING PARITY

`pyfly new` is the equivalent of the Spring Initializr (`start.spring.io`). The `web` and `data-relational` features are the PyFly counterparts of the `spring-boot-starter-web` and `spring-boot-starter-data-jpa` starters — naming a feature pulls in exactly the dependencies and auto-configuration that feature needs, and nothing more.

The application class

Two files turn a package into a PyFly application. The first, `app.py`, declares the application itself: which packages to scan for components and which framework tiers to switch on.

`lumen/app.py`

```
from __future__ import annotations

from pyfly.core import pyfly_application
from pyfly.starters.domain import enable_domain_stack

@EnableDomainStack
@pyfly_application(
    name="lumen",
    version="1.0.0",
    description="Lumen – a DDD digital-wallet service built on the PyFly framework.",
    scan_packages=[
        "lumen.models.repositories",
        "lumen.core.services.wallets",
        "lumen.web.controllers",
    ],
)
class LumenApplication:
    pass
```

Listing 0.1 — The application class

`@pyfly_application` marks the class as a PyFly app and `scan_packages` tells the dependency-injection container where to look for the components you will declare — your repositories, services, command/query handlers, and controllers.

`@enable_domain_stack` switches on the domain tiers we will lean on later: CQRS, the transactional engine, the relational data layer, and events.

The ASGI entry point

The second file, `main.py`, is what an ASGI server actually imports and serves. It bootstraps PyFly — loads configuration, scans your packages, and builds the application context — then hands the resulting web application to Starlette.

`lumen/main.py`

```
from __future__ import annotations

from lumen.app import LumenApplication
from pyfly.core import PyFlyApplication
from pyfly.web.adapters.starlette import create_app

# Bootstrap: load config, scan packages, build the DI context.
_pyfly = PyFlyApplication(LumenApplication)

app = create_app(
    title="lumen",
    version="1.0.0",
    description="Lumen — a DDD digital-wallet service built on the PyFly framework.",
    context=_pyfly.context,
)
```

Listing 0.2 — The ASGI entry point

NOTE

The real `samples/lumen/main.py` adds a `lifespan` hook and a `/static` mount. Those are refinements you will meet in Chapter 4; the two essentials — `PyFlyApplication(LumenApplication)` to bootstrap, and `create_app(...)` to build the web app — are exactly what you see here.

Configuration

PyFly reads `pyfly.yaml` from the project root. Create one that names the app, sets the HTTP port, and switches on the tiers we need. Everything is nested under a top-level `pyfly` key.

`pyfly.yaml`

```
pyfly:
  app:
    name: lumen
    version: 1.0.0
  server:
    # App on 8080; the actuator + admin default to the management port 9090.
    port: 8080
  cqrs:
    enabled: true
  transactional:
    enabled: true
  eda:
    provider: memory          # in-memory event bus, no broker needed
  data:
    relational:
      enabled: true
      url: "sqlite+aiosqlite:///./lumen.db"
      ddl-auto: create        # create tables on startup
```

Listing 0.3 — `pyfly.yaml`

A few keys are worth a moment now and a chapter later. `pyfly.server.port` is the application's HTTP port — `8080` by default, exactly like Spring's `server.port`. `data.relational` points at a SQLite file (`lumen.db`) and `ddl-auto: create` tells the framework to create the database schema on startup, so there is no migration step to run for this tour. `eda.provider: memory` gives us an in-process event bus.

⚠ WARNING

If you are coming from an older PyFly, note that the port key is `pyfly.server.port` (env override `PYFLY_SERVER_PORT`). The old `pyfly.web.port` / `PYFLY_WEB_PORT` were removed — set the port under `pyfly.server` from now on.

Run it

Even with no endpoints yet, the application boots. Start it:

```
uv run pyfly run --server uvicorn
```

You will see the PyFly banner, structured startup logs, and a line telling you the server is listening on `http://0.0.0.0:8080`. The `--server uvicorn` flag selects the Uvicorn server (it comes with `pyfly[web]`); for development, add `--reload` to restart automatically when you edit a file.

PyFly also exposes a **health endpoint** so orchestrators can tell the app is alive. Actuator endpoints and the admin dashboard run on a *separate management port*, `9090` by default, which keeps operational endpoints off your public application port. In another terminal:

```
curl -s localhost:9090/actuator/health
```

```
{"status": "UP"}
```

❗ TWO PORTS, ON PURPOSE

The application serves your API on `8080` ; the **management port** `9090` serves `/actuator/health` , `/actuator/info` , and the admin dashboard. This is Spring Boot's `management.server.port` behaviour. The management port is *open and unauthenticated by default* — fine on a private network, but in production you would set `pyfly.management.security.enabled: true` to lock it down, or `pyfly.management.server.port: -1` to disable management endpoints entirely. By default only `health` and `info` are exposed over HTTP; expose more (metrics, env, ...) with `pyfly.management.endpoints.web.exposure.include` .

Stop the server with `Ctrl-C` . The shell is empty, but the foundation is live: the DI container builds, the server starts, and health reporting works. Now we give it something to do.

Step 3 — The first slice of the domain

We start where DDD says to start: with the model. Two objects carry the whole domain — `Money` , a value object for exact amounts, and `Wallet` , the aggregate that owns the balance.

Money — a value object

`Money` is the textbook *value object*: it has no identity, two instances with the same amount and currency are interchangeable, and it never changes. We store amounts as integer **minor units** (cents) plus an ISO-4217 currency code, so arithmetic is exact — `Money(1050, EUR)` is €10.50, and there is no floating-point rounding to worry about.

First the tiny currency enum it depends on, under `interfaces/enums/v1/` :

`lumen/interfaces/enums/v1/currency.py`

```
from __future__ import annotations

from enum import StrEnum


class Currency(StrEnum):
    """ISO-4217 currency codes Lumen wallets can hold."""

    EUR = "EUR"
    USD = "USD"
    GBP = "GBP"
```

Listing 0.4 — The Currency enum

Now `Money` itself, under `models/entities/v1/`. It builds on `pyfly.domain.ValueObject` and is a frozen dataclass, so equality is structural and instances are immutable. Arithmetic returns *new* `Money` objects and refuses to mix currencies.

lumen/models/entities/v1/money.py

```

from __future__ import annotations

from dataclasses import dataclass

from lumen.interfaces.enums.v1.currency import Currency
from pyfly.domain import BusinessRuleViolation, ValueError

@dataclass(frozen=True)
class Money(ValueObject):
    """An exact monetary amount in a single currency (minor units)."""

    amount: int
    currency: Currency

    def __post_init__(self) → None:
        if not isinstance(self.amount, int) or isinstance(self.amount, bool):
            raise BusinessRuleViolation(
                "money-amount-integer", "amount must be an integer number of
minor units"
            )

    @classmethod
    def zero(cls, currency: Currency) → Money:
        """The additive identity for *currency* (a zero balance)."""
        return cls(amount=0, currency=currency)

    def add(self, other: Money) → Money:
        self._assert_same_currency(other)
        return Money(amount=self.amount + other.amount,
currency=self.currency)

    def subtract(self, other: Money) → Money:
        self._assert_same_currency(other)
        return Money(amount=self.amount - other.amount,
currency=self.currency)

```

```

@property
def is_positive(self) → bool:
    return self.amount > 0

@property
def is_negative(self) → bool:
    return self.amount < 0

@property
def major_units(self) → float:
    """The amount as a major-unit decimal (cents / 100)."""
    return round(self.amount / 100, 2)

def _assert_same_currency(self, other: Money) → None:
    if self.currency is not other.currency:
        raise BusinessRuleViolation(
            "money-currency-mismatch",
            f"cannot combine {self.currency.value} with
{other.currency.value}",
        )

```

Listing 0.5 — The Money value object

`BusinessRuleViolation` is the framework's signal that a domain rule was broken — here, "amounts are whole minor units" and "you cannot add euros to dollars". Notice there is no HTTP, no database, no framework wiring: a value object is pure domain.

Wallet — the aggregate

The `Wallet` is the *aggregate root*: the object that owns the invariant. State only changes through intent-revealing methods (`open`, `deposit`, `withdraw`), each of which protects the rule **balance never goes negative** and records a *domain event* describing what happened. Built on `pyfly.domain.AggregateRoot`, it raises events with `raise_event(...)`, which we will drain and publish in Step 9.

lumen/models/entities/v1/wallet_entity.py

```

from __future__ import annotations

from dataclasses import dataclass
from datetime import UTC, datetime

from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.money import Money
from pyfly.domain import AggregateRoot, BusinessRuleViolation, DomainEvent

@dataclass(frozen=True)
class WalletOpened(DomainEvent):
    wallet_id: str = ""
    owner_id: str = ""
    currency: str = ""

@dataclass(frozen=True)
class FundsDeposited(DomainEvent):
    wallet_id: str = ""
    amount: int = 0
    currency: str = ""
    balance: int = 0

class Wallet(AggregateRoot[str]):
    """Wallet aggregate root - owns the ``balance ≥ 0`` invariant."""

    __slots__ = ("owner_id", "balance", "created_at")

    def __init__(
        self,
        id: str,
        owner_id: str,
        balance: Money,
        created_at: datetime | None = None,
    ) → None:

```

```

    super().__init__(id)
    self.owner_id = owner_id
    self.balance = balance
    self.created_at = created_at or datetime.now(UTC)

    @property
    def currency(self) → Currency:
        return self.balance.currency

    @classmethod
    def open(cls, wallet_id: str, owner_id: str, currency: Currency) →
Wallet:
        """Open a new, empty wallet; raises :class:`WalletOpened`."""
        if not owner_id.strip():
            raise BusinessRuleViolation("wallet-owner-required", "owner_id is
required")
        wallet = cls(id=wallet_id, owner_id=owner_id,
balance=Money.zero(currency))
        wallet.raise_event(
            WalletOpened(wallet_id=wallet_id, owner_id=owner_id,
currency=currency.value)
        )
        return wallet

    def deposit(self, amount: Money) → None:
        """Credit *amount* to the balance; raises :class:`FundsDeposited`."""
        self._assert_currency(amount)
        if not amount.is_positive:
            raise BusinessRuleViolation("wallet-deposit-positive", "deposit
amount must be > 0")
        self.balance = self.balance.add(amount)
        assert self.id is not None
        self.raise_event(
            FundsDeposited(
                wallet_id=self.id,
                amount=amount.amount,
                currency=amount.currency.value,
                balance=self.balance.amount,
            )

```

```

    )

    def _assert_currency(self, amount: Money) → None:
        if amount.currency is not self.balance.currency:
            raise BusinessRuleViolation(
                "wallet-currency-mismatch",
                f"wallet holds {self.balance.currency.value}, got
{amount.currency.value}",
            )

```

Listing 0.6 — The Wallet aggregate and its domain events

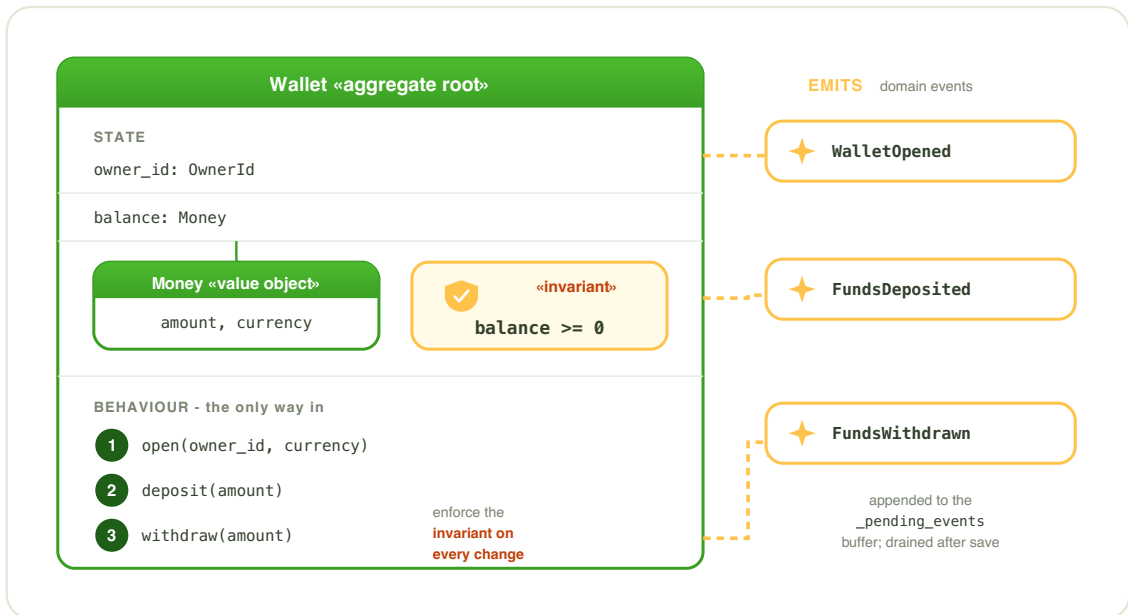


Figure 0.1 — The Wallet aggregate owns its invariant; all state changes go through its methods.

NOTE

The finished `Wallet` in `samples/lumen` also has a `withdraw` method and a `FundsWithdrawn` event, which follow the same shape — we have left them out here to keep this first listing short. Chapter 6 builds the full aggregate.

Run it

You do not need the server running to exercise the domain. Open a Python REPL inside the project and drive the model directly:

```
uv run python
```

```
>>> from lumen.interfaces.enums.v1.currency import Currency
>>> from lumen.models.entities.v1.money import Money
>>> from lumen.models.entities.v1.wallet_entity import Wallet
>>> w = Wallet.open("wlt-1", "alice", Currency.EUR)
>>> w.deposit(Money(1500, Currency.EUR))
>>> w.balance.amount, w.balance.currency.value
(1500, 'EUR')
>>> w.deposit(Money(100, Currency.USD))      # wrong currency → rejected
pyfly.domain.exceptions.BusinessRuleViolation: wallet holds EUR, got USD
```

The aggregate enforces its own rules in pure Python — no infrastructure required.

Step 4 — Persist it

A wallet that lives only in memory is not much use. We need to save it. PyFly's data layer gives you a *Spring-Data-style repository* over SQLAlchemy, and because we configured SQLite there is no database to install.

The persistence row

The aggregate is rich; the row it persists as is flat. We map `Wallet` onto a `WalletEntity` — one row per wallet, with the balance split into an integer column (`balance_minor`) and a currency column. It inherits the framework's declarative `Base`, which lets it keep the aggregate's own string id as its primary key.

lumen/models/entities/v1/wallet_orm.py

```

from __future__ import annotations

from datetime import UTC, datetime

from sqlalchemy import String
from sqlalchemy.orm import Mapped, mapped_column

from pyfly.data.relational.sqlalchemy import Base

class WalletEntity(Base):
    """One persisted wallet row, keyed by the aggregate's own string id."""

    __tablename__ = "wallets"

    id: Mapped[str] = mapped_column(String(64), primary_key=True)
    owner_id: Mapped[str] = mapped_column(String(255), nullable=False,
index=True)
    currency: Mapped[str] = mapped_column(String(3), nullable=False)
    balance_minor: Mapped[int] = mapped_column(nullable=False, default=0)
    created_at: Mapped[datetime] = mapped_column(default=lambda:
datetime.now(UTC))

```

Listing 0.7 — The WalletEntity persistence row

Because the class subclasses `Base`, importing it registers the `wallets` table; with `ddl-auto: create` the framework creates that table on startup.

The repository

Instead of hand-writing SQL, you subclass the framework's generic `Repository[Entity, IdType]`. That single declaration tells the framework the entity type and the primary-key type, and in return you get the full async repository surface for free — `save`, `find_by_id`, `find_all`, `count`, `delete`, paging, and more — with the database session injected for you.

lumen/models/repositories/wallet_repository.py

```

from __future__ import annotations

from lumen.models.entities.v1.wallet_orm import WalletEntity
from pyfly.container import repository
from pyfly.data.relational.sqlalchemy import Repository

@repository
class WalletRepository(Repository[WalletEntity, str]):
    """CRUD for :class:`WalletEntity`, plus a convenience upsert."""

    async def find_by_owner_id(self, owner_id: str) → list[WalletEntity]:
        """All wallets owned by *owner_id* (derived query stub)."""
        ...

    async def upsert(self, entity: WalletEntity) → WalletEntity:
        """Insert *entity*, or update the row with the same id."""
        session = self._require_session()
        merged = await session.merge(entity)
        await session.flush()
        return merged

```

Listing 0.8 — A Spring-Data-style repository

Two things here are pure Spring Data. `find_by_owner_id` is a **derived query** — its body is an elided stub (`...`), and at startup the framework parses the method *name* and compiles a real `SELECT ... WHERE owner_id = :owner_id` for you. `upsert` is a small convenience over `session.merge` so a handler can persist a wallet whether it is new or already exists, with a single call.

The `@repository` decorator registers the class as a managed component — a *bean* in the DI container — so it can be injected into the handlers we write next.

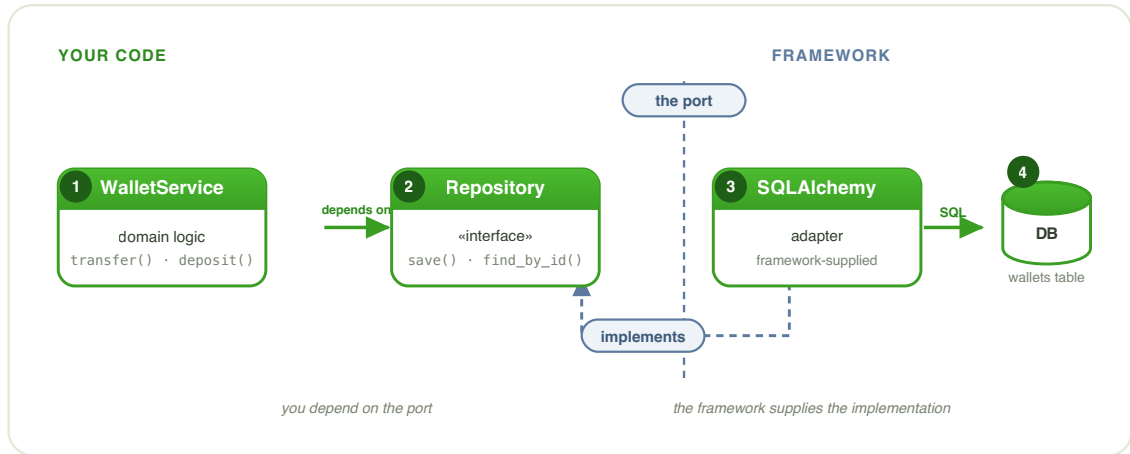


Figure 0.2 — A framework Repository turns a typed declaration into a full CRUD surface.

🍃 SPRING PARITY

`Repository[WalletEntity, str]` is the direct analogue of Spring Data's `JpaRepository<WalletEntity, String>`. You declare the interface; the framework provides the implementation. Derived queries (`findByOwnerId` in Spring, `find_by_owner_id` here) are parsed from the method name in exactly the same way.

Step 5 — A write path with CQRS

Now we wire the model to the repository through a *command*. PyFly uses **CQRS** — Command Query Responsibility Segregation — which means writes flow through one path (commands) and reads through another (queries). A command is a small, immutable object describing intent; a handler executes it.

The command

`OpenWallet` carries the data needed to open a wallet and validates itself before anything runs. It is a frozen dataclass extending `Command[str]` — the `str` says "this command, when handled, produces a wallet id".

`lumen/core/services/wallets/open_wallet_command.py`

```
from __future__ import annotations

from dataclasses import dataclass

from lumen.interfaces.enums.v1.currency import Currency
from pyfly.cqrs import Command, ValidationResult

@dataclass(frozen=True)
class OpenWallet(Command[str]):
    """Open a new wallet. Returns the generated wallet id."""

    owner_id: str
    currency: Currency

    async def validate(self) → ValidationResult: # type: ignore[override]
        if not self.owner_id.strip():
            return ValidationResult.failure("owner_id", "Owner id is
required")
        return ValidationResult.success()
```

Listing 0.9 — The OpenWallet command

The handler

The handler is where the work happens: generate an id, create the `Wallet` aggregate, persist it through the repository, then drain and publish the aggregate's events. It runs inside `@transactional()`, which opens a unit of work, commits on success, and rolls back on failure. The repository and the event publisher are injected by the container — you only declare them in `__init__`.

lumen/core/services/wallets/open_wallet_handler.py

```

from __future__ import annotations

from uuid import uuid4

from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker

from lumen.core.mappers.wallet_mapper import to_entity
from lumen.core.services.wallets.event_publishing import publish_domain_events
from lumen.core.services.wallets.open_wallet_command import OpenWallet
from lumen.models.entities.v1.wallet_entity import Wallet
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.data.relational.sqlalchemy import transactional
from pyfly.eda import EventPublisher

@command_handler
@service
class OpenWalletHandler(CommandHandler[OpenWallet, str]):
    """Open a new, empty wallet."""

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events
        self._session_factory = session_factory

    @transactional()
    async def do_handle(self, command: OpenWallet) → str: # type:
        ignore[override]
        wallet_id = f"wlt-{uuid4()}"

```

```

wallet = Wallet.open(
    wallet_id=wallet_id,
    owner_id=command.owner_id,
    currency=command.currency,
)
await self._repository.upsert(to_entity(wallet))
await publish_domain_events(self._events, wallet.clear_events())
return wallet_id

```

Listing 0.10 — The OpenWallet handler

The handler calls `to_entity(wallet)` — a small mapper that flattens the aggregate into a `WalletEntity` row. Create it under `core/mappers/`. (We will add the read-side projection it also needs in the next step.)

`lumen/core/mappers/wallet_mapper.py`

```

from __future__ import annotations

from lumen.models.entities.v1.wallet_entity import Wallet
from lumen.models.entities.v1.wallet_orm import WalletEntity

def to_entity(wallet: Wallet) → WalletEntity:
    """Flatten a :class:`Wallet` aggregate into a persistable row."""
    assert wallet.id is not None
    return WalletEntity(
        id=wallet.id,
        owner_id=wallet.owner_id,
        currency=wallet.currency.value,
        balance_minor=wallet.balance.amount,
        created_at=wallet.created_at,
    )

```

Listing 0.11 — Mapping the aggregate to its row

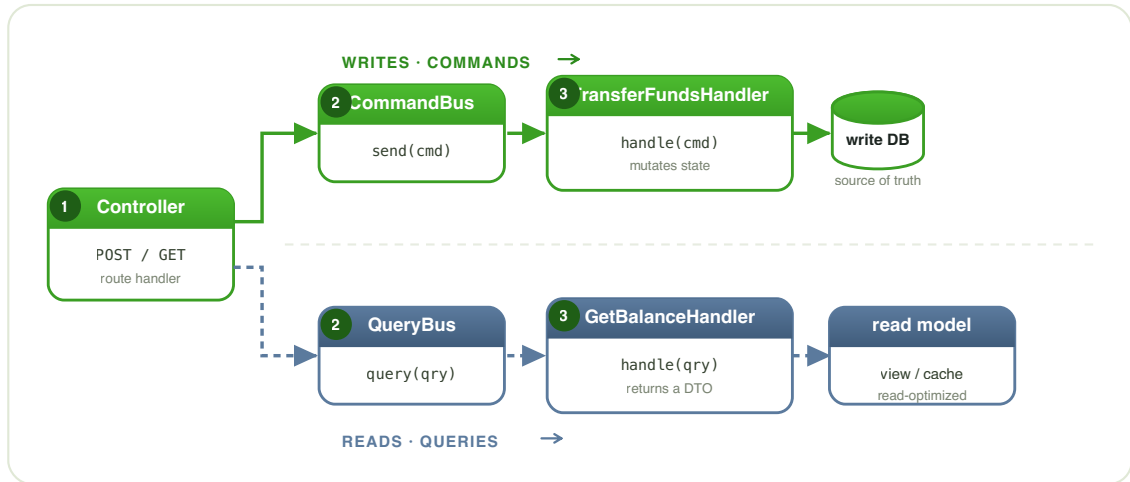


Figure 0.3 — A command flows through the bus to its handler; queries take a separate path.

🌿 SPRING PARITY

`@command_handler + @service` registers a handler the command bus dispatches to — much like a Spring `@Service` whose method handles a request. `@transactional()` is the PyFly counterpart of Spring's `@Transactional`: it manages the unit of work so the persist either fully commits or fully rolls back.

Step 6 — A read path

Reads take the other lane. A *query* asks a question; a *query handler* answers it, typically projecting a database row onto a small, purpose-built DTO. We will read just the balance.

The DTO and the query

The balance response is a tiny Pydantic model under `interfaces/dtos/v1/`:

lumen/interfaces/dtos/v1/balance_dto.py

```

from __future__ import annotations

from pydantic import BaseModel

from lumen.interfaces.enums.v1.currency import Currency


class BalanceDto(BaseModel):
    """Lightweight balance projection for the balance endpoint."""

    id: str
    currency: Currency
    balance_minor: int
    balance: float

```

Listing 0.12 — The BalanceDto response model

The query carries just the wallet id and declares it returns a `BalanceDto` or `None` :

lumen/core/services/wallets/get_balance_query.py

```

from __future__ import annotations

from dataclasses import dataclass

from lumen.interfaces.dtos.v1.balance_dto import BalanceDto
from pyfly.cqrs import Query


@dataclass(frozen=True)
class GetBalance(Query[BalanceDto | None]):
    """Look up just the balance of a wallet by its identifier."""

    wallet_id: str

```

Listing 0.13 — The GetBalance query

The handler and the projection

Add the read-side mapper to `wallet_mapper.py` — a small function that projects a row onto the DTO, computing the major-unit balance:

`lumen/core/mappers/wallet_mapper.py`

```
from __future__ import annotations

from lumen.interfaces.dtos.v1.balance_dto import BalanceDto
from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.wallet_orm import WalletEntity


def entity_to_balance_dto(entity: WalletEntity) → BalanceDto:
    """Project a persisted row onto the lightweight balance DTO."""
    return BalanceDto(
        id=entity.id,
        currency=Currency(entity.currency),
        balance_minor=entity.balance_minor,
        balance=round(entity.balance_minor / 100, 2),
    )
```

Listing 0.14 — Projecting a row onto the balance DTO

The query handler loads the row by id and projects it — returning `None` when there is no such wallet:

lumen/core/services/wallets/get_balance_handler.py

```

from __future__ import annotations

from lumen.core.mappers.wallet_mapper import entity_to_balance_dto
from lumen.core.services.wallets.get_balance_query import GetBalance
from lumen.interfaces.dtos.v1.balance_dto import BalanceDto
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cqrs import QueryHandler, query_handler

@query_handler
@service
class GetBalanceHandler(QueryHandler[GetBalance, BalanceDto | None]):
    def __init__(self, repository: WalletRepository) → None:
        super().__init__()
        self._repository = repository

    async def do_handle(self, query: GetBalance) → BalanceDto | None: #
type: ignore[override]
        entity = await self._repository.find_by_id(query.wallet_id)
        return entity_to_balance_dto(entity) if entity is not None else None

```

Listing 0.15 — The GetBalance handler

Notice the asymmetry CQRS gives you: the write side rehydrates the full aggregate to protect invariants; the read side touches only the columns the balance view needs and never constructs an aggregate at all. Each side is shaped for its own job.

Step 7 — Expose it over HTTP

The domain works, it persists, and it has write and read paths. Now we put a web edge on it. A **controller** maps HTTP requests onto commands and queries and dispatches them through the bus — it holds no business logic of its own.

First the request DTO for opening a wallet, under `interfaces/dtos/v1/`:

`lumen/interfaces/dtos/v1/open_wallet_request.py`

```
from __future__ import annotations

from pydantic import BaseModel, Field

from lumen.interfaces.enums.v1.currency import Currency


class OpenWalletRequest(BaseModel):
    """Wallet-opening request payload."""

    owner_id: str = Field(min_length=1, max_length=64,
description="Identifier of the wallet owner")
    currency: Currency = Field(default=Currency.EUR, description="ISO-4217
currency the wallet holds")
```

Listing 0.16 — The OpenWalletRequest payload

Now the controller, under `web/controllers/`. The DI container injects the command and query buses; each handler builds a command or query and `await`s it on the bus. Parameter annotations declare where data comes from: `Valid[Body[...]]` binds and validates the JSON body, `PathVar[str]` binds a URL segment.

lumen/web/controllers/wallet_controller.py

```

from __future__ import annotations

from lumen.core.services.wallets.get_balance_query import GetBalance
from lumen.core.services.wallets.open_wallet_command import OpenWallet
from lumen.interfaces.dtos.v1.balance_dto import BalanceDto
from lumen.interfaces.dtos.v1.open_wallet_request import OpenWalletRequest
from pyfly.container import rest_controller
from pyfly.cqrs import DefaultCommandBus, DefaultQueryBus
from pyfly.kernel import ResourceNotFoundException
from pyfly.web import Body, PathVar, Valid, get_mapping, post_mapping,
request_mapping

@rest_controller
@request_mapping("/api/v1/wallets")
class WalletController:
    """Digital-wallet REST API: open a wallet, read its balance."""

    def __init__(self, commands: DefaultCommandBus, queries: DefaultQueryBus)
→ None:
        self._commands = commands
        self._queries = queries

    @post_mapping("", status_code=201)
    async def open_wallet(self, request: Valid[Body[OpenWalletRequest]]) →
dict[str, str]:
        wallet_id = await self._commands.send(
            OpenWallet(owner_id=request.owner_id, currency=request.currency)
        )
        return {"wallet_id": wallet_id}

    @get_mapping("/{wallet_id}/balance")
    async def wallet_balance(self, wallet_id: PathVar[str]) → BalanceDto:
        result = await self._queries.query(GetBalance(wallet_id=wallet_id))
        if result is None:
            raise ResourceNotFoundException(
                f"Wallet {wallet_id!r} not found",

```

```

        code="WALLET_NOT_FOUND",
        context={"wallet_id": wallet_id},
    )
    return result

```

Listing 0.17 — The wallet controller

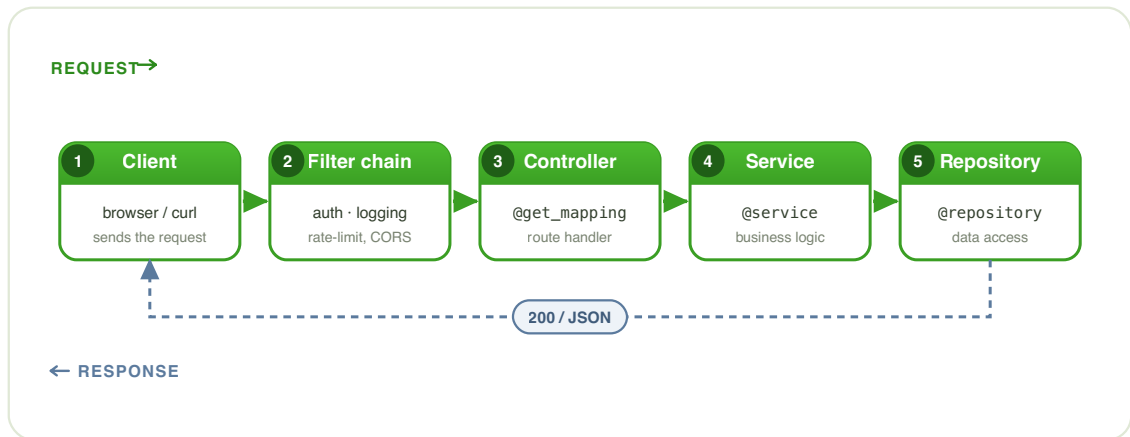


Figure 0.4 — A request binds to a handler, which dispatches a command or query through the bus.

Run it

Start the server:

```
uv run pyfly run --server uvicorn
```

In another terminal, open a wallet:

```
curl -s -X POST localhost:8080/api/v1/wallets \
  -H 'content-type: application/json' \
  -d '{"owner_id":"alice","currency":"EUR"}'
```

```
{"wallet_id":"wlt-7d2c1a9e- ..."}
```

Read the balance back (substitute the id you got above):

```
curl -s localhost:8080/api/v1/wallets/wlt-7d2c1a9e-.../balance
```

```
{"id":"wlt-7d2c1a9e-...", "currency":"EUR", "balance_minor":0, "balance":0.0}
```

A freshly opened wallet has a zero balance — exactly what `Money.zero(EUR)` produced back in the aggregate's `open` factory. The request travelled from HTTP, through the command bus, into the handler, through the repository, to SQLite, and back out the read path. That is the whole arc, end to end.

INTERACTIVE DOCS FOR FREE

While the server runs, open `http://localhost:8080/docs` in a browser. PyFly generated an OpenAPI document and a Swagger UI from your controller and DTOs — you can try the endpoints right there, no extra code.

Step 8 — Prove it with a test

Running `curl` by hand is fine once; a test proves it forever. PyFly is designed to be testable without a running server — you can dispatch commands and queries straight through the buses. Write a test under `tests/`:

tests/test_quickstart.py

```

from __future__ import annotations

import pytest
from lumen.core.services.wallets.get_balance_query import GetBalance
from lumen.core.services.wallets.open_wallet_command import OpenWallet
from lumen.interfaces.enums.v1.currency import Currency

from pyfly.cqrs import DefaultCommandBus, DefaultQueryBus

@pytest.mark.asyncio
async def test_open_wallet_then_read_balance(
    command_bus: DefaultCommandBus,
    query_bus: DefaultQueryBus,
) → None:
    wallet_id = await command_bus.send(OpenWallet(owner_id="alice",
currency=Currency.EUR))
    assert wallet_id.startswith("wlt-")

    balance = await query_bus.query(GetBalance(wallet_id=wallet_id))
    assert balance is not None
    assert balance.balance_minor == 0
    assert balance.currency is Currency.EUR

```

Listing 0.18 — An end-to-end test through the buses

The `command_bus` and `query_bus` parameters are *fixtures*: they boot the application context once and hand you wired buses, the same components the controller uses in production. (The finished `samples/lumen/tests/conftest.py` defines these fixtures; copy it when you build your own suite — Chapter 16 explains it in full.)

Run it

```
uv run --extra dev pytest -q
```

```
. [100%]
1 passed in 0.42s
```

Green. You now have a wallet feature that is not just running but *verified* — the same test runs in CI on every change.

SPRING PARITY

Dispatching through the buses in a test mirrors Spring Boot's slice tests: you exercise real wired beans without standing up the HTTP server. The `command_bus / query_bus` fixtures are the PyFly equivalent of an injected Spring `ApplicationContext` in an `@SpringBootTest`.

Step 9 — A taste of events

The aggregate has been recording domain events all along — `WalletOpened`, `FundsDeposited` — and the handler drains them with `wallet.clear_events()` and publishes them. So far nothing has *listened*. Let's add a small listener that reacts.

First, the publishing bridge the handler already imports. It turns each drained domain event into a payload and publishes it on the event bus, under `core/services/wallets/`:

lumen/core/services/wallets/event_publishing.py

```

from __future__ import annotations

import dataclasses
from collections.abc import Iterable
from typing import Any

from lumen.core.services.listeners.wallet_audit_listener import
WALLET_EVENTS_DESTINATION
from pyfly.domain import DomainEvent
from pyfly.eda import EventPublisher

def _to_payload(event: DomainEvent) → dict[str, Any]:
    """Flatten a frozen-dataclass domain event into a JSON-friendly dict."""
    payload: dict[str, Any] = dataclasses.asdict(event)
    payload.setdefault("event_type", event.event_type)
    return payload

async def publish_domain_events(publisher: EventPublisher, events:
Iterable[DomainEvent]) → None:
    """Publish each drained domain event on the wallet events channel."""
    for event in events:
        await publisher.publish(
            destination=WALLET_EVENTS_DESTINATION,
            event_type=event.event_type,
            payload=_to_payload(event),
        )

```

Listing 0.19 — Publishing drained domain events

Now the listener, under `core/services/listeners/`. It is a plain `@service` whose method is stamped `@event_listener`; at startup PyFly discovers it and subscribes it to the bus — no wiring by hand. Here it keeps a tiny in-memory audit log and a running total per wallet.


```
lumen/core/services/listeners/wallet_audit_listener.py
```

```
from __future__ import annotations

from pyfly.container import service
from pyfly.eda import EventEnvelope, event_listener

# The logical channel the wallet handlers publish domain events to.
WALLET_EVENTS_DESTINATION = "wallet.events"

@service
class WalletAuditListener:
    """In-memory audit log + running-total projection over wallet events."""

    def __init__(self) → None:
        self._running_totals: dict[str, int] = {}

    @event_listener(event_types=["WalletOpened", "FundsDeposited"])
    async def on_wallet_event(self, envelope: EventEnvelope) → None:
        payload = dict(envelope.payload)
        wallet_id = str(payload.get("wallet_id", ""))
        if envelope.event_type == "WalletOpened":
            self._running_totals.setdefault(wallet_id, 0)
        elif envelope.event_type == "FundsDeposited":
            amount = int(payload.get("amount", 0))
            self._running_totals[wallet_id] =
self._running_totals.get(wallet_id, 0) + amount

    def running_total(self, wallet_id: str) → int:
        """Net funds for *wallet_id*, in minor units."""
        return self._running_totals.get(wallet_id, 0)
```

Listing 0.20 — A domain-event listener

Add the listeners package to `scan_packages` in `app.py` so the container discovers it:

```
scan_packages=[
    "lumen.models.repositories",
    "lumen.core.services.wallets",
    "lumen.core.services.listeners",  # ← add this
    "lumen.web.controllers",
],
```

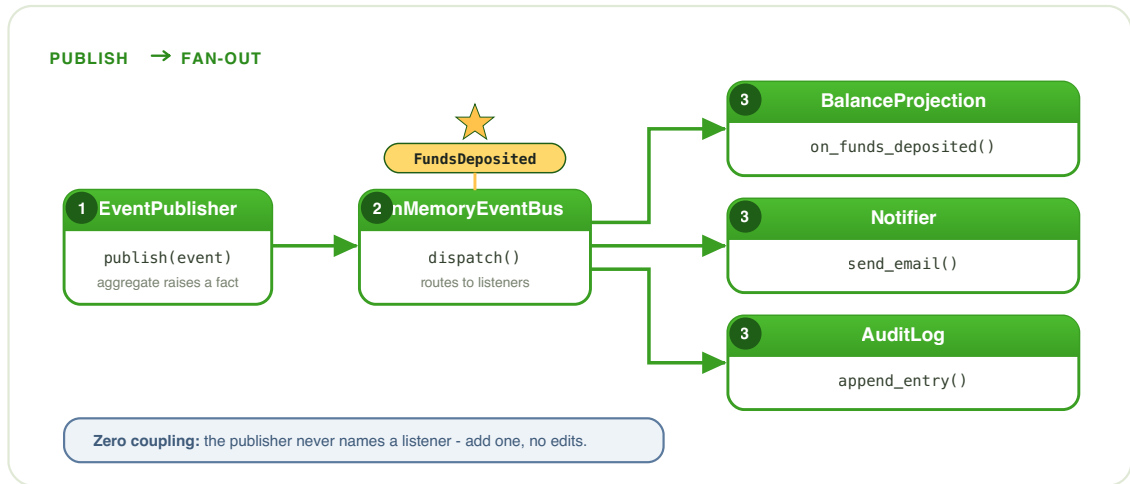


Figure 0.5 — A handler publishes domain events; listeners subscribe and react, decoupled from the command.

Run it

Open a wallet, then deposit, and the listener's running total updates as a side effect of those commands — without the command knowing the listener exists. That decoupling is the whole point of events: you add reactions (audit logs, notifications, projections) without touching the code that triggered them.

 SPRING PARITY

`@event_listener` is the PyFly counterpart of Spring's `@EventListener`. Publishing through an `EventPublisher` and subscribing with a stamped method is the same publish/subscribe model as Spring's `ApplicationEventPublisher` and `@EventListener`-annotated beans.

✓ What you built

You just built — and tested — a real vertical slice of a service: a domain model, a database, a write path, a read path, an HTTP edge, and an event reaction. Every one of those was a *preview*. The rest of the book takes each one apart and rebuilds it properly, with the reasoning, the alternatives, and the production details.

Here is the map from what you just did to the chapter that goes deep:

In this Quick Start you...	Goes deep in
Saw the container build and inject your beans (<code>@repository</code> , <code>@service</code>)	Chapter 2 — Dependency Injection & the Application Context
Configured the app with <code>pyfly.yaml</code> and the management port	Chapter 3 — Configuration, Profiles & Secrets
Exposed an HTTP API with <code>@rest_controller</code> , binding, and validation	Chapter 4 — Your First HTTP API
Persisted with a framework <code>Repository</code> over SQLAlchemy/SQLite	Chapter 5 — Persistence & the Repository Pattern
Modelled <code>Money</code> , <code>Wallet</code> , invariants, and domain events	Chapter 6 — Domain-Driven Design
Split writes and reads with commands, queries, and the bus	Chapter 7 — CQRS: Commands & Queries
Published and reacted to domain events with <code>@event_listener</code>	Chapter 8 — Domain Events & Event-Driven Architecture
(Next) rebuilt state from an event log	Chapter 9 — Event Sourcing the Ledger
(Next) called other services and split the monolith	Chapters 11–12 — HTTP Clients, the BFF & Sagas
(Next) secured, observed, and shipped it	Chapters 14–18 — Security, Observability, Testing & Production

When you are ready for the *why* behind all of it, turn the page to Chapter 1.

Try it yourself

If you want to keep moving on your own first, three small extensions build directly on what you have:

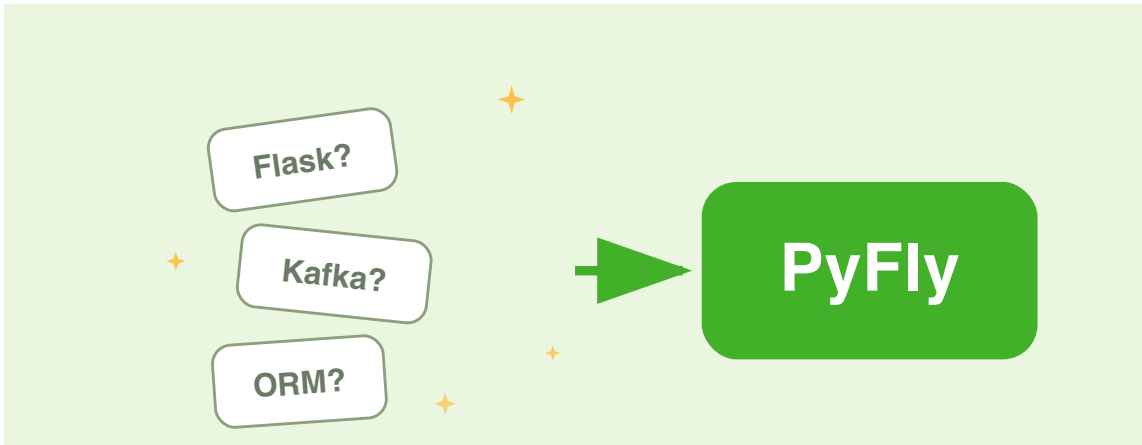
1. **Add a deposit endpoint.** You already have the `FundsDeposited` event and the aggregate's `deposit` method. Add a `DepositFunds` command + handler (model them on `OpenWallet`), a `POST /{wallet_id}/deposit` route, and watch the listener's running total climb.
2. **Add a `withdraw` path** that refuses to overdraw — the aggregate's `balance ≥ 0` invariant should reject it, and your handler should surface that as a clean error. Add a `FundsWithdrawn` domain event that mirrors the `FundsDeposited` event shown in Listing 0.6.
3. **Write a test for the listener**, asserting that opening a wallet and depositing leaves the expected running total — proving the event path end to end.

PART I

Foundations

CHAPTER 1

Why PyFly?



By the end of this chapter you will have installed PyFly, scaffolded the **Lumen** wallet service, and run it locally — with structured logging, a live health endpoint, and interactive API docs already working, all without a single line of boilerplate.

The cohesion problem

Picture your first day on a new Python microservice. Before you write one line of business logic, you face two weeks of architectural choices.

Which web framework do you reach for? FastAPI, Flask, Starlette, Django — each is reasonable, each introduces its own idioms. Which ORM? SQLAlchemy (sync or async?), Tortoise, Beanie? How do you wire dependencies — dependency-injector,

python-inject, or a hand-rolled factory module? How do you manage configuration — pydantic-settings, python-dotenv, dynaconf? And how should the project be laid out? Every team invents its own answer.

You eventually assemble a bespoke stack, glue it together with good intentions, and ship. Six months later a second team starts a new service — and makes entirely different choices. Now you have two codebases with incompatible conventions, different testing strategies, different deployment patterns, and no shared understanding of how anything works.

Python gives you infinite choice. What it does not give you is cohesion.

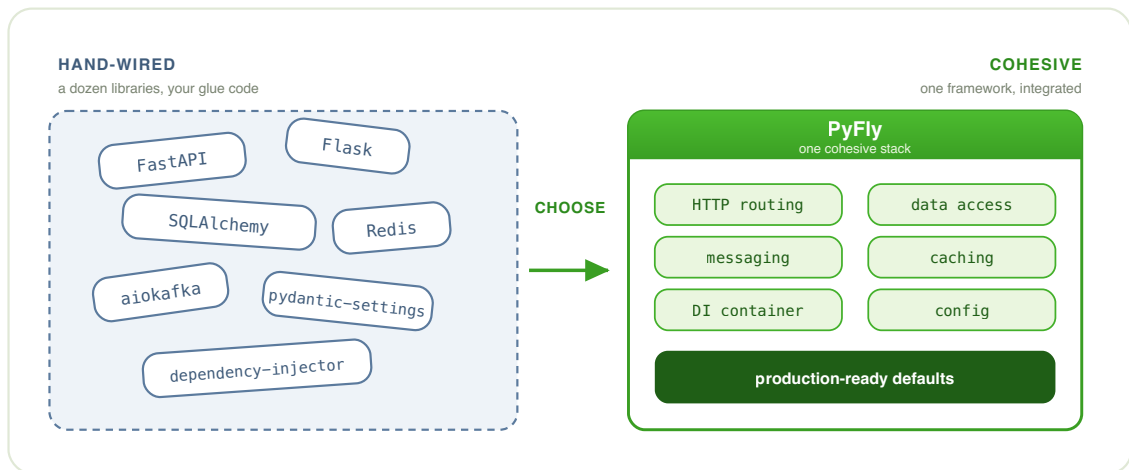


Figure 1.1 — Infinite choice, no cohesion.

The stack-assembly problem is not a skills failure — it is a tooling gap. Java developers solved it years ago with Spring Boot: one opinionated framework that makes sensible choices for you, lets you override what matters, and enforces a consistent idiom across every service. PyFly brings that same discipline to Python.

What is PyFly?

The solution to the cohesion problem is not to ban choice — it is to make one good set of choices and package them as a framework. That is precisely what PyFly does.

PyFly is a **cohesive, full-stack, async-native framework** for building production-grade Python applications — microservices, monoliths, and libraries alike. It makes the stack decisions for you: dependency injection, HTTP routing, database access, messaging, caching, security, observability — all integrated, all consistent, all with production-ready defaults from the very first `pyfly run`.

Under the hood, PyFly delegates to battle-tested async libraries in the Python ecosystem — Starlette for HTTP, SQLAlchemy (async) for relational data, structlog for logging, Pydantic for validation — but you never import them directly. You depend on **PyFly's ports** (Python `Protocol` classes), and the DI container wires the concrete adapters at startup. Swap PostgreSQL for MongoDB, or Kafka for RabbitMQ, without touching a single line of business logic.

PyFly is the **official Python implementation of the Firefly Framework**, a battle-tested enterprise platform originally built for Java (40+ modules in production). It brings the same programming model to Python 3.12+, not as a port but as a native reimplementation designed around `async/await` and type hints.

Its architecture rests on four layers — foundation, application, infrastructure, and integration — each composed of focused modules that interlock cleanly.

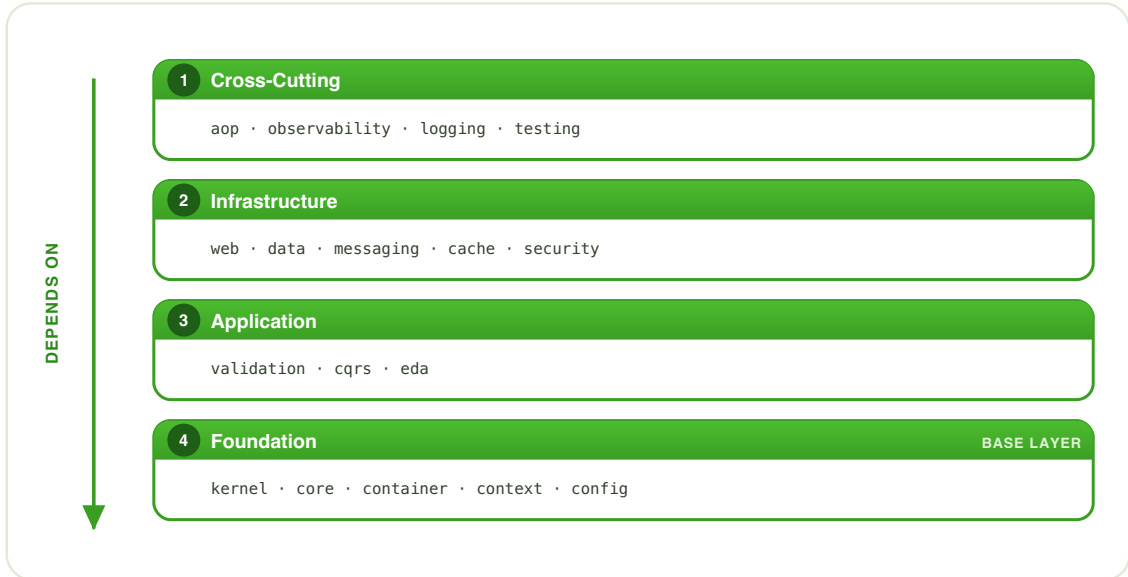


Figure 1.2 — PyFly's four module layers.

Every layer follows the same **hexagonal architecture** principle: your code lives in the centre and depends only on ports; adapters live at the edges and can be swapped without disturbing the core. This design lets the framework grow with you — start with a simple REST service and graduate to CQRS, event-driven messaging, or saga orchestration without rewriting the foundation you built on day one.

🌿 SPRING PARITY

If you are coming from Spring Boot, PyFly will feel like home almost immediately. `@pyfly_application` is your `@SpringBootApplication`. `@service`, `@rest_controller`, and `@repository` are the exact stereotypes you know. Constructor-injection from type hints mirrors `@Autowired` with no XML or reflection magic. The `pyfly.yaml` configuration hierarchy (defaults → profile → env vars) maps directly to `application.yaml` + profiles. A **Spring parity** callout like this one appears throughout the book wherever the concepts align closely enough to save you the mental translation work.

Installing PyFly

To make these ideas concrete, you will build **Lumen** — a fintech wallet platform — throughout the book. Lumen starts simply: a REST API that records ledger entries and reports wallet balances. By the final chapter it will span multiple services communicating over events, with sagas coordinating cross-service transfers. Starting with a well-structured skeleton saves you the pain of retrofitting architecture later, so PyFly's scaffolder generates that structure for you up front.

We will take this one short step at a time. Nothing here assumes prior PyFly experience — if you can run a command in a terminal, you can follow along.

NEW TERM: UV

`uv` is a fast Python package manager and project runner (think `pip` + `virtualenv` + a task runner, rolled into one binary). PyFly recommends it, and every command in this book that starts with `uv run` simply means "run this inside the project's virtual environment". If you have only ever used `pip`, you lose nothing — `uv` reads the same standard `pyproject.toml`.

Step 1 — Check your prerequisites. PyFly targets Python 3.12+ and uses `uv`. Confirm both are installed:

terminal

```
python --version
# Python 3.12.0 or later

uv --version
# uv 0.5.0 or later
```

Listing 1.1 — Verify prerequisites

If either command is missing or reports an older version, install it before continuing (uv's install instructions are one short script on its website). Everything else PyFly needs, it pulls in for you.

Step 2 — Scaffold the project. A single command generates a complete, production-shaped project directory:

terminal

```
# web-api archetype, web feature
uv run pyfly new lumen --archetype web-api --features web
```

Listing 1.2 — Scaffold the Lumen wallet service

NEW TERM: SCAFFOLD / ARCHETYPE

To *scaffold* is to generate a ready-made project skeleton so you start from working code instead of an empty folder. An *archetype* is the template that scaffolding uses — `web-api` is the REST-service template. The `--features web` flag adds the ASGI server dependency (the piece that actually accepts HTTP requests). `uv run pyfly new` calls PyFly's archetype registry and writes the whole directory in one shot.

Step 3 — Move into the project and install dependencies.

terminal

```
cd lumen

# Install dependencies (including the pyfly CLI and ASGI server)
uv sync
```

Listing 1.3 — Enter the project and sync dependencies

`uv sync` reads the declared dependencies from `pyproject.toml` and installs them into a project-local virtual environment — including the `pyfly` command itself (available via the `cli` extra). The first sync downloads packages; subsequent syncs are near-instant because uv caches everything.

💡 RUN IT: CONFIRM THE CLI WORKS

From the project root, run `uv run pyfly --help`. You should see the PyFly command list (`new` , `run` , and friends). If that prints, your toolchain is healthy and you are ready to inspect the generated project.

💡 TIP

Run `pyfly new` **without arguments** to enter interactive mode. It walks you through archetype and feature selection with arrow-key navigation — handy when you want to pre-select extras like relational data or messaging support.

The scaffolder prints the generated layout as a tree:

```
Created web-api project

lumen-demo/
├── .env.example
├── .gitignore
├── Dockerfile
├── README.md
├── pyfly.yaml
├── pyproject.toml
├── src/lumen_demo/__init__.py
├── src/lumen_demo/app.py
├── src/lumen_demo/controllers/__init__.py
├── src/lumen_demo/controllers/health_controller.py
├── src/lumen_demo/controllers/todo_controller.py
├── src/lumen_demo/main.py
├── src/lumen_demo/models/__init__.py
├── src/lumen_demo/models/todo.py
├── src/lumen_demo/repositories/__init__.py
├── src/lumen_demo/repositories/todo_repository.py
├── src/lumen_demo/services/__init__.py
├── src/lumen_demo/services/todo_service.py
├── tests/__init__.py
├── tests/conftest.py
└── tests/test_todo_service.py
```

Next steps:

```
cd lumen-demo
uv sync --group dev
pyfly run --reload
```

The apparent simplicity is deliberate. `pyproject.toml` gives you a standards-compliant project from day one — no `setup.py` legacy debt. `pyfly.yaml` is the single source of truth for every runtime setting, from log level to database URL, so configuration never scatters across a dozen files. The `src/` layout prevents the `lumen` package from

being accidentally importable without an explicit install, catching import-path bugs early. The scaffolder also generates runnable sample controllers, services, and repositories so you have working code to study immediately rather than an empty shell.

Two files that matter

Understanding the scaffold's entry-point structure early will save you confusion later. The scaffold generates two files that work together: `app.py` declares the application and `main.py` exposes the ASGI `app` that the server imports. Each has a distinct responsibility.

NEW TERM: ASGI

ASGI (Asynchronous Server Gateway Interface) is the standard contract between a Python web server and an async web application — the modern, async successor to WSGI. You do not have to implement it; PyFly hands the server a ready-made ASGI `app` object. Just remember: the *server* (Uvicorn or Granian) speaks ASGI to your *app*.

Step 1 — Open the application declaration. Look at `src/lumen/app.py`:

lumen/app.py

```

from pyfly.core import pyfly_application

@pyfly_application(
    name="lumen-demo",
    scan_packages=[
        "lumen_demo.controllers",
        "lumen_demo.services",
        "lumen_demo.repositories",
    ],
)
class Application:
    pass

```

Listing 1.4 — Application declaration (@pyfly_application)

Short as it is, every parameter pulls its weight:

- **name** — the service identity. It appears in every structured log event, in the OpenAPI title, and in health-check payloads — an unambiguous handle when you are reading aggregated logs from a dozen services.
- **scan_packages** — the instruction that drives the whole DI system. PyFly walks every listed module and registers any class decorated with `@service`, `@rest_controller`, `@repository`, or `@configuration` in the `ApplicationContext`. Add a new service class anywhere in those packages and it is wired automatically; no explicit registration code required.

The `Application` class body is intentionally empty. You never instantiate it yourself — `PyFlyApplication` does that during startup: it loads configuration from `pyfly.yaml`, configures structured logging, prints the startup banner, scans packages, initialises the `ApplicationContext`, and logs startup timing.

📘 NEW TERMS: DI AND THE APPLICATIONCONTEXT

Dependency injection (DI) means a class declares what it needs in its constructor and the framework supplies those collaborators, rather than the class building them itself. The *ApplicationContext* (often just "the container") is the registry that holds every wired object — PyFly calls each one a *bean*, the same term Spring uses. `scan_packages` is what populates it.

Step 2 — Open the ASGI entry point. Now look at `src/lumen/main.py`:

lumen/main.py

```

from collections.abc import AsyncIterator
from contextlib import asynccontextmanager

from starlette.applications import Starlette

from pyfly.core import PyFlyApplication
from pyfly.web.adapters.starlette import create_app

from lumen_demo.app import Application

# Bootstrap: load config, scan packages, build DI context
_pyfly = PyFlyApplication(Application)

@asynccontextmanager
async def _lifespan(app: Starlette) → AsyncIterator[None]:
    """Manage application startup and shutdown lifecycle."""
    _pyfly._route_metadata = getattr(
        app.state, "pyfly_route_metadata", []
    )
    _pyfly._docs_enabled = getattr(
        app.state, "pyfly_docs_enabled", False
    )
    _pyfly._host = str(_pyfly.config.get("pyfly.server.host", "0.0.0.0"))
    _pyfly._port = int(_pyfly.config.get("pyfly.server.port", 8080))
    await _pyfly.startup()
    yield
    await _pyfly.shutdown()

app = create_app(
    title="lumen-demo",
    version="0.1.0",
    context=_pyfly.context,
    lifespan=_lifespan,
)

```

Listing 1.5 — ASGI entry point (`main.py`)

This is the file that `pyfly run` (and any ASGI server like Uvicorn) discovers. The module-level `app` object is a Starlette application: `create_app` mounts every `@rest_controller` found in the DI context and wires the lifespan hook so startup and shutdown happen cleanly. The `@pyfly_application` decorator alone does **not** create the ASGI `app` — `main.py` is the explicit bridge between the DI world (`app.py`) and the HTTP world.

WHAT JUST HAPPENED

Two small files split one job in two. `app.py` answers *what is this application and where does its code live* (name, version, scan packages). `main.py` answers *how does a web server reach it* — it bootstraps PyFly into an `ApplicationContext`, then exposes a single ASGI `app` object. When you later add a controller or service, you edit neither file: you just drop the new class into one of the scanned packages and PyFly wires it on the next boot.

Project files

The other two files you will edit most often are `pyproject.toml` and `pyfly.yaml`. Each plays a distinct role: `pyproject.toml` manages dependencies, while `pyfly.yaml` owns every runtime knob.

Step 1 — Read the dependency manifest. `pyproject.toml` records the project's dependencies. Notice the extras:

lumen/pyproject.toml

```

[project]
name = "lumen-demo"
version = "0.1.0"
description = "lumen-demo - built with PyFly"
requires-python = "≥ 3.12"
dependencies = [
    "pyfly[web]",
]

[dependency-groups]
dev = [
    "pytest ≥ 8.0",
    "pytest-asyncio ≥ 0.23",
    "pytest-cov ≥ 5.0",
    "ruff ≥ 0.3",
    "mypy ≥ 1.8",
]

```

Listing 1.6 — pyproject.toml (key sections)

NEW TERM: EXTRAS

An *extra* is an optional bundle of dependencies named in square brackets — `pyfly[web]` means "install PyFly plus its `web` optional group". Extras keep your install lean: you pull in a database driver, a CLI, or a faster server only when you actually use it.

The `web` extra in `pyfly[web]` bundles Uvicorn together with the ASGI adapter. The `cli` extra (add it when you need the `pyfly` command outside of `uv run`) provides the command-line tooling. Other extras — `data-relational`, `granian` — are opt-in; you add them only when a feature needs them. The Lumen wallet sample, for example, declares `pyfly[cli,web,data-relational]` because it connects to SQLite.

Step 2 — Read the runtime configuration. `pyfly.yaml` is where all runtime settings live:

`lumen/pyfly.yaml`

```
pyfly:
  app:
    name: lumen-demo
    version: 0.1.0
    module: lumen_demo.main:app

  server:
    port: 8080
    type: "auto"
    event-loop: "auto"
    workers: 0

  actuator:
    endpoints:
      enabled: true

  admin:
    enabled: true

  logging:
    level:
      root: INFO
```

Listing 1.7 — pyfly.yaml (scaffold default)

The `module` key tells `pyfly run` exactly where the ASGI `app` lives — `lumen_demo.main:app`. The application listens on `pyfly.server.port` (default `8080`); server type, actuator endpoints, and log level all have sensible defaults too, so you override only what you actually need to change.

🍃 SPRING PARITY

`pyfly.server.port` is the direct counterpart of Spring Boot's `server.port`, right down to the `8080` default. The matching environment-variable override is `PYFLY_SERVER_PORT` (Spring's `SERVER_PORT`). Application identity follows the same pattern: `pyfly.app.name` and `pyfly.app.version` mirror `spring.application.name`.

⚠️ RENAMED KEY: USE SERVER.PORT, NOT WEB.PORT

Earlier PyFly releases used `pyfly.web.port` (and the `PYFLY_WEB_PORT` environment variable) for the listen port. Both were **removed** in v26.06.102 in favour of `pyfly.server.port` / `PYFLY_SERVER_PORT`. If you copy an old config and the port seems ignored, this is almost always why — rename the key under `server:` and the override takes effect.

📄 DEFAULT SERVER: GRANIAN VS UVICORN

PyFly's default ASGI server is **Granian**, a high-performance Rust-based server. The scaffold's `pyfly[web]` dependency bundles **Uvicorn** instead (lighter install). The two are interchangeable: pass `--server uvicorn` on the command line, or add `pyfly[granian]` to your dependencies to use Granian. The Lumen wallet sample runs with `--server uvicorn` for this reason.

Your first run

With the project in place, start the server.

Step 1 — Start the server. From the project root, run:

terminal

```
uv run pyfly run --server uvicorn
```

Listing 1.8 — Run the server with uv

❗ WHY `--SERVER UVICORN`

PyFly's default server is Granian, but the `pyfly[web]` extra ships Uvicorn (a lighter install). Passing `--server uvicorn` tells PyFly to use the server you actually have. If you ever see an error about Granian not being installed, this flag is the fix — or add `pyfly[granian]` and drop the flag.

After a moment you will see the PyFly ASCII banner followed by a stream of structured startup events:

```

      -----
----- .-.-/ ----\ | ---.-.
\___ < | |\ _\ |< | |
| |> >__ | | | | |\___ |
| _// ____| | | |____/ ____|
|_| V                \ /

```

```

:: PyFly Framework :: (v26.06.110) (Python 3.13.13)
Copyright 2026 Firefly Software Foundation. | Apache License 2.0
2026-06-07 20:34:32,442 [INFO] pyfly.core: starting_application |
    app=lumen-demo version=0.1.0 python=3.13.13 pid=72300
2026-06-07 20:34:32,442 [INFO] pyfly.core: runtime_environment |
    os=Darwin os_version=25.5.0 arch=arm64 cpus=11
2026-06-07 20:34:32,442 [INFO] pyfly.core: loaded_config |
    source=pyfly-defaults.yaml (framework defaults)
2026-06-07 20:34:32,442 [INFO] pyfly.core: loaded_config |
    source=pyfly.yaml
2026-06-07 20:34:32,442 [INFO] pyfly.core: scanned_package |
    package=lumen-demo.controllers beans_found=1
2026-06-07 20:34:32,580 [INFO] pyfly.core: bean_summary |
    total=127 services=6 repositories=2 controllers=4
2026-06-07 20:34:32,580 [INFO] pyfly.core: mapped_endpoints | count=5
2026-06-07 20:34:32,580 [INFO] pyfly.core: request_mapping |
    method=POST path=/api/v1/wallets handler=open_wallet
2026-06-07 20:34:32,580 [INFO] pyfly.core: api_documentation |
    swagger_ui=http://0.0.0.0:8080/docs
2026-06-07 20:34:32,580 [INFO] pyfly.core: management_server |
    url=http://0.0.0.0:9090 endpoints=actuator + admin
2026-06-07 20:34:32,580 [INFO] pyfly.core: admin_dashboard |
    url=http://0.0.0.0:9090/admin
2026-06-07 20:34:32,581 [INFO] pyfly.core: server_started |
    server=uvicorn host=0.0.0.0 port=8080 workers=1

```


Read these log lines as a story of what the framework did on your behalf. Each event has a name rather than a raw message so log aggregators can filter by key:

1. `starting_application` — PyFly announces itself with the service name and version you declared, giving every log aggregator a clean filter from the very first event.
2. `runtime_environment` — OS, architecture, and CPU count are emitted once on boot, making it easy to correlate behaviour with the environment later.
3. `loaded_config` (x2) — configuration is layered: the framework ships `pyfly-defaults.yaml` with safe production defaults; your `pyfly.yaml` overrides only what you need. Activate a profile (e.g., `--profile prod`) and you will see a third entry.
4. `scanned_package` — the DI container found one `@rest_controller` in the web controllers package. `beans_found` grows as you add services and repositories.
5. `bean_summary` — the total DI context size, including framework-internal beans. Even with 127 beans registered, PyFly boots in well under a second.
6. `management_server` — the actuator endpoints and admin dashboard are served on a *separate* management port (default `9090`), independent of the app's `8080`.
7. `admin_dashboard` — the exact URL of the live admin UI, `http://0.0.0.0:9090/admin`.
8. `server_started` — Uvicorn is accepting connections on the application port.

WHAT JUST HAPPENED

You ran one command and PyFly did the work of an entire boilerplate sprint: it loaded layered configuration, scanned your packages, wired the DI container, mapped your HTTP routes, published interactive API docs, brought up a separate management server with health checks and an admin dashboard, and started accepting requests — all logged as named, machine-filterable events. Leave this server running in its terminal; you will call it from a second terminal in the next steps. Press `Ctrl+C` when you want to stop it.

 SPRING PARITY

Serving actuator and the admin dashboard on a dedicated port mirrors Spring Boot's `management.server.port`. In PyFly the key is `pyfly.management.server.port` (default `9090`). Set it equal to `pyfly.server.port` for single-port behaviour, or set it to `-1` to disable the management endpoints entirely. By default this port is **open and unauthenticated** (also Spring parity) — fine for local development, but in production secure it with `pyfly.management.security.enabled: true`.

Two endpoints are already live before you have written a single line of application logic. Open a *second* terminal (leave the server running in the first) and try the health check.

Step 2 — Check the live health endpoint. It lives on the management port, `9090`:

terminal

```
curl -s localhost:9090/actuator/health
```

Listing 1.9 — Health check (management port 9090)

 NEW TERM: ACTUATOR

The *actuator* is PyFly's built-in operations layer: a small set of HTTP endpoints that report on the running app — `/actuator/health`, `/actuator/info`, and (when you expose them) metrics, environment, and more. It is the same concept, and the same `/actuator` base path, as Spring Boot Actuator. By default only `health` and `info` are exposed over HTTP; widen that with `pyfly.management.endpoints.web.exposure.include`.

You should see a JSON document with a top-level `"status": "UP"` and a `components` map — one entry per health indicator the framework registered:

```
{
  "status": "UP",
  "components": {
    "cache_health_indicator": {
      "status": "UP",
      "details": {"adapter": "InMemoryCache", "latencyMs": 0.01}
    },
    "cqrs_health_indicator": {
      "status": "UP",
      "details": {"command_handlers": 3, "query_handlers": 2}
    },
    "eda_health": {
      "status": "UP",
      "details": {"adapter": "InMemoryEventBus"}
    },
    "db_health_indicator": {
      "status": "UP",
      "details": {"database": "sqlite"}
    }
  }
}
```

Step 3 — Open a wallet. Now call the first business endpoint. Unlike the actuator, this lives on the *application* port, **8080**:

terminal

```
curl -s -X POST localhost:8080/api/v1/wallets \
  -H 'content-type: application/json' \
  -d '{"owner_id":"u-1","currency":"EUR"}'
```

Listing 1.10 — Open a wallet (application port 8080)

The response is the new wallet's identifier (your UUID will differ):

```
{"wallet_id": "wlt-c5bbb2a7-dd49-4321-932e-e4c6bfa5cc2c"}
```

WHAT JUST HAPPENED

That single `curl` travelled through the whole stack the framework built for you: the JSON body was validated against the `OpenWalletRequest` model, the `WalletController` turned it into an `OpenWallet` command, the command bus dispatched it to its handler, a `Wallet` aggregate was created and persisted, and the new id came back as JSON. You will build each of those layers yourself in later chapters — for now, notice that it already works end to end.

Step 4 — Explore the interactive docs. Open `http://localhost:8080/docs` in your browser for the interactive Swagger UI; `http://localhost:8080/redoc` gives you ReDoc; the raw OpenAPI spec is at `http://localhost:8080/openapi.json`. The docs live on the application port alongside your API.

NOTE

All three doc URLs are emitted in the boot log (`api_documentation` entries) so you never have to remember them. The PyFly Admin Dashboard — a live view of beans, endpoints, health indicators, and recent log events — opens at `http://localhost:9090/admin` on the management port (its exact URL is the `admin_dashboard` boot-log line).

Step 5 — Run the generated tests. The scaffold ships a working test suite so you can confirm the toolchain end to end. Install the dev tools, then run `pytest`:

terminal

```
uv sync --group dev
uv run --group dev pytest -q
```

Listing 1.11 — Run the test suite

i EXPECTED OUTPUT

You should see a short run of dots (or **PASSED** lines) and a green summary like **3 passed in 0.42s** — the exact count depends on the archetype. Any green summary means PyFly, your virtual environment, and pytest are all wired correctly. (The Lumen wallet sample you study throughout the book ships a far larger suite, but it runs the same way:

```
uv run --group dev pytest -q .)
```

That completes the loop: install, scaffold, run, call, and test — all before writing a line of your own logic.

✓ What you built

In under five minutes you installed PyFly, scaffolded a production-shaped service, and ran it locally. The table below summarises what Lumen already delivers — entirely from the framework — before you have written a single line of application logic.

Capability	How you got it
Structured JSON logging with runtime metadata	Framework default — emitted on every boot
Interactive API docs (Swagger UI + ReDoc)	Built-in; always on, opt out via <code>pyfly.yaml</code>
Health endpoint (<code>/actuator/health</code> on port 9090)	<code>actuator.endpoints.enabled: true</code> in scaffold
Profile-aware configuration layering	<code>pyfly.yaml</code> + <code>--profile</code> flag
Auto-discovery DI container	<code>scan_packages</code> in <code>@pyfly_application</code>
Startup timing, bean summary, endpoint map	Emitted as structured log events on every boot
Admin Dashboard (<code>/admin</code> on management port 9090)	Enabled in the scaffold by default

That is the PyFly promise: sensible decisions made for you, conventions consistent across every service, and production-ready defaults from the very first run.

Try it yourself

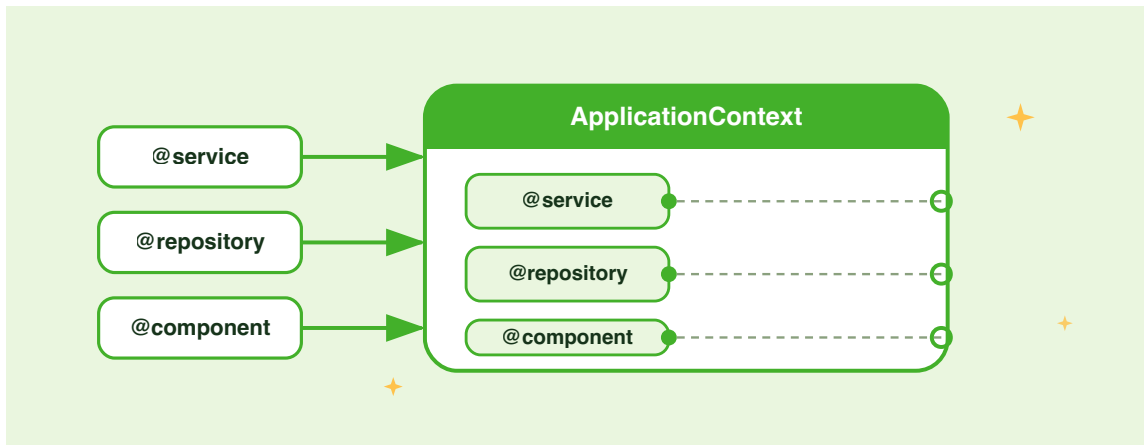
- 1. Rename the service.** Open `pyfly.yaml` and change `app.name` from `lumen-demo` to `lumen-wallet`. Also update the `name` parameter in `@pyfly_application`. Re-run with `uv run pyfly run --server uvicorn` and confirm the new name appears in the `starting_application` log line.
- 2. Explore the live docs.** With the server running, open `http://localhost:8080/docs`. Notice the service name and version at the top of the Swagger UI. Then navigate to `http://localhost:8080/redoc` and compare the two doc renderers.

Check the health response at `http://localhost:9090/actuator/health` (remember: the actuator lives on the management port, not the application port) and note which health indicators are already reporting. Open `http://localhost:9090/admin` to see the same information in the live dashboard.

3. **Switch to Granian.** Add `pyfly[granian]` to the `dependencies` list in `pyproject.toml`, run `uv sync`, then start the server without the `--server uvicorn` flag. Compare the `server_started` log line — the `server` field should now read `granian`.
4. **Map files to responsibilities.** Look at the two entry-point files `app.py` and `main.py`. Write a one-sentence description of what each one is responsible for. Where does the DI container come from? Where does the ASGI `app` object live? Which one would you edit to change the scan packages?
5. **Collapse to a single port.** Add `pyfly.management.server.port: 8080` under the `server`-sibling `management:` section in `pyfly.yaml` (so it equals `pyfly.server.port`). Re-run the server and watch the boot log: the `management_server` line disappears and `http://localhost:8080/actuator/health` now responds on the application port. Then try `pyfly.management.server.port: -1`, re-run, and confirm the actuator and admin routes are gone entirely.

CHAPTER 2

Dependency Injection & the Application Context



In the previous chapter you gave Lumen its application entry point and watched the container start. Now you will declare Lumen's first real components — a `WalletRepository` that subclasses the framework's Spring-Data-style `Repository`, a `WalletEntity` that maps the persistence row, and a CQRS handler that depends on both — and let PyFly wire them together from nothing but type hints. No factories, no manual `new`, no glue code.

This chapter is hands-on. We will build each piece in small, numbered steps, and after every milestone there is a **Run it** checkpoint that shows the exact command to type and the output you should see on screen. If your output matches, you are on track; if it does not, the gap tells you precisely what to fix. You are following along inside `samples/lumen` with PyFly **v26.6.110** installed (`uv sync` from the Lumen directory pulls it in).

i NEW TERM: CONTAINER

Throughout the chapter we talk about *the container*. A **container** is simply the object PyFly creates at startup that knows how to build, connect, and hand out all your application's objects. You never construct it yourself — when you run the app, PyFly stands the container up, fills it with your beans, and keeps it alive until shutdown. Think of it as a smart warehouse that both stores your objects and assembles them on demand.

Before a single line of Lumen code appears, it is worth pausing on *why* that matters. In a conventional Python project you would write something like:

```
handler = DepositFundsHandler(  
    repository=InMemoryWalletRepository(),  
    events=InMemoryEventBus(),  
)
```

somewhere near the startup path. That one line seems harmless, but it locks every decision — which repository class, which event bus — at the point of construction. Swap the repository for a Postgres adapter and you must find every construction site. Add a test double and you need to restructure the wiring. **Dependency injection** inverts this relationship: classes *declare* what they need, and the container *decides* what to provide. The result is code that is open to extension but closed to modification — the `DepositFundsHandler` you write today will accept a production database adapter in Part II without a single change to its source.

Stereotypes: declaring your beans

Before the container can wire anything, it needs to know which classes to manage. A **bean** is any object the container creates, wires, and owns. You make a class visible to the container by applying a **stereotype decorator** — a thin annotation that registers the class and signals its architectural role.

NEW TERMS: BEAN AND STEREOTYPE

A **bean** is just an instance the container owns — it builds it, supplies its dependencies, and (usually) keeps a single shared copy around. A **stereotype** is the decorator you put on a class to say "container, this is yours, please manage it." The word *stereotype* is borrowed from Spring; it simply means "a labelled role." Putting `@service` on a class is the whole act of registration — there is no separate config file to edit.

PyFly ships five stereotypes:

Decorator	Meaning
<code>@service</code>	Business-logic layer: domain operations, use-case orchestration.
<code>@component</code>	Generic managed bean with no specific architectural role.
<code>@repository</code>	Data-access layer: databases, external storage, ports.
<code>@configuration</code>	Configuration class that can contain <code>@bean</code> factory methods.
<code>@rest_controller</code>	HTTP layer: handles requests and returns JSON responses.

All five stereotypes are **container-equivalent**: they share the same internal `_make_stereotype()` factory and accept the same optional keyword arguments (`name`, `scope`, `profile`, `condition`). The meaningful differences are the `__pyfly_stereotype__` label — used by the web layer to discover controllers and by

the context to find `@configuration` classes — and the architectural clarity each name brings to readers of your code. Choosing `@repository` over `@component` costs nothing technically but tells every future reader exactly what the class is for.

Both bare and parenthesised forms work:

```
@service          # bare – all defaults
class SimpleService:
    pass

@service(name="wallet_svc")  # with keyword args
class NamedService:
    pass
```

The scan_packages bootstrap

The container only discovers beans in packages it has been told to scan. In `lumen/app.py`, `@pyfly_application` lists every subpackage the container should introspect for stereotype declarations:

lumen/app.py

```

from pyfly.core import pyfly_application
from pyfly.starters.domain import enable_domain_stack

@EnableDomainStack
@pyfly_application(
    name="lumen",
    version="1.0.0",
    description=(
        "Lumen – a DDD digital-wallet service"
        " built on the PyFly framework."
    ),
    scan_packages=[
        "lumen.models.repositories",
        "lumen.core.services.wallets",
        "lumen.core.services.transfers",
        "lumen.core.services.listeners",
        "lumen.web.controllers",
    ],
)
class LumenApplication:
    pass

```

Listing 2.1 — Application entry point with `scan_packages`

How it works. `@pyfly_application` registers `LumenApplication` as the application root and seeds the container with framework auto-configurations. `scan_packages` is the exact list of Python package paths the container walks at startup, collecting every class decorated with a stereotype. Any package not listed here is invisible to the container — the most common source of "why is my bean not found?" confusion when adding new subpackages. `@enable_domain_stack` activates the CQRS, transactional engine, event sourcing, relational data, and rule-engine auto-configurations in a single line.

Read that listing as four decisions:

- **Step 1 — name the application.** `name="lumen"` and `version="1.0.0"` become the identity the startup banner and the `/actuator/info` endpoint report. (These are the *application's* name and version; the framework version is separate — v26.6.110 here.)
- **Step 2 — describe it.** `description=...` is human-facing metadata surfaced in the generated API docs.
- **Step 3 — list the packages to scan.** Every entry in `scan_packages` is a dotted Python path the container will import and inspect. If a class with a stereotype lives in a package *not* on this list, the container never sees it.
- **Step 4 — enable the stack.** `@enable_domain_stack` switches on the domain-tier auto-configurations so the CQRS buses, the transactional session, and the relational data layer all exist before your beans are wired.

💡 **TIP: SCAN THE PACKAGE, NOT THE CLASS**

`scan_packages` entries are *package* paths (`lumen.web.controllers`), never individual module or class paths. The container walks the package and discovers every stereotype-decorated class inside it. When you add a new handler under `lumen.core.services.wallets` , it is picked up automatically — no edit to `scan_packages` needed. You only touch this list when you introduce a brand-new subpackage.

Run it. From the Lumen project root, boot the application and watch the container assemble itself:

```
cd samples/lumen
uv run pyfly run --server uvicorn
```

You should see the banner followed by structured startup lines — the container reporting exactly what it scanned and wired:

```
:: PyFly Framework :: (v26.6.110) (Python 3.12.13)

pyfly.core: starting_application | app=lumen version=1.0.0
pyfly.core: scanned_package | package=lumen.models.repositories beans_found=1
pyfly.core: scanned_package | package=lumen.core.services.wallets
beans_found=7
pyfly.core: scanned_package | package=lumen.core.services.transfers
beans_found=2
pyfly.core: scanned_package | package=lumen.core.services.listeners
beans_found=1
pyfly.core: scanned_package | package=lumen.web.controllers beans_found=1
pyfly.core: bean_summary | total=137 services=10 repositories=1 controllers=4
configurations=19
pyfly.core: server_started | server=uvicorn host=0.0.0.0 port=8080 workers=1
pyfly.core: application_started | app=lumen startup_time_s=0.143
beans_initialized=137
```

The `scanned_package` lines are `scan_packages` doing its job: one line per entry, each reporting how many beans it found. The final `application_started` line — the equivalent of Spring Boot's "Started Application in N seconds" — is your signal that the context booted cleanly. Press `Ctrl-C` to stop the server.

NEW TERM: THE MANAGEMENT PORT

Two ports appear at startup. Your application's HTTP API listens on `pyfly.server.port` (default **8080**). Operational endpoints — the health check, info, and the admin dashboard — listen separately on the **management port** `pyfly.management.server.port` (default **9090**), which is open and unauthenticated by default. You will not touch either in this chapter, but it is worth knowing why two ports light up. (The older `pyfly.web.port` key was removed in v26.6.102; always use `pyfly.server.port` now.)

What just happened. A single command did a lot. PyFly loaded `pyfly.yaml`, imported each package in `scan_packages`, found every stereotype-decorated class, asked the container to build them in dependency order, and reported the totals — 137 beans, of which most are framework auto-configuration beans and only a handful are yours so far. From this point on, the rest of the chapter is about *adding* to that bean count: an entity, a repository, and a command handler, each discovered by exactly this mechanism.

🌿 SPRING PARITY

`scan_packages` is the equivalent of Spring's `@ComponentScan(basePackages = {...})`. The semantics are identical: list every subpackage you want the framework to introspect, and it will register everything it finds. The `application_started` log line mirrors Spring Boot's "Started Application in N seconds (process running for M)" startup summary.

The entity and the repository

Lumen stores wallets in a relational database. Two classes carry this responsibility: `WalletEntity` (the persistence row) and `WalletRepository` (the data-access bean). We will build them in order: first the row shape, then the data-access bean that reads and writes it.

📄 NEW TERM: ENTITY

An **entity** is the in-database shape of one record — here, one wallet, stored as one row in a `wallets` table. It is deliberately plain: just typed columns, no behaviour. (Lumen keeps the *rich* domain object, the `Wallet` aggregate, separate; you will meet it in Chapter 6. For now, the entity is simply how a wallet is written to and read from the database.)

The entity. `WalletEntity` is a plain SQLAlchemy-mapped class that inherits the framework's `Base`. Build it field by field:

- **Step 1 — inherit `Base`.** Subclassing PyFly's declarative `Base` is what enrolls the class in the ORM's metadata so the framework can create its table.
- **Step 2 — name the table.** `__tablename__ = "wallets"` is the SQL table this class maps to.
- **Step 3 — declare the primary key.** `id` is a `str` column marked `primary_key=True` — the wallet keeps its own domain id (`wlt-...`) rather than a generated number.
- **Step 4 — declare the remaining columns.** `owner_id`, `currency`, `balance_minor` (the balance in minor units — cents — so there is never a floating-point rounding bug), and a `created_at` timestamp.

lumen/models/entities/v1/wallet_orm.py

```

from __future__ import annotations

from datetime import UTC, datetime

from sqlalchemy import String
from sqlalchemy.orm import Mapped, mapped_column

from pyfly.data.relational.sqlalchemy import Base

class WalletEntity(Base):
    """One persisted wallet row, keyed by the aggregate's own id."""

    __tablename__ = "wallets"

    id: Mapped[str] = mapped_column(
        String(64), primary_key=True
    )
    owner_id: Mapped[str] = mapped_column(
        String(255), nullable=False, index=True
    )
    currency: Mapped[str] = mapped_column(String(3), nullable=False)
    balance_minor: Mapped[int] = mapped_column(
        nullable=False, default=0
    )
    created_at: Mapped[datetime] = mapped_column(
        default=lambda: datetime.now(UTC)
    )

```

Listing 2.2a — *WalletEntity*: the persistence row

Inheriting `Base` (PyFly's declarative base) registers the `wallets` table in `Base.metadata`; the framework's engine lifecycle creates it on startup. No further wiring is needed.

Run it. With `ddl-auto: create` set in `pyfly.yaml`, the framework builds the schema from your mapped entities the moment the app boots. Start the app again and look for the data-layer lines:

```
uv run pyfly run --server uvicorn
```

```
pyfly.data.relational.auto_configuration: Initializing database schema (ddl-
auto=create)
pyfly.data.relational.auto_configuration: Database schema initialized (1
tables)
```

`1 tables` is your `wallets` table — created purely because `WalletEntity` inherits `Base`. There is no migration script to run and no `CREATE TABLE` to write by hand. (Lumen uses SQLite by default, so the database is just a `lumen.db` file in the project directory.)

NEW TERM: REPOSITORY

A **repository** is the object your code talks to when it wants to load or save entities. Instead of writing SQL, you call methods like `find_by_id` or `save`. PyFly's `Repository` base class *generates* those methods for you from the entity and key types, so the repository you write is mostly empty — you declare it, and the framework fills in the implementation. (CRUD, used below, just stands for Create, Read, Update, Delete — the four basic data operations.)

The repository. `WalletRepository` subclasses the framework's generic `Repository[WalletEntity, str]`. The two type arguments tell the framework the *entity type* (`WalletEntity`) and the *primary-key type* (`str`); from that it generates and injects a full async CRUD surface — `find_by_id`, `save`, `find_all`, `find_all(pageable)`, `delete`, `delete_by_id`, `count`, and more — backed by the transactional `AsyncSession` from the relational auto-configuration:

```
lumen/models/repositories/wallet_repository.py
```

```
from __future__ import annotations

from lumen.models.entities.v1.wallet_orm import WalletEntity
from pyfly.container import repository
from pyfly.data import Page, Pageable
from pyfly.data.relational.sqlalchemy import (
    Repository,
    Specification,
)

def balance_at_least(min_minor: int) → Specification[WalletEntity]:
    """Reusable predicate: wallets with balance ≥ min_minor."""
    return Specification(
        lambda root, q: q.where(root.balance_minor ≥ min_minor)
    )

@repository
class WalletRepository(Repository[WalletEntity, str]):
    """CRUD + derived + specification queries for WalletEntity."""

    # Derived query – compiled from the name by the post-processor
    async def find_by_owner_id(
        self, owner_id: str
    ) → list[WalletEntity]:
        """All wallets owned by owner_id (derived query stub)."""
        ...

    # Specification query – composable predicate + pagination
    async def find_rich(
        self, min_minor: int, pageable: Pageable
    ) → Page[WalletEntity]:
        """Page of wallets with balance ≥ min_minor."""
        return await self.find_all_by_spec_paged(
            balance_at_least(min_minor), pageable
        )
```

```
# Upsert: one call for INSERT or UPDATE
async def upsert(self, entity: WalletEntity) → WalletEntity:
    """Persist entity whether the row is new or already exists."""
    session = self._require_session()
    merged = await session.merge(entity)
    await session.flush()
    return merged
```

Listing 2.2b — `WalletRepository`: framework `Repository` subclass

How it works. `@repository` tells the container to manage `WalletRepository` as a DI bean. The framework reads `Repository[WalletEntity, str]` at startup, generates the CRUD implementation internally, and registers the class — you inject `WalletRepository` directly by type anywhere in the application. There is no hand-written port interface and no separate adapter to maintain: **the framework supplies and injects the implementation; you depend on the repository class itself by type.**

The three extra methods show the extension points the framework exposes on top of the inherited CRUD. Look at them one at a time:

- **Step 1 — a derived query.** `find_by_owner_id` is a **derived query**: the `RepositoryBeanPostProcessor` (a startup component that edits beans after they are built) parses the method *name* and compiles a real `SELECT ... WHERE owner_id = :owner_id`. You write only the stub body (`...`); the framework supplies the SQL. The naming convention is the API — `find_by_<column>` becomes `WHERE <column> = ?`.
- **Step 2 — a Specification query.** `find_rich` composes a reusable `Specification` predicate (here, `balance_at_least`) and runs it with pagination and sorting via the inherited `find_all_by_spec_paged`. Specifications are how you build a composable, type-safe `WHERE` clause when a method name would get unwieldy.

- **Step 3 — an upsert.** `upsert` is a thin convenience over `session.merge` so a command handler can persist an entity whether it is new (INSERT) or already exists (UPDATE) with a single call. Because the wallet owns its own id, both cases key on the same primary key.

What just happened. You declared a repository whose body is almost entirely empty, yet it now exposes a complete async CRUD surface plus one derived query, one specification query, and an upsert. The `@repository` stereotype registered it as a bean; the framework read the `Repository[WalletEntity, str]` base, generated the implementation, and made it injectable by type. You wrote intent; PyFly wrote the plumbing.

💡 **TIP: CONFIRM THE REPOSITORY IS REGISTERED**

Re-run `uv run pyfly run --server uvicorn` and look at the `bean_summary` line: `repositories=1`. That single registered repository is your `WalletRepository`. If you ever add a second repository and it does not appear in this count, the usual cause is that its package is missing from `scan_packages`.

🍃 **SPRING PARITY**

`@service`, `@component`, `@repository`, and `@configuration` map directly to Spring's `@Service`, `@Component`, `@Repository`, and `@Configuration`. `@rest_controller` mirrors `@RestController`. `Repository[E, ID]` mirrors Spring Data's `JpaRepository<E, ID>`: declare the entity and key types; the framework generates and injects the full implementation. Derived query methods (names like `find_by_owner_id`) compile to SQL at startup — the same mechanism as Spring Data's query derivation from method names.

Constructor injection

With the repository declared, you need a handler that uses it. That is where the container's most important capability becomes visible: you never call constructors yourself. You declare what a class *needs* as `__init__` parameters with type annotations, and the container fills them in automatically. This is **constructor injection**, and it is the recommended approach for all mandatory dependencies.

NEW TERM: INJECTION

Injection is the act of the container *handing* a class the objects it depends on, rather than the class building them itself. With *constructor* injection, you simply list the dependencies as typed `__init__` parameters; the container reads those type hints and passes in matching beans when it builds the object. The class never says *how* to obtain its dependencies — only *what* it needs.

The mental model is a simple wishlist: list the types you need; the container delivers the right instances. If a dependency does not exist at startup, you get a clear `NoSuchBeanError` immediately — not a cryptic `AttributeError` three call frames deep at runtime.

Stacking handler decorators on `@service`

In Lumen's CQRS design, every write-side handler carries two decorators:

`@command_handler` (or `@query_handler`) **stacked on** `@service`. The pattern is non-negotiable: `@service` registers the class as a bean; the CQRS decorator adds only routing metadata (`__pyfly_command_type__` or `__pyfly_query_type__`) so the command/query bus can dispatch to the right handler. Without `@service`, the container never sees the class and the command bus raises `CommandHandlerNotFoundException` at dispatch time (the query bus raises `QueryHandlerNotFoundException` for a missing query handler).

Before reading the listing, here is the shape of what you are about to write, step by step:

- **Step 1 — register the bean.** Put `@service` directly on the class. This is the line that makes the container manage it.
- **Step 2 — add routing metadata.** Stack `@command_handler` above `@service`. It reads `CommandHandler[DepositFunds, int]` and records "this bean handles `DepositFunds` commands."
- **Step 3 — declare dependencies in `__init__`.** List the repository, the event publisher, and the session factory as typed parameters. This single signature is the complete wiring specification — the container reads it and supplies all three.
- **Step 4 — write the business logic in `do_handle`.** Wrap it in `@transactional()` so the whole load-mutate-save sequence is one committed unit of work.

The `DepositFundsHandler` shows the pattern in full:

```
lumen/core/services/wallets/deposit_funds_handler.py
```

```
from __future__ import annotations

from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker

from lumen.core.mappers.wallet_mapper import to_aggregate, to_entity
from lumen.core.services.wallets.deposit_funds_command import (
    DepositFunds,
)
from lumen.core.services.wallets.event_publishing import (
    publish_domain_events,
)
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.data.relational.sqlalchemy import transactional
from pyfly.domain import AggregateNotFound
from pyfly.eda import EventPublisher

@command_handler
@service
class DepositFundsHandler(CommandHandler[DepositFunds, int]):
    """Credit funds to an existing wallet; returns new balance."""

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events
        self._session_factory = session_factory

    @transactional()
```



```

async def do_handle( # type: ignore[override]
    self, command: DepositFunds
) → int:
    entity = await self._repository.find_by_id(
        command.wallet_id
    )
    if entity is None:
        raise AggregateNotFound("Wallet", command.wallet_id)

    wallet = to_aggregate(entity)
    wallet.deposit(
        Money(amount=command.amount, currency=wallet.currency)
    )
    await self._repository.upsert(to_entity(wallet))

    await publish_domain_events(
        self._events, wallet.clear_events()
    )
    return wallet.balance.amount

```

Listing 2.3 — *DepositFundsHandler*: *@command_handler* + *@service* stacked

How it works. Five decisions are visible in this listing:

- `@service` registers the class as a singleton bean. Without it, the container never sees the class.
- `@command_handler` (applied above `@service`, so it runs *after* registration) reads the first generic argument of `CommandHandler[DepositFunds, int]` and records that this bean handles `DepositFunds` commands.
- The `__init__` signature is the complete wiring specification: `repository: WalletRepository` — the framework-generated CRUD bean; `events: EventPublisher` — resolved by the CQRS auto-configuration; `session_factory: async_sessionmaker[AsyncSession]` — the shared connection factory provided by

the relational auto-configuration. All three are resolved by type;

`DepositFundsHandler` never imports a concrete class.

- `@transactional()` on `do_handle` wraps the entire body in a single committed unit of work. The decorator opens a session from `session_factory`, binds it to the repository for the duration of the call, and commits on success (or rolls back on error).
- The business logic follows the standard CQRS/DDD sequence: load the entity, rehydrate the aggregate via the mapper, mutate through domain methods that enforce invariants, persist via `upsert`, drain and publish the events. The wallet is saved *before* events are published, so any listener that queries the repository finds the updated record.

A read-side handler uses the same stacking pattern, only with `@query_handler` and `QueryHandler`:

```
from pyfly.container import service
from pyfly.cqrs import QueryHandler, query_handler
from lumen.models.repositories.wallet_repository import WalletRepository

@query_handler
@service
class GetWalletHandler(QueryHandler[GetWallet, WalletDto | None]):
    def __init__(self, repository: WalletRepository) → None:
        super().__init__()
        self._repository = repository

    async def do_handle(
        self, query: GetWallet
    ) → WalletDto | None:
        entity = await self._repository.find_by_id(query.wallet_id)
        return entity_to_dto(entity) if entity is not None else None
```

The container resolves dependencies **recursively**. When it constructs `DepositFundsHandler` it also constructs `WalletRepository` (the framework-generated CRUD bean), the `EventPublisher`, and the `async_sessionmaker` — none of which the handler needs to know about.

What just happened. You declared a handler that needs three collaborators and wrote not one line of wiring code. The container read the `__init__` type hints, built each dependency (and *their* dependencies, recursively), and handed the finished object to whoever asks for `DepositFundsHandler`. Swapping the in-memory event bus for Kafka later will not touch this class at all — it asks for an `EventPublisher` and is content with whatever the container provides.

Run it. The surest way to confirm the whole graph wires together is the integration test that boots the *real* application context and drives a deposit through the command bus. From the Lumen project root:

```
uv run --extra dev pytest tests/test_app_context_integration.py -q
```

```
.  
[100%]  
1 passed in 0.19s
```

That single dot is the container proving itself: it scanned the packages, generated the repository, resolved the `EventPublisher` and session factory, constructed `DepositFundsHandler` with all three injected, and ran an `open → deposit → withdraw → reload` lifecycle end to end. If a dependency were missing, this test would fail at *startup* with a `NoSuchBeanError` long before any assertion ran — which is exactly the early, loud failure the container is designed to give you.

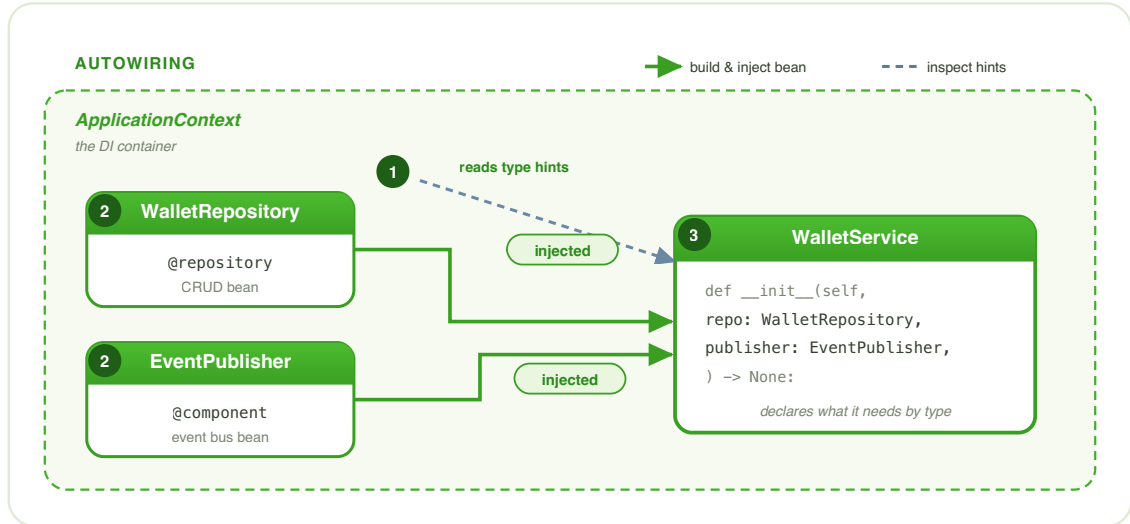


Figure 2.1 — The container injects dependencies from type hints.

🌿 SPRING PARITY

Constructor injection in PyFly is functionally identical to Spring's `@Autowired` constructor injection. In modern Spring you do not even write `@Autowired` — the framework infers injection from the single constructor, just as PyFly reads `__init__` type hints. The mental model is the same: declare what you need, let the container provide it.

💡 TIP

Prefer constructor injection for mandatory dependencies. It makes them visible in the class signature, lets you write plain-Python unit tests without a container (`handler = DepositFundsHandler(repo=MockRepo(), events=MockBus())`), and prevents accidental missing-dependency bugs at startup rather than at runtime.

The Container and the ApplicationContext

PyFly's DI system has two layers, and understanding the boundary between them will save you real debugging time. One layer handles object graphs; the other handles the full application lifecycle. Conflating them is a common source of confusion.

Container (from `pyfly.container`) is the low-level DI engine. It stores **Registration** objects, resolves types by constructor hints, manages scopes, applies **@primary** disambiguation, and handles **Qualifier**-based named lookups. It has no lifecycle awareness — it is a pure "give me a **T**" machine.

ApplicationContext (from `pyfly.context`) is the high-level orchestrator. It wraps **Container** and adds the full startup sequence: profile filtering, condition evaluation, **@configuration** / **@bean** processing, **BeanPostProcessor** weaving, **@post_construct** / **@pre_destroy** hooks, event publishing, and auto-configuration. You interact with the **ApplicationContext** in application code; the raw **Container** is an implementation detail, accessible via `ctx.container` as an escape hatch.

Think of it this way: **Container** is the factory floor — it knows how to build things. **ApplicationContext** is the production manager — it decides what gets built, in what order, and what happens when the factory opens or closes.

Resolution rules

When the container needs to resolve a type **T**, it applies four rules in strict priority order:

1. **Direct registration** — if **T** is registered directly, resolve it.
2. **Interface binding** — if **T** is a **Protocol** or ABC with exactly one bound implementation, resolve that implementation.

3. **@primary disambiguation** — if multiple implementations are bound, the one decorated with `@primary` wins.
4. **Error** — `NoSuchBeanError` when nothing matches; `NoUniqueBeanError` when multiple candidates exist with no `@primary`.

Step 4 is deliberately loud. A missing or ambiguous dependency is a configuration error, and surfacing it at startup rather than burying it in a runtime traceback is one of the container's key guarantees.

@primary

`@primary` resolves ambiguity when several beans satisfy the same interface. Place it on the implementation you want as the default. This arises whenever you have a `@runtime_checkable Protocol` port with more than one registered adapter — a common pattern for swappable infrastructure (cache store, message bus, notification channel).

For example, suppose your application defines a `CacheStore` protocol and ships two adapters — an in-process one for local development and a Redis one for production:

```

from pyfly.container import repository, primary

@primary
@Repository
class InMemoryCacheStore(CacheStore):
    """Default: active in development and tests."""
    ...

@Repository
class RedisCacheStore(CacheStore):
    """Production cache – activated by profile or condition."""
    ...

```

Without `@primary`, resolving `CacheStore` with two registered implementations raises:

```

NoUniqueBeanError: Multiple beans of type 'CacheStore' found
but none is marked @primary
Candidates: ['InMemoryCacheStore', 'RedisCacheStore']

```

The message names every competing candidate so you can make a deliberate decision rather than guessing which one the container would have picked. Moving `@primary` from one adapter to the other is the only change needed to switch the application's backing store — nothing in the service code changes.

Note that Lumen's own `WalletRepository` is a framework `Repository` subclass, so only one bean is registered and no `@primary` is needed. `@primary` is relevant whenever you hand-roll a port/adaptor pair with multiple adapter implementations.

@order

The container initialises singleton beans eagerly during startup, but some beans genuinely must be ready before others — a security filter that must wrap every inbound request, or a schema migrator that must run before any repository is touched. `@order` gives you explicit control over initialisation sequence.

Lower values are resolved first during the eager startup pass. The constants `HIGHEST_PRECEDENCE` (`-(2**31)`) and `LOWEST_PRECEDENCE` (`2**31 - 1`) mark the extremes:

```
from pyfly.container import order, HIGHEST_PRECEDENCE, service

@order(HIGHEST_PRECEDENCE)
@service
class SecurityInitializer:
    """Must be ready before any other service."""
    ...
```

`@order` affects singleton resolution during startup, the sequence in which `BeanPostProcessor` instances run, and the ordering of `get_beans_of_type()` results.

Qualifier — named bean resolution

Type-based injection covers most scenarios. Occasionally, though, you genuinely need a particular *instance* rather than any satisfying implementation — the classic case being a `@configuration` class that produces two beans of the same type (say, a primary and a read-replica database connection) where a downstream service must receive a specific one.

Select a specific bean by name with `Annotated[T, Qualifier("name")]`:


```

from typing import Annotated
from pyfly.container import Qualifier, service

@service
class ReportService:
    def __init__(
        self,
        db: Annotated[object, Qualifier("analytics_db")],
    ) → None:
        self.db = db  # receives the bean named "analytics_db"

```

The container calls `resolve_by_name("analytics_db", expected_type=T)` and verifies assignability — a mistyped name pointing at the wrong type raises `NoSuchBeanError` with a clear message rather than silently injecting the wrong object.

Bean factories: @configuration and @bean

Stereotype decorators work beautifully for classes you own, but not every dependency is a class you control. Third-party clients need constructor arguments known only at runtime; related beans share configuration state; some families of beans are most clearly expressed as a single factory. For all of these situations, PyFly provides the `@configuration` / `@bean` pattern — explicit factory code that still participates fully in the container's resolution and lifecycle machinery.

i NEW TERMS: @CONFIGURATION AND @BEAN

A `@configuration` class is a place to put **factory methods**. A `@bean` method is one such factory: the container *calls* it during startup and registers whatever it returns as a bean. You reach for this pattern when a dependency cannot simply be stereotyped — for example a third-party object you do not own, or one that needs custom construction. The method's **return type annotation** is what the container uses to register the result, so it is mandatory.

A `@configuration` class acts as a factory. Its `@bean` methods are called during the startup sequence, and each method's return value is registered as a bean whose type comes from the method's return annotation. Read the listing below in two steps:

- **Step 1 — mark the class `@configuration`.** This tells the context to scan it for `@bean` methods before any stereotype beans are built.
- **Step 2 — write a `@bean` method with a return annotation.**
`event_publisher(self) → EventPublisher` constructs an `InMemoryEventBus` and — because the annotation says `EventPublisher` — registers it as an `EventPublisher`. Anything that asks for an `EventPublisher` now receives this instance.

lumen/infra_config.py

```

from pyfly.container import configuration, bean
from pyfly.eda import EventPublisher, InMemoryEventBus

@configuration
class LumenInfraConfig:
    """Wires infrastructure beans that require explicit construction."""

    @bean
    def event_publisher(self) → EventPublisher:
        """In-memory event bus – replace with Kafka adapter in production."""
        return InMemoryEventBus()

```

Listing 2.4 — Producing an EventPublisher bean via @configuration

How it works. `@configuration` tells the context to scan `LumenInfraConfig` for `@bean` methods during startup, before any stereotype beans are constructed. The return annotation `EventPublisher` is the key: the context reads it and registers the produced `InMemoryEventBus` instance as an `EventPublisher`, not as an `InMemoryEventBus`. That distinction matters — when `DepositFundsHandler` later asks for an `EventPublisher`, it receives the `InMemoryEventBus` instance without knowing or caring about the concrete type.

Swapping to a Kafka adapter for production means replacing `InMemoryEventBus()` with `KafkaEventPublisher(settings.kafka_url)` in a single method. The rest of the codebase is untouched.

NOTE: HOW LUMEN ACTUALLY GETS ITS EVENTPUBLISHER

The listing above shows the `@configuration` / `@bean` pattern you would write to hand-build a bean. Lumen itself does *not* need this for its event bus: setting `eda.provider: memory` in `pyfly.yaml` asks the framework's EDA auto-configuration to register an `EventPublisher` bean for you (the same `InMemoryEventBus` you see in the `events_is InMemoryEventBus` wiring at startup). That is why `DepositFundsHandler` can simply ask for an `EventPublisher` — the auto-configuration already supplied one. Reach for `@bean` when you need a bean the framework does *not* provide out of the box.

`@bean` methods can also declare parameters; the container resolves them automatically:

```
@configuration
class MessagingConfig:

    @bean
    def audited_publisher(self, base: EventPublisher) → EventPublisher:
        """Wrap the base publisher with audit logging."""
        return AuditingEventPublisher(base)
```

@bean parameters

Parameter	Default	Description
<code>name</code>	method name	Bean name for named resolution.
<code>scope</code>	<code>Scope.SINGLETON</code>	Lifecycle scope of the produced bean.
<code>primary</code>	<code>False</code>	Mark this the primary candidate for its interface.
<code>profile</code>	<code>""</code>	Only create the bean when the profile expression matches.

NOTE

The return type annotation on a `@bean` method is **mandatory**. The context reads it to know which interface type to register the produced bean under. Omitting it will cause the bean to be unreachable by type.

Scopes

Every bean has a **scope** that controls how long its instance lives. Getting scope right is less about performance and more about correctness: sharing a stateful object designed for single-use produces race conditions; re-creating a singleton on every resolution wastes resources and defeats caching. The `Scope` enum defines three values that cover the vast majority of real-world needs.

`Scope.SINGLETON` (default) — one instance is created on first resolution and reused for the life of the application. Singletons are instantiated eagerly during `ApplicationContext.start()`, sorted by `@order`. Almost all application beans should be singletons.

`Scope.TRANSIENT` — a fresh instance is created on every resolution. Use this for stateful, non-shareable objects:

lumen/contexts.py

```

from pyfly.container import component, Scope

@Component(scope=Scope.TRANSIENT)
class TransferContext:
    """Carries state for a single wallet transfer operation."""

    def __init__(self) → None:
        self.steps: list[str] = []
        self.rolled_back: bool = False

```

Listing 2.5 — A transient bean for per-operation context

How it works. `TransferContext` accumulates the steps of a multi-hop transfer so that a saga can roll them back in reverse order if anything fails. Sharing a single instance across concurrent requests would blend their state; `Scope.TRANSIENT` ensures every resolution produces a fresh, empty `TransferContext`. The container still manages the class — injecting it, profiling it, post-processing it — but never caches the result.

`Scope.REQUEST` — scoped to a single HTTP request. A new instance is created when a request arrives and discarded when it completes. Use this for web-layer beans that carry request-specific state, such as the current authenticated user.

```

from pyfly.container import component, Scope

@Component(scope=Scope.REQUEST)
class CurrentUser:
    user_id: str = ""
    roles: list[str] = []

```

A quick rule of thumb:

- **SINGLETON** — the bean is stateless, or its state is safe to share across all callers (connection pools, caches, service objects).
 - **TRANSIENT** — the bean accumulates per-operation state that must not bleed between operations (sagas, builders, context carriers).
 - **REQUEST** — the bean carries per-HTTP-request state that must be isolated between concurrent requests (authenticated user, request-scoped trace ID).
-

Lifecycle and conditions

Construction and wiring are only half the story. Real infrastructure beans need to *act* after they are built — reserving a thread pool, pre-loading a cache, subscribing to a message queue — and they need to *undo* those actions cleanly on shutdown. PyFly gives you two lifecycle hooks for this, plus a family of conditional decorators that control whether a bean participates in the container at all.

@post_construct and @pre_destroy

Once the container constructs a bean and injects all its dependencies, you often need one-time initialisation — opening a connection pool, warming a cache, registering a listener. Mark a method `@post_construct` and the context calls it after construction completes. Both synchronous and `async` methods are supported:

lumen/wallet_audit_listener.py

```

from pyfly.container import service
from pyfly.context import post_construct, pre_destroy
import logging

logger = logging.getLogger(__name__)

@service
class WalletAuditListenerWithLifecycle:
    def __init__(self) → None:
        self._entries: list[dict] = []

    @post_construct
    async def on_start(self) → None:
        logger.info("wallet_audit_listener_ready")

    @pre_destroy
    async def on_stop(self) → None:
        logger.info("wallet_audit_listener_shutting_down")

```

Listing 2.6 — Lifecycle hooks on a @service bean

How it works. `on_start` fires *after* the constructor returns and all injected dependencies are set — making it safe to issue repository queries, open connections, or publish an application event. The `async` keyword works without any extra setup: the context calls `await on_start()` when it detects a coroutine, and falls back to a direct call for synchronous methods.

`@pre_destroy` is the counterpart, called during `ApplicationContext.stop()` before the bean is discarded. Beans are destroyed in **reverse** initialisation order, so a listener started after the repository is stopped before it.

Run it. Add the bean above to a scanned package, then start and stop the app to watch both hooks fire. The `@post_construct` line appears during the startup pass; pressing `Ctrl-C` triggers the `@pre_destroy` line:

```
uv run pyfly run --server uvicorn
```

```
... wallet_audit_listener_ready          # @post_construct, during startup
^C
... shutting_down | app=lumen
... wallet_audit_listener_shutting_down  # @pre_destroy, during shutdown
```

Seeing `..._ready` before `application_started` confirms the hook runs as part of the eager startup pass; seeing `..._shutting_down` after the `Ctrl-C` confirms the symmetric teardown.

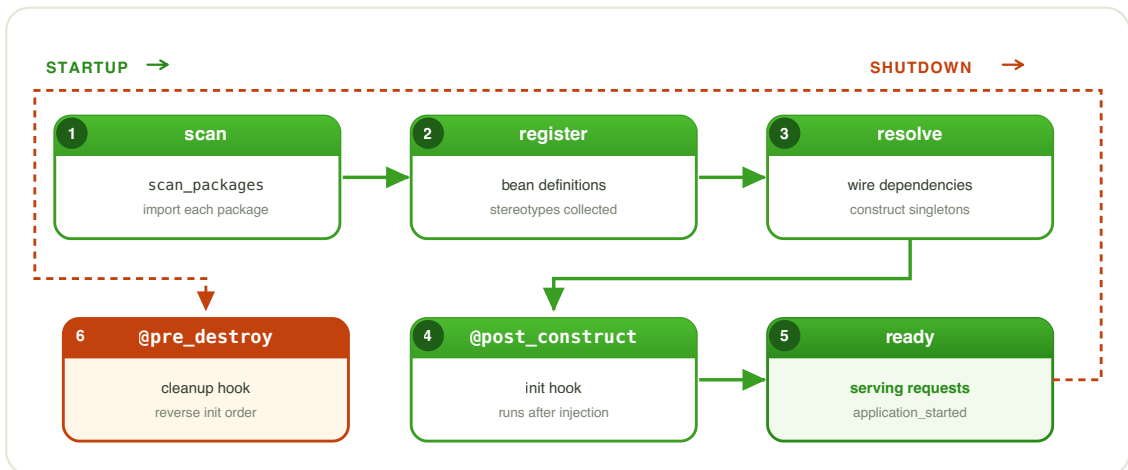


Figure 2.2 — A bean's lifecycle.

Conditional beans

Conditions answer a powerful question: *should this bean exist at all, given the current environment?* They are how the same codebase works in development (cheap in-memory adapters), in CI (Testcontainers), and in production (real infrastructure) — without a single `if` statement in your service code.

Conditional decorators are evaluated in a two-pass strategy during

`ApplicationContext.start()`:

Pass 1 (before user `@configuration` is processed) evaluates: -

`@conditional_on_property(key, having_value="...")` — the config key must exist and optionally match a value. - `@conditional_on_class("module.name")` — the Python module must be importable. - The `condition` callable on a stereotype decorator.

Pass 2 (after user `@configuration` is processed) evaluates: -

`@conditional_on_bean(SomeType)` — only register if another bean of that type already exists. - `@conditional_on_missing_bean(SomeType)` — only register if no bean of that type exists yet.

The two-pass design is deliberate. Pass 1 conditions depend on external facts — configuration files and installed packages — that are knowable before any beans are constructed. Pass 2 conditions depend on *which beans got registered*, information available only after Pass 1 settles. Processing them in order ensures each condition evaluates against a stable, predictable view of the container.

The most powerful pattern is **"default with override"** — ship a fallback that automatically yields to any user-provided implementation:

lumen/notifications.py

```

from pyfly.container import service
from pyfly.context import conditional_on_missing_bean, conditional_on_property
import logging

logger = logging.getLogger(__name__)

class NotificationPort:
    async def send(self, owner_id: str, message: str) → None:
        ...

@conditional_on_property("lumen.smtp.host")
@service
class SmtplibNotificationAdapter:
    """Real email sender – only active when SMTP is configured."""

    async def send(self, owner_id: str, message: str) → None:
        logger.info("smtp_send owner=%s", owner_id)

@conditional_on_missing_bean(NotificationPort)
@service
class LoggingNotificationFallback:
    """Log-only fallback – active whenever no real sender is wired."""

    async def send(self, owner_id: str, message: str) → None:
        logger.info("notification_fallback owner=%s", owner_id)

```

Listing 2.7 — Default-with-override using @conditional_on_missing_bean

How it works. Read the two classes as a chain of intent. `SmtplibNotificationAdapter` activates only when `lumen.smtp.host` is present in configuration, keeping development environments free of half-configured mail clients.

`LoggingNotificationFallback` activates whenever no real `NotificationPort` is registered — in practice, any environment where SMTP is not configured. The fallback does not check *why* the real adapter is absent; it simply fills the gap.

Any handler that injects `NotificationPort` therefore always receives *something* — no `NoSuchBeanError`, no `None` guard. In development and CI you get structured log output; in production you get real email. The choice is made entirely in configuration, with no code change and no branch in service logic.

💡 TIP

The `@conditional_on_missing_bean` / `@conditional_on_property` pair is how all of PyFly's own auto-configuration works. Every subsystem (cache, messaging, HTTP client) ships a default bean that backs off automatically the moment you register your own implementation.

✓ What you built

Lumen now has a `WalletEntity` mapped to the `wallets` table, a `WalletRepository` that subclasses the framework's `Repository[WalletEntity, str]` (giving a full async CRUD surface, a derived query, and a specification query), and a `DepositFundsHandler` wired to the repository, the event publisher, and the session factory — all by type hints alone. You saw why `@command_handler` and `@query_handler` **must be stacked on `@service`** — the CQRS decorators add routing metadata, but `@service` is what registers the bean. You saw that the framework auto-generates and injects the `Repository` implementation so you depend on the repository class itself by type, with no hand-written port/adaptor pair required. You also saw how `@primary` resolves ambiguity when two adapters compete for a hand-rolled port, how

`@post_construct` / `@pre_destroy` bracket a bean's life, and how `@conditional_on_missing_bean` enables defaults that automatically yield to real implementations.

The through-line is consistent: you declare intent with decorators and type hints; PyFly provides the instances. That separation lets you test each class in isolation, swap adapters without touching business logic, and drive the full application configuration from a YAML file — all of which become essential as Lumen grows through the rest of the book.

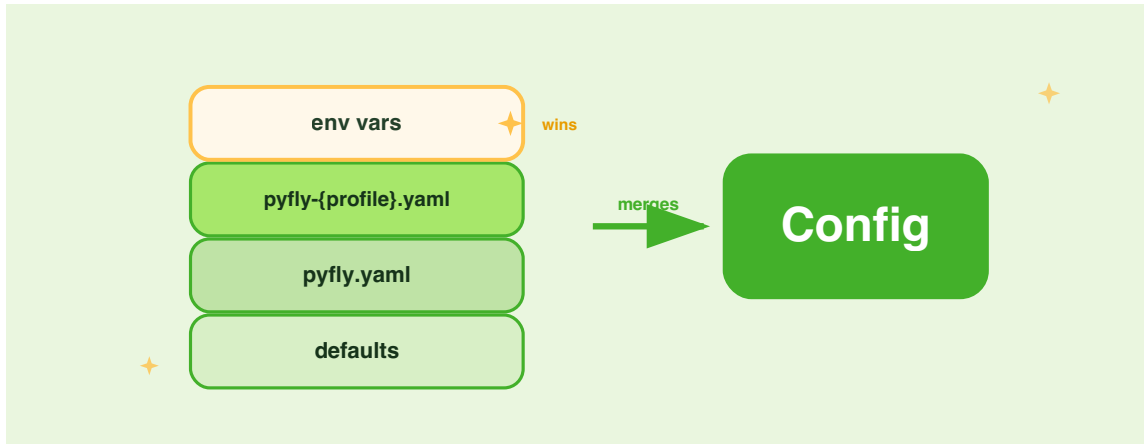
Try it yourself

1. **Practice `@primary` with a hand-rolled port.** Define a `CacheStore` protocol with a single `async def get(key)` method. Register two `@repository` adapters — an in-process one and a stub "remote" one — both inheriting `CacheStore`. Start the application and observe the `NoUniqueBeanError`. Then add `@primary` to the in-process adapter and watch startup succeed. Next, try injecting the stub by name: annotate a constructor parameter as `Annotated[CacheStore, Qualifier("remote_cache")]` after registering it with `@repository(name="remote_cache")`.
2. **Add a `@post_construct` that logs startup metadata.** Extend `WalletAuditListener` with an `async def on_ready(self)` method decorated with `@post_construct`. Inside it, log the class name of any injected dependency. Run `pyfly run --reload`, start the server, and confirm the log line appears after the framework's own startup messages.

3. Make a bean conditional on a property. Add a `WalletAuditService` decorated with `@conditional_on_property("lumen.audit.enabled", having_value="true")`. Open `pyfly.yaml` and omit the key. Verify the service is absent from the bean list at startup. Then add `lumen.audit.enabled: "true"` to `pyfly.yaml` and re-run — confirm it appears. This is exactly how you gate optional subsystems without touching service code.

CHAPTER 3

Configuration, Profiles & Secrets



Lumen now has wired services — a `WalletService` backed by a repository, an event publisher, and a full DI lifecycle. The trouble is that every environment where Lumen runs — your laptop, a shared staging cluster, a hardened production deployment — needs different settings: different ports, different database URLs, different log verbosity, different secrets. Baking those differences into code is fragile; scattering them across a dozen `os.environ` calls is unreadable and impossible to audit.

This chapter shows how PyFly solves that with a single `pyfly.yaml`, a four-layer precedence system, environment-specific overlay files, and strongly-typed configuration classes. By the end, Lumen will have a clean configuration story that scales from `pyfly run` on your laptop to a containerised production deploy — without touching a line of business logic.

pyfly.yaml: your single source of settings

Every non-trivial application has at least two audiences for its configuration: a developer who wants verbose logs and a relaxed local database, and a production system that demands structured JSON logs, a real connection pool, and no debug mode. The naive solution — `if os.getenv("ENV") == "prod":` scattered through a dozen files — quickly becomes impossible to audit. PyFly's answer is one canonical YAML (or TOML) file that holds everything your application knows about itself, with separate mechanisms for what changes between environments.

PyFly auto-discovers this file in your project root. The framework checks candidates in order — `pyfly.yaml`, `pyfly.toml`, `config/pyfly.yaml`, `config/pyfly.toml` — and loads the first one it finds.

NEW TERM: AUTO-DISCOVERY

"Auto-discovery" simply means you do not have to tell PyFly where the config file is. You drop `pyfly.yaml` in your project root and the framework finds it on startup. No path argument, no registration call.

Let us look at Lumen's base `pyfly.yaml` one section at a time, then read it as a whole.

Step 1 — Identify the service. The first block names the application and gives it a version. These two values flow into the startup banner, the health endpoint, and trace metadata, so every part of the system reports the same identity:

```
pyfly:
  app:
    name: lumen
    version: 1.0.0
```


Step 2 — Pick the listen port. PyFly's business API listens on `pyfly.server.port`.

The default is `8080`, so this line is technically redundant — it is written out for clarity. (If you have read older PyFly material that mentioned `pyfly.web.port`, note that key was removed in v26.06.102; use `pyfly.server.port` now.)

```
server:  
  port: 8080
```

Step 3 — Turn on the domain features Lumen uses. The remaining blocks switch on observability, CQRS, the transactional engine, event sourcing, caching, the in-memory event bus, and the relational data layer. Each block is one feature; you enable what you need and leave the rest at their framework defaults.

Here is the complete file. Notice that `pyfly:` is the only top-level key — everything Lumen tells the framework lives under it:

pyfly.yaml

```

pyfly:
  app:
    name: lumen
    version: 1.0.0
  banner:
    mode: console
  server:
    # App on 8080; actuator + admin default to the management port 9090.
    port: 8080
  observability:
    metrics:
      enabled: true
    tracing:
      enabled: false
  cqrs:
    enabled: true
  transactional:
    enabled: true
    persistence:
      provider: in-memory
  event sourcing:
    enabled: true
  cache:
    provider: in-memory
    # Event-Driven Architecture: the in-memory bus (no broker needed).
    # Setting the provider registers the EventPublisher bean that the
    # wallet command handlers publish domain events through and that the
    # @event_listener audit projection auto-subscribes to.
  eda:
    provider: memory
    # Relational data layer (SQLAlchemy + SQLite via aiosqlite). The
    # framework creates the schema on startup (ddl-auto=create) and backs
    # the WalletRepository (a framework Repository[WalletEntity, str]) that
    # the command handlers persist through inside @transactional().
  data:
    relational:
      enabled: true

```

```
url: "sqlite+aiosqlite:///./lumen.db"
ddl-auto: create
```

Listing 3.1 — Lumen's base configuration file

RUN IT

From the Lumen project root, start the app and watch which file the framework loads:

```
cd samples/lumen
uv sync
uv run pyfly run
```

The startup banner prints the framework version, then PyFly logs each configuration source it merged (one `loaded_config` line per layer) before the application binds to port `8080`:

```
:: PyFly Framework :: (v26.06.110) (Python 3.12.13)
Copyright 2026 Firefly Software Foundation. | Apache License 2.0
no_active_profiles message=No active profiles set, falling back to
default
loaded_config source=pyfly-defaults.yaml (framework defaults)
loaded_config source=.../samples/lumen/pyfly.yaml
Uvicorn running on http://0.0.0.0:8080
```

Leave it running; you will hit it with `curl` shortly. Press `Ctrl+C` to stop.

Three things are worth noting. First, the `pyfly:` top-level key is reserved exclusively for framework settings — web server, observability, CQRS, EDA, data access, and profiles all live there. Your own application keys go under a different top-level name (such as `lumen:`). Second, `pyfly.app.name` and `pyfly.app.version` identify the service throughout — startup banner, health endpoints, and trace metadata all read these values. Third, the `pyfly.data.relational.*` block configures the SQLAlchemy/aiosqlite layer; `url`, `ddl-auto`, and `enabled` are its three core keys.

Nested keys map directly to dot-notation access through `Config.get()`. To read a nested value, you join the key path with dots — `pyfly.server.port` walks `pyfly:` then `server:` then `port:`:

```
config.get("pyfly.app.name")           # "lumen"
config.get("pyfly.server.port")        # 8080
config.get("pyfly.data.relational.url") # "sqlite+aiosqlite:///./lumen.db"
config.get("pyfly.eda.provider")       # "memory"
```

`Config.get()` uses **relaxed segment matching**: `ddl-auto` and `ddl_auto` resolve to the same key. Your YAML can use kebab-case (the conventional YAML style) and your Python code can use snake_case — no need to remember which form you used in the file.

PyFly uses `PyYAML` (`yaml.safe_load`) for YAML parsing; native YAML types are preserved. The integer `8080` in YAML arrives as a Python `int` — no string parsing required.

WHAT JUST HAPPENED

You wrote one YAML file, and PyFly turned it into a typed, queryable configuration object. The `pyfly:` block told the framework which features to switch on (data layer, CQRS, event bus); `Config.get("...")` reads any value back out by its dotted path; and the relaxed matching means you never have to remember whether you wrote `ddl-auto` or `ddl_auto`. That single object is the source of truth the rest of this chapter builds on.

TIP

You can also use TOML if your team prefers INI-like syntax with strict typing. Rename the file to `pyfly.toml` and use TOML table syntax — `[pyfly.web]`, `[pyfly.data.relational]` — instead of YAML nesting. Every feature described in this chapter works identically with both formats.

How configuration is layered

A single file works well with one environment. Real projects have three or four — development, test, staging, production — and the differences between them are usually small: a database URL here, a log level there. Duplicating the entire file for each environment is a maintenance burden; the first time someone updates the port in one file and forgets the others, you have a configuration drift bug.

PyFly avoids this by layering four configuration sources, each deeply merged on top of the previous. Later layers always win:

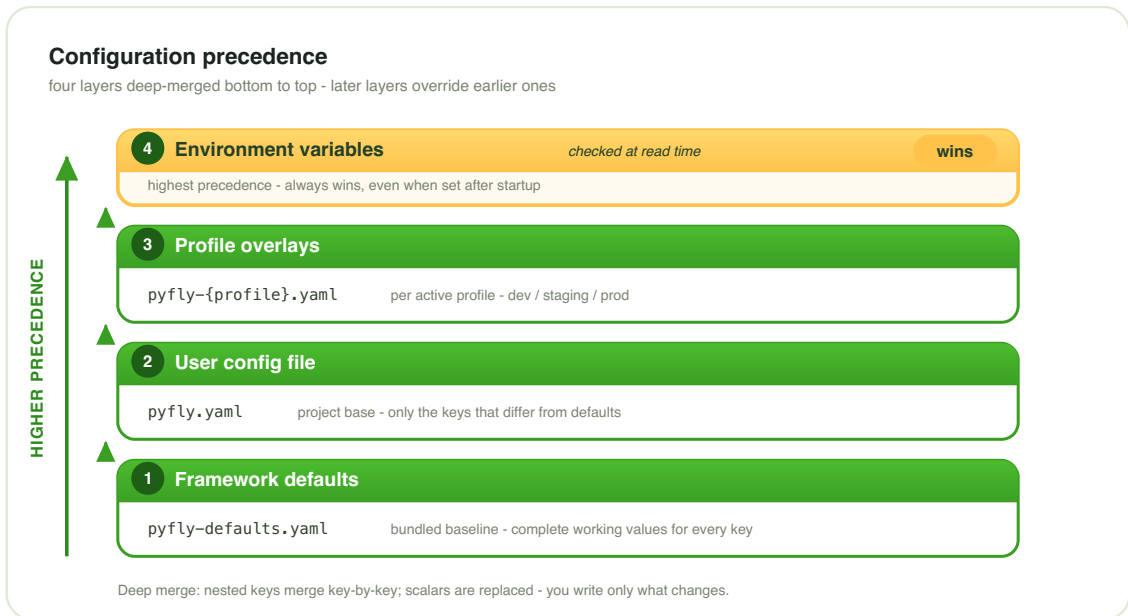


Figure 3.1 — Configuration precedence (later layers win).

Layer 1 — Framework defaults. The bundled `pyfly-defaults.yaml` inside the `pyfly.resources` package provides a sensible default for every key the framework reads. You never edit this file — it is loaded via `importlib.resources` and works correctly in packaged distributions. The framework always starts from a complete, working baseline.

Layer 2 — User configuration file. Your `pyfly.yaml` (or `pyfly.toml`). Include only the keys that differ from the framework defaults. In Listing 3.1, `pyfly.server.port: 8080` matches the default — it is included for clarity, not necessity.

Layer 3 — Profile overlay files. For each active profile, PyFly looks for a file named `pyfly-{profile}.yaml` alongside the base file and deep-merges it in. Profile overlays contain only the keys that change.

Layer 4 — Environment variables. Checked at **read time** on every `Config.get()` call, not baked in at startup. This means an env var set after the application starts still wins — the right behaviour for container deployments where secrets are injected at runtime. Environment variables always override everything else.

Deep merge, not replacement

Layers combine through a recursive deep merge (`Config._deep_merge()`). Nested dictionaries are merged key-by-key; scalar values are replaced. The distinction matters in practice: without deep merge, a production overlay that changes only `pyfly.server.port` would wipe out the `host` key sitting alongside it in the same `server:` section — and the unrelated `web.docs` block in a sibling section. With deep merge, you write only what you mean to change.

To make this concrete, consider a base file and a production overlay:

```
# pyfly.yaml (base)
pyfly:
  server:
    port: 8080
    host: "0.0.0.0"
  web:
    docs:
      enabled: true
```

```
# pyfly-prod.yaml (overlay)
pyfly:
  server:
    port: 443
```

After merge, the effective configuration is:

```
pyfly:
  server:
    port: 443           # overridden by prod overlay
    host: "0.0.0.0"     # preserved from base
  web:
    docs:
      enabled: true     # preserved from base (sibling section untouched)
```

Only the keys that differ appear in the overlay. Everything else is preserved from the layer below.

🍃 SPRING PARITY

This four-layer model maps directly to Spring Boot's configuration hierarchy:

`application.yaml` (defaults embedded in the jar) → your `application.yaml` → `application-{profile}.yaml` → environment variables. The deep-merge behaviour, the env-var-always-wins rule, and the early profile resolution step are all deliberate parity decisions.

Profiles

The layering system provides the mechanism for varying configuration between environments. Profiles provide the vocabulary to name those environments and activate them cleanly, without any `if/else` logic in your application code.

A **profile** is a named environment variant — `dev`, `test`, `staging`, `prod`. Activating a profile loads an overlay file and can conditionally include or exclude beans.

NEW TERM: OVERLAY FILE

An "overlay" is a small YAML file that contains *only the keys that change* for one environment. PyFly merges it on top of the base `pyfly.yaml`. You never repeat the full configuration — you just state the differences, and the deep merge from the previous section fills in everything else.

Activating profiles

PyFly must know which profiles are active *before* loading the full configuration, because it needs to know which overlay files to merge. This **early profile resolution** follows a deliberate priority order:

1. `PYFLY_PROFILES_ACTIVE` **environment variable** — highest priority; comma-separated for multiple profiles.
2. `pyfly.profiles.active` **in the base config file** — fallback when the env var is not set.
3. **Passed programmatically** — via `Config.from_file("pyfly.yaml", active_profiles=["prod"])`.

In production, override with an env var — no code change, no file edit:

```
PYFLY_PROFILES_ACTIVE=prod uv run pyfly run
```


Profile overlay files

For each active profile `{name}`, PyFly looks for `pyfly-{name}.yaml` next to the base file. We will add three overlays to Lumen, each containing only the keys that differ from the base. Build them one at a time.

Step 1 — The development overlay. Create `pyfly-dev.yaml` alongside `pyfly.yaml`. Dev wants the loudest possible feedback loop: full tracebacks, every SQL query echoed to the terminal, and framework internals visible at `DEBUG`.

`pyfly-dev.yaml`

```
pyfly:
  web:
    debug: true
  data:
    relational:
      echo: true
  logging:
    level:
      root: "DEBUG"
```

Listing 3.2 — Development overlay: verbose logging, debug mode

Three keys cover everything the dev environment needs: debug mode for detailed tracebacks, SQL echo so every query appears in the terminal, and `DEBUG` log level so framework internals are visible. Everything else comes unchanged from the base file. Note that `echo` lives under `pyfly.data.relational.*`, consistent with the base file structure.

Step 2 — The test overlay. Create `pyfly-test.yaml`. The test environment wants the opposite of dev: quiet. Silence the banner so test output stays readable, turn off real persistence (unit tests mock the repository), and raise the log threshold so passing tests print nothing.

`pyfly-test.yaml`

```
pyfly:
  banner:
    mode: "OFF"
  data:
    relational:
      enabled: false
  logging:
    level:
      root: "WARNING"
```

Listing 3.3 — Test overlay: in-memory SQLite, silent banner

The test overlay silences the startup banner so test output stays clean, disables data persistence (unit tests mock the repository layer), and raises the log threshold to `WARNING` so passing tests produce no noise.

Step 3 — The production overlay. Create `pyfly-prod.yaml`. Production flips many switches at once: a real PostgreSQL URL, structured JSON logs, the interactive docs turned off, and a quiet banner.

`pyfly-prod.yaml`

```
pyfly:
  server:
    port: 443
  web:
    debug: false
    docs:
      enabled: false
  data:
    relational:
      enabled: true
      url: "postgresql+asyncpg://prod-db:5432/lumen"
  logging:
    level:
      root: "WARNING"
    format: "json"
  banner:
    mode: "OFF"
```

Listing 3.4 — Production overlay: real database, JSON logging, docs off

The production overlay makes several deliberate choices. It disables the interactive API docs — you do not want a live Swagger UI on a production endpoint. It switches logging to `json` format so aggregators like Datadog or CloudWatch can parse structured fields rather than scraping human-readable text. It points `pyfly.data.relational.url` at the real PostgreSQL instance. It sets `pyfly.server.port: 443`, though in practice you will override this with `PYFLY_SERVER_PORT` from your deployment pipeline so no topology details enter the repository.

 RUN IT

With the three overlay files in place, activate the dev profile and watch the merge happen. The `PYFLY_PROFILES_ACTIVE` environment variable tells PyFly which overlays to load before it reads the rest of the config:

```
PYFLY_PROFILES_ACTIVE=dev uv run pyfly run
```

The startup log now lists the dev overlay among the merged sources, and because dev sets `pyfly.data.relational.echo: true`, every SQL statement appears in the terminal as soon as the app touches the database:

```
active_profiles profiles=['dev']
loaded_config source=pyfly-defaults.yaml (framework defaults)
loaded_config source=.../samples/lumen/pyfly.yaml
loaded_config source=.../samples/lumen/pyfly-dev.yaml (profile: dev)
INFO sqlalchemy.engine.Engine BEGIN (implicit)
```

Stop the app, run it again *without* the env var, and the dev overlay disappears from the source list — the base file's quieter defaults take over. That is the whole profile mechanism in one experiment: one env var swaps an entire layer in and out.

 TIP

Multiple profiles are comma-separated in the env var and are applied in order, so the last profile wins on conflicts: `PYFLY_PROFILES_ACTIVE=prod,metrics` first applies `pyfly-prod.yaml`, then `pyfly-metrics.yaml`. Use this to compose cross-cutting concerns — a `metrics` profile can enable Prometheus scraping without duplicating your entire prod config.

Profile-scoped beans

Sometimes the difference between environments is not a value but whether a component exists at all. A seed loader that populates test wallets must never run in production. A verbose audit logger that records every request field is useful in development but a compliance risk in prod.

The `profile` parameter on any stereotype controls when a bean participates in the container. The expression supports negation and comma-separated OR:

```
from pyfly.container import service

@service(profile="dev")
class DevSeedLoader:
    """Seeds the database with test wallets – only in dev."""
    ...

@service(profile="!prod")
class VerboseAuditLogger:
    """Detailed audit logging – active everywhere except prod."""
    ...
```

NEW TERM: BEAN

A "bean" is any object the DI container creates and manages for you — a `@service`, `@repository`, `@command_handler`, and so on (the term comes straight from Spring). "Profile-scoped" means the container only creates the bean when a matching profile is active.

`Environment.accepts_profiles()` evaluates profile expressions during the first pass of `ApplicationContext.start()`. Beans whose expression does not match the active set are removed before any resolution takes place — never instantiated, never wired, never present in the container. The result is a container that is structurally different per environment, with no `if` statement in your application code.

WHAT JUST HAPPENED

Profiles gave you two independent tools. The *overlay files* (`pyfly-{name}.yaml`) change configuration *values* per environment. The `profile=` parameter on a stereotype changes which *beans exist* per environment. Both are driven by the same active-profile set — set once via `PYFLY_PROFILES_ACTIVE` — so a single env var reshapes both your settings and your object graph, with zero conditional logic in your business code.

Type-safe settings with `@config_properties`

String-key lookups like `config.get("pyfly.data.relational.url")` work for occasional reads, but they do not scale. Each call is an isolated read with no type information — you must remember to call `float()` on the result, and a typo in a key surfaces at the first request in production, not at startup. For anything beyond a handful of scattered values, the right approach is to group related settings into a typed Python class that is populated once at startup and injected wherever it is needed.

`@config_properties` solves exactly this by binding a config section to a typed Python dataclass.

i NEW TERM: BINDING

"Binding" means copying values out of the config tree into the fields of a typed object. A Python `@dataclass` is a class whose fields are declared with type hints (`url: str` , `pool_size: int`); after binding, you read `props.pool_size` and get a real `int` , not a string you have to convert yourself.

Declaring a properties class

Decorate a `@dataclass` with `@config_properties(prefix="...")` . The `prefix` identifies the config section to bind; field names must match the keys under that section (kebab/snake interchangeable).

Step 1 — Write the class. Here is the framework's own `RelationalProperties` , which binds the `pyfly.data.relational.*` block:

pyfly/config/properties/data.py

```
from dataclasses import dataclass

from pyfly.core.config import config_properties

@config_properties(prefix="pyfly.data.relational")
@dataclass
class RelationalProperties:
    """Typed binding for pyfly.data.relational.*"""

    enabled: bool = False
    url: str = "sqlite+aiosqlite:///pyfly.db"
    echo: bool = False
    pool_size: int = 5
```

Listing 3.5 — `RelationalProperties`: typed settings for the data layer

The decorator sets `__pyfly_config_prefix__` on the class and marks it as an injectable bean. Field types must be `int`, `float`, `bool`, or `str` for automatic coercion; more complex types are left as-is.

Notice that each field carries a default value matching the framework's built-in `pyfly-defaults.yaml`. This is intentional: the class is self-documenting, and it can be constructed and used in unit tests without any YAML file on disk — just instantiate `RelationalProperties()` and you get the development defaults.

Step 2 — Apply the pattern to your own settings. The same decorator works for application-level configuration. Here is how a `WalletProperties` class would look for Lumen's business rules:

```
from dataclasses import dataclass
from pyfly.core import config_properties

@config_properties(prefix="lumen.wallet")
@dataclass
class WalletProperties:
    daily_transfer_limit: float = 10_000.0
    default_currency: str = "USD"
```

Step 3 — Add the matching YAML. Put the block under the `lumen:` top-level key (outside `pyfly:`, since this is your application's own namespace, not the framework's) and the framework binds it automatically — no special registration required:

```
lumen:
  wallet:
    daily-transfer-limit: 10000.0
    default-currency: USD
```


Binding and injecting

Call `config.bind(PropertiesClass)` to produce a populated, typed instance. `Config` is registered as a singleton bean, so you can inject it into any service and bind from there:

`lumen/wallet_service.py`

```
from pyfly.container import service
from pyfly.core import Config
from pyfly.config.properties import RelationalProperties
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.eda import EventPublisher

@service
class WalletService:
    def __init__(
        self,
        repo: WalletRepository,
        events: EventPublisher,
        config: Config,
    ) → None:
        self.repo = repo
        self.events = events
        self.db: RelationalProperties = config.bind(
            RelationalProperties
        )

    async def transfer(
        self, from_id: str, to_id: str, amount: float
    ) → dict:
        if not self.db.enabled:
            raise RuntimeError("Relational layer not enabled")
        # ... perform transfer using self.db.url for diagnostics ...
        return {"from": from_id, "to": to_id, "amount": amount}
```

Listing 3.6 — Injecting `RelationalProperties` via `config.bind()`

When the DI container starts Lumen, `WalletService.__init__` receives the shared `Config` singleton and immediately calls `config.bind(RelationalProperties)`. That call resolves the bound values once, at startup, and stores them in `self.db` — a plain Python dataclass with real types. From that point on, `transfer()` reads `self.db.enabled` as a `bool`, with full IDE autocompletion and no parsing code anywhere.

`config.bind()` works through five steps:

1. Reads `__pyfly_config_prefix__` from the class.
2. Calls `effective_section(prefix)` — a resolved copy of the subtree with `${...}` placeholders expanded, environment-variable overrides applied, and env-only keys injected.
3. Matches section keys to dataclass fields using relaxed (kebab/snake interchangeable) lookup.
4. Applies type coercion for fields whose values arrived as strings (for example, from environment variables).
5. Constructs the dataclass with the gathered kwargs; fields absent from config use their dataclass defaults.

The critical detail in step 2 is that `effective_section()` applies the full four-layer stack — defaults, file, profile overlay, env vars — before the dataclass is constructed. By the time `bind()` finishes, `RelationalProperties` reflects whatever the production overlay or a runtime env var says, not just the base YAML.

WHAT JUST HAPPENED

You replaced scattered `config.get("...")` string lookups with a single typed object. `config.bind(RelationalProperties)` reads the whole `pyfly.data.relational.*` section once, applies the four-layer precedence, coerces strings to the right types, and hands back a plain dataclass. From then on your service reads `self.db.enabled` as a `bool` with IDE autocompletion — and a typo in a field name is caught at startup, not at request time in production.

RUN IT

You can prove the env-var layer reaches a bound property with a tiny check. The base `pyfly.yaml` sets `pyfly.data.relational.enabled: true`; here we override it from the environment. From the Lumen project root:

```
PYFLY_DATA_RELATIONAL_ENABLED=false uv run python -c "
from pyfly.core import Config
from pyfly.config.properties import RelationalProperties
db = Config.from_file('pyfly.yaml').bind(RelationalProperties)
print('enabled =', db.enabled, type(db.enabled).__name__)
"
```

Expected output — the env var wins over the YAML and the string `"false"` is coerced to a `bool`:

```
enabled = False bool
```

Injecting individual values with Value

For isolated settings that do not warrant a full properties class, PyFly provides a `Value` descriptor. Declare it as a class-level field and the DI container resolves it at bean creation time — exactly like Spring Boot's `@Value("${...}")`:

```

from pyfly.container import service
from pyfly.core import Value

@service
class WalletService:
    # Resolved from pyfly.app.name in the merged config.
    app_name: str = Value("${pyfly.app.name}")
    # Falls back to 10000 when the key is absent.
    transfer_limit: float = Value(
        "${lumen.wallet.daily-transfer-limit:10000}"
    )

```

`Value("${key}")` raises `KeyError` at startup when the key is missing and no default is provided — a fail-fast guarantee that keeps missing-config bugs out of production.

`Value("${key:default}")` uses the colon-delimited default when the key is absent.

Type coercion

Native YAML types arrive correctly typed — integers, booleans, and floats need no coercion. When a value arrives from an environment variable (always a string) and the target field has a non-string type, `bind()` coerces automatically:

Target type	Coercion rule
<code>int</code>	<code>int(value)</code>
<code>float</code>	<code>float(value)</code>
<code>bool</code>	<code>value.lower() in ("true", "1", "yes")</code>
<code>str</code>	no coercion needed

The `bind()` dataclass path treats `"true"`, `"1"`, and `"yes"` as `True`; the read-time `get()` /env override path additionally accepts `"on"`.

Calling `bind()` on a class not decorated with `@config_properties` raises `ValueError` immediately — a clear fail-fast signal at startup rather than a silent wrong-value bug at request time.

🌿 SPRING PARITY

`@config_properties` is PyFly's answer to Spring Boot's `@ConfigurationProperties`. The mental model is identical: annotate a POJO (here, a dataclass) with a prefix, and the framework binds the matching config section to it with full type coercion. `Value("${...}")` maps to Spring's `@Value("${...}")` — same expression syntax, same fail-fast-on-missing guarantee. The combination of `pyfly.yaml` + profile overlays + `@config_properties` + `Value` maps to `application.yaml` + `application-{profile}.yaml` + `@ConfigurationProperties` + `@Value` — same concepts, Pythonic idioms.

Environment variables & secrets

Files are the right home for configuration that varies by environment but is safe to commit — ports, log levels, database hostnames. They are the wrong home for secrets: passwords, API keys, signing tokens, and database credentials must never enter source control. The fourth layer of the configuration stack exists specifically to receive these values at deploy time, from a secrets manager or a CI/CD pipeline, without any of them touching the file system.

Environment variables are the fourth and highest-priority layer. PyFly checks them on every `Config.get()` call — at read time, not at startup — so they always win, even when set after the process begins.

Naming convention

Every dot-notation config key maps to a `PYFLY_`-prefixed environment variable through a mechanical three-step transformation:

1. Strip the `pyfly.` prefix (if present).
2. Replace dots (`.`) and hyphens (`-`) with underscores (`_`).
3. Uppercase the result and prefix with `PYFLY_`.

Config key	Environment variable
<code>pyfly.app.name</code>	<code>PYFLY_APP_NAME</code>
<code>pyfly.server.port</code>	<code>PYFLY_SERVER_PORT</code>
<code>pyfly.management.server.port</code>	<code>PYFLY_MANAGEMENT_SERVER_PORT</code>
<code>pyfly.web.debug</code>	<code>PYFLY_WEB_DEBUG</code>
<code>pyfly.data.relational.url</code>	<code>PYFLY_DATA_RELATIONAL_URL</code>
<code>pyfly.data.relational.pool-size</code>	<code>PYFLY_DATA_RELATIONAL_POOL_SIZE</code>
<code>pyfly.logging.level.root</code>	<code>PYFLY_LOGGING_LEVEL_ROOT</code>
<code>pyfly.eda.provider</code>	<code>PYFLY_EDA_PROVIDER</code>
<code>pyfly.profiles.active</code>	<code>PYFLY_PROFILES_ACTIVE</code>

For application-specific keys that do not start with `pyfly.`, the full dotted path is transformed the same way (no prefix stripping):

```
lumen.wallet.daily-transfer-limit
→ PYFLY_LUMEN_WALLET_DAILY_TRANSFER_LIMIT
```

The rule is consistent: when you need to tell a Kubernetes operator which env var controls a given setting, the answer is always "apply the three-step transformation" rather than hunting through framework source code.

Env vars always win

Activating the production profile and overriding the database URL for a specific container instance is a single command:

```
PYFLY_PROFILES_ACTIVE=prod \
  PYFLY_DATA_RELATIONAL_URL="postgresql+asyncpg://rds-prod:5432/lumen" \
  PYFLY_SERVER_PORT=8080 \
  uv run pyfly run
```

Here, `PYFLY_SERVER_PORT=8080` overrides the prod overlay's `port: 443`. The precedence stack resolves like this:

1. Framework defaults → `port: 8080`
2. Base config → `port: 8080` (unchanged)
3. Prod overlay → `port: 443`
4. Env var → `port: 8080` (wins)

The effective port is `8080`. This is a useful pattern during a staged migration: keep `port: 443` in the overlay as the intended production default, then use a temporary env var to hold the service on `8080` for a traffic-splitting experiment. When the experiment ends, remove the env var and the overlay takes over — no file edits needed.

Keeping secrets out of files

Config files must never contain credentials, API keys, or signing secrets. The `pyfly-defaults.yaml` ships with a placeholder JWT secret (`"change-me-in-production"`) that exists only to keep the framework runnable out of the box. Replace it before going to production:

```
PYFLY_SECURITY_JWT_SECRET="$(vault kv get -field=jwt_secret secret/lumen)"
```

⚠ NEVER COMMIT SECRETS

Do not put passwords, API keys, database credentials, or JWT secrets in `pyfly.yaml` , `pyfly-prod.yaml` , or any file that enters source control. Use environment variables sourced from a secret manager (HashiCorp Vault, AWS Secrets Manager, Kubernetes Secrets, or similar). The env-var layer exists precisely to receive these at deploy time, not at development time.

A note on env-only keys

`Config.bind()` also handles values that exist *only* as environment variables — no matching entry in any YAML file. `effective_section()` injects these env-only keys into the bound section so `bind()` sees the same value that `get()` would. Add a new field to a `@config_properties` class, set it exclusively via an env var in your deployment pipeline, and it is populated correctly even when the YAML files have not been updated yet:

```
# No YAML entry for pyfly.data.relational.echo?
# Set it exclusively via env var – bind() still picks it up.
PYFLY_DATA_RELATIONAL_ECHO=true uv run pyfly run
```

This is a practical escape hatch during incremental rollouts: the deploying team can inject a new value before the YAML file is updated and reviewed, and the application picks it up without a code change.

⚠ MULTI-WORD FIELD NAMES AND ENV-ONLY INJECTION

Env-only injection treats each underscore in a `PYFLY_*` name as a path separator, so `PYFLY_DATA_RELATIONAL_POOL_SIZE` is read as the nested path `pool` → `size`, not the flat field `pool_size`. For a single-word field like `echo` this is unambiguous and `bind()` injects it cleanly. For a multi-word field such as `pool_size`, give the key a real home in your YAML (even just `pool-size: 5`) so the env var overrides an existing leaf instead of relying on env-only injection. Read-time `config.get("pyfly.data.relational.pool-size")` always returns the env value regardless, because `get()` maps the whole dotted key in one step.

✓ What you built

Lumen now has a clean configuration story across three environments. A `pyfly.yaml` holds the shared baseline — `pyfly.app.name`, `pyfly.eda.provider`, `pyfly.data.relational.*`, and every other framework knob Lumen uses. `pyfly-dev.yaml`, `pyfly-test.yaml`, and `pyfly-prod.yaml` hold only the per-environment deltas. Activating a profile takes a single env var (`PYFLY_PROFILES_ACTIVE=prod`). Typed settings live in `@config_properties` dataclasses, bound at startup with full type coercion — so services read typed fields rather than calling `float(os.environ.get(...))` in scattered service code. Individual values inject cleanly via `Value("${key}")`, which fails fast at startup when the key is missing. Secrets stay in environment variables, never in files.

The four-layer stack — defaults → file → profile overlay → env vars — gives you one mental model that works from `pyfly run` on your laptop to a locked-down container with secrets injected at deploy time, without touching a line of business logic.

Application and management ports

PyFly separates the **application** port from the **management** port, mirroring Spring Boot's `server.port` / `management.server.port`. Out of the box the business API listens on `pyfly.server.port` (**8080**) while the actuator endpoints (`/actuator/*`) and the admin dashboard (`/admin`) are served on a dedicated `pyfly.management.server.port` (**9090**). This keeps health checks, Prometheus scraping, and the admin console off the public port — you expose only 8080 to the internet and reach 9090 from inside the cluster.

Key	Env var	Default	Purpose
<code>pyfly.server.port</code>	<code>PYFLY_SERVER_PORT</code>	8080	Application HTTP port
<code>pyfly.server.host</code>	<code>PYFLY_SERVER_HOST</code>	0.0.0.0	Application bind address
<code>pyfly.management.server.port</code>	<code>PYFLY_MANAGEMENT_SERVER_PORT</code>	9090	Management (actuator + admin) port
<code>pyfly.management.server.address</code>	<code>PYFLY_MANAGEMENT_SERVER_ADDRESS</code>	app host	Management bind address

The management port is a second **in-process** listener — not extra worker processes — sharing the same event loop and beans, so it works with any server adapter (Granian, Uvicorn, Hypercorn). Two values change the topology: set `pyfly.management.server.port` **equal** to the app port to serve everything on a single port (the pre- `v26.06.102` behaviour), or set it to `-1` to disable the management web endpoints entirely.

⚠ THE MANAGEMENT PORT IS OPEN BY DEFAULT

For Spring Boot parity, the management port (9090) is **open and unauthenticated** by default — it is meant to live on an internal network protected by network isolation, not by the app's login filters. Before you expose it anywhere reachable, secure it with

`pyfly.management.security.enabled: true`, which applies the app's security, session, and CSRF filters to the management port too. By default only the `health` and `info` actuator endpoints are exposed over HTTP; widen that with

`pyfly.management.endpoints.web.exposure.include` (for example `"health,info,metrics,prometheus"`).

i RUN IT

Start Lumen and hit the management port directly. The health endpoint lives on 9090, not on the application's 8080:

```
uv run pyfly run          # in one terminal
curl http://localhost:9090/actuator/health # in another
```

Expected output — a small JSON document reporting the service is up:

```
{"status": "UP"}
```

The same path on the app port (`curl http://localhost:8080/actuator/health`) will not answer, because the actuator listens only on the management port.

🌿 SPRING PARITY

`pyfly.server.port` $\equiv$ `Spring server.port`, `pyfly.server.host` $\equiv$ `server.address`, and `pyfly.management.server.port` $\equiv$ `management.server.port`. Setting a distinct management port runs the actuator on its own connector, exactly as Spring Boot does.

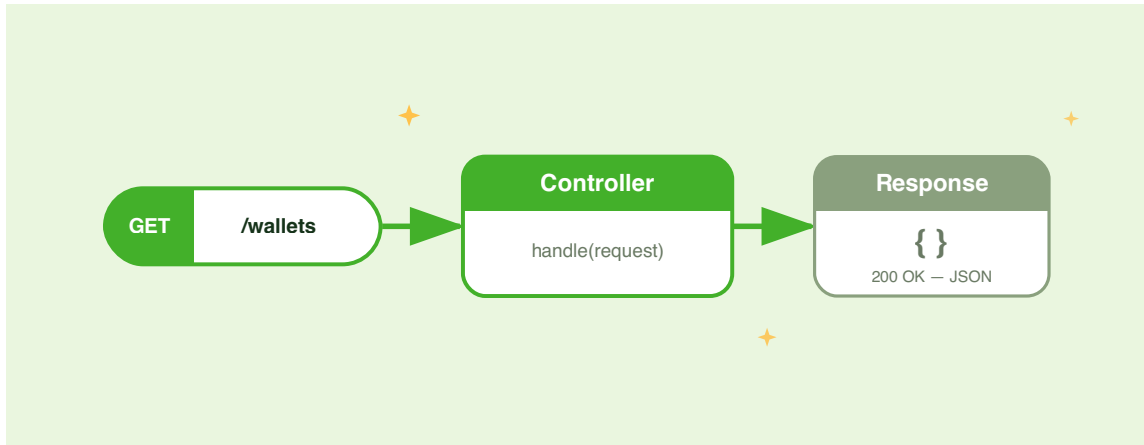
`pyfly.management.security.enabled` and `pyfly.management.endpoints.web.exposure.include` mirror Spring's management security and `management.endpoints.web.exposure.include` settings.

Try it yourself

- 1. Add a staging overlay.** Create `pyfly-staging.yaml` with a PostgreSQL URL for a shared test database under `pyfly.data.relational.url`, `pyfly.data.relational.enabled: true`, and logging at `INFO`. Activate it with `PYFLY_PROFILES_ACTIVE=staging uv run pyfly run` and verify from the startup log that the staging source was loaded. Compare the effective configuration to what the prod overlay would produce.
- 2. Bind a new typed property and use it.** Add a `max_wallets_per_owner: int = 5` field to a new `WalletProperties` class decorated with `@config_properties(prefix="lumen.wallet")`, and a matching `lumen.wallet.max-wallets-per-owner: 5` key in `pyfly.yaml` (outside the `pyfly:` block). Inject `Config` into `WalletService`, call `config.bind(WalletProperties)`, and add a guard in `open_wallet` that raises `ValueError` when the owner already holds the maximum number of wallets. Write a quick test that overrides the limit to `1` by setting `PYFLY_LUMEN_WALLET_MAX_WALLETS_PER_OWNER=1` and verifying the error fires on the second wallet.
- 3. Override a value via an env var and observe precedence.** Set `PYFLY_SERVER_PORT=9090` before starting Lumen. Check the startup log and confirm the server binds to `9090`, not the `8080` in `pyfly.yaml`. Then unset the env var and restart — the port should revert to `8080`. This exercise makes the read-time nature of env-var resolution concrete: the env var always wins, and removing it immediately restores the file value without any code change.

CHAPTER 4

Your First HTTP API



Lumen has wired services, a clean configuration story, and a lifecycle that spans development through production. The only thing missing is a way for the outside world to talk to it. This chapter closes Part I by turning the wallet domain into a clean, validated REST API — one with automatic OpenAPI documentation, structured error responses that clients can trust, and the framework-managed conventions you have come to expect from the rest of PyFly.

i HTML APPLICATIONS AND MODEL ADMINISTRATION

Appendix E follows the runnable catalog: `ModelAndView`, `Form[T]`, `reverse`, `static_url`, HTML errors, and CRUD over existing entities/documents. Data administration always requires authentication; a separate listener also requires `pyfly.management.security.enabled: true`.

Controllers and route mappings

Every web framework must answer two questions: how does a request find the right handler, and how does that handler get the dependencies it needs? Frameworks that answer these questions with separate mechanisms force you to maintain a router file, a DI glue file, and a documentation scaffold in three different places. PyFly collapses all three concerns into a single class.

A **controller** in PyFly is an ordinary Python class that the DI container manages and the web layer routes requests into. Two decorators mark it: `@rest_controller` from `pyfly.container` registers it as a bean and sets its stereotype; `@request_mapping` from `pyfly.web` sets the URL prefix inherited by every handler in the class.

A few terms in that paragraph will recur throughout the chapter, so let us pin them down once. A **bean** is simply an object the DI container creates and hands out for you — you never call `WalletController()` yourself; the framework constructs it and keeps a single shared instance. A **stereotype** is a label the framework stamps on a class so it knows *what kind* of bean it is — `@rest_controller` stamps "this is a web controller", which is the cue the startup machinery uses to go looking for routes inside it. A **handler** is one `async def` method on the controller that answers one kind of request. With those three words in hand, the rest of the chapter reads as plain English.

Route handlers are plain `async def` methods, each decorated with `@get_mapping`, `@post_mapping`, `@put_mapping`, `@patch_mapping`, or `@delete_mapping`. Every mapping decorator accepts an optional relative path and an optional `status_code`. The full URL is the base path from `@request_mapping` concatenated with the relative path from the method decorator.

A pre-CQRS reading example

Chapter 7 introduces the full CQRS command/query bus that Lumen uses in production. Here in Part I the web-layer mechanics are taught on the same wallet domain, backed by a simple in-memory store instead of the bus. The controller structure, imports, and decorator shapes are *identical* to Chapter 7; only the dispatch target changes.

lumen/web/controllers/wallet_controller.py

```

from __future__ import annotations

from pyfly.container import rest_controller
from pyfly.kernel import ResourceNotFoundException
from pyfly.web import (
    Body,
    PathVar,
    QueryParam,
    Valid,
    get_mapping,
    post_mapping,
    request_mapping,
)

from lumen.interfaces.dtos.v1.balance_dto import BalanceDto
from lumen.interfaces.dtos.v1.deposit_request import DepositRequest
from lumen.interfaces.dtos.v1.open_wallet_request import OpenWalletRequest
from lumen.interfaces.dtos.v1.wallet_dto import WalletDto

# -----
# In-memory store (replaced by a database repository in Chapter 5)
# -----
_wallets: dict[str, WalletDto] = {}

@rest_controller
@request_mapping("/api/v1/wallets")
class WalletController:
    """Digital-wallet REST API: open, deposit, inspect.

    In Part I the controller holds a minimal in-memory store so you can
    focus on the web-layer mechanics – decorators, binding, validation, and
    error handling – without persistence or CQRS machinery. Chapter 7
    replaces the store with DefaultCommandBus / DefaultQueryBus dispatching.
    """

```



```

@post_mapping("", status_code=201)
async def open_wallet(
    self, request: Valid[Body[OpenWalletRequest]]
) → dict[str, str]:
    import uuid
    wallet_id = str(uuid.uuid4())
    wallet = WalletDto(
        id=wallet_id,
        owner_id=request.owner_id,
        currency=request.currency,
        balance_minor=0,
        balance=0.0,
        created_at=__import__("datetime").datetime.now(
            __import__("datetime").timezone.utc
        ),
    )
    _wallets[wallet_id] = wallet
    return {"wallet_id": wallet_id}

@get_mapping("/{wallet_id}")
async def get_wallet(self, wallet_id: PathVar[str]) → WalletDto:
    result = _wallets.get(wallet_id)
    if result is None:
        raise ResourceNotFoundException(
            f"Wallet {wallet_id!r} not found",
            code="WALLET_NOT_FOUND",
            context={"wallet_id": wallet_id},
        )
    return result

@get_mapping("/{wallet_id}/balance")
async def get_balance(self, wallet_id: PathVar[str]) → BalanceDto:
    wallet = _wallets.get(wallet_id)
    if wallet is None:
        raise ResourceNotFoundException(
            f"Wallet {wallet_id!r} not found",
            code="WALLET_NOT_FOUND",
            context={"wallet_id": wallet_id},
        )

```

```

        return BalanceDto(
            id=wallet.id,
            currency=wallet.currency,
            balance_minor=wallet.balance_minor,
            balance=wallet.balance,
        )

    @post_mapping("/{wallet_id}/deposit")
    async def deposit(
        self,
        wallet_id: PathVar[str],
        request: Valid[Body[DepositRequest]],
    ) → dict[str, int | str]:
        wallet = _wallets.get(wallet_id)
        if wallet is None:
            raise ResourceNotFoundException(
                f"Wallet {wallet_id!r} not found",
                code="WALLET_NOT_FOUND",
                context={"wallet_id": wallet_id},
            )
        new_balance = wallet.balance_minor + request.amount
        _wallets[wallet_id] = wallet.model_copy(
            update={
                "balance_minor": new_balance,
                "balance": new_balance / 100,
            }
        )
        return {"wallet_id": wallet_id, "balance_minor": new_balance}

    @get_mapping("")
    async def list_wallets(
        self,
        owner_id: QueryParam[str] = None,
    ) → list[WalletDto]:
        wallets = list(_wallets.values())
        if owner_id is not None:
            wallets = [w for w in wallets if w.owner_id == owner_id]
        return wallets

```

Listing 4.1 — *WalletController* using real PyFly web decorators

Four design choices in this listing are worth examining.

`@rest_controller` does two things at once: it registers `WalletController` as a singleton bean in the DI container and sets the `__pyfly_stereotype__` marker that `ControllerRegistrar` uses to discover and mount routes at startup. Pairing it with `@request_mapping("/api/v1/wallets")` means every method-level decorator inherits that prefix — you write the base path once.

This version has no `__init__` with injected collaborators. The in-memory `_wallets` dictionary is a module-level store that suffices for Part I. Chapter 5 introduces repositories; Chapter 7 shows the production pattern: a constructor that receives `DefaultCommandBus` and `DefaultQueryBus` from the DI container, dispatching commands and queries through the bus rather than reading `_wallets` directly.

Each handler returns a Pydantic model (`WalletDto`, `BalanceDto`) or a plain `dict`. The framework serialises the return value to JSON and sets the `Content-Type` header — the handler never builds a response object. The `status_code=201` argument to `@post_mapping` produces a 201 Created on success; all other handlers default to 200.

All five mapping decorators accept the same two parameters:

Parameter	Default	Description
<code>path</code>	<code>""</code>	Relative path appended to the base. Use <code>{name}</code> for path variables.
<code>status_code</code>	<code>200</code>	HTTP status code for a successful response.

`@post_mapping("", status_code=201)` maps `POST /api/v1/wallets` and returns 201 on success. `@get_mapping("/{wallet_id}")` maps `GET /api/v1/wallets/{wallet_id}`. Paths are concatenated at startup; duplicate or trailing slashes are normalised automatically.

Building the controller, step by step

If you are typing this in from scratch, the listing above lands all at once. Here is the same controller assembled in the order you would actually build it, so each decorator has a job to do before the next one arrives.

Step 1 — Create the file and the class. Make `src/lumen/web/controllers/wallet_controller.py` and define an empty class decorated with the two class-level decorators. This is enough for PyFly to discover the controller at startup, even before it has a single route.

```
from pyfly.container import rest_controller
from pyfly.web import request_mapping

@rest_controller
@request_mapping("/api/v1/wallets")
class WalletController:
    """Digital-wallet REST API: open, deposit, inspect."""
```

Step 2 — Add the first handler. Give the class one `async def` method and mark it with a mapping decorator. `@post_mapping("", status_code=201)` maps `POST /api/v1/wallets` — the empty path means "the base path with nothing appended" — and promises a `201 Created` on success.

Step 3 — Add the remaining handlers. Repeat the pattern: one `async def` per route, each with its own mapping decorator and relative path. The full set in Listing 4.1 gives you open, fetch, balance, deposit, and list.

Step 4 — Wire the store. The module-level `_wallets` dictionary is the only "database" Part I needs. Each handler reads from and writes to it directly; Chapter 5 swaps it for a real repository without touching a single decorator.

 NOTE

Notice what you did *not* write: no router file mapping URLs to functions, no registration call in `main.py`, no manual OpenAPI entry. The decorators are the registration. At startup `ControllerRegistrar` finds every `@rest_controller` bean and mounts its routes for you.

 RUN IT

Start the server and confirm the routes are live. From the project root:

```
uv run pyfly run --server uvicorn
```

The boot banner reports the framework version and the bound port:

```
:: PyFly Framework :: (v26.06.110) (Python 3.13.13)
```

In a second terminal, open a wallet and read it back:

```
curl -s -X POST localhost:8080/api/v1/wallets \
  -H 'Content-Type: application/json' \
  -d '{"owner_id": "alice", "currency": "EUR"}'
```

You should see a **201** body with the generated id:

```
{"wallet_id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890"}
```

Copy that id and fetch the wallet:

```
curl -s localhost:8080/api/v1/wallets/a1b2c3d4-e5f6-7890-abcd-ef1234567890
```

```
{"id": "a1b2c3d4-...", "owner_id": "alice", "currency": "EUR",
  "balance_minor": 0, "balance": 0.0, "created_at": "2026-06-15T10:30:00+00:00"}
```

What just happened. Two class-level decorators registered the controller and fixed its URL prefix; one method-level decorator turned an `async def` into a live route; the framework did the routing, JSON serialisation, and status-code handling. You wrote business intent, not plumbing.

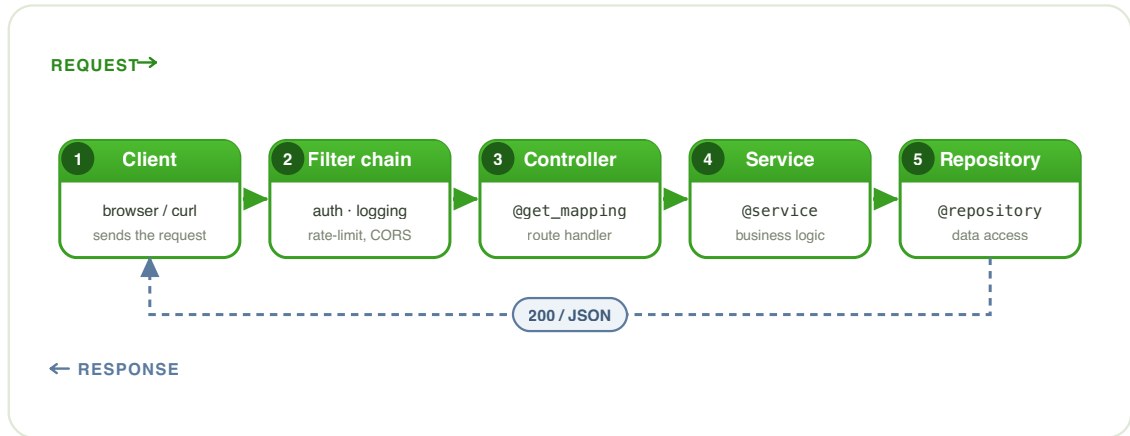


Figure 4.1 — How a request flows to your handler.

🌿 SPRING PARITY

`@rest_controller` + `@request_mapping` + `@get_mapping` / `@post_mapping` is a direct translation of Spring's `@RestController` + `@RequestMapping` + `@GetMapping` / `@PostMapping`. Handler methods return values directly (not `ResponseEntity`) and the framework converts them to JSON — exactly the pattern Spring encourages with `@ResponseBody` on `@RestController`.

Binding request data

A request carries data in several places at once: the URL path identifies the resource, the query string carries filters and pagination, the body carries the payload, and headers carry metadata. Most frameworks address these with separate mechanisms, each with its own conventions. PyFly unifies them under a single idea: **generic type annotations on handler parameters declare where data comes from.**

This approach makes handler signatures self-documenting. The parameter list of any handler tells you exactly which parts of the request it reads and what types it expects — without opening a router file or consulting the docs.

In plain terms, **binding** is the framework copying a piece of the incoming request into one of your handler's parameters, converting it to the type you asked for along the way. You declare *what you want and where it comes from* with a type annotation; PyFly does the extracting, parsing, and type-coercion before your method body runs.

The `ParameterResolver` inspects each handler signature at startup and builds a resolution plan, so there is zero overhead per request for introspection. Five binding types cover every part of an HTTP request:

PathVar[T] — path variables

Extracts a named segment from the URL path. The parameter name must match a `{placeholder}` in the route.

```
@get_mapping("/{wallet_id}")
async def get_wallet(self, wallet_id: PathVar[str]) → WalletDto:
    ...

@get_mapping("/{wallet_id}/transactions/{txn_id}")
async def get_transaction(
    self,
    wallet_id: PathVar[str],
    txn_id: PathVar[str],
) → dict:
    ...
```

`PathVar` coerces the raw string segment to `T` automatically. `PathVar[int]`, `PathVar[float]`, and `PathVar[UUID]` all work — the coercion calls `int(value)`, `float(value)`, and `UUID(value)` respectively.

QueryParam[T] — query parameters

Extracts a value from the query string, with support for defaults and optional values.

```
@get_mapping("")
async def list_wallets(
    self,
    owner_id: QueryParam[str] = None,
    page: QueryParam[int] = 1,
    size: QueryParam[int] = 20,
) → List[WalletDto]:
    ...
```

A parameter is **required** when it has no Python default and its type does not admit `None`. A missing required `QueryParam` raises `InvalidRequestException` (HTTP 400). To make a parameter optional, give it a default value or annotate it `QueryParam[str | None]`.

RUN IT

With the server running and at least one wallet opened, exercise the `list_wallets` handler — first with no filter, then with the optional `owner_id` query parameter:

```
curl -s 'localhost:8080/api/v1/wallets'
curl -s 'localhost:8080/api/v1/wallets?owner_id=alice'
```

The first returns every wallet; the second returns only Alice's. Because `owner_id` has a default of `None`, omitting it is perfectly valid — no 400. The path variable behaves the same way in reverse: ask for a wallet id that does not exist and you get a clean `404`, which the next section dissects.

Body[T] — request body

Deserialises the JSON (or XML) request body. When `T` is a Pydantic `BaseModel`, `model_validate_json()` is called automatically.

```
@post_mapping("", status_code=201)
async def open_wallet(
    self, request: Valid[Body[OpenWalletRequest]]
) → dict[str, str]:
    ...
```

Header[T] and Cookie[T]

Extract values from request headers and cookies. For headers, the parameter name is converted from `snake_case` to `kebab-case` automatically:

```
@get_mapping("/me")
async def get_my_wallets(
    self,
    x_api_key: Header[str],
    session_id: Cookie[str | None],
) → list[WalletDto]:
    ...
```

`x_api_key: Header[str]` reads the `x-api-key` header. A missing required header or cookie raises `InvalidRequestException` (HTTP 400), the same as a missing query parameter.

TIP

All five binding types follow the same **required vs optional** rule: no default + non- `None` type = required (HTTP 400 when absent); any default or `T | None` = optional. The rule is uniform across `QueryParam`, `Header`, and `Cookie` — you learn it once, it applies everywhere.

What just happened. You learned the whole binding vocabulary as five parallel annotations — `PathVar`, `QueryParam`, `Body`, `Header`, `Cookie` — that all read like English in a handler signature and all share one required-vs-optional rule. The framework reads the annotation, pulls the value from the right place, coerces it to your type, and hands you a ready-to-use argument. There is nothing else to wire.

Validation with Valid[T]

Binding tells the framework *where* data comes from. Validation tells it *what that data must look like* before your handler ever sees it. Without a layer that intercepts bad input early, validation logic scatters into service methods, manual `if` blocks litter business code, and different handlers produce inconsistent error responses depending on where they happen to catch the problem.

PyFly solves this at the type level. Pydantic `BaseModel` gives you field-level constraints for free. `Valid[T]` is PyFly's marker that converts a Pydantic `ValidationError` into a **structured 422 response** instead of letting it bubble up to a 500.

A quick gloss before the code. A **DTO** — Data Transfer Object — is a small class that describes the *shape* of data crossing the wire: what fields a request must carry, or what fields a response will return. Lumen's DTOs are plain Pydantic models, so the field declarations double as validation rules. **Validation** is the act of checking incoming data against those rules and rejecting it cleanly if it does not fit — before any of your handler code runs.

Pydantic DTOs for Lumen

The request and response DTOs used in Lumen's wallet API live under `lumen/interfaces/dtos/v1/` — one file per DTO. The directory name encodes a convention worth noting: `interfaces` holds the contracts the outside world sees, and `v1` versions them so a future `v2` payload shape can live alongside the old one without breaking existing clients. Here they are in full.

`lumen/interfaces/dtos/v1/open_wallet_request.py`

```
from __future__ import annotations

from pydantic import BaseModel, Field

from lumen.interfaces.enums.v1.currency import Currency


class OpenWalletRequest(BaseModel):
    """Wallet-opening request payload."""

    owner_id: str = Field(
        min_length=1,
        max_length=64,
        description="Identifier of the wallet owner",
    )
    currency: Currency = Field(
        default=Currency.EUR,
        description="ISO-4217 currency the wallet holds",
    )
```

Listing 4.2a — `OpenWalletRequest`: wallet-opening payload

`lumen/interfaces/dtos/v1/deposit_request.py`

```
from __future__ import annotations

from pydantic import BaseModel, Field

class DepositRequest(BaseModel):
    """Deposit/withdrawal request payload.

    Shared by POST /{id}/deposit and POST /{id}/withdraw – both move a
    positive amount of money in the wallet's own currency.
    """

    amount: int = Field(
        gt=0,
        description="Amount in minor units (cents); must be positive",
    )
```

Listing 4.2b — *DepositRequest: deposit/withdrawal payload*

```
lumen/interfaces/dtos/v1/wallet_dto.py
```

```
from __future__ import annotations

from datetime import datetime

from pydantic import BaseModel

from lumen.interfaces.enums.v1.currency import Currency


class WalletDto(BaseModel):
    """Full wallet representation returned to clients.

    ``balance_minor`` is in minor units (cents); ``balance`` is the same
    value rendered as a major-unit decimal for human-friendly display.
    """

    id: str
    owner_id: str
    currency: Currency
    balance_minor: int
    balance: float
    created_at: datetime
```

Listing 4.2c — *WalletDto: full wallet response*

```
lumen/interfaces/dtos/v1/balance_dto.py
```

```
from __future__ import annotations

from pydantic import BaseModel

from lumen.interfaces.enums.v1.currency import Currency


class BalanceDto(BaseModel):
    """Lightweight balance projection for the balance endpoint."""

    id: str
    currency: Currency
    balance_minor: int
    balance: float
```

Listing 4.2d — BalanceDto: lightweight balance projection

These are pure Pydantic models — PyFly adds nothing to them. Four design decisions are worth unpacking.

`OpenWalletRequest.owner_id` uses `Field(min_length=1, max_length=64)`. The lower bound prevents phantom wallets from empty-string owner IDs that would silently pollute your data; the upper bound keeps identifiers within a sensible column width when the database layer arrives in Chapter 5.

`currency` is a `Currency` enum (a `StrEnum` with `EUR`, `USD`, `GBP`). Using an enum rather than a raw string means Pydantic rejects `"XYZ"` at deserialisation time — you never validate the currency code yourself. `Field(default=Currency.EUR)` provides a sensible default so callers can omit the field for EUR wallets.

`DepositRequest.amount` is an `int` with `Field(gt=0)`. Storing money in minor units avoids floating-point rounding: `1050` means €10.50 for an EUR wallet. The `gt=0` constraint makes a zero or negative deposit a 422 client error, not a business-logic decision — the constraint lives in the type, and Pydantic enforces it before your handler runs.

`WalletDto` and `BalanceDto` are response models. Returning a typed Pydantic model instead of a plain `dict` lets the framework generate accurate OpenAPI response schemas and gives clients a machine-readable contract.

Using `Valid[T]` in a handler

Wrap `Body[T]` in `Valid` to opt into structured 422 errors on validation failure:

```
@post_mapping("", status_code=201)
async def open_wallet(
    self, request: Valid[Body[OpenWalletRequest]]
) → dict[str, str]:
    ...

@post_mapping("/{wallet_id}/deposit")
async def deposit(
    self,
    wallet_id: PathVar[str],
    request: Valid[Body[DepositRequest]],
) → dict[str, int | str]:
    ...
```

`Valid[Body[OpenWalletRequest]]` tells the resolver two things: bind from the request body (`Body`) and run Pydantic validation before the handler executes (`Valid`). When the body fails validation, the resolver catches the `ValidationError` and raises a `ValidationException` with `code="VALIDATION_ERROR"` and a `context.errors` array containing each field-level detail.

What the client sees on failure

To see validation in action, send a `POST /api/v1/wallets` with an empty `owner_id`:

```
POST /api/v1/wallets
Content-Type: application/json

{"owner_id": ""}
```

RUN IT

With the server running, send the bad payload and watch for the `422`:

```
curl -s -w '\nHTTP %{http_code}\n' -X POST localhost:8080/api/v1/wallets \
-H 'Content-Type: application/json' \
-d '{"owner_id": ""}'
```

The `-w '\nHTTP %{http_code}\n'` flag prints the status line after the body, so you can confirm it is `HTTP 422` — not the `201` a valid request returns, and not a `500`. The body is the structured envelope shown below. Try a second variant — `-d '{"owner_id": "alice", "currency": "XYZ"}'` — to see the `Currency` enum reject an unknown code with the same envelope shape.

The response is HTTP 422:

```

{
  "error": {
    "message": "Validation failed: owner_id: String should have at least 1
character",
    "code": "VALIDATION_ERROR",
    "status": 422,
    "path": "/api/v1/wallets",
    "timestamp": "2026-06-07T10:30:00+00:00",
    "transaction_id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
    "context": {
      "errors": [
        {
          "type": "string_too_short",
          "loc": ["owner_id"],
          "msg": "String should have at least 1 character",
          "input": "",
          "ctx": {"min_length": 1}
        }
      ]
    }
  }
}

```

Every field error carries a `type` (machine-readable), a `loc` (path to the failing field), a `msg` (human-readable), and an `input` (the rejected value). API consumers parse this array deterministically — no scraping of error strings.

The difference between bare `Body[T]` and `Valid[Body[T]]` is exactly this:

Annotation	On validation failure
<code>Body[T]</code>	Raw Pydantic <code>ValidationError</code> propagates — may become a 500 without extra handling
<code>Valid[Body[T]]</code>	Caught, converted to <code>ValidationException</code> , always produces a structured 422

Use `Valid[Body[T]]` for every endpoint that accepts user input.

SPRING PARITY

`Valid[Body[T]]` maps directly to Spring's `@Valid + @RequestBody` combination on a `@RestController` method. In Spring you write `@PostMapping public ResponseEntity create(@Valid @RequestBody OpenWalletRequest body)`; in PyFly you write `async def open_wallet(self, request: Valid[Body[OpenWalletRequest]])`. The 422 response shape (field-level errors with location paths) mirrors Spring Boot 3's `MethodArgumentNotValidException` payload.

What just happened. The validation rules never left the DTO. `Field(min_length=1)` on `owner_id`, the `Currency` enum, and `Field(gt=0)` on `amount` are the entire specification — and wrapping the body in `Valid` turned any breach of those rules into a predictable, machine-readable 422 before your handler ran. You wrote constraints once, on the data; the framework enforced them everywhere the data arrives.

Errors that clients can trust

A well-designed API fails loudly, consistently, and informatively. Clients must never parse exception stack traces or guess what went wrong from a generic 500. The challenge is achieving this without scattering HTTP-specific logic through your service code — the HTTP status code is an infrastructure concern, not a business one.

PyFly's exception hierarchy is the backbone of its error story. Every exception in the tree carries three things: a human-readable `message`, a machine-readable `code`, and an optional `context` dict for debugging detail. The web layer's global exception handler maps each subclass to the correct HTTP status code automatically — you `raise`, the framework responds.

The exception tree

```

PyFlyException
├── BusinessException → 400 (catch-all)
│   ├── ValidationException → 422
│   ├── ResourceNotFoundException → 404
│   ├── ConflictException → 409
│   ├── InvalidRequestException → 400
│   └── ...
├── SecurityException → 403
│   ├── UnauthorizedException → 401
│   └── ForbiddenException → 403
├── InfrastructureException → 502 (catch-all)
│   ├── ServiceUnavailableException → 503
│   ├── CircuitBreakerException → 503
│   └── ...

```

The hierarchy is intentionally shallow. `BusinessException` covers anything that is the caller's fault; `InfrastructureException` covers anything that is the system's fault. Subclasses pin the status code. When a new domain error does not fit an existing subclass, extend the nearest parent and the status code comes for free.

Import them from `pyfly.kernel`:

```

from pyfly.kernel import (
    ResourceNotFoundException,
    ConflictException,
    ValidationException,
    InvalidRequestException,
)

```

Raise them from handler code without worrying about HTTP:

```
raise ResourceNotFoundException(
    f"Wallet {wallet_id!r} not found",
    code="WALLET_NOT_FOUND",
    context={"wallet_id": wallet_id},
)
```

The global handler catches it, maps it to 404, and emits a structured JSON response:

```
{
  "error": {
    "message": "Wallet 'w-999' not found",
    "code": "WALLET_NOT_FOUND",
    "status": 404,
    "path": "/api/v1/wallets/w-999",
    "timestamp": "2026-06-07T10:30:00+00:00",
    "transaction_id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
    "context": {
      "wallet_id": "w-999"
    }
  }
}
```

💡 RUN IT

Ask for a wallet that was never opened and watch the framework turn your `raise` into a clean `404`:

```
curl -s -w '\nHTTP %{http_code}\n' localhost:8080/api/v1/wallets/w-999
```

The status line reads `HTTP 404` and the body is the envelope above, with `"code": "WALLET_NOT_FOUND"` and your `context` carried through verbatim. You never wrote a status code in `get_wallet` — `ResourceNotFoundException` maps to 404 for you. Note the `transaction_id` in the response; copy it and `grep` your server log to find the exact request.

The `transaction_id` is free: the `TransactionIdFilter` assigns a UUID to every request and threads it through all error responses. Clients log it; support uses it to find the corresponding server log entry. A single ID is all that is needed to reconstruct what happened.

What just happened. Your handler expressed a domain fact — "this wallet does not exist" — by raising a typed exception with a message, a code, and some context. The web layer's global handler did the HTTP translation: it picked the status code from the exception's class, wrapped everything in the standard error envelope, and stamped a `transaction_id`. HTTP concerns stayed out of your business code entirely.

RFC 7807

The default error envelope — `{"error": {...}}` — is PyFly's own format. If your team prefers the IETF standard, set `pyfly.web.problem-details.enabled: true` in `pyfly.yaml`. With that flag on, the same `ResourceNotFoundException` produces an `application/problem+json` response with `type`, `title`, `status`, `detail`, and `instance` as the standard RFC 7807 members, plus `code` and `transactionId` as PyFly extension members. Both modes use the same exception hierarchy and status mapping.

Content negotiation & OpenAPI

JSON and XML

Returning a `dict` or a Pydantic model is not quite the end of the story. Somewhere between your handler's `return` statement and the bytes the client receives, the framework decides on a wire format. Rather than hardcoding JSON, PyFly runs the

return value through an ordered `HttpMessageConverter` chain — important for enterprise APIs that must serve XML partners or negotiate the lightest format for mobile clients.

JSON is the default. When no `Accept` header is sent, the response is `application/json`. When the client sends `Accept: application/xml`, the XML converter takes over and serialises the same return value as XML, with no change to your handler code:

```
GET /api/v1/wallets/w-001   Accept: application/json
→ {"id": "w-001", ...}

GET /api/v1/wallets/w-001   Accept: application/xml
→ <response><id>w-001</id>... </response>
```

The same negotiation applies to inbound data: a `Body[T]` or `Valid[Body[T]]` parameter accepts both `Content-Type: application/json` and `Content-Type: application/xml` request bodies. JSON is the fallback when no `Content-Type` header is present.

Auto-generated documentation

Manually maintained API specs drift. As routes change, parameters are renamed, and new models are added, hand-written specs fall behind the code. PyFly eliminates this entirely by generating documentation from the same metadata that drives routing — the spec is always in sync because it is the same source.

As soon as Lumen starts, three documentation endpoints are live at no cost:

Endpoint	Purpose
<code>/docs</code>	Swagger UI — interactive, try-it-now documentation
<code>/redoc</code>	ReDoc — clean, two-panel reference documentation
<code>/openapi.json</code>	Raw OpenAPI 3.0 specification

The `OpenAPIGenerator` introspects `ControllerRegistrar`'s route metadata — every path, method, path variable, query parameter, and request/response schema (from Pydantic model introspection) — and assembles the spec at startup. You never write the spec by hand. The docs endpoints live on the **application** port (8080) alongside your API; they are on by default (`pyfly.web.docs.enabled: true`). Disable them in production with `pyfly.web.docs.enabled: false` in `pyfly.yaml` .

NOTE

Do not confuse the docs endpoints with the **admin dashboard**. `/docs` , `/redoc` , and `/openapi.json` describe *your* API and serve on the app port (8080). The PyFly Admin Dashboard (`/admin`) and the actuator health endpoints (`/actuator/*`) describe the *running process* and serve on the separate **management** port (`pyfly.management.server.port` , default 9090), introduced in Chapter 3. They are two different listeners with two different audiences.

 RUN IT

With the server running, fetch the raw spec and confirm your routes are in it:

```
curl -s localhost:8080/openapi.json | head -c 200
```

You will see the OpenAPI 3.0 header and the start of the `paths` map. Then open `http://localhost:8080/docs` in a browser. You will see `POST /api/v1/wallets`, `GET /api/v1/wallets/{wallet_id}`, `POST /api/v1/wallets/{wallet_id}/deposit`, and the others — each with the correct request and response schemas derived from your Pydantic models, and the `owner_id` query parameter on `list_wallets` already documented with its type and default. Click "Try it out" on `POST /api/v1/wallets` to open a real wallet straight from the browser.

What just happened. You did not write a line of API documentation, yet a complete, interactive, always-accurate spec appeared. The same route and model metadata that drives request handling also drives the docs, so the two can never drift apart.

The server underneath

Lumen now has routes, bindings, validation, and documentation. The last question is what actually listens on port 8080. The answer matters: different servers make different trade-offs in throughput, HTTP version support, OS compatibility, and ecosystem tooling. Locking an application to a single server at the framework level forces you to accept those trade-offs permanently.

An **ASGI server** is the process that actually accepts TCP connections, parses HTTP, and calls your application — the layer between the operating system's socket and your handlers. PyFly does not hardcode one. At startup, `ServerAutoConfiguration` runs a cascading selection based on what is installed:

Priority	Server	Characteristic
1st	Granian	Rust/tokio-powered; fastest single-worker throughput
2nd	Uvicorn	Ecosystem standard; best tooling support
3rd	Hypercorn	Native HTTP/2 and HTTP/3

All three start through the same `ApplicationServerPort` protocol, so your code is completely unaware of which one is running. Override with `pyfly.server.type:` `uvicorn` in `pyfly.yaml` or with the `--server` CLI flag:

```
pyfly run --server uvicorn --reload      # development: auto-reload
pyfly run --server granian --workers 4  # production: multi-worker
```

RUN IT

For day-to-day development, run with auto-reload so the server restarts on every save:

```
uv run pyfly run --reload
```

PyFly logs the chosen server and the bound port at startup. Because `--reload` requires a built-in file watcher, PyFly selects **Uvicorn** for reload mode regardless of the cascade order. Edit a handler, save, and watch the log report the restart — then re-run any `curl` from earlier and see your change live without stopping the process.

The event loop is pluggable too: `uvloop` (Linux/macOS) and `winloop` (Windows) are selected automatically when installed, delivering a 2–4× throughput improvement over the `asyncio` default. Install them with `uv add "pyfly[web-fast]"`.

 TIP

For development, `pyfly run --reload` is all you need — it picks the best available server and event loop automatically. For production, pass an explicit positive worker count to scale across cores — `pyfly run --server granian --workers 4`, as in the example above. A `0` or negative `--workers` value resolves to a single worker, so multi-worker is always an explicit opt-in. CLI flags always override `pyfly.yaml`.

✓ What you built

Part I is complete.

In four chapters you went from an empty scaffold to a production-shaped service.

Lumen now **boots** (`@pyfly_application`, startup banner, structured logging), is **wired** (services and repositories connected through constructor injection with no glue code), **configured** (four-layer `pyfly.yaml` + profile overlays + env-var secrets, typed `WalletProperties`), and **serves** — a validated REST API at `/api/v1/wallets` with `PathVar`, `QueryParam`, and `Valid[Body[T]]` body binding; structured 422 errors from Pydantic constraints; domain-error-to- status mapping from the exception hierarchy; typed response models (`WalletDto`, `BalanceDto`) that drive OpenAPI schema generation; and a pluggable ASGI server running underneath.

Every part of this stack follows the same hexagonal principle you have seen throughout: your code depends on ports and decorators; the framework wires the adapters. Swap the in-memory store for a PostgreSQL adapter in Chapter 5, replace direct dispatch with a full CQRS bus in Chapter 7, or enable XML responses — none of it requires touching the controller's decorator structure or the DTO shapes.

Part II takes Lumen further: persistent data with SQLAlchemy, domain events, resilience with circuit breakers, and security with JWT. The foundations you built here carry forward intact.

Try it yourself

Each exercise is small and self-contained. After every change, restart with `uv run pyfly run --reload` and re-run the suggested `curl` to confirm the behaviour. If you have the dev dependencies installed, you can also run the project's test suite at any point to make sure nothing regressed:

```
uv run --extra dev pytest
```

You should see a row of passing dots and a `passed` summary line.

1. **Add a `DELETE /api/v1/wallets/{wallet_id}` endpoint.** Remove the wallet from `_wallets` and return 204 No Content. Raise `ResourceNotFoundException` if the wallet does not exist. Decorate with `@delete_mapping("/{wallet_id}", status_code=204)` — PyFly converts a `None` return with `status_code=204` into a 204 response with no body. Verify with `curl -X DELETE http://localhost:8080/api/v1/wallets/{id}`.
2. **Add currency filtering to `list_wallets`.** Add a `currency: QueryParam[str] = None` parameter and filter `_wallets.values()` when it is not `None`. Test with `GET /api/v1/wallets?currency=EUR` and confirm only EUR wallets are returned; confirm `GET /api/v1/wallets` without the parameter returns all wallets. Then make the parameter required by removing the default — observe the 400 response when you omit it from the request.

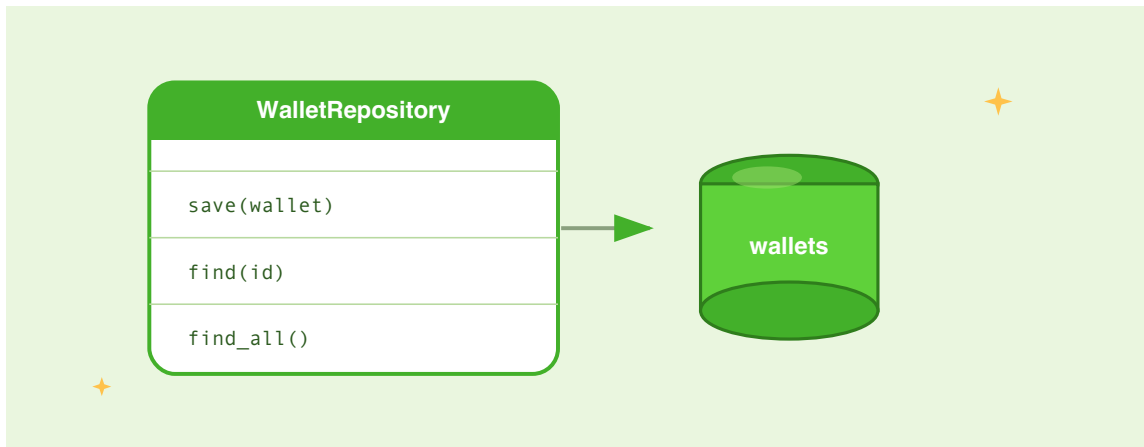
3. **Raise a domain-specific subclass of `ResourceNotFoundException`**. Create a `WalletNotFoundError` that subclasses `ResourceNotFoundException` and carry an extra `currency` field in `context`. Raise it from `get_wallet` and `get_balance` instead of bare `ResourceNotFoundException`. Verify the JSON error response includes the extra context field without any change to the global handler — the hierarchy maps it to 404 automatically.

PART II

Modeling & Persisting the Domain

CHAPTER 5

Persistence & the Repository Pattern



Lumen has a wallet API that works — but every wallet disappears the moment you restart the process. It is time to make wallets durable.

The naïve approach is to scatter SQLAlchemy `select()` and `session.commit()` calls through the command handlers. PyFly offers something far better: a **Spring-Data-style repository layer**. You declare an interface — `class WalletRepository(Repository[WalletEntity, str])` — and the framework *implements it for you*. Full async CRUD comes for free. Query methods are derived from their **names**. Pagination, sorting, composable filters, and read projections are first-class. There is no hand-written adapter and no SQL in the application code.

This chapter rebuilds Lumen's persistence on that layer, exactly as the running sample does it: the SQLAlchemy entity, the repository with its derived and specification queries, `Page / Pageable / Sort`, projections for read views, and the transaction seam that keeps the `Wallet` aggregate intact. Everything here runs against a real SQLite file with zero external infrastructure — the sample's 41 tests are green on it. This chapter targets PyFly **v26.6.110**.

We will build the persistence layer one piece at a time, and at every milestone there is a **Run it** box with the exact command to type and the output you should see. If you are following along in the Lumen sample, work from the project root (`samples/lumen`) where `pyfly.yaml` and `pyproject.toml` live; every command below assumes that directory.

i RUN IT: SEE THE PROBLEM FIRST

Before adding persistence, it is worth feeling the gap it closes. Open a wallet, then stop and restart the process — with the in-memory store from Part I the wallet is gone. The rest of this chapter makes it survive that restart.

```
# Terminal 1 – start the app
uv run pyfly run --server uvicorn
# ... startup banner, then: Uvicorn running on http://0.0.0.0:8080

# Terminal 2 – open a wallet, then read it back
curl -s -X POST localhost:8080/api/v1/wallets \
  -H 'content-type: application/json' \
  -d '{"owner_id": "alice", "currency": "EUR"}'
```

You get back the new id, then the balance read confirms it exists:

```
{"wallet_id": "wlt-7f3c..."}
```

Now press **Ctrl+C** in Terminal 1, start the app again, and ask for that wallet's balance. Before this chapter, the row was never written to disk, so it is gone:

```
{"detail": "Wallet 'wlt-7f3c...' not found", "code": "WALLET_NOT_FOUND"}
```

By the end of this chapter the same sequence returns the balance after a restart.

The repository, in one sentence

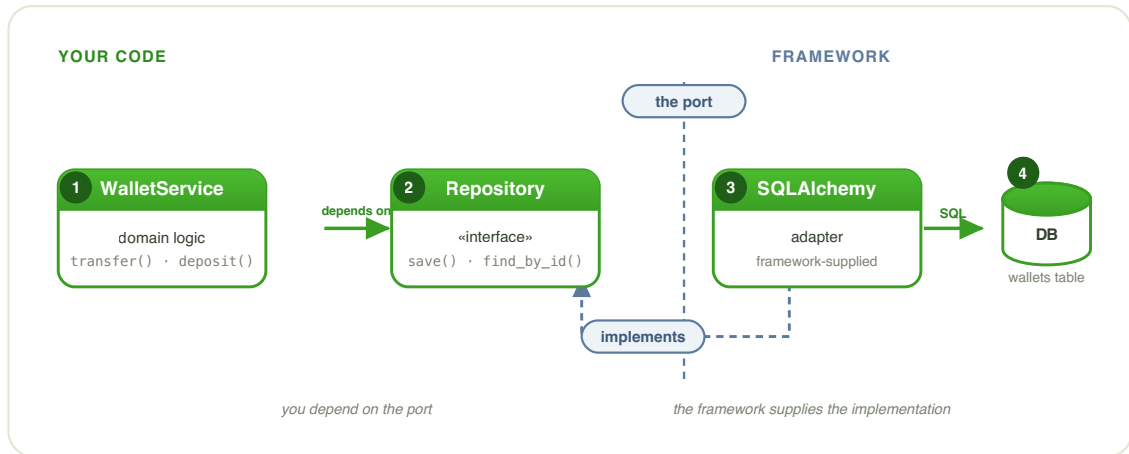


Figure 5.1 — Your code depends on the repository; the framework supplies the SQLAlchemy implementation behind it.

A PyFly repository is a class that subclasses the generic `Repository[Entity, ID]` and is marked with the `@repository` stereotype. That is the whole declaration. From the two type parameters the framework learns the **entity type** and the **primary-key type**, and from there it provides a complete async data-access surface — `save`, `find_by_id`, `find_all`, `delete / delete_by_id`, `count`, `exists_by_id`, plus pagination and specification queries — with the database `AsyncSession` injected for you.

This is the Repository pattern as Spring Data popularised it, translated to idiomatic async Python. You write *what* you want (the method) and the framework writes *how* (the SQL).

 SPRING PARITY

`Repository[T, ID]` is PyFly's `JpaRepository<T, ID>`. Subclassing it to inherit CRUD, deriving queries from method names, `Pageable / Page`, `Specification`, and interface projections are all carried over almost name-for-name from Spring Data JPA. If you have written a Spring interface `OrderRepository` extends `JpaRepository<Order, UUID>`, you already know the shape of this chapter.

The entity: one row per wallet

Before a repository can store anything, it needs an **entity** — the on-disk shape of a wallet, one flat row per aggregate. (An *entity*, in this layer, is just a Python class that maps to one database table; each instance is one row.) PyFly entities are ordinary SQLAlchemy 2.0 models built on a declarative base the framework exports.

We will build it field by field. Create the file `src/lumen/models/entities/v1/wallet_orm.py` and add the pieces in order.

Step 1 — import the framework's declarative base. Every entity inherits from a *declarative base*: a SQLAlchemy class that records each table you define so the framework can create them all on startup. PyFly exports one as `Base`:

```
from pyfly.data.relational.sqlalchemy import Base
```

Step 2 — name the table and declare the columns. Subclass `Base`, set `__tablename__`, and write one typed attribute per column. The full entity is short:

lumen/models/entities/v1/wallet_orm.py

```

from __future__ import annotations

from datetime import UTC, datetime

from sqlalchemy import String
from sqlalchemy.orm import Mapped, mapped_column

from pyfly.data.relational.sqlalchemy import Base

class WalletEntity(Base):
    """One persisted wallet row, keyed by the aggregate's string id."""

    __tablename__ = "wallets"

    id: Mapped[str] = mapped_column(String(64), primary_key=True)
    owner_id: Mapped[str] = mapped_column(
        String(255), nullable=False, index=True
    )
    currency: Mapped[str] = mapped_column(String(3), nullable=False)
    balance_minor: Mapped[int] = mapped_column(nullable=False, default=0)
    created_at: Mapped[datetime] = mapped_column(
        default=lambda: datetime.now(UTC)
    )

```

Listing 5.1 — *WalletEntity*: the SQLAlchemy persistence row

The `Mapped[T]` / `mapped_column(...)` syntax is SQLAlchemy 2.0 style: each type annotation drives both the Python attribute type and the generated column DDL (the `CREATE TABLE` statement), so every column has a single source of truth. Because `WalletEntity` subclasses `Base`, importing this module registers the `wallets` table in `Base.metadata` — the registry of every table the base knows about — and the framework's engine lifecycle then creates it on startup.

❗ WHAT JUST HAPPENED

You wrote one class, no SQL. The five typed attributes became five columns; `primary_key=True` marked `id` as the key; `index=True` on `owner_id` will speed up the "wallets owned by X" query you build later; `nullable=False` and `default=...` set the constraints. Importing this module is enough for the framework to know the table exists — you never call `CREATE TABLE` yourself.

Two design choices are worth calling out.

Base, not BaseEntity. PyFly ships two declarative bases. `BaseEntity` gives you a surrogate **UUID** primary key plus four audit columns (`created_at` , `updated_at` , `created_by` , `updated_by`) populated automatically — the right default for most tables. Lumen deliberately uses plain `Base` instead, because the `Wallet` aggregate already owns its identity: a string id of the form `wlt-...`. Inheriting `Base` lets the row keep that **string** primary key, so the row and the aggregate share one identity rather than the row inventing a second, surrogate one.

Integer minor units. Amounts live in `balance_minor` as integer cents, never as a float. Floating-point columns accumulate rounding error over millions of transactions; integer arithmetic stays exact. A balance of `2500` means €25.00 — the major-unit decimal is computed only at the edges, for display.

💡 REACH FOR BASEENTITY BY DEFAULT

Unless your aggregate owns a natural key the way `Wallet` does, prefer `class Order(BaseEntity)`. You get a UUID PK and audit columns for free, and the `AuditingEntityListener` fills `created_by` / `updated_by` from the security context on every insert and update. Lumen is the exception, not the rule.

The repository: CRUD for free

Now the centrepiece. Lumen's `WalletRepository` subclasses `Repository[WalletEntity, str]` — entity type `WalletEntity`, primary-key type `str` — and is registered with `@repository`. (A *stereotype* like `@repository` is a class decorator that tells the framework's container, "manage an instance of this for me." That managed instance is a **bean** — an object the container creates once and hands to anything that asks for it. *Dependency injection*, or *DI*, is the container doing that handing-over for you, so a handler never constructs its own repository.) That single declaration is enough for the framework to supply the entire CRUD surface — *CRUD* being the four basic table operations: Create, Read, Update, Delete.

Create `src/lumen/models/repositories/wallet_repository.py` in two steps.

Step 1 — import what the declaration needs. Three imports: your entity, the `@repository` stereotype, and the generic `Repository` base.

```
from lumen.models.entities.v1.wallet_orm import WalletEntity
from pyfly.container import repository
from pyfly.data.relational.sqlalchemy import Repository
```

Step 2 — subclass the generic base and mark it `@repository`. The two type parameters carry all the wiring:

```
lumen/models/repositories/wallet_repository.py
```

```
from __future__ import annotations

from lumen.models.entities.v1.wallet_orm import WalletEntity
from pyfly.container import repository
from pyfly.data import Page, Pageable
from pyfly.data.relational.sqlalchemy import Repository, Specification

@repository
class WalletRepository(Repository[WalletEntity, str]):
    """CRUD + derived + specification queries for WalletEntity.

    The @repository stereotype registers this as a DI bean. The
    framework reads the entity/PK types from the
    Repository[WalletEntity, str] base and injects the shared
    AsyncSession.
    """

    # (query methods follow – see the next sections)
```

Listing 5.2 — *WalletRepository*: subclass the framework repository

There is no `__init__`, no SQL, and no adapter class. With just that declaration, any handler that injects a `WalletRepository` can already call:

Method	Returns	What it does
<code>save(entity)</code>	<code>T</code>	Insert or update; flushes + refreshes
<code>find_by_id(id)</code>	<code>T None</code>	Load by primary key
<code>find_all(**filters)</code>	<code>list[T]</code>	All rows, optional equality filters
<code>find_all(sort)</code>	<code>list[T]</code>	All rows in a given <code>Sort</code> order
<code>find_all(pageable)</code>	<code>Page[T]</code>	Paged + sorted query (see below)
<code>stream_all(sort)</code>	<code>AsyncIterator[T]</code>	Stream every row (the <code>Flux<T></code> analogue)
<code>delete(entity)</code>	<code>None</code>	Delete a given entity
<code>delete_by_id(id)</code>	<code>None</code>	Delete by primary key (no-op if absent)
<code>delete_all(entities=None)</code>	<code>None</code>	Delete the given entities (or all rows)
<code>delete_all_by_id(ids)</code>	<code>None</code>	Delete many rows by primary key
<code>count()</code>	<code>int</code>	Count every row in the table
<code>exists_by_id(id)</code>	<code>bool</code>	Whether a row with this id exists
<code>save_all(entities)</code>	<code>list[T]</code>	Bulk insert/update
<code>find_all_by_id(ids)</code>	<code>list[T]</code>	Load many rows by primary key
<code>find_all_by_spec(spec)</code>	<code>list[T]</code>	Rows matching a <code>Specification</code>

Method	Returns	What it does
<code>find_all_by_spec_paged(...)</code>	<code>Page[T]</code>	Paged + sorted <code>Specification</code> query

That is more than enough for most entities. Lumen adds three methods of its own on top — a derived query, a specification query, and an upsert — which the next sections build up.

🍃 SPRING PARITY

This inherited surface is exactly Spring Data's repository hierarchy, carried over name-for-name and stable as of PyFly **v26.6.110**: `CrudRepository` → `ReactiveSortingRepository` → `PagingAndSortingRepository`. `save / save_all`, `find_by_id`, `find_all`, `exists_by_id`, `count`, and the `delete*` family map to their Spring equivalents; `find_all(pageable)` → `Page[T]` is `findAll(Pageable)`, and `find_all_by_spec*` is the `JpaSpecificationExecutor`. If you know `JpaRepository<T, ID>`, you already know this table.

How the framework knows the types

When you write `Repository[WalletEntity, str]`, the base class's `__init_subclass__` hook inspects `__orig_bases__` at class-definition time and pulls the entity type (`WalletEntity`) and id type (`str`) out of the generic parameters. (`__init_subclass__` is a Python hook that runs once, automatically, when a subclass is *defined* — so this happens at import time, before any object is created.) The `AsyncSession` — the database connection-and-transaction handle every query runs through — is then supplied as an injected dependency by the relational auto-configuration. Nothing is passed manually — the type parameters *are* the wiring.

📘 RUN IT: CONFIRM THE REPOSITORY WIRES UP

The fastest proof that the entity and repository are sound is the test suite, which exercises the repository against a real SQLite file with no server. From the Lumen project root:

```
uv run --extra dev pytest tests/test_sql_wallet_repository.py -q
```

You should see every repository test pass:

```
..... [100%]
6 passed in 0.30s
```

These tests construct the repository directly and drive `upsert`, `find_by_id`, `count`, the derived query, and the specification path — the same methods this chapter builds. If they are green, your entity columns and the `Repository[WalletEntity, str]` declaration are correct.

Derived queries: the method name is the query

CRUD covers lookups by primary key. Real applications also need to query by other columns — "all wallets owned by this customer." In most frameworks you would write the SQL by hand. In PyFly you declare a **stub** — a method with no body, just `...` — and let the framework compile the query *from the method name*.

Step 1 — declare the stub. Add a method to `WalletRepository`. The *name* describes the query; the body is literally `...`:

```
lumen/models/repositories/wallet_repository.py
```

```
@repository
class WalletRepository(Repository[WalletEntity, str]):

    # derived query: compiled from the method name by the post-processor
    async def find_by_owner_id(
        self, owner_id: str
    ) → list[WalletEntity]:
        """All wallets owned by *owner_id* (derived query stub)."""
        ...
```

Listing 5.3 — A derived query: declared as a stub, compiled from its name

Step 2 — let the framework fill in the body. You write no more code. At startup a `BeanPostProcessor` — the `RepositoryBeanPostProcessor` — does the work. (A `BeanPostProcessor` is a hook the container runs over every bean just after it is created; this one specialises in repositories.) It scans the repository, spots that `find_by_owner_id` is a stub, parses the **name** into a parsed query, and replaces the stub with a real implementation that runs `SELECT ... FROM wallets WHERE owner_id = :owner_id`. Calling `await repo.find_by_owner_id("alice")` now returns exactly the rows for that owner.

WHAT JUST HAPPENED

You declared a method and got a working query — the framework read the *intent* from the name `find_by_owner_id` and wrote the SQL for you. The naming is not magic; it follows a grammar, covered next. The key idea: in this layer you describe *what* you want by how you name the method, and the post-processor supplies the *how*.

The grammar is the Spring Data convention. A method name is a **prefix** followed by a **subject** built from field names, operators, connectors, and an optional ordering clause:

Part	Tokens
Prefix	<code>find_by</code> · <code>count_by</code> · <code>exists_by</code> · <code>delete_by</code>
Connectors	<code>_and_</code> · <code>_or_</code>
Operators	<code>_greater_than</code> · <code>_less_than</code> · <code>_between</code> · <code>_in</code> · <code>_like</code> · <code>_containing</code> · <code>_is_null</code> · <code>_is_not_null</code>
Ordering	<code>_order_by_<field>_<asc\\desc></code>

Each clause consumes the matching number of method arguments (equality and the comparisons take one; `_between` takes two; `_is_null` / `_is_not_null` take none). A few examples on a hypothetical orders repository:

```

@repository
class OrderRepository(Repository[Order, UUID]):
    async def find_by_status(self, status: str) → list[Order]: ...

    async def find_by_customer_id_and_status(
        self, customer_id: str, status: str
    ) → list[Order]: ...

    async def find_by_total_greater_than(
        self, min_total: float
    ) → list[Order]: ...

    async def find_by_total_between(
        self, low: float, high: float
    ) → list[Order]: ...

    async def count_by_status(self, status: str) → int: ...

    async def exists_by_customer_id(self, customer_id: str) → bool: ...

    async def find_by_status_order_by_created_at_desc(
        self, status: str
    ) → list[Order]: ...

```

The prefix decides the *shape* of the result: `find_by` returns a list, `count_by` returns an `int`, `exists_by` returns a `bool`, and `delete_by` issues a `DELETE` and returns the number of rows removed. You never write the SQL; you name the method and annotate the return type.

💡 WHEN A NAME WOULD GET SILLY, USE @QUERY

Derived names are perfect up to two or three predicates. Past that they become unreadable. For anything more complex, drop a

`@query("SELECT w FROM WalletEntity w WHERE ...")` decorator (JPQL-like, or `native=True` for raw SQL) on the stub and write the query explicitly. Same stub-plus-decorator pattern; you just supply the query text instead of encoding it in the name.

Pagination: Page, Pageable, and Sort

A list endpoint should never return *every* wallet. *Pagination* is the standard fix: return one fixed-size **page** of rows at a time, plus enough metadata for the client to ask for the next one. PyFly's pagination types — `Pageable` (what page, what size, what sort), `Sort` (the ordering), and `Page[T]` (the slice plus metadata) — are inherited straight from the CRUD surface via `find_all(pageable)`.

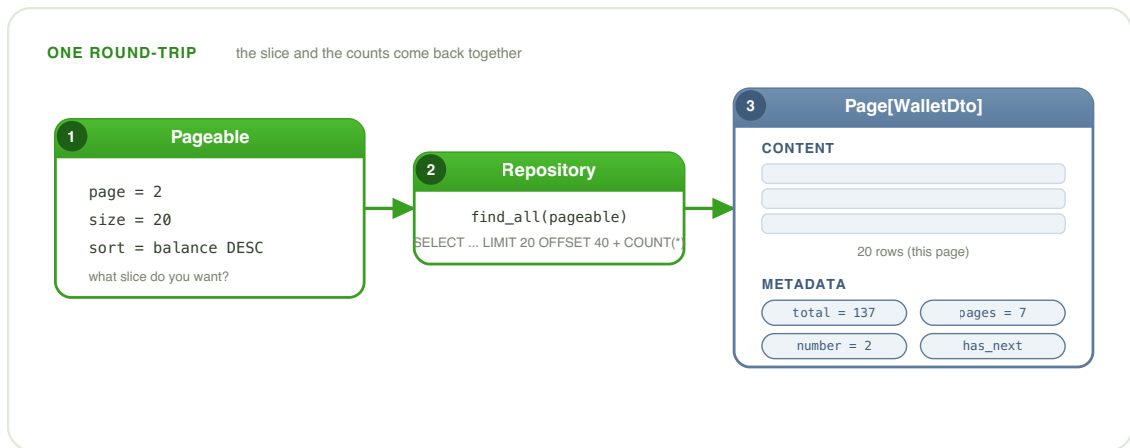


Figure 5.2 — One round-trip: a `Pageable` (page, size, sort) goes into `find_all`, and a `Page` comes back carrying the row slice plus the total counts the client needs to render pagination controls.

There are three small pieces to assemble: the handler that calls `find_all(pageable)`, the `Page[T]` it returns, and the controller that builds the `Pageable` from the request. We will take them in that order.

Lumen's `ListWallets` query handler is the whole story in three lines:

lumen/core/services/wallets/list_wallets_handler.py

```

@query_handler
@service
class ListWalletsHandler(
    QueryHandler[ListWallets, Page[WalletDto]]
):
    def __init__(self, repository: WalletRepository) → None:
        super().__init__()
        self._repository = repository

    async def do_handle( # type: ignore[override]
        self, query: ListWallets
    ) → Page[WalletDto]:
        page = await self._repository.find_all(
            query.pageable
        )
        return page.map(entity_to_dto)

```

Listing 5.4 — *Paginating with find_all(pageable), then mapping the page*

`find_all(pageable)` does three things in one call: it counts the total number of matching rows, applies the `Pageable` 's sort, and slices the result with `LIMIT / OFFSET`. It hands back a `Page[WalletEntity]`. The handler then calls `page.map(entity_to_dto)` to turn each row into a `WalletDto` **without losing the pagination metadata** — `.map` carries `total`, `page`, `size`, and the rest across to the new page.

A `Page[T]` exposes everything a client needs to render a pager:

Member	Meaning
<code>items</code>	The rows on this page (<code>list[T]</code>)
<code>total</code>	Total matching rows across all pages
<code>page</code>	Current page number (1-based)
<code>size</code>	Maximum items per page
<code>total_pages</code>	<code>ceil(total / size)</code>
<code>has_next</code>	Whether a next page exists
<code>has_previous</code>	Whether a previous page exists
<code>map(fn)</code>	Transform items, preserving metadata

The `Pageable` itself is built at the edge — the controller turns `?page=&size=` query params into a `Pageable` with a shared newest-first `Sort` :

`lumen/web/controllers/wallet_controller.py`

```
#: Newest-first ordering shared by the list endpoints.
_NEWEST_FIRST = Sort.by("created_at").descending()

@get_mapping("")
async def list_wallets(
    self, page: QueryParam[int] = 1, size: QueryParam[int] = 20
) → PageDto[WalletDto]:
    """A page of wallets, newest first."""
    result = await self._queries.query(
        ListWallets(pageable=Pageable.of(page, size, _NEWEST_FIRST))
    )
    return PageDto.from_page(result)
```

Listing 5.5 — Building a `Pageable` from query params (controller)

`Sort.by("created_at").descending()` names the column and the direction;
`Pageable.of(page, size, sort)` packages it with the page coordinates. The handler returns a framework `Page`, and the controller folds it into a serialisable `PageDto` — a plain Pydantic mirror of the page — so `GET /api/v1/wallets?page=1&size=20` returns JSON like `{"items": [...], "total": 42, "page": 1, "total_pages": 3, "has_next": true, ...}`.

❗ RUN IT: PAGE THROUGH THE WALLETS

With the relational layer turned on (the "Turning it on" section below switches it on; the Lumen sample ships it already enabled), open a couple of wallets, then ask for the first page. From a running app:

```
curl -s 'localhost:8080/api/v1/wallets?page=1&size=20'
```

The response carries the rows *and* the pager metadata — note `total`, `page`, `total_pages`, and `has_next` alongside `items`:

```
{
  "items": [
    {
      "id": "wlt-...",
      "owner_id": "alice",
      "currency": "EUR",
      "balance_minor": 0,
      "balance": 0.0,
      "created_at": "..."
    }
  ],
  "total": 1,
  "page": 1,
  "size": 20,
  "total_pages": 1,
  "has_next": false,
  "has_previous": false
}
```

Those are exactly the `Page[T]` members from the table above, serialised by `PageDto`. The client renders a pager straight from this shape — no extra count query needed.

Specifications: composable, reusable filters

Derived queries answer fixed questions. Sometimes you want a **reusable predicate** — a `WHERE` condition you can name once and reuse — that you compose at the call site: "wallets with at least this balance," combined freely with other conditions. That is what a `Specification` is: a small object wrapping a `WHERE` fragment, composable with `&` (AND), `|` (OR), and `~` (NOT).

We build it in two steps: a factory that *returns* a `Specification`, then a repository method that *runs* it.

Step 1 — write a factory that returns a `Specification`. It takes the parameter (the minimum balance) and returns a predicate object. **Step 2 — add a repository method that runs it** through the inherited `find_all_by_spec_paged`. Lumen does both in one file:

lumen/models/repositories/wallet_repository.py

```

def balance_at_least(min_minor: int) → Specification[WalletEntity]:
    """Wallets whose balance is at least *min_minor*.

    Returned as a Specification, so it composes via & / | / ~ and
    runs through find_all_by_spec / find_all_by_spec_paged.
    """
    return Specification(
        lambda root, q: q.where(root.balance_minor ≥ min_minor)
    )

@repository
class WalletRepository(Repository[WalletEntity, str]):

    async def find_rich(
        self, min_minor: int, pageable: Pageable
    ) → Page[WalletEntity]:
        """A page of wallets with balance ≥ min_minor."""
        return await self.find_all_by_spec_paged(
            balance_at_least(min_minor), pageable
        )

```

Listing 5.6 — A Specification factory and a method that runs it paged

A `Specification` wraps a callable `(root, q) → q` — given the entity class (`root`) and a SQLAlchemy `Select`, it returns the statement with a predicate added.

`balance_at_least(1000)` yields the predicate `balance_minor ≥ 1000`. Because specifications compose with Python operators, you can build arbitrarily complex filters from small pieces:

```

rich = balance_at_least(1000)
in_eur = Specification(
    lambda root, q: q.where(root.currency == "EUR")
)
rich_eur = rich & in_eur           # AND
rich_or_eur = rich | in_eur       # OR
not_rich = ~rich                  # NOT

```

You run a specification two ways. `find_all_by_spec(spec)` returns every matching row as a list; `find_all_by_spec_paged(spec, pageable)` applies the predicate, counts the matches, sorts, and slices — returning a `Page[T]`. `find_rich` uses the paged form, so the rich-wallets endpoint is itself paginated. The handler mirrors the list handler exactly, mapping rows to DTOs:

lumen/core/services/wallets/list_rich_wallets_handler.py

```

@query_handler
@service
class ListRichWalletsHandler(
    QueryHandler[ListRichWallets, Page[WalletDto]]
):
    def __init__(self, repository: WalletRepository) → None:
        super().__init__()
        self._repository = repository

    async def do_handle( # type: ignore[override]
        self, query: ListRichWallets
    ) → Page[WalletDto]:
        page = await self._repository.find_rich(
            query.min_minor, query.pageable
        )
        return page.map(entity_to_dto)

```

Listing 5.7 — The rich-wallets handler runs the Specification path

`GET /api/v1/wallets/rich?min_minor=1000&page=1&size=20` now returns a page of wallets at or above €10.00, newest first.

RUN IT: FILTER TO THE RICH WALLETS

Open one wallet and deposit €25.00 into it (2500 minor units), open another and leave it empty, then ask for wallets with at least €10.00:

```
curl -s 'localhost:8080/api/v1/wallets/rich?
min_minor=1000&page=1&size=20'
```

Only the funded wallet comes back, and `total` counts just the matches — the empty wallet is filtered out by the `Specification`:

```
{ "items": [ { "id": "wlt-...", "balance_minor": 2500, "balance":
25.0, [...] },
  "total": 1, "page": 1, "total_pages": 1, "has_next": false }
```

The same predicate (`balance_at_least`) that runs here can be combined with others using `&`, `|`, and `~` — that is the payoff of writing the filter as a `Specification` instead of a one-off query.

FILTERS WITHOUT LAMBDAS

For the common case — equality and a handful of comparisons — you don't even need to write a lambda. `FilterOperator.gte("balance_minor", 1000) & FilterOperator.eq("currency", "EUR")` produces the same composable `Specification` from static factory methods, and `FilterUtils.by(currency="EUR")` builds one from keyword arguments (Query-by-Example). Lumen uses an explicit lambda here because the intent reads clearly; both styles produce a `Specification` you can pass to the same repository methods.

Projections: read only the columns you need

The balance endpoint does not need the whole row — just the id, currency, and a computed balance. PyFly supports **interface projections**, Spring Data's idea of declaring the subset of fields a read-view wants and letting the framework copy exactly those. (A *projection* is a read-only view onto a subset of an entity's columns — you name the few fields you care about, and the framework copies only those, leaving the rest of the row unread.)

Building one takes three pieces: the projection class, the mapper that knows how to fill it, and the handler that uses it. We take them in order.

Step 1 — declare the projection. A projection is a class marked `@projection`. In Lumen it is a concrete dataclass:

lumen/interfaces/dtos/v1/balance_dto.py

```

from dataclasses import dataclass

from pyfly.data import projection

@projection
@dataclass
class BalanceView:
    """Projection: just the fields the balance view needs.

    id, currency and balance_minor are copied straight from the
    WalletEntity; balance is a computed major-unit decimal supplied
    by a registered transform on the mapper.
    """

    id: str
    currency: str
    balance_minor: int
    balance: float

```

Listing 5.8 — BalanceView: a @projection of just the balance fields

Step 2 — register the projection on a Mapper. A `Mapper` is the framework helper that copies entity fields onto a projection. It reads those four fields off a `WalletEntity` and constructs the view. Three (`id`, `currency`, `balance_minor`) are copied straight across; the fourth (`balance`, the major-unit decimal) is supplied by a *transform* — a small function registered against a field name that computes a value the entity does not store directly:

lumen/core/mappers/wallet_mapper.py

```

from pyfly.data import Mapper

_mapper = Mapper()
_mapper.register_projection(
    WalletEntity,
    BalanceView,
    transforms={"balance": lambda e: round(e.balance_minor / 100, 2)},
)

def entity_to_balance_dto(entity: WalletEntity) → BalanceDto:
    """Project a row onto the balance DTO via the projection."""
    view = _mapper.project(entity, BalanceView)
    return BalanceDto(
        id=view.id,
        currency=Currency(view.currency),
        balance_minor=view.balance_minor,
        balance=view.balance,
    )

```

Listing 5.9 — Registering and running the projection via Mapper

Step 3 — use the projection from a read handler. `Mapper.project(entity, BalanceView)` reads only the declared fields, applies the `balance` transform, and returns a `BalanceView`. The query handler then loads the row by id and projects it:

lumen/core/services/wallets/get_balance_handler.py

```

@query_handler
@service
class GetBalanceHandler(QueryHandler[GetBalance, BalanceDto | None]):
    def __init__(self, repository: WalletRepository) → None:
        super().__init__()
        self._repository = repository

    async def do_handle( # type: ignore[override]
        self, query: GetBalance
    ) → BalanceDto | None:
        entity = await self._repository.find_by_id(query.wallet_id)
        return (
            entity_to_balance_dto(entity)
            if entity is not None
            else None
        )

```

Listing 5.10 — The balance read handler: find by id, then project

i RUN IT: READ JUST THE BALANCE

The balance endpoint returns only the four projected fields, not the whole row. Against a running app with a funded wallet:

```
curl -s localhost:8080/api/v1/wallets/wlt-.../balance
```

```
{"id": "wlt-...", "currency": "EUR", "balance_minor": 2500, "balance": 25.0}
```

No `owner_id`, no `created_at` — the projection declared four fields, so four fields are read and returned. `balance` (the major-unit `25.0`) is the computed transform; the rest are copied straight from the row.

⚠ A PROJECTION MUST BE INSTANTIABLE

Spring lets a projection be a bare interface and returns a runtime proxy. Python has no such proxy, so a PyFly projection must be a **concrete** type the mapper can construct — here, a `@dataclass`. Marking a `Protocol` `@projection` will not work: a `Protocol` cannot be instantiated, and `Mapper.project` has nothing to build. Use a dataclass (or any plain class with matching fields) and you are safe.

Transactions and the aggregate seam

The repository surface is clean — but two honest subtleties decide whether your writes actually survive. Both come from how the framework manages the session, and Lumen handles both deliberately.

save() flushes; it does not commit

This is the single most important thing to understand about the data layer. Two database verbs are easy to confuse. To **flush** is to send the pending SQL (the `INSERT` / `UPDATE`) to the database so it is visible to *this* connection's later reads — but still inside an open transaction that can be undone. To **commit** is to make those changes permanent and visible to everyone. A flush without a commit is rolled back when the session closes.

The framework uses **one shared** `AsyncSession`, and `Repository.save()` calls `session.add()` followed by `session.flush()` and `session.refresh()` — it **flushes**, making the write visible *within* the current session, but it never **commits**. If nothing commits, the write is rolled back when the session closes and the wallet does not survive a restart. (This is exactly the disappearing-wallet problem from the opening **Run it** box.)

The commit happens at the **unit-of-work boundary**. (A *unit of work* is one all-or-nothing batch of changes: either every write in it commits together, or — if anything fails — none of them do.) You declare that boundary with `@transactional()`. A handler that writes decorates its `do_handle` with `@transactional()`, injects the `async_sessionmaker` — the factory that hands out sessions — as `self._session_factory`, and the decorator opens a unit of work, swaps that transactional session onto the repository for the call, **commits on success**, and rolls back on failure:

lumen/core/services/wallets/open_wallet_handler.py

```

@command_handler
@service
class OpenWalletHandler(CommandHandler[OpenWallet, str]):
    """Open a new, empty wallet."""

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events
        # @transactional resolves the unit-of-work session from here.
        self._session_factory = session_factory

    @transactional()
    async def do_handle( # type: ignore[override]
        self, command: OpenWallet
    ) → str:
        wallet_id = f"wlt-{uuid4()}"
        wallet = Wallet.open(
            wallet_id=wallet_id,
            owner_id=command.owner_id,
            currency=command.currency,
        )
        await self._repository.upsert(to_entity(wallet))

        await publish_domain_events(
            self._events, wallet.clear_events()
        )
        return wallet_id

```

Listing 5.11 — A write handler: *@transactional()* commits the unit of work

`@transactional()` (imported from `pyfly.data.relational.sqlalchemy`) resolves the `async_sessionmaker` from `self._session_factory`, runs the body inside a `session.begin()` block, and commits at the end. Drop the decorator and the `upsert` would only flush — the wallet would never reach disk. The read handlers earlier in this chapter need no `@transactional`: a read makes no changes to commit.

WHAT JUST HAPPENED

The rule in one line: **reads need nothing; writes need `@transactional()` . `save` / `upsert` only *flush***, so a write handler must run inside a unit of work that commits. The decorator does three things for you — opens the transaction, hands the repository the right session for the call, and commits (or rolls back on an exception). This is why the opening wallet survived once persistence was on, and why removing the decorator would silently lose it.

upsert, not save, for an aggregate that owns its id

Notice the handler calls `self._repository.upsert(...)`, not `save(...)`. (*Upsert* is the portmanteau verb for "insert if new, update if already present" — one call for both cases.) That is the second subtlety. The framework's `save()` issues `session.add()`, which SQLAlchemy treats as a **pending INSERT**. But the `Wallet` aggregate generates its *own* primary key up front (`wlt-...`), so by the time a deposit or withdrawal persists an already-loaded wallet, a row with that id already exists — and a second **INSERT** on the same primary key raises `IntegrityError`.

The fix is `session.merge`, which inserts when the id is new and updates when it already exists. Lumen wraps it in an `upsert` convenience method:

lumen/models/repositories/wallet_repository.py

```

@repository
class WalletRepository(Repository[WalletEntity, str]):

    async def upsert(self, entity: WalletEntity) → WalletEntity:
        """Insert *entity* or update the existing row with the same id.

        Uses session.merge so a freshly-mapped entity carrying the
        aggregate's id persists whether or not a row already exists –
        the aggregate owns its primary key, so identity is never
        ambiguous. Flushes so the write is visible in the current
        unit of work; the surrounding @transactional commits it.
        """
        session = self._require_session()
        merged = await session.merge(entity)
        await session.flush()
        return merged

```

Listing 5.12 — upsert: one call for both INSERT and UPDATE

`_require_session()` is the inherited accessor that returns the active session (the transactional one, once `@transactional` has swapped it in). `merge` keys on the primary key, so both the first write (open) and every later write (deposit, withdraw) take the same code path with no `IntegrityError`. For entities whose ids are database-generated, `save` is the natural choice; for an aggregate that owns its id, `upsert` is.

WHAT JUST HAPPENED

Two questions decide every write: *did this row commit?* and *did this write insert or update?* `@transactional()` answers the first (it commits the unit of work); `upsert` / `merge` answers the second (one code path for both INSERT and UPDATE, because the aggregate owns its id). Get both right and a wallet you open, then deposit into, then read back after a restart returns the correct balance — which is exactly what the repository test below asserts against a *fresh* engine.

The aggregate ↔ entity mapper seam

There is one more boundary, and it is a feature, not an accident. Lumen keeps two distinct types:

- `Wallet` — the DDD *aggregate root* from Chapter 6. It owns the `balance ≥ 0` invariant, exposes intent-revealing methods (`open` , `deposit` , `withdraw`), and raises domain events. It knows nothing about SQLAlchemy.
- `WalletEntity` — the *persistence row*. It is a flat SQLAlchemy model with columns and no behaviour.

A small mapper bridges them — one pure function each way:

lumen/core/mappers/wallet_mapper.py

```

def to_entity(wallet: Wallet) → WalletEntity:
    """Flatten a Wallet aggregate into a persistable row."""
    assert wallet.id is not None
    return WalletEntity(
        id=wallet.id,
        owner_id=wallet.owner_id,
        currency=wallet.currency.value,
        balance_minor=wallet.balance.amount,
        created_at=wallet.created_at,
    )

def to_aggregate(entity: WalletEntity) → Wallet:
    """Rehydrate a Wallet aggregate from a persistence row."""
    currency = Currency(entity.currency)
    return Wallet(
        id=entity.id,
        owner_id=entity.owner_id,
        balance=Money(amount=entity.balance_minor, currency=currency),
        created_at=entity.created_at,
    )

```

Listing 5.13 — The aggregate ↔ row mapper

The write side calls `to_entity` before `upsert`; the read side either rehydrates with `to_aggregate` (when a command needs the rich aggregate) or projects straight to a DTO (when a query only needs data). Keeping the row separate from the aggregate means persistence concerns — column types, nullability, the merge dance — never leak into the domain model, and the domain's invariants never constrain the table schema. The repository stores rows; the mapper is the seam that keeps the aggregate pure.

i REHYDRATION SKIPS THE FACTORY

`to_aggregate` calls the `Wallet` **constructor** directly, never the `Wallet.open` factory. The factory is for *new* wallets: it validates inputs and raises a `WalletOpened` event. A row loaded from the database already represents a valid, committed wallet — re-running the factory would re-fire that event and re-check rules that passed long ago. The constructor sets fields quietly, producing a `Wallet` indistinguishable from a freshly opened one but with no spurious events.

Turning it on

Activating the relational layer is configuration, not code — three keys in `pyfly.yaml`. Add a `data.relational` block:

`pyfly.yaml`

```
pyfly:
  data:
    relational:
      enabled: true
      url: "sqlite+aiosqlite:///./lumen.db"
      ddl-auto: create
```

Listing 5.14 — Relational data layer configuration

`enabled: true` activates the relational auto-configuration, which builds the async SQLAlchemy engine and the `async_sessionmaker`, registers the `AsyncSession` and `session_factory` beans the repository and handlers inject, and installs the `RepositoryBeanPostProcessor` that compiles your derived-query stubs. `url` is a standard SQLAlchemy connection string — SQLite via `aiosqlite` here for zero-infrastructure development, `postgresql+asyncpg://...` in production. `ddl-auto:`

`create` runs `Base.metadata.create_all` on startup, so the `wallets` table (discovered because `WalletEntity` subclasses `Base`) is built automatically the first time the app boots.

The dependency footprint is tiny: `pyfly[data-relational]` pulls in `sqlalchemy[asyncio]` and `aiosqlite`, and nothing else. No database server, no driver install — which is exactly why the sample runs anywhere.

RUN IT: THE DISAPPEARING WALLET, FIXED

Re-run the opening experiment, now with persistence on. Open a wallet, stop the app, start it again, and read the balance back:

```
# Terminal 1
uv run pyfly run --server uvicorn

# Terminal 2 - open a wallet
curl -s -X POST localhost:8080/api/v1/wallets \
  -H 'content-type: application/json' \
  -d '{"owner_id": "alice", "currency": "EUR"}'
# → {"wallet_id": "wlt-..."}

# Ctrl+C in Terminal 1, then start it again, then:
curl -s localhost:8080/api/v1/wallets/wlt-.../balance
```

This time the wallet survives — its row was committed to `lumen.db` on disk:

```
{"id": "wlt-...", "currency": "EUR", "balance_minor": 0, "balance": 0.0}
```

Look in the project directory and you will see the `lumen.db` SQLite file the engine created on first boot, with the `wallets` table inside it. The whole chapter comes down to this: the wallet outlives the process.

💡 SCHEMA LIFECYCLE IN PRODUCTION

`ddl-auto: create` is right for development and samples: it creates missing tables and leaves existing ones alone. In production set `ddl-auto: none` and manage the schema with a migration tool (Alembic), which generates versioned scripts from the diff between `Base.metadata` and the live database. The application code does not change — only the `ddl-auto` setting and the migration pipeline.

Proving it works

Because the repository is an ordinary class, you can test it directly against a real SQLite file — no application context, no HTTP. Lumen's repository test creates a temp database, runs `Base.metadata.create_all`, and exercises the surface end to end, including the `RepositoryBeanPostProcessor` that compiles the derived query (the same processor the live `ApplicationContext` runs):

```
lumen/tests/test_sql_wallet_repository.py
```

```
def _make_repo(session: AsyncSession) → WalletRepository:
    repo = WalletRepository(WalletEntity, session)
    # Mirror the ApplicationContext: compile derived-query stubs.
    RepositoryBeanPostProcessor().after_init(repo, "walletRepository")
    return repo
```

```
@pytest.mark.asyncio
```

```
async def test_derived_find_by_owner_id(sqlite_factory) → None:
    factory, _ = sqlite_factory
    async with factory() as session:
        repo = _make_repo(session)
        await repo.upsert(_entity("wlt-1", "alice", 100))
        await repo.upsert(_entity("wlt-2", "alice", 200))
        await repo.upsert(_entity("wlt-3", "bob", 300))
        await session.commit()

        owned = await repo.find_by_owner_id("alice")
        assert sorted(w.id for w in owned) == ["wlt-1", "wlt-2"]
        assert await repo.find_by_owner_id("nobody") == []
```

```
@pytest.mark.asyncio
```

```
async def test_specification_find_rich_paged_and_sorted(
    sqlite_factory,
) → None:
    factory, _ = sqlite_factory
    async with factory() as session:
        repo = _make_repo(session)
        # age_days drives created_at for newest-first ordering.
        await repo.upsert(_entity("wlt-poor", "a", 50, age_days=3))
        await repo.upsert(_entity("wlt-mid", "b", 1000, age_days=2))
        await repo.upsert(_entity("wlt-rich", "c", 5000, age_days=1))
        await session.commit()

        # balance_minor ≥ 1000, newest first, page size 1.
        newest_first = Sort.by("created_at").descending()
```

```

page = await repo.find_rich(1000, Pageable.of(1, 1, newest_first))
assert page.total == 2 # mid + rich
assert page.total_pages == 2
assert page.has_next is True
assert [w.id for w in page.items] == ["wlt-rich"]

# The bare predicate also works through find_all_by_spec.
rich = await repo.find_all_by_spec(balance_at_least(5000))
assert [w.id for w in rich] == ["wlt-rich"]

```

Listing 5.15 — Testing CRUD, the derived query, and the Specification path

The first test drives the derived query: three wallets in, two owners out, and `find_by_owner_id("alice")` returns exactly the two — proof that the framework compiled `WHERE owner_id = :owner_id` from the method name. The second drives the `Specification` path: it asserts the threshold filter (`total == 2`, only mid and rich match `>= 1000`), the newest-first sort (`wlt-rich` is the newest of the two), the page metadata (`total_pages == 2`, `has_next`), and that the same `balance_at_least` predicate also runs unpaged through `find_all_by_spec`.

The fixture mirrors what the framework does at startup — build the engine, run `Base.metadata.create_all` inside a `begin()` block so the DDL commits, hand back a session factory — so the test exercises the exact table the application creates. Other tests in the same file prove `upsert` round-trips through a *fresh* engine (durability across reconnect) and that `find_all(pageable)` counts and slices a five-wallet table correctly.

📄 RUN IT: PROVE THE WHOLE LAYER GREEN

Run the repository test file end to end. From the Lumen project root:

```
uv run --extra dev pytest tests/test_sql_wallet_repository.py -v
```

Each named test reports **PASSED** — CRUD round-trip, durability across reconnect, the derived query, the specification path, and pagination:

```
tests/
test_sql_wallet_repository.py::test_upsert_inserts_then_updates_and_persi
sts PASSED
tests/test_sql_wallet_repository.py::test_find_by_id_unknown_returns_none
PASSED
tests/test_sql_wallet_repository.py::test_derived_find_by_owner_id PASSED
tests/
test_sql_wallet_repository.py::test_specification_find_rich_paged_and_sor
ted PASSED
tests/
test_sql_wallet_repository.py::test_find_all_pageable_counts_and_pages
PASSED
6 passed in 0.31s
```

Run the whole suite (`uv run --extra dev pytest -q`) to confirm the rest of Lumen still passes alongside the persistence layer.

🍃 SPRING PARITY

Constructing the repository directly against a real in-process database mirrors Spring's `@DataJpaTest` slice, which boots an H2 database and the JPA layer in isolation to test repositories without the full context. `Base.metadata.create_all` is the analogue of `spring.jpa.hibernate.ddl-auto=create`, and running `RepositoryBeanPostProcessor` by hand stands in for the Spring proxy that materialises derived queries on a `JpaRepository` at startup.

✓ What you built

Lumen now persists wallets through PyFly's Spring-Data-style repository layer:

- **Entity** — `WalletEntity(Base)`, a SQLAlchemy 2.0 row with a string primary key (the aggregate's own id) and integer minor-unit balances.
- **Repository** — `WalletRepository(Repository[WalletEntity, str])`, marked `@repository`. The framework supplies full async CRUD (`save`, `find_by_id`, `find_all`, `delete` / `delete_by_id`, `delete_all` / `delete_all_by_id`, `count`, `exists_by_id`, `save_all`, `find_all_by_id`, `stream_all`, pagination, specifications) with no hand-written adapter.
- **Derived query** — `find_by_owner_id`, declared as a `...` stub and compiled from its name by the `RepositoryBeanPostProcessor`.
- **Pagination** — `find_all(pageable)` returning a `Page[T]` with `total` / `total_pages` / `has_next`, mapped to DTOs with `Page.map`, exposed at `GET /api/v1/wallets`.
- **Specification** — `balance_at_least(n)` composed with `&` / `|` / `~` and run via `find_all_by_spec_paged`, exposed at `GET /api/v1/wallets/rich`.
- **Projection** — `@projection BalanceView`, a concrete dataclass the `Mapper` projects rows onto for the balance read view.
- **Transactions** — write handlers decorated `@transactional()` (because `save` / `upsert` only *flush*), using `upsert` / `session.merge` for an aggregate that owns its id, with the aggregate ↔ entity mapper keeping the domain model pure.

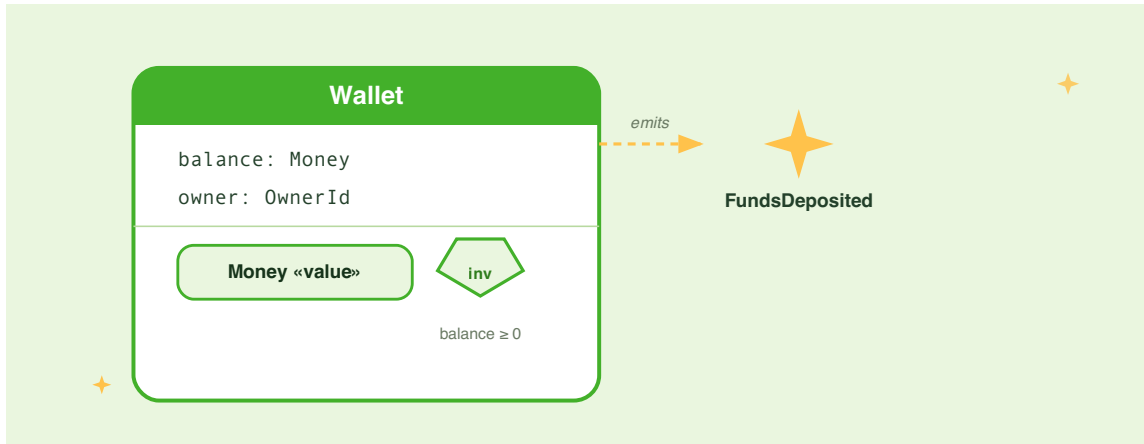
You wrote interfaces and stubs; the framework wrote the SQL. That is the payoff of the repository pattern.

Try it yourself

- 1. Add a derived counter.** Declare `async def count_by_currency(self, currency: str) → int: ...` on `WalletRepository` (body `...`). Write a test that upserts wallets in two currencies and asserts the count for each — confirming the `count_by` prefix compiles to `SELECT COUNT(*) ... WHERE currency = :currency` with no SQL on your part.
- 2. Compose two specifications.** Define a second factory `in_currency(code: str) → Specification[WalletEntity]` (predicate `currency == code`), then add a repository method that runs `balance_at_least(min_minor) & in_currency(code)` through `find_all_by_spec_paged`. Test that it returns only rich wallets in the chosen currency, newest first.
- 3. Trace the transaction boundary.** Temporarily change `OpenWalletHandler.do_handle` to call `self._repository.save(to_entity(wallet))` instead of `upsert`, open the same wallet twice in one test, and observe the `IntegrityError`. Restore `upsert`. Then remove the `@transactional()` decorator, open a wallet, and assert it does **not** survive a fresh-engine reconnect — proving that without the unit-of-work commit, `flush` alone is not durability.
- 4. Project a different view.** Add an `@projection OwnerView` dataclass with just `id` and `owner_id`, register it on a `Mapper`, and write a handler-free test that loads a `WalletEntity` and projects it — verifying that only the two declared columns are read and the rest of the row is ignored.

CHAPTER 6

Domain-Driven Design



Lumen's wallet feature works. Deposits land, balances update, and the database persists everything across restarts. But look closely at the service layer and you will notice something uncomfortable: the overdraft check lives in the service method, not in the wallet itself. The currency validation is a property comparison scattered across a handful of `if`-statements. Nothing stops a future developer — or a future you at 11 pm — from bypassing those guards by calling `repo.save(entity)` directly.

Domain-Driven Design addresses this by making the model responsible for its own rules. Data stops being a passive bag of values that any caller can mutate; it becomes an object with opinions — one that enforces its invariants, announces what happened, and cooperates with the persistence layer while remaining free of any database import.

This chapter promotes the wallet into a proper DDD aggregate: a `Money` value object, a `Wallet` aggregate root that guards the overdraft rule and the currency-match rule, and a set of `DomainEvent`s emitted every time the wallet changes state. A thin mapper converts between the rich domain model and the flat persistence record — neither side needs to know the shape of the other.

Entities and value objects

Before you can build a model that enforces its own rules, you need a vocabulary for the two fundamentally different kinds of objects that appear in every domain.

Think about what makes two wallets distinct. Even if two wallets happen to hold exactly one hundred euros, they are still separate wallets belonging to separate owners — you care *which one* you have. Now think about the amount itself. One hundred euros is one hundred euros; the exact Python object that holds the value is irrelevant. If a deposit adds fifty euros to a wallet's balance, you do not want to update the existing amount in place — you want to derive a brand-new amount that records the result. Mutating in place invites aliasing bugs where two parts of the code unknowingly share a reference to the same object and see each other's changes.

JARGON, IN PLAIN LANGUAGE

A few words recur throughout this chapter. An **invariant** is a rule that must always hold — for a wallet, "the balance is never negative." An **aggregate** is a small cluster of objects that change together and must stay consistent as a group. An **aliasing bug** happens when two pieces of code accidentally hold the *same* object and one mutates it, surprising the other. Keep these three in mind; the rest of the chapter is largely about preventing the last one and guaranteeing the first inside the second.

DDD names these two roles **entities** and **value objects**, and PyFly's `pyfly.domain` module makes them first-class concepts:

Concept	PyFly base	Equality	Mutation
<code>Entity[TID]</code>	<code>Entity</code>	Identity — equal only when <code>id</code> matches	Allowed through owned methods
<code>ValueObject</code>	<code>ValueObject</code>	Structural — all fields compared	Forbidden; <code>replace(**changes)</code> creates a new instance

Transient entities (those with `id=None`) compare equal only by Python's object identity, so you can safely put entities in sets and dicts without worrying about hash collisions from unsaved objects.

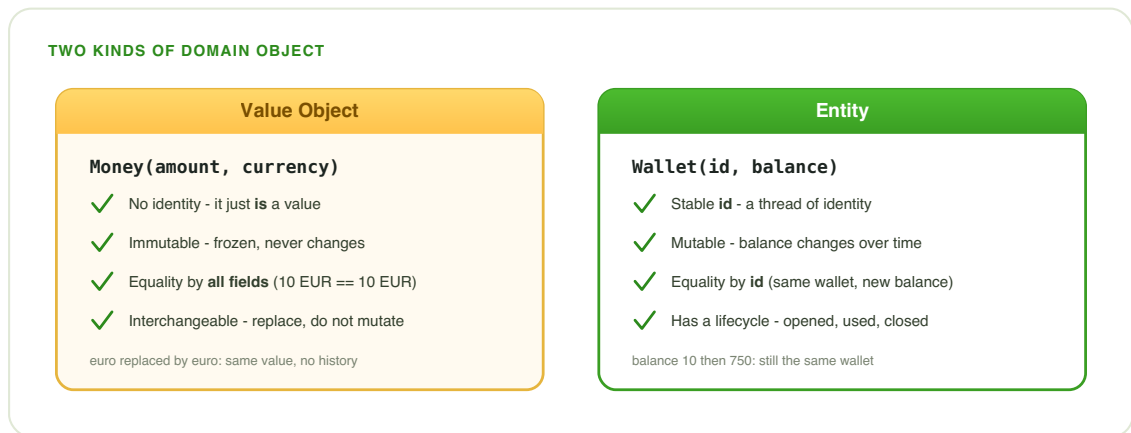


Figure 6.1 — Two kinds of domain object: a Money value object (no identity, immutable, compared by its fields) versus a Wallet entity (a stable identity, mutable, compared by id).

Money is the textbook value object. An amount of one hundred euros is not a specific object you track over time; it is a value. Two separate `Money(100, "EUR")` instances are equal. A deposit does not mutate the existing amount — it produces a new one, leaving the original untouched and the model free of hidden side-effects.

Here is the `Money` value object for Lumen:

```
lumen/models/entities/v1/money.py
```

```
from __future__ import annotations

from dataclasses import dataclass

from lumen.interfaces.enums.v1.currency import Currency
from pyfly.domain import BusinessRuleViolation, ValueError

@dataclass(frozen=True)
class Money(ValueObject):
    """An exact monetary amount in a single currency.

    ``amount`` is in minor units (e.g. cents): ``Money(1050,
    Currency.EUR)`` is €10.50. Arithmetic returns new ``Money``
    instances and refuses to mix currencies.
    """

    amount: int
    currency: Currency

    def __post_init__(self) → None:
        if not isinstance(self.amount, int) or isinstance(self.amount, bool):
            raise BusinessRuleViolation(
                "money-amount-integer",
                "amount must be an integer number of minor units",
            )

    @classmethod
    def zero(cls, currency: Currency) → Money:
        """The additive identity for *currency* (a zero balance)."""
        return cls(amount=0, currency=currency)

    def add(self, other: Money) → Money:
        """Return ``self + other``; both must share a currency."""
        self._assert_same_currency(other)
        return Money(amount=self.amount + other.amount,
            currency=self.currency)
```

```

def subtract(self, other: Money) → Money:
    """Return `self - other`; both must share a currency."""
    self._assert_same_currency(other)
    return Money(amount=self.amount - other.amount,
currency=self.currency)

@property
def is_positive(self) → bool:
    return self.amount > 0

@property
def is_negative(self) → bool:
    return self.amount < 0

@property
def major_units(self) → float:
    """The amount rendered as a major-unit decimal (cents / 100)."""
    return round(self.amount / 100, 2)

def _assert_same_currency(self, other: Money) → None:
    if self.currency is not other.currency:
        raise BusinessRuleViolation(
            "money-currency-mismatch",
            f"cannot combine {self.currency.value} "
            f"with {other.currency.value}",
        )

def __str__(self) → str:
    return f"{self.major_units:.2f} {self.currency.value}"

```

Listing 6.1 — Money: an immutable value object with currency-aware arithmetic

How it works. The amount is stored in **minor units** — integer cents, pence, or whatever the currency's smallest denomination is — to eliminate floating-point rounding entirely. Financial calculations that use `float` are a chronic source of off-by-one-cent bugs that only surface in production, usually during reconciliation. Storing €10.50 as

`amount=1050` keeps all arithmetic exact. `__post_init__` rejects non-integer amounts immediately with `BusinessRuleViolation("money-amount-integer")`, so a stray float like `10.5` never silently enters the model.

The `currency` field uses the `Currency` enum (`Currency.EUR`, `Currency.USD`, `Currency.GBP`) rather than a bare string, ruling out typos at construction time. Currency comparisons inside `_assert_same_currency` use Python's identity check (`is`) — raising `BusinessRuleViolation("money-currency-mismatch")` if they differ — so the error surfaces exactly where the mistake was made.

Both `add` and `subtract` return a *new* `Money` instance rather than modifying `self`, a direct consequence of `frozen=True`. This immutability guarantee means the aggregate holding a `Money` value can never be partially updated: either the whole replacement succeeds or the old value remains in place. The `is_positive` and `is_negative` properties expose the sign without leaking the raw integer. `major_units` converts to a decimal for display, and `__str__` formats as `"10.50 EUR"` via `currency.value`.

Build it step by step. If you are creating `Money` from scratch, here is the order to write it in, and why each line matters.

Step 1 — Declare the two fields and freeze the class. Add `amount: int` and `currency: Currency` to a class decorated with `@dataclass(frozen=True)` that subclasses `ValueObject`. `frozen=True` is what makes the object immutable and gives you structural equality for free: two `Money` instances with the same amount and currency are now `==`.

Step 2 — Guard the constructor. Add `__post_init__` to reject anything that is not a plain integer. The `isinstance(self.amount, bool)` check is deliberate — in Python `True` is an `int`, and you do not want `Money(True, ...)` slipping through.

Step 3 — Add the `zero` factory. A wallet opens at a zero balance, so

`Money.zero(currency)` reads better at the call site than `Money(0, currency)`.

Step 4 — Add `add` and `subtract`, each routing through `_assert_same_currency` first. Returning a brand-new `Money` (never mutating `self`) is what prevents the aliasing bug from the section opener.

Step 5 — Add the display helpers: `is_positive`, `is_negative`, `major_units`, and `__str__`.

Run it. `Money` has no dependency on the framework runtime or a database, so you can exercise every rule from a Python prompt. Start one with `uv run python` from the `samples/lumen` directory and type:

```
>>> from lumen.models.entities.v1.money import Money
>>> from lumen.interfaces.enums.v1.currency import Currency
>>> ten_fifty = Money(1050, Currency.EUR)
>>> str(ten_fifty)
'10.50 EUR'
>>> ten_fifty.add(Money(450, Currency.EUR))
Money(amount=1500, currency=<Currency.EUR: 'EUR'>)
>>> ten_fifty.add(Money(100, Currency.USD))
Traceback (most recent call last):
...
pyfly.domain.exceptions.BusinessRuleViolation: cannot combine EUR with USD
>>> Money(10.5, Currency.EUR)
Traceback (most recent call last):
...
pyfly.domain.exceptions.BusinessRuleViolation: amount must be an integer number of minor units
```

The two tracebacks are the point: the model refuses a cross-currency sum and a float amount *at the moment you make the mistake*, not three layers deeper during reconciliation.

Lumen ships these exact behaviours as tests. Run them with:

```
uv run --extra dev pytest tests/test_money.py -q
```

and you should see all six pass:

```
..... [100%]
6 passed in 0.00s
```

What just happened. You built a value object that is impossible to misuse: it cannot be mutated, it cannot mix currencies, and it cannot hold a float. Every other piece of the wallet model will lean on these guarantees, which is why `Money` comes first.

MINOR UNITS VS DECIMAL

Storing money as integer cents is one convention; another is Python's `decimal.Decimal` with a fixed scale. Both are valid. What matters is picking one and sticking to it within the bounded context. For Lumen, integer minor units keep the model free of import-time precision configuration, and `__post_init__` enforces the constraint with `BusinessRuleViolation("money-amount-integer")` so a float never silently enters the model.

SPRING PARITY

`ValueObject` mirrors the `@ValueObject` / `@Embeddable` cluster in Spring's JPA ecosystem and the `ValueObject` marker interface from Spring Modulith. The `frozen=True` dataclass maps to Java's `record` type introduced in Java 16 — immutable, value-based equality, concise syntax. jMolecules's `@ValueObject` annotation carries the same intent.

The aggregate root

`Money` solves the representation problem — amounts are now immutable and currency-aware. But Lumen still needs something to *own* the wallet's balance and decide when a deposit or withdrawal is permitted. That is the role of the **aggregate root**.

An entity becomes an aggregate root when it owns a cluster of related objects and acts as the single point of entry for all changes within that cluster. The aggregate root is the **consistency boundary**: no external code reaches inside and mutates an inner object directly. All changes flow through the root's methods, which enforce the rules. This is the design that prevents the 11 pm bypass described in the chapter introduction — once every change must flow through the root, there is no back-channel.

`AggregateRoot[TID]` extends `Entity[TID]` with one addition: an internal buffer of **pending domain events**. Every state-changing method calls `self.raise_event(event)` to record what happened. When the repository saves the aggregate, the application service drains that buffer with `clear_events()` and publishes the events to the event bus. You will see the full publish cycle in the Domain events section; for now, focus on the aggregate itself.

Here is the `Wallet` aggregate root:

```
lumen/models/entities/v1/wallet_entity.py
```

```
from __future__ import annotations

from dataclasses import dataclass
from datetime import UTC, datetime

from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.money import Money
from pyfly.domain import AggregateRoot, BusinessRuleViolation, DomainEvent
```

```
# — Domain events
```

```
@dataclass(frozen=True)
class WalletOpened(DomainEvent):
    wallet_id: str = ""
    owner_id: str = ""
    currency: str = ""
```

```
@dataclass(frozen=True)
class FundsDeposited(DomainEvent):
    wallet_id: str = ""
    amount: int = 0
    currency: str = ""
    balance: int = 0
```

```
@dataclass(frozen=True)
class FundsWithdrawn(DomainEvent):
    wallet_id: str = ""
    amount: int = 0
    currency: str = ""
    balance: int = 0
```

```
# — Aggregate root
```

```

class Wallet(AggregateRoot[str]):
    """Wallet aggregate root - owns the ``balance ≥ 0`` invariant."""

    __slots__ = ("owner_id", "balance", "created_at")

    def __init__(
        self,
        id: str,
        owner_id: str,
        balance: Money,
        created_at: datetime | None = None,
    ) → None:
        super().__init__(id)
        self.owner_id = owner_id
        self.balance = balance
        self.created_at = created_at or datetime.now(UTC)

    @property
    def currency(self) → Currency:
        return self.balance.currency

# — Factory method —————

    @classmethod
    def open(cls, wallet_id: str, owner_id: str, currency: Currency) →
Wallet:
    """Open a new, empty wallet; raises WalletOpened."""
    if not owner_id.strip():
        raise BusinessRuleViolation(
            "wallet-owner-required", "owner_id is required"
        )
    wallet = cls(
        id=wallet_id,
        owner_id=owner_id,
        balance=Money.zero(currency),
    )
    wallet.raise_event(

```

```

        WalletOpened(
            wallet_id=wallet_id,
            owner_id=owner_id,
            currency=currency.value,
        )
    )
    return wallet

# — Behaviour —————

def deposit(self, amount: Money) → None:
    """Credit *amount* to the balance; raises FundsDeposited."""
    self._assert_currency(amount)
    if not amount.is_positive:
        raise BusinessRuleViolation(
            "wallet-deposit-positive",
            "deposit amount must be > 0",
        )
    self.balance = self.balance.add(amount)
    assert self.id is not None
    self.raise_event(
        FundsDeposited(
            wallet_id=self.id,
            amount=amount.amount,
            currency=amount.currency.value,
            balance=self.balance.amount,
        )
    )

def withdraw(self, amount: Money) → None:
    """Debit *amount*; refuses to overdraw. Raises FundsWithdrawn."""
    self._assert_currency(amount)
    if not amount.is_positive:
        raise BusinessRuleViolation(
            "wallet-withdrawal-positive",
            "withdrawal amount must be > 0",
        )
    remaining = self.balance.subtract(amount)
    if remaining.is_negative:

```

```

        raise BusinessRuleViolation(
            "wallet-insufficient-funds",
            f"cannot withdraw {amount}; balance is {self.balance}",
        )
    self.balance = remaining
    assert self.id is not None
    self.raise_event(
        FundsWithdrawn(
            wallet_id=self.id,
            amount=amount.amount,
            currency=amount.currency.value,
            balance=self.balance.amount,
        )
    )

# — Helpers —————

def _assert_currency(self, amount: Money) → None:
    if amount.currency is not self.balance.currency:
        raise BusinessRuleViolation(
            "wallet-currency-mismatch",
            f"wallet holds {self.balance.currency.value}, "
            f"got {amount.currency.value}",
        )

```

Listing 6.2 — *Wallet: the aggregate root that owns balance and enforces its rules*

How it works. The aggregate boundary is enforced at three levels. First, `__slots__` locks the attribute set, and `balance` and `owner_id` are deliberately public-but-owned: only the aggregate's own methods (`deposit` , `withdraw`) mutate them, while the `currency` property is a read-only convenience that delegates to `balance.currency` . Second, the factory classmethod `open` is the sole legitimate way to create a new wallet: the caller supplies the `wallet_id` (so the application layer controls ID generation), `open` validates that `owner_id` is non-blank, initializes the balance with `Money.zero(currency)` , and immediately queues `WalletOpened` . Using a factory

rather than calling `__init__` directly ensures the opening event is *never* forgotten, even in a test fixture. Third, the domain events — `WalletOpened`, `FundsDeposited`, `FundsWithdrawn` — are frozen dataclasses. `FundsDeposited` and `FundsWithdrawn` carry a `balance` field (the post-operation balance in minor units), so a subscriber never needs to call back into the aggregate to learn the current state.

Build it step by step. The `Wallet` class has more moving parts than `Money`, so here is the order to assemble it.

Step 1 — Define the three event classes first (`WalletOpened`, `FundsDeposited`, `FundsWithdrawn`), each a `@dataclass(frozen=True)` subclass of `DomainEvent`. They have to exist before the aggregate can reference them. We cover events in depth in the next section but two; for now they are just the records the wallet will emit.

Step 2 — Declare the aggregate. Subclass `AggregateRoot[str]` (the `[str]` says the id is a string), set `__slots__` to lock the attribute names, and write `__init__` to store `owner_id`, `balance`, and `created_at`. Call `super().__init__(id)` first so the base class sets up the id and the internal event buffer.

Step 3 — Add the `currency` read-only property that delegates to `self.balance.currency`. This is a convenience that keeps callers from reaching two levels deep into `wallet.balance.currency`.

Step 4 — Write the `open` factory classmethod. It validates `owner_id`, builds the wallet with a `Money.zero(currency)` balance, and calls `self.raise_event(WalletOpened(...))`. Making `open` a classmethod — rather than expecting callers to use `__init__` plus a manual event — guarantees the opening event is never forgotten.

Step 5 — Write `deposit` and `withdraw`. Each one validates first (currency match, positive amount, and for withdraw, sufficient funds), *then* mutates `self.balance`, *then* raises its event. Order matters: never mutate before every check has passed, or you can leave the wallet in a half-changed state.

Step 6 — Add the private `_assert_currency` helper that both transitions share.

RAISE_EVENT IS NOT RAISING AN EXCEPTION

Despite the name, `raise_event` does not throw anything. It *appends* an event to an internal buffer on the aggregate (`AggregateRoot._pending_events`). Nothing is published at that moment. A later step — the application service, after a successful save — drains the buffer with `clear_events()` and hands the events to the event bus. Think of `raise_event` as "make a note that this happened", not "abort".

Run it. Like `Money`, the `Wallet` aggregate is pure Python — no database needed.

From `uv run python`:

```
>>> from lumen.models.entities.v1.wallet_entity import Wallet
>>> from lumen.models.entities.v1.money import Money
>>> from lumen.interfaces.enums.v1.currency import Currency
>>> w = Wallet.open("wlt-1", "owner-1", Currency.EUR)
>>> w.deposit(Money(1000, Currency.EUR))
>>> w.withdraw(Money(400, Currency.EUR))
>>> str(w.balance)
'6.00 EUR'
>>> [e.event_type for e in w.pending_events()]
['WalletOpened', 'FundsDeposited', 'FundsWithdrawn']
>>> w.withdraw(Money(9999, Currency.EUR))
Traceback (most recent call last):
...
pyfly.domain.exceptions.BusinessRuleViolation: cannot withdraw 99.99 EUR; balance is 6.00 EUR
```


Notice the three events sitting in the buffer after the successful operations, and notice that the overdraft attempt raised *before* touching the balance — `pending_events()` still shows three, not four. Lumen's test suite asserts exactly this:

```
uv run --extra dev pytest tests/test_wallet_aggregate.py -q
```

```
..... [100%]  
6 passed in 0.0Xs
```

What just happened. The wallet now owns its rules. There is no way to overdraw it, no way to feed it the wrong currency, and no way to change its state without leaving an event behind that records what happened. The service layer no longer has to remember any of that.

The diagram below shows the complete picture: state, invariants, and the events the wallet emits.

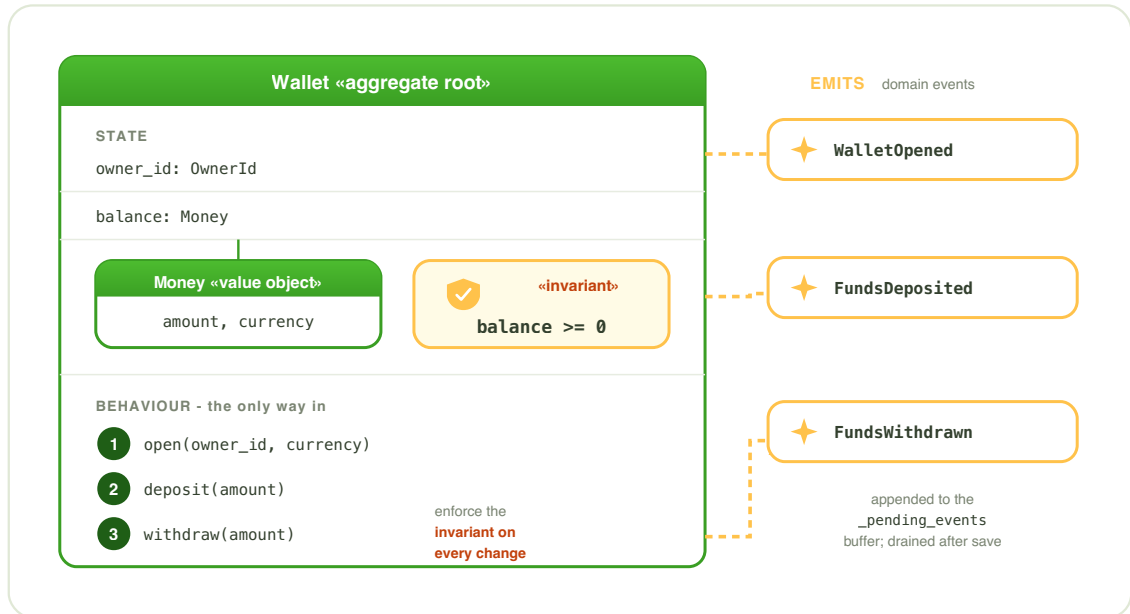


Figure 6.2 — The Wallet aggregate: state, invariants, and the events it emits.

🌿 SPRING PARITY

`AggregateRoot[str]` maps to `jMolecules's org.jmolecules.ddd.types.AggregateRoot<A, ID>` and to Spring Data's `AbstractAggregateRoot<A>`, which offers the same `registerEvent()` / `@DomainEvents` / `@AfterDomainEventPublication` mechanism. The pattern is identical in spirit: the aggregate accumulates events in a buffer; the repository drains them after a successful save; a `DomainEventPublisher` dispatches them. PyFly's `raise_event + clear_events` is the Python equivalent of `registerEvent + @AfterDomainEventPublication`.

Protecting invariants

The aggregate root is only valuable if the rules it enforces are genuinely unreachable by any other path. That is what **invariant** means in DDD: a condition the model must uphold regardless of how it is called, who calls it, or how many services exist in the application. An invariant is not a suggestion — it is a constraint that cannot be violated because the model exposes no mechanism to do so.

Lumen's `Wallet` has three invariants:

1. The balance must never go below zero (no overdraft).
2. Funds can only be deposited or withdrawn in the wallet's native currency.
3. Deposit and withdrawal amounts must be strictly positive.

All three are enforced inside the aggregate methods. The framework exception for this is `BusinessRuleViolation` from `pyfly.domain`. It takes two required arguments: a stable machine-readable `rule` slug and a human-readable `message`. Lumen's slugs — `"wallet-insufficient-funds"`, `"wallet-currency-mismatch"`, `"wallet-deposit-positive"`, `"wallet-withdrawal-positive"` — are kebab-cased labels that travel in the RFC 7807 response body and in structured log fields.

`BusinessRuleViolation` extends `pyfly.kernel.BusinessException`, so the RFC 7807 problem-details mapper from Chapter 4 translates it automatically into an HTTP 422 response — no extra handler required.

WHAT RFC 7807 MEANS HERE

RFC 7807 is the web standard for "problem details" — a small, predictable JSON shape (`type`, `title`, `status`, `detail`, plus your own fields) that an API returns when something goes wrong. PyFly's mapper turns any `BusinessException` into one of these automatically, so the `rule` slug you set on `BusinessRuleViolation` ends up in the response body where a client can read it without parsing English prose.

See the invariant hold. The promise of an invariant is that a *failed* operation leaves the model exactly as it was. You can prove that from `uv run python`:

```
>>> from lumen.models.entities.v1.wallet_entity import Wallet
>>> from lumen.models.entities.v1.money import Money
>>> from lumen.interfaces.enums.v1.currency import Currency
>>> from pyfly.domain import BusinessRuleViolation
>>> w = Wallet.open("wlt-1", "owner-1", Currency.EUR)
>>> w.deposit(Money(500, Currency.EUR))
>>> w.clear_events()           # drain the open + deposit events
[WalletOpened(...), FundsDeposited(...)]
>>> try:
...     w.withdraw(Money(501, Currency.EUR))
... except BusinessRuleViolation as exc:
...     print(exc.rule)
...
wallet-insufficient-funds
>>> str(w.balance)           # unchanged – the rule fired before any
mutation
'5.00 EUR'
>>> w.pending_events()       # and no event was queued for the failed
attempt
[]
```

That is the whole guarantee in three lines: the rule slug is stable and machine-readable, the balance did not move, and no event leaked for an operation that never happened.

⚠ KEEP INVARIANTS IN THE MODEL, NOT THE SERVICE

Moving the overdraft check back into `WalletService` creates two problems. First, any code that calls `repo.save(entity)` directly bypasses the check entirely. Second, you end up duplicating the rule across every path that modifies a wallet — the service, a background job, an admin command. When the rule changes — say, the product team introduces a configurable overdraft buffer — there is exactly one place to update: the aggregate method. That is the whole point.

The difference between a service-level guard and an aggregate invariant is enforceability. A service guard is a convention; an aggregate invariant is a physical constraint enforced by encapsulation. To make that concrete, here is what the service-level approach looks like and why it is fragile:

`lumen/wallet_service_before.py`

```
# DO NOT DO THIS — rules that belong in the model
from pyfly.container import service

@service
class WalletServiceBefore:

    async def withdraw(self, wallet_id: str, amount: float) → None:
        # Rule lives here — but anyone calling repo.save directly skips it
        wallet = {"id": wallet_id, "balance": 50.0, "currency": "EUR"}
        if wallet["balance"] < amount:
            raise ValueError("Insufficient funds")
        wallet["balance"] -= amount
        # ... save
```

Listing 6.3 — Before: business rules scattered across the service (fragile)

And here is what the service looks like after the model takes ownership:

lumen/wallet_service_after.py

```

from pyfly.container import service
from pyfly.domain import AggregateNotFound

from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.money import Money

@service
class WalletService:

    async def withdraw(
        self,
        wallet_id: str,
        amount_cents: int,
        currency: Currency,
    ) → None:
        # The service orchestrates; the aggregate decides.
        wallet = await self._repo.find(wallet_id)
        if wallet is None:
            raise AggregateNotFound("Wallet", wallet_id)
        wallet.withdraw(Money(amount=amount_cents, currency=currency))
        await self._repo.save(wallet)
        # Events are drained and published by the repository/service boundary

```

Listing 6.4 — After: the service delegates to the aggregate

How it works. The after version reads as an instruction: `wallet.withdraw(...)` means "ask the wallet to withdraw". The service does not know — or care — what that entails. It trusts the aggregate to either succeed or raise a `BusinessRuleViolation`. That thin-orchestrator pattern has a practical benefit for team workflows: a new developer can implement a `transfer` endpoint without reading `WalletService` at all. The constraints live in `Wallet` — one place to look.

The `rule` slug on `BusinessRuleViolation` matters too. Strings like `"wallet-insufficient-funds"` and `"wallet-currency-mismatch"` travel in the RFC 7807 response body, where client code can match them without parsing free-text messages. They also appear in structured log fields, making production alerts straightforward to write.

AGGREGATENOTFOUND

`AggregateNotFound` is the second domain exception in `pyfly.domain`. Raise it when the requested aggregate does not exist — it maps to a 404 problem-details response via the same RFC 7807 handler. The constructor takes the aggregate type name and the ID: `AggregateNotFound("Wallet", wallet_id)`.

Domain events

Your aggregate now enforces its invariants and controls all state changes. But Lumen will eventually need to react to those changes: update an audit log, send a push notification, trigger fraud detection, publish a ledger entry. The tempting solution is to put those side-effects directly inside `deposit` and `withdraw`. That couples the domain model to infrastructure — suddenly your wallet needs to know about Kafka topics and email templates, and every unit test drags in a broker connection.

Domain events cut that coupling. A domain event records something that *happened* inside the aggregate — past tense, immutable fact. The aggregate does not know what will be done with the fact; it only records it. Downstream consumers — event listeners, projectors, notification services — subscribe and react in their own context, without the aggregate ever depending on them.

`DomainEvent` from `pyfly.domain` is a frozen-dataclass base that auto-populates three fields when an instance is created:

- `event_id` — a UUID v4 that uniquely identifies this occurrence.
- `occurred_at` — a UTC timestamp at the moment of construction.
- `event_type` — a property that defaults to the subclass's class name (`"WalletOpened"`, `"FundsDeposited"`, `"FundsWithdrawn"`).

You already saw the three wallet events defined in Listing 6.2. Here they are in isolation with an explicit look at what you get from the base:


```
lumen/models/entities/v1/wallet_entity.py
```

```
from dataclasses import dataclass
from pyfly.domain import DomainEvent

@dataclass(frozen=True)
class WalletOpened(DomainEvent):
    wallet_id: str = ""
    owner_id: str = ""
    currency: str = ""

@dataclass(frozen=True)
class FundsDeposited(DomainEvent):
    wallet_id: str = ""
    amount: int = 0
    currency: str = ""
    balance: int = 0 # balance after the deposit, in minor units

@dataclass(frozen=True)
class FundsWithdrawn(DomainEvent):
    wallet_id: str = ""
    amount: int = 0
    currency: str = ""
    balance: int = 0 # balance after the withdrawal, in minor units

# Each event carries event_id (UUID), occurred_at (UTC datetime),
# and event_type (class name) – all set by DomainEvent.__post_init__.

def demonstrate_event_fields() → None:
    evt = FundsDeposited(
        wallet_id="w-1",
        amount=5000,
        currency="EUR",
        balance=15000,
    )
```

```

print(evt.event_id)      # e.g. "3fa85f64-5717-4562-b3fc-2c963f66afa6"
print(evt.occurred_at)   # e.g. datetime(2026, 6, 7, 9, 30, 0,
tzinfo=UTC)
print(evt.event_type)     # "FundsDeposited"

```

Listing 6.5 — Domain events and the fields `DomainEvent` provides automatically

How it works. Each event class declares only the fields unique to that occurrence. Everything else comes from `DomainEvent`: a UUID `event_id` for idempotent processing, `occurred_at` for the audit trail, and `event_type` — the class name — for consumer routing without inspecting the Python class. All fields default to zero or empty string so that the `frozen=True` dataclass machinery can provide keyword-argument construction without requiring positional arguments.

Notice that `FundsDeposited` carries both `amount` (the transaction) and `balance` (the post-operation balance, in minor units). A subscriber updating a read-model balance needs no callback into the aggregate or the database — everything is in the event. That self-contained design keeps consumers simple and eliminates extra round-trips.

READ-MODEL, IN PLAIN LANGUAGE

A **read-model** is a separate, query-optimised copy of data shaped for displaying — a dashboard total, a search index, a cache. Because `FundsDeposited` already carries the post-operation `balance`, a service that maintains such a copy can apply the event blindly without ever loading the wallet again. We build read-models properly in a later chapter; here, just note why putting `balance` in the event pays off.

Run it. You can see the three auto-populated fields on any event from `uv run python :`

```

>>> from lumen.models.entities.v1.wallet_entity import FundsDeposited
>>> evt = FundsDeposited(wallet_id="w-1", amount=5000, currency="EUR",
balance=15000)
>>> evt.event_type
'FundsDeposited'
>>> evt.event_id          # a fresh UUID, set by DomainEvent.__post_init__
'3fa85f64-5717-4562-b3fc-2c963f66afa6'
>>> evt.occurred_at      # a UTC timestamp, also set automatically
datetime.datetime(2026, 6, 16, 9, 30, 0, tzinfo=datetime.timezone.utc)

```

You declared four fields; you got seven, because `DomainEvent` contributes `event_id`, `occurred_at`, and the `event_type` property. That is the whole appeal of subclassing it — every event is self-identifying with zero extra code.

The event lifecycle spans two phases. Inside the aggregate: when `wallet.deposit(amount)` succeeds, it calls `self.raise_event(FundsDeposited(...))`, appending the event to a private buffer in `AggregateRoot`. Nothing is published yet. At the service boundary: after the repository saves the aggregate and the transaction commits, the application service drains the buffer and publishes. This *save-first, publish-after* sequence guarantees that an event is never dispatched for a change that failed to persist. Listing 6.6 shows that boundary in full:

lumen/wallet_application_service.py

```

import uuid

from pyfly.container import service
from pyfly.eda import EventPublisher

from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.money import Money
from lumen.models.entities.v1.wallet_entity import Wallet

@service
class WalletApplicationService:

    def __init__(
        self,
        repo: object,          # typed as WalletRepository in practice
        events: EventPublisher,
    ) → None:
        self._repo = repo
        self._events = events

    async def open_wallet(self, owner_id: str, currency: Currency) → str:
        wallet_id = str(uuid.uuid4())
        wallet = Wallet.open(
            wallet_id=wallet_id, owner_id=owner_id, currency=currency
        )
        await self._repo.save(wallet)
        for event in wallet.clear_events():
            await self._events.publish(event)
        return wallet_id

    async def deposit(
        self,
        wallet_id: str,
        amount_cents: int,
        currency: Currency,
    ) → None:

```

```

wallet = await self._repo.find(wallet_id)
wallet.deposit(Money(amount=amount_cents, currency=currency))
await self._repo.save(wallet)
for event in wallet.clear_events():
    await self._events.publish(event)

```

Listing 6.6 — Draining domain events after a successful save

How it works. `open_wallet` calls `Wallet.open`, which queues a `WalletOpened` event internally; after the `save`, `wallet.clear_events()` returns that one event and `publish` dispatches it. `deposit` follows the same three-step pattern: load, mutate, save — then drain. The `for event in wallet.clear_events():` loop is intentionally explicit rather than hidden inside the repository, because the application service is the right place to decide *when* publishing happens — after the transaction boundary, not before.

The publish cycle, step by step. Every command method in the application service follows the same four-beat rhythm. Learn it once and every future use-case (transfer, refund, freeze) writes itself:

Step 1 — Load (or create). For a new wallet, call the `Wallet.open` factory; for an existing one, `await self._repo.find(wallet_id)`.

Step 2 — Mutate through a behaviour method. Call `wallet.deposit(...)` or `wallet.withdraw(...)`. This is where the invariants run and the events get queued in the aggregate's buffer.

Step 3 — Save. `await self._repo.save(wallet)`. Nothing has been published yet, on purpose — if the save fails, no event escapes.

Step 4 — Drain and publish. `for event in wallet.clear_events(): await self._events.publish(event)`. This runs only after the save succeeds, so an event is never dispatched for a change that did not persist.

What just happened. The aggregate decides *what* happened and records it; the application service decides *when* the world hears about it. Keeping those two responsibilities apart is why the wallet stays free of any broker, queue, or transaction code — and why the publish order is "save first, publish after."

💡 EVENT ORDERING

`raise_event` appends to the buffer in call order. `clear_events` drains and clears it, returning events in the same order. If a single aggregate method raises multiple events (a batch operation, for example), they arrive at the event bus in the order they were raised — oldest first.

Domain vs persistence

With the domain model and its events in place, one tension remains: how does the `Wallet` aggregate reach the database?

The tempting shortcut is to annotate `Wallet` directly with SQLAlchemy `Mapped[]` fields and a `__tablename__`. That merges two concerns that change at very different rates: business rules evolve with the product; column definitions evolve with the schema. Mixing them means a schema change forces you to touch the aggregate, and a rule change risks accidentally breaking a column mapping. It also drags SQLAlchemy into every unit test.

The alternative is two models that coexist without knowing about each other — and a thin mapper that converts between them:

Model	Contains	Knows about
<code>Wallet</code>	Business rules, domain events, invariants	Nothing outside <code>pyfly.domain</code>
<code>WalletEntity</code>	Five columns: <code>id</code> , <code>owner_id</code> , <code>currency</code> , <code>balance_minor</code> , <code>created_at</code>	Only SQLAlchemy + <code>pyfly.data</code>

`Wallet` is pure Python: no `Mapped[]` annotations, no `__tablename__`. You can instantiate it in a unit test with two lines and exercise every invariant without a database connection. `WalletEntity` is pure persistence: it subclasses `Base` from `pyfly.data.relational.sqlalchemy`, carries SQLAlchemy 2.0 typed columns, and knows nothing about domain rules or events. The framework's `Repository[WalletEntity, str]` (Chapter 5) stores and retrieves rows; a thin mapper converts between the row and the aggregate on every crossing.

Listing 6.7 shows `WalletEntity` — the ORM row — followed by the two mapper functions that cross the boundary:

lumen/models/entities/v1/wallet_orm.py

```

from __future__ import annotations

from datetime import UTC, datetime

from sqlalchemy import String
from sqlalchemy.orm import Mapped, mapped_column

from pyfly.data.relational.sqlalchemy import Base

class WalletEntity(Base):
    """One persisted wallet row, keyed by the aggregate's own string id."""

    __tablename__ = "wallets"

    id: Mapped[str] = mapped_column(String(64), primary_key=True)
    owner_id: Mapped[str] = mapped_column(
        String(255), nullable=False, index=True
    )
    currency: Mapped[str] = mapped_column(String(3), nullable=False)
    balance_minor: Mapped[int] = mapped_column(nullable=False, default=0)
    created_at: Mapped[datetime] = mapped_column(
        default=lambda: datetime.now(UTC)
    )

```

Listing 6.7 — *WalletEntity*: the SQLAlchemy persistence row

lumen/core/mappers/wallet_mapper.py

```

from __future__ import annotations

from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.money import Money
from lumen.models.entities.v1.wallet_entity import Wallet
from lumen.models.entities.v1.wallet_orm import WalletEntity


def to_entity(wallet: Wallet) → WalletEntity:
    """Flatten a Wallet aggregate into a persistable row."""
    assert wallet.id is not None
    return WalletEntity(
        id=wallet.id,
        owner_id=wallet.owner_id,
        currency=wallet.currency.value,
        balance_minor=wallet.balance.amount,
        created_at=wallet.created_at,
    )


def to_aggregate(entity: WalletEntity) → Wallet:
    """Rehydrate a Wallet aggregate from a persistence row."""
    currency = Currency(entity.currency)
    return Wallet(
        id=entity.id,
        owner_id=entity.owner_id,
        balance=Money(amount=entity.balance_minor, currency=currency),
        created_at=entity.created_at,
    )

```

Listing 6.8 — wallet_mapper: pure functions that cross the domain/persistence boundary

How it works. `WalletEntity` subclasses `Base` rather than carrying any domain logic, so importing it registers the `wallets` table in `Base.metadata`; the framework's `EngineLifecycle` creates the table on startup when `ddl-auto=create` is set. The primary key is the aggregate's own string id (`wlt-...`) — not a surrogate — so the row and the `Wallet` share one identity and no translation is needed.

`to_entity` writes `wallet.balance.amount` (an integer) directly into `balance_minor` — no float conversion. `to_aggregate` reconstructs a `Currency` enum from the stored ISO-4217 string via `Currency(entity.currency)`, then builds a `Money` from the raw integer minor-unit value. The `created_at` field is preserved on the round-trip so rehydrated aggregates carry their original timestamp. There is no floating-point boundary crossing anywhere.

Build the mapper step by step. A mapper is just two pure functions. There is no framework magic to wire up; you write them and call them at the right moments.

Step 1 — Write `to_entity(wallet)`. Read each piece of the aggregate and copy it into the flat row: `wallet.id`, `wallet.owner_id`, `wallet.currency.value` (the ISO string, not the enum), `wallet.balance.amount` (the raw integer minor units), and `wallet.created_at`. The `assert wallet.id is not None` makes the precondition explicit — you only persist wallets that already have an id.

Step 2 — Write `to_aggregate(entity)`, the reverse. Rebuild the `Currency` enum from the stored string with `Currency(entity.currency)`, wrap the integer back into a `Money`, and construct a `Wallet`. Passing `created_at=entity.created_at` preserves the original timestamp across the round-trip.

Step 3 — Call them at the boundary, not inside the aggregate. The command handler calls `to_entity` just before `repo.save(...)` and `to_aggregate` just after `repo.find(...)`. The `Wallet` class never imports the row, and the row never imports the `Wallet`.

Run it. The round-trip is pure Python — no live database required to prove it preserves every field. From `uv run python`:

```
>>> from lumen.models.entities.v1.wallet_entity import Wallet
>>> from lumen.models.entities.v1.money import Money
>>> from lumen.interfaces.enums.v1.currency import Currency
>>> from lumen.core.mappers import wallet_mapper
>>> w = Wallet.open("wlt-1", "owner-1", Currency.EUR)
>>> w.deposit(Money(2500, Currency.EUR))
>>> row = wallet_mapper.to_entity(w)           # aggregate → flat row
>>> (row.id, row.currency, row.balance_minor)
('wlt-1', 'EUR', 2500)
>>> back = wallet_mapper.to_aggregate(row)     # flat row → aggregate
>>> str(back.balance), back.owner_id
('25.00 EUR', 'owner-1')
```

The integer `2500` crosses both directions untouched — no float ever appears at the persistence boundary, which is the whole reason `Money` stores minor units.

What just happened. You kept two models that never import each other and bridged them with two tiny functions. A schema change now touches only `WalletEntity`; a rule change touches only `Wallet`. Neither can break the other by accident.

The mapper is intentionally narrow. It does not enforce rules — `Wallet.__init__` and the behaviour methods do that. It does not publish events — the application service does that. It only translates shape.

The repository. The application service never interacts with `WalletEntity` directly. Instead a command handler calls `wallet_mapper.to_entity(wallet)` before persisting and `wallet_mapper.to_aggregate(entity)` after loading, while the framework `WalletRepository(Repository[WalletEntity, str])` handles all SQL.

Chapter 5 covers `Repository` in full; the key point here is that `Wallet` itself never imports SQLAlchemy — the aggregate stays free of persistence concerns across both sides of the mapper boundary.

SPRING PARITY

This two-model-plus-mapper structure is the Python equivalent of the pattern advocated in Vaughn Vernon's *Implementing Domain-Driven Design* for Spring: a `WalletJpaEntity` annotated with `@Entity` (the persistence row), a `Wallet` domain object (the aggregate), and a `WalletAssembler` or MapStruct-generated mapper that translates between them. Spring Data JPA's `JpaRepository<WalletJpaEntity, String>` corresponds to PyFly's `Repository[WalletEntity, str]`. The structure is identical; the boilerplate is less.

Specifications for business rules

The aggregate guards state-changing operations well — you cannot overdraw a wallet or deposit the wrong currency. But not every rule is about mutation. Some rules are eligibility checks: "before we show this user the withdrawal button, is the wallet in an operable state?" or "out of ten thousand wallets, which ones qualify for the loyalty bonus?" These are read-only predicates, and encoding them as aggregate methods would clutter `Wallet` with query logic unrelated to state transitions.

The **Specification pattern** solves this cleanly. A specification is a named, reusable predicate: a single `is_satisfied_by(obj) → bool` method wrapped in an object that composes with others using Boolean operators. Because each rule is its own class, you can name rules clearly, reuse them across services, and combine them at runtime based on context — something an `if` chain cannot do.

`Specification[T]` from `pyfly.domain` is a composable in-memory predicate. Subclass it, implement `is_satisfied_by`, and combine instances with `&` (and), `|` (or), and `~` (not). A specification is also directly callable, so you can pass it to Python's built-in `filter` without any adapter code.

TWO KINDS OF SPECIFICATION

`pyfly.domain.Specification` is the in-memory predicate used inside domain services. `pyfly.data.relational.sqlalchemy.Specification` (Chapter 5) is the database-aware query predicate that pushes the rule down into SQL. The two coexist. Domain specifications are for business logic; data specifications are for queries.

Here is a specification that expresses the "eligible for withdrawal" rule:

lumen/domain/specs.py

```

from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.wallet_entity import Wallet
from pyfly.domain import Specification

class HasPositiveBalance(Specification[Wallet]):
    """The wallet has at least one cent remaining."""

    def is_satisfied_by(self, wallet: Wallet) → bool:
        return wallet.balance.is_positive

class IsInCurrency(Specification[Wallet]):
    """The wallet holds a specific currency."""

    def __init__(self, currency: Currency) → None:
        self._currency = currency

    def is_satisfied_by(self, wallet: Wallet) → bool:
        return wallet.balance.currency is self._currency

# Compose: a wallet is eligible for withdrawal if it has a positive
# balance in the requested currency.
def eligible_for_withdrawal(currency: Currency) → Specification[Wallet]:
    return HasPositiveBalance() & IsInCurrency(currency)

# Use as a predicate:
def filter_eligible(
    wallets: list[Wallet],
    currency: Currency,
) → list[Wallet]:
    spec = eligible_for_withdrawal(currency)
    return list(filter(spec, wallets))

```

Listing 6.9 — EligibleForWithdrawal: a composable domain Specification

How it works. `HasPositiveBalance` delegates to `wallet.balance.is_positive` (a property, no parentheses). `IsInCurrency` uses identity comparison (`is`) because `Currency` is a `StrEnum` with singleton members. The `eligible_for_withdrawal` factory combines them with `&`, producing a composite whose `is_satisfied_by` returns `True` only when both checks pass. Because `Specification` implements `__call__`, you pass the composite directly to `filter()` — no lambda wrapper needed.

Build a specification step by step.

Step 1 — Subclass `Specification[Wallet]` and implement the single required method, `is_satisfied_by(self, wallet) → bool`. That is the entire contract: return `True` or `False`, never raise.

Step 2 — If the rule needs a parameter (like a currency to match), take it in `__init__` and store it, as `IsInCurrency` does. A parameterless rule like `HasPositiveBalance` skips this.

Step 3 — Compose with the Boolean operators. `HasPositiveBalance() & IsInCurrency(currency)` builds a new specification whose `is_satisfied_by` is true only when both halves are. `|` is or, `~` is not.

Step 4 — Use it as a predicate. Because `Specification` implements `__call__`, the composite *is* a callable, so you can hand it straight to Python's built-in `filter()` with no lambda in between.

Run it. Specifications are plain objects. Drop the classes from Listing 6.9 into a module (say `src/lumen/domain/specs.py`) and try them from `uv run python :`

```

>>> from lumen.models.entities.v1.wallet_entity import Wallet
>>> from lumen.models.entities.v1.money import Money
>>> from lumen.interfaces.enums.v1.currency import Currency
>>> from lumen.domain.specs import eligible_for_withdrawal
>>> empty = Wallet.open("wlt-1", "owner-1", Currency.EUR)
>>> funded = Wallet.open("wlt-2", "owner-2", Currency.EUR)
>>> funded.deposit(Money(1000, Currency.EUR))
>>> usd = Wallet.open("wlt-3", "owner-3", Currency.USD)
>>> usd.deposit(Money(1000, Currency.USD))
>>> spec = eligible_for_withdrawal(Currency.EUR)
>>> spec(funded), spec(empty), spec(usd)
(True, False, False)
>>> [w.id for w in filter(spec, [empty, funded, usd])]
['wlt-2']

```

Only the funded EUR wallet satisfies both halves of the composite, so `filter` keeps just that one.

What just happened. You expressed a read-only business rule as a named, reusable object instead of an inline `if`. It composes with other rules at runtime, it passes straight into `filter`, and it is trivial to unit-test in isolation — and crucially, it lives *outside* the aggregate, where eligibility checks belong.

The key design discipline: a specification is a *predicate*, not a *guard*. It returns `True` or `False` and never raises. Aggregate invariants (overdraft, currency mismatch) belong inside `deposit` and `withdraw` because they must *prevent* a state change. Specifications belong in services and query handlers because they *select* or *classify* — they never mutate.

Specifications are especially useful where rules combine dynamically: an admin search that adds filters based on the operator's role, or a batch job that partitions a list into eligible and ineligible wallets. Each class has exactly one method and no side-effects, making isolated unit tests trivial.

💡 SPECIFICATION.OF FOR QUICK LAMBDAS

For one-off predicates that do not need a class, use the factory method: `spec = Specification.of(lambda w: w.balance.amount ≥ 1000, name="minimum-balance")`. It composes with `&`, `|`, and `~` the same way as a full subclass.

✓ What you built

Lumen's wallet is now a first-class domain model.

`Money` is a frozen `ValueObject` that stores amounts as integer minor units, enforces currency homogeneity via `_assert_same_currency`, and is replaced rather than mutated. `__post_init__` rejects floats immediately. `is_positive` and `is_negative` are properties; `major_units` and `__str__` handle display.

`Wallet(AggregateRoot[str])` is the consistency boundary. Its factory `open` and behaviour methods `deposit` / `withdraw` enforce all three invariants — no overdraft, no cross-currency operations, no non-positive amounts — by raising `BusinessRuleViolation` with a stable rule slug. Every state change queues a domain event (`WalletOpened`, `FundsDeposited`, `FundsWithdrawn`); the post-operation balance is recorded in each event so subscribers need no callback. After a successful save the application service drains events with `clear_events()` and hands them to `EventPublisher`.

The persistence layer sees only `WalletEntity` (five columns, no domain logic). `to_aggregate` in `wallet_mapper` rehydrates the row into a `Wallet`, and `to_entity` flattens it back; the framework `WalletRepository(Repository[WalletEntity, str])`

handles all SQL without the aggregate ever importing SQLAlchemy.

`Specification[Wallet]` gives you a composable, callable predicate for eligibility checks that live outside the aggregate boundary.

The controller is untouched. The service shrank. The rules are enforced by the object that owns them.

Try it yourself

1. Add a `transfer_to` method and a `DailyLimit` value object. Add a `DailyLimit(ValueObject)` with `max_amount: int` and `currency: Currency`, decorated with `@dataclass(frozen=True)`. Then add `Wallet.transfer_to(target: Wallet, amount: Money) → None`. The method should call `self.withdraw(amount)` and `target.deposit(amount)` in sequence, raising `BusinessRuleViolation("transfer-currency-mismatch", ...)` if the two wallets hold different currencies. Because `Wallet` uses `__slots__`, add `"_frozen"` to the tuple if you also do exercise 2. Verify that both wallets each accumulate a `FundsWithdrawn` / `FundsDeposited` event, respectively, and that a transfer between mismatched wallets raises before modifying either balance.
2. Add a `WalletFrozen` event and a `freeze()` behaviour. Define `WalletFrozen(DomainEvent)` with `wallet_id: str = ""` and `reason: str = ""`. Add `"frozen"` to `__slots__` and a `frozen: bool` attribute (default `False`) to `Wallet.__init__`. Add a `freeze(reason: str) → None` method that sets `self.frozen = True` and calls `self.raise_event(WalletFrozen(...))`. Guard `deposit` and `withdraw` with `if self.frozen: raise`

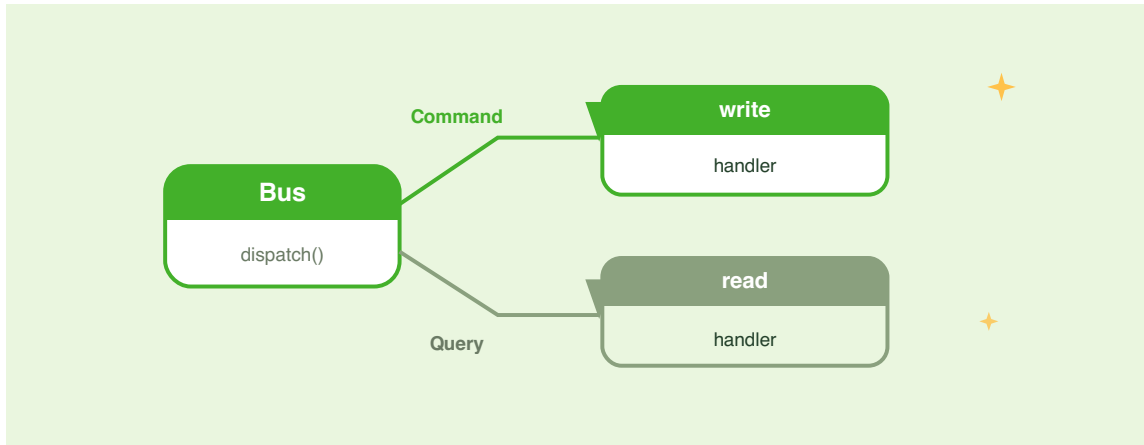
`BusinessRuleViolation("wallet-frozen", ...)` at the top of each method. Then write an event listener using `@event_listener` from `pyfly.eda` that logs a structured warning every time a `WalletFrozen` event is published.

3. Express a business rule as a Specification. Write a

`MinimumBalance(Specification[Wallet])` that checks whether a wallet's balance is at or above a threshold amount (in minor units) passed to its `__init__`. Combine it with `IsInCurrency` from Listing 6.9 using `&` to produce a `premium_eligible(currency: Currency, threshold: int)` factory function. Call `list(filter(premium_eligible(Currency.EUR, 50000), wallets))` over a list of test wallets and assert that only wallets with at least 500.00 EUR (50 000 cents) appear in the result.

CHAPTER 7

CQRS: Commands & Queries



Lumen's wallet is now a first-class citizen of the domain. The `Wallet` aggregate enforces its own invariants, emits domain events, and persists through a clean repository boundary. The controller, though, still calls `WalletApplicationService` directly — one method per operation, reads and writes sharing the same code path. That design is fine at small scale, but it shows friction as the system grows. The team wants to cache wallet balances, maintain a single audit trail for every write, add authorization rules to specific operations, and test each piece of logic in complete isolation from the others.

CQRS — Command Query Responsibility Segregation — addresses all of this by drawing a bright line between the two things a service can do: *change state* and *read state*. Writes become **commands**: strongly typed, named, immutable messages that flow through a `CommandBus`. Reads become **queries**: equally typed messages that flow

through a `QueryBus`. Each bus runs a fixed pipeline — validation, authorization, execution, then (for commands) domain event publishing. Your handler implements exactly one intent; the bus handles everything else.

By the end of this chapter Lumen's controller dispatches commands and queries instead of calling the service directly. `OpenWallet`, `DepositFunds`, and `WithdrawFunds` travel the command path; `GetWallet`, `GetBalance`, `ListWallets`, and `ListRichWallets` travel the query path. The `Wallet` aggregate you built in Chapter 6 remains untouched — CQRS does not replace the domain model; it is the delivery mechanism for instructions to it.

NEW JARGON, IN PLAIN TERMS

A **bus** here is not hardware — it is a single object you hand a message to, and it figures out which handler should run. A **handler** is one small class that does the work for exactly one message type. A **DTO** (data transfer object) is a plain shape — id, owner, balance — that you put on the wire as JSON; it is deliberately separate from your rich domain object. A **projection** is a read-only slice of your data shaped for one specific view. You will meet each of these as we build, one piece at a time, so do not worry if they feel abstract right now.

This chapter is built around PyFly **v26.6.110**, and every listing is taken verbatim from the Lumen sample under `samples/lumen/src/lumen`. We will go gently: build the commands first, then their handlers, then the queries, then wire everything into the controller — running the app and the tests at each milestone so you can see the pieces come alive before the next one is added.

Why separate reads from writes

Picture Lumen at the end of Chapter 6. `WalletController` calls `WalletApplicationService.credit(wallet_id, amount)`. That call mutates state, but nothing in the method signature makes that obvious. The team wants to add a balance cache. Where does it go? Inside `credit`? In a decorator around the service? The question reveals the problem: a single service method is asked to serve two masters — the write path, which must always touch the database, and the read path, which should avoid it whenever possible. Bolting caching onto a write method is awkward at best and dangerous at worst.

Writes and reads have fundamentally different shapes. A write carries intent and data: "deposit 1 500 minor units into wallet wlt-001". A read carries a question: "what is the current balance of wallet wlt-001?" The first must reach the database every time. The second is repeatable — asking twice should return the same answer without doubling the database load. Funnelling both through the same method conflates concerns that scale differently, test differently, and need different cross-cutting behaviour.

The deeper benefit is **clarity of intent**. When a teammate reads `wallet_service.credit(wallet_id, amount)`, they must inspect the implementation to know whether it is safe to call twice, whether it publishes events, and whether it is idempotent. When they read `DepositFunds(wallet_id=..., amount=...)`, the intent is unambiguous — and if the intent turns out to be wrong, you rename the command, not the service signature.

Three concrete benefits matter for Lumen:

Independent scaling. Reads typically outnumber writes by an order of magnitude or more. Once the two paths are separate, the bus can cache query results without touching the write path. You can route queries to a read replica and commands to the primary database with a configuration change, not a code change.

Focused handlers. Each handler implements exactly one operation.

`DepositFundsHandler` loads a wallet, drives its domain behaviour, persists it, and drains events — nothing more. `GetBalanceHandler` loads one wallet and returns a lightweight projection — nothing more. Because handlers are plain Python classes with injected dependencies, you can unit-test each in complete isolation from the HTTP layer.

Centralized cross-cutting concerns. Validation, authorization, and distributed tracing are implemented once in the bus pipeline and apply uniformly to every handler — no boilerplate in the handler itself. Adding per-operation authorization later is a matter of overriding `authorize()` on the command; the bus ensures it runs before `do_handle` is ever reached.

Commands and command handlers

Before writing a single line of handler code, name your system's intentions. In Lumen's wallet domain three things can happen: a wallet can be opened, funds can be deposited, and funds can be withdrawn. Each is a **command** — a named, immutable message that expresses one intent. The bus delivers it; the handler acts on it; the domain aggregate enforces the rules. Commands are not method calls dressed up as objects: they are explicit contracts that live in your codebase as first-class citizens.

A command is a frozen dataclass that inherits from `Command[R]`, where `R` is the type the handler returns. The generic parameter is documentation and a type-checker hint; the bus does not enforce it at runtime.

WHAT IS `COMMAND[R]` ?

The `[R]` in `Command[R]` is a *generic type parameter* — a placeholder for "whatever this command returns". `OpenWallet(Command[str])` says "sending me gives you back a `str`" (the new wallet id). `DepositFunds(Command[int])` says "sending me gives you back an `int`" (the new balance). Your editor and type checker use this to catch mistakes; at runtime the bus simply returns whatever the handler returned.

Lumen's commands live in three separate files under `lumen/core/services/wallets/`, one per intent. We will build them one at a time.

Step 1 — Write the `OpenWallet` command. Create `open_wallet_command.py`. It carries the two facts needed to open a wallet — who owns it and which currency it holds — and a `validate()` hook that rejects a blank owner before the bus ever looks for a handler.

lumen/core/services/wallets/open_wallet_command.py

```

from __future__ import annotations

from dataclasses import dataclass

from lumen.interfaces.enums.v1.currency import Currency
from pyfly.cqrs import Command, ValidationResult

@dataclass(frozen=True)
class OpenWallet(Command[str]):
    """Open a new wallet. Returns the generated wallet id."""

    owner_id: str
    currency: Currency

    async def validate(self) → ValidationResult: # type: ignore[override]
        if not self.owner_id.strip():
            return ValidationResult.failure(
                "owner_id", "Owner id is required"
            )
        return ValidationResult.success()

```

Listing 7.1 — OpenWallet: a frozen command with built-in validation

Step 2 — Write the `DepositFunds` and `WithdrawFunds` commands. Each carries a `wallet_id` to target and an `amount` in minor units, and validates that the id is present and the amount is positive. They are deliberately near-identical twins — same shape, opposite direction.

```
lumen/core/services/wallets/deposit_funds_command.py
```

```
from __future__ import annotations

from dataclasses import dataclass

from pyfly.cqrs import Command, ValidationResult

@dataclass(frozen=True)
class DepositFunds(Command[int]):
    """Deposit ``amount`` minor units. Returns the new balance."""

    wallet_id: str
    amount: int

    async def validate(self) → ValidationResult: # type: ignore[override]
        if not self.wallet_id.strip():
            return ValidationResult.failure(
                "wallet_id", "Wallet id is required"
            )
        if self.amount ≤ 0:
            return ValidationResult.failure(
                "amount", "Deposit amount must be > 0"
            )
        return ValidationResult.success()
```

Listing 7.2 — *DepositFunds: amount in minor units, no currency field*

lumen/core/services/wallets/withdraw_funds_command.py

```

from __future__ import annotations

from dataclasses import dataclass

from pyfly.cqrs import Command, ValidationResult

@dataclass(frozen=True)
class WithdrawFunds(Command[int]):
    """Withdraw ``amount`` minor units. Returns the new balance."""

    wallet_id: str
    amount: int

    async def validate(self) → ValidationResult: # type: ignore[override]
        if not self.wallet_id.strip():
            return ValidationResult.failure(
                "wallet_id", "Wallet id is required"
            )
        if self.amount ≤ 0:
            return ValidationResult.failure(
                "amount", "Withdrawal amount must be > 0"
            )
        return ValidationResult.success()

```

Listing 7.3 — *WithdrawFunds*: same shape as *DepositFunds*

Four design choices are baked into every command:

- `frozen=True` makes the dataclass immutable the moment it is constructed. Fields cannot be accidentally mutated in one layer of the pipeline before reaching another, and immutable messages are hashable by default — useful when storing or comparing them in tests.

- `validate()` is an async hook the bus calls before dispatching the handler.
`OpenWallet.validate` checks that `owner_id` is not blank;
`DepositFunds.validate` and `WithdrawFunds.validate` check that the amount is positive. These pre-conditions belong on the command — they require no database lookup and do not belong in the domain aggregate. The aggregate enforces invariants that need loaded state (overdraft, currency match); commands enforce invariants knowable from the fields alone. Keeping these two layers separate means the aggregate is never called with structurally wrong data.
- **No currency field** on `DepositFunds` or `WithdrawFunds`. The wallet's own currency is the only valid currency for a deposit or withdrawal, and the repository resolves that once the aggregate is loaded. Carrying a currency on the command would invite mismatches; the aggregate enforces the invariant from its own state.
- **Imperative-mood naming:** `DepositFunds`, not `WalletDeposit` or `DepositFundsCommand`. This makes the command log read like a business audit trail — a sequence of things that *happened* — rather than a list of technical operations.

WHAT JUST HAPPENED

You now have three small files, each describing *one thing the system can do*, with no logic beyond a couple of field checks. There are no handlers yet — these commands do nothing on their own. That is the point: a command is an envelope, not the worker who opens it. Next you will write the workers (the handlers) that actually carry out each intent.

Implementing a command handler

A command handler inherits from `CommandHandler[C, R]` and implements exactly one method: `do_handle`. You write the *what*; the bus wraps it with the *how*.

Both decorators on every handler are required. `@command_handler` registers the class with the `HandlerRegistry` by introspecting the first generic type argument — no manual registration needed. `@service` wires the handler into PyFly's DI container so constructor arguments are resolved and injected automatically at startup. The order matters: `@command_handler` on top, `@service` directly below. Without `@service`, the DI container never instantiates the class and the bus cannot find the handler; without `@command_handler`, the registry never maps the command type to the class. Omitting either decorator is a silent failure — the bus raises "no handler found" at dispatch time.

`@transactional()` turns `do_handle` into a committed unit of work. Command handlers inject `session_factory: async_sessionmaker[AsyncSession]` and store it as `self._session_factory`. When `@transactional()` runs `do_handle` it opens a fresh session from that factory, swaps it onto the repository for the duration of the call, commits on success, and rolls back on any exception. Without `@transactional()` the framework's shared session only flushes — the write survives within the request but is never committed to the database.

FLUSH VS. COMMIT, IN PLAIN TERMS

A **flush** pushes your pending changes into the database connection so later queries in the *same* session can see them — but they are still inside an open transaction that can be rolled back. A **commit** makes them permanent. Without `@transactional()` your deposit would flush (visible mid-request) but never commit (gone after the request). The decorator is what makes the change stick.

Step 3 — Write `OpenWalletHandler`. Now build the worker for the first command. Create `open_wallet_handler.py`, stack `@command_handler` over `@service`, inject the repository, the event publisher, and the session factory, and implement the single `do_handle` method.

lumen/core/services/wallets/open_wallet_handler.py

```

from __future__ import annotations

from uuid import uuid4

from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker

from lumen.core.mappers.wallet_mapper import to_entity
from lumen.core.services.wallets.event_publishing import publish_domain_events
from lumen.core.services.wallets.open_wallet_command import OpenWallet
from lumen.models.entities.v1.wallet_entity import Wallet
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.data.relational.sqlalchemy import transactional
from pyfly.eda import EventPublisher

@command_handler
@service
class OpenWalletHandler(CommandHandler[OpenWallet, str]):
    """Open a new, empty wallet."""

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events
        # @transactional resolves the unit-of-work session from here.
        self._session_factory = session_factory

    @transactional()
    async def do_handle(self, command: OpenWallet) → str: # type:
        ignore[override]

```

```

wallet_id = f"wlt-{uuid4()}"
wallet = Wallet.open(
    wallet_id=wallet_id,
    owner_id=command.owner_id,
    currency=command.currency,
)
await self._repository.upsert(to_entity(wallet))
await publish_domain_events(self._events, wallet.clear_events())
return wallet_id

```

Listing 7.4 — *OpenWalletHandler: @transactional() unit of work + upsert*

Walk through `do_handle` step by step. `f"wlt-{uuid4()}"` generates a stable prefixed identifier. `Wallet.open(...)` calls the factory, which enforces the non-empty owner pre-condition and buffers a `WalletOpened` event. `to_entity(wallet)` maps the aggregate to a flat `WalletEntity` row. `repository.upsert(...)` calls `session.merge` — a single call that inserts if no row exists or updates if one does — then flushes. Using `upsert` instead of `save` avoids an `IntegrityError` on the primary key: the aggregate owns its id, so both INSERT and UPDATE key on the same stable string. `wallet.clear_events()` drains the buffer and `publish_domain_events` forwards each event to the EDA bus. The `@transactional()` decorator commits the session on the way out. The handler returns the wallet ID, which flows back to the controller as the `send` return value.

Note the constructor requirement: `super().__init__()` is mandatory on `CommandHandler`. Skip it and the base-class bookkeeping — correlation context, lifecycle hooks — is never initialized. The repository, `EventPublisher`, and `session_factory` are all injected by the DI container from type hints; no factory configuration is needed.

Step 4 — Write the deposit and withdrawal handlers. These two add one move that `OpenWalletHandler` did not need: they *load* an existing wallet before acting on it. The shape is the same in both, differing only in whether they call `wallet.deposit(...)` or `wallet.withdraw(...)`.


```
lumen/core/services/wallets/deposit_funds_handler.py
```

```
from __future__ import annotations

from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker

from lumen.core.mappers.wallet_mapper import to_aggregate, to_entity
from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.core.services.wallets.event_publishing import publish_domain_events
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cQRS import CommandHandler, command_handler
from pyfly.domain import AggregateNotFound
from pyfly.data.relational.sqlalchemy import transactional
from pyfly.eda import EventPublisher

@command_handler
@service
class DepositFundsHandler(CommandHandler[DepositFunds, int]):
    """Credit funds to an existing wallet; returns the new balance."""

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events
        self._session_factory = session_factory

    @transactional()
    async def do_handle(self, command: DepositFunds) → int: # type:
        ignore[override]
        entity = await self._repository.find_by_id(command.wallet_id)
        if entity is None:
```

```
        raise AggregateNotFound("Wallet", command.wallet_id)

    wallet = to_aggregate(entity)
    wallet.deposit(Money(amount=command.amount, currency=wallet.currency))
    await self._repository.upsert(to_entity(wallet))

    await publish_domain_events(self._events, wallet.clear_events())
    return wallet.balance.amount
```

Listing 7.5 — *DepositFundsHandler*: *find_by_id* → *to_aggregate* → *act* → *upsert*

```
lumen/core/services/wallets/withdraw_funds_handler.py
```

```
from __future__ import annotations

from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker

from lumen.core.mappers.wallet_mapper import to_aggregate, to_entity
from lumen.core.services.wallets.event_publishing import publish_domain_events
from lumen.core.services.wallets.withdraw_funds_command import WithdrawFunds
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cQRS import CommandHandler, command_handler
from pyfly.domain import AggregateNotFound
from pyfly.data.relational.sqlalchemy import transactional
from pyfly.eda import EventPublisher

@command_handler
@service
class WithdrawFundsHandler(CommandHandler[WithdrawFunds, int]):
    """Debit funds from an existing wallet; returns the new balance."""

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events
        self._session_factory = session_factory

    @transactional()
    async def do_handle(self, command: WithdrawFunds) → int: # type:
        ignore[override]
        entity = await self._repository.find_by_id(command.wallet_id)
        if entity is None:
```

```

        raise AggregateNotFound("Wallet", command.wallet_id)

    wallet = to_aggregate(entity)
    wallet.withdraw(Money(amount=command.amount,
currency=wallet.currency))
    await self._repository.upsert(to_entity(wallet))

    await publish_domain_events(self._events, wallet.clear_events())
    return wallet.balance.amount

```

Listing 7.6 — *WithdrawFundsHandler*: identical pattern, overdraft refused by the aggregate

`DepositFundsHandler` and `WithdrawFundsHandler` follow the classic pattern: **find** → **to_aggregate** → **act** → **to_entity** → **upsert** → **drain**. `repository.find_by_id` returns the flat `WalletEntity` row; `to_aggregate(entity)` rehydrates the rich domain object so the aggregate's invariants are in scope. `Money` is constructed from the command's `amount` and the *wallet*'s currency — never a currency from the command itself — because the wallet owns that invariant. If `wallet.withdraw` refuses (balance would go negative), it raises `BusinessRuleViolation`, which propagates as HTTP 422 without a single line of error-handling code in the handler.

Notice what is absent: no try/except blocks, no logging calls, no tracing setup. All of that belongs to the bus pipeline. The handler is a pure expression of business intent.

❗ WHAT JUST HAPPENED

The write side is now complete: three commands and three handlers. Sending `OpenWallet` creates and persists a fresh wallet; sending `DepositFunds` or `WithdrawFunds` loads one, drives its domain behaviour, and saves it. The `@command_handler` + `@service` stack means PyFly discovers and wires these at startup — you never call them directly, and you never register them by hand.

Run it — confirm the write side works end to end. The Lumen sample already ships a test that exercises the full command path. From the `samples/lumen` directory, run just that test:

terminal

```
uv run --extra dev pytest tests/test_cqrs_flow.py::test_full_wallet_lifecycle -q
```

Listing 7.4a — Exercise the command path

You should see a single passing test:

```
1 passed in 0.42s
```

That one test opens a wallet, deposits 1 500 minor units, withdraws 500, and asserts the balance lands at 1 000 — proving `OpenWalletHandler`, `DepositFundsHandler`, and `WithdrawFundsHandler` all commit through the bus. If you see `0 items collected`, you are not in the `samples/lumen` directory; `cd` there first. If you see `no handler found`, double-check that both decorators are present on each handler — that is the single most common cause.

The entity↔aggregate mapper

Command handlers do not interact with the repository through the domain aggregate. They interact through a flat `WalletEntity` row — the persistence shape the framework `Repository[WalletEntity, str]` understands — and use `wallet_mapper` to translate between the two worlds:

```
# Aggregate → row (before upsert)
to_entity(wallet)      # Wallet → WalletEntity

# Row → aggregate (after find_by_id)
to_aggregate(entity)   # WalletEntity → Wallet
```

This separation keeps the aggregate free of SQLAlchemy annotations and the repository free of domain logic. The mapper is a single module; changing the storage schema touches one file, not every handler.

Sending a command

The `CommandBus` is the single entry point for all writes. PyFly's auto-configuration registers a `DefaultCommandBus` as a singleton in the DI container; declare it as a constructor argument and the framework injects it. Sending a command is a single awaited call:

```
from pyfly.cqrs import DefaultCommandBus
from lumen.core.services.wallets.open_wallet_command import OpenWallet
from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.interfaces.enums.v1.currency import Currency

wallet_id: str = await command_bus.send(
    OpenWallet(owner_id="u-1", currency=Currency.EUR)
)
balance: int = await command_bus.send(
    DepositFunds(wallet_id=wallet_id, amount=1500)
)
```

`send` is a coroutine — always `await` it. The return value is whatever `do_handle` returned: a `str` wallet ID for `OpenWallet`, and the new balance as an `int` (minor units) for `DepositFunds` and `WithdrawFunds`. If anything in the pipeline fails —

validation, authorization, or the handler itself — the exception wraps in `CommandProcessingException` and propagates out of `send`, where the global error handler maps it to the appropriate HTTP status code.

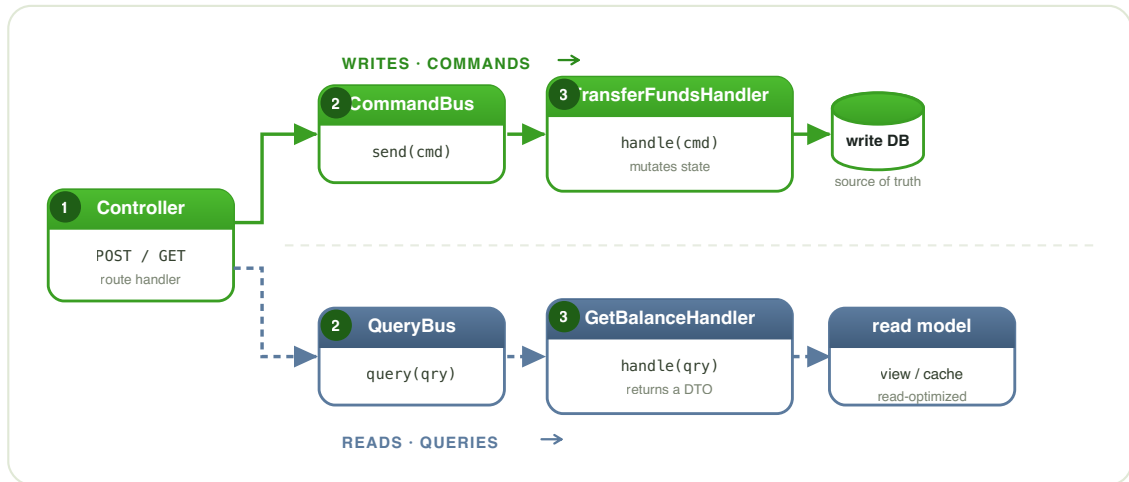


Figure 7.1 — Commands flow to the write model; queries to the read model.

🌿 SPRING PARITY

`CommandBus.send(command)` is the Python equivalent of Axon Framework's `CommandGateway.send(command)` or `CommandGateway.sendAndWait(command)`. Each command handler class corresponds to a method annotated with `@CommandHandler` in Axon, or a `@MessageHandler` in Spring Modulith's `ApplicationEventPublisher` model. The `@command_handler` decorator is PyFly's counterpart of `@CommandHandler`: it registers the handler with the registry by introspecting the generic type parameter, exactly as Axon resolves handler methods by parameter type. The `@service` stacking mirrors the fact that in Spring every `@CommandHandler` bean is also a Spring `@Component` — registration and injection are inseparable. The `@transactional()` decorator maps directly to Spring's `@Transactional`: both open a unit-of-work session, commit on success, and roll back on any exception — so `upsert` (backed by `session.merge`) is the Python analogue of `repository.save()` inside a `@Transactional` method.

Queries and query handlers

Commands travel one direction: into the write model. Queries are the return journey — they ask the system for a projection of current state and expect an answer, not a side effect.

A **query** is a frozen dataclass that inherits from `Query[R]`, where `R` is the result type. Like commands, queries are immutable messages, but they carry no intent to change state. `query_bus.query(GetBalance(...))` loads fresh data from the repository and returns a typed DTO. Queries do not need `@transactional()` — reads do not mutate state, so there is nothing to commit or roll back.

Queries return **read DTOs** rather than domain aggregates. The separation is deliberate. If `GetWalletHandler` returned a `Wallet` aggregate, the API layer would be coupled to every field on the aggregate — a change to the domain model could silently break the API contract. A dedicated `WalletDto` Pydantic model projects exactly the fields the HTTP response needs. Add a field to `Wallet` ? The projection changes only if you explicitly include it in the DTO. Remove a field from `Wallet` ? The projection compiles until you clean it up.

The read side mirrors the write side step for step — query message, then query handler — but with two simplifications: no `@transactional()` (nothing to commit) and no event publishing (nothing changed).

Step 5 — Write the single-lookup queries. Create `get_wallet_query.py` and `get_balance_query.py`. Both carry only a `wallet_id`; what differs is what they promise to return.


```
lumen/core/services/wallets/get_wallet_query.py
```

```
from __future__ import annotations

from dataclasses import dataclass

from lumen.interfaces.dtos.v1.wallet_dto import WalletDto
from pyfly.cqrs import Query

@dataclass(frozen=True)
class GetWallet(Query[WalletDto | None]):
    """Look up a wallet by its identifier."""

    wallet_id: str
```

Listing 7.7 — *GetWallet*: a single-lookup query returning a full *WalletDto*

```
lumen/core/services/wallets/get_balance_query.py
```

```
from __future__ import annotations

from dataclasses import dataclass

from lumen.interfaces.dtos.v1.balance_dto import BalanceDto
from pyfly.cqrs import Query

@dataclass(frozen=True)
class GetBalance(Query[BalanceDto | None]):
    """Look up just the balance of a wallet by its identifier."""

    wallet_id: str
```

Listing 7.8 — *GetBalance*: a lighter query returning only the balance projection

Both queries carry only `wallet_id`. `GetWallet` returns a `WalletDto` — the full representation including `id`, `owner_id`, `currency`, `balance_minor`, `balance`, and `created_at`. `GetBalance` returns a `BalanceDto` — a lighter projection that omits `owner_id` and `created_at`. A balance poll does not need the owner; leaving those fields out saves bandwidth and avoids accidentally exposing account ownership in a response that callers may log. Keeping the two queries separate means you can tune each independently — caching, authorization, or a dedicated read store — without touching the other.

Step 6 — Write the single-lookup query handlers. The query handlers live under the same `wallets/` package as the commands. **The same `@query_handler` + `@service` stacking applies:** `@query_handler` registers the class with the handler registry; `@service` wires it into the DI container. Both decorators are required for the same reasons as on command handlers. Notice how much smaller these are than the command handlers — no session factory, no event publisher, just the repository and a one-line projection.

lumen/core/services/wallets/get_wallet_handler.py

```

from __future__ import annotations

from lumen.core.mappers.wallet_mapper import entity_to_dto
from lumen.core.services.wallets.get_wallet_query import GetWallet
from lumen.interfaces.dtos.v1.wallet_dto import WalletDto
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cqrs import QueryHandler, query_handler

@query_handler
@service
class GetWalletHandler(QueryHandler[GetWallet, WalletDto | None]):
    def __init__(self, repository: WalletRepository) → None:
        super().__init__()
        self._repository = repository

    async def do_handle(self, query: GetWallet) → WalletDto | None: # type:
        ignore[override]
        entity = await self._repository.find_by_id(query.wallet_id)
        return entity_to_dto(entity) if entity is not None else None

```

Listing 7.9 — GetWalletHandler: find_by_id → entity_to_dto → return

lumen/core/services/wallets/get_balance_handler.py

```

from __future__ import annotations

from lumen.core.mappers.wallet_mapper import entity_to_balance_dto
from lumen.core.services.wallets.get_balance_query import GetBalance
from lumen.interfaces.dtos.v1.balance_dto import BalanceDto
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cqrs import QueryHandler, query_handler

@query_handler
@service
class GetBalanceHandler(QueryHandler[GetBalance, BalanceDto | None]):
    def __init__(self, repository: WalletRepository) → None:
        super().__init__()
        self._repository = repository

    async def do_handle(self, query: GetBalance) → BalanceDto | None: #
        type: ignore[override]
        entity = await self._repository.find_by_id(query.wallet_id)
        return entity_to_balance_dto(entity) if entity is not None else None

```

Listing 7.10 — *GetBalanceHandler: @projection view via Mapper.project*

Both handlers delegate projection to `wallet_mapper` — the single module that owns the DTO shape. `entity_to_dto` fills in all six fields of `WalletDto` directly from the row. `entity_to_balance_dto` takes a different path: it calls `Mapper.project(entity, BalanceView)` against a `@projection`-marked interface that declares exactly the four fields the balance endpoint needs, with a registered transform that computes `balance` (major units) from `balance_minor`. The mapper copies only those declared fields — a read-side equivalent of Spring Data's interface projections. Neither handler touches the Pydantic model directly; a field rename touches one file.

Paged and specification queries

The read side does not stop at single-resource lookups. Production systems need lists with pagination metadata and the ability to filter by runtime predicates. The framework handles both through the `Repository` base class.

PAGEABLE AND SPECIFICATION, IN PLAIN TERMS

A `Pageable` bundles three things a list endpoint needs: which page you want, how big each page is, and how to sort. A `Specification` is a reusable, composable filter — think of it as a `WHERE` clause you can build as an object and combine with `&` (and), `|` (or), and `~` (not) before it ever touches SQL. Both come from the framework's data layer; you do not write the SQL yourself.

Step 7 — Write the list queries and handlers. `ListWallets` wraps a `Pageable` (page number, size, sort) and asks the repository for a counted, sorted, limited slice. `ListRichWallets` adds a `min_minor` threshold and runs it through a composable `Specification`. Build the two query messages first, then their handlers.

```
lumen/core/services/wallets/list_wallets_query.py
```

```
from __future__ import annotations

from dataclasses import dataclass

from lumen.interfaces.dtos.v1.wallet_dto import WalletDto
from pyfly.data import Page, Pageable
from pyfly.cqrs import Query

@dataclass(frozen=True)
class ListWallets(Query[Page[WalletDto]]):
    """List wallets, one page at a time."""

    pageable: Pageable
```

Listing 7.11 — *ListWallets: a Pageable-carrying query*

```
lumen/core/services/wallets/list_rich_wallets_query.py
```

```
from __future__ import annotations

from dataclasses import dataclass

from lumen.interfaces.dtos.v1.wallet_dto import WalletDto
from pyfly.data import Page, Pageable
from pyfly.cqrs import Query

@dataclass(frozen=True)
class ListRichWallets(Query[Page[WalletDto]]):
    """List wallets whose balance is at least ``min_minor``, paged."""

    min_minor: int
    pageable: Pageable
```

Listing 7.12 — *ListRichWallets: adds a balance threshold for Specification filtering*

```
lumen/core/services/wallets/list_wallets_handler.py
```

```
from __future__ import annotations

from lumen.core.mappers.wallet_mapper import entity_to_dto
from lumen.core.services.wallets.list_wallets_query import ListWallets
from lumen.interfaces.dtos.v1.wallet_dto import WalletDto
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cqrs import QueryHandler, query_handler
from pyfly.data import Page

@query_handler
@service
class ListWalletsHandler(QueryHandler[ListWallets, Page[WalletDto]]):
    def __init__(self, repository: WalletRepository) → None:
        super().__init__()
        self._repository = repository

    async def do_handle(self, query: ListWallets) → Page[WalletDto]:
        # type: ignore[override]
        page = await self._repository.find_all(query.pageable)
        return page.map(entity_to_dto)
```

Listing 7.13 — ListWalletsHandler: find_all(pageable) + Page.map

lumen/core/services/wallets/list_rich_wallets_handler.py

```

from __future__ import annotations

from lumen.core.mappers.wallet_mapper import entity_to_dto
from lumen.core.services.wallets.list_rich_wallets_query import
ListRichWallets
from lumen.interfaces.dtos.v1.wallet_dto import WalletDto
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cqrs import QueryHandler, query_handler
from pyfly.data import Page

@query_handler
@service
class ListRichWalletsHandler(QueryHandler[ListRichWallets, Page[WalletDto]]):
    def __init__(self, repository: WalletRepository) → None:
        super().__init__()
        self._repository = repository

    async def do_handle(self, query: ListRichWallets) → Page[WalletDto]: #
type: ignore[override]
        page = await self._repository.find_rich(query.min_minor,
query.pageable)
        return page.map(entity_to_dto)

```

Listing 7.14 — ListRichWalletsHandler: Specification + find_all_by_spec_paged

`find_all(pageable)` is inherited from the framework `Repository` base. It counts the total rows, applies the `Pageable` 's sort, and slices with `LIMIT / OFFSET` — returning a `Page[WalletEntity]` that carries `items`, `total`, `page`, `size`, `total_pages`, `has_next`, and `has_previous`. `Page.map(entity_to_dto)` transforms the items without touching the metadata. The controller wraps the result in a `PageDto` for the wire.

`find_rich` is defined on `WalletRepository` itself and delegates to the inherited `find_all_by_spec_paged`. It constructs a `Specification` — a composable `WHERE` predicate — and passes it alongside the `Pageable`. The framework appends the `WHERE` clause, the sort, and the `LIMIT / OFFSET`, then executes a count query for the total. The handler calls `repo.find_rich(query.min_minor, query.pageable)` and maps the page exactly as before.

Executing a query goes through `QueryBus.query`:

```
from pyfly.cqrs import DefaultQueryBus
from lumen.core.services.wallets.get_balance_query import GetBalance

balance_dto = await query_bus.query(GetBalance(wallet_id="wlt-001"))
```

The return value is whatever `do_handle` returned — a `BalanceDto` or `None`. `None` means the wallet was not found. The controller is responsible for translating that into HTTP 404, keeping HTTP concerns out of the handler.

QUERIES RETURN NONE, NOT EXCEPTIONS

Query handlers return `None` when the resource is not found rather than raising `AggregateNotFound`. This is a deliberate convention: a query that finds nothing is not an error — it is an answer. The controller turns a `None` result into a 404 response, keeping the HTTP concern out of the handler.

WHAT JUST HAPPENED

Both sides of CQRS now exist. Three commands and three handlers change state; four queries and four handlers read it. None of them know about HTTP, and none of them are registered by hand — the decorators do that at startup. The only thing missing is the HTTP boundary that turns a web request into a message and a message result into a web response. That is the controller, which you build next.

Run it — confirm every handler is registered. Before touching the controller, prove the bus discovered all your handlers. Start the app, then ask the CQRS health indicator how many handlers it found. From the `samples/lumen` directory:

terminal

```
uv run pyfly run --server uvicorn
```

Listing 7.14a — Start Lumen

In a second terminal, query the actuator health endpoint. In v26.6.110 the actuator lives on its own management port, **9090** by default — not the app's 8080:

terminal

```
curl -s localhost:9090/actuator/health | python -m json.tool
```

Listing 7.14b — Count the registered handlers

Look for the `cqrs_health_indicator` block. With every command and query handler from this chapter in place it reports three command handlers and four query handlers:

```
"cqrs_health_indicator": {  
  "status": "UP",  
  "details": {"command_handlers": 3, "query_handlers": 4}  
}
```

If a count is lower than you expect, a handler is missing one of its two decorators and the registry never mapped it. Stop the server with `Ctrl-C` when you are done.

📄 THE ACTUATOR LIVES ON ITS OWN PORT NOW

In PyFly v26.6.110 the business API and the actuator run on **separate** ports — Spring-style. Your wallet endpoints listen on `pyfly.server.port` (default **8080**), while actuator endpoints and the admin dashboard listen on `pyfly.management.server.port` (default **9090**), which is open and unauthenticated by default. Lumen keeps the defaults, so health is at `localhost:9090/actuator/health` and the wallet API is at `localhost:8080/api/v1/wallets`. The default actuator HTTP exposure is `health,info`; expose more via `pyfly.management.endpoints.web.exposure.include`. Lock the management port down in production with `pyfly.management.security.enabled: true`, or disable it entirely with `pyfly.management.server.port: -1`.

Wiring the bus into the controller

The controller is the system's HTTP boundary. Its only job is to translate an HTTP request into a domain message and map the result back to an HTTP response. Everything in between belongs to the bus and the handlers — and that boundary becomes much cleaner once the controller dispatches commands and queries instead of calling service methods directly.

Before CQRS, `WalletController` held a reference to `WalletApplicationService` and called methods on it directly. Every time the service interface changed — a new parameter, a renamed method, a different return type — the controller had to change too. With CQRS, the controller knows one thing: what message to send.

The controller injects `DefaultCommandBus` and `DefaultQueryBus` by type — the **concrete bus classes**, not abstract protocols. This is the correct import:

```
from pyfly.cqrs import DefaultCommandBus, DefaultQueryBus
```

Why concrete classes? PyFly's CQRS auto-configuration registers exactly one instance of each bus in the DI container. Injecting by the concrete type is unambiguous — no protocol dispatch, and the type checker sees the full `send` / `query` surface. Using a protocol alias would require an explicit container binding; the concrete type works out of the box.

Route ordering: why the single-resource handlers are named `wallet_*`

The framework registers a controller's routes in **alphabetical method-name order**; Starlette's router then applies first-registered-wins matching. This means a literal segment like `/rich` must be registered *before* the path variable `/ {wallet_id}` — otherwise every `GET /api/v1/wallets/rich` request would match the variable route and look for a wallet whose id is the string `"rich"`.

The collection handlers are named `list_wallets` and `list_rich_wallets`; the single-resource handlers are named `wallet_detail` and `wallet_balance`. Alphabetically, `l` sorts before `w`, so the collection routes (`GET /`, `GET /rich`) are always registered ahead of the parameterised routes (`GET / {wallet_id}`, `GET / {wallet_id}/balance`). If you rename `wallet_detail` to something that sorts before `list_*`, the `/rich` route will silently break.

Step 8 — Wire the buses into the controller. Replace the old `WalletApplicationService` dependency with the two buses, then turn each endpoint into a one-liner: build a command or query from the request, dispatch it, return the result. Here is the complete controller.

lumen/web/controllers/wallet_controller.py

```

from __future__ import annotations

from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.core.services.wallets.get_balance_query import GetBalance
from lumen.core.services.wallets.get_wallet_query import GetWallet
from lumen.core.services.wallets.list_rich_wallets_query import
ListRichWallets
from lumen.core.services.wallets.list_wallets_query import ListWallets
from lumen.core.services.wallets.open_wallet_command import OpenWallet
from lumen.core.services.wallets.withdraw_funds_command import WithdrawFunds
from lumen.interfaces.dtos.v1.balance_dto import BalanceDto
from lumen.interfaces.dtos.v1.deposit_request import DepositRequest
from lumen.interfaces.dtos.v1.open_wallet_request import OpenWalletRequest
from lumen.interfaces.dtos.v1.page_dto import PageDto
from lumen.interfaces.dtos.v1.wallet_dto import WalletDto
from pyfly.container import rest_controller
from pyfly.cqrs import DefaultCommandBus, DefaultQueryBus
from pyfly.data import Pageable, Sort
from pyfly.kernel import ResourceNotFoundException
from pyfly.web import (
    Body, PathVar, QueryParam, Valid,
    get_mapping, post_mapping, request_mapping,
)

#: Newest-first ordering shared by the list endpoints.
_NEWEST_FIRST = Sort.by("created_at").descending()

@rest_controller
@request_mapping("/api/v1/wallets")
class WalletController:
    """Digital-wallet REST API: open, deposit, withdraw, list, inspect."""

    def __init__(
        self, commands: DefaultCommandBus, queries: DefaultQueryBus
    ) → None:
        self._commands = commands

```

```

        self._queries = queries

# --- commands -----

@post_mapping("", status_code=201)
async def open_wallet(
    self, request: Valid[Body[OpenWalletRequest]]
) → dict[str, str]:
    wallet_id = await self._commands.send(
        OpenWallet(owner_id=request.owner_id, currency=request.currency)
    )
    return {"wallet_id": wallet_id}

@post_mapping("/{wallet_id}/deposit")
async def deposit(
    self,
    wallet_id: PathVar[str],
    request: Valid[Body[DepositRequest]],
) → dict[str, int | str]:
    balance = await self._commands.send(
        DepositFunds(wallet_id=wallet_id, amount=request.amount)
    )
    return {"wallet_id": wallet_id, "balance_minor": balance}

@post_mapping("/{wallet_id}/withdraw")
async def withdraw(
    self,
    wallet_id: PathVar[str],
    request: Valid[Body[DepositRequest]],
) → dict[str, int | str]:
    balance = await self._commands.send(
        WithdrawFunds(wallet_id=wallet_id, amount=request.amount)
    )
    return {"wallet_id": wallet_id, "balance_minor": balance}

# --- paged / specification queries (registered before /{wallet_id}) --

@get_mapping("")
async def list_wallets(

```

```

        self, page: QueryParam[int] = 1, size: QueryParam[int] = 20
    ) → PageDto[WalletDto]:
        result = await self._queries.query(
            ListWallets(pageable=Pageable.of(page, size, _NEWEST_FIRST))
        )
        return PageDto.from_page(result)

@get_mapping("/rich")
async def list_rich_wallets(
    self,
    min_minor: QueryParam[int] = 0,
    page: QueryParam[int] = 1,
    size: QueryParam[int] = 20,
) → PageDto[WalletDto]:
    result = await self._queries.query(
        ListRichWallets(
            min_minor=min_minor,
            pageable=Pageable.of(page, size, _NEWEST_FIRST),
        )
    )
    return PageDto.from_page(result)

# --- single-wallet queries (named wallet_* so they sort after list_*) -

@get_mapping("/{wallet_id}")
async def wallet_detail(self, wallet_id: PathVar[str]) → WalletDto:
    result = await self._queries.query(GetWallet(wallet_id=wallet_id))
    if result is None:
        raise ResourceNotFoundException(
            f"Wallet {wallet_id!r} not found",
            code="WALLET_NOT_FOUND",
            context={"wallet_id": wallet_id},
        )
    return result

@get_mapping("/{wallet_id}/balance")
async def wallet_balance(self, wallet_id: PathVar[str]) → BalanceDto:
    result = await self._queries.query(
        GetBalance(wallet_id=wallet_id)
    )

```

```

    )
    if result is None:
        raise ResourceNotFoundException(
            f"Wallet {wallet_id!r} not found",
            code="WALLET_NOT_FOUND",
            context={"wallet_id": wallet_id},
        )
    return result

```

Listing 7.15 — *WalletController: DefaultCommandBus + DefaultQueryBus + paged list endpoints*

Compare the constructor to its pre-CQRS form. Before, the controller took `WalletApplicationService` — a concrete class whose method signatures leaked business logic into the HTTP layer. Now it takes `DefaultCommandBus` and `DefaultQueryBus` — two opaque channels. The controller knows *what* to send; it knows nothing about *how* the message is processed.

Look at `open_wallet`. Before, it called `self._service.open_wallet(owner_id=..., currency=...)` — a positional contract that breaks whenever the service grows a new parameter. Now it constructs `OpenWallet(owner_id=request.owner_id, currency=request.currency)` — a named, immutable object whose fields are its own API. Add a field to the command? The controller stays the same until you choose to populate it.

The request DTOs (`OpenWalletRequest`, `DepositRequest`) are Pydantic models in `lumen/interfaces/dtos/v1/`. `OpenWalletRequest` validates `owner_id` length and constrains `currency` to the `Currency` enum. `DepositRequest` is shared by both the deposit and withdraw endpoints — both move a positive `amount` in the wallet's own currency. Field-level constraints in those DTOs are enforced by `Valid[Body[...]]` before the handler is ever called.

The paged list endpoints (`list_wallets` , `list_rich_wallets`) build a `Pageable` from the query-string parameters, dispatch the query through the bus, and wrap the resulting `Page[WalletDto]` in a `PageDto` for the wire. `PageDto` is a Pydantic model that mirrors all the `Page` metadata fields — `total` , `total_pages` , `has_next` , `has_previous` — so clients get consistent pagination envelopes without a custom serializer.

The `wallet_detail` and `wallet_balance` methods show the only remaining HTTP concern in the controller: translating a `None` query result into a 404 via `ResourceNotFoundException` . That mapping belongs here because 404 is an HTTP status code and the handler deliberately has no HTTP knowledge. Return types are declared as `WalletDto` and `BalanceDto` — Pydantic models the framework serializes to JSON automatically.

💡 LET THE BUS RAISE

You do not need to catch `CommandProcessingException` or `QueryProcessingException` in the controller unless you want to customize the error shape. The global exception handler maps `AggregateNotFound` to 404 and `BusinessRuleViolation` to 422 — the same as before. The bus exceptions propagate those originals transparently.

Run it — drive the full HTTP path. The vertical slice is complete: HTTP request → command/query → bus → handler → domain → repository → response. Prove it from the outside. Start the app (`uv run pyfly run --server uvicorn`), then in a second terminal open a wallet:

terminal

```
curl -s -X POST localhost:8080/api/v1/wallets \
-H 'content-type: application/json' \
-d '{"owner_id":"u-1","currency":"EUR"}'
```

Listing 7.15a — Open a wallet over HTTP

The `open_wallet` endpoint dispatches `OpenWallet` and returns the generated id:

```
{"wallet_id": "wlt-c5bbb2a7-dd49-4321-932e-e4c6bfa5cc2c"}
```

Copy that id, deposit into it, then read the balance back:

terminal

```
curl -s -X POST localhost:8080/api/v1/wallets/
wlt-c5bbb2a7-dd49-4321-932e-e4c6bfa5cc2c/deposit \
-H 'content-type: application/json' -d '{"amount":1500}'

curl -s localhost:8080/api/v1/wallets/
wlt-c5bbb2a7-dd49-4321-932e-e4c6bfa5cc2c/balance
```

Listing 7.15b — Deposit, then read the balance

The deposit echoes the new balance in minor units; the balance query returns the

`BalanceDto` :

```
{"wallet_id": "wlt-c5bbb2a7-dd49-4321-932e-e4c6bfa5cc2c", "balance_minor":
1500}
{"wallet_id": "wlt-c5bbb2a7-dd49-4321-932e-e4c6bfa5cc2c", "balance_minor": 150
0, "balance": 15.0}
```

Finally, confirm the route-ordering decision from earlier pays off — list the wallets and the "rich" subset:

terminal

```
curl -s 'localhost:8080/api/v1/wallets?page=1&size=20'
curl -s 'localhost:8080/api/v1/wallets/rich?min_minor=1000'
```

Listing 7.15c — Both list routes resolve correctly

Both return a `PageDto` envelope (`items` , `total` , `total_pages` , `has_next` , `has_previous`). The `/rich` call resolves to `list_rich_wallets` , *not* to `wallet_detail` looking for a wallet whose id is the literal string `"rich"` — exactly because `list_*` sorts before `wallet_*` . If `/rich` ever returns a 404, that ordering has been broken; revisit the method-name rule above.

⚠ USE A REAL ID

The wallet id above is illustrative — yours will differ on every `open_wallet` call. Paste the id returned by your own `POST /api/v1/wallets` into the deposit and balance URLs, or you will get a 404.

The handler pipeline

A single `send` or `query` call triggers more than just the handler. Understanding the pipeline tells you where to put each cross-cutting concern — and, just as importantly, where *not* to put it.

❗ WHAT IS A 'PIPELINE'?

A **pipeline** is just a fixed sequence of steps the bus runs around your handler — like an assembly line. Your message enters one end, passes through validation and authorization, gets handled, and (for commands) has its events published, before the result comes back out. You write only the one step in the middle (`do_handle`); the bus owns the rest, identically for every message.

The pipeline is defined once, in the bus, and applies uniformly to every handler. You never write pipeline logic inside a handler. The order is strict:

Step	Where it is defined	Applies to	Failure result
Business pre-condition validation	<code>validate()</code> hook on the message	Commands + Queries	<code>CqrsValidationException</code> (HTTP 422)
Authorization	<code>authorize()</code> hook on the message	Commands + Queries	<code>AuthorizationException</code> (HTTP 403)
Handler execution	<code>do_handle()</code>	Commands + Queries	Domain exceptions (4xx/5xx)
Domain event publishing	Bus pipeline (post-handler)	Commands only	—
Correlation ID cleanup	Bus pipeline (finally block)	Commands + Queries	—

Validation

Without a structured validation step, every handler would open with its own guard clauses: check this field is not blank, check that amount is positive. That logic would be duplicated across handlers and tested only through integration paths. Centralizing validation in the message itself solves both problems.

The bus invokes `validate()` before looking up the handler. If validation fails, the bus raises `CqrsValidationException` without ever reaching the handler.

The validation hook is also the right place for cross-field pre-conditions knowable from the fields alone — too simple for the domain aggregate, too application-specific for the request model:

```

@dataclass(frozen=True)
class DepositFunds(Command[int]):
    wallet_id: str
    amount: int

    async def validate(self) → ValidationResult:
        if not self.wallet_id.strip():
            return ValidationResult.failure(
                "wallet_id", "Wallet id is required"
            )
        if self.amount ≤ 0:
            return ValidationResult.failure(
                "amount", "Deposit amount must be > 0"
            )
        return ValidationResult.success()

```

Authorization

Once a message is structurally valid, the bus asks: is the caller *allowed* to perform this operation? Authorization answers before any database access happens — more efficient and safer, since you never load sensitive data only to discard it because the caller lacked permission.

Both commands and queries expose an `authorize()` hook. Return `AuthorizationResult.success()` to allow execution, or `AuthorizationResult.failure(resource, message)` to deny it. The bus raises `AuthorizationException` on denial, mapping to HTTP 403 via the global error handler.

A clean rule of thumb: use `authorize()` on the command for **operation-level** checks — who is allowed to call this command at all — and leave **resource-level** decisions (can this caller access *this specific* wallet?) to the handler, which has the loaded aggregate in scope:

lumen/cqrs/commands_auth.py

```

from __future__ import annotations
from dataclasses import dataclass

from pyfly.cqrs.authorization.types import AuthorizationResult
from pyfly.cqrs import Command

@dataclass(frozen=True)
class CloseWallet(Command[None]):
    """Close a wallet. Only internal service accounts may do this."""
    wallet_id: str
    requested_by: str

    async def authorize(self) → AuthorizationResult:
        internal_accounts = {"ops-service", "compliance-bot"}
        if self.requested_by not in internal_accounts:
            return AuthorizationResult.failure(
                "wallet",
                "Only internal service accounts may close wallets",
            )
        return AuthorizationResult.success()

```

Listing 7.16 — Authorization hook on a command

`CloseWallet.authorize` checks a known set of internal service accounts. If `requested_by` is not in the set, authorization fails before the handler is called. The set would normally come from a configuration value or a token claim injected at the controller boundary — it is hardcoded here for readability. The key point is that the check lives inside the command, not scattered across handler code.

Distributed tracing

When one HTTP request triggers multiple commands — and each command may call downstream services — you need a way to stitch all the logs and spans together. That is what `CorrelationContext` provides.

Both buses set a correlation ID at the start of every pipeline execution. If the message carries an ID already (set via `command.set_correlation_id(id)`), that ID is used; otherwise a new UUID is generated. The prior ID is always restored in a `finally` block, so nested command dispatches within the same request do not clobber the outer trace.

`CorrelationContext` propagates across `await` chains via Python's `contextvars` — no need to thread the ID manually through every function argument. For cross-service propagation, serialize the context to outgoing headers and restore it on the receiving side:

```
from pyfly.cqrs.tracing.correlation import CorrelationContext

# On the sending side
headers = CorrelationContext.create_context_headers()
# {"X-Correlation-ID": "...", "X-Trace-ID": "...", "X-Span-ID": "..."}

# On the receiving side
CorrelationContext.extract_context_from_headers(headers)
```

The three headers — `X-Correlation-ID`, `X-Trace-ID`, and `X-Span-ID` — follow W3C Trace Context naming, so they are compatible with OpenTelemetry-instrumented infrastructure out of the box.

💡 WHERE TO PUT CROSS-CUTTING LOGIC

The bus pipeline is the right home for concerns that apply to *all* operations: validation, authorization, tracing, and metrics. The handler is the right home for concerns specific to *one* operation: loading the aggregate, driving behaviour, saving, draining events. If you find yourself adding a try/except to every handler, or copying the same pre-condition check into multiple handlers, it belongs in the pipeline — either as a `validate()` hook on the command or as a bus-level service. The pipeline scales uniformly; handler boilerplate does not.

✓ What you built

Part II is complete. Lumen now has a full vertical slice from HTTP to domain and back — one built on architectural decisions that will scale without rewriting.

In Chapter 5 you gave the system persistence: a `WalletRepository` subclassing `Repository[WalletEntity, str]` — the framework's Spring-Data-style generic repository that provides `find_by_id`, `find_all(pageable)`, `find_all_by_spec_paged`, and more out of the box, with the `AsyncSession` injected by relational auto-configuration. In Chapter 6 you promoted the wallet to a proper DDD aggregate: `Money` as an immutable value object, `Wallet(AggregateRoot[str])` as the consistency boundary enforcing the overdraft, currency-match, and positive-amount invariants, with `WalletOpened`, `FundsDeposited`, and `FundsWithdrawn` domain events buffered in the aggregate and drained to the event bus after a successful save.

In this chapter you separated the write model from the read model. `OpenWallet`, `DepositFunds`, and `WithdrawFunds` are frozen, validated command messages that flow through `DefaultCommandBus` — a pipeline that runs validation, authorization, handler execution, domain event publishing, and distributed tracing automatically for every command. Each command handler carries `@transactional()` on `do_handle`:

the decorator opens a committed unit of work from `self._session_factory`, swaps the session onto the repository, commits on success, and rolls back on failure. Persistence goes through `repository.upsert` — backed by `session.merge` — so INSERT and UPDATE share a single code path keyed on the aggregate's own id.

`GetWallet` and `GetBalance` are query messages that flow through `DefaultQueryBus` — the same pipeline without the event-publishing step, and without `@transactional()` because reads do not commit. `GetBalanceHandler` projects through a `@projection`-marked `BalanceView` interface and `Mapper.project`, copying only the declared fields and applying a registered major-unit transform. `ListWallets` and `ListRichWallets` round out the query side: `find_all(pageable)` returns a counted, sorted, offset-limited `Page[WalletEntity]`; `find_all_by_spec_paged` runs a composable `Specification` predicate on top of the same pagination machinery. Both use `Page.map(entity_to_dto)` to project items without touching the metadata.

Each handler carries the `@command_handler` + `@service` (or `@query_handler` + `@service`) stack: the first decorator registers the class by introspecting its generic type argument; the second wires it into the DI container so constructor dependencies are injected automatically.

`WalletController` no longer knows about the service layer. It injects `DefaultCommandBus` and `DefaultQueryBus`, builds a command or query from the HTTP request, dispatches it, and either returns the result or raises a domain exception. Single-resource handler methods are named `wallet_detail` and `wallet_balance` — a deliberate choice so they sort alphabetically *after* the collection methods `list_wallets` and `list_rich_wallets`, ensuring the literal `/rich` segment is registered before the `/ {wallet_id}` variable route.

Adding a new command now means three things: define a frozen dataclass, implement one `do_handle` decorated with `@command_handler` + `@service` and annotated with `@transactional()`, and add one endpoint that calls `self._commands.send`. The pipeline applies automatically.

Run it — the whole chapter, in one command. From the `samples/lumen` directory, run the CQRS flow tests one last time to confirm every piece you built this chapter still hangs together:

terminal

```
uv run --extra dev pytest tests/test_cqrs_flow.py -q
```

Listing 7.17 — Verify the full CQRS slice

All five scenarios pass — the happy-path lifecycle, the not-found query, the rejected overdraft, the rejected non-positive deposit, and the rejected deposit to an unknown wallet:

```
5 passed in 0.61s
```

Those last four cases are worth pausing on: each one exercises the *pipeline*, not the handler. The overdraft is refused by the aggregate and surfaces as `CommandProcessingException`; the non-positive deposit never reaches a handler at all because `validate()` rejects it first. You wrote zero error-handling code to get any of that.

Try it yourself

- Trace the full lifecycle in the test suite.** Open `samples/lumen/tests/test_cqrs_flow.py` and run it against a real database using Testcontainers (Chapter 11). The test `test_full_wallet_lifecycle` opens a wallet, deposits 1 500 minor units, withdraws 500, then queries both `GetWallet` and `GetBalance`. Step through it with a debugger: confirm that `wallet.clear_events()` drains the `FundsDeposited` and `FundsWithdrawn` events after each `upsert` call, and that `GetWallet` returns a `WalletDto` with `balance_minor = 1000` and `balance = 10.0`.
- Observe `upsert` vs `save`.** In a test, call `DepositFunds` twice on the same wallet without `@transactional()` and observe the `IntegrityError`. Then restore `@transactional()` and verify both deposits commit. Open `WalletRepository.upsert` and trace how `session.merge` resolves the primary-key conflict that a plain `INSERT` would raise.
- Add a `ListByOwner` query.** Define `ListByOwner(Query[list[WalletDto]])` with an `owner_id: str` field. Implement `ListByOwnerHandler` — decorated with `@query_handler` + `@service` — that calls `WalletRepository.find_by_owner_id(query.owner_id)` (the derived query stub already exists) and maps the result list with `entity_to_dto`. Add a `GET /api/v1/wallets/by-owner/{owner_id}` endpoint to `WalletController`. Ensure the new endpoint method name sorts before `wallet_detail` so Starlette matches the literal `/by-owner/...` segment first.
- Add authorization to `WithdrawFunds`.** Extend `WithdrawFunds` with an `initiated_by: str` field. Override `authorize()` to return `AuthorizationResult.failure("withdraw", "Initiator is required")` when

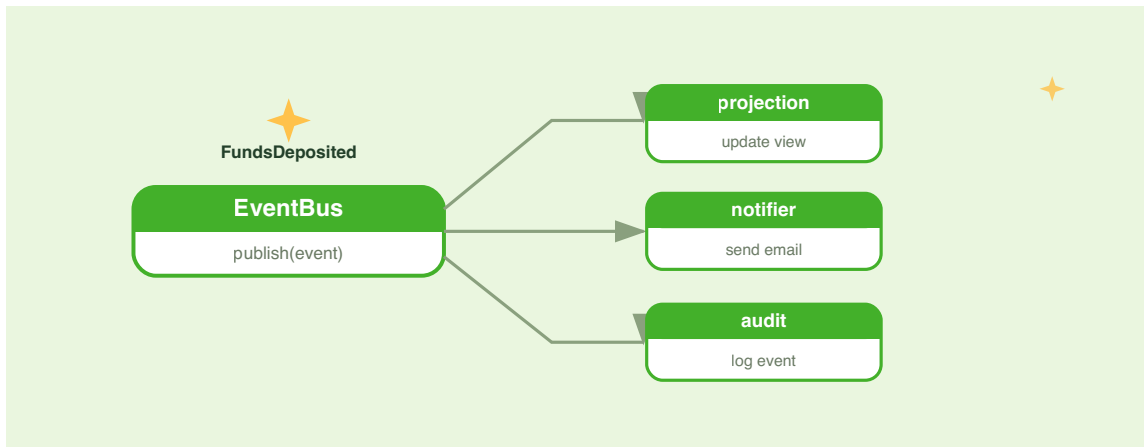
`initiated_by` is blank, and `AuthorizationResult.success()` otherwise. Update `WithdrawFundsHandler.do_handle` to record `command.initiated_by` in the `FundsWithdrawn` event payload. Write a test that calls `await WithdrawFunds(wallet_id="wlt-1", amount=100, initiated_by="").authorize()` and asserts that the result denies authorization.

PART III

Event-Driven Architecture

CHAPTER 8

Domain Events & Event-Driven Architecture



Lumen's wallet saves correctly, validates rigorously, and dispatches every write through a typed command. But look at what the command handlers do after `repo.add`: nothing. The domain events that the **Wallet** aggregate buffers — **WalletOpened**, **FundsDeposited**, **FundsWithdrawn** — are drained and discarded. The bus pipeline that Chapter 7 promised would "publish domain events" has nowhere to send them yet.

The gap matters in practice. Lumen needs a balance read model that stays in sync without reloading the aggregate on every query, a welcome notification when a new wallet opens, and an immutable audit trail of every financial movement for compliance. All three features depend on knowing *that something happened* — not who requested it or how it was handled.

Domain events are the answer. Instead of having a command handler call the notification service directly — or coupling the audit log to the repository — you publish the event: a concise, timestamped, immutable record of a fact. Every interested party subscribes independently. The handler that saved the wallet does not need to know what the auditor does, or even that an auditor exists. You can add a new subscriber months later without touching a single line of existing handler code.

This chapter builds the reaction side of Lumen's architecture. You will wire `EventPublisher` into the command handlers, introduce the `publish_domain_events` bridge that drains the aggregate's buffer and forwards each event to the bus, and write a `WalletAuditListener` that subscribes using `@event_listener` and maintains two in-memory projections: an immutable audit trail and a running deposit total. By the end of the chapter the write path and the read infrastructure are fully decoupled — each side evolves without the other noticing.

We will build this gradually, one small piece at a time. Each feature comes with a numbered walkthrough, the exact command to run, and the output you should expect to see. If you have followed along from Chapter 7 you already have the wallet command handlers and the `Wallet` aggregate in place; this chapter is verified against PyFly v26.6.110 and the Lumen sample under `samples/lumen`.

i JARGON, IN PLAIN LANGUAGE

A handful of terms recur in this chapter. A **domain event** is a small, frozen record of a business fact that already happened — for example "funds were deposited." A **publisher** is the thing that announces such facts; a **listener** (or *subscriber*) is code that reacts to them. A **bus** is the in-between switchboard that carries each announcement from the publisher to every interested listener. A **projection** is a read model that a listener builds up from events — here, an audit trail and a running total. A **port** is an interface (a Python `Protocol`) that your code depends on so the concrete implementation behind it can be swapped without changing callers. Keep these five in mind; the rest of the chapter is mostly about connecting them.

Two kinds of events

Before touching any code, it is worth being precise about what "event" means in PyFly. The framework uses the word for two distinct things, and confusing them leads to the wrong bus, the wrong subscription API, and subtle runtime surprises.

Application events (`pyfly.context.events`) are framework lifecycle notifications.

`ContextRefreshedEvent` fires when the DI container finishes wiring;

`ApplicationReadyEvent` fires when the HTTP server begins accepting connections;

`ContextClosedEvent` fires during shutdown. The `ApplicationEventBus` dispatches them to subscribers matched by Python class type — they are infrastructure plumbing for bootstrapping, deliberately separate from any business concept.

Domain events (`pyfly.eda`) are business-level facts: *a wallet was opened, funds were deposited, a transfer was completed*. The `EventPublisher` port wraps each payload in an `EventEnvelope` and routes it by the domain event class name —

"WalletOpened", "FundsDeposited", "FundsWithdrawn" — so listeners subscribe to named business facts rather than implementation details. Domain events are the subject of this chapter.

The distinction shapes what you can do with each kind. The `ApplicationEventBus` dispatches to callables keyed by class; `InMemoryEventBus` routes by class name and can be swapped for a Kafka-backed adapter without touching subscriber code. The rule is simple: use lifecycle events for infrastructure bootstrapping, domain events for everything with business meaning.

APPLICATION EVENTS ARE STILL USEFUL

If you need to warm a cache as soon as the application is ready, `@app_event_listener` on `ApplicationReadyEvent` is the right tool. The two systems coexist; you can use both in the same service.

Publishing events

The `EventPublisher` port

The first question a new team member usually asks is: "which class do I import to fire an event?" The answer is deliberately not a class — it is a protocol. `EventPublisher` is a **port** in the hexagonal-architecture sense: any code that needs to publish an event depends on this interface, and the bus implementation is injected from outside. That design decision is what lets you run `InMemoryEventBus` locally today and swap in a Kafka adapter in production without touching a single handler.

The protocol exposes two methods:

```

from pyfly.eda import EventPublisher

class EventPublisher(Protocol):
    def subscribe(self, event_type_pattern: str, handler: EventHandler) → None
    e: ...

    async def publish(
        self,
        destination: str,
        event_type: str,
        payload: dict,
        headers: dict[str, str] | None = None,
    ) → None: ...

```

`publish` wraps your data in an `EventEnvelope` before delivery — you never construct the envelope yourself. `subscribe` registers handlers programmatically, though in practice you will use the `@event_listener` decorator instead, because it lets the `ApplicationContext` wire subscriptions automatically at startup.

The bus bean exists only when `pyfly.eda.provider` is configured. For Lumen, `pyfly.yaml` sets it to `memory`:

`pyfly.yaml`

```

pyfly:
  eda:
    provider: memory
    # ... other keys omitted for brevity

```

Listing 8.0 — Enabling the in-memory EDA bus in `pyfly.yaml`

Without this line the `EventPublisher` bean is not registered and any handler that declares `events: EventPublisher` in its constructor will fail to start.

Let us turn that bus on before we use it.

Step 1 — Open Lumen's `pyfly.yaml`. Find the top-level `pyfly:` block. You will already see keys such as `app`, `server`, and `data` from the previous chapters.

Step 2 — Add the `eda` block. Insert the two lines shown in Listing 8.0 as a child of `pyfly:`, at the same indentation level as `server:` and `data:`. Indentation is significant in YAML, so line them up exactly.

Step 3 — Save the file. That is all the configuration the in-memory bus needs — no broker, no connection string, no extra dependency.

 RUN IT

Start the application and confirm it boots cleanly with the bus registered:

```
uv run pyfly run --server uvicorn
```

In the startup log you should see the application come up on the default app port and the management endpoints on the separate management port:

```
INFO starting_application name=lumen version=1.0.0
INFO uvicorn running on http://127.0.0.1:8080
INFO management endpoints on http://127.0.0.1:9090
```

If instead the process exits with an error mentioning that an `EventPublisher` dependency could not be resolved, the `eda` block is missing or mis-indented — return to Step 2.

WHERE THE APP AND THE DASHBOARD LIVE

As of v26.6.110 the application listens on `pyfly.server.port` (default `8080`), while the actuator and admin dashboard run on a **separate** management port, `pyfly.management.server.port` (default `9090`). The management port is open and unauthenticated by default; set `pyfly.management.security.enabled: true` to lock it down, or `pyfly.management.server.port: -1` to disable management endpoints entirely. None of this affects the EDA bus — it is purely in-process — but it is worth knowing which port is which when you start poking at the running app later in the chapter.

What just happened. You flipped a single configuration switch and PyFly registered an `EventPublisher` bean for you. From now on any `@service` that asks for `events: EventPublisher` in its constructor receives the in-memory bus automatically. Nothing publishes anything yet — that comes next — but the plumbing is in place.

The EventEnvelope

Every domain event reaches its listeners wrapped in an `EventEnvelope`. Think of it as the metadata layer that transforms a bare Python dictionary into a traceable, auditable, first-class fact. It is a frozen dataclass — immutable once created — that pairs the payload with the context every listener needs.

Field	Type	Default	Description
<code>event_type</code>	<code>str</code>	required	The domain event class name, e.g. <code>"FundsDeposited"</code> . Used for routing.
<code>payload</code>	<code>dict[str, Any]</code>	required	The event data.
<code>destination</code>	<code>str</code>	required	Logical channel or topic, e.g. <code>"wallet.events"</code> .
<code>event_id</code>	<code>str</code>	auto UUID	Unique ID for this event instance.
<code>timestamp</code>	<code>datetime</code>	<code>datetime.now(UTC)</code>	UTC creation time.
<code>headers</code>	<code>dict[str, str]</code>	<code>{}</code>	Arbitrary metadata: correlation IDs, trace context, and so on.

Three fields deserve particular attention. `event_id` is a stable UUID generated by the bus at publish time — your **idempotency key** for exactly-once semantics, available in every listener with no extra work. `timestamp` records when the fact was observed, not when the listener processes it, so it stays accurate even if a listener runs with a delay. `headers` carries cross-cutting concerns such as distributed trace IDs — metadata that has nothing to do with the business payload but matters enormously for observability. Because the envelope is frozen, handlers can safely pass it across async boundaries without defensive copying.

`event_type` holds the **class name** of the domain event — `"WalletOpened"`, `"FundsDeposited"`, or `"FundsWithdrawn"` — not a dot-separated path. Listeners subscribe by those same class names, so the subscription contract is defined by the domain model, not by string conventions invented outside it.

What just happened. Do not be put off by the six-field table — in everyday code you touch only two of them. When you publish you supply `event_type`, `payload`, and `destination`; the bus fills in `event_id`, `timestamp`, and `headers` for you. When you react, you read `envelope.event_type` to know *what* happened and `envelope.payload` to know the details. The other three fields are there when you need them (idempotency, ordering, tracing) and quietly out of the way when you do not.

The domain events in the Wallet aggregate

The `Wallet` aggregate raises typed, frozen-dataclass domain events. Each event's class name becomes its routing `event_type` on the bus:

lumen/models/entities/v1/wallet_entity.py

```

from __future__ import annotations

from dataclasses import dataclass
from datetime import UTC, datetime

from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.money import Money
from pyfly.domain import AggregateRoot, BusinessRuleViolation, DomainEvent


@dataclass(frozen=True)
class WalletOpened(DomainEvent):
    wallet_id: str = ""
    owner_id: str = ""
    currency: str = ""


@dataclass(frozen=True)
class FundsDeposited(DomainEvent):
    wallet_id: str = ""
    amount: int = 0
    currency: str = ""
    balance: int = 0


@dataclass(frozen=True)
class FundsWithdrawn(DomainEvent):
    wallet_id: str = ""
    amount: int = 0
    currency: str = ""
    balance: int = 0


class Wallet(AggregateRoot[str]):
    """Wallet aggregate root — owns the ``balance ≥ 0`` invariant."""

    def deposit(self, amount: Money) → None:

```

```

        """Credit *amount*; raises FundsDeposited."""
        self._assert_currency(amount)
        self.balance = self.balance.add(amount)
        self.raise_event(
            FundsDeposited(
                wallet_id=self.id,
                amount=amount.amount,
                currency=amount.currency.value,
                balance=self.balance.amount,
            )
        )
    # ... open() and withdraw() follow the same pattern

```

Listing 8.1 — Domain events raised by the Wallet aggregate

`DomainEvent` is a base frozen dataclass. Its `event_type` property returns `type(self).__name__` — the class name — which is exactly what `EventPublisher` uses as the routing key. `raise_event` buffers the event in the aggregate; the command handler drains that buffer by calling `wallet.clear_events()` after a successful persist.

The publish bridge

Rather than repeating the drain loop in every command handler, Lumen extracts it into a single `publish_domain_events` coroutine. The bridge serialises each drained event with `dataclasses.asdict`, then calls `publisher.publish` with the class name as `event_type` and `"wallet.events"` as the logical channel:

lumen/core/services/wallets/event_publishing.py

```

from __future__ import annotations

import dataclasses
from collections.abc import Iterable
from typing import Any

from lumen.core.services.listeners.wallet_audit_listener import (
    WALLET_EVENTS_DESTINATION,
)
from pyfly.domain import DomainEvent
from pyfly.eda import EventPublisher

def _to_payload(event: DomainEvent) → dict[str, Any]:
    """Flatten a frozen-dataclass domain event into a dict."""
    payload: dict[str, Any] = dataclasses.asdict(event)
    payload.setdefault("event_type", event.event_type)
    return payload

async def publish_domain_events(
    publisher: EventPublisher, events: Iterable[DomainEvent]
) → None:
    """Publish each drained domain event on the wallet events channel.

    The envelope's ``event_type`` is the domain event class name
    (``WalletOpened`` / ``FundsDeposited`` / ``FundsWithdrawn``).
    """
    for event in events:
        await publisher.publish(
            destination=WALLET_EVENTS_DESTINATION,
            event_type=event.event_type,
            payload=_to_payload(event),
        )

```

Listing 8.2 — `publish_domain_events` bridges drained events to the EDA bus

`WALLET_EVENTS_DESTINATION` is the constant `"wallet.events"` defined in `wallet_audit_listener.py` and shared by publisher and listener so the channel name cannot drift. `event.event_type` is the class-name property on `DomainEvent`: `"WalletOpened"`, `"FundsDeposited"`, or `"FundsWithdrawn"`.

Wiring the publisher into the command handlers

In Chapter 7 the command handlers loaded aggregates, drove domain behaviour, and saved — leaving buffered events on the floor. Now you close that gap. Inject an `EventPublisher` alongside the `WalletRepository` and an `async_sessionmaker`, decorate `do_handle` with `@transactional()`, and after `repo.upsert(...)` drain the aggregate's buffer and publish each event through the bridge.

The `@transactional()` decorator (from `pyfly.data.relational.sqlalchemy`) opens a dedicated `AsyncSession` from the injected `async_sessionmaker`, binds it to the repository for the call, commits on success, and rolls back on failure. That means the load → mutate → save sequence is one committed unit of work, and no event is published unless the row actually lands in the database.

Here is the change, broken into the four edits you will make to `DepositFundsHandler`.

Step 1 — Add the publisher to the constructor. Alongside the existing `repository` parameter, accept `events: EventPublisher` and store it as `self._events`. Type it as the *protocol* `EventPublisher`, never as `InMemoryEventBus` — that is what keeps the handler ignorant of which bus is running.

Step 2 — Accept the session factory. Add `session_factory: async_sessionmaker[AsyncSession]` and store it as `self._session_factory`. The `@transactional()` decorator looks for exactly this attribute name to open its unit of work, so the name matters.

Step 3 — Decorate `do_handle` with `@transactional()`. This wraps the whole load-mutate-save sequence in one committed transaction.

Step 4 — Drain and publish after the save. As the last step inside `do_handle`, after `self._repository.upsert(...)`, call `await publish_domain_events(self._events, wallet.clear_events())`. `wallet.clear_events()` returns the buffered events *and* empties the buffer, so they are never published twice.

Putting those four edits together gives the updated `DepositFundsHandler`:

lumen/core/services/wallets/deposit_funds_handler.py

```

from __future__ import annotations

from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker

from lumen.core.mappers.wallet_mapper import to_aggregate, to_entity
from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.core.services.wallets.event_publishing import publish_domain_events
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.domain import AggregateNotFound
from pyfly.data.relational.sqlalchemy import transactional
from pyfly.eda import EventPublisher

@command_handler
@service
class DepositFundsHandler(CommandHandler[DepositFunds, int]):
    """Credit funds to an existing wallet; returns the new balance."""

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events
        self._session_factory = session_factory

    @transactional()
    async def do_handle(self, command: DepositFunds) → int:
        entity = await self._repository.find_by_id(command.wallet_id)
        if entity is None:
            raise AggregateNotFound("Wallet", command.wallet_id)

```

```

wallet = to_aggregate(entity)
wallet.deposit(Money(amount=command.amount, currency=wallet.currency))
await self._repository.upsert(to_entity(wallet))

await publish_domain_events(self._events, wallet.clear_events())
return wallet.balance.amount

```

Listing 8.3 — *DepositFundsHandler*: *@transactional* unit-of-work, then publish

Three design decisions are worth noting. First, `events: EventPublisher` is typed as the protocol, not as `InMemoryEventBus` — the DI container injects whichever implementation is registered, so the handler never knows or cares which bus is active. Second, the publish call sits *after* `self._repository.upsert(...)` inside the `@transactional()` unit of work: if the save fails the decorator rolls back before `publish_domain_events` is reached, so listeners never see a fact that never persisted. Third, the handler works with the ORM entity directly via `to_aggregate` / `to_entity` mappers — the aggregate is rehydrated from the row, mutated, and mapped back to a row before the upsert. If the publish fails after a successful save you have an at-least-once delivery challenge — Chapter 10 addresses that with transactional outbox patterns. For now, the in-memory bus never fails.

📘 RUN IT

With the application running (`uv run pyfly run --server uvicorn`), open a wallet and deposit into it from a second terminal:

```
# Open a wallet - returns its id
curl -s -X POST http://localhost:8080/api/v1/wallets \
  -H 'content-type: application/json' \
  -d '{"owner_id": "u-1", "currency": "EUR"}'
# {"wallet_id": "wlt-..."}

# Deposit 1500 minor units (15.00 EUR) into that wallet
curl -s -X POST http://localhost:8080/api/v1/wallets/wlt-.../deposit \
  -H 'content-type: application/json' \
  -d '{"amount": 1500}'
# {"wallet_id": "wlt-...", "balance": 1500}
```

The HTTP response confirms the balance, but the more interesting evidence is in the application log: because the deposit published a `FundsDeposited` event and the audit listener (which you will build in the next section) reacts to it, you will see a `wallet_audit_observed` log line for `event_type=FundsDeposited`. No listener yet? Then the publish happens silently — which is exactly the point: the handler does not know whether anyone is listening.

What just happened. The command handler now does one extra thing after saving: it drains the events the aggregate buffered and hands them to the bus. The crucial ordering is *save first, publish second*, all inside one transaction. If the database write rolls back, the events are never published, so a listener can never observe a fact that did not actually persist. The handler gained four lines and zero new knowledge — it still has no idea what, if anything, will react.

The `OpenWalletHandler` follows the same pattern:

lumen/core/services/wallets/open_wallet_handler.py

```

from __future__ import annotations

from uuid import uuid4

from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker

from lumen.core.mappers.wallet_mapper import to_entity
from lumen.core.services.wallets.event_publishing import publish_domain_events
from lumen.core.services.wallets.open_wallet_command import OpenWallet
from lumen.models.entities.v1.wallet_entity import Wallet
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.data.relational.sqlalchemy import transactional
from pyfly.eda import EventPublisher

@command_handler
@service
class OpenWalletHandler(CommandHandler[OpenWallet, str]):
    """Open a new, empty wallet."""

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events
        # @transactional resolves the unit-of-work session from here.
        self._session_factory = session_factory

    @transactional()
    async def do_handle(self, command: OpenWallet) → str:
        wallet_id = f"wlt-{uuid4()}"

```

```

    wallet = Wallet.open(
        wallet_id=wallet_id,
        owner_id=command.owner_id,
        currency=command.currency,
    )
    await self._repository.upsert(to_entity(wallet))

    await publish_domain_events(self._events, wallet.clear_events())
    return wallet_id

```

Listing 8.4 — *OpenWalletHandler: @transactional, upsert, then publish WalletOpened*

`return wallet_id` comes *after* the publish call — the handler fulfils its contract only once every fact produced by the operation has been dispatched. The wallet id is generated locally with `uuid4()` rather than delegated to a repository method; `to_entity` maps the aggregate to a row before the upsert so the framework `Repository` receives the ORM model it expects.

The `@publish_result` shortcut

In simpler services where a method's return value *is* the event payload — common in code that has not adopted the full aggregate pattern — `@publish_result` removes the manual publish call entirely:

lumen/eda/publish_result_example.py

```

from pyfly.eda import publish_result
from pyfly.eda.adapters.memory import InMemoryEventBus

bus = InMemoryEventBus()

@publish_result(bus, destination="wallet.events",
event_type="FundsTransferred")
async def transfer_funds(source_id: str, target_id: str, amount: int) → dict:
    # Business logic omitted – the returned dict IS the event payload.
    return {
        "source_id": source_id,
        "target_id": target_id,
        "amount": amount,
    }

```

Listing 8.5 — `@publish_result` auto-publishes the method's return value

When `transfer_funds` returns, the decorator intercepts the result and calls `bus.publish` with it as the payload — no boilerplate loop needed. `destination` and `event_type` are fixed at decoration time, keeping the business function clean. `@publish_result` also accepts an optional `condition` predicate: the event is published only when the result satisfies the test, which is useful for conditional workflows where not every successful execution should broadcast.

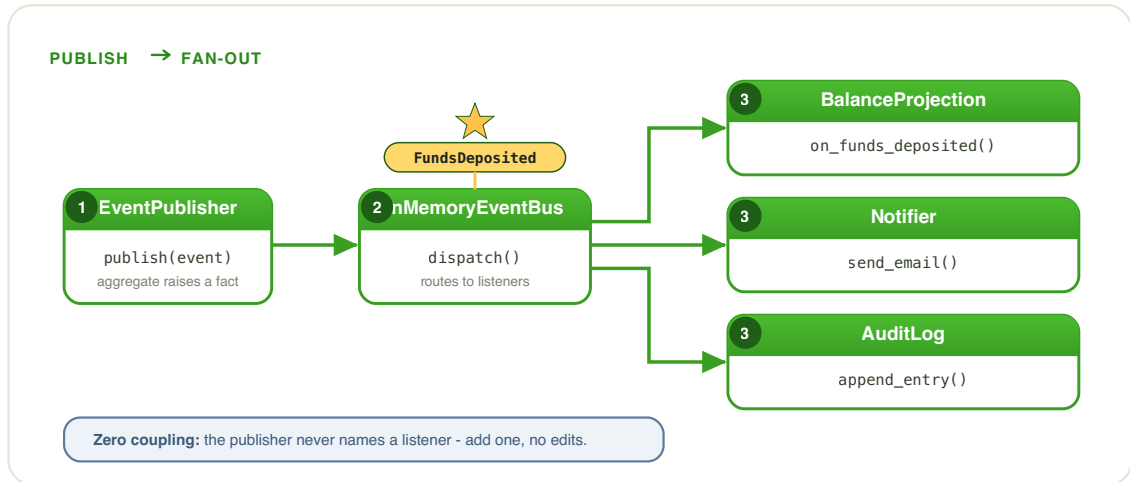


Figure 8.1 — One publisher, many independent listeners.

🌿 SPRING PARITY

`EventPublisher` is PyFly's counterpart of Spring's `ApplicationEventPublisher`. Calling `publisher.publish(...)` is equivalent to `applicationEventPublisher.publishEvent(event)`. The `@event_listener` decorator (next section) mirrors Spring's `@EventListener` for synchronous, same-transaction reactions. `@publish_result` achieves what Spring developers often wire manually with AOP `@AfterReturning` advice.

Reacting with `@event_listener`

Publishing an event is only half the picture. An event that nobody reacts to is just a log entry. The value of the event-driven model lies in the *reactions* it enables — independent behaviours that activate in response to the same published fact, each unaware of the others.

PyFly's `@event_listener` decorator is the simplest way to register a reaction.

Decorate any async method with the class names it cares about, and

`ApplicationContext` wires the subscription during startup — no bus reference needed at decoration time.

```
from pyfly.eda import event_listener, EventEnvelope

@event_listener(event_types=["FundsDeposited"])
async def on_funds_deposited(envelope: EventEnvelope) → None:
    ...
```

`event_types` accepts exact class names. Listeners inside a `@service` class receive an `EventEnvelope` as their sole argument. Because matching happens at the bus level — not inside your function — a single listener method can subscribe to multiple event types in one declaration.

WalletAuditListener

Lumen's production listener is `WalletAuditListener`. It subscribes to all three wallet domain events and maintains two in-memory projections: an ordered **audit trail** and a **running net-deposit total** per wallet.

We will assemble it in small pieces. Read the four steps first, then study the full listing below — it is the same code, shown whole.

Step 1 — Declare a plain `@service` class. A listener is just a service bean with some state. Give it an `__init__` that initialises the two projections: `self._entries: list[AuditEntry] = []` for the audit trail and `self._running_totals: dict[str, int] = {}` for the per-wallet totals.

Step 2 — Write the reaction method. Add an `async def on_wallet_event(self, envelope: EventEnvelope) → None`. It receives one `EventEnvelope` and nothing else.

Step 3 — Stamp it with `@event_listener` . Decorate the method with `@event_listener(event_types=["WalletOpened", "FundsDeposited", "FundsWithdrawn"])` . One method can subscribe to all three class names in a single declaration. This decorator does not subscribe immediately — it *stamps* the method with metadata that `ApplicationContext` finds at startup and uses to auto-subscribe it to the `EventPublisher` bean.

Step 4 — Project the event. Inside the method, append an `AuditEntry` for every event, then branch on `envelope.event_type` to adjust the running total: set it to zero on `WalletOpened` , add on `FundsDeposited` , subtract on `FundsWithdrawn` . Expose read accessors (`entries` , `entries_for` , `running_total`) so other code can query the projections.

lumen/core/services/listeners/wallet_audit_listener.py

```

from __future__ import annotations

import logging
from dataclasses import dataclass
from datetime import datetime

from pyfly.container import service
from pyfly.eda import EventEnvelope, event_listener

logger = logging.getLogger(__name__)

WALLET_EVENTS_DESTINATION = "wallet.events"

@dataclass(frozen=True)
class AuditEntry:
    """One observed domain event, captured for the audit trail."""

    event_type: str
    wallet_id: str
    event_id: str
    occurred_at: datetime
    payload: dict[str, object]

@service
class WalletAuditListener:
    """In-memory audit log + running-total projection over wallet events."""

    def __init__(self) → None:
        self._entries: list[AuditEntry] = []
        self._running_totals: dict[str, int] = {}

    @event_listener(
        event_types=["WalletOpened", "FundsDeposited", "FundsWithdrawn"]
    )
    async def on_wallet_event(self, envelope: EventEnvelope) → None:

```

```

"""Project every wallet domain event into the read models."""
payload = dict(envelope.payload)
wallet_id = str(payload.get("wallet_id", ""))

self._entries.append(
    AuditEntry(
        event_type=envelope.event_type,
        wallet_id=wallet_id,
        event_id=str(payload.get("event_id", envelope.event_id)),
        occurred_at=envelope.timestamp,
        payload=payload,
    )
)

if envelope.event_type == "WalletOpened":
    self._running_totals.setdefault(wallet_id, 0)
elif envelope.event_type == "FundsDeposited":
    amount = int(payload.get("amount", 0))
    self._running_totals[wallet_id] = (
        self._running_totals.get(wallet_id, 0) + amount
    )
elif envelope.event_type == "FundsWithdrawn":
    amount = int(payload.get("amount", 0))
    self._running_totals[wallet_id] = (
        self._running_totals.get(wallet_id, 0) - amount
    )

logger.info(
    "wallet_audit_observed",
    extra={"event_type": envelope.event_type, "wallet_id": wallet_id},
)

@property
def entries(self) → list[AuditEntry]:
    """A snapshot of the audit log, in observation order."""
    return list(self._entries)

def entries_for(self, wallet_id: str) → list[AuditEntry]:
    """The audit entries recorded for one wallet."""

```

```

    return [e for e in self._entries if e.wallet_id == wallet_id]

    def running_total(self, wallet_id: str) → int:
        """Net funds (deposited minus withdrawn) for wallet_id, minor
        units."""
        return self._running_totals.get(wallet_id, 0)

```

Listing 8.6 — *WalletAuditListener: audit trail + running-total projection*

Here is what the listener does, step by step.

`@event_listener(event_types=["WalletOpened", "FundsDeposited", "FundsWithdrawn"])` tells `ApplicationContext` to subscribe `on_wallet_event` to those three class names. Because the class is a `@service` bean, PyFly discovers it at startup and wires the subscriptions automatically — you never call `bus.subscribe` by hand.

`on_wallet_event` receives an `EventEnvelope`. `envelope.event_type` is the class name of the raised domain event. `envelope.payload` is the dict produced by `dataclasses.asdict` in the publish bridge, so its keys match the dataclass field names exactly — `wallet_id`, `amount`, `currency`, `balance`.

The method appends an `AuditEntry` for every event, then branches on `event_type` to update the running total. Notice what is absent: no import of the `Wallet` aggregate, no repository call, no knowledge of how the deposit was processed. The projection reacts purely to the published fact.

What just happened. You wrote a self-contained reaction. The `@event_listener` decorator is the entire wiring story — there is no `bus.subscribe(...)` call anywhere in this file. At startup, `ApplicationContext` scans your `@service` beans, finds the stamped `on_wallet_event` method, and subscribes it to the bus for each of the three class names. The listener and the command handlers never reference each other; they are connected only through the events they agree to name.

 ENVELOPE METADATA IN PROJECTIONS

`envelope.timestamp` gives you the authoritative event time — when the fact was recorded, not when the listener ran. Store it in your read model and you get a cheap `occurred_at` column for free, with no clock skew between writer and reader.

Testing the listener end-to-end

The test suite exercises the full publish-and-receive path with no mocks. The `conftest` wires a shared `InMemoryEventBus`, mirrors the `@event_listener` discovery step by subscribing `on_wallet_event` to each declared class name, and registers real command handlers that share the same bus reference:

```
# tests/conftest.py (abbreviated)
from pyfly.eda.adapters.memory import InMemoryEventBus

@pytest_asyncio.fixture
async def event_bus() → InMemoryEventBus:
    yield InMemoryEventBus()

@pytest_asyncio.fixture
async def audit_listener(event_bus: InMemoryEventBus) → WalletAuditListener:
    listener = WalletAuditListener()
    method = listener.on_wallet_event
    for pattern in method.__pyfly_event_patterns__:
        event_bus.subscribe(pattern, method)
    yield listener
```

With that wiring in place, the test sends real commands and asserts on the listener's read models:

lumen/tests/test_event_listener.py

```

from __future__ import annotations

import pytest
from lumen.core.services.listeners import WalletAuditListener
from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.core.services.wallets.open_wallet_command import OpenWallet
from lumen.core.services.wallets.withdraw_funds_command import WithdrawFunds
from lumen.interfaces.enums.v1.currency import Currency

from pyfly.cqrs import DefaultCommandBus

@pytest.mark.asyncio
async def test_listener_observes_wallet_events(
    command_bus: DefaultCommandBus,
    audit_listener: WalletAuditListener,
) → None:
    wallet_id = await command_bus.send(
        OpenWallet(owner_id="u-1", currency=Currency.EUR)
    )
    await command_bus.send(DepositFunds(wallet_id=wallet_id, amount=1500))
    await command_bus.send(WithdrawFunds(wallet_id=wallet_id, amount=400))

    entries = audit_listener.entries_for(wallet_id)
    assert [e.event_type for e in entries] == [
        "WalletOpened",
        "FundsDeposited",
        "FundsWithdrawn",
    ]

    deposited = entries[1]
    assert deposited.payload["amount"] == 1500
    assert deposited.payload["currency"] == "EUR"
    assert deposited.payload["balance"] == 1500

```

```
# running_total = deposited - withdrawn
assert audit_listener.running_total(wallet_id) == 1100
```

Listing 8.7 — End-to-end test: commands publish, listener projects

The test proves the full chain: `OpenWalletHandler` → `publish_domain_events` → `InMemoryEventBus` → `WalletAuditListener.on_wallet_event` → `audit_listener.entries_for(...)`. No mocks, no fakes — the production code path runs as written.

RUN IT

Run the event-listener test from the Lumen project root:

```
uv run --extra dev pytest tests/test_event_listener.py -q
```

You should see the suite pass:

```
...
[100%]
3 passed in 0.XXs
```

The three tests cover the happy path (open, deposit, withdraw, then assert the audit trail and running total), the empty-projection case before any command runs, and the negative case where an overdrawing withdrawal raises and therefore publishes *no* event — so it must not appear in the audit log. If a test fails with a missing-attribute error on `__pyfly_event_patterns__`, the `@event_listener` decorator is not applied to `on_wallet_event`; revisit Step 3 of the previous section.

What makes this design compelling is that adding the listener required zero changes to the command handlers, the `Wallet` aggregate, or any repository.

`DepositFundsHandler` has no idea a projection exists. Both sides are entirely independent — each is a consequence of the same published fact, connected only by the bus.

When listeners fail: error strategies

A misbehaving listener raises a pointed question: should the failure stop the entire delivery chain, or should the bus continue notifying the remaining listeners? The right answer depends on the listener's role. PyFly gives you explicit control rather than imposing a single policy.

By default, `InMemoryEventBus` invokes listeners sequentially and propagates any exception — the right behaviour for development, where a failing listener should surface loudly. In production you usually need finer control.

`ErrorStrategy` is an enum that governs how the bus behaves when a listener raises:

```
from pyfly.eda import ErrorStrategy
```

Member	Value	Behaviour
<code>IGNORE</code>	<code>"IGNORE"</code>	Silently swallow the exception. Processing continues with the next handler.
<code>LOG_AND_CONTINUE</code>	<code>"LOG_AND_CONTINUE"</code>	Log the error then continue. The safest default for non-critical listeners.
<code>RETRY</code>	<code>"RETRY"</code>	Re-attempt delivery. Retry count and back-off are configured separately.
<code>DEAD_LETTER</code>	<code>"DEAD_LETTER"</code>	Move the failed event to a dead-letter destination for later inspection.
<code>FAIL_FAST</code>	<code>"FAIL_FAST"</code>	Propagate the exception immediately. No further handlers are invoked.

💡 MATCH STRATEGY TO LISTENER CRITICALITY

An audit listener should use `LOG_AND_CONTINUE` — a broken audit logger must not halt a financial transaction. A projection that drives query responses might warrant `RETRY` to ensure the read model stays consistent. A notifier can tolerate `IGNORE` since a missed welcome email is not a data integrity issue.

⚠️ SIDE EFFECTS AND IDEMPOTENCY

If a listener performs a side effect — writing a database row, sending an email — and the bus retries delivery after a transient failure, the effect can run more than once. Design listeners to be idempotent: write a row only if the `event_id` has not already been recorded, send an email only if the welcome flag is not already set. The `envelope.event_id` (a stable UUID generated by the bus) is your idempotency key.

In-memory today, a broker tomorrow

`InMemoryEventBus` is the out-of-the-box implementation — the default `EventPublisher` the `ApplicationContext` provides. It runs entirely in-process: `publish` is a direct async call, there is no serialisation, and undelivered events vanish if the process dies. For local development, integration tests, and monoliths that need no cross-process delivery that is perfectly acceptable.

Understanding how the in-memory bus works internally makes it easier to reason about behaviour at the edges — and to appreciate exactly what changes when you swap in a broker.

```
from pyfly.eda.adapters.memory import InMemoryEventBus

bus = InMemoryEventBus()

bus.subscribe("FundsDeposited", my_handler)

await bus.publish(
    destination="wallet.events",
    event_type="FundsDeposited",
    payload={"wallet_id": "w-001", "amount": 5000, "currency": "EUR", "balance": 5000},
)
```

A `publish` call executes four steps in sequence:

1. Wraps the arguments in an `EventEnvelope` with a generated `event_id` and a UTC `timestamp`.
2. Iterates every registered `(pattern, handler)` pair.
3. For each pair where `fnmatch.fnmatch(event_type, pattern)` is `True`, calls the handler with the envelope.
4. Handlers run sequentially in subscription order.

Subscriptions use Python's `fnmatch`, so `"Funds*"` matches both `"FundsDeposited"` and `"FundsWithdrawn"`, and `"*"` matches everything. The sequential invocation in step 4 makes listener order deterministic — useful in tests — but it also means a slow listener delays all subsequent ones. Broker-backed adapters typically dispatch in parallel; keep that difference in mind when reasoning about throughput.

Because every listener in Lumen depends on the `EventPublisher` *protocol*, not on `InMemoryEventBus` directly, the implementation swaps without touching a single listener. Chapter 10 introduces Kafka and RabbitMQ adapters; switching either in is a configuration change — `WalletAuditListener` keeps working without modification.

INMEMORYEVENTBUS AND TESTING

`InMemoryEventBus` is also the right tool for tests. Inject a fresh `InMemoryEventBus` as a fixture, subscribe a capturing handler, exercise your command handler, and assert on the `EventEnvelope` objects the handler received — including `event_type`, `payload`, `event_id`, and `timestamp`. No mocking, no fakes, just the real bus with controlled inputs.

✓ What you built

Part III is open.

This chapter closed the loop Chapter 7 began. `Wallet` raised domain events in Chapter 6; the command handlers published them here; `WalletAuditListener` reacts to those facts without knowing anything about the command path that triggered them.

The architecture is genuinely event-driven within a single process. Here is a quick reference to each piece:

Piece	Role
<code>EventPublisher</code>	Port — a protocol any bus implementation fulfils
<code>InMemoryEventBus</code>	Default adapter — in-process, zero config; activated by <code>pyfly.eda.provider: memory</code>
<code>EventEnvelope</code>	Carries payload + <code>event_id</code> , <code>timestamp</code> , <code>destination</code> , <code>headers</code>
<code>@event_listener(event_types=[...])</code>	Subscription decorator — class names; context wires it at startup
<code>publish_domain_events</code>	Bridge — drains <code>wallet.clear_events()</code> , serialises with <code>dataclasses.asdict</code> , calls <code>publisher.publish</code>
<code>ErrorStrategy</code>	Controls failure handling: <code>IGNORE</code> , <code>LOG_AND_CONTINUE</code> , <code>RETRY</code> , <code>DEAD_LETTER</code> , <code>FAIL_FAST</code>

Three principles carry forward into the rest of Part III: **save before you publish** — listeners must never see uncommitted facts; **design listeners for idempotency** — retries must be safe; **depend on the port, not the adapter** — the bus can be swapped without touching listener code.

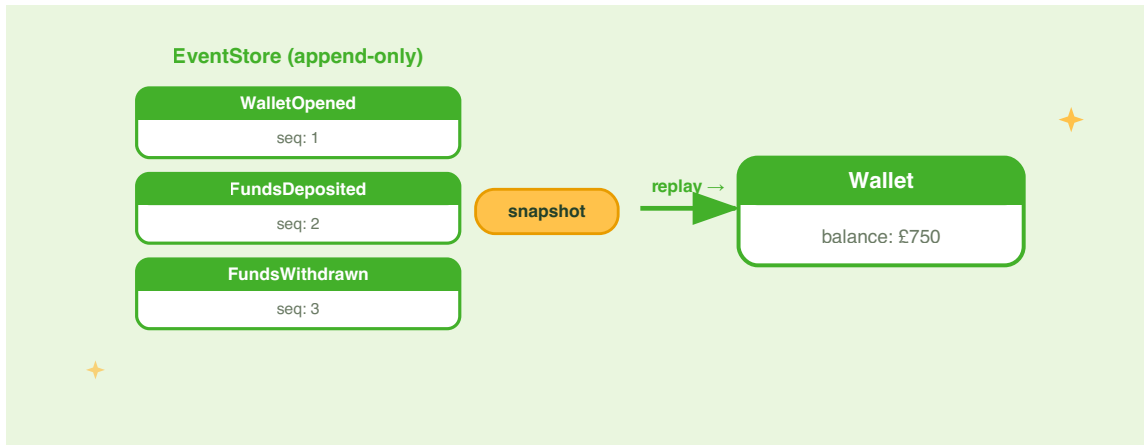
Chapter 9 pushes the event idea further. Instead of maintaining a separate read model alongside a mutable aggregate, you store the events themselves as the system of record — event sourcing the ledger so that every historical balance is computable from first principles.

Try it yourself

1. **Add a `FraudDetector` listener.** Create a `FraudDetector` service that subscribes to `"FundsDeposited"` using `@event_listener(event_types=["FundsDeposited"])`. If the `amount` in the payload exceeds `1_000_000` (ten thousand euros in minor units), log a warning that includes the `envelope.event_id`, `envelope.timestamp`, and the `wallet_id` from the payload. Verify it fires by publishing a `FundsDeposited` event directly to an `InMemoryEventBus` in a unit test, and assert that the warning was triggered.
2. **Extend `WalletAuditListener` with per-event filtering.** Add a method `entries_by_type(self, event_type: str) → list[AuditEntry]` that returns only the entries with a matching `event_type`. Write a test that opens a wallet, makes two deposits and one withdrawal, and asserts that `entries_by_type("FundsDeposited")` returns exactly two entries.
3. **Observe error strategy behaviour.** Create a listener whose handler raises `RuntimeError("failure")` unconditionally. Register it alongside a list-appending capturing handler on an `InMemoryEventBus`. Configure `ErrorStrategy.LOG_AND_CONTINUE` and confirm the capturing handler still receives the event despite the failure. Then switch to `ErrorStrategy.FAIL_FAST` and confirm the capturing handler does *not* receive the event.

CHAPTER 9

Event Sourcing the Ledger



Chapter 8 left one structural gap unresolved. The `WalletAuditListener` keeps a fast read model by reacting to domain events — but the wallet's canonical state is still a row in the `wallets` table, holding a single `balance` column. Every time the balance changes, the previous value is gone forever. If a compliance auditor asks "what was the balance of wallet `w-001` at 14:32 on 3 March?", the honest answer is: you cannot know. The database remembers only the present.

Event sourcing turns this design inside out. Instead of storing the *current* state and discarding each change, you store the *sequence of changes* and derive the current state by replaying them. The `wallets` table disappears. In its place is an **event stream**: an append-only log of every `LedgerOpened`, `Credited`, and `Debited` event the ledger has ever produced. The balance at any point in time is a pure function of the events up to that moment. You can rewind to 14:32 on any date because you have a complete record of everything that happened between then and now.

A financial ledger is the ideal domain for event sourcing. Accountants have understood for centuries that a ledger's authority comes from its entries, not from a running total at the foot of the column. The running total is a *derived fact*; the entries are the *source of truth*. PyFly's `pyfly.eventsourcing` module brings that accounting intuition into code: aggregates emit domain events, an `EventStore` records them immutably, a repository replays the stream to reconstruct state, and a `ProjectionRunner` builds read models on top.

This chapter builds the `LedgerAccount` aggregate — a purpose-built event-sourced domain object that sits alongside the Chapter 6 state-stored `Wallet`. You will see every component of the `pyfly.eventsourcing` module, and how the event store, snapshotting, projections, and the transactional outbox work together to give Lumen a ledger that is both auditable and performant.

From state to events

The clearest way to grasp the shift is to compare what the database looks like in each model.

In the **state-storage model**, the database holds only the current state of the wallet:

wallet_id	owner_id	balance_cents	currency	updated_at
w-001	u-42	8500	EUR	2026-03-03 17:11

Every `deposit` and `withdraw` overwrites `balance_cents`. The history is gone. You know the wallet holds 85.00 EUR right now; you cannot know how it got there.

In the **event-storage model**, the database holds the event stream:

stream_id	seq	event_type	payload	occurred_at
led-001	1	LedgerOpened	<code>{"currency": "EUR", "owner_id": "u-42"}</code>	2026-03-01 09:00
led-001	2	Credited	<code>{"amount": 10000, "balance": 10000}</code>	2026-03-01 09:01
led-001	3	Debited	<code>{"amount": 1500, "balance": 8500}</code>	2026-03-03 17:11

The current balance is still 85.00 EUR — but now you can read every decision that led to it. An auditor, a regulator, or a fraud investigator can replay the stream from any offset and see exactly what happened and when.

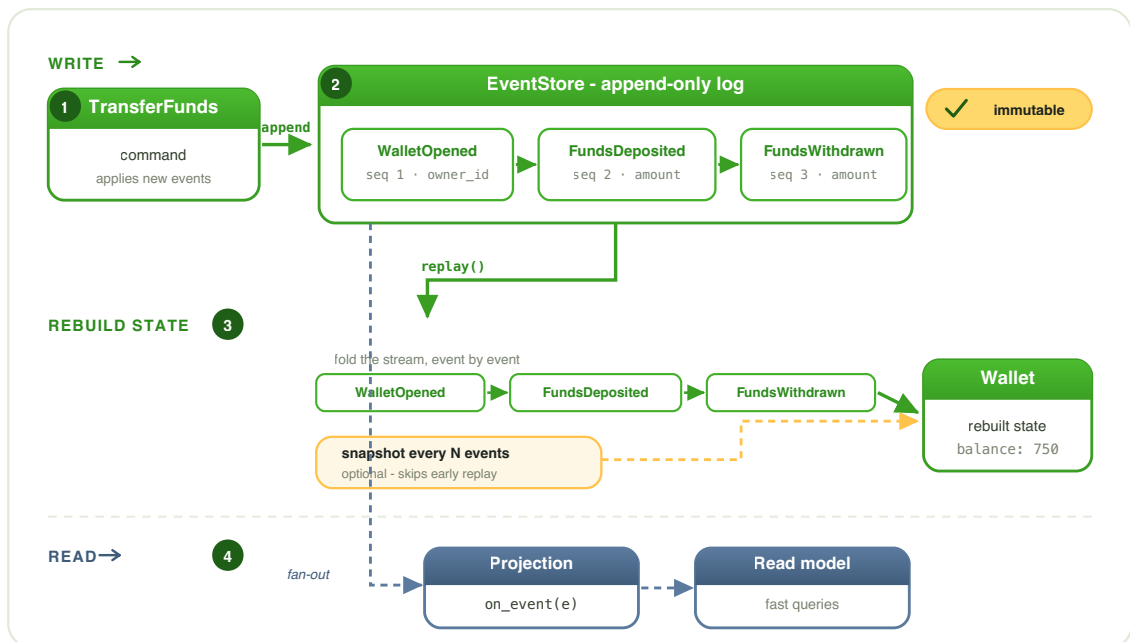


Figure 9.1 — State storage vs event storage: one model keeps a snapshot of the present; the other keeps every fact that led to it.

The trade-off is real. Event storage makes reads more expensive by default — you must replay the stream to compute the current balance — and it demands discipline around schema evolution (events are immutable; you cannot rename a field after the fact). Both concerns have solutions in PyFly: **snapshots** accelerate replay of long streams, and **upcasters** translate old event shapes to new ones during load. You will see both before the end of this chapter.

EVENTS AS THE SYSTEM OF RECORD

Event sourcing is not the same as event-driven architecture. Chapter 8 used EDA: the aggregate stored its state normally and published domain events as a side effect. Event sourcing goes further: the events *are* the state. There is no separate `balance` column to keep in sync — the balance is computed by the repository every time it loads the aggregate.

A separate base for event-sourced aggregates

Before writing a line of ledger code, one naming distinction matters. Chapter 6 built

`Wallet` on `pyfly.domain.AggregateRoot` — the state-stored base class.

`LedgerAccount` uses a **different** base class: `pyfly.eventsourcing.AggregateRoot`.

The two live in separate packages and are deliberately unrelated.

Concern	Chapter 6 <code>Wallet</code>	Chapter 9 <code>LedgerAccount</code>
Base class	<code>pyfly.domain.AggregateRoot</code>	<code>pyfly.eventsourcing.AggregateRoot</code>
Domain event	<code>pyfly.domain.DomainEvent</code>	<code>pyfly.eventsourcing.DomainEvent</code>
State lives in	a database row	the event stream
Repository	<code>WalletRepository</code> (R2DBC)	<code>LedgerAccountRepository</code> (<code>EventSourcedRepository</code>)

`pyfly.eventsourcing.AggregateRoot` supplies the event-sourcing machinery through four members:

- `when(EventType, handler)` — registers a handler for a given event class. The handler receives `(aggregate, event)` as two arguments and performs the mutation — a lambda, a free function, or a one-liner delegating to a private method.
- `apply(event)` — routes a brand-new event through its registered handler and queues it for the event store. Both happen atomically: in-memory state updates immediately, with no round-trip to the store.
- `replay(event_type, event)` — re-runs a persisted event through the same handler *without* re-queuing it. The repository calls this during load to rebuild state from the stored stream.
- `version` — an integer counter incremented after each dispatched event; the store uses it as the optimistic-concurrency token.

Dispatch order is: registered `when()` handler first; then a method named `on_{event_type}` if one exists on the aggregate; if neither is found, `EventHandlerException` is raised. A missing handler would silently corrupt reconstructed state, so the aggregate fails loudly rather than proceeding.

⚠ TWO-ARG HANDLER — THE MOST COMMON TRAP

Every handler registered with `when()` is called as `handler(aggregate, event)` — two arguments. A bound method like `self._on_opened` has signature `(self, event)`, which makes it a one-arg callable from the outside. Passing a bound method directly causes a `TypeError` at runtime. The pattern used throughout this chapter is a one-liner lambda — `lambda agg, evt: agg._on_opened(evt)` — which is correctly two-arg and keeps the real logic in a private method where it can be unit-tested and type-checked independently.

The event-sourced aggregate

In Chapter 6, `Wallet` held `_balance: Money` as direct Python state — `deposit` added to it, `withdraw` subtracted from it. In the event-sourced version the aggregate never mutates its own fields directly. Every state change is mediated by a domain event: the behaviour method *applies* the event, the event handler *updates the fields*, and the `EventStore` persists the event. On load, the repository replays all stored events through the same handlers, rebuilding in-memory state event by event.

This two-step indirection — apply, then handle — is the core mechanic of event sourcing. It enforces a strict discipline: every state transition is recorded exactly once as an event, and the aggregate's current state is always provable from its history.

❗ JARGON: AGGREGATE, HANDLER, FOLD

An **aggregate** is a single consistency boundary — one object that owns a set of related state and the rules that keep it valid. `LedgerAccount` is an aggregate: its balance and its overdraft rule live together. A **handler** (sometimes called an *apply-handler*) is the small function that takes one event and updates the aggregate's fields. A **fold** is the functional-programming term for walking a list and accumulating a result one element at a time — replaying a stream of events into a balance is exactly a fold over the event list. You will see "fold" used as a synonym for "the handler runs over each event."

Zero-arg constructor. `LedgerAccount.__init__` takes no arguments. This is required because `EventSourcedRepository` calls the factory as `LedgerAccount()` and then assigns `.id` before replaying the stream. Never construct a new ledger by passing arguments to `__init__` — call the `open` classmethod instead.

We will build the aggregate in four moves, then run the unit tests to prove each invariant holds. Read the full listing first, then walk the steps below it.

Here are the domain events and the aggregate:

```
lumen/models/entities/v1/ledger_account.py
```

```
from __future__ import annotations

from dataclasses import dataclass

from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.money import Money
from pyfly.domain import BusinessRuleViolation
from pyfly.eventsourcing import AggregateRoot, DomainEvent

# --- Domain events – the durable facts of the ledger -----

@dataclass
class LedgerOpened(DomainEvent):
    """The ledger was opened for an owner in a single currency."""
    account_id: str = ""
    owner_id: str = ""
    currency: str = ""

@dataclass
class Credited(DomainEvent):
    """Money moved *into* the ledger (a deposit / inbound transfer leg)."""
    account_id: str = ""
    amount: int = 0
    currency: str = ""
    balance: int = 0

@dataclass
class Debited(DomainEvent):
    """Money moved *out of* the ledger (withdrawal / outbound transfer
leg)."""
    account_id: str = ""
    amount: int = 0
    currency: str = ""
    balance: int = 0
```



```

# --- Event-sourced aggregate root -----

class LedgerAccount(AggregateRoot):
    """An event-sourced money-movement ledger.

    Zero-arg constructible so it can serve as the repository's factory –
    EventSourcedRepository calls LedgerAccount() then assigns .id before
    replaying the stream. Use the open() classmethod to create new ledgers.
    """

    def __init__(self) → None:
        super().__init__()
        self.owner_id: str = ""
        self.currency: Currency = Currency.EUR
        self.balance: Money = Money.zero(Currency.EUR)
        # Register apply-handlers. _dispatch calls handler(aggregate, event).
        # Use a lambda so the callable is two-arg; delegate to a private
        # method to keep the real logic type-checked and unit-testable.
        self.when(LedgerOpened, lambda agg, evt: agg._on_opened(evt))
        self.when(Credited, lambda agg, evt: agg._on_credited(evt))
        self.when(Debited, lambda agg, evt: agg._on_debited(evt))

# --- factory -----

@classmethod
def open(
    cls, account_id: str, owner_id: str, currency: Currency
) → "LedgerAccount":
    """Open a new empty ledger; appends LedgerOpened."""
    if not owner_id.strip():
        raise BusinessRuleViolation(
            "ledger-owner-required", "owner_id is required"
        )
    account = cls()
    account.id = account_id
    account.apply(
        LedgerOpened(

```

```

        account_id=account_id,
        owner_id=owner_id,
        currency=currency.value,
    )
)
return account

# --- commands (validate invariants, then apply) -----

def credit(self, amount: Money) → None:
    """Record money entering the ledger; appends Credited."""
    self._assert_currency(amount)
    if not amount.is_positive:
        raise BusinessRuleViolation(
            "ledger-credit-positive", "credit amount must be > 0"
        )
    new_balance = self.balance.add(amount)
    self.apply(
        Credited(
            account_id=self.id,
            amount=amount.amount,
            currency=amount.currency.value,
            balance=new_balance.amount,
        )
    )

def debit(self, amount: Money) → None:
    """Record money leaving; refuses to overdraw. Appends Debited."""
    self._assert_currency(amount)
    if not amount.is_positive:
        raise BusinessRuleViolation(
            "ledger-debit-positive", "debit amount must be > 0"
        )
    remaining = self.balance.subtract(amount)
    if remaining.is_negative:
        raise BusinessRuleViolation(
            "ledger-insufficient-funds",
            f"cannot debit {amount}; balance is {self.balance}",
        )

```

```

    self.apply(
        Debited(
            account_id=self.id,
            amount=amount.amount,
            currency=amount.currency.value,
            balance=remaining.amount,
        )
    )

# --- apply-handlers (pure folds, shared by apply + replay) -----

def _on_opened(self, event: object) → None:
    self.owner_id = event.owner_id          # type: ignore[attr-defined]
    self.currency = Currency(event.currency) # type: ignore[attr-defined]
    self.balance = Money.zero(self.currency)

def _on_credited(self, event: object) → None:
    self.balance = Money(
        event.balance, Currency(event.currency) # type: ignore[attr-
defined]
    )

def _on_debited(self, event: object) → None:
    self.balance = Money(
        event.balance, Currency(event.currency) # type: ignore[attr-
defined]
    )

# --- helpers -----

def _assert_currency(self, amount: Money) → None:
    if amount.currency is not self.currency:
        raise BusinessRuleViolation(
            "ledger-currency-mismatch",
            f"ledger holds {self.currency.value}, "
            f"got {amount.currency.value}",
        )

```

Listing 9.1 — *LedgerAccount*: an event-sourced aggregate that derives its balance from replay

How it works. `__init__` registers three `when()` handlers — one per event class — each as a two-arg lambda delegating to a private method. No arithmetic happens inside the handlers; the behaviour methods (`credit`, `debit`) own all validation and compute the new state before constructing the event. The handler simply *applies* the already-computed result.

`account.apply(Credited(...))` does two things atomically: appends the event to the pending-events buffer (so the repository can persist it) and immediately dispatches it to the `Credited` handler (so `balance` updates in memory). The aggregate's in-memory state is therefore always consistent with its pending events, even before a save.

The factory method `open` calls `apply(LedgerOpened(...))` rather than setting fields directly. That is intentional: if you loaded this aggregate from its event stream, `LedgerOpened` would pass through exactly the same `_on_opened` handler and produce identical results. Write path and replay path are the same code — that symmetry is the correctness guarantee of event sourcing.

The `version` counter starts at zero and increments after each dispatched event. After `open`, `account.version = 1`; after one credit, `account.version = 2`. You will see this number again when the `EventStore` enforces optimistic concurrency.

Building it step by step. The listing is dense the first time through. Here is the same code as four deliberate moves — the order in which you would actually write it.

Step 1 — Define the durable facts. Write the three event dataclasses

(`LedgerOpened`, `Credited`, `Debited`) extending `pyfly.eventsourcing.DomainEvent`. Give every field a default value (`account_id: str = ""`, `amount: int = 0`). The defaults are not cosmetic: the repository rebuilds these dataclasses from a stored payload, and a dataclass with required fields could not be reconstructed when an old event predates a newer field.

Step 2 — Set up the empty aggregate. Write `__init__` with no parameters. Initialise the fields to neutral defaults (`owner_id = ""` , `balance = Money.zero(Currency.EUR)`) and register one `when()` handler per event class, each as a two-arg lambda delegating to a private method. At this point the aggregate is a blank slate that knows how to react to events but has not seen any.

Step 3 — Add the factory and the commands. Write the `open` classmethod and the `credit` / `debit` methods. Each one validates its invariants first (currency match, positive amount, no overdraft), computes the resulting balance, and only then calls `self.apply(SomeEvent(...))`. Validation lives in the command; the handler never validates.

Step 4 — Write the pure folds. Write `_on_opened` , `_on_credited` , `_on_debited` . Each reads fields off the event and assigns them to the aggregate — no arithmetic, no validation, no exceptions. Because these same three methods run on both the write path and the replay path, keeping them dumb is what guarantees that a reloaded ledger equals the live one.

💡 WHY VALIDATION BELONGS IN THE COMMAND, NOT THE HANDLER

The handler runs again every time the aggregate is loaded from history. If validation lived in the handler, you would be re-checking the overdraft rule against data that already passed it years ago — and worse, a rule that has since *changed* could reject a historical event that was perfectly valid when it happened. Validate once, on the write that produces the event; trust the event forever after.

`pending_events()` returns the events queued since the last save. The unit tests drive the aggregate in isolation — no repository needed — which makes invariant verification straightforward:

tests/test_ledger_event_sourcing.py

```

def test_open_emits_ledger_opened_and_starts_empty() → None:
    account = LedgerAccount.open(
        "led-1", owner_id="owner-1", currency=Currency.EUR
    )
    assert account.id == "led-1"
    assert account.owner_id == "owner-1"
    assert account.currency is Currency.EUR
    assert account.balance == Money.zero(Currency.EUR)
    # apply() queued the event for the store and bumped the version.
    [event] = account.pending_events()
    assert isinstance(event, LedgerOpened)
    assert event.account_id == "led-1"
    assert event.currency == "EUR"
    assert account.version == 1

def test_credit_then_debit_track_the_balance() → None:
    account = LedgerAccount.open("led-2", owner_id="o", currency=Currency.EUR)
    account.credit(Money(1000, Currency.EUR))
    account.debit(Money(400, Currency.EUR))
    assert account.balance == Money(600, Currency.EUR)
    kinds = [type(e).__name__ for e in account.pending_events()]
    assert kinds == ["LedgerOpened", "Credited", "Debited"]
    assert account.version == 3

def test_debit_cannot_overdraw() → None:
    account = LedgerAccount.open("led-3", owner_id="o", currency=Currency.EUR)
    account.credit(Money(500, Currency.EUR))
    with pytest.raises(BusinessRuleViolation) as exc:
        account.debit(Money(501, Currency.EUR))
    assert exc.value.rule == "ledger-insufficient-funds"
    # Invariant held: balance unchanged, no Debited event queued.
    assert account.balance == Money(500, Currency.EUR)
    assert [type(e).__name__ for e in account.pending_events()] == [
        "LedgerOpened",

```

```
"Credited",
]
```

Listing 9.2 — Unit tests: aggregate in isolation, commands and invariants

Run it. These three tests need no database and no event store — the aggregate stands entirely on its own. Run just this file's unit tests from the Lumen project root:

```
uv run --extra dev pytest tests/test_ledger_event_sourcing.py -q -k "open or
credit or debit or currency"
```

You should see the in-isolation tests pass:

```
.... [100%]
4 passed, 7 deselected in 0.05s
```

The four dots are the four aggregate-only tests (the three shown above plus a currency-mismatch check); the seven deselected tests are the store-backed ones, which we run later. If you see a `TypeError` about the number of arguments, you have almost certainly hit the bound-method trap from the warning above — passing `self._on_opened` directly to `when()` instead of wrapping it in a two-arg lambda.

What just happened. You built a domain object that records *what changed* instead of *what is*. `open` emitted a `LedgerOpened` fact and bumped the version to 1. `credit` and `debit` each validated their rule, computed the new balance, and emitted a fact — so a credit followed by a debit left three facts queued and the version at 3. When a rule failed (the overdraft test), the command raised before calling `apply`, so no faulty event ever entered the buffer. The aggregate's in-memory balance and its list of pending events stayed perfectly in step, and nothing touched a database yet.

💡 `ON_{EVENT_TYPE}` AS AN ALTERNATIVE

Instead of `when()` lambdas, you can define a method on the aggregate named after the event's `event_type` — which is the event **class name**, so `on_LedgerOpened(self, evt)` (matching the `LedgerOpened` dataclass) is discovered automatically. Use `when()` for concise one-liners and named methods for handlers that need multiple statements or local variables. The dispatch order is: `when()` handler first; then the `on_{event_type}` method; then `EventHandlerException` if neither exists.

🍃 `SPRING PARITY`

`AggregateRoot + apply() + when()` is PyFly's equivalent of Axon Framework's `@Aggregate + AggregateLifecycle.apply(event) + @EventSourcingHandler`. Axon uses annotation-driven handler discovery (`@EventSourcingHandler`); PyFly uses `when()` registration or `on_*` method convention. The replaying mechanic — load events from the store, call the same handlers, rebuild state — is identical in both frameworks.

The EventStore

The aggregate knows how to produce and replay events. The `EventStore` knows how to persist and retrieve them. These are deliberately separate concerns: the aggregate is pure business logic with no I/O; the event store is pure I/O with no business logic.

The `EventStore` protocol exposes two core operations:

- `append(aggregate_id, aggregate_type, events, *, expected_version)` — persists a batch of events for a stream. Raises `ConcurrencyError` if the stream's actual version does not match `expected_version`.

- `load(aggregate_id, *, after_sequence=0)` — returns the ordered sequence of `StoredEventEnvelope` objects from the first event (or from `after_sequence`) to the most recent.

`InMemoryEventStore` is the out-of-the-box implementation. Like `InMemoryEventBus` in Chapter 8, it runs entirely in-process with no I/O — ideal for development and tests. A production deployment swaps in a PostgreSQL- or EventStoreDB-backed adapter.

`EventSourcedRepository` wraps the `EventStore` and handles the full save/load cycle. Application code never calls the store directly; it calls `repo.save(aggregate)` and `repo.load(aggregate_id)`, and the repository handles the rest.

All imports come from two locations:

```
# Core event-sourcing types – all in the base package
from pyfly.eventsourcing import (
    AggregateRoot,
    DomainEvent,
    EventStore,
    InMemoryEventStore,
    SnapshotStore,
    InMemorySnapshotStore,
    StoredEventEnvelope,
)
# The generic repository lives in the .repository submodule
from pyfly.eventsourcing.repository import EventSourcedRepository
```

The LedgerAccountRepository

You subclass `EventSourcedRepository` for two reasons: to pass the concrete factory and snapshot store through a single well-named constructor, and — optionally — to override `_envelope_to_event` so that replayed events are real typed dataclasses rather than the generic attribute-bag the base class produces.

JARGON: ENVELOPE AND ATTRIBUTE-BAG

A `StoredEventEnvelope` is the *wire shape* the store persists: it wraps the event's `payload` (a plain dict) together with bookkeeping fields the store needs — `aggregate_id`, `aggregate_type`, `sequence`, `event_type`, `event_id`, and `occurred_at`. Think of it as the addressed envelope around the letter. An **attribute-bag** is the generic stand-in object the base repository creates on load if you do not override anything: a nameless object with the payload's keys set as attributes. It works — the handlers only read attributes — but it is not your real `Credited` dataclass, so it cannot be type-checked or `isinstance`-tested. The override below trades that anonymity for the real thing.

Build the repository in three small steps.

Step 1 — Map class names back to dataclasses. The store records each event's `event_type` as the class name string (`"Credited"`). To rebuild the real dataclass on load, you need a lookup from that string to the class. That is `_EVENT_TYPES` — one entry per event type.

Step 2 — Subclass and forward the constructor.

`LedgerAccountRepository.__init__` takes the `store` (and an optional `snapshots` store) and forwards everything to `super().__init__`, supplying `factory=LedgerAccount` so the base class knows how to make a blank aggregate.

Step 3 — Override `_envelope_to_event` for typed replay. Look the `event_type` up in `_EVENT_TYPES`, filter the stored payload down to fields the dataclass actually declares (using `__dataclass_fields__`), and construct the real event. Fall back to the base class's generic hydration for any event type you do not recognise.

```
lumen/models/repositories/ledger_repository.py
```

```
from __future__ import annotations

from typing import ClassVar

from lumen.models.entities.v1.ledger_account import (
    Credited,
    Debited,
    LedgerAccount,
    LedgerOpened,
)

from pyfly.eventsourcing import (
    DomainEvent,
    EventStore,
    SnapshotStore,
    StoredEventEnvelope,
)

from pyfly.eventsourcing.repository import EventSourcedRepository

# Map a stored event_type (the event class name) back to its dataclass.
_EVENT_TYPES: dict[str, type[DomainEvent]] = {
    LedgerOpened.__name__: LedgerOpened,
    Credited.__name__: Credited,
    Debited.__name__: Debited,
}

class LedgerAccountRepository(EventSourcedRepository[LedgerAccount]):
    """Loads/saves LedgerAccount aggregates via the event store."""

    SNAPSHOT_INTERVAL: ClassVar[int] = 100

    def __init__(
        self,
        store: EventStore,
        *,
        snapshots: SnapshotStore | None = None,
    ) → None:
```

```

super().__init__(
    store,
    factory=LedgerAccount,
    snapshots=snapshots,
    snapshot_interval=self.SNAPSHOT_INTERVAL,
)

@staticmethod
def _envelope_to_event(envelope: StoredEventEnvelope) → object:
    """Rebuild the concrete event dataclass from a stored payload.

    Overrides the base-class generic hydration so that replayed events
    are the same dataclasses the aggregate applied on the write side.
    Unknown fields are ignored for forward-compatibility.
    """
    event_cls = _EVENT_TYPES.get(envelope.event_type)
    if event_cls is None:
        # Fall back to generic hydration for unrecognised event types.
        return EventSourcedRepository._envelope_to_event(envelope)
    field_names = {
        f.name for f in event_cls.__dataclass_fields__.values()
    }
    kwargs = {
        k: v for k, v in envelope.payload.items() if k in field_names
    }
    return event_cls(**kwargs)

```

Listing 9.3 — *LedgerAccountRepository: typed replay via `_envelope_to_event`*

How it works. `super().__init__(store, factory=LedgerAccount, ...)` tells the base repository to call `LedgerAccount()` — no arguments — to create a blank aggregate, assign `.id`, then replay the stream. `factory` accepts any zero-arg callable; passing the class itself (`factory=LedgerAccount`) is equivalent to `factory=lambda: LedgerAccount()`.

The `_envelope_to_event` override looks up the stored `event_type` string in `_EVENT_TYPES`, uses `__dataclass_fields__` to filter the payload to known fields, and reconstructs the real dataclass. Unknown fields are silently dropped — forward-compatibility in practice: if a future event version adds a field the old handler does not recognise, the ledger keeps replaying instead of crashing. The base-class fallback at the bottom handles any event type the repository does not know about.

❗ JARGON: FORWARD-COMPATIBILITY

A reader is **forward-compatible** when it can ingest data written by a *newer* version of itself without breaking — it simply ignores parts it does not understand. Dropping unknown payload fields is exactly that: a v2 writer can add a `reference_code` to `Credited`, and a still-running v1 reader keeps folding the event correctly instead of crashing on the surprise field.

Run it. A focused test exercises this override directly — it hands the repository a hand-built envelope and asserts it gets back a real `Credited` dataclass:

```
uv run --extra dev pytest tests/test_ledger_event_sourcing.py -q -k
typed_replay
```

```
. [100%]
1 passed, 10 deselected in 0.05s
```

What just happened. You wired the persistence boundary without writing a single line of SQL or storage code. The base `EventSourcedRepository` already knows how to drain pending events on `save` and replay them on `load`; your subclass added only two things — a zero-arg factory so it can build a blank ledger, and a typed `_envelope_to_event` so the events it replays are the very same dataclasses the aggregate emitted on the write side. That symmetry is the whole point: write and replay run identical code.

Save, load, and the replay proof

This is the moment the whole chapter has been building toward: proving that a ledger reloaded from nothing but its stored events equals the live ledger that produced them. The test below does it in two halves.

Step 1 — Write a ledger and persist it. Open a ledger, credit it, debit it, then call `repo.save(account)`. The repository drains the three pending events into the store and clears the aggregate's buffer.

Step 2 — Reload from a clean slate and compare. Make a *brand-new* repository and call `Load("acct-1")`. Nothing in this second half shares memory with the first object. The recovered ledger's balance can only be correct if it was recomputed by replaying the stored stream — which is exactly the proof we want.

The following test demonstrates the complete save-and-load cycle:

tests/test_ledger_event_sourcing.py

```

@pytest.mark.asyncio
async def test_balance_survives_reload_by_replay(
    event_store: InMemoryEventStore,
) → None:
    # 1. Open + credit + debit, then persist the pending events.
    account = LedgerAccount.open(
        "acct-1", owner_id="owner-7", currency=Currency.EUR
    )
    account.credit(Money(2500, Currency.EUR)) # +25.00
    account.debit(Money(1000, Currency.EUR)) # -10.00 → 15.00
    assert account.balance == Money(1500, Currency.EUR)

    repo = LedgerAccountRepository(event_store)
    await repo.save(account)
    # After committing, the aggregate has no pending events left.
    assert account.pending_events() == []

    # 2. Reconstruct from a *fresh* repository + aggregate. Nothing here
    #     carries the in-memory object's state – load() rebuilds the
    #     LedgerAccount purely by replaying the stored event stream.
    fresh_repo = LedgerAccountRepository(event_store)
    recovered = await fresh_repo.load("acct-1")

    assert recovered is not None
    assert recovered is not account
    assert recovered.owner_id == "owner-7"
    assert recovered.currency is Currency.EUR
    # The load-by-replay proof: balance recomputed from events, not stored.
    assert recovered.balance == Money(1500, Currency.EUR)
    # Three events were folded in: LedgerOpened, Credited, Debited.
    assert recovered.version == 3
    # A reconstructed aggregate has nothing pending – it was not "changed".
    assert recovered.pending_events() == []

```

Listing 9.4 — Headline test: balance survives a reload by replay

How it works. `repo.save(account)` calls `store.append("acct-1", "LedgerAccount", pending_events, expected_version=0)`. The three events — `LedgerOpened`, `Credited`, `Debited` — are serialised into `StoredEventEnvelope` objects and written to the stream in order; the pending-events buffer on the aggregate is then cleared.

`fresh_repo.load("acct-1")` calls `store.load("acct-1")`, receives the three envelopes, constructs a blank `LedgerAccount()` via the factory, assigns `.id = "acct-1"`, and passes each envelope through `_envelope_to_event` before replaying it. The `_on_opened`, `_on_credited`, and `_on_debited` handlers run in sequence; after all three, `recovered.balance.amount` is `1500` — the same value the live aggregate held, computed without any shared state between the two objects.

The test uses *two independent repository instances* sharing the same in-memory store — `repo` for the write, `fresh_repo` for the read. This proves that replay, not in-process object identity, is the source of truth.

Run it. Run the headline test plus the raw-stream inspection test that follows it:

```
uv run --extra dev pytest tests/test_ledger_event_sourcing.py -q -k
"reload_by_replay or immutable_event_stream"
```

```
.. [100%]
2 passed, 9 deselected in 0.05s
```

What just happened. You proved event sourcing actually works end to end. The first ledger lived only in memory; after `save`, its facts lived only in the store. A second, unrelated ledger object then rebuilt the exact same 15.00 EUR balance by folding those facts back through the same handlers — no stored balance column anywhere in sight.

The version landed on 3 (one event per `LedgerOpened` / `Credited` / `Debited`), and the reconstructed ledger had nothing pending because loading is not the same as changing.

The event store also exposes raw envelopes for inspection — useful for audits and tests:

`tests/test_ledger_event_sourcing.py`

```
@pytest.mark.asyncio
async def test_store_holds_the_immutable_event_stream(
    event_store: InMemoryEventStore,
) → None:
    account = LedgerAccount.open(
        "acct-2", owner_id="o", currency=Currency.GBP
    )
    account.credit(Money(800, Currency.GBP))
    account.debit(Money(300, Currency.GBP))
    await LedgerAccountRepository(event_store).save(account)

    envelopes = await event_store.load("acct-2")
    assert [e.event_type for e in envelopes] == [
        "LedgerOpened", "Credited", "Debited"
    ]
    assert [e.sequence for e in envelopes] == [1, 2, 3]
    assert all(e.aggregate_type == "LedgerAccount" for e in envelopes)
    # The Debited payload carries the post-debit running balance.
    assert envelopes[2].payload["balance"] == 500
    assert await event_store.latest_version("acct-2") == 3
```

Listing 9.5 — The store holds the immutable event stream

i FACTORY ARGUMENT

`EventSourcedRepository` accepts a `factory` callable that produces a blank aggregate instance. The factory must return an aggregate in its initial `__init__` state — no events applied — because the repository will apply the full history itself. Passing a factory that constructs an already-mutated aggregate (for example, one that calls `open()` internally) will corrupt the replay: `_on_opened` would run twice, the second time overwriting the fields that the real events already set.

Optimistic concurrency

Two concurrent requests — a credit from the mobile app and an automated fee debit from a background job — can load the same ledger at the same version, each apply their own change, and then attempt to save. Without a concurrency guard, one save silently wins, the other's events are lost, sequence numbers collide, and the resulting balance is wrong.

Optimistic concurrency prevents this. Before appending new events, the `EventStore` compares the stream's *current* version against the *expected* version the repository recorded at load time. If they match, the append proceeds and the version advances. If they do not match — because another writer already appended — `ConcurrencyError` is raised and the losing request must retry from a fresh load.

ⓘ JARGON: OPTIMISTIC VS PESSIMISTIC LOCKING

Pessimistic locking assumes conflict is likely, so it grabs a lock up front — no other writer can touch the row until you release it. *Optimistic* concurrency assumes conflict is rare, so it takes no lock at all: it lets both writers proceed and only checks, at save time, whether anyone else changed the stream first. The loser retries. For most ledgers, two simultaneous writers to the *same* account are uncommon, so the optimistic strategy avoids the cost of locking on the overwhelmingly common no-conflict path.

The `expected_version` is passed implicitly by the repository: it records the version at which it loaded the aggregate and supplies it to the store on save. You never manage version numbers in application code.

The version progression is deterministic: after a save-and-reload, further writes advance the version without conflict:

tests/test_ledger_event_sourcing.py

```

@pytest.mark.asyncio
async def test_continues_appending_after_a_reload(
    event_store: InMemoryEventStore,
) → None:
    repo = LedgerAccountRepository(event_store)
    account = LedgerAccount.open(
        "acct-3", owner_id="o", currency=Currency.EUR
    )
    account.credit(Money(1000, Currency.EUR))
    await repo.save(account)

    # Reload, mutate again, save again – version must advance, no conflict.
    reloaded = await repo.load("acct-3")
    assert reloaded is not None
    assert reloaded.version == 2
    reloaded.debit(Money(250, Currency.EUR))
    await repo.save(reloaded)

    final = await repo.load("acct-3")
    assert final is not None
    assert final.balance == Money(750, Currency.EUR)
    assert final.version == 3
    assert await event_store.latest_version("acct-3") == 3

```

Listing 9.6 — Continuing to append after a reload: optimistic lock advances correctly

How it works. After the first save the stream is at version 2. When loaded,

`reloaded.version == 2`. `repo.save(reloaded)` appends with `expected_version=2`; the store advances to 3 and succeeds. The `final` load replays all three events and confirms the correct balance.

Run it.

```
uv run --extra dev pytest tests/test_ledger_event_sourcing.py -q -k  
continues_appending
```

```
. [100%]  
1 passed, 10 deselected in 0.05s
```

What just happened. A reloaded ledger remembered its version, so the next save lined up cleanly behind the events already in the stream — no conflict, sequence numbers in order, balance correct. This is the *happy path* of optimistic concurrency: one writer at a time. The moment two writers race, the second one's `expected_version` would no longer match and the store would raise `ConcurrencyError` instead — which is the case the warning below tells you how to handle.

⚠ ALWAYS HANDLE CONCURRENCYERROR

When two writers race, the losing save raises `ConcurrencyError`. Your application service must catch it and decide what to do: retry the full load-mutate-save cycle (appropriate for low-contention writes), or surface a 409 Conflict to the caller (appropriate when the caller should re-submit with fresh data). Never silently swallow the error — a swallowed concurrency error leaves the stream in an inconsistent state.

Snapshots

Event sourcing trades write simplicity for read cost. Loading a ledger that has recorded 10 000 money movements means replaying 10 000 events every time the aggregate is needed. For most ledgers the stream stays short; for high-frequency accounts it can grow prohibitively long.

Snapshots address this. A snapshot is a serialised checkpoint of the aggregate's state at a specific version. On load, the repository looks for the most recent snapshot first, deserialises the state directly to that version, then replays only the events that arrived after it. A snapshot at version 9 000 reduces a 10 000-event replay to 1 000 events.

`InMemorySnapshotStore` stores snapshots in memory. Pass it to `EventSourcedRepository` alongside the event store via the `snapshots` keyword — exactly as `LedgerAccountRepository.__init__` accepts it:

```
store = InMemoryEventStore()
snapshots = InMemorySnapshotStore()
repo = LedgerAccountRepository(store, snapshots=snapshots)
```

The repository decides when to snapshot automatically using `snapshot_interval` (default `100`, set by `LedgerAccountRepository.SNAPSHOT_INTERVAL`). After every `save`, it checks whether the aggregate's new version **crosses** a multiple of `snapshot_interval`:

```
# crosses_interval is True when this batch pushes the stream past a
# 100-event boundary – e.g., version 95 → 105 crosses the 100 mark.
crossed = (
    (aggregate.version // snapshot_interval)
    > (previous_version // snapshot_interval)
)
```

This interval-crossing logic (rather than exact divisibility) handles the common case where a single save batch straddles the threshold. A bulk import that adds 10 events takes the version from 95 to 105 and correctly triggers a snapshot, even though neither 95 nor 105 is exactly divisible by 100.

The snapshot seam is harmless when the stream is shorter than the interval — the following test proves it:

tests/test_ledger_event_sourcing.py

```

@pytest.mark.asyncio
async def test_snapshot_store_round_trips_the_ledger() → None:
    """With a snapshot store wired, reload still yields the right state.

    The ledger's stream is far shorter than the snapshot interval, so this
    proves the snapshot seam is harmless and the repository falls back
    to a full replay.
    """
    store = InMemoryEventStore()
    snapshots = InMemorySnapshotStore()
    repo = LedgerAccountRepository(store, snapshots=snapshots)

    account = LedgerAccount.open(
        "acct-4", owner_id="o", currency=Currency.EUR
    )
    account.credit(Money(5000, Currency.EUR))
    await repo.save(account)

    recovered = await repo.load("acct-4")
    assert recovered is not None
    assert recovered.balance == Money(5000, Currency.EUR)

```

Listing 9.7 — Snapshot store wired: reload still yields correct state

How it works. After the save, the repository checks: does `version 2 // 100 > 0 // 100`? No — the snapshot threshold has not been crossed, so no snapshot is taken. The next `load` performs a full replay of the two events and returns the correct balance. Once a ledger does cross a 100-event boundary, the repository serialises the aggregate's state into a snapshot envelope. The next load finds the snapshot, deserialises directly to that version, and then asks the event store for events with a sequence number greater than the snapshot version — reducing replay cost to the delta only.

Run it.


```
uv run --extra dev pytest tests/test_ledger_event_sourcing.py -q -k
snapshot_store_round_trips
```

```
. [100%]
1 passed, 10 deselected in 0.05s
```

What just happened. You wired a snapshot store into the repository and the ledger still reloaded correctly — which is exactly what should happen for a short stream. Because the two events never crossed the 100-event interval, no snapshot was written and the load fell back to a plain full replay. Snapshots are pure optimization: wiring one in costs nothing for short streams and quietly pays off once an account becomes high-frequency. You can prove the correctness of your ledger with the snapshot store both present and absent — the answer is identical.

💡 SNAPSHOT INTERVAL IN PRODUCTION

A `snapshot_interval` of 100 is the default and a sensible starting point. For high-frequency ledgers you might lower it; for accounts that only change a few times a day, a higher interval reduces snapshot-storage cost. Snapshots are an optimization, not a correctness requirement — removing them leaves the system correct but slower.

Boot wiring and auto-configuration

The `pyfly.eventsourcing` module ships in the **base** `pyfly` package — no extra dependency. Enabling it takes two steps: set `pyfly.eventsourcing.enabled: true` in `pyfly.yaml` and annotate the application with `@enable_domain_stack`. PyFly's auto-configuration then registers `event_store` and `snapshot_store` beans automatically.

❗ JARGON: BEAN AND AUTO-CONFIGURATION

A **bean** is just an object the framework creates once and hands to anything that asks for it — the same dependency-injection idea you met in Chapter 2. **Auto-configuration** is a class of bean-producing factory methods that PyFly activates *only when a condition is met* — here, the `@conditional_on_property("pyfly.eventsourcing.enabled", having_value="true")` guard. Flip that one config flag on, and the `event_store` and `snapshot_store` beans appear in the container; leave it off, and the event-sourcing machinery stays dormant. You never call the factory yourself.

The test suite confirms this directly:

tests/test_ledger_event_sourcing.py

```
def test_auto_configuration_registers_event_store_beans() → None:
    """enable_domain_stack activates this auto-config when
    pyfly.eventsourcing.enabled=true (set in pyfly.yaml), registering
    the in-memory event/snapshot stores the ledger repository depends on."""
    from pyfly.core.config import Config
    from pyfly.eventsourcing.auto_configuration import (
        EventSourcingAutoConfiguration,
    )

    auto = EventSourcingAutoConfiguration()
    cfg = Config() # empty config → providers default to "memory"
    assert isinstance(auto.event_store(cfg), InMemoryEventStore)
    assert isinstance(auto.snapshot_store(cfg), InMemorySnapshotStore)
```

Listing 9.8 — Auto-configuration registers the event store beans

Run it.

```
uv run --extra dev pytest tests/test_ledger_event_sourcing.py -q -k
auto_configuration
```

```
. [100%]
1 passed, 10 deselected in 0.05s
```

What just happened. With the `enabled` flag on, the auto-config produced the two in-memory stores the ledger repository depends on — without you instantiating either. Note that the factory methods take a `Config` argument: that is how they read `pyfly.eventsourcing.store.provider` to decide between the `memory` default and a SQL-backed adapter. Passing an empty `Config()` leaves them on `memory`, which is what the test asserts.

In application code, the repository is wired through dependency injection:

```
# pyfly.yaml
pyfly:
  eventsourcing:
    enabled: true
```

```
# In a service or handler — the beans are injected automatically
@Component
class LedgerService:
    def __init__(
        self,
        event_store: EventStore,
        snapshot_store: SnapshotStore,
    ) → None:
        self._repo = LedgerAccountRepository(
            event_store, snapshots=snapshot_store
        )
```

Projections

The event store is the system of record, but most application queries — "what is the current balance?", "show all ledgers for owner u-42", "which accounts are above 1 000 EUR?" — must not replay event streams on every read. They should hit a pre-computed read model: a table optimised for queries, kept in sync by a background process that consumes the event stream.

That background process is a **projection**. A projection subscribes to the event stream and updates a read model each time a relevant event arrives. PyFly provides

`FunctionProjection` and `ProjectionRunner` in `pyfly.eventsourcing.projection`:

❗ JARGON: READ MODEL, PROJECTION, WRITE MODEL

The **write model** is the side you have built so far — the aggregate and its event stream, optimised for *recording changes correctly*. A **read model** is a separate, query-shaped copy of the data, optimised for *answering questions fast* (one row per ledger with its current balance, say). A **projection** is the process that keeps a read model in sync by replaying the event stream into it. This separation — one model for writes, another for reads — is the **CQRS** pattern you met in Chapter 7, now backed by an event stream instead of a relational table.

- `FunctionProjection(name, handler_fn)` — wraps an async function that receives one `StoredEventEnvelope` and updates the read model.
- `ProjectionRunner(projection, store)` — drives the projection by iterating the `EventStore` in sequence-number order and calling the handler for each envelope.

A projection comes together in three pieces. **Step 1** — write the handler: an `async` function that takes one envelope and updates the read model (here a plain `dict`; in production a database table). **Step 2** — wrap it:

`FunctionProjection("balance_ledger", handler)` turns the bare function into a

named projection. **Step 3** — drive it: `ProjectionRunner(projection, store)` connects the projection to the event store and, when started, feeds it every stored envelope in sequence order.

Here is a `BalanceLedgerProjection` that builds a balance read model from the event stream:

```
lumen/event sourcing/balance_projection.py
```

```
from __future__ import annotations

import asyncio

from pyfly.event sourcing import InMemoryEventStore
from pyfly.event sourcing.projection import FunctionProjection,
ProjectionRunner

# The in-process read model – in production, replace with a DB table.
_balance_store: dict[str, dict] = {}

async def _handle_envelope(envelope: object) → None:
    """Update the balance read model for each ledger event."""
    event_type: str = getattr(envelope, "event_type", "")
    payload: dict = getattr(envelope, "payload", {})

    if event_type == "LedgerOpened":
        _balance_store[payload["account_id"]] = {
            "account_id": payload["account_id"],
            "owner_id": payload.get("owner_id", ""),
            "balance_cents": 0,
            "currency": payload.get("currency", ""),
        }
    elif event_type in ("Credited", "Debited"):
        account_id = payload["account_id"]
        if account_id in _balance_store:
            _balance_store[account_id]["balance_cents"] = (
                payload["balance"]
            )

def build_projection(store: InMemoryEventStore) → ProjectionRunner:
    projection = FunctionProjection("balance_ledger", _handle_envelope)
    return ProjectionRunner(projection, store)
```

```

async def demo_projection(store: InMemoryEventStore) → None:
    runner = build_projection(store)
    # start() launches a background polling task and returns immediately;
    # the read model is populated asynchronously as the loop drains the
    # store. Poll until the projection has caught up, then stop the runner.
    await runner.start()
    try:
        for _ in range(50):
            if "led-001" in _balance_store:
                break
            await asyncio.sleep(0.05)
    finally:
        await runner.stop()

    balance = _balance_store.get("led-001", {})
    print(f"Balance read model: {balance}")

```

Listing 9.9 — *BalanceLedgerProjection*: a read model built from the event stream

How it works. `FunctionProjection("balance_ledger", _handle_envelope)` wraps the async handler. `ProjectionRunner(projection, store)` links it to the `InMemoryEventStore`. `await runner.start()` launches a background polling task and returns *immediately* — it does not block until the store is drained. The task loops on `store.stream_all(...)`, calling `_handle_envelope` for each new envelope in order and advancing a cursor (`_last_event_id`) so it never re-processes an event. Because population happens asynchronously, the demo polls `_balance_store` until the projection has caught up, then calls `await runner.stop()` to halt the loop before reading the result. Only after the projection has processed the envelopes does `_balance_store` reflect the current state of every ledger in the store.

The projection is intentionally stateless — it reads only `envelope.event_type` and `envelope.payload`. No aggregate is loaded; no repository is called. The read model is cheap to rebuild: stop the runner, clear `_balance_store`, call `start()` again. This rebuild-from-history property is unique to event sourcing — state-storage models have already discarded the history.

In production, `_handle_envelope` would write to a real database (PostgreSQL, Redis, Elasticsearch). The `ProjectionRunner` would persist a cursor in a checkpointing table so restarts continue from the last processed event instead of replaying everything from the beginning. The projection pattern is identical regardless of the underlying storage.

PROJECTIONS VS CHAPTER 8 LISTENERS

Chapter 8's `BalanceProjection` (Listing 8.4) was an `@event_listener` subscriber on the `InMemoryEventBus` — it reacted to events as they were published. This chapter's `BalanceLedgerProjection` reads directly from the `EventStore` — it can replay history from the beginning, catch up to the present, and continue consuming future events. Both keep a balance read model; the event-store projection is rebuildable from history; the bus listener is not.

The transactional outbox

Consider the `repo.save(account)` call in Listing 9.4: three events are appended to the event store. Now suppose those events also need to reach an external broker — Kafka, RabbitMQ, another microservice. The naive approach is to call `broker.publish(envelope)` immediately after `store.append(...)`. But what if the process crashes between the append and the publish? The events are in the store, but the broker never received them. The downstream service never learned about the credit.

The **transactional outbox** pattern solves this. Instead of publishing directly, you enqueue the event into an *outbox* — a durable intermediary. The outbox persists the event alongside the aggregate's events in the same store operation. A separate background worker (the *relay*) drains the outbox and forwards each event to the broker with at-least-once semantics. If the relay crashes, it restarts and retries from the last unacknowledged event.

PyFly's `TransactionalOutbox` lives in `pyfly.eventsourcing`. It accepts a `publish` coroutine and a `max_attempts` limit, and exposes two methods:

- `enqueue(envelope)` — adds an event envelope to the outbox for delivery.
- `start()` — starts the background relay loop that calls `publish(envelope)` for each queued item, retrying up to `max_attempts` times on failure.

lumen/eventsourcing/outbox_demo.py

```

from __future__ import annotations

from pyfly.eventsourcing import (
    InMemoryEventStore,
    InMemorySnapshotStore,
    TransactionalOutbox,
)
from pyfly.eventsourcing.repository import EventSourcedRepository

from lumen.models.entities.v1.ledger_account import LedgerAccount
from lumen.models.entities.v1.money import Money
from lumen.interfaces.enums.v1.currency import Currency

# Simulated broker: collect published envelopes for inspection.
_published: list = []

async def _broker_publish(envelope: object) → None:
    _published.append(envelope)

async def demo_outbox() → None:
    store = InMemoryEventStore()
    repo = LedgerAccountRepository(store)
    outbox = TransactionalOutbox(publish=_broker_publish, max_attempts=5)
    await outbox.start()

    account = LedgerAccount.open("led-004", "u-11", Currency.EUR)
    account.credit(Money(5000, Currency.EUR))
    await repo.save(account)

    # Enqueue the stored envelopes into the outbox.
    for envelope in await store.load("led-004"):
        await outbox.enqueue(envelope)

```

```
# The relay has delivered all envelopes to the broker.
assert len(_published) == 2 # LedgerOpened + Credited
```

Listing 9.10 — TransactionalOutbox: reliable at-least-once delivery to a broker

How it works. The outbox holds envelopes in a durable queue. `_broker_publish` is the delivery function — replace it with your Kafka or RabbitMQ producer.

`max_attempts=5` means the relay retries a failing delivery up to five times before dead-lettering the envelope.

The critical guarantee: the outbox is drained independently of the request that created the events. If the process crashes after `repo.save(account)` but before the outbox finishes flushing, the next restart picks up from where it left off and completes the delivery. The aggregate state in the event store is already correct; only the broker-side delivery was interrupted.

⚠ AT-LEAST-ONCE, NOT EXACTLY-ONCE

The outbox guarantees that every event reaches the broker *at least once*. If the relay delivers an event and then crashes before marking it as acknowledged, the event is delivered again on restart. Your broker consumers — and downstream services — must be idempotent: use the `envelope.event_id` as a deduplication key. Chapter 10 shows how Kafka and RabbitMQ consumer adapters handle deduplication automatically.

The transactional outbox is the bridge between event sourcing and event-driven messaging. Chapter 10 picks up exactly here, introducing Kafka producers and RabbitMQ exchanges and showing how to configure the relay for reliable delivery to each.

🍃 SPRING PARITY

The transactional outbox pattern is well known in the Spring ecosystem under the same name. Spring Modulith's `EventPublicationRegistry` and Spring's `@TransactionalEventListener(phase = AFTER_COMMIT)` approximate the same guarantee — the event is recorded durably before being dispatched. Axon Server's event store fulfils a similar role for Axon-based applications: events are written to the store first, and projection groups / event processors consume them with at-least-once guarantees from the stored log. PyFly's `TransactionalOutbox` is the portable equivalent of that pattern, without requiring a dedicated event server.

Advanced: upcasting and multi-tenancy

Two concerns appear in every long-lived event-sourced system. This section introduces them briefly; full treatment is beyond the scope of this book.

Upcasting

Events are immutable. Once a `Credited` event is written to the stream you cannot go back and add a `reference_code` field. But product requirements change: three months from now the finance team will need a reference code on every credit for reconciliation. New events include it; old events do not.

An **upcaster** is a function that transforms an old event shape into the current shape during replay. The `EventStore` calls it transparently — the aggregate never sees the old shape. You register upcasters per event type and per version:

```
# Conceptual – upcaster API varies by adapter
def upcast_credited_v1(payload: dict) → dict:
    payload.setdefault("reference_code", "LEGACY")
    return payload
```

The upcaster runs when the `EventStore` loads an event whose schema version is lower than the current one. Old data becomes readable without a migration; new data is written in the current schema.

Multi-tenancy

When multiple tenants share the same event store, events must be scoped by tenant so one tenant can never read or replay another's stream. PyFly gives you two seams for this; neither lives on `EventSourcedRepository` (its constructor takes only `store`, `factory`, `snapshots`, and `snapshot_interval` — there is no `tenant_id` parameter).

The first seam is on the **envelope itself**. `StoredEventEnvelope` carries a dedicated `tenant_id` field, and `StoredEventEnvelope.of(...)` accepts it as a keyword argument:

```
envelope = StoredEventEnvelope.of(
    aggregate_id="led-001",
    aggregate_type="LedgerAccount",
    sequence=0,
    event=Credited(...),
    tenant_id="tenant-A",
)
```

The SQL adapter persists this as a `tenant_id` column on `pyfly_event_store`, so a tenant-aware store can filter every query by it. This keeps the `aggregate_id` clean while still partitioning the log per tenant.

The second seam is a **pattern you implement yourself**: prefix every `aggregate_id` with the tenant identifier — `"tenant-A::led-001"` rather than `"led-001"` — when you call `repo.save` and `repo.load`. A thin tenant-aware wrapper around the repository can apply and strip the prefix so application code never sees it:

```

class TenantLedgerRepository:
    def __init__(self, repo: LedgerAccountRepository, tenant_id: str) → None:
        self._repo = repo
        self._prefix = f"{tenant_id}::"

    async def load(self, account_id: str) → LedgerAccount | None:
        return await self._repo.load(self._prefix + account_id)

```

Projections must scope their read models similarly — typically by including `tenant_id` as a column in the read-model table and filtering on it at query time.

CHOOSING EVENT SOURCING

Event sourcing adds operational complexity — upcasters, snapshot management, projection rebuild procedures, outbox relay monitoring. Choose it deliberately for domains where auditability and time-travel queries are first-class requirements: financial ledgers, medical records, supply-chain logs. For CRUD-heavy domains where the current state is all that matters, state storage is simpler and sufficient.

Run the whole chapter

You ran each piece in isolation as you built it. Now run the complete ledger suite to confirm everything still passes together:

```
uv run --extra dev pytest tests/test_ledger_event_sourcing.py -q
```

```

.....
[100%]
11 passed in 0.06s

```

Eleven dots: the four aggregate-isolation tests, the headline reload-by-replay proof, the raw-stream inspection, the typed-replay override, the post-reload concurrency check, the snapshot round-trip, the unknown-ledger lookup, and the auto-configuration wiring check. With those green, the `LedgerAccount` aggregate and its repository are complete and correct.

✓ What you built

Lumen's `LedgerAccount` is now a fully event-sourced ledger that coexists with the Chapter 6 state-stored `Wallet`.

The two aggregate bases are separate by design: `Wallet` extends `pyfly.domain.AggregateRoot` and stores its balance in a database row; `LedgerAccount` extends `pyfly.eventsourcing.AggregateRoot` and derives its balance from an immutable event stream. The distinction is not cosmetic — the event-sourcing machinery (`apply`, `replay`, `when`) lives only on the eventsourcing base class.

The three domain events — `LedgerOpened`, `Credited`, `Debited` — are `dataclass`es extending `pyfly.eventsourcing.DomainEvent`, each with default field values so the repository can reconstruct them from a stored payload. Handlers are registered with `when()` as two-arg lambdas delegating to private methods; the bound-method trap to avoid is that a bound method is already one-arg from the outside, which causes a `TypeError` at dispatch time.

`LedgerAccountRepository` is a thin subclass of `EventSourcedRepository[LedgerAccount]` that passes the zero-arg factory and overrides `_envelope_to_event` to hydrate concrete dataclasses on replay. The

headline test confirmed it: after opening, crediting, and debiting a ledger, a brand-new repository and aggregate reconstructed the correct balance purely by replaying the stored stream — no stored balance column, no shared state.

The `version` counter drives the optimistic-concurrency guard: the store rejects any `append` whose `expected_version` does not match the stream's actual version, forcing the losing writer to retry from a fresh load. `InMemorySnapshotStore` proved harmless when the stream is shorter than the snapshot interval and will accelerate replay once a ledger crosses the threshold.

`BalanceLedgerProjection` used `FunctionProjection` and `ProjectionRunner` to maintain a fast balance read model from the raw event stream — one that can be rebuilt from history at any time, unlike the bus-listener approach from Chapter 8.

`TransactionalOutbox` then connected the event store to the broker world, enqueueing events for at-least-once delivery and retrying on failure so no fact is silently lost between the store and downstream consumers. Chapter 10 picks up exactly there, introducing the Kafka and RabbitMQ adapters that the outbox relay targets.

Try it yourself

1. **Replay to a point in time.** Apply ten credits of 100 cents each to a `LedgerAccount`. Then load the aggregate manually by calling `await store.load("led-X")`, filter envelopes to those with `sequence ≤ 5`, and replay only those through a fresh `LedgerAccount()`. Assert that the resulting balance equals 400 cents (four credits after the open) rather than 1 000 cents (ten credits). This is the "time-travel query" that state-storage models cannot provide.

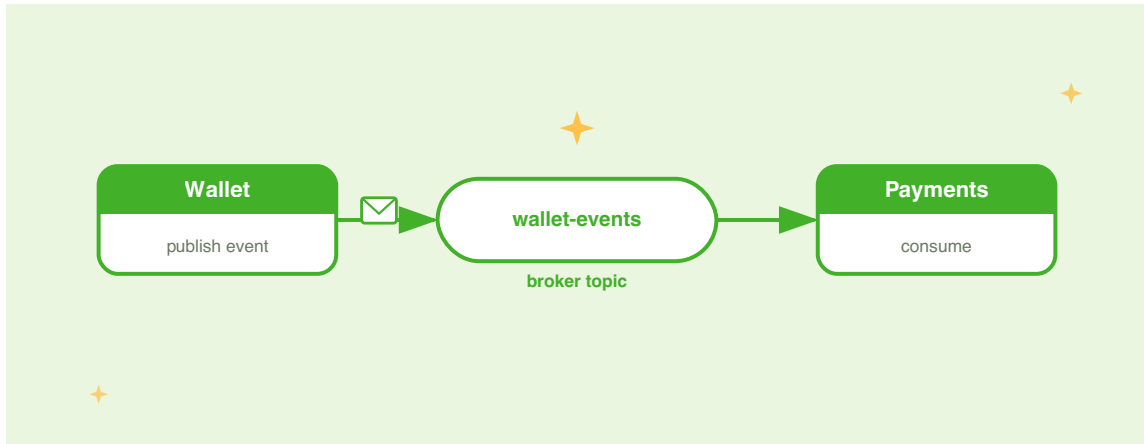
2. Implement a `Transferred` event and dual-aggregate save. Add a

`Transferred(DomainEvent)` with fields `source_id: str`, `target_id: str`, `amount: int`, `currency: str`. Add a `transfer_to(target: LedgerAccount, amount: Money) → None` method to `LedgerAccount` that calls `self.debit(amount)` and `target.credit(amount)` in sequence, then applies a `Transferred` event on `self`. Wire a `demo_transfer` coroutine that opens two ledgers, credits 10 000 cents into the first, transfers 3 000 cents to the second, saves both aggregates independently, reloads both from the store, and asserts that the balances are 7 000 and 3 000 cents respectively.

3. Add an `OwnerLedgerProjection`. Write a second `FunctionProjection` named `owner_ledger` whose handler maintains a `dict[str, list[dict]]` mapping each `owner_id` to a chronological list of transaction records. Each record should include `event_type`, `amount` (from the payload for `Credited` / `Debited` events), and the envelope's sequence number. Feed it the same `InMemoryEventStore`, open three ledgers for the same owner, perform a mix of credits and debits, start the projection runner, and assert that the owner's transaction list has the correct number of entries in the correct order.

CHAPTER 10

Messaging with Kafka & RabbitMQ



Lumen's wallet service is genuinely event-driven. Commands flow through typed handlers; domain events publish to an in-process bus; listeners react independently without coupling to the write path. In Chapter 8 you built `WalletAuditListener` — a service that reacts to `WalletOpened`, `FundsDeposited`, and `FundsWithdrawn` events without knowing the command handlers. In Chapter 9 you went further, storing those events *as the source of truth* so every historical balance is computable from first principles.

There is one boundary neither chapter crossed: the network. `InMemoryEventBus` lives inside the Python process. The moment another service — a future `PaymentsService` that settles transfers, or a `NotificationsService` that sends push alerts — needs to react to Lumen's facts, you need a **message broker**: an independent infrastructure component that stores events durably, routes them to subscribers in other processes, and replays them when a consumer restarts after a crash.

This chapter takes Lumen's event-driven foundation across that boundary. You will see how PyFly wraps the complexity of Apache Kafka and RabbitMQ behind a single clean abstraction — `MessageBrokerPort` — so that application code never knows which broker is running beneath it. You will publish Lumen's wallet events to real topics, consume them with the `@message_listener` decorator, choose the right serialisation format for your schema-evolution requirements, handle poisoned messages with dead-letter queues built into the decorator, and protect your service against broker outages with circuit breakers and retries.

By the end of the chapter Lumen's integration events flow across process boundaries, ready for the Part IV services that will consume them.

We will build this gradually, one piece at a time. Each feature comes with a numbered walkthrough, the exact command to run, and the output you should expect to see. If you have followed along from Chapter 8 you already have `EventPublisher` wired into the wallet command handlers and the `WalletAuditListener` reacting in-process; this chapter is verified against PyFly v26.6.110 and the Lumen sample under `samples/Lumen`. Nothing here requires a running Kafka or RabbitMQ cluster to follow along: PyFly ships an in-memory broker that satisfies the same contract, so you can read, run, and test every listing before you ever touch Docker.

i JARGON, IN PLAIN LANGUAGE

A handful of terms recur in this chapter. A **message broker** is a separate server (Kafka or RabbitMQ) that stores messages and hands them to other processes. A **topic** is a named channel on the broker; publishers write to it and subscribers read from it. A **producer** (or *publisher*) puts messages onto a topic; a **consumer** (or *listener*) pulls them off and reacts. A **consumer group** is a label that lets several copies of the same service share the work, so each message is handled once. A **serialiser** turns a Python object into the raw `bytes` the broker stores; a **deserialiser** turns those bytes back into an object on the other side. A **dead-letter queue** (DLQ) is a holding area for messages that could not be processed. An **adapter** is the concrete broker driver hiding behind the `MessageBrokerPort` interface. Keep these eight in mind; the rest of the chapter is mostly about connecting them.

One abstraction, many brokers

Why an abstraction matters

Before writing a line of Kafka or RabbitMQ code, it is worth asking: why does PyFly introduce an abstraction layer at all? `aiokafka` and `aio-pika` both expose perfectly usable async APIs. The answer is the same reason you depend on `EventPublisher` rather than `InMemoryEventBus` — the abstraction is what lets you swap infrastructure without touching business logic.

Without an abstraction, every service that produces or consumes a message imports Kafka-specific or RabbitMQ-specific types. Switching brokers — or running Kafka in production and an in-memory broker in CI — means changing import paths, constructor signatures, and consumer-loop boilerplate across every affected file. With `MessageBrokerPort`, the swap is a YAML change. The listeners and publishers that make up your business logic never change.

The abstraction pays dividends in testing too. `InMemoryMessageBroker` satisfies the port protocol. Inject it wherever `MessageBrokerPort` is expected and write fast, deterministic tests with no Docker dependency. Chapter 16 makes this concrete.

The MessageBrokerPort protocol

`MessageBrokerPort` is a `@runtime_checkable Protocol`. Use it as a type hint throughout your code; call `isinstance(obj, MessageBrokerPort)` at runtime if you need to verify that an injected bean satisfies the contract.

The protocol defines four methods:

```
from pyfly.messaging import MessageBrokerPort

class MessageBrokerPort(Protocol):
    async def publish(
        self,
        topic: str,
        value: bytes,
        *,
        key: bytes | None = None,
        headers: dict[str, str] | None = None,
    ) → None: ...

    async def subscribe(
        self,
        topic: str,
        handler: MessageHandler,
        group: str | None = None,
    ) → None: ...

    async def start(self) → None: ...

    async def stop(self) → None: ...
```

The four methods:

`publish` sends a single message to the named topic. `value` is raw bytes — the protocol deliberately leaves serialisation to you; encode the payload before calling `publish` and decode it inside your handler. `key` and `headers` are keyword-only so callers cannot accidentally transpose them. `key` drives Kafka partition assignment for ordering guarantees; RabbitMQ ignores it. `headers` carry cross-cutting metadata such as `event-type` and correlation IDs.

`subscribe` registers an async `MessageHandler` for a topic. The optional `group` parameter maps to Kafka consumer groups and RabbitMQ competing-consumer queues. Deploy three instances of a service, all subscribing with the same `group`, and only one instance processes each message. Omit `group` for broadcast semantics — every subscriber receives every message — which is useful for analytics that need a copy of every event.

`start` creates connections and begins consuming. Register all subscriptions *before* calling `start`, then call it once during application startup.

`stop` drains in-flight messages and closes connections cleanly. PyFly's application lifecycle calls `stop` automatically during shutdown, so you rarely need to invoke it manually.

The Message dataclass

Every handler receives a `Message` — a frozen dataclass carrying the full envelope of a received message:

```
from pyfly.messaging import Message

msg = Message(
    topic="wallet.events",
    value=b'{"wallet_id": "w-001", "amount": 5000}',
    key=b"w-001",
    headers={"event-type": "FundsDeposited"},
)
```

Field	Type	Default	Description
topic	str	required	The topic or queue the message arrived on.
value	bytes	required	The raw payload. You decode it inside your handler.
key	bytes None	None	Partition or routing key. Kafka uses it for partition assignment.
headers	dict[str, str]	{}	String metadata attached by the publisher.

The dataclass is frozen: once the broker hands you a `Message` its fields are immutable — safe to pass across async boundaries without defensive copying, and immune to accidental mutation inside handlers.

Kafka vs RabbitMQ — choosing the right broker

Before diving into configuration, it helps to understand where each broker fits. The table below summarises the key trade-offs; neither choice is universally correct.

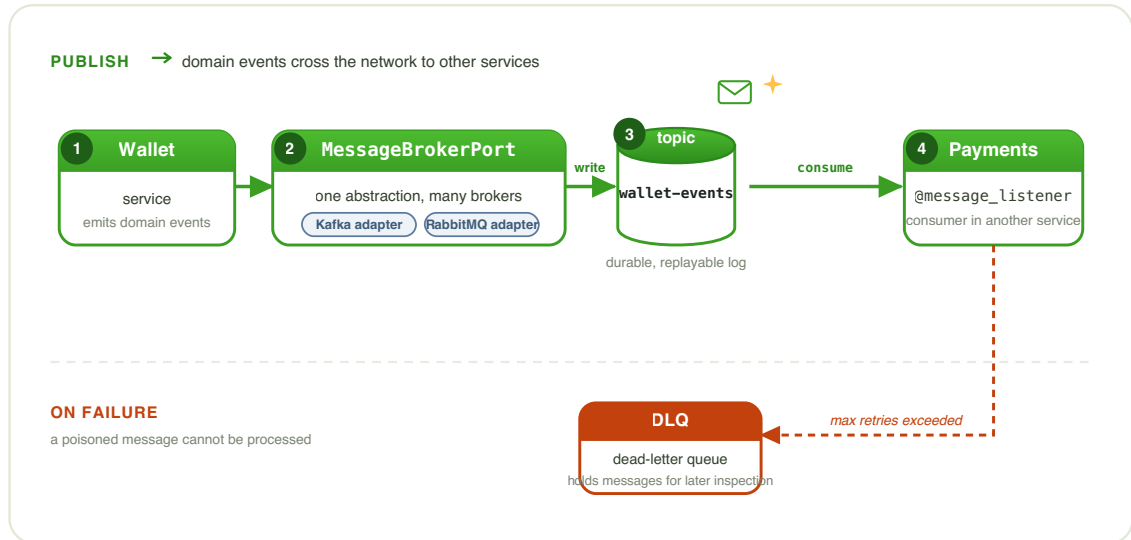


Figure 10.1 — MessageBrokerPort sits between application code and the broker adapters.

Dimension	Apache Kafka	RabbitMQ
Model	Distributed commit log; consumers maintain their own offset	Message broker; messages are removed from the queue after acknowledgement
Retention	Configurable (days to indefinite); consumers can replay from any offset	Messages are removed on delivery; dead-letter queues for failed messages
Throughput	Millions of messages/second; optimised for streaming	Tens of thousands/second; optimised for task routing
Ordering	Guaranteed within a partition (keyed producers)	Guaranteed within a single queue (FIFO)
Consumer groups	Native partition-level load balancing	Competing-consumer queues; one message per consumer
Schema evolution	Works well with Avro/Protobuf + Schema Registry	Works well with JSON; schema coupling is a user concern
When to choose	Event streaming, audit logs, replay, high throughput	Task queues, RPC patterns, per-message routing with complex bindings
PyFly extra	<code>uv add "pyfly[kafka]"</code>	<code>uv add "pyfly[rabbitmq]"</code>

For Lumen, Kafka is the natural fit: wallet events form an ordered stream per wallet, are worth replaying when a new consumer comes online, and will eventually feed high-throughput analytics. The examples in this chapter show both adapters interchangeably — from your code's perspective, the choice is a configuration detail.

i INSTALLING BOTH ADAPTERS

If you want to support either broker in one install, `uv add "pyfly[eda]"` pulls in both `aiokafka` and `aio-pika`. The auto-configuration then selects Kafka if `aiokafka` is importable, RabbitMQ if `aio-pika` is importable, and falls back to the in-memory broker if neither is present.

Configuring the adapters

Wiring a broker into Lumen is a configuration task, not a coding one. You add one extra to the project, add a `pyfly.messaging` block to `pyfly.yaml`, and PyFly does the rest: it constructs the right adapter and registers it under the `MessageBrokerPort` bean so anything that asks for that port gets the running broker injected. We will start with the broker that needs no infrastructure at all, then graduate to Kafka and RabbitMQ.

i TURN MESSAGING ON WITH ONE KEY

In v26.6.110 the messaging subsystem only wires itself up when the `pyfly.messaging.provider` key is **present** in your configuration. No key means no `MessageBrokerPort` bean — a deliberate "off by default" so that an app with no messaging needs pull no broker libraries. Once the key is set, its value (`"memory"`, `"kafka"`, `"rabbitmq"`, or `"auto"`) selects the adapter. The companion keys live under the same block: `pyfly.messaging.kafka.bootstrap-servers` and `pyfly.messaging.rabbitmq.url`.

Start with the in-memory broker

Lumen's `pyfly.yaml` already runs on the in-memory EDA bus from Chapter 8. To bring the *messaging* abstraction online without standing up Docker, add a single `provider` line.

Step 1 — Enable the in-memory broker. Open `pyfly.yaml` and add a `messaging` block under `pyfly`:

```
pyfly:
  messaging:
    provider: "memory"
```

Step 2 — Add a listener so there is something to wake up. Drop the standalone listener from Listing 10.4 (a few pages on) into `src/lumen/messaging/payments_consumer.py`. At startup PyFly discovers the stamped function and subscribes it for you — you write no `subscribe()` call.

RUN IT

Start the app. The in-memory broker needs no external server, so this works on a laptop with nothing installed:

```
uv run pyfly run
```

The boot banner reports the framework version and the bound port (`pyfly.server.port` , `8080` by default in v26.6.110):

```
:: PyFly Framework :: (v26.06.110) (Python 3.13.13)
app=lumen version=1.0.0 ... started_in=0.42s port=8080
```

Nothing else is printed yet — no message has been published. That is expected: the broker is running and the listener is subscribed, waiting. The next sections give it events to carry.

What just happened. One YAML line switched the `MessageBrokerPort` bean from "not present" to a working in-memory broker, and the framework auto-subscribed your `@message_listener` to it during startup. No Kafka, no RabbitMQ, no Docker — yet the *exact same code* will run against a real broker once you change that one line to `"kafka"`. That swap-without-recompile property is the whole point of the abstraction.

Kafka

When you are ready for a real broker, add `pyfly[kafka]` to your project and point the provider at Kafka.

Step 1 — Install the Kafka extra. This pulls in `aiokafka`, the async driver PyFly's `KafkaAdapter` wraps:

```
uv add "pyfly[kafka]"
```

Step 2 — Declare the broker in `pyfly.yaml`. Switch the provider and list your brokers:

```
pyfly:
  messaging:
    provider: "kafka"
    kafka:
      bootstrap-servers: "kafka-1:9092,kafka-2:9092"
```

That is all PyFly needs to auto-configure a `KafkaAdapter` and register it as the `MessageBrokerPort` bean. (`bootstrap-servers` is a comma-separated list of `host:port` pairs — the addresses of one or more brokers in the cluster; the client discovers the rest from any one of them.) For most services the YAML is sufficient; if you need advanced producer options, construct the adapter manually as a `@bean` inside a `@configuration` class.

RabbitMQ

```
pyfly:
  messaging:
    provider: "rabbitmq"
    rabbitmq:
      url: "amqp://user:password@rabbitmq-host:5672/"
```

`RabbitMQAdapter` uses a durable direct exchange named `"pyfly"` by default. To customise the exchange name, construct the adapter manually:

`lumen/messaging/config.py`

```
from pyfly.container import configuration, bean
from pyfly.messaging import MessageBrokerPort
from pyfly.messaging.adapters.rabbitmq import RabbitMQAdapter

@Configuration
class BrokerConfig:
    """Wire up the message broker bean."""

    @bean
    def broker(self) → MessageBrokerPort:
        return RabbitMQAdapter(
            url="amqp://user:password@rabbitmq-host:5672/",
            exchange_name="lumen-events",
        )
```

Listing 10.1 — Custom RabbitMQ exchange name via `@bean`

How it works. `@configuration` marks the class as a factory that the DI container calls during startup. `@bean` on `broker` tells the container to call `broker()` once, cache the result, and inject it wherever `MessageBrokerPort` is requested. Any `@service` that declares `MessageBrokerPort` in its constructor receives this instance automatically — no import of `RabbitMQAdapter` required in the consumer class.

Auto-detection

When `provider` is `"auto"`, PyFly probes installed packages in order and picks the first broker it finds:

Priority	Library checked	Adapter selected
1	<code>aiokafka</code>	<code>KafkaAdapter</code>
2	<code>aio_pika</code>	<code>RabbitMQAdapter</code>
3	<i>(fallback)</i>	<code>InMemoryMessageBroker</code>

`provider: "memory"` is different from `"auto"`: it *always* selects the in-memory broker regardless of what is installed, which is exactly what you want in tests. An explicit `provider: "kafka"` or `"rabbitmq"` skips probing entirely and demands that adapter's library be present.

The practical pattern is per-environment YAML. Set `provider: "memory"` in `pyfly-test.yaml` and `provider: "kafka"` in `pyfly-prod.yaml`, and every test and production run uses the appropriate adapter without code changes.

💡 RUN IT

You can confirm which adapter PyFly selected without sending a single message. With messaging enabled, start the app and look for the broker line in the startup log:

```
uv run pyfly run
```

```
pyfly.messaging provider=memory broker=InMemoryMessageBroker started
```

Change `provider` to `"kafka"` (with `pyfly[kafka]` installed and a broker reachable) and restart; the same line now reports `broker=KafkaAdapter`. The business code that publishes and consumes did not change — only the YAML did.

Publishing integration events

From in-process events to integration events

In Chapter 8, Lumen's command handlers drained the `Wallet` aggregate's event buffer with `wallet.clear_events()` and published each domain event through `EventPublisher`. `WalletAuditListener` subscribed using `@event_listener` and reacted within the same process.

The **integration event** pattern crosses the process boundary. Where a *domain event* describes what happened inside an aggregate — a private fact, available to same-process listeners — an integration event is a sanitised, public representation of the same fact: designed for external consumers, stable across versions, and serialised to bytes for transport over a broker.

For Lumen, the integration event for a deposit carries only what an external consumer needs: the wallet identifier, the amount in minor units, the currency code, and the resulting balance. It does not expose the aggregate's internal implementation details.

How Lumen drains events to the broker

Lumen separates the publish bridge from the command handlers so every handler publishes events identically. `publish_domain_events` (in `lumen/core/services/wallets/event_publishing.py`) iterates the drained events, converts each frozen dataclass to a dict, and calls `EventPublisher.publish`:

```
# lumen/core/services/wallets/event_publishing.py (real Lumen code)
from pyfly.eda import EventPublisher
from pyfly.domain import DomainEvent

async def publish_domain_events(
    publisher: EventPublisher,
    events: Iterable[DomainEvent],
) → None:
    for event in events:
        payload = dataclasses.asdict(event)
        payload.setdefault("event_type", event.event_type)
        await publisher.publish(
            destination="wallet.events",
            event_type=event.event_type,    # "WalletOpened" /
            "FundsDeposited" / ...
            payload=payload,
        )
```

The `EventPublisher.publish` signature is:

```
async def publish(
    self,
    destination: str,
    event_type: str,
    payload: dict[str, Any],
    headers: dict[str, str] | None = None,
) → None: ...
```

`destination` is the logical channel name (`"wallet.events"`). `event_type` is the domain event class name — `"WalletOpened"`, `"FundsDeposited"`, or `"FundsWithdrawn"` — which is exactly what `@event_listener` subscribers filter on.

Every command handler wires in `EventPublisher` via the constructor and calls `publish_domain_events` after persisting:

lumen/core/services/wallets/deposit_funds_handler.py

```

from __future__ import annotations

from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.core.services.wallets.event_publishing import publish_domain_events
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.domain import AggregateNotFound
from pyfly.eda import EventPublisher

@command_handler
@service
class DepositFundsHandler(CommandHandler[DepositFunds, int]):
    """Credit funds to an existing wallet; returns the new balance."""

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events

    async def do_handle(self, command: DepositFunds) → int:
        wallet = await self._repository.find(command.wallet_id)
        if wallet is None:
            raise AggregateNotFound("Wallet", command.wallet_id)

        wallet.deposit(Money(
            amount=command.amount,      # integer minor units (e.g. 5000 =
€50.00)
            currency=wallet.currency,
        ))
        await self._repository.add(wallet)

```

```

    # Drain pending events and forward them to the EDA bus.
    await publish_domain_events(
        self._events, wallet.clear_events()
    )
    return wallet.balance.amount

```

Listing 10.2 — *DepositFundsHandler drains events via EventPublisher*

Key design decisions:

`events: EventPublisher` is the port, not the adapter. The DI container injects whichever bus is configured — in-memory in tests, a broker-backed bus in production. This handler never mentions Kafka or RabbitMQ.

`command.amount` is the deposit in integer minor units (e.g. `5000` means €50.00 for a EUR wallet). The `FundsDeposited` domain event records the same `amount` field, plus `currency` (a string like `"EUR"`) and `balance` (the new balance in minor units).

`wallet.clear_events()` drains the aggregate's pending events list and returns them. Calling it *after* `repository.add` ensures the events describe a fact that persisted. Publishing before saving would create phantom events — facts about things that never happened.

The domain events raised during a deposit are instances of:

```

@dataclass(frozen=True)
class FundsDeposited(DomainEvent):
    wallet_id: str = ""
    amount: int = 0      # integer minor units
    currency: str = ""   # e.g. "EUR"
    balance: int = 0     # new balance after deposit, minor units

```

When `publish_domain_events` publishes this event, `event_type` is the class name `"FundsDeposited"` — *not* a dotted string like `"wallet.fundsdeposited"`.

Publishing an integration event directly to the broker

When a separate service running in a different process needs to receive Lumen's wallet events, the EDA bus must be backed by a real broker adapter. The payload flowing over the wire is the same dict the in-process listeners see. A dedicated `OutboxRelay` (covered in the resilience section) or a broker-backed `EventPublisher` handles the transport.

It helps to see the publish in its smallest possible form first. The next listing is a plain `async` function — no class, no decorator — that takes a `MessageBrokerPort`, builds the payload, and calls `publish`. Build it in three moves:

Step 1 — Encode the payload to bytes. `MessageBrokerPort.publish` only ever sees `bytes`, so the function serialises the event with `json.dumps(...).encode()`. The `.encode()` turns the JSON string into UTF-8 bytes the broker can store verbatim.

Step 2 — Choose a partition key. Passing `key=wallet_id.encode()` tells Kafka to route every message for a given wallet to the same partition, which preserves their order. (RabbitMQ ignores the key, so including it is harmless either way.)

Step 3 — Attach the event-type header. `headers={"event-type": "FundsDeposited"}` lets a consumer decide whether it cares about this message *before* deserialising the body — cheap routing.

lumen/messaging/deposit_publisher.py

```

from __future__ import annotations

import json

from pyfly.messaging import MessageBrokerPort

async def publish_deposit_event(
    broker: MessageBrokerPort,
    wallet_id: str,
    amount: int,
    currency: str,
    balance: int,
) → None:
    """Encode a FundsDeposited integration event and publish to the topic."""
    payload = json.dumps({
        "wallet_id": wallet_id,
        "amount": amount,          # integer minor units
        "currency": currency,      # e.g. "EUR"
        "balance": balance,        # new balance, minor units
        "event_type": "FundsDeposited",
    }).encode()

    await broker.publish(
        "wallet.events",
        payload,
        key=wallet_id.encode(),
        headers={"event-type": "FundsDeposited"},
    )

```

Listing 10.3 — Publishing a wallet integration event to a Kafka topic

Key design decisions:

`broker: MessageBrokerPort` is the port, not the adapter. The DI container injects whichever adapter is configured — Kafka in production, the in-memory broker in tests.

`key=wallet_id.encode()` is the routing key. On Kafka, all messages sharing the same key land on the same partition, delivering them to consumers in publication order — critical for a ledger where deposit before withdraw must be preserved. On RabbitMQ the key is ignored (routing uses the exchange binding), so this field is safe to include regardless of which broker is running.

`headers={"event-type": "FundsDeposited"}` uses the domain event class name — not a dotted path like `"wallet.fundsdeposited"`. Consumers can inspect the event type without decoding the payload, which is useful for routing and filtering without full deserialisation.

What just happened. You crossed the process boundary. The same `FundsDeposited` fact that `WalletAuditListener` consumed in-process in Chapter 8 is now bytes on a topic, addressable by any service that connects to the broker — and the function that put it there names no broker, only the `MessageBrokerPort` port. Swap the configured adapter and this code is unchanged.

⚠ PUBLISH AFTER SAVE, NOT BEFORE

Always drain and publish events *after* `repository.add(wallet)`. If the save fails, no message reaches the broker and external consumers never see a fact that never persisted. The transactional outbox pattern (where the outbox row and the aggregate row are written in the same database transaction) provides the stronger atomic guarantee for production; direct publishing as shown here is a reasonable starting point for simpler services.

Consuming events with `@message_listener`

The problem with polling

Before brokers, services reacted to another service's state changes by polling a shared database or a REST endpoint. Polling adds latency (the reaction waits until the next poll interval), wastes resources (most polls find nothing new), and couples consumer to producer at the API level. A message listener eliminates all three problems: the broker pushes the event as soon as it is available, idle connections consume negligible CPU, and the consumer depends only on the message schema — not on the producer's internal API.

Declarative listeners with `@message_listener`

`@message_listener` is the declarative subscription decorator. Decorate any async function or method with the topic it should consume, and PyFly wires the subscription during application startup — no bus reference, no `subscribe()` call, no lifecycle management in your code.

The decorator signature is:

```
def message_listener(  
    topic: str,  
    group: str | None = None,  
    *,  
    retries: int = 0,  
    retry_delay: float = 0.0,  
    dead_letter_topic: str | None = None,  
) → ...: ...
```

Parameter	Type	Default	Description
<code>topic</code>	<code>str</code>	required	The topic to listen on.
<code>group</code>	<code>str</code> <code>None</code>	<code>None</code>	Consumer group name.
<code>retries</code>	<code>int</code>	<code>0</code>	Times to re-invoke the handler on failure.
<code>retry_delay</code>	<code>float</code>	<code>0.0</code>	Base delay (seconds) between retries — attempt N waits <code>retry_delay * N</code> .
<code>dead_letter_topic</code>	<code>str</code> <code>None</code>	<code>None</code>	When set, a message still failing after <code>retries</code> is re-published here.

The first listener is a free-standing function — the simplest shape. Build it in three moves:

Step 1 — Write an async function that takes a `Message`. Every listener receives one argument: the frozen `Message` envelope (`topic` , `value` , `key` , `headers`). The function must be `async` because the broker awaits it.

Step 2 — Decorate it with the topic and group.

`@message_listener(topic="wallet.events", group="payments-service")` is the entire subscription. There is no `subscribe()` call to write and no bus to import — the decorator stamps the function with metadata the framework reads at startup.

Step 3 — Decode inside the handler. The body checks the `event-type` header, then `json.loads(msg.value)` turns the raw bytes back into a dict. The handler decides what it cares about; here it reacts only to `FundsDeposited`.

lumen/messaging/payments_consumer.py

```

from __future__ import annotations

import json

from pyfly.messaging import Message, message_listener

@message_listener(topic="wallet.events", group="payments-service")
async def on_wallet_event(msg: Message) → None:
    """React to every wallet event published to the topic."""
    event_type = msg.headers.get("event-type", "unknown")
    payload = json.loads(msg.value)

    if event_type == "FundsDeposited":
        wallet_id: str = payload["wallet_id"]
        amount: int = payload["amount"]          # minor units
        currency: str = payload["currency"]
        print(
            f"[Payments] Deposit received: "
            f"wallet={wallet_id} "
            f"amount={amount} {currency}"
        )

```

Listing 10.4 — @message_listener on a standalone function

How it works. The decorator stores six metadata attributes on the wrapped function —

`__pyfly_message_listener__ = True`, plus `__pyfly_listener_topic__`, `__pyfly_listener_group__`, `__pyfly_listener_retries__`, `__pyfly_listener_retry_delay__`, and `__pyfly_listener_dlq__`. During startup the framework scans all registered beans, finds functions carrying `__pyfly_message_listener__ = True`, and calls `broker.subscribe(topic, handler, group)` automatically. You never call `subscribe()` manually.

`group="payments-service"` places the consumer in a consumer group. Scale to multiple instances of the payments service and only one processes each message — the broker distributes load across the group. Omit `group` for broadcast semantics where every instance receives every message.

Inside the handler, `msg.headers.get("event-type", "unknown")` inspects the envelope metadata before touching the payload. The header value is the domain event class name — `"FundsDeposited"`, `"WalletOpened"`, or `"FundsWithdrawn"` — matching what Lumen sets on the publisher side.

💡 RUN IT

With `provider: "memory"` set (no Docker), this is the full publish → consume round-trip in one place. Save the snippet below as `roundtrip.py` and run it with `uv run python roundtrip.py`:

```
import asyncio, json
from pyfly.messaging.adapters.memory import InMemoryMessageBroker
from lumen.messaging.payments_consumer import on_wallet_event

async def main() → None:
    broker = InMemoryMessageBroker()
    await broker.subscribe(
        "wallet.events", on_wallet_event, group="payments-service"
    )
    await broker.start()
    await broker.publish(
        "wallet.events",
        json.dumps({
            "wallet_id": "w-001", "amount": 5000,
            "currency": "EUR", "balance": 5000,
        }).encode(),
        headers={"event-type": "FundsDeposited"},
    )
    await asyncio.sleep(0.1)  # let the listener run
    await broker.stop()

asyncio.run(main())
```

The listener prints the line it built from the decoded payload:

```
[Payments] Deposit received: wallet=w-001 amount=5000 EUR
```

Inside a running app you would *not* write this wiring by hand — the `@message_listener` decorator and the configured broker bean do the `subscribe / start / stop` for you. This standalone script just makes the round-trip visible in isolation.

What just happened. A message you published to `wallet.events` arrived at a function you never explicitly connected to anything. The decorator carried the topic and group; the broker (here in-memory, in production Kafka) did the delivery. That is the consume side of the same abstraction you used to publish — and the function body is broker-agnostic from top to bottom.

Listeners on service classes

When a listener needs collaborators — a repository, another service — declare it as a method on a `@service` class. PyFly injects the dependencies through the constructor and wires the listener subscription after the bean is initialised. The shape changes only slightly from the standalone version:

Step 1 — Make the class a `@service`. This registers it in the DI container so the framework can both inject its constructor and discover its listener method.

Step 2 — Declare collaborators in the constructor. Here `smtp_client` stands in for an email or push service; the container supplies it. Listing 10.4's free function had nowhere to keep such a dependency — that is the reason to reach for a class.

Step 3 — Decorate a *method* with `@message_listener`. The signature gains `self`, but otherwise the decorator and body are identical to the function form. Because the bean is created first, `self._smtp` is ready by the time a message arrives.

lumen/messaging/notifications_consumer.py

```

from __future__ import annotations

import json

from pyfly.container import service
from pyfly.messaging import Message, message_listener

@service
class WalletNotificationConsumer:
    """Sends push notifications when wallet events arrive via the broker."""

    def __init__(self, smtp_client: object) → None:
        # smtp_client would be an injected email/push service.
        self._smtp = smtp_client

    @message_listener(topic="wallet.events", group="notifications-service")
    async def on_wallet_event(self, msg: Message) → None:
        event_type = msg.headers.get("event-type", "unknown")

        if event_type ≠ "WalletOpened":
            return

        payload = json.loads(msg.value)
        owner_id: str = payload.get("owner_id", "")
        wallet_id: str = payload.get("wallet_id", "")
        currency: str = payload.get("currency", "")

        print(
            f"[Notification] Welcome {owner_id}! "
            f"Your {currency} wallet {wallet_id} is ready."
        )

```

Listing 10.5 — @message_listener on a @service method with dependencies

How it works. `@service` registers `WalletNotificationConsumer` in the DI container. The constructor receives `smtp_client` through injection. After the bean is created, the framework detects `on_wallet_event` carrying `__pyfly_message_listener__ = True` and registers it as a bound-method listener — `self` is already captured, so every invocation has full access to `self._smtp`.

The early return on `event_type != "WalletOpened"` is a filtering guard. A single topic (`wallet.events`) carries multiple event types, so each listener filters for the ones it cares about. This is simpler than maintaining a separate topic per event type, though for very high-volume streams, topic-per-type is a legitimate design trade-off.

💡 CONSUMER GROUP SEMANTICS AT A GLANCE

Two services with *different* group names each receive every message — the broker delivers a copy to each group. Two *instances* of the same service sharing the *same* group name share the load — each message goes to exactly one instance. Use different groups for fanout (payments and notifications both need the event); use the same group for horizontal scaling (three instances of the payments service share the work).

Serialisation and schema evolution

Why bytes, and why this matters

`MessageBrokerPort.publish` accepts raw `bytes`. That is a deliberate choice. A broker adaptor that forced a single serialisation format would be convenient for simple cases and painful for everything else — schema evolution, multi-language consumers, compliance requirements, and throughput constraints all push in different directions. By leaving serialisation to you, PyFly stays out of the way.

Three formats are worth knowing: JSON for simplicity, Avro for schema-registry-backed evolution, and Protobuf for performance-critical or multi-language environments:

Format	Human-readable	Schema enforcement	Schema evolution	Multi-language	PyFly encoding
JSON	Yes	Optional	Manual (consumer discipline)	Universal	<code>json.dumps(...).encode()</code>
Avro	No	Yes (via registry)	First-class (<code>BACKWARD</code> / <code>FORWARD</code> / <code>FULL</code>)	Good	<code>fastavro</code> library
Protobuf	No	Yes (<code>.proto</code> files)	First-class (field numbering)	Excellent	<code>protobuf</code> library

JSON — start here

Serialisation is just the act of turning an in-memory object into a flat sequence of bytes you can store or send, and *deserialisation* is the reverse. The three formats below differ only in how compact those bytes are and how strictly they police the shape of the data.

JSON is the right default. It requires no tooling beyond the standard library, every language can parse it, and the payload is readable in broker monitoring UIs. The encoding pattern is two lines:

```
import json

payload: bytes = json.dumps({
    "wallet_id": "w-001",
    "amount": 5000,          # integer minor units (€50.00)
    "currency": "EUR",
    "balance": 10000,       # new balance, minor units
    "event_type": "FundsDeposited",
}).encode()

await broker.publish("wallet.events", payload)
```

Decoding in the consumer:

```
data: dict = json.loads(msg.value)
```

JSON's weakness is that the schema is unenforced. If a publisher adds a required field and the consumer has not been updated, the consumer breaks silently. For Lumen's internal events where producer and consumer are deployed together, this is manageable. For events shared with external teams or long-lived topics, you need stronger guarantees.

Avro — schema-registry-backed evolution

Avro schemas are JSON documents describing the shape of a message. A Schema Registry (Confluent's is the most common, but open-source alternatives exist) stores those schemas and enforces compatibility rules when producers register new versions. The `fastavro` library encodes and decodes the binary payload. The publish path is the same `broker.publish(...)` you already know; only the encoding step changes:

Step 1 — Declare the schema once. `WALLET_DEPOSITED_SCHEMA` lists each field and its Avro type (`string` , `long`). It is a module-level constant so it is written once, not per message.

Step 2 — Compile it once. `fastavro.parse_schema(...)` is called at import time and the result cached in `_PARSED`. Parsing on every publish would be wasted work on the hot path.

Step 3 — Encode and publish. `fastavro.schemaless_writer` serialises the record into a `BytesIO` buffer; `buf.getvalue()` hands the bytes to `broker.publish` exactly as the JSON path did.


```
lumen/messaging/avro_publisher.py
```

```
from __future__ import annotations

import io

import fastavro # type: ignore[import]

from pyfly.messaging import MessageBrokerPort

WALLET_DEPOSITED_SCHEMA = {
    "type": "record",
    "name": "FundsDeposited",
    "namespace": "lumen.wallet",
    "fields": [
        {"name": "wallet_id", "type": "string"},
        {"name": "amount", "type": "long", # integer minor units
        {"name": "currency", "type": "string"},
        {"name": "balance", "type": "long", # new balance, minor units
    ],
}

_PARSED = fastavro.parse_schema(WALLET_DEPOSITED_SCHEMA)

async def publish_deposit_avro(
    broker: MessageBrokerPort,
    wallet_id: str,
    amount: int,
    currency: str,
    balance: int,
) → None:
    """Encode a FundsDeposited event with Avro and publish to the topic."""
    record = {
        "wallet_id": wallet_id,
        "amount": amount, # integer minor units
        "currency": currency,
        "balance": balance,
    }
```

```

buf = io.BytesIO()
fastavro.schemaless_writer(buf, _PARSED, record)

await broker.publish(
    "wallet.events",
    buf.getvalue(),
    headers={"content-type": "avro/binary",
            "event-type": "FundsDeposited"},
)

```

Listing 10.6 — Publishing a wallet event with Avro encoding

How it works. `fastavro.parse_schema` compiles the JSON schema document once at module load time — never parse it inside the publish function or you pay the compilation cost on every call. `fastavro.schemaless_writer` serialises the record into the `BytesIO` buffer without embedding the schema in each message (the registry provides the schema on the consumer side). `buf.getvalue()` extracts the bytes for `broker.publish`.

The `headers={"content-type": "avro/binary", "event-type": "FundsDeposited"}` headers signal to consumers that Avro decoding is required and carry the event type for routing — consistent with the JSON convention.

Protobuf — performance and polyglot

Protocol Buffers compile a `.proto` file into a generated class. They produce smaller messages than JSON or Avro, and the generated code is available in every major language — making Protobuf the right choice when the consumer is a Go or Java service.

```

# Assumes a generated class lumen_pb2.FundsDeposited
from lumen_pb2 import FundsDeposited # type: ignore[import]

event = FundsDeposited(
    wallet_id="w-001",
    amount=5000,          # integer minor units
    currency="EUR",
    balance=10000,
)
payload: bytes = event.SerializeToString()

await broker.publish(
    "wallet.events",
    payload,
    headers={"content-type": "application/protobuf",
            "event-type": "FundsDeposited"},
)

```

Decoding in the consumer follows the mirror pattern:

```

from lumen_pb2 import FundsDeposited # type: ignore[import]

event = FundsDeposited()
event.ParseFromString(msg.value)

```

💡 START WITH JSON, MIGRATE WHEN YOU FEEL THE PAIN

The correct progression for most teams is: JSON first (fast to ship, easy to debug); add Avro when multiple teams own different sides of a topic and schema drift becomes a real coordination cost; switch to Protobuf when binary size or multi-language interop is a hard requirement. Because PyFly's `publish` and `@message_listener` accept raw bytes, you can change serialization format without changing the broker API calls — just swap the encoding and decoding steps.

When delivery fails: dead-letter queues

The inevitable bad message

Even a well-designed consumer will eventually encounter a message it cannot process. A downstream database may be unavailable. The payload may violate an assumption the consumer relied on. A transient network error may interrupt a third-party API call mid-handler. The question is not whether a consumer will fail — it is what happens when it does.

Without a dead-letter strategy, a failed consumer either drops the message (data loss) or re-queues it indefinitely, creating an infinite retry loop that blocks all subsequent messages — a *poison pill*. A **dead-letter queue** (DLQ) is the structured answer: a separate topic or queue where messages that cannot be processed after a configurable number of attempts are parked for inspection and manual reprocessing.

Decorator-native retry and DLQ

In PyFly, retry and dead-letter routing are built into `@message_listener` — no try/except scaffolding, no manual publish to the DLQ. Declare `retries` and `dead_letter_topic` directly on the decorator. You add resilience by adding *arguments*, not code:

Step 1 — Start from a normal listener. The handler body in the next listing is an ordinary consumer that decodes the payload and calls a worker (`_charge`).

Step 2 — Add `retries` (and optionally `retry_delay`). `retries=3` tells the framework to re-invoke the handler up to three more times if it raises. `retry_delay=0.5` spaces those attempts out with linear back-off.

Step 3 — Add `dead_letter_topic` . When all retries are exhausted, the framework republishes the message there instead of letting the exception crash the consumer. You write neither the retry loop nor the DLQ publish.

lumen/messaging/resilient_consumer.py

```

from __future__ import annotations

import json
import logging

from pyfly.container import service
from pyfly.messaging import Message, message_listener

logger = logging.getLogger(__name__)

@service
class ResilientWalletConsumer:
    """Processes wallet events with built-in retry and DLQ fallback."""

    @message_listener(
        topic="wallet.events",
        group="payments-dlq",
        retries=3,
        retry_delay=0.5,          # waits 0.5s, 1.0s, 1.5s (linear)
        dead_letter_topic="wallet.events.DLQ",
    )
    async def on_wallet_event(self, msg: Message) → None:
        payload = json.loads(msg.value)
        event_type = msg.headers.get("event-type", "unknown")
        logger.info(
            "Processing event: type=%s wallet=%s",
            event_type,
            payload.get("wallet_id"),
        )
        # Any unhandled exception triggers a retry; after 3 retries
        # the original message is forwarded to wallet.events.DLQ.
        await self._charge(payload)

    async def _charge(self, payload: dict) → None:
        raise NotImplementedError("replace with real payment logic")

```

Listing 10.7 — Retry and DLQ wired through `@message_listener`

The three parameters:

`retries=3` re-invokes `on_wallet_event` up to three more times after the first failure. Retries are appropriate for *transient* failures (a single database node restarting); keep the count low and let the DLQ handle sustained failures.

`retry_delay=0.5` applies linear back-off: attempt 1 waits 0.5 s, attempt 2 waits 1.0 s, attempt 3 waits 1.5 s. With `retry_delay=0.0` (the default), retries are immediate.

`dead_letter_topic="wallet.events.DLQ"` is the safety net. When all retries are exhausted, the framework re-publishes the original message to the DLQ topic, preserving the original `value` and `key`, and adds two diagnostic headers:

Header	Value
<code>x-original-topic</code>	The topic the message was originally consumed from.
<code>x-exception</code>	The exception class name (e.g. <code>RuntimeError</code>).

The exception is then swallowed so the consumer keeps running — the message is parked, not lost, and the next message on the topic is processed normally.

 RUN IT

You can watch a poisoned message land in the DLQ without a real broker. At wiring time the framework wraps every listener with `pyfly.messaging.error_handling.wrap_listener` — the same helper does the retrying and dead-lettering. Drive it directly so the flow is visible. Save this as `dlq_demo.py` and run `uv run python dlq_demo.py`:

```
import asyncio, json
from pyfly.messaging.adapters.memory import InMemoryMessageBroker
from pyfly.messaging.error_handling import wrap_listener
from pyfly.messaging.types import Message

async def always_fails(msg: Message) → None:
    raise RuntimeError("downstream unavailable")

async def main() → None:
    broker = InMemoryMessageBroker()
    await broker.start()

    async def show_dlq(msg: Message) → None:
        print("DLQ:", msg.headers["x-original-topic"],
              msg.headers["x-exception"])
    await broker.subscribe("wallet.events.DLQ", show_dlq)

    handler = wrap_listener(
        always_fails, broker,
        retries=2, dead_letter_topic="wallet.events.DLQ",
    )
    await handler(Message(
        topic="wallet.events",
        value=json.dumps({"wallet_id": "w-001"}).encode(),
    ))
    await broker.stop()

asyncio.run(main())
```


After two retries the wrapper gives up and re-publishes to the DLQ, where your monitor prints the diagnostic headers:

```
DLQ: wallet.events RuntimeException
```

The handler returned normally — the exception was swallowed, not propagated — so in a real app the consumer would simply move on to the next message.

Monitoring the DLQ

Subscribe to the DLQ topic like any other listener to observe and alert on dead-lettered messages:

lumen/messaging/dlq_monitor.py

```

from __future__ import annotations

import json
import logging

from pyfly.messaging import Message, message_listener

logger = logging.getLogger(__name__)

@message_listener(topic="wallet.events.DLQ", group="dlq-monitor")
async def on_dead_letter(msg: Message) → None:
    """Log every message that failed all retries."""
    original = msg.headers.get("x-original-topic", "unknown")
    exc = msg.headers.get("x-exception", "unknown")
    payload = json.loads(msg.value) if msg.value else {}
    logger.warning(
        "DLQ message: original_topic=%s exception=%s wallet=%s",
        original,
        exc,
        payload.get("wallet_id"),
    )

```

Listing 10.8 — Subscribing to the DLQ topic

⚠ DESIGN CONSUMERS FOR IDEMPOTENCY

A consumer that reaches the DLQ retry limit has consumed the message. If an operator later replays the DLQ message, the consumer will process it again. Without idempotency, that double-processing can corrupt data — crediting a wallet twice, sending a duplicate notification. Use the message's stable identifier as an idempotency key: before processing, check whether that ID has already been recorded in a `processed_events` table, and skip the work if it has. The check-and-record step should be in the same database transaction as the business write.

Resilience: circuit breakers and retries

Protecting Lumen from a broker outage

A healthy broker is not guaranteed. Network partitions, rolling upgrades, and resource exhaustion can all make the broker temporarily unavailable. If the command handler calls `broker.publish(...)` and the broker is down, you face two bad choices without a resilience layer: fail the entire command (refusing to deposit funds because the broker is unreachable) or silently drop the event (the deposit succeeds but the integration event is lost).

Neither is acceptable. The transactional outbox (Chapter 9) is the atomic solution — the event is captured in the database and a relay publishes it asynchronously, so a broker outage adds only latency, not data loss. Alongside the outbox, **circuit breakers** and **retries** protect the relay and any broker-calling code from cascading failures.

A **circuit breaker** is the electrical metaphor made into code: after too many failures in a row it "trips" and stops letting calls through for a cool-down period, so a struggling broker is not hammered by thousands of doomed reconnection attempts. A **retry** is the complementary tactic — try the same call again a few times, because many failures are momentary.

PyFly's resilience module (`pyfly.resilience`) provides both primitives. The circuit breaker opens after a configurable failure threshold and blocks calls to the broker during a cool-down period, preventing a thundering-herd reconnection storm. The retry decorator handles transient errors with configurable back-off. You apply them as a pair of decorators on the publish method:

Step 1 — Write the plain publish method. `forward` does one thing: encode the record and call `self._broker.publish(...)`. No resilience logic lives in the body.

Step 2 — Wrap it in `@circuit_breaker`. Pass a shared `CircuitBreaker` instance so the failure count accumulates across calls, not per call. When the broker is down, the breaker trips and fails fast.

Step 3 — Wrap that in `@retry` on the outside. Decorator order matters: `@retry` sits above `@circuit_breaker`, so all of one call's retry attempts happen before the breaker registers a single failure. Keep `max_attempts` low and let the breaker absorb sustained outages.

lumen/messaging/resilient_publisher.py

```

from __future__ import annotations

import json
import logging

from pyfly.container import service
from pyfly.messaging import MessageBrokerPort
from pyfly.resilience import CircuitBreaker, circuit_breaker, retry

logger = logging.getLogger(__name__)

@service
class OutboxRelay:
    """
    Drains pending outbox records and forwards them to the broker.
    Applies retry and circuit-breaker protection on every publish call.
    """

    def __init__(self, broker: MessageBrokerPort) → None:
        self._broker = broker

    @retry(max_attempts=3, delay=1.0, backoff=2.0)
    @circuit_breaker(CircuitBreaker(failure_threshold=5, recovery_timeout=30))
    async def forward(
        self,
        topic: str,
        payload: dict,
        event_type: str,
    ) → None:
        """Forward a single outbox record to the broker."""
        await self._broker.publish(
            topic,
            json.dumps(payload).encode(),
            headers={"event-type": event_type},
        )
        logger.info(

```

```

        "Event forwarded: topic=%s event-type=%s",
        topic,
        event_type,
    )

```

Listing 10.9 — Resilient broker publishing with retry and circuit breaker

The two decorators:

`@retry(max_attempts=3, delay=1.0, backoff=2.0)` wraps `forward` in a retry loop of up to three attempts. After the first failure it waits `delay` seconds (1 s); after the second it waits `delay * backoff` (2 s) — the wait grows as `delay * backoff ** attempt`. If the third attempt still fails, the exception propagates. Retries suit *transient* failures (a single broker node restarting); they are counterproductive for *permanent* failures (a misconfigured topic). Keep `max_attempts` low and let the circuit breaker handle sustained outages.

`@circuit_breaker(CircuitBreaker(failure_threshold=5, recovery_timeout=30))` guards `forward` with a shared `CircuitBreaker` instance that tracks consecutive failures across all calls. When the count reaches `failure_threshold`, the circuit *opens* and subsequent calls fail immediately with `CircuitBreakerException` rather than attempting to reach an unreachable broker — preventing a reconnection storm. After `recovery_timeout` seconds the circuit enters a *half-open* state: the next call is allowed through as a probe. If it succeeds, the circuit closes; if it fails, it re-opens. Decorator order matters: `@retry` is the outer decorator, so all three attempts of one logical call happen before the circuit breaker registers a single failure.

🍃 SPRING PARITY

`MessageBrokerPort` with `KafkaAdapter` is PyFly's counterpart of Spring Kafka's `KafkaTemplate` (publishing) and `@KafkaListener` (consuming). `RabbitMQAdapter` mirrors Spring AMQP's `RabbitTemplate` and `@RabbitListener`. The lifecycle model is the same: register listeners before starting the container, and the framework manages the consumer threads. Dead-letter queues in Spring Kafka are configured via `DeadLetterPublishingRecoverer` on the `DefaultErrorHandler`; in Spring AMQP via `RabbitListenerContainerFactory` with a `MessageRecoverer`. PyFly implements the same pattern declaratively through `@message_listener(retries=..., dead_letter_topic=...)` rather than requiring broker-specific container configuration. The `@retry` and `@circuit_breaker` decorators mirror Resilience4j's `@Retryable` and `@CircuitBreaker` annotations used with Spring's messaging infrastructure.

✓ What you built

Part III is complete.

Lumen is now fully event-driven, event-sourced, and broker-connected. Here is where each chapter left things.

Chapter 8 introduced the two-bus model: `ApplicationEventBus` for framework lifecycle events, `InMemoryEventBus` for domain events. `EventPublisher` was wired into the command handlers so every aggregate mutation produced a fact that independent listeners — `WalletAuditListener` among them — could react to without knowing each other. Subscriptions use `@event_listener(event_types=["WalletOpened", "FundsDeposited", "FundsWithdrawn"])`; handlers receive an `EventEnvelope` whose `event_type` is the domain event class name.

Chapter 9 replaced the mutable aggregate-plus-read-model approach with event sourcing. Every financial movement is an immutable event appended to the ledger. The current balance is computed by replaying the event stream. `EventEnvelope` became the unit of storage, and snapshots kept replay times bounded.

This chapter crossed the network boundary. `MessageBrokerPort` is the single abstraction in front of Kafka, RabbitMQ, or the in-memory broker. Swapping adapters is a configuration change — no business code changes. `@message_listener` gives declarative, zero-boilerplate subscriptions on both standalone functions and `@service` methods. The `retries` and `dead_letter_topic` parameters handle poisoned messages without manual try/except scaffolding. Payloads were encoded as JSON bytes, with Avro and Protobuf available when schema enforcement or binary efficiency matters more than simplicity. `@retry` and `@circuit_breaker` protect the publish path from transient and sustained broker failures.

The domain events flowing through all three chapters are:

Event class	Fields
<code>WalletOpened</code>	<code>wallet_id</code> , <code>owner_id</code> , <code>currency</code>
<code>FundsDeposited</code>	<code>wallet_id</code> , <code>amount</code> , <code>currency</code> , <code>balance</code>
<code>FundsWithdrawn</code>	<code>wallet_id</code> , <code>amount</code> , <code>currency</code> , <code>balance</code>

`amount` and `balance` are always integer minor units (e.g. `5000` for €50.00). `currency` is a string value from the `Currency` StrEnum (`"EUR"`, `"USD"`, `"GBP"`). The `event_type` header value is always the class name — `"FundsDeposited"` — never a dotted path.

Three principles carry forward into Part IV:

- **Depend on the port, not the adapter.** `MessageBrokerPort` is injected; `KafkaAdapter` is a configuration detail.
- **Design consumers for idempotency.** Brokers deliver *at least once*. Guard against duplicate processing with a stable message identifier.
- **Capture events atomically.** The transactional outbox ensures an event is never lost even when the broker is unavailable at write time.

Part IV introduces the `PaymentsService` and `NotificationsService`. Both subscribe to `wallet.events`. The adapter and configuration choices made in this chapter are all they need to start receiving Lumen's facts the moment they connect.

Try it yourself

RUN IT

Each exercise below ends in a test. Because `provider: "memory"` needs no broker, you can run them with nothing installed beyond the dev extra. From the Lumen project root:

```
uv run --extra dev pytest tests/test_messaging.py -q
```

A green run looks like:

```
...
[100%]
3 passed in 0.XXs
```

If a test fails with a missing-attribute error such as `__pyfly_message_listener__`, the `@message_listener` decorator is not applied to your handler — recheck Step 2 of "Declarative listeners."

- 1. Swap the adapter in one line.** Start with `provider: "memory"` in `pyfly.yaml` and add the `@message_listener` from Listing 10.4. Write an integration test that publishes a `FundsDeposited` message with `amount=5000` and `currency="EUR"` and asserts the listener receives it. Then switch `provider: "kafka"` in the YAML and confirm the same test (with a Testcontainers-managed Kafka broker) passes without changing the listener or the test assertion.
- 2. Add a DLQ monitor.** Create a second `@message_listener` on topic `wallet.events.DLQ` with group `dlq-monitor`. It should log the `x-original-topic` and `x-exception` headers along with the decoded payload. Write a test that simulates a failing consumer by raising `RuntimeError` inside the handler,

configures `retries=2` and `dead_letter_topic="wallet.events.DLQ"`, and confirms the DLQ monitor receives the message with `x-original-topic: "wallet.events"`.

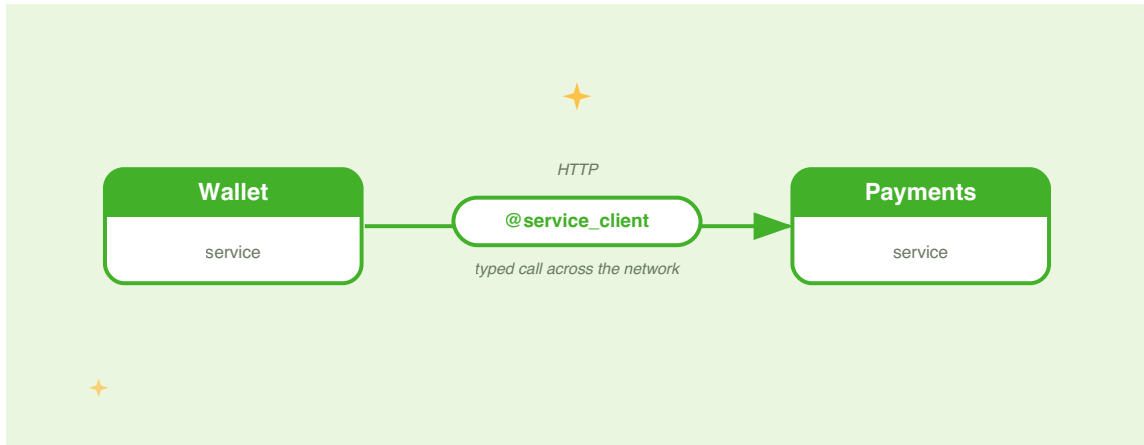
- 3. Evolve the schema with Avro.** Start with the `WALLET_DEPOSITED_SCHEMA` from Listing 10.6. Add an optional `note` field with a default of `None` (Avro union `["null", "string"]`, default `null`). Confirm that a consumer compiled against the original schema can still decode a message encoded with the new schema — this is a *backward-compatible* change. Then try adding a required field without a default and observe the `SchemaParseException` the registry would raise, illustrating why defaults are mandatory for safe evolution.

PART IV

Into Microservices

CHAPTER 11

Splitting the Monolith: HTTP Clients & the BFF



When Lumen was a single service, every capability lived in the same process. The wallet, the balance check, and the payment processing all ran together — straightforward to test, simple to deploy, and perfectly adequate until the team needed to ship features at different cadences. Then came the hard conversation about splitting.

The promise of a microservice split is real: teams own their services independently, scale them separately, and deploy without coordinating a shared release window. But every split introduces a problem the monolith never had — the network. What was a local function call becomes an HTTP request that can time out, fail halfway, or land on an overwhelmed service. That network boundary is not a deployment detail; it is a first-class engineering concern.

This chapter introduces `PaymentsService`, a second service that Lumen's Wallet service calls to settle transfers. Instead of hand-rolling `httpx` sessions and threading circuit-breaker logic through every handler, you will define the Payments client as an ordinary Python class — a typed, declarative interface that PyFly fills in at startup. By the end of the chapter you will also see how a **BFF (Backend for Frontend)** tier sits in front of both services and composes their capabilities into a single, user-journey-focused API.

 NEW JARGON, IN PLAIN LANGUAGE

A few terms recur throughout this chapter. **Service-to-service call** means one of your services making an HTTP request to another of your services (Wallet calling Payments), as opposed to a request coming from a browser. A **declarative client** is a client you *describe* rather than *implement*: you write the method signatures and let the framework fill in the HTTP plumbing. A **circuit breaker** is a safety switch that stops calling a remote service after it has failed too many times in a row. A **BFF** (Backend for Frontend) is a thin service whose only job is to call other services and reshape their answers for one particular app. We will build each of these one piece at a time.

This chapter is built around `httpx` and the `pyfly.client` package, both of which ship with the framework. Everything here runs against PyFly v26.6.110. You do not need to install anything new — if you have been following along with Lumen, the client tooling is already on your path.

Why split (and why it hurts)

The monolith comfort zone

A monolith is not an architectural mistake — it is an architectural starting point. Lumen began as one service because one service was right: one team, one deployment pipeline, one set of concerns to reason about. The database transaction that writes a wallet row and publishes a domain event in the same unit of work was not a compromise; it was the optimal choice.

The pressure to split usually arrives from outside the architecture. Payments needs a separate compliance audit trail. Risk scoring needs a specialist team with access to a private data source. Settlement processing demands throughput an order of magnitude higher than balance reads. Any one of these is a good reason to extract a service — and none of them erases the fact that the rest of the system still needs to call the extracted service across a network boundary.

The cost of the network

Network calls fail in ways that local function calls do not. A method on a local object either returns a value or raises an exception. An HTTP call to a remote service can time out (the remote is slow), refuse the connection (the remote is down), return a transient 503 (the remote is overloaded), or succeed only on the third attempt. In a monolith these failure modes are irrelevant; in a distributed system they are your baseline.

The naive fix — use `httpx` directly with `try/except` around every call — works for one call site but does not scale. You end up with circuit- breaker logic duplicated across every service client, retry delays hardcoded in handlers, and timeout values scattered through `pyfly.yaml` fragments that nobody owns. When Payments introduces a new endpoint, every caller must remember to add all the resilience scaffolding again.

PyFly's typed HTTP client eliminates that duplication. You declare what the remote service looks like — its endpoints, paths, and parameter shapes. PyFly generates the implementation at startup, wires in a circuit breaker and retry policy from `pyfly.yaml`, and registers the bean in the container so any handler that needs it can declare it as a constructor argument. Resilience is applied once, consistently, in the right layer.

A typed service client

Declarative over imperative

The core insight behind PyFly's client module is that service-to-service contracts are better expressed as types than as procedural HTTP logic. When you describe `PaymentsClient` as a class with typed method signatures, you get a Python interface that any IDE can navigate, any type checker can verify, and any test can mock — without ever importing `httpx` in the code that uses it.

Two decorators define that contract:

Decorator	Resilience built in	Use for
<code>@service_client</code>	Circuit breaker + retry	Production service-to-service calls
<code>@http_client</code>	None	Lightweight clients, testing, internal tooling

Use `@service_client` whenever the target is another microservice. Reserve `@http_client` for internal utilities and test doubles.

Defining the Payments client

The Payments service exposes two endpoints: one to create a payment instruction and one to retrieve a payment by identifier. Defining the client means writing the class. Let us build it one decision at a time.

Step 1 — Create the file. Add `src/lumen/sdk/payments_client.py` alongside the existing `client.py`. The `sdk` package is where Lumen keeps the code that *talks to* services, so this is the natural home for a service client.

Step 2 — Decorate the class. Put `@service_client(base_url=...)` on a plain class. That single decorator is what turns an ordinary class into a PyFly-managed HTTP client: it records the base URL, switches on the resilience features, and registers the class as a bean so the container can inject it later.

Step 3 — Declare one method per endpoint. Each method gets a verb decorator (`@post`, `@get`, `@patch`, `@delete`) carrying the path. The method body is just `...` — an *ellipsis*, Python's literal for "nothing here yet." You never write the request code; PyFly writes it for you at startup.

The full client looks like this.

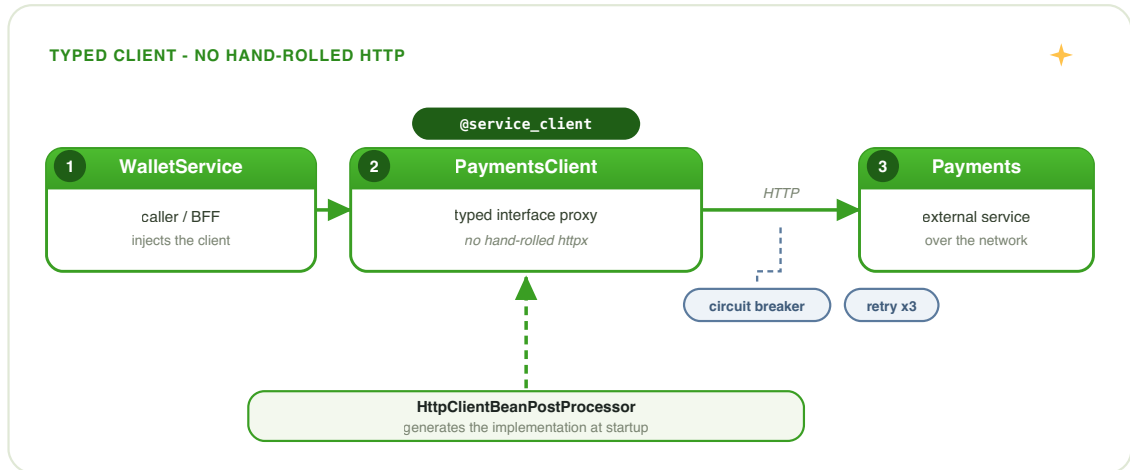


Figure 11.1 — The PyFly declarative client pipeline. You write the interface; `HttpClientBeanPostProcessor` generates the implementation.

lumen/sdk/payments_client.py

```

from __future__ import annotations

from pyfly.client import (
    delete,
    get,
    patch,
    post,
    service_client,
)

@service_client(
    base_url="http://payments-service:8080",
    circuit_breaker=True,
    retry=3,
    circuit_breaker_failure_threshold=5,
    circuit_breaker_recovery_timeout=60.0,
    retry_base_delay=1.0,
)
class PaymentsClient:
    """Typed HTTP client for the Payments service.

    Method stubs are replaced with real HTTP implementations by
    HttpClientBeanPostProcessor at application startup. Declare
    this class as a constructor argument to have it injected.
    """

    @post("/payments")
    async def create_payment(self, body: dict) → dict:
        """POST /payments – submit a payment instruction."""
        ...

    @get("/payments/{payment_id}")
    async def get_payment(self, payment_id: str) → dict:
        """GET /payments/:payment_id – fetch a payment by ID."""
        ...

```

```

@patch("/payments/{payment_id}/cancel")
async def cancel_payment(self, payment_id: str) → dict:
    """PATCH /payments/:payment_id/cancel – cancel pending."""
    ...

@delete("/payments/{payment_id}")
async def delete_payment(self, payment_id: str) → None:
    """DELETE /payments/:payment_id – remove a completed record."""
    ...

```

Listing 11.1 — Typed Payments client with `@service_client`

How it works — the declaration pipeline:

`@service_client(base_url=...)` stamps metadata attributes on the class and registers it as a singleton bean in the PyFly container — the same `__pyfly_injectable__ = True` mechanism that `@service` uses. The `base_url` is stored as `__pyfly_http_base_url__`; the resilience options land in `__pyfly_resilience__`.

The verb decorators — `@post("/payments")`, `@get("/payments/{payment_id}")`, and the others — each attach two attributes to their method: `__pyfly_http_method__` (the HTTP verb string) and `__pyfly_http_path__` (the path template). The method body itself becomes a stub that raises `NotImplementedError` and should never be called directly.

At startup, `HttpClientBeanPostProcessor.after_init()` inspects every bean. When it finds a class with `__pyfly_http_client__ = True`, it creates an `HttpxClientAdapter` for `base_url`, scans every method for `__pyfly_http_method__`, and replaces each stub with a real async implementation. That implementation uses `inspect.signature()` to bind the caller's arguments, interpolates path variables (`{payment_id}` → the actual value), separates remaining

parameters into query strings or a JSON body, and calls `client.request()`. Responses with status ≥ 400 raise typed exceptions; successful responses return `response.json()`.

Path variable interpolation is positional: any parameter whose name matches a `{placeholder}` in the path template is substituted. For `get_payment(self, payment_id: str)`, calling `client.get_payment("pay-123")` sends `GET /payments/pay-123`. For `create_payment(self, body: dict)`, calling `client.create_payment({"amount": 5000})` sends `POST /payments` with the dict serialised as the JSON body. Parameters named `body` on POST/PUT/PATCH methods are always treated as the JSON request body; all other non-path parameters on GET/DELETE become query-string parameters.

WHAT JUST HAPPENED

You wrote four method *signatures* and zero lines of HTTP code. The `@service_client` decorator tagged the class for the container; the verb decorators tagged each method with a verb and a path. At startup, a behind-the-scenes component called `HttpClientBeanPostProcessor` reads those tags and quietly replaces every `...` stub with a working async method that builds the URL, sends the request, and turns the response back into Python. From the caller's point of view, `await client.get_payment("pay-123")` looks exactly like calling a local method — the network is hidden.

Run it — confirm the stubs are wired. Until the application context starts, those `...` bodies raise `NotImplementedError` on purpose, so you cannot just call the method in a bare script. The honest way to prove the client works is a tiny test that starts a context (or wires the post-processor) and inspects the generated method. The quickest smoke check is to confirm the metadata the decorators stamped on:

```
uv run python -c "from lumen.sdk.payments_client import PaymentsClient; \
print(PaymentsClient.__pyfly_http_base_url__); \
print(PaymentsClient.create_payment.__pyfly_http_method__, \
PaymentsClient.create_payment.__pyfly_http_path__)"
```

Expected output:

```
http://payments-service:8080
POST /payments
```

If you see the base URL and `POST /payments`, the decorators applied correctly and the post-processor has everything it needs to generate the real implementation when the app boots.

🍃 SPRING PARITY

`@service_client` with `@get` / `@post` / `@put` / `@delete` / `@patch` is PyFly's counterpart of Spring Cloud OpenFeign's `@FeignClient` with `@GetMapping` / `@PostMapping` etc. In Feign you annotate an interface; in PyFly you annotate a class with stub methods — the intent is identical. Both frameworks generate the HTTP implementation at startup time, inject the bean through the DI container, and support circuit breakers (Feign via Resilience4j; PyFly via the built-in `CircuitBreaker`). The key difference is that Feign works on Java interfaces while PyFly works on ordinary Python classes, which means you can add helper methods alongside the stub methods — useful for response-shaping logic that belongs inside the client class itself.

Injecting the client into a handler

Because `PaymentsClient` is a singleton bean, any `@service` or `@command_handler` can declare it as a constructor argument. PyFly's container injects it through the same autowiring path used for repositories and domain services.

i BEAN AND AUTOWIRING, BRIEFLY

A **bean** is just an object the framework creates and manages for you — you never call its constructor yourself. **Autowiring** is how the container hands a bean to whatever needs it: when a class lists `payments: PaymentsClient` in its `__init__`, PyFly notices the type, finds the matching bean, and passes it in automatically. You declare the dependency by *type*; the container does the lookup.

Lumen's Wallet service already applies the `WalletRepository` and `Money` value object pattern from earlier chapters. When the wallet must call Payments to settle a withdrawal, the handler follows that same pattern in three steps: load the wallet, withdraw through the aggregate, then call the external service.

Step 1 — Declare the dependency. Add `payments: PaymentsClient` to the handler's constructor and stash it on `self`. That is the only wiring you write; the container supplies the live client.

Step 2 — Do the local work first. Load the wallet and call `wallet.withdraw(...)`, then persist it, so the wallet's own state is settled before any network call happens.

Step 3 — Call the remote service. `await self._payments.create_payment(...)` reads like a local method call. The resilience and HTTP details are already baked into the injected client.

Here is the handler.

lumen/core/services/wallets/settle_transfer_handler.py

```

from __future__ import annotations

from lumen.core.services.wallets.settle_transfer_command import (
    SettleTransfer,
)
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository
from lumen.sdk.payments_client import PaymentsClient
from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.domain import AggregateNotFound

@command_handler
@service
class SettleTransferHandler(CommandHandler[SettleTransfer, dict]):
    """Withdraw from the wallet and submit a payment instruction."""

    def __init__(
        self,
        repository: WalletRepository,
        payments: PaymentsClient,
    ) → None:
        super().__init__()
        self._repository = repository
        self._payments = payments

    async def do_handle(self, command: SettleTransfer) → dict:
        wallet = await self._repository.find(command.wallet_id)
        if wallet is None:
            raise AggregateNotFound("Wallet", command.wallet_id)

        wallet.withdraw(
            Money(amount=command.amount, currency=wallet.currency)
        )
        await self._repository.add(wallet)

```



```

    payment = await self._payments.create_payment({
        "wallet_id": command.wallet_id,
        "amount": command.amount,
        "currency": wallet.currency.value,
        "reference": command.reference,
    })
    return payment

```

Listing 11.2 — *CommandHandler injecting PaymentsClient*

How it works — the injection path:

`payments: PaymentsClient` in the constructor is resolved by the container at startup. `HttpClientBeanPostProcessor` wires `PaymentsClient` before `SettleTransferHandler` is instantiated, so the injected bean is fully operational. The handler calls `await self._payments.create_payment(...)` exactly as if it were a local async method. Connection pooling, header propagation, and error mapping are all invisible to the handler.

`wallet.withdraw(Money(...))` runs before the network call, so the wallet state is committed before Payments is contacted. If Payments is temporarily unavailable, the retry and circuit breaker — described in the next section — handle recovery transparently, without any code in the handler.

Run it — exercise the handler with a fake client. Because the handler depends on the type `PaymentsClient`, you can substitute a stand-in in a test without any network. Drop this into `tests/test_settle_transfer.py` and run it:

```
uv run --extra dev pytest tests/test_settle_transfer.py -q
```

A passing run prints something like:

```
1 passed in 0.12s
```

The point of the test is the substitution: you pass a hand-made object in place of the real `PaymentsClient`, assert the handler called `create_payment` with the expected body, and never open a socket. That is the practical payoff of declaring the dependency by type — the handler neither knows nor cares whether the client on the other end is real or faked.

Resilience on the wire

Why the client layer is the right place for resilience

Resilience logic inside a handler mixes business concerns with infrastructure plumbing. A handler that catches `httpx.ConnectError` and implements its own backoff loop is doing two things at once: settling a transfer *and* managing HTTP failure modes. Those responsibilities belong in separate layers.

`@service_client` moves the **circuit breaker** and **retry policy** to the client layer, where they belong. You configure them once on the decorator, and every method on the client inherits them uniformly. The handler code stays focused on the business operation.

Circuit breaker

A circuit breaker monitors every call to the remote service. When `failure_threshold` consecutive calls fail, the circuit **opens**: subsequent calls are rejected immediately with `CircuitBreakerException` rather than waiting for a timeout. This prevents a single slow or unavailable service from blocking the event loop and exhausting connection pools across every caller.

After `circuit_breaker_recovery_timeout` seconds, the circuit enters **half-open**: one probe request is admitted. If it succeeds, the circuit closes and normal operation resumes. If it fails, the circuit re-opens and the recovery timer resets.

`@service_client` wires the breaker automatically. If you need it standalone:

`lumen/sdk/standalone_breaker.py`

```
from __future__ import annotations

from datetime import timedelta

from pyfly.client import CircuitBreaker
from pyfly.kernel.exceptions import CircuitBreakerException

breaker = CircuitBreaker(
    failure_threshold=3,
    recovery_timeout=timedelta(seconds=30),
)

async def call_with_breaker(client, payment_id: str) → dict:
    """Fetch a payment through a standalone circuit breaker."""
    try:
        return await breaker.call(
            client.get_payment,
            payment_id,
        )
    except CircuitBreakerException:
        return {"status": "unavailable", "payment_id": payment_id}
```

Listing 11.3 — Using `CircuitBreaker` standalone

How it works: `CircuitBreaker.__init__` accepts `failure_threshold` (default 5) and `recovery_timeout` as a `timedelta` (default 30 s). `breaker.call(func, *args)` executes `func(*args)` inside the breaker: on success it resets the failure count; on failure it increments the count and flips the state to `OPEN` once the threshold is reached. The state transitions `CLOSED → HALF_OPEN` are computed lazily with `time.monotonic()` — there is no background timer.

`CircuitBreakerException` is never counted as a failure. It signals that the circuit is already open, so re-raising it without recording another failure prevents the recovery timeout from resetting indefinitely.

❗ THREE STATES, IN PLAIN LANGUAGE

Think of the breaker as a light switch with three positions. **Closed** is normal — calls flow through. **Open** means "stop trying" — calls fail instantly without touching the network, which spares your service from waiting on something that is clearly down. **Half-open** is "let me test the water" — after the recovery timeout, the breaker lets exactly one call through; if it works, the switch goes back to closed, and if it fails, it snaps open again.

Run it — watch a breaker open and recover. You can drive the breaker by hand from a REPL with no real service involved. Start one with `uv run python` from `samples/lumen` and try:

```

uv run python -c "
import asyncio
from datetime import timedelta
from pyfly.client import CircuitBreaker, CircuitState
from pyfly.kernel.exceptions import CircuitBreakerException

async def boom():
    raise RuntimeError('payments down')

async def main():
    cb = CircuitBreaker(failure_threshold=2,
recovery_timeout=timedelta(seconds=30))
    for _ in range(2):
        try:
            await cb.call(boom)
        except RuntimeError:
            pass
    print('state after 2 failures:', cb.state.name)
    try:
        await cb.call(boom)
    except CircuitBreakerException:
        print('open circuit rejected the call without calling boom')

asyncio.run(main())
"

```

Expected output:

```

state after 2 failures: OPEN
open circuit rejected the call without calling boom

```

The second line is the whole point: once the circuit is open, the breaker raises `CircuitBreakerException` *immediately* instead of running `boom` again. Drop `recovery_timeout` to `timedelta(seconds=0)` and re-run — the next read of `cb.state` reports `HALF_OPEN` and the third call admits a probe (so `boom` runs again) rather than being rejected. That is the breaker letting the service prove it has recovered.

Retry policy

Transient failures — a momentary latency spike, a rolling restart, a brief connection reset — do not need a circuit breaker; they need a second attempt. `RetryPolicy` provides exponential backoff with configurable exception filtering:

`lumen/sdk/standalone_retry.py`

```
from __future__ import annotations

from datetime import timedelta

from pyfly.client import RetryPolicy

policy = RetryPolicy(
    max_attempts=3,
    base_delay=timedelta(milliseconds=500),
    retry_on=(ConnectionError, TimeoutError),
)

async def resilient_fetch(client, payment_id: str) → dict:
    """Fetch a payment with retry on transient network errors."""
    return await policy.execute(
        client.get_payment,
        payment_id,
    )
```

Listing 11.4 — Using `RetryPolicy` standalone

How it works: `RetryPolicy.__init__` accepts `max_attempts` (default 3, counting the first attempt), `base_delay` (default 1 s), and `retry_on` — a tuple of exception types. The backoff formula is `base_delay * (2 ** attempt)`: for `base_delay=0.5 s`, the

delays are 0.5 s, 1 s, 2 s. Only exceptions matching `retry_on` trigger a retry; others propagate immediately. This matters: you do not want to retry a 404 (the resource does not exist) or a 422 (the request is semantically invalid).

BACKOFF, IN PLAIN LANGUAGE

Backoff means waiting a little longer before each retry instead of hammering the remote service the instant it fails. *Exponential* backoff doubles the wait each time, so a service that needs a moment to recover gets more breathing room with every attempt while a healthy one is retried almost immediately.

Run it — see retry recover from a transient error. Simulate a call that fails twice and then succeeds:

```
uv run python -c "
import asyncio
from datetime import timedelta
from pyfly.client import RetryPolicy

attempts = {'n': 0}
async def flaky():
    attempts['n'] += 1
    if attempts['n'] < 3:
        raise ConnectionError('reset')
    return 'ok'

async def main():
    policy = RetryPolicy(max_attempts=3, base_delay=timedelta(milliseconds=1),
                        retry_on=(ConnectionError,))
    print('result:', await policy.execute(flaky))
    print('attempts:', attempts['n'])

asyncio.run(main())
"
```

Expected output:

```
result: ok
attempts: 3
```

Three attempts, one success. Change `retry_on` to `(TimeoutError,)` and the very first `ConnectionError` propagates instead — proof that only the exceptions you list are retried.

When `@service_client` enables both features, the post-processor wraps them in the correct order: circuit breaker *outside*, retry *inside*. A single logical call attempts up to `max_attempts` retries before the circuit breaker records one failure. An open circuit rejects the call immediately, bypassing the retry loop entirely.

Typed error exceptions

When the remote service returns a 4xx or 5xx response, the generated method raises a typed exception instead of returning the error payload as if it were a success. The exception hierarchy lives in `pyfly.client.exceptions` — import the classes you want to catch from there (for example, `from pyfly.client.exceptions import ServiceNotFoundException`):

Status	Exception class	retryable
400	<code>ServiceValidationException</code>	False
401 / 403	<code>ServiceAuthenticationException</code>	False
404	<code>ServiceNotFoundException</code>	False
409	<code>ServiceConflictException</code>	False
422	<code>ServiceUnprocessableEntityException</code>	False
429	<code>ServiceRateLimitException</code>	True
5xx	<code>ServiceUnavailableException</code>	True

All exceptions extend `ServiceClientException` (itself an `InfrastructureException`). The `retryable` flag on `ServiceRateLimitException` and `ServiceUnavailableException` tells the post-processor which exceptions to pass to the retry policy. 4xx validation errors and 404s are never retried.

WHAT JUST HAPPENED

The three resilience pieces fit together like nested boxes. The **typed exceptions** classify *what kind* of failure occurred — a 404 is not retryable, a 503 is. The **retry policy** uses that classification to decide whether to try again. The **circuit breaker** sits outside the retry loop and counts sustained failures so it can stop calling a service that is genuinely down. You did not write any of this glue: `@service_client` assembled it the moment you set `circuit_breaker=True` and `retry=3`.

Configuring defaults in `pyfly.yaml`

Per-service overrides on `@service_client` always take precedence. Setting process-wide defaults in `pyfly.yaml` lets new clients inherit sensible values without repeating them on every decorator:

pyfly.yaml

```

pyfly:
  client:
    timeout: 10
    retry:
      max-attempts: 3
      base-delay: 1.0
    circuit-breaker:
      failure-threshold: 5
      recovery-timeout: 30

```

Listing 11.5 — Client resilience defaults in pyfly.yaml

Key	Description	Default
<code>pyfly.client.timeout</code>	Request timeout in seconds	<code>30</code>
<code>pyfly.client.retry.max-attempts</code>	Total attempts including first	<code>3</code>
<code>pyfly.client.retry.base-delay</code>	Base delay in seconds	<code>1.0</code>
<code>pyfly.client.circuit-breaker.failure-threshold</code>	Consecutive failures to open	<code>5</code>
<code>pyfly.client.circuit-breaker.recovery-timeout</code>	Seconds before probing	<code>30</code>

`ClientAutoConfiguration` reads `pyfly.client.timeout` at startup and passes it to `HttpClientAdapter`. The `retry` and `circuit-breaker` sub-maps are forwarded as `default_retry` and `default_circuit_breaker` to `HttpClientBeanPostProcessor`.

Any value set directly on

`@service_client(circuit_breaker_failure_threshold=...)` overrides the default.

❗ PRECEDENCE, IN PLAIN LANGUAGE

Two layers can set these knobs: the global `pyfly.yaml` defaults and the per-client decorator arguments. The decorator always wins. Think of `pyfly.yaml` as the house style every new client inherits, and the decorator as the place to override that style for one demanding upstream.

Run it — confirm the config is read. After adding the block to `pyfly.yaml`, read the values back through PyFly's `Config` to be sure the keys are spelled correctly (a common cause of "my timeout is being ignored" is a typo). Run this from the project root, where `pyfly.yaml` lives:

```
uv run python -c "
from pyfly.core.config import Config
cfg = Config.from_sources('.')
print('timeout:', cfg.get('pyfly.client.timeout'))
print('cb:', cfg.get('pyfly.client.circuit-breaker'))
"
```

Expected output once Listing 11.5 is in place:

```
timeout: 10
cb: {'failure-threshold': 5, 'recovery-timeout': 30}
```

Before you add the block, `timeout` reports `30` — the framework default from `pyfly-defaults.yaml` — which confirms the override is what changed it. If a key comes back as `None`, check the indentation in `pyfly.yaml`: YAML nesting is whitespace-sensitive, and a misaligned `circuit-breaker:` silently lands under the wrong parent.

 SET PER-SERVICE TIMEOUTS LOW

The default `timeout: 30` is conservative. In production, each service should carry a `pyfly.yaml` override tuned to its SLA. A payments call that should complete in 500 ms should have `timeout: 2` — not 30 s — so a slow Payments instance fails fast and the circuit breaker can open before threads pile up.

Auth, discovery, and deduplication

Propagating identity downstream

When the Wallet service calls Payments, it often needs to carry the caller's identity — a JWT or an internal service token — so that Payments can enforce its own authorisation rules. The `headers` parameter is treated specially by the post-processor: when a stub method declares `headers: dict`, the value is forwarded as HTTP request headers, not serialised as a query string.

 JWT, IN PLAIN LANGUAGE

A **JWT** (JSON Web Token) is a signed string that travels in the `Authorization` header and proves who the caller is. When Wallet forwards the caller's JWT to Payments, Payments can re-check it and apply its own rules — the identity is carried across the network boundary rather than re-established from scratch.

To forward headers, you add a single optional parameter. There is no new decorator and no special configuration:

- **Add `headers: dict | None = None` to the method.** The name `headers` is the magic word — the post-processor recognises it and routes its value to HTTP headers instead of the query string.

- **Pass a dict at the call site.** The handler supplies `headers={"Authorization": f"Bearer {token}"}`, and PyFly attaches it to the outgoing request.

`lumen/sdk/payments_client_auth.py`

```
from __future__ import annotations

from pyfly.client import get, post, service_client

@service_client(
    base_url="http://payments-service:8080",
    circuit_breaker=True,
    retry=3,
)
class AuthenticatedPaymentsClient:
    """Payments client that forwards caller identity on each request."""

    @post("/payments")
    async def create_payment(
        self,
        body: dict,
        headers: dict | None = None,
    ) → dict:
        """POST /payments – body is the JSON payload; headers forwarded."""
        ...

    @get("/payments/{payment_id}")
    async def get_payment(
        self,
        payment_id: str,
        headers: dict | None = None,
    ) → dict:
        """GET /payments/:payment_id – headers are forwarded."""
        ...
```

Listing 11.6 — Forwarding auth headers per-call

How it works: The post-processor checks whether a parameter named `headers` is present in the bound arguments and is a `dict`. When both conditions hold, it extracts the value from the query-parameter pool and forwards it as HTTP request headers. The handler passes the incoming `Authorization` header (or a freshly minted service token) as `headers={"Authorization": f"Bearer {token}"}`.

`HttpClientAdapter` also calls `inject_headers(headers)` on every request, propagating the W3C `traceparent` and `tracestate` headers from the current observability context so distributed traces stitch across service boundaries without any application-level work.

SERVICE-TO-SERVICE IDENTITY PATTERNS

For internal services on a trusted network, a shared secret in an `X-Internal-Token` header is the simplest approach. For zero-trust architectures, consider mTLS (mutual TLS at the infrastructure layer) or a service mesh that injects identity certificates. For user-delegated calls, forward the original JWT. Whatever pattern you choose, the `headers` parameter gives you a clean injection point in the declarative client.

Service discovery

When `base_url` is a static string like `http://payments-service:8080`, you rely on DNS-based discovery — a Kubernetes `Service` or a Consul record resolves `payments-service` to the correct cluster IP. This is the recommended starting point and sufficient for most deployments.

For environments that need dynamic URL resolution (multiple environments behind the same client class, feature-flagged routing), supply the URL via configuration instead:

pyfly.yaml

```
pyfly:
  client:
    timeout: 10

services:
  payments:
    base-url: "${PAYMENTS_SERVICE_URL:http://payments-service:8080}"
```

Listing 11.7 — Per-environment base URL in pyfly.yaml

A thin factory bean reads the config key and constructs the post-processor with a custom factory that injects the resolved URL. The client class itself does not change — only the factory does.

Request deduplication

Financial operations must be idempotent at the HTTP layer. If `create_payment` is called, times out, and is retried, Payments must not create two payment records. The standard mechanism is an `Idempotency-Key` header: a stable, caller-chosen identifier — typically the command's UUID — that Payments uses to detect and deduplicate repeated requests.

lumen/core/services/wallets/settle_transfer_idempotent.py

```

from __future__ import annotations

from lumen.core.services.wallets.settle_transfer_command import (
    SettleTransfer,
)
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository
from lumen.sdk.payments_client_auth import AuthenticatedPaymentsClient
from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.domain import AggregateNotFound

@command_handler
@service
class SettleTransferIdempotentHandler(
    CommandHandler[SettleTransfer, dict]
):
    """Withdraw funds and submit payment with idempotency key."""

    def __init__(
        self,
        repository: WalletRepository,
        payments: AuthenticatedPaymentsClient,
    ) → None:
        super().__init__()
        self._repository = repository
        self._payments = payments

    async def do_handle(self, command: SettleTransfer) → dict:
        wallet = await self._repository.find(command.wallet_id)
        if wallet is None:
            raise AggregateNotFound("Wallet", command.wallet_id)

        wallet.withdraw(
            Money(amount=command.amount, currency=wallet.currency)
        )

```



```

await self._repository.add(wallet)

idempotency_key = str(command.transfer_id)
return await self._payments.create_payment(
    body={
        "wallet_id": command.wallet_id,
        "amount": command.amount,
        "currency": wallet.currency.value,
    },
    headers={"Idempotency-Key": idempotency_key},
)

```

Listing 11.8 — Idempotency-Key forwarded via headers parameter

How it works: `command.transfer_id` is the stable identifier for this business operation, determined before the command reaches the handler. If the handler is called again for the same command — from a retry, a re-delivery, or a dead-letter replay — it passes the same `Idempotency-Key`. Payments stores the key alongside the created payment record and returns the existing record when the key has been seen before, rather than creating a second payment. That deduplication is a server-side concern; the client's job is simply to forward the key consistently.

The experience tier: the BFF

Why the frontend cannot talk to both services directly

When a mobile app or web frontend needs to display a wallet summary that includes pending payment instructions, it faces a choice: call Wallet for the balance, call Payments for the pending list, and merge the results in the client — or talk to a single API that does the merging server-side. The first option incurs two round trips, exposes

each service's internal shape to the client, and forces the client to implement retry and error handling for two independent failure domains. The second option is the **BFF pattern**.

A **Backend for Frontend** is a lightweight service in the *experience tier* that composes responses from multiple domain services into a shape tailored to a specific frontend's needs. It handles response aggregation, field renaming to match client conventions, and caching of composed results. It never touches the database directly — it depends entirely on domain service clients.

Building the Lumen BFF

The Lumen SDK already ships a `LumenClient` in `lumen/sdk/client.py` that wraps a raw `httpx.AsyncClient` — it takes an `httpx.AsyncClient` in its constructor and calls `self._http.get(...)` / `self._http.post(...)` by hand, raising on error with `response.raise_for_status()`. That is the *imperative* style: useful, explicit, and entirely your responsibility to keep resilient. In the BFF tier you use PyFly's declarative `@service_client` instead — the calling code looks the same, but the circuit breaker and retry are built in automatically.

We will build the BFF in four small pieces, each its own file:

Step 1 — A `WalletClient` that knows how to reach the Wallet service. **Step 2 — A `PaymentsClient`** with one extra endpoint the BFF needs. **Step 3 — A `WalletSummaryService`** that calls both and merges the results. **Step 4 — A thin controller** that exposes the merged view.

Start with the wallet-side client.

lumen_bff/sdk/wallet_client.py

```

from __future__ import annotations

from pyfly.client import get, service_client

@service_client(
    base_url="http://wallet-service:8080",
    circuit_breaker=True,
    retry=3,
)
class WalletClient:
    """Typed HTTP client for the Lumen Wallet service."""

    @get("/api/v1/wallets/{wallet_id}")
    async def get_wallet(self, wallet_id: str) → dict:
        """GET /api/v1/wallets/:wallet_id - fetch a wallet."""
        ...

    @get("/api/v1/wallets/{wallet_id}/balance")
    async def get_balance(self, wallet_id: str) → dict:
        """GET /api/v1/wallets/:wallet_id/balance - current balance."""
        ...

```

Listing 11.9 — WalletClient for the BFF tier

The paths mirror the real Lumen controller — `@request_mapping("/api/v1/wallets")` with `@get_mapping("/{wallet_id}")` — so the BFF client matches exactly what the Wallet service exposes.

The Payments service needs a `list_pending` endpoint that lets the BFF query pending records by wallet:

lumen_bff/sdk/payments_client_bff.py

```

from __future__ import annotations

from pyfly.client import get, post, service_client

@service_client(
    base_url="http://payments-service:8080",
    circuit_breaker=True,
    retry=3,
)
class PaymentsClient:
    """Typed HTTP client for Payments (BFF edition)."""

    @post("/payments")
    async def create_payment(self, body: dict) → dict:
        """POST /payments – submit a payment instruction."""
        ...

    @get("/payments/{payment_id}")
    async def get_payment(self, payment_id: str) → dict:
        """GET /payments/:payment_id – fetch a payment by ID."""
        ...

    @get("/payments")
    async def list_pending(self, wallet_id: str) → list:
        """GET /payments?wallet_id=... – list payments for a wallet."""
        ...

```

Listing 11.10 — Extended PaymentsClient for the BFF

The BFF service then composes the wallet balance with the pending payments list into a single response.

ASYNCIO.GATHER, IN PLAIN LANGUAGE

`asyncio.gather(a, b)` starts coroutines `a` and `b` at the same time and waits for both to finish, returning their results as a list. For a BFF this means two upstream calls overlap instead of queuing one after the other. Adding `return_exceptions=True` changes one thing: instead of the whole `gather` blowing up when one call fails, the failed call's exception is handed back as that call's *result*, so you can inspect it and still use the successful one.

lumen_bff/application/bff_service.py

```

from __future__ import annotations

import asyncio

from lumen_bff.sdk.payments_client_bff import PaymentsClient
from lumen_bff.sdk.wallet_client import WalletClient
from pyfly.container import service

@service
class WalletSummaryService:
    """Composes wallet balance and pending payments into one view.

    Calls both domain services concurrently using asyncio.gather so the
    total latency is max(wallet_latency, payments_latency) rather than
    their sum.
    """

    def __init__(
        self,
        wallet: WalletClient,
        payments: PaymentsClient,
    ) → None:
        self._wallet = wallet
        self._payments = payments

    async def get_summary(self, wallet_id: str) → dict:
        """Return a unified summary for the given wallet."""
        wallet_data, pending = await asyncio.gather(
            self._wallet.get_wallet(wallet_id),
            self._payments.list_pending(wallet_id),
            return_exceptions=True,
        )

        balance_minor: int = 0
        if isinstance(wallet_data, dict):
            balance_minor = wallet_data.get("balance_minor", 0)

```

```

pending_list: list = []
if isinstance(pending, list):
    pending_list = pending

return {
    "wallet_id": wallet_id,
    "balance_minor": balance_minor,
    "pending_payments": pending_list,
}

```

Listing 11.11 — BFF service composing Wallet + Payments

How it works — the composition pattern:

`asyncio.gather(...)` fires both upstream calls concurrently. The wallet and payments calls run in parallel, so the composite latency is bounded by the slower of the two rather than their sum — at 50 ms per service, sequential calls cost 100 ms while concurrent calls cost roughly 55 ms.

`return_exceptions=True` is critical for a BFF. Without it, a single upstream failure raises an exception and the caller receives nothing. With it, a failed coroutine returns its exception object as the result instead of propagating it. The service inspects each result with `isinstance(wallet_data, dict)` and degrades gracefully — returning a partial response with a zero balance or an empty payment list rather than an HTTP 500. The BFF should make that decision explicit in its response shape, for example by including an `"errors"` key listing degraded fields.

The field name `balance_minor` follows Lumen's convention: amounts are stored as integer minor units (cents) and the field is named `balance_minor` throughout — in `WalletDto`, in deposit/withdraw responses, and here in the BFF summary.

Each `@service_client` wrapper on `WalletClient` and `PaymentsClient` handles retries and circuit breaking for its upstream call independently. If Payments is circuit-open, the wallet balance still appears; only the pending payments list is empty.

The BFF controller

The BFF exposes its composed response through a standard PyFly web handler. The controller is intentionally thin — its sole job is to delegate to the service:

`lumen_bff/web/controllers/summary_controller.py`

```
from __future__ import annotations

from lumen_bff.application.bff_service import WalletSummaryService
from pyfly.container import rest_controller
from pyfly.web import get_mapping, request_mapping

@rest_controller
@request_mapping("/api/v1/wallets")
class WalletSummaryController:
    """Experience-tier controller for the wallet summary view."""

    def __init__(self, summary: WalletSummaryService) → None:
        self._summary = summary

    @get_mapping("/{wallet_id}/summary")
    async def get_wallet_summary(self, wallet_id: str) → dict:
        """GET /api/v1/wallets/:wallet_id/summary"""
        return await self._summary.get_summary(wallet_id)
```

Listing 11.12 — BFF controller

How it works: The BFF controller imports no domain models and touches no repositories — it depends only on `WalletSummaryService`, which in turn depends only on typed client interfaces. The dependency chain is controller → BFF service →

declarative clients → remote HTTP. Each layer is independently testable: the controller with a mock service, the service with mock clients, and the clients with a mock

`HttpClientPort`.

Run it — call the composed endpoint. With the BFF application running (`uv run pyfly run --server uvicorn`) and the Wallet and Payments services reachable, hit the summary route with a wallet id you have already created:

```
curl -s http://localhost:8080/api/v1/wallets/wal-123/summary
```

Expected shape (values depend on your data):

```
{"wallet_id": "wal-123", "balance_minor": 5000, "pending_payments": []}
```

The single response merges two upstream services. To see the graceful degradation in action, stop the Payments service and call again — the `balance_minor` still comes back from Wallet, and `pending_payments` falls back to `[]` rather than the whole request returning a 500. That is `return_exceptions=True` doing its job.

DEFAULT APP PORT

PyFly serves the application on `pyfly.server.port`, which defaults to `8080` (matching Spring's `server.port`). Override it with the `PYFLY_SERVER_PORT` environment variable or the `pyfly.server.port` config key. Note that the actuator and admin dashboard run on a *separate* management port — `pyfly.management.server.port`, default `9090` — so health and info endpoints never collide with your API routes.

BFF SCOPE AND TEAM OWNERSHIP

A BFF is scoped to one frontend or one user journey — not one per microservice. Lumen might have a `lumen-mobile-bff` and a `lumen-web-bff`, each composing the same domain services but returning shapes optimised for their respective clients. The BFF is owned by the frontend team, not the domain team. Domain services expose stable contracts; BFFs adapt those contracts to client-specific shapes without coupling the domain services to any particular frontend's conventions.

SPRING PARITY

The BFF pattern in PyFly mirrors the Spring Boot API Gateway / BFF approach where a thin Spring Boot application aggregates responses from multiple microservices. In the reactive Spring stack, `Mono.zip()` provides the same concurrent aggregation that `asyncio.gather()` does in Python. The `@FeignClient` in the BFF corresponds to `@service_client` in PyFly; the Spring `WebClient` approach of chaining `.flatMap()` calls corresponds to PyFly's `asyncio.gather()` + `isinstance` error handling. The team-ownership model — BFF owned by the frontend team, domain services owned by domain teams — is identical.

✓ What you built

You started this chapter with a single Lumen service and finished with an architecture that scales in multiple dimensions. Here is what changed and why it matters.

You extracted `PaymentsService`. Payments now runs in its own process with its own deployment pipeline and its own data store. Wallet handlers know nothing about how Payments stores payment records or which database engine it uses — all they see is the typed interface that `PaymentsClient` exposes.

You declared the client, not the implementation. `@service_client` with `@post`, `@get`, `@patch`, and `@delete` stubs gave you a typed interface that IDEs navigate and type checkers verify. `HttpClientBeanPostProcessor` generated the HTTP implementation at startup, read the resilience configuration from `pyfly.yaml`, and registered the bean for injection anywhere.

You made the network resilient. `circuit_breaker=True` and `retry=3` on the decorator wrapped every method with a shared circuit breaker and retry policy — circuit breaker outside, retry inside — so a sustained Payments outage opens the circuit fast while transient errors recover automatically. Typed exceptions (`ServiceNotFoundException`, `ServiceUnavailableException`) give callers a clean signal without exposing raw HTTP status codes.

You introduced the BFF tier. `WalletSummaryService` composes two upstream calls with `asyncio.gather`, returns a partial response when one service is degraded, and exposes a single contract to the frontend. The BFF absorbs each domain service's independent release cycle and shields the frontend from their internal shapes.

Three principles carry through the rest of Part IV:

- **Depend on the typed client, not on `httpx` directly.** The declaration is your contract; the implementation is a framework detail.
- **Resilience belongs in the client layer.** Configure it once on `@service_client`; every handler that uses the client inherits it.
- **BFFs compose; domain services provide.** Domain services own stable, fine-grained contracts; BFFs own the coarse-grained compositions that specific frontends need.

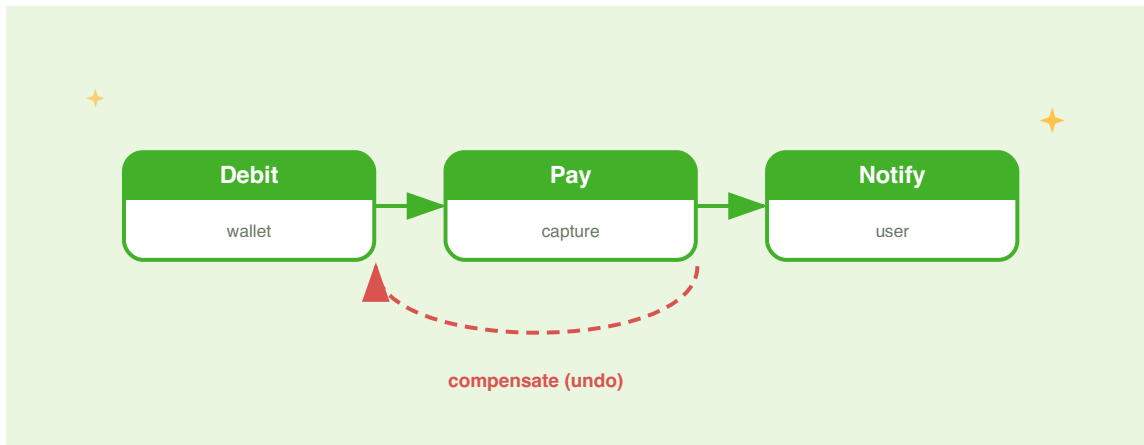


Try it yourself

- 1. Add a fourth endpoint and verify path interpolation.** Extend `PaymentsClient` with a `@get("/payments")` method that accepts `wallet_id: str` and `status: str = "pending"` as parameters. Call it from a test and assert that the generated HTTP request is `GET /payments?wallet_id=abc&status=pending`. Verify that changing the default to `status="completed"` and calling the method without a `status` argument sends `status=completed` in the query string.
- 2. Test the BFF with degraded upstream services.** Write a unit test for `WalletSummaryService.get_summary` that mocks `WalletClient.get_wallet` to succeed and `PaymentsClient.list_pending` to raise `ServiceUnavailableException` (import it with `from pyfly.client.exceptions import ServiceUnavailableException`). Assert that the method returns a dict with the correct `balance_minor` and an empty `pending_payments` list — confirming that the partial-response fallback works and a single upstream failure does not propagate as an exception to the BFF's caller. Run it with `uv run --extra dev pytest tests/test_wallet_summary.py -q` and look for `1 passed`.
- 3. Tune circuit breaker thresholds for a brittle upstream.** Suppose Payments has a known flakiness window during its nightly batch run: it returns 503 on roughly 20 % of requests for about 10 seconds before stabilising. Configure `PaymentsClient` with `circuit_breaker_failure_threshold=2` and `circuit_breaker_recovery_timeout=15.0` and write a test using a mock `HttpClientPort` that simulates two consecutive failures followed by success. Assert that the third call (the probe after the recovery timeout) succeeds and that the circuit transitions back to `CLOSED`.

CHAPTER 12

Distributed Transactions: Sagas, Workflows & TCC



Chapter 10 sent Lumen's wallet events across process boundaries through Kafka. Chapter 11 split the application into co-operating services and showed how to call them over HTTP. Both steps unlocked scale and team autonomy, but they also exposed a new class of danger: multiple aggregates — or multiple services — may all need to change state as part of one business operation, with no distributed ACID transaction to protect you.

Picture a Lumen wallet transfer. You debit the source wallet, then credit the destination. If the source is debited and the credit fails — wrong currency, missing wallet — the source owner loses money with nothing deposited on the other side. You cannot wrap two independent repository calls in a single `BEGIN ... COMMIT` when each aggregate owns its own consistency boundary, and two-phase commit across independent aggregates is operationally fragile.

The answer is **eventual consistency with explicit compensation**. Each step commits to its own store independently, and you design a recovery path — a **compensating transaction** — for every step that could succeed before a later one fails. When the whole sequence succeeds you have your business result; when any step fails, the engine walks back the completed steps in reverse, calling each compensation to restore consistent state. This chapter shows you how to build that with PyFly's `pyfly.transactional` module.

You will model the money transfer as an **orchestrated saga** — a central class that declares each step and its compensation, organised in a DAG (directed acyclic graph) so the engine can run independent steps in parallel. You will then explore compensation in depth, the **Workflow** pattern for long-running or human-in-the-loop flows, and **TCC (Try-Confirm-Cancel)** as a reservation-based alternative. A closing section shows how pluggable persistence lets the engine survive a process crash and automatically resume stale executions.

The problem with distributed writes

Before writing any code, make the failure modes concrete.

Two aggregates, no safety net

Lumen's wallet transfer operates on two `Wallet` aggregates stored in the same PostgreSQL schema but treated as independent domain objects — each is loaded, mutated, and saved in its own round trip. The two steps are:

1. **Debit the source** — withdraw `amount` from the source `Wallet` (enforces `balance ≥ 0`).

2. Credit the destination — deposit `amount` into the destination `Wallet` (enforces currency match).

In a monolith, both writes could share one database transaction. In the Lumen domain service each step is an independent repository call. A currency mismatch on the destination, or a missing wallet ID, causes step 2 to fail after step 1 has already committed — leaving the source wallet debited and the destination unchanged. The user loses money.

Retrying the whole operation is unsafe: you might debit the source twice. Silently skipping the failed step leaves balances inconsistent. You need a principled pattern that commits each step independently and rolls back every completed step consistently on failure.

Eventual consistency and compensation

A **saga** decomposes the operation into a sequence of local transactions, each committing to its own store independently. When a step fails, the engine runs **compensating transactions** in reverse order for every completed step.

Compensations are not database rollbacks; they are *semantic undos* — new forward operations that reverse the effect. "Re-credit the source wallet" is a new deposit operation that restores the original balance, not a rollback.

SAGAS ARE EVENTUALLY CONSISTENT

A saga does not give you serializability or isolation. Between the moment the source wallet is debited and the moment the destination wallet is credited, another request could read the source wallet and see a balance that is lower than it will ultimately be. This is the trade-off you accept when you choose to operate across independent aggregates without a distributed lock. Sagas give you *consistency in the end* — either all forward steps committed or all are compensated — not *consistency at every point*.

An orchestrated saga

PyFly's `pyfly.transactional` module provides the `@saga` and `@saga_step` decorators. You declare one class per saga, annotate each method as a step with its compensation, and declare the dependency ordering. The engine discovers the class through the DI container, builds a validated DAG at startup, and drives execution asynchronously.

NEW TERM: ORCHESTRATION

Orchestration means one central component — here, PyFly's `SagaEngine` — decides the order in which steps run and what to do when one fails. The alternative, *choreography*, has each service react to events with no central conductor. This chapter uses orchestration because it makes the recovery path explicit and easy to test: the engine owns the rules, your saga class just declares the steps.

We will build the transfer saga in four moves: turn the engine on, declare the saga class, look at the DAG the engine builds from it, then call the engine from a service. Take them one at a time.

Enabling the engine

The transactional engine is activated by the `@enable_domain_stack` starter decorator on your application class — and that single decorator is all you need. No extra YAML is required. In Lumen:

Add the starter decorator. Open your application class and stack

`@enable_domain_stack` above `@pyfly_application`. A *starter* decorator is PyFly's way of switching on a whole feature area (here, the transactional engine and its DI wiring) without you registering each component by hand — the Spring equivalent is an `@EnableXXX` annotation.

lumen/app.py

```

from pyfly.core import pyfly_application
from pyfly.starters.domain import enable_domain_stack

@enable_domain_stack
@pyfly_application(
    name="lumen",
    scan_packages=[
        "lumen.models.repositories",
        "lumen.core.services.transfers",
        # ... other packages
    ],
)
class LumenApplication:
    pass

```

Listing 12.1 — Enabling the transactional engine via the domain stack

Turning it off, or on under a narrower starter. `@enable_domain_stack` already sets `pyfly.transactional.enabled: true` for you, so the engine is live as soon as the decorator is on the class. The same property is the knob you reach for in two situations:

- **To switch the engine off** under the domain stack, set it to `false` in `application.yaml` — the auto-configuration is gated on the value being exactly `"true"`, so anything else disables it.
- **To switch it on under a narrower starter** such as `@enable_core_stack` (which does *not* include the transactional engine), add the property yourself:

```

pyfly:
  transactional:
    enabled: true

```

❗ RUN IT: CONFIRM THE ENGINE WIRED UP

Start the app on its default port (`pyfly.server.port` is `8080` in v26.6.110) and watch the startup log:

```
uv run pyfly run
```

Among the startup lines you should see the transactional components register, for example:

```
INFO pyfly.starters.domain domain stack enabled: transactional engine
active
INFO pyfly.transactional registered saga 'money-transfer' (2 steps)
INFO pyfly.server Uvicorn running on http://0.0.0.0:8080
```

If you explicitly set `transactional.enabled: false` (or enable only the core stack without adding the property), the saga line never appears and `SagaEngine.execute(...)` later raises `ValueError: Saga 'money-transfer' is not registered`. Seeing the `registered saga` line is your proof the wiring worked.

How it works: `@enable_domain_stack` merges `DOMAIN_STACK_PROPERTIES` into the active config, and that dict already contains `pyfly.transactional.enabled: "true"` — so the decorator both registers *and* activates the engine in one move. The auto-configuration, `TransactionalEngineAutoConfiguration`, is guarded by `@conditional_on_property("pyfly.transactional.enabled", having_value="true")`; because the starter set the value to `"true"`, the condition matches and the auto-configuration wires every engine component — `SagaEngine`, `TccEngine`, `WorkflowEngine`, `SagaRegistry`, `InMemoryPersistenceAdapter`, and `LoggerEventsAdapter` — into the DI container. The `OrchestrationBeanPostProcessor` then scans every bean produced at startup: any bean carrying `__pyfly_saga__` metadata is registered into `SagaRegistry` automatically. You never call `registry.register_from_bean()` in production code.

WHAT JUST HAPPENED

One small change — a single decorator — gave you a fully wired saga engine.

`@enable_domain_stack` both declared the components and turned them on (it sets `pyfly.transactional.enabled: true` for you), and a startup bean post-processor found your saga classes and registered them for you. From here on you only write saga classes and call `SagaEngine.execute(...)`; the plumbing is done.

Declaring the transfer saga

Lumen's wallet transfer is a two-step saga: debit the source wallet, then credit the destination. If the credit fails — wrong currency or missing wallet — the engine compensates by re-crediting the source, returning both balances to their original values.

NEW TERM: COMPENSATION

A *compensation* (or *compensating transaction*) is the undo for a step. It is not a database rollback — by the time you compensate, the original write has already committed to its own store. Instead it is a *new forward operation* that semantically reverses the effect. The undo for "debit the source" is not `ROLLBACK`; it is "deposit the same amount back into the source". Every step that changes state needs a matching compensation.

Build it in three moves. **Step 1** — declare the class and stack the decorators. **Step 2** — write the forward steps and their compensation. **Step 3** — wire the parameters with injection markers. The complete file is below; we then walk each move.

lumen/core/services/transfers/money_transfer_saga.py

```

from __future__ import annotations

from dataclasses import dataclass
from typing import Annotated

from lumen.core.mappers.wallet_mapper import to_aggregate, to_entity
from lumen.core.services.transfers.transfer_request import TransferRequest
from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.domain import AggregateNotFound
from pyfly.transactional.saga.annotations import (
    FromStep,
    Input,
    saga,
    saga_step,
)
from pyfly.transactional.saga.core.context import SagaContext

MONEY_TRANSFER_SAGA = "money-transfer"

@dataclass(frozen=True)
class DebitResult:
    wallet_id: str
    amount: int
    currency: Currency
    balance: int

@saga(name=MONEY_TRANSFER_SAGA)
@service
class MoneyTransferSaga:
    """Debit source wallet, credit destination; compensate on failure."""

    def __init__(self, repository: WalletRepository) → None:

```

```

self._repository = repository

# -- Step 1: debit the source -----

@saga_step(id="debit-source", compensate="recredit_source")
async def debit_source(
    self,
    request: Annotated[TransferRequest, Input()],
    ctx: SagaContext,
) → DebitResult:
    entity = await self._repository.find_by_id(
        request.source_wallet_id
    )
    if entity is None:
        raise AggregateNotFound("Wallet", request.source_wallet_id)
    wallet = to_aggregate(entity)
    wallet.withdraw(
        Money(amount=request.amount, currency=request.currency)
    )
    await self._repository.upsert(to_entity(wallet))
    wallet.clear_events()
    return DebitResult(
        wallet_id=request.source_wallet_id,
        amount=request.amount,
        currency=request.currency,
        balance=wallet.balance.amount,
    )

async def recredit_source(
    self,
    debit: Annotated[DebitResult, FromStep("debit-source")],
) → int:
    """Compensation: put the money back. Receives the forward step's
    result via FromStep – NOT the saga input."""
    entity = await self._repository.find_by_id(debit.wallet_id)
    if entity is None:
        raise AggregateNotFound("Wallet", debit.wallet_id)
    wallet = to_aggregate(entity)
    wallet.deposit(

```

```

        Money(amount=debit.amount, currency=debit.currency)
    )
    await self._repository.upsert(to_entity(wallet))
    wallet.clear_events()
    return wallet.balance.amount

# -- Step 2: credit the destination -----

@saga_step(id="credit-destination", depends_on=["debit-source"])
async def credit_destination(
    self,
    request: Annotated[TransferRequest, Input()],
    ctx: SagaContext,
) → int:
    entity = await self._repository.find_by_id(
        request.destination_wallet_id
    )
    if entity is None:
        raise AggregateNotFound(
            "Wallet", request.destination_wallet_id
        )
    wallet = to_aggregate(entity)
    wallet.deposit(
        Money(amount=request.amount, currency=request.currency)
    )
    await self._repository.upsert(to_entity(wallet))
    wallet.clear_events()
    return wallet.balance.amount

```

Listing 12.2 — MoneyTransferSaga: debit → credit, with compensation

How it works — step by step:

Step 1 — the decorator stack. `@saga(name=MONEY_TRANSFER_SAGA)` stamps `__pyfly_saga__` on the class with the saga name. The decorator only attaches metadata — it does not wrap the class or create a proxy. **The critical requirement** is that `@saga` must be stacked *on top of* `@service`. The `@service` annotation causes

the DI container to instantiate and scan the bean at startup; the

`OrchestrationBeanPostProcessor.after_init()` hook then sees `__pyfly_saga__` on the bean and calls `SagaRegistry.register_from_bean()`. Without `@service`, the class is never scanned and the saga cannot be executed by name.

⚠️ DECORATOR ORDER IS NOT OPTIONAL

Read the stack top-to-bottom: `@saga` is *above* `@service`. Swap them — `@service` above `@saga` — and the bean still registers with DI, but the saga metadata is applied to the already-wrapped object, the post-processor never finds it, and `execute("money-transfer")` fails with `ValueError: Saga 'money-transfer' is not registered`. If you hit that error, check the decorator order first.

Step 2 — the step methods. `@saga_step` attaches `__pyfly_saga_step__` metadata directly to the async method — no wrapper, no proxy — so `inspect.iscoroutinefunction` keeps returning `True` and the engine correctly `await`s the call. The `compensate="recredit_source"` parameter names the *method on the same class* to invoke when rolling back this step. Omitting `depends_on` (or passing `[]`) means the step can run as soon as the engine starts.

Repository interaction — the load-mutate-save cycle. Each step follows the same three-phase pattern, using the framework

```
WalletRepository(Repository[WalletEntity, str]):
```

1. `find_by_id(id)` — loads the raw `WalletEntity` row from the database.
2. `to_aggregate(entity)` — rehydrates the rich `Wallet` aggregate from that row; the aggregate enforces all invariants (`balance ≥ 0`, currency match).
3. Mutate — call `wallet.withdraw(...)` or `wallet.deposit(...)` on the aggregate, letting it raise `BusinessRuleViolation` if an invariant is broken before any write occurs.

4. `upsert(to_entity(wallet))` — flattens the mutated aggregate back to a `WalletEntity` and calls `session.merge + flush`, so the write is visible to subsequent steps in the same `AsyncSession` without committing.

Because saga steps share one `AsyncSession`, `upsert` flushes so each step sees the previous step's write; the surrounding application boundary owns the final commit.

Step 3 — wire the parameters. Parameter injection uses `typing.Annotated` with **marker instances**, not bare classes:

- `Annotated[TransferRequest, Input()]` — `Input()` is an instance (note the parentheses); bare `Input` without `()` does not resolve.
- `Annotated[DebitResult, FromStep("debit-source")]` — reads the result that step `"debit-source"` stored in `SagaContext` when it completed.
- `ctx: SagaContext` — injected by type; no `Annotated` marker needed.

The resolver inspects type hints at runtime via `typing.get_type_hints(func, include_extras=True)`.

Compensation methods do not receive the saga input. `recredit_source` takes `Annotated[DebitResult, FromStep("debit-source")]` — the value the forward step returned — not the `TransferRequest`. It re-loads the entity via `find_by_id`, rehydrates the aggregate, deposits the original amount back, and upserts — the same load-mutate-save cycle as the forward steps. Compensations always read from `ctx.step_results` via `FromStep`, never from the original input.

WHAT JUST HAPPENED

You wrote one class that holds the whole transfer story: a forward step to debit, its compensation to re-credit, and a second forward step to credit. The decorators told the engine *what each method is* (a step, a compensation) and *how they connect* (`compensate=` , `depends_on=`). You did not write any orchestration loop or try/except rollback logic — that is the engine's job. Your code only describes the business operation and its undo.

The step DAG

NEW TERM: DAG

A *DAG* — directed acyclic graph — is a set of steps connected by "must-run-before" arrows, with no cycles (no step can, directly or indirectly, depend on itself). The engine reads your `depends_on` declarations, builds this graph, and sorts it into *layers*: everything in layer 0 has no unmet dependencies and runs first; layer 1 runs once layer 0 finishes; and so on. Steps in the same layer are independent, so the engine runs them at the same time. A cycle would make the layering impossible, so the engine rejects it at startup rather than at run time.

The two steps form a linear chain:

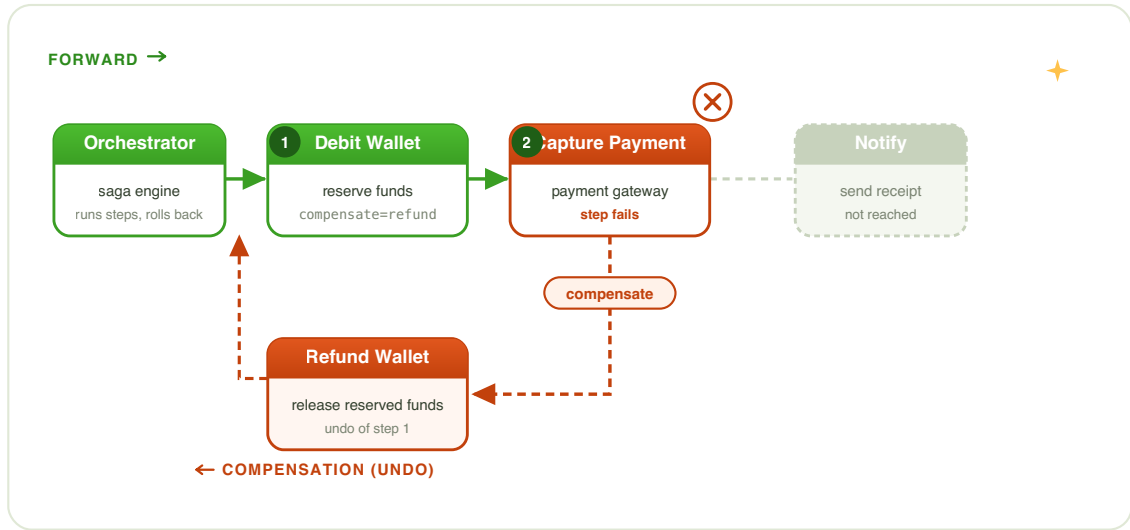


Figure 12.1 — DAG for MoneyTransferSaga: steps run in topological-layer order; independent steps in a layer run with `asyncio.gather`.

```

Layer 0:  debit-source
         |
Layer 1:  credit-destination
    
```

Because `credit-destination` depends on `debit-source`, they run sequentially. A more complex saga — a fraud check and a KYC check that are independent of each other but both feed a capture step — would place the two checks in the same layer and run them concurrently with `asyncio.gather`.

Executing the saga

The saga class only *describes* the operation. To *run* it you need a thin service that injects the engine and calls it by name.

Step 1 — inject `SagaEngine`. Declare a `@service` whose constructor takes a `SagaEngine` parameter; the DI container hands you the auto-configured engine. **Step 2 — call `execute`** with the saga name and the input payload. **Step 3 — fold the `SagaResult`** into a small, JSON-friendly dict for the caller.

lumen/core/services/transfers/transfer_service.py

```

from __future__ import annotations

from typing import Any

from lumen.core.services.transfers.money_transfer_saga import
MONEY_TRANSFER_SAGA
from lumen.core.services.transfers.transfer_request import TransferRequest
from pyfly.container import service
from pyfly.transactional.saga.core.result import SagaResult
from pyfly.transactional.saga.engine.saga_engine import SagaEngine

@service
class TransferService:
    """Run the money-transfer saga and report the outcome."""

    def __init__(self, saga_engine: SagaEngine) → None:
        self._saga_engine = saga_engine

    async def transfer(self, request: TransferRequest) → dict[str, Any]:
        result: SagaResult = await self._saga_engine.execute(
            saga_name=MONEY_TRANSFER_SAGA,
            input_data=request,
        )

        if result.success:
            debit = result.result_of("debit-source")
            return {
                "status": "completed",
                "correlation_id": result.correlation_id,
                "source_balance": debit.balance,
                "destination_balance": result.result_of("credit-destination"),
            }

        return {
            "status": "failed",
            "correlation_id": result.correlation_id,

```

```

    "failed_steps": list(result.failed_steps().keys()),
    "compensated_steps": list(result.compensated_steps().keys()),
    "error": str(result.error),
}

```

Listing 12.3 — Executing the money transfer saga

How it works: `saga_engine.execute()` resolves `MoneyTransferSaga` from the registry by name, creates a `SagaContext` with an auto-generated UUID `correlation_id`, and starts executing layers. On success, `SagaResult.success` is `True` and `result_of("debit-source")` returns the `DebitResult` the forward step produced. On failure, `result.failed_steps()` returns a dict of step ID to `StepOutcome` for every step that exhausted its retries; `result.compensated_steps()` returns the steps that were successfully rolled back.

`SagaResult` is an immutable frozen dataclass. Its key members:

- `result.success` — `True` when every forward step completed.
- `result.result_of(step_id)` — the value that step returned, or `None`.
- `result.failed_steps()` — dict of step ID → `StepOutcome` for failed steps.
- `result.compensated_steps()` — dict of step ID → `StepOutcome` for compensated steps.
- `result.correlation_id` — UUID to correlate logs and traces across services.
- `result.error` — the exception that stopped the saga, or `None` on success.

📄 RUN IT: THE HAPPY PATH AND THE COMPENSATED PATH

Expose `TransferService.transfer` behind an HTTP route (Chapter 11 covered controllers) and exercise both outcomes against the running app on `pyfly.server.port` (`8080`).

A valid transfer between two existing same-currency wallets returns the completed summary:

```
curl -s -X POST http://localhost:8080/transfers \
  -d '{"source_wallet_id":"w-1","destination_wallet_id":"w-2","amount":500,"currency":"EUR"}'
```

```
{
  "status": "completed",
  "correlation_id": "8f3c...",
  "source_balance": 9500,
  "destination_balance": 10500
}
```

Now point the transfer at a destination wallet that does not exist. `credit-destination` raises `AggregateNotFound`, the engine compensates `debit-source`, and the response reports exactly that:

```
curl -s -X POST http://localhost:8080/transfers \
  -d '{"source_wallet_id":"w-1","destination_wallet_id":"does-not-exist","amount":500,"currency":"EUR"}'
```

```
{
  "status": "failed",
  "correlation_id": "1a2b...",
  "failed_steps": ["credit-destination"],
  "compensated_steps": ["debit-source"],
  "error": "Wallet 'does-not-exist' not found"
}
```

The key observation: re-read `w-1` afterwards and its balance is back to `9500` — `debit-source` was rolled back by `recredit_source`. A failed transfer leaves both wallets exactly as they started.

🌱 SPRING PARITY

`@saga / @saga_step` mirror `@Saga / @SagaStep` in the Java `fireflyframework-transactional-engine` library. The decorator-stack rule (`@saga` on `@service`) mirrors the Java rule that `@Saga` must be on a `@Service`-annotated class so the `WorkflowBeanPostProcessor` can discover it. The parameter-injection markers (`Input()`, `FromStep("id")`) map directly to `@Input` and `@FromStep` in the Java version. The async model differs: Java uses Project Reactor (`Mono<T>`) while PyFly uses native `async/await` with `asyncio.gather` for parallel layers.

Compensation in depth

The happy path is straightforward: every step succeeds and the saga commits. The real design challenge is the unhappy path. Understanding what happens on failure — and why compensation must be designed carefully — is what separates a reliable saga from a brittle one.

What runs on failure

When a step fails after all retries, the engine enters *compensation mode*. It inspects `SagaContext` for every step whose status is `DONE`, then calls their compensation methods in reverse completion order under the default `STRICT_SEQUENTIAL` policy. In `MoneyTransferSaga`, a missing destination wallet causes `credit-destination` to raise `AggregateNotFound`. The engine then compensates the step that already completed:

```
Forward path: debit-source ✓ → credit-destination ×  
Compensation: recredit_source (for debit-source)
```

The net effect: the source wallet is restored to its original balance and the destination wallet was never touched — as if the transfer never happened.

Compensation methods receive their arguments through the same injection system as forward steps. `Annotated[DebitResult, FromStep("debit-source")]` reads the `DebitResult` that `debit_source` stored in the context when it completed — so you always compensate with the actual committed data, never an approximation.

❗ RUN IT: PROVE COMPENSATION IN A TEST

You do not need a running server to verify the unhappy path — a fast unit test against the engine is enough, and it is the kind of test you will write for every saga. Drive

`TransferService.transfer` with a destination wallet that does not exist, then assert the compensation ran:

```
async def test_failed_transfer_compensates_the_debit(transfer_service):
    result = await transfer_service.transfer(
        TransferRequest(
            source_wallet_id="w-1",
            destination_wallet_id="does-not-exist",
            amount=500,
            currency=Currency.EUR,
        )
    )
    assert result["status"] == "failed"
    assert result["failed_steps"] == ["credit-destination"]
    assert result["compensated_steps"] == ["debit-source"]
```

Run just this test (the `--extra dev` group installs pytest; Chapter 16 covers fixtures in depth):

```
uv run --extra dev pytest -q -k compensates
```

Expected output:

```
1 passed in 0.05s
```

A green test here is your guarantee that a broken transfer never leaves money missing.

Compensation policies

Five policies govern how the engine runs compensations. Set the global default in YAML or override it per execution:

```

pyfly:
  transactional:
    saga:
      compensation_policy: STRICT_SEQUENTIAL

```

Policy	Behaviour	Use when
STRICT_SEQUENTIAL	Reverse order, one at a time. Stops on first compensation error.	Ordering matters; partial rollback is unacceptable.
GROUPED_PARALLEL	Reverses topology layers; compensates each layer in parallel.	You want speed without violating the dependency structure.
RETRY_WITH_BACKOFF	Reverse order with exponential backoff. Continues if retries succeed.	Transient network failures are likely during compensation.
CIRCUIT_BREAKER	Tracks consecutive failures; opens after 3 and skips the rest.	Avoid cascading failures; manual recovery handles skipped steps.
BEST_EFFORT_PARALLEL	All compensations simultaneously; errors logged, never raised.	Speed critical; separate reconciliation handles partial failures.

⚠ COMPENSATION MUST BE IDEMPOTENT

The engine may call a compensation method more than once. If the engine crashes between calling `void_payment` and persisting the compensation result, it will call `void_payment` again on restart. Your compensation methods must be safe to call multiple times with the same arguments. For payment voids, this means the `PaymentsService` should treat a double-void as a no-op (return success if already voided, do not raise). Design compensation *before* designing the forward step — idempotency is not an afterthought.

Per-step compensation configuration

You can override retry count and timeout for a compensation step without changing forward-step behaviour:

lumen/transfer/transfer_saga_hardened.py

```

from pyfly.container import service
from pyfly.transactional.saga.annotations import saga, saga_step

@saga(name="money-transfer-hardened")
@service
class HardenedTransferSaga:

    @saga_step(
        id="debit-wallet",
        compensate="refund_wallet",
        depends_on=[],
        retry=3,
        backoff_ms=200,
        timeout_ms=5_000,
        compensation_retry=5,
        compensation_backoff_ms=1_000,
        compensation_timeout_ms=8_000,
        compensation_critical=True,
    )
    async def debit_wallet(self, *args: object) → None: ...

    async def refund_wallet(self, *args: object) → None: ...

```

Listing 12.4 — Per-step compensation retry and timeout

How it works: `compensation_retry=5` gives the compensation five attempts of its own, independent of the three forward-step retries. `compensation_critical=True` means that if the compensation exhausts all its retries and still fails, the engine raises that exception — surfacing the *compensation failure* as an observable error rather than silently swallowing it.

External compensation steps

When compensation logic is complex enough to warrant its own class, or when it lives in a different module, move it out entirely:

`lumen/transfer/compensation_steps.py`

```
from typing import Annotated

from pyfly.container import service
from pyfly.transactional.saga.annotations import (
    FromStep,
    compensation_step,
)

from lumen.core.services.transfers.money_transfer_saga import DebitResult
from lumen.models.repositories.wallet_repository import WalletRepository

@compensation_step(saga="money-transfer", for_step_id="debit-source")
@service
class SourceRecreditCompensation:

    def __init__(self, repository: WalletRepository) → None:
        self._repository = repository

    async def execute(
        self,
        debit: Annotated[DebitResult, FromStep("debit-source")],
    ) → None:
        """External alternative to the inline recredit_source method."""
        ...
```

Listing 12.5 — External compensation step class

The `SagaRegistry` discovers `@compensation_step` classes at startup alongside `@saga` classes and wires them into their step definitions automatically. The `for_step_id` parameter must match the step's `id` string exactly.

Workflows and signals

`@saga` is the right tool when all steps are known upfront and the operation completes within minutes. Some business processes are inherently longer — a loan approval waiting on a compliance officer, a multi-step onboarding blocked on an email click, a payment that needs a cooldown before settling. These fit the **Workflow** pattern.

How workflows differ from sagas

	Saga	Workflow
Duration	Seconds to minutes	Minutes to days
Waiting	Retries only	Signals, timers, child workflows
Human-in-the-loop	No	Yes (<code>@wait_for_signal</code>)
State persistence	Per-saga checkpoint	After every layer
DAG fan-out	Yes (parallel layers)	Yes + gate primitives

Declaring a workflow

lumen/transfer/approval_workflow.py

```

from __future__ import annotations

from pyfly.container import service
from pyfly.transactional.core.model import TriggerMode
from pyfly.transactional.workflow.annotations import (
    compensation_step,
    on_workflow_complete,
    on_workflow_error,
    wait_for_signal,
    workflow,
    workflow_query,
    workflow_step,
)

@workflow(
    id="large-transfer-approval",
    trigger_mode=TriggerMode.SYNC,
    timeout_ms=86_400_000,    # 24 hours
    max_retries=1,
)

@service
class LargeTransferWorkflow:
    """High-value transfers require a compliance officer to approve."""

    @workflow_step(id="enrich-request", depends_on=[])
    async def enrich_request(self, payload: dict) → dict:
        return {**payload, "risk_score": 0.12}

    @workflow_step(
        id="compliance-review",
        depends_on=["enrich-request"],
        compensatable=True,
        compensation_method="release_review",
        timeout_ms=82_800_000,
    )
    @wait_for_signal("approved", timeout_ms=82_800_000)

```



```

async def compliance_review(self) → None:
    """Suspends until a compliance officer delivers the signal."""

    @compensation_step(for_step="compliance-review")
    async def release_review(self) → None:
        """Called if the workflow is cancelled during review."""

    @workflow_step(
        id="settle-transfer",
        depends_on=["compliance-review"],
    )
    async def settle_transfer(self, payload: dict) → dict:
        return {"settled": True}

    @workflow_query(name="status")
    async def get_status(self, ctx: object) → str:
        return str(getattr(ctx, "status", "UNKNOWN"))

    @on_workflow_complete
    async def on_done(self, ctx: object) → None:
        pass # emit audit event

    @on_workflow_error
    async def on_error(self, ctx: object, err: Exception) → None:
        pass # alert on-call

```

Listing 12.6 — LargeTransferWorkflow: signal-driven approval for high-value transfers

How it works: The decorator stack follows the same rule as sagas: `@workflow` on top of `@service`. `@workflow(id=...)` takes keyword-only arguments — `id` is required; all others are optional. `@wait_for_signal("approved", timeout_ms=82_800_000)` stacks on top of `@workflow_step` and tells the engine to suspend at that step until a signal named `"approved"` is delivered. The engine persists the `ExecutionContext` to the configured `ExecutionPersistenceProvider`; if the process restarts, it re-hydrates the context and resumes from the last completed layer.

`@compensation_step(for_step="compliance-review")` uses the keyword argument `for_step` (not positional) and registers `release_review` as the compensation handler for the `compliance-review` step.

`@workflow_query(name="status")` marks a method as a read-side query handler — callable while the workflow is suspended without advancing execution.

Driving the workflow engine

lumen/transfer/approval_controller.py

```
from __future__ import annotations

from pyfly.container import service
from pyfly.transactional.workflow.engine import WorkflowEngine
from pyfly.transactional.workflow.result import WorkflowResult

@service
class TransferApprovalService:

    def __init__(self, workflow_engine: WorkflowEngine) → None:
        self._wf = workflow_engine

    async def request_large_transfer(self, payload: dict) → str:
        result: WorkflowResult = await self._wf.start(
            "large-transfer-approval",
            input=payload,
        )
        # Returns immediately; workflow is now suspended at compliance-review.
        return result.correlation_id

    async def approve(self, correlation_id: str, reviewer_id: str) → None:
        await self._wf.deliver_signal(
            correlation_id,
            "approved",
            payload={"by": reviewer_id},
        )

    async def check_status(self, correlation_id: str) → str:
        return await self._wf.query(correlation_id, "status")
```

Listing 12.7 — Starting a workflow and delivering a signal

How it works: `workflow_engine.start(workflow_id, input=payload)` runs the first layer (`enrich-request`) synchronously, then suspends at `compliance-review` because of `@wait_for_signal` . It returns a `WorkflowResult` immediately with a `correlation_id` — the caller stores this ID and polls later. When `deliver_signal()` is called, the workflow resumes and `settle-transfer` runs to completion.

`WorkflowResult` carries: `workflow_id` , `correlation_id` , `status` (an `ExecutionStatus` enum), `duration_ms` , `step_results` (dict), and `variables` . The boolean `result.successful` is `True` when `status` is `COMPLETED` or `CONFIRMED` .

The programmatic builder

When you need to construct a workflow dynamically — from a database configuration or a rules engine — use `WorkflowBuilder` :

lumen/transfer/dynamic_workflow.py

```

from pyfly.transactional.workflow.builder import WorkflowBuilder
from pyfly.transactional.workflow.definition import WorkflowDefinition

async def enrich_fn(payload: dict) → dict:
    return {**payload, "enriched": True}

async def settle_fn(payload: dict) → dict:
    return {"settled": True}

definition: WorkflowDefinition = (
    WorkflowBuilder("simple-transfer")
    .step("enrich", enrich_fn, depends_on=[])
    .wait_signal(
        "await-approval",
        "approved",
        depends_on=["enrich"],
        timeout_ms=3_600_000,
    )
    .step(
        "settle",
        settle_fn,
        depends_on=["await-approval"],
    )
    .build()
)

```

Listing 12.8 — Building a workflow programmatically

`WorkflowBuilder.step(step_id, handler, *, depends_on, timeout_ms, max_retries, ...)` accepts a callable and keyword arguments for dependencies, timeouts, and retries. `wait_signal(step_id, signal, *, depends_on, timeout_ms)`

inserts a signal-gate step without a real handler — it creates an internal no-op coroutine that the engine replaces with signal-wait logic. `build()` returns a `WorkflowDefinition` you register directly with `WorkflowEngine`.

TCC: Try-Confirm-Cancel

The saga pattern runs steps forward and compensates backwards. **TCC (Try-Confirm-Cancel)** takes a different approach: all participants first *tentatively reserve* their resources without committing (Try), then either all *commit* those reservations (Confirm) or all *release* them (Cancel). This gives you strong all-or-nothing semantics across participants without a distributed lock.

TCC suits scenarios where each participant can cheaply hold a reservation — for example, pre-authorising a payment card hold rather than immediately charging it.

The three phases

1. **Try** — every participant reserves resources. Reservations are visible internally but not final. If any Try fails, participants that succeeded cancel their reservations.
2. **Confirm** — if all Try phases succeed, the coordinator instructs every participant to commit its reservation.
3. **Cancel** — if any Try phase fails, the coordinator instructs every participant that completed Try to release its reservation.

Declaring a TCC transaction

lumen/transfer/transfer_tcc.py

```

from __future__ import annotations

from typing import Annotated

from pyfly.container import service
from pyfly.transactional.tcc.annotations import (
    FromTry,
    cancel_method,
    confirm_method,
    tcc,
    tcc_participant,
    try_method,
)
from pyfly.transactional.tcc.core.context import TccContext

from lumen.wallet.service import WalletService
from lumen.payments.service import PaymentsService

@tcc(
    name="wallet-transfer",
    timeout_ms=30_000,
    retry_enabled=True,
    max_retries=3,
    backoff_ms=500,
)
@service
class WalletTransferTcc:
    """Reserve funds and payment in lockstep; confirm or cancel together."""

    @tcc_participant(id="wallet-hold", order=1, timeout_ms=5_000)
    class WalletParticipant:

        def __init__(self, wallet_svc: WalletService) → None:
            self._wallet = wallet_svc

        @try_method(timeout_ms=4_000, retry=2, backoff_ms=200)

```



```

    async def try_hold(
        self,
        request: object,
        ctx: TccContext,
    ) → str:
        """Tentatively hold funds – does not debit yet."""
        return await self._wallet.hold_funds(
            wallet_id=getattr(request, "sender_id", ""),
            amount=getattr(request, "amount_cents", 0),
        ) # returns a hold_id

    @confirm_method(timeout_ms=5_000, retry=3)
    async def confirm_hold(
        self,
        hold_id: Annotated[str, FromTry()],
        ctx: TccContext,
    ) → None:
        await self._wallet.commit_hold(hold_id)

    @cancel_method(timeout_ms=3_000, retry=2)
    async def cancel_hold(
        self,
        hold_id: Annotated[str, FromTry()],
    ) → None:
        await self._wallet.release_hold(hold_id)

    @tcc_participant(id="payment-auth", order=2, timeout_ms=8_000)
    class PaymentParticipant:

        def __init__(self, payments_svc: PaymentsService) → None:
            self._payments = payments_svc

    @try_method(timeout_ms=6_000, retry=2, backoff_ms=300)
    async def try_auth(
        self,
        request: object,
        ctx: TccContext,
    ) → str:
        return await self._payments.pre_authorise(

```

```

        amount=getattr(request, "amount_cents", 0),
    ) # returns auth_id

@confirm_method(timeout_ms=8_000, retry=3)
async def confirm_auth(
    self,
    auth_id: Annotated[str, FromTry()],
    ctx: TccContext,
) → None:
    await self._payments.capture_auth(auth_id)

@cancel_method(timeout_ms=4_000, retry=2)
async def cancel_auth(
    self,
    auth_id: Annotated[str, FromTry()],
) → None:
    await self._payments.void_auth(auth_id)

```

Listing 12.9 — *WalletTransferTcc: Try-Confirm-Cancel for payment reservation*

How it works: `@tcc_participant(order=1)` tells the TCC engine to run `WalletParticipant`'s Try phase before `PaymentParticipant`'s — lower `order` means earlier. `FromTry()` is TCC's equivalent of `FromStep`: it injects the value returned by the same participant's `@try_method` into its `@confirm_method` and `@cancel_method`.

The engine runs all participants' Try phases in `order` sequence. If every Try succeeds, it runs all Confirm methods. If any Try fails, it runs Cancel for every participant that completed its Try — again in declared order. An `optional=True` participant that fails its Try does not trigger a global Cancel; its failure is logged and skipped.

Executing a TCC transaction

lumen/transfer/tcc_service.py

```

from __future__ import annotations

from typing import Any

from pyfly.container import service
from pyfly.transactional.tcc.core.result import TccResult
from pyfly.transactional.tcc.engine.tcc_engine import TccEngine

from lumen.core.services.transfers.transfer_request import TransferRequest

@service
class TccTransferService:

    def __init__(self, tcc_engine: TccEngine) → None:
        self._engine = tcc_engine

    async def transfer(self, req: TransferRequest) → dict[str, Any]:
        result: TccResult = await self._engine.execute(
            tcc_name="wallet-transfer",
            input_data=req,
        )

        if result.success:
            hold_id = result.result_of("wallet-hold")
            return {
                "status": "confirmed",
                "hold_id": hold_id,
                "correlation_id": result.correlation_id,
            }

        failed = result.failed_participants()
        return {
            "status": "cancelled",
            "failed": list(failed.keys()),
        }

```

```
"error": str(result.error),
}
```

Listing 12.10 — Executing a TCC transaction

TCC vs Saga: choosing the right pattern

Use this table to choose between the two approaches:

Question	Saga	TCC
Steps run independently?	Yes — each commits locally	No — all Try phases must succeed first
Needs compensation logic?	Yes, per step	No — Cancel handles rollback
Resource reservation needed?	No	Yes — participants hold resources during Try
Best for	Long sequential operations	Short all-or-nothing locks

Persistence: surviving a crash

The engine stores saga and TCC state through the `TransactionalPersistencePort` protocol. The default adapter keeps state in memory — fast for development, but lost on process restart. Production deployments swap in a durable adapter.

How state flows

Every time a step completes — successfully or otherwise — the engine calls:

1. `persistence_port.update_step_status(correlation_id, step_id, status)` — record the step outcome.
2. `persistence_port.mark_completed(correlation_id, successful)` — record the saga's final result.

On startup, `SagaRecoveryService` queries `persistence_port.get_stale(before)` to find executions that started but never completed. For each stale saga still in `IN_FLIGHT` status, it marks the saga `FAILED` and emits lifecycle events so observability systems can alert on-call.

Configuration

```
pyfly:
  transactional:
    saga:
      persistence_enabled: true
      recovery_enabled: true
      recovery_interval_seconds: 60
      stale_threshold_seconds: 600
      cleanup_older_than_hours: 24
```

With `recovery_enabled: true`, the framework runs

`SagaRecoveryService.recover_stale()` on a background task every `recovery_interval_seconds` seconds. Sagas last updated more than `stale_threshold_seconds` seconds ago are considered stuck, marked failed, and surfaced for manual investigation or automatic retry.

Implementing a custom persistence adapter

To persist to a real database, implement `TransactionalPersistencePort` and register your implementation as a `@bean` or `@component`. The auto-configuration detects your bean at startup and uses it in preference to `InMemoryPersistenceAdapter`:

lumen/infra/persistence/saga_postgres_adapter.py

```

from __future__ import annotations

from datetime import datetime
from typing import Any

from pyfly.container import component
from pyfly.transactional.shared.ports.outbound import (
    TransactionalPersistencePort,
)

@component
class SagaPostgresAdapter(TransactionalPersistencePort):

    async def persist_state(self, state: dict[str, Any]) → None:
        # INSERT INTO saga_executions ...
        ...

    async def get_state(
        self, correlation_id: str
    ) → dict[str, Any] | None:
        # SELECT * FROM saga_executions WHERE ...
        ...

    async def update_step_status(
        self,
        correlation_id: str,
        step_id: str,
        status: str,
    ) → None: ...

    async def mark_completed(
        self, correlation_id: str, successful: bool
    ) → None: ...

    async def get_in_flight(self) → list[dict[str, Any]]:
        return []

```



```

async def get_stale(
    self, before: datetime
) → list[dict[str, Any]]:
    return []

async def cleanup(self, older_than: datetime) → int:
    return 0

async def is_healthy(self) → bool:
    return True

```

Listing 12.11 — Skeleton of a PostgreSQL persistence adapter

💡 USE SAGARECOVERYSERVICE IN INTEGRATION TESTS

In tests that simulate a crash, create a `SagaRecoveryService` with an `InMemoryPersistenceAdapter`, run a saga to a midpoint, manually mark it stale, then call `await recovery.recover_stale(stale_threshold_seconds=0)`. Assert that `SagaResult.success` is `False` and the right steps are marked as failed. This gives you confidence in your recovery logic without spinning up a real database.

The programmatic saga builder

When you need to build a saga from dynamic configuration — loading step definitions from a rules database or a configuration file — `SagaBuilder` gives you the full fluent API without any decorators:

lumen/transfer/dynamic_saga.py

```

from __future__ import annotations

from pyfly.transactional.saga.registry.saga_builder import SagaBuilder
from pyfly.transactional.saga.core.result import SagaResult


async def debit_fn(req: object, ctx: object) → str:
    return "debit-ref-001"


async def capture_fn(req: object, ctx: object) → str:
    return "txn-001"


async def refund_fn(result: str) → None:
    pass    # undo debit


saga_def = (
    SagaBuilder("dynamic-transfer")
    .step("debit")
    .handler(debit_fn)
    .compensate(refund_fn)
    .retry(3)
    .backoff_ms(200)
    .timeout_ms(5_000)
    .jitter(enabled=True, factor=0.3)
    .add()
    .step("capture")
    .handler(capture_fn)
    .depends_on("debit")
    .retry(2)
    .backoff_ms(500)
    .add()
    .layer_concurrency(5)

```

```
.build()
)
```

Listing 12.12 — Building a saga programmatically with SagaBuilder

How it works: Each `.step(step_id)` call returns a `StepBuilder`. Chain configuration methods —

`.handler()`, `.compensate()`, `.depends_on()`, `.retry()`, `.backoff_ms()`, `.timeout_ms()`, `.j`

— then call `.add()` to finalise the step and return the parent `SagaBuilder`. `.build()`

runs the same DAG validation as the decorator path: missing handlers, nonexistent

`depends_on` references, and cycles all raise `SagaValidationError` immediately at registration time.

✓ What you built

You began with a concrete problem: a wallet transfer across two independent aggregates cannot use a single database transaction. You declared

`MoneyTransferSaga` by stacking `@saga` on `@service`, causing the `OrchestrationBeanPostProcessor` to register it into the auto-configured `SagaEngine`

at startup. Each step uses `Annotated[T, Input()]` and `Annotated[T, FromStep("step-id")]` marker instances — not bare classes — for parameter

injection; `ctx: SagaContext` is injected by type. The compensation method

`recredit_source` does not receive the saga input; it pulls the forward step's

`DebitResult` via `FromStep("debit-source")`. When `credit-destination` raises

`AggregateNotFound`, the engine automatically runs `recredit_source`, leaving both balances unchanged.

You explored compensation in depth: five policies ranging from strict sequential to best-effort parallel, per-step compensation retries and timeouts, and the non-negotiable requirement that all compensations be idempotent. You saw how `@workflow(id=...)` `@service` and `@wait_for_signal` suspend a long-running workflow until a human delivers a signal, and how `WorkflowResult.successful` reports the final state. You walked through TCC as a reservation-based alternative that locks resources across all participants before committing any. Finally you wired a custom `TransactionalPersistencePort` and configured `SagaRecoveryService` to detect and surface stale executions after a crash.

Key concepts to carry forward:

- `@saga on @service` — the decorator stack that makes a class both a DI bean and a registered saga; `@saga` alone is not enough.
- **Marker instances** — `Input()`, `FromStep("step-id")` must be instances (with parentheses), not bare classes.
- **Compensation via `FromStep`** — compensation methods receive the forward step's result, never the saga input.
- `SagaEngine.execute(saga_name, input_data)` — the single call that returns `SagaResult` with `.success`, `.result_of()`, `.failed_steps()`, `.compensated_steps()`.
- `@workflow(id=...) @service` / `@wait_for_signal` — signal-driven, long-running alternative; check `WorkflowResult.successful` for the final state.
- `@tcc on @service` / `@tcc_participant` — reservation-based coordination; `FromTry()` (instance) injects the try-result into confirm and cancel methods.
- `TransactionalPersistencePort` — implement and register this protocol to give the engine durable state and crash recovery.

Try it yourself

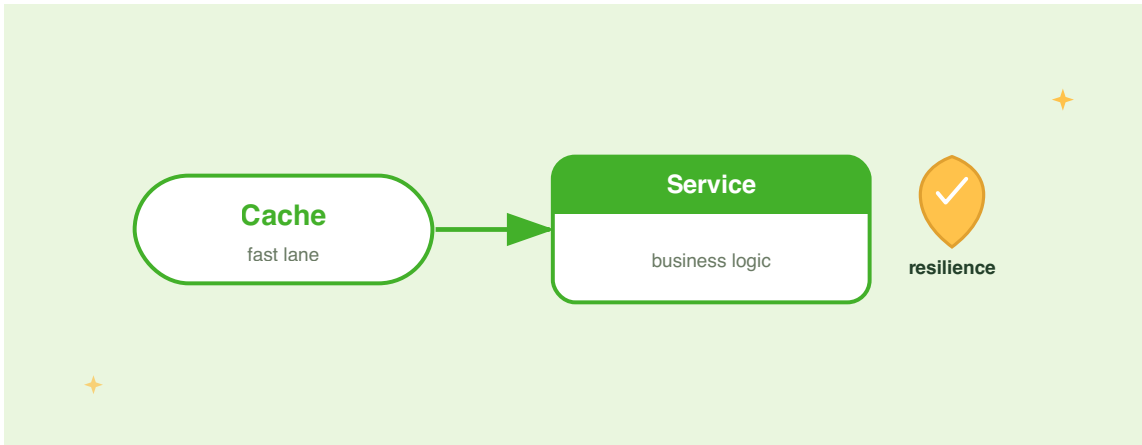
Exercise 1 — Parallel balance validation. Add a `validate-source` step to `MoneyTransferSaga` that checks the source wallet has sufficient funds, without performing the debit. The step should run *in parallel* with nothing (no `depends_on`), and `debit-source` should depend on it. Extend `credit-destination` to depend on `debit-source` as before. Verify the topology via `SagaRegistry.get("money-transfer")` in a test and assert that `definition.steps["debit-source"].depends_on == ["validate-source"]`.

Exercise 2 — Compensation error handler. Change `MoneyTransferSaga`'s global compensation policy to `RETRY_WITH_BACKOFF` in `application.yaml`. Then deliberately make `recredit_source` raise `RuntimeError` on the first call and succeed on the second. Write a pytest test using `AsyncMock` on `WalletRepository` that verifies the saga eventually compensates successfully and `result.compensated_steps()` contains `"debit-source"`.

Exercise 3 — Custom persistence. Implement `TransactionalPersistencePort` backed by a plain Python `dict` that logs every call. Register it as a `@service` and write a test that runs `TransferService`, then calls `get_state(correlation_id)` on your adapter and asserts the recorded `status` is `"COMPLETED"`. Extend the test to simulate a stale saga by manually setting `status = "IN_FLIGHT"` and a past `started_at`, then assert `SagaRecoveryService.recover_stale(stale_threshold_seconds=0)` returns `1`.

CHAPTER 13

Caching & Resilience



In Chapter 11 you split Lumen into separate services and taught its wallet handler to call a downstream `AccountService` over HTTP. In Chapter 12 you added a `DepositSaga` to coordinate multi-step operations across service boundaries, with compensating transactions ready to fire when any step goes wrong.

Those two chapters introduced a new class of problem: latency and failure propagation. Every HTTP hop to `AccountService` is a round trip that could be slow on a busy network, and every call to Lumen's own database competes with concurrent saga participants. In a distributed system, failures are not exceptional events — they are scheduled maintenance. `AccountService` will be upgraded mid-traffic. Redis will hiccup. A payment gateway will spike to three-second response times during peak settlement.

Without protection, Lumen propagates those failures upstream. A slow `AccountService` ties up coroutines, blocking wallet reads for unrelated users. A brief Redis outage wipes cached balances and sends every request straight to the database, multiplying load at exactly the wrong moment.

This chapter makes Lumen **fast** and **fault-tolerant**. The first half covers PyFly's declarative caching layer — `@cacheable`, `@cache_put`, and `@cache_evict` — and shows how to back them with an in-process `InMemoryCache` for development and a shared `RedisCacheAdapter` with automatic failover for production. The second half layers in the resilience toolkit: a token-bucket **rate limiter** that caps inbound traffic, a semaphore **bulkhead** that isolates concurrency, a **time limiter** that cancels hanging coroutines, a **fallback** that degrades gracefully, and **retry** and **circuit-breaker** patterns that protect outbound calls. A closing section shows how to stack all of them in the right order.

By the end of the chapter, every hot path in Lumen will be cached and every outbound dependency wrapped in a resilience fence.

WHAT YOU WILL BUILD, IN PLAIN TERMS

This chapter introduces a lot of vocabulary — *cache*, *token bucket*, *bulkhead*, *circuit breaker*. Do not let the jargon intimidate you. Every one of these is a small, self-contained tool that you bolt onto a function with a single decorator. We will introduce each tool one at a time, build it into Lumen step by step, run it, and watch what changes. By the end you will have a mental checklist: *is this read hot? cache it; is this call going over the network? fence it*. The version of PyFly used throughout is **v26.6.110** — every command and config key below matches that release.

Caching the read path

Why cache wallet reads?

NEW TERM: CACHE

A *cache* is a small, fast holding area where you keep the answer to an expensive question so you can hand it back instantly the next time someone asks. The first time Lumen computes wallet `w-001`'s balance it stores the result in the cache; subsequent reads return that stored copy without re-running the database query. The trade-off — and there is always a trade-off — is that the stored copy can be slightly out of date. The rest of this section is about keeping that staleness inside acceptable bounds.

Lumen's most frequent operation is the balance query: "what is wallet `w-001`'s current balance?" Under normal load that query hits the read replica. Under heavy load it competes with deposit commands, saga participants, and snapshot writes. A cached balance costs one Redis lookup — one co-located network round trip — compared with a full SQL query that the read replica must also parse, plan, and execute.

The economics are compelling, but caching introduces a correctness concern: the cached balance may lag the committed balance by up to the TTL. For Lumen, a five-second stale balance is an acceptable trade-off for normal query traffic. When a deposit completes, the handler invalidates the cache entry immediately, so the next balance read reflects the change. Updates that go through the saga use `@cache_put` to refresh the cached value as a side-effect of the write, eliminating any visible staleness window.

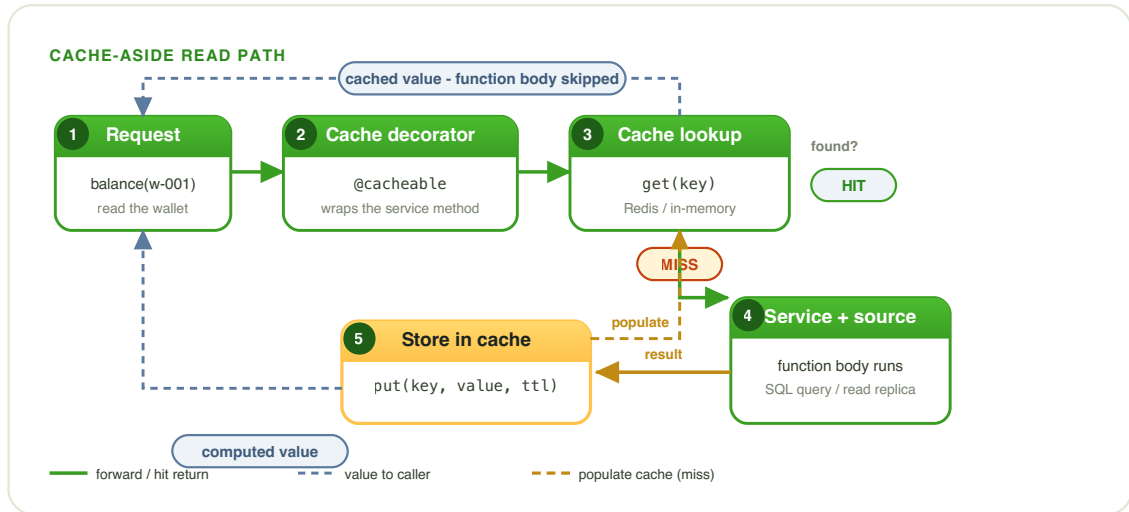


Figure 13.1 — Cache decorators sit in front of the service layer. On a hit the function body never executes; on a miss it runs and the result is stored.

The cache abstraction

PyFly's cache layer follows the hexagonal principle you have seen throughout the book: business logic depends on a `CacheAdapter` protocol, not on any specific backend.

Concrete implementations — `InMemoryCache` for development and `RedisCacheAdapter` for production — are wired in through the DI container. Swapping backends requires no changes to business logic.

The `CacheAdapter` protocol defines the full contract:

Method	Returns	Description
<code>get(key)</code>	<code>Any</code> <code> </code> <code>None</code>	Return the cached value, or <code>None</code> if absent or expired.
<code>put(key, value, ttl=None)</code>	<code>None</code>	Store a value; <code>ttl</code> is a <code>timedelta</code> or <code>None</code> for no expiry.
<code>evict(key)</code>	<code>bool</code>	Remove one key; returns <code>True</code> if it existed.
<code>exists(key)</code>	<code>bool</code>	Check presence without fetching the value.
<code>clear()</code>	<code>None</code>	Flush the entire cache.
<code>start()</code>	<code>None</code>	Called once at application startup.
<code>stop()</code>	<code>None</code>	Called once at application shutdown.

Both `InMemoryCache` and `RedisCacheAdapter` implement this contract.

`InMemoryCache` stores entries in an `OrderedDict` with lazy TTL expiry and optional LRU bounding; it is ideal for single-process development and test suites because it has no external dependencies. `RedisCacheAdapter` wraps a `redis.asyncio.Redis` client, serialises values to JSON before storage, and delegates TTL management to Redis itself — expired keys disappear server-side with zero cleanup overhead on your side.

Setting up a cache backend

We will wire up two backends: an in-process one for development and a shared Redis-backed one for production. Take them one at a time.

Step 1 — Pick a development backend. For development, a single import is all you need:

`lumen/cache/config_dev.py`

```
from pyfly.cache.adapters.memory import InMemoryCache

wallet_cache = InMemoryCache(max_size=1000)
```

Listing 13.1 — *InMemoryCache* for development

`max_size=1000` bounds the LRU eviction window: once the cache holds 1,000 entries, the least-recently-used entry is dropped to make room. Pass `None` (the default) to leave the cache unbounded and rely entirely on TTLs.

i NEW TERM: LRU AND TTL

LRU stands for *least-recently-used* — when the cache is full, the entry nobody has touched for the longest is the one evicted to make room. *TTL* stands for *time-to-live* — how long an entry stays valid before it expires on its own. `InMemoryCache` supports both: `max_size` caps how many entries it holds (LRU); each `put` can carry a TTL that ages the entry out.

Step 2 — Pick a production backend. For production, point `RedisCacheAdapter` at a `redis.asyncio.Redis` client, and wrap it with an in-memory fallback so a Redis hiccup never takes Lumen down:

lumen/cache/config_prod.py

```

import redis.asyncio as aioredis

from pyfly.cache import CacheAdapter, CacheManager
from pyfly.cache.adapters.memory import InMemoryCache
from pyfly.cache.adapters.redis import RedisCacheAdapter
from pyfly.container import bean, configuration

@configuration
class CacheConfig:

    @bean
    def wallet_cache(self) → CacheAdapter:
        client = aioredis.from_url("redis://localhost:6379/0")
        primary = RedisCacheAdapter(client)
        fallback = InMemoryCache(max_size=500)
        return CacheManager(primary=primary, fallback=fallback)

```

Listing 13.2 — RedisCacheAdapter for production

How it works: `CacheManager` wraps a primary Redis backend and an in-memory fallback. Every write goes to both caches, keeping the fallback warm. On reads, the manager tries Redis first; if Redis raises an exception it logs a `WARNING` and falls back to the in-process store silently. When Redis recovers, new writes immediately repopulate it — no manual intervention required. The `@bean` method tells PyFly's DI container to create a singleton and inject it wherever `CacheAdapter` is declared as a dependency.

What just happened. You now have one `CacheAdapter` interface and two ways to satisfy it. In development you hand the DI container an `InMemoryCache`; in production you hand it a `CacheManager` that fronts Redis and quietly falls back to memory when

Redis is unreachable. Every handler in the rest of this chapter asks for `cache:` `CacheAdapter` in its constructor and never knows or cares which one it got — that is the hexagonal payoff.

💡 AUTO-CONFIGURATION

You do not have to write the `@configuration` class at all. Add the following to `pyfly.yaml` and PyFly's `CacheAutoConfiguration` builds a `CacheAdapter` bean for you at startup:

```
pyfly:
  cache:
    enabled: true          # required to switch the subsystem on
    provider: redis       # redis | postgres | memory | auto
    redis:
      url: redis://localhost:6379/0
    max-size: 1000        # used by the memory provider
```

With `provider: redis` (or `auto`, which detects an installed `redis.asyncio`) the auto-config wires a `RedisCacheAdapter` pointed at `pyfly.cache.redis.url`. It registers that single adapter as the `CacheAdapter` bean — it does **not** add the in-memory failover layer. When you want Redis *plus* the transparent in-process fallback shown in Listing 13.2, declare the `CacheManager` yourself in a `@configuration` class as above. The auto-config also backs off entirely if you have already defined your own `CacheAdapter` bean (`@conditional_on_missing_bean`), so the two approaches never collide.

@cacheable — skip execution on a hit

`@cacheable` is the most common decorator. On the first call it executes the function body and stores the return value. On every subsequent call with the same key it returns the stored value *without executing the function body at all*.

Lumen's `GetBalanceHandler` is a natural fit: balance reads are frequent, cheap to cache, and tolerate a few seconds of staleness. We will add caching to it in three small moves.

Step 1 — Accept the cache. Add a `cache: CacheAdapter` parameter to the constructor. PyFly's DI container sees the type and injects whichever backend you wired up in the previous section.

Step 2 — Move the real work into a private method. Rename the body that hits the database to `_fetch`. This is the function the cache will wrap.

Step 3 — Wrap it at construction time. Inside `__init__`, set `self.do_handle = cacheable(...)(self._fetch)`. We wrap inside `__init__` (rather than as a `@cacheable` decorator on the method) for one reason: the `backend=cache` argument only exists once `cache` has been injected, and that does not happen until `__init__` runs.

The handler receives `CacheAdapter` through its constructor — injected by PyFly — and wraps `do_handle` at construction time:

lumen/core/services/wallets/get_balance_handler.py

```

from datetime import timedelta

from lumen.core.mappers.wallet_mapper import entity_to_balance_dto
from lumen.core.services.wallets.get_balance_query import GetBalance
from lumen.interfaces.dtos.v1.balance_dto import BalanceDto
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.cache import CacheAdapter, cacheable
from pyfly.container import service
from pyfly.cqrs import QueryHandler, query_handler

@query_handler
@service
class GetBalanceHandler(QueryHandler[GetBalance, BalanceDto | None]):
    """Return a cached :class:`BalanceDto`; bypass the DB on a hit."""

    def __init__(
        self,
        repository: WalletRepository,
        cache: CacheAdapter,
    ) → None:
        super().__init__()
        self._repository = repository
        # Wrap do_handle at construction time so `cache` is in scope.
        self.do_handle = cacheable(
            backend=cache,
            key="wallet:balance:{query.wallet_id}",
            ttl=timedelta(seconds=5),
        )(self._fetch)

    async def _fetch(
        self, query: GetBalance
    ) → BalanceDto | None:
        entity = await self._repository.find_by_id(query.wallet_id)
        return entity_to_balance_dto(entity) if entity is not None else None

```

Listing 13.3 — @cacheable on GetBalanceHandler

📄 KEY TEMPLATE AND SELF

The `key` template `"wallet:balance:{query.wallet_id}"` uses Python's `str.format` syntax. PyFly binds the actual call arguments with `inspect.signature(func).bind(*args, **kwargs)`, then calls `key.format(**bound.arguments)`. Because `_fetch` is wrapped inside `__init__`, the first positional argument is `query` — so `{query.wallet_id}` expands to the wallet id. Calling with `GetBalance(wallet_id="wlt-001")` produces the cache key `"wallet:balance:wlt-001"`. The mapper function `entity_to_balance_dto` goes through `Mapper.project` against the `@projection`-marked `BalanceView` interface, copying only the fields the balance view declares and computing `balance` from `balance_minor`.

`ttl=timedelta(seconds=5)` means the cache entry expires five seconds after it is written. After expiry, the next call re-executes the function body and refreshes the entry. A TTL of `None` (the default) means the entry never expires — appropriate only for truly immutable data.

Null caching: When the function returns `None`, PyFly still stores the entry and records that the key *exists*. A subsequent call finds the key and returns `None` without touching the database. This prevents cache-penetration attacks where an adversary floods requests for non-existent keys, each of which would otherwise fall through to the database.

condition and unless: Both `@cache` and `@cacheable` accept optional predicates. `condition` is a callable with the same signature as the decorated function; if it returns `False`, caching is bypassed for that call. `unless` is a callable that receives the *result*; if it returns `True`, the result is returned but not stored. Both are keyword-only:


```
cacheable(  
    backend=cache,  
    key="wallet:balance:{query.wallet_id}",  
    ttl=timedelta(seconds=5),  
    condition=lambda query: not query.wallet_id.startswith("test-"),  
    unless=lambda result: result is None,  
) (self._fetch)
```

Run it — prove the second read skips the database

The cleanest way to *see* a cache hit is a unit test that counts how many times the repository is called. Use a real `InMemoryCache` (no Redis needed) and a tiny stub repository:

```
tests/cache/test_get_balance_cache.py
```

```
from datetime import timedelta

import pytest

from pyfly.cache import cacheable
from pyfly.cache.adapters.memory import InMemoryCache

class _CountingRepo:
    """Stub repository that records how many times it is queried."""

    def __init__(self) → None:
        self.calls = 0

    async def find_by_id(self, wallet_id: str) → dict:
        self.calls += 1
        return {"wallet_id": wallet_id, "balance_minor": 500}

@pytest.mark.asyncio
async def test_second_read_is_a_cache_hit() → None:
    repo = _CountingRepo()
    cache = InMemoryCache(max_size=10)

    fetch = cacheable(
        backend=cache,
        key="wallet:balance:{wallet_id}",
        ttl=timedelta(seconds=5),
    )(repo.find_by_id)

    first = await fetch("wlt-001") # miss → runs the repo
    second = await fetch("wlt-001") # hit → repo NOT called again

    assert first == second
    assert repo.calls == 1 # the body ran exactly once
```

Listing 13.3a — A test that proves the second read is a hit

Run just this test:

```
$ uv run --extra dev pytest tests/cache/test_get_balance_cache.py -q
.
[100%]
1 passed in 0.04s
```

The single `.` and `1 passed` confirm it: the second call returned the cached value and `repo.calls` stayed at `1`, so the database was touched exactly once across two reads. That is a cache hit, demonstrated rather than asserted in prose.

What just happened. You wrapped a plain async function with `cacheable`, backed it with an `InMemoryCache`, and confirmed that identical keys short-circuit the body. In `GetBalanceHandler` the wrapped function is `_fetch` and the backend is the injected `CacheAdapter`, but the mechanics are exactly what you just ran.

🍃 SPRING PARITY

`@cacheable` mirrors Spring's `@Cacheable`. The `key` template uses Python's `str.format` syntax instead of SpEL, but the semantics — skip-on-hit, store-on-miss, `condition`, `unless` — are identical. `@cache` is a lower-level alias that behaves the same way; use whichever name reads better in your codebase.

@cache_put — always execute, always store

`@cacheable` is for reads: it short-circuits the function when the cache already holds a value. `@cache_put` is for writes: it *always* executes the function and *always* stores the result. Use it when the function is the source of truth — a command handler that modifies the wallet and must keep the cache current.

`DepositFundsHandler` is the canonical example. After a deposit succeeds, the new balance must be visible to the next read without waiting for the TTL to expire. The wiring mirrors what you did for `@cacheable`, with one critical detail to watch:

Step 1 — Accept the cache in the constructor, exactly as before.

Step 2 — Move the deposit logic into `_deposit` and keep its `@transactional()` decorator so the write still commits as one unit of work.

Step 3 — Wrap with `cache_put`, reusing the *same* key shape. The deposit handler must write to the very cache slot the balance reader looks up. `@cacheable` uses `"wallet:balance:{query.wallet_id}"`; here the argument is named `command`, so the template is `"wallet:balance:{command.wallet_id}"`. Different parameter names, but both resolve to `wallet:balance:wlt-001`.

Wrapping `do_handle` with `@cache_put` refreshes the cache entry atomically with the write:

```
lumen/core/services/wallets/deposit_funds_handler.py
```

```
from datetime import timedelta

from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker

from lumen.core.mappers.wallet_mapper import to_aggregate, to_entity
from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.core.services.wallets.event_publishing import publish_domain_events
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.cache import CacheAdapter, cache_put
from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.data.relational.sqlalchemy import transactional
from pyfly.domain import AggregateNotFound
from pyfly.eda import EventPublisher
```

```
@command_handler
```

```
@service
```

```
class DepositFundsHandler(CommandHandler[DepositFunds, int]):
```

```
    """Credit funds to an existing wallet; returns the new balance
    in minor units and refreshes the cached balance entry."""
```

```
    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
        cache: CacheAdapter,
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events
        self._session_factory = session_factory
        # Wrap at construction time so `cache` is in scope.
        self.do_handle = cache_put(
            backend=cache,
```

```

        key="wallet:balance:{command.wallet_id}",
        ttl=timedelta(seconds=5),
    )(self._deposit)

    @transactional()
    async def _deposit(self, command: DepositFunds) → int:
        entity = await self._repository.find_by_id(command.wallet_id)
        if entity is None:
            raise AggregateNotFound("Wallet", command.wallet_id)

        wallet = to_aggregate(entity)
        wallet.deposit(Money(amount=command.amount, currency=wallet.currency))
        await self._repository.upsert(to_entity(wallet))
        await publish_domain_events(self._events, wallet.clear_events())
        return wallet.balance.amount

```

Listing 13.4 — `@cache_put` refreshes the cache on a deposit

How it works: `@cache_put` awaits the wrapped function, then calls `backend.put(resolved_key, result, ttl=ttl)`. Because the function always runs, the cached value after a `DepositFunds` command is the freshly committed balance — not a stale pre-deposit snapshot. `_deposit` runs inside `@transactional()`, so the `find_by_id → to_aggregate → mutate → upsert` sequence is committed as one unit of work before the cache is refreshed. The next `@cacheable` read in `GetBalanceHandler` picks up this fresh value without touching the database.

❗ CACHE KEY MUST MATCH

The `@cache_put` key `"wallet:balance:{command.wallet_id}"` must match the `@cacheable` key `"wallet:balance:{query.wallet_id}"` when both resolve to the same wallet id. Mismatched keys mean the deposit writes to a different cache slot than the balance read looks up — staleness returns.

Decorator	Function executes?	On hit
<code>@cacheable</code> / <code>@cache</code>	Only on a miss	Returns cached value
<code>@cache_put</code>	Always	Replaces cached value with fresh result

@cache_evict — remove after deletion

When you close a wallet or roll back a transaction, the associated cache entry must be removed. `@cache_evict` runs the function body first, then removes the named key — or clears the entire cache when `all_entries=True`.

lumen/core/services/wallets/close_wallet_handler.py

```

from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.cache import CacheAdapter, cache_evict
from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.data.relational.sqlalchemy import transactional

@command_handler
@service
class CloseWalletHandler(CommandHandler["CloseWallet", None]):
    """Close a wallet and evict its cached balance entry."""

    def __init__(
        self,
        repository: WalletRepository,
        cache: CacheAdapter,
    ) → None:
        super().__init__()
        self._repository = repository
        self.do_handle = cache_evict(
            backend=cache,
            key="wallet:balance:{command.wallet_id}",
        )(self._close)

    @transactional()
    async def _close(self, command) → None:
        entity = await self._repository.find_by_id(command.wallet_id)
        if entity is not None:
            await self._repository.delete(entity)

```

Listing 13.5 — @cache_evict after removing a wallet

To flush every cached balance at once — useful for an administrative reset — pass

```
all_entries=True:
```



```
self.do_handle = cache_evict(
    backend=cache,
    all_entries=True,
)(self._reset_all)
```

How it works: The function body runs first — `repository.delete(entity)` removes the row before eviction, so a failure does not prematurely drop the cache entry. Then either `backend.evict(resolved_key)` removes one key or `backend.clear()` flushes everything. With `CacheManager`, the evict propagates to both primary and fallback caches so no stale entry lingers in either tier.

`all_entries=True` is a blunt instrument reserved for administrative resets. In normal operation, prefer targeted eviction by key.

Invalidation strategy

A coherent strategy matches each operation to the right decorator:

Operation	Decorator	Rationale
<code>GetBalance</code> query	<code>@cacheable</code>	Skip DB on hit; 5 s TTL bounds staleness
<code>DepositFunds</code> command	<code>@cache_put</code>	Refresh the cache entry atomically with the write
<code>WithdrawFunds</code> command	<code>@cache_put</code>	Same — keep the post-withdrawal balance warm
Close wallet	<code>@cache_evict</code>	Remove the entry; the next read rebuilds it from DB
Admin truncate	<code>@cache_evict(all_entries=True)</code>	Bulk reset; full cache flush is correct

⚠️ ASYNC REQUIREMENT

All three decorators require the wrapped function to be declared `async`. Cache adapters are fully async (they `await` backend operations), so a synchronous target will fail with a `TypeError` at decoration time — PyFly raises the error immediately so you catch the mistake at startup rather than at runtime.

Resilience patterns

Why protection matters

📄 NEW TERM: RESILIENCE

Resilience here means the system keeps serving requests it *can* serve even when a dependency it relies on is slow, overloaded, or down. The tools in this section do not make the downstream faster — they stop one sick dependency from making the whole of Lumen sick. Each tool is, again, a decorator you stack on the function that makes the risky call.

Caching makes the happy path fast. Resilience patterns protect Lumen when the happy path is unavailable. Without protection, a slow `AccountService` triggers a cascade:

1. Requests from wallet handlers pile up, each waiting on an HTTP response.
2. Lumen's asyncio event loop — single-threaded by default — processes pending tasks in order; a backlog of slow HTTP calls delays every other operation.
3. Memory and open file-descriptors climb as coroutines stack up.
4. Lumen becomes unavailable to requests that have nothing to do with `AccountService`.

Four complementary patterns break this cascade before it starts:

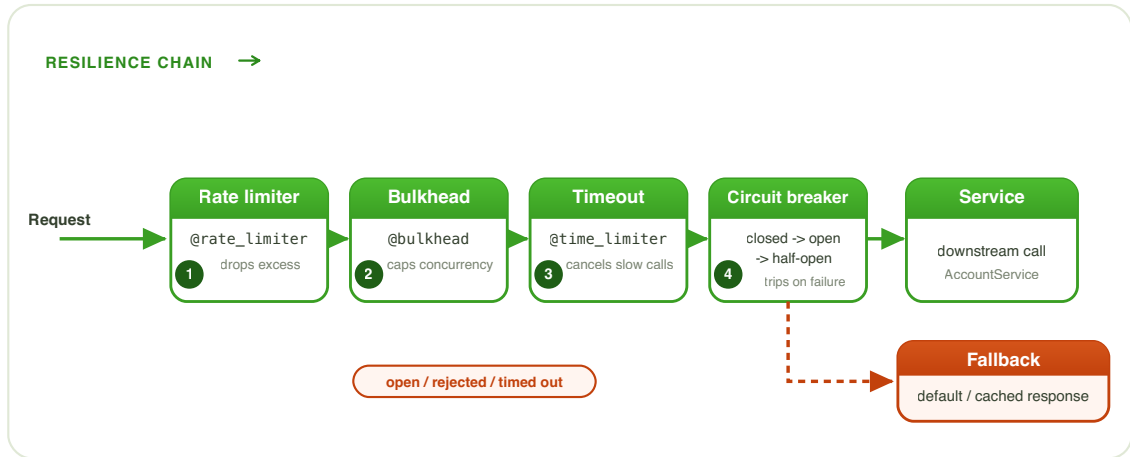


Figure 13.2 — Four resilience layers guard the outbound call. Rate limiter drops excess traffic before it enters the system; bulkhead limits concurrency; timeout cancels slow operations; fallback provides a safe response when all else fails.

Pattern	Protects against	Fail-fast or wait?
Rate limiter	Traffic spikes overwhelming the downstream	Fail-fast (reject excess)
Bulkhead	Too many concurrent calls tying up resources	Fail-fast (reject over limit)
Time limiter	Hanging calls that never return	Cancels after timeout
Fallback	Any failure reaching the caller	Returns degraded value

All four are in `pyfly.resilience`:

```

from pyfly.resilience import (
    RateLimiter, rate_limiter,
    Bulkhead, bulkhead,
    time_limiter,
    fallback,
)

```

Rate limiter — token bucket

`RateLimiter` uses a **token bucket**: the bucket holds up to `max_tokens` tokens and refills at `refill_rate` tokens per second. Each call consumes one token. When the bucket is empty, `RateLimitException` is raised immediately — no queuing, no waiting.

`lumen/resilience/rate_example.py`

```
from pyfly.resilience import RateLimiter, rate_limiter

# Sustained: 20 calls/s; burst: up to 40
account_limiter = RateLimiter(max_tokens=40, refill_rate=20.0)

@rate_limiter(account_limiter)
async def fetch_account(account_id: str) → dict:
    # This body is reached only when a token is available.
    ...
```

Listing 13.6 — Token-bucket rate limiter on account lookups

How it works: `@rate_limiter(limiter)` calls `await limiter.acquire()` before every invocation. `acquire()` refills the bucket based on elapsed wall-clock time (using `time.monotonic()`), then atomically checks and decrements the token count under a `threading.Lock` — not an asyncio lock — so that both async tasks and sync callers share the same count without races. If fewer than 1.0 tokens remain, `RateLimitException` propagates to the caller.

The token-bucket shape allows controlled bursting: a service that typically sees 10 calls per second can absorb a burst of 40 calls immediately (drawing on saved tokens), then sustains 20 calls per second afterwards. Fixed-window rate limiters cannot express this nuance.

Run it — watch the bucket run dry

A small script makes the behaviour concrete. Create a limiter with a tiny bucket, call past its capacity, and observe the rejection:

scratch/rate_demo.py

```
import asyncio

from pyfly.kernel.exceptions import RateLimitException
from pyfly.resilience import RateLimiter, rate_limiter

# 3 tokens, refilling slowly so the burst is what we observe.
limiter = RateLimiter(max_tokens=3, refill_rate=1.0)

@rate_limiter(limiter)
async def ping(n: int) → str:
    return f"ok-{{n}}"

async def main() → None:
    for n in range(5):
        try:
            print(await ping(n))
        except RateLimitException:
            print(f"rejected-{{n}}")

asyncio.run(main())
```

Listing 13.6a — Draining the token bucket on purpose

Run it directly:

```
$ uv run python scratch/rate_demo.py
ok-0
ok-1
ok-2
rejected-3
rejected-4
```

The first three calls each spend a token; the fourth and fifth arrive with an empty bucket and are rejected immediately with `RateLimitException` — no queuing, no waiting. Slow the loop down (or raise `refill_rate`) and the rejections disappear because tokens refill between calls.

What just happened. You did not change the function `ping` at all — you decorated it. The decorator inserted an `await limiter.acquire()` before every call, and `acquire()` raised when the bucket was empty. This is the shape every resilience tool in this chapter takes: a decorator that guards the call without the function body knowing it exists.

Multiple functions sharing one `RateLimiter` instance enforce a *global* rate across all of them — useful for capping total traffic to a downstream service regardless of which internal method initiates the call.

Bulkhead — concurrency isolation

NEW TERM: BULKHEAD

The name comes from shipbuilding: a ship's hull is divided into sealed compartments (*bulkheads*) so that a breach in one does not flood the whole vessel. A software bulkhead caps how many calls to one dependency can run at once, so a flood of slow calls to `AccountService` cannot consume every coroutine and sink unrelated requests.

`Bulkhead` is a semaphore: it limits the number of calls *in-flight at the same time*. Calls beyond `max_concurrent` are rejected immediately with `BulkheadException`.

lumen/resilience/bulkhead_example.py

```

from pyfly.resilience import Bulkhead, bulkhead

# At most 5 concurrent calls to AccountService
account_bulkhead = Bulkhead(max_concurrent=5)

@bulkhead(account_bulkhead)
async def fetch_account(account_id: str) → dict:
    ...

```

Listing 13.7 — Bulkhead limiting concurrent account service calls

How it works: The decorator acquires a permit (`_acquire_slot`) before entering the function and releases it (`_release_slot`) in a `finally` block, so the slot is always returned even when the function raises. Slots are tracked by a single lock-guarded integer counter shared by async and sync call paths, so one `Bulkhead` instance safely decorates a mix of coroutines and regular functions.

This fail-fast behaviour is intentional: when 5 concurrent calls are in-flight and a 6th arrives, rejecting it immediately lets the caller retry or invoke a fallback — far better than queuing it indefinitely and causing cascading backpressure.

💡 MONITORING BULKHEAD UTILIZATION

`account_bulkhead.available_slots` returns the number of free permits at any moment. Expose this in a health endpoint or feed it to your observability stack to detect persistent saturation before it becomes an outage.

Time limiter — enforcing a deadline

A slow downstream is sometimes worse than a crashed one: indefinitely blocking calls consume resources without bound. `@time_limiter` cancels the coroutine if it does not complete within a `timedelta`:

`lumen/resilience/timeout_example.py`

```
from datetime import timedelta

from pyfly.resilience import time_limiter

@time_limiter(timeout=timedelta(seconds=2))
async def fetch_account(account_id: str) → dict:
    ...
```

Listing 13.8 — 2-second deadline on account lookup

How it works: Internally, `time_limiter` calls `asyncio.wait_for(func(*args, **kwargs), timeout=timeout_seconds)`. When the deadline passes, `asyncio.wait_for` cancels the underlying task, causing any `await` inside the function to raise `asyncio.CancelledError`. The decorator catches `TimeoutError` and re-raises it as `OperationTimeoutException` with a descriptive message:

```
OperationTimeoutException: fetch_account exceeded timeout of 2.0s
```

Resources acquired inside the timed function should be guarded with `try/finally` so they are released even on cancellation:


```

@time_limiter(timeout=timedelta(seconds=2))
async def fetch_account(account_id: str) → dict:
    conn = await pool.acquire()
    try:
        return await conn.execute(query, account_id)
    finally:
        await pool.release(conn)

```

Fallback — graceful degradation

`@fallback` is the safety net at the outermost layer: it catches exceptions and returns an alternative response rather than propagating the error to the caller. Lumen's balance summary endpoint can return a degraded response — last known balance, marked as potentially stale — rather than an HTTP 500 when `AccountService` is down.

Two modes are available. The first returns a **static value**:

`lumen/resilience/fallback_static.py`

```

from pyfly.resilience import fallback

@fallback(fallback_value={"balance_minor": 0, "source": "fallback"})
async def fetch_account(account_id: str) → dict:
    ...

```

Listing 13.9 — Static fallback value

The second invokes a **fallback method** that receives the original arguments plus the exception:

lumen/resilience/fallback_method.py

```

from pyfly.cache import CacheAdapter
from pyfly.resilience import fallback

_cache: CacheAdapter # injected elsewhere

async def account_from_cache(
    account_id: str,
    exc: Exception = None,
) → dict:
    cached = await _cache.get(f"account:{account_id}")
    if cached:
        return {**cached, "source": "cache"}
    return {"account_id": account_id, "balance_minor": 0, "source":
"fallback"}

@fallback(fallback_method=account_from_cache)
async def fetch_account(account_id: str) → dict:
    ...

```

Listing 13.10 — Fallback method with cached data

How it works: When the primary function raises one of the exception types listed in `on` (default: all `Exception` subclasses), the decorator calls `fallback_method(*args, exc=exc, **kwargs)`. The `exc` keyword argument carries the caught exception so the fallback can log it, inspect its type, or return different values for different failure modes. If the fallback method returns a coroutine, PyFly awaits it automatically. Narrow the exception filter with `on=(OperationTimeoutException, CircuitBreakerException)` to let programming errors propagate normally.

⚠️ FALLBACK METHOD SIGNATURE

The fallback method must accept `exc` as a keyword argument. PyFly passes the caught exception as `exc=<exception>`. If your fallback method's signature does not include `exc`, you will see a `TypeError` with a clear message at the first failure — not at decoration time.

Retry and circuit breaker

@retry — bounded re-attempts with backoff

Network errors are often transient: a packet is lost, a connection pool is momentarily exhausted, a downstream pod restarts. `@retry` re-invokes the decorated function up to `max_attempts` times with exponential backoff between attempts.

`max_attempts` is the only positional argument; every other parameter is keyword-only:

`lumen/resilience/retry_example.py`

```
from pyfly.resilience import retry

@retry(
    max_attempts=3,
    delay=0.1,
    backoff=2.0,
    max_delay=2.0,
    exceptions=(IOError, TimeoutError),
)
async def fetch_account(account_id: str) → dict:
    ...
```

Listing 13.11 — Retry with exponential backoff

How it works: The decorator executes the function, catches exceptions matching `exceptions`, sleeps `delay * backoff ** attempt` seconds (capped at `max_delay`), and tries again. On the final attempt it re-raises the last exception. The sleep uses `await asyncio.sleep(...)` for async functions and `time.sleep(...)` for sync functions — the same implementation handles both. The `jitter` parameter adds randomisation to avoid thundering-herd retries when many instances restart simultaneously.

Parameter	Default	Description
<code>max_attempts</code>	<code>3</code>	Total attempts including the first (≥ 1). Positional.
<code>delay</code>	<code>0.0</code>	Base sleep in seconds before the first retry. Keyword-only.
<code>backoff</code>	<code>1.0</code>	Multiplier applied to <code>delay</code> each attempt. Keyword-only.
<code>max_delay</code>	<code>None</code>	Cap on per-attempt sleep. <code>None</code> means no cap. Keyword-only.
<code>jitter</code>	<code>0.0</code>	Randomisation fraction <code>[0, 1]</code> applied to each wait. Keyword-only.
<code>exceptions</code>	<code>(Exception,)</code>	Exception types that trigger a retry; others propagate immediately. Keyword-only.

⚠️ IDEMPOTENCY IS YOUR RESPONSIBILITY

`@retry` will call the function body multiple times. If the operation is not idempotent — if calling it twice has a different effect than calling it once — you can apply changes more than once. Wallet deposits are not safe to retry naively: retrying a failed deposit could credit the same amount twice. Wrap non-idempotent operations in an idempotency key check (store the operation ID before executing; skip if the ID already exists) or limit `exceptions` to errors that are definitely pre-execution (connection errors, timeouts during the request phase) rather than post-execution ambiguity.

@circuit_breaker — fast failure under sustained outage**📖 NEW TERM: CIRCUIT BREAKER**

Borrowed from electrical wiring: a circuit breaker *trips* (opens) when too much current flows, cutting the circuit before the wiring overheats. A software circuit breaker trips after too many failures, cutting off calls to a failing dependency so you stop hammering it — and so your own callers fail fast instead of waiting on calls that are doomed to error anyway.

Retrying a genuinely unavailable service amplifies load at exactly the moment that service most needs relief. The circuit-breaker pattern solves this: after a threshold of consecutive failures the circuit **opens** and subsequent calls are rejected immediately — without attempting the remote call — until a recovery timeout elapses.

PyFly's circuit breaker has three states:

State	Behaviour
CLOSED	Normal operation. Every call goes through; failures are counted.
OPEN	All calls raise <code>CircuitBreakerException</code> immediately, without network I/O.
HALF_OPEN	After <code>recovery_timeout</code> seconds, a limited probe call is admitted. If it succeeds the circuit closes; if it fails the circuit reopens.

`@circuit_breaker` takes a `CircuitBreaker` **instance** — not keyword arguments. Construct the `CircuitBreaker` separately and pass it in:

`lumen/resilience/cb_example.py`

```
from pyfly.resilience import CircuitBreaker, circuit_breaker

account_cb = CircuitBreaker(
    failure_threshold=5,
    recovery_timeout=30.0,
    expected=(IOError, TimeoutError),
)

@circuit_breaker(account_cb)
async def fetch_account(account_id: str) → dict:
    ...
```

Listing 13.12 — Circuit breaker around `AccountService`

How it works: Before each call, `breaker.before_call()` checks the current state. If OPEN, it raises `CircuitBreakerException` immediately. If HALF_OPEN and the probe budget is exhausted, it also raises. Otherwise the call proceeds. On success, `breaker.on_success()` resets the consecutive-failure counter (or, in HALF_OPEN, closes the circuit once enough probes succeed). On failure, `breaker.on_failure()` increments the counter and opens the circuit when `failure_threshold` is reached.

Only exceptions in `expected` trip the breaker. Business exceptions — `ValueError`, `PermissionError` — propagate normally without affecting the circuit state.

CircuitBreaker constructor parameters (`failure_rate_threshold`, `window_size`, and `half_open_max_calls` are keyword-only):

Parameter	Default	Description
<code>failure_threshold</code>	<code>5</code>	Consecutive failures that trip the circuit.
<code>recovery_timeout</code>	<code>30.0</code>	Seconds in OPEN before moving to HALF_OPEN.
<code>expected</code>	<code>(Exception,)</code>	Exception types that count as failures.
<code>failure_rate_threshold</code>	<code>None</code>	Switch to windowed-rate mode when set (e.g. <code>0.5</code>).
<code>window_size</code>	<code>10</code>	Outcome window size for rate-based tripping.
<code>half_open_max_calls</code>	<code>1</code>	Probe calls required to close from HALF_OPEN.

The `failure_rate_threshold` and `window_size` parameters switch from consecutive-count mode to windowed-rate mode, matching Resilience4j's COUNT_BASED sliding window. Set `failure_rate_threshold=0.5` and `window_size=10` to open the circuit when more than half of the last 10 calls fail.

 SPRING PARITY

`@retry` mirrors Spring Retry's `@Retryable` (with `maxAttempts`, `backoff`, `include`). `CircuitBreaker` mirrors Resilience4j's `CircuitBreaker` (failure threshold, recovery timeout, CLOSED/OPEN/HALF_OPEN state machine, half-open probe calls, expected-exception filter). PyFly does not use the Resilience4j Java library — it is a pure-Python re-implementation with the same semantics.

Configuring resilience from `pyfly.yaml`

So far you have constructed every `RateLimiter`, `Bulkhead`, and `CircuitBreaker` in code. That is perfect for a single gateway, but operations teams usually want to tune these thresholds *without a code change* — bump a timeout, widen a rate limit — and they want one obvious place to read the current settings. PyFly v26.6.110 ships a config-driven `ResilienceRegistry` for exactly this, giving parity with Resilience4j's named-registry model.

Step 1 — Declare named instances in `pyfly.yaml`. Each entry under `pyfly.resilience.*` becomes a named instance. Names are yours to choose; group them by the downstream they protect:


```

pyfly:
  resilience:
    circuit-breaker:
      account-api:
        failure-threshold: 5
        recovery-timeout: 30s
        # or switch to windowed-rate mode:
        # failure-rate-threshold: 0.5
        # window-size: 10
    rate-limiter:
      account-api:
        max-tokens: 50
        refill-rate: 20.0
    bulkhead:
      account-api:
        max-concurrent: 8
    time-limiter:
      account-api:
        timeout: 2s

```

Durations accept friendly suffixes — `30s`, `500ms`, `1m`, `2h` — or a bare number read as seconds. Keys use kebab-case (`failure-threshold`); PyFly's relaxed binding accepts snake_case too.

Step 2 — Inject the registry and look instances up by name.

PyFly's `ResilienceAutoConfiguration` registers a single `ResilienceRegistry` bean built from those keys (it is always on, and returns an empty registry when no keys are present). Ask for it in any `@service` constructor:

lumen/account/gateway_configured.py

```

from pyfly.container import service
from pyfly.resilience import (
    ResilienceRegistry,
    bulkhead,
    circuit_breaker,
    rate_limiter,
)

@service
class AccountGateway:

    def __init__(self, http_client, registry: ResilienceRegistry) → None:
        self._http = http_client
        # Look up the named instances declared in pyfly.yaml.
        cb = registry.circuit_breaker("account-api")
        rl = registry.rate_limiter("account-api")
        bh = registry.bulkhead("account-api")

        # Wrap the real call with the config-driven instances.
        guarded = circuit_breaker(cb)(self._raw_get)
        guarded = bulkhead(bh)(guarded)
        self.get_account = rate_limiter(rl)(guarded)

    async def _raw_get(self, account_id: str) → dict:
        resp = await self._http.get(f"/accounts/{account_id}")
        return resp.json()

```

Listing 13.12a — Pulling resilience instances from the registry

What just happened. The thresholds now live in configuration, not in Python literals. A `CircuitBreaker` named `account-api` is materialised once at startup and shared by everything that looks it up — so the failure counts and OPEN/CLOSED state are *global*

across all callers of that name, exactly like a shared in-code instance. Looking up an unknown name raises `KeyError` with the list of available names, so a typo fails loudly at startup rather than silently creating an unprotected path.

💡 TIME LIMITER RETURNS A TIMEDELTA

`registry.time_limiter("account-api")` returns the configured `timedelta`, not a decorator — feed it straight into `time_limiter(timeout=registry.time_limiter("account-api"))`. The other three accessors (`circuit_breaker`, `rate_limiter`, `bulkhead`) return the instance you pass to the matching decorator.

🌿 SPRING PARITY

The `ResilienceRegistry` mirrors Resilience4j's `CircuitBreakerRegistry`, `RateLimiterRegistry`, and `BulkheadRegistry` — named instances declared in configuration and looked up at runtime. Spring Boot's `resilience4j.circuitbreaker.instances.<name>.*` becomes `pyfly.resilience.circuit-breaker.<name>.*`; the property names line up one-for-one.

Composing the layers

Decorator order

PyFly's resilience decorators compose by stacking. Python applies decorators bottom-up at decoration time but executes them top-down at call time. The recommended order, outermost to innermost:

```

@fallback          ← 1. Catch any exception; return degraded response
@rate_limiter      ← 2. Reject excess traffic before it acquires resources
@bulkhead          ← 3. Limit concurrency of rate-limited calls
@time_limiter      ← 4. Cancel if execution takes too long
async def func()   ← 5. The actual operation

```

This ordering ensures:

1. **Fallback** catches exceptions from every inner layer — including `RateLimitException`, `BulkheadException`, and `OperationTimeoutException` — so the caller always receives a usable response.
2. **Rate limiter** drops excess requests before they consume a bulkhead slot, preventing a traffic flood from exhausting the concurrency budget.
3. **Bulkhead** limits how many rate-permitted calls run concurrently, protecting the downstream from overload.
4. **Time limiter** applies only to actual execution; when it fires, the bulkhead `finally` block releases the slot correctly.

Add `@retry` and `@circuit_breaker` on the innermost side — wrapping only the actual I/O call — so the fallback absorbs their exceptions and the rate limiter and bulkhead account for retried calls correctly:

```

@fallback
@rate_limiter
@bulkhead
@time_limiter
@circuit_breaker(account_cb)
@retry(max_attempts=2, delay=0.05, backoff=2.0, exceptions=(IOError,))
async def fetch_account(account_id: str) → dict: ...

```

With `@retry` below `@time_limiter`, the timeout budget covers the entire retry sequence, not each individual attempt. To bound each attempt independently, move `@time_limiter` below `@retry`.

Putting it all together — Lumen's account gateway

Here is the full pattern assembled into a realistic `AccountGateway` that Lumen's wallet handlers use to look up account information:

lumen/account/gateway.py

```

from datetime import timedelta

from pyfly.cache import CacheAdapter, cacheable
from pyfly.container import service
from pyfly.kernel.exceptions import CircuitBreakerException,
OperationTimeoutException
from pyfly.resilience import (
    Bulkhead,
    CircuitBreaker,
    RateLimiter,
    bulkhead,
    circuit_breaker,
    fallback,
    rate_limiter,
    retry,
    time_limiter,
)

_limiter = RateLimiter(max_tokens=50, refill_rate=20.0)
_bh = Bulkhead(max_concurrent=8)
_cb = CircuitBreaker(
    failure_threshold=5,
    recovery_timeout=30.0,
    expected=(IOError, TimeoutError),
)

DEGRADED = {"status": "degraded", "balance_minor": None}

@service
class AccountGateway:

    def __init__(self, http_client, cache: CacheAdapter) → None:
        self._http = http_client
        self._cache = cache

    @cacheable(

```

```

        backend=None, # pass self._cache at runtime (see note below)
        key="account:{account_id}",
        ttl=timedelta(seconds=30),
    )
    @fallback(
        fallback_value=DEGRADED,
        on=(OperationTimeoutException, CircuitBreakerException, IOError),
    )
    @rate_limiter(_limiter)
    @bulkhead(_bh)
    @time_limiter(timeout=timedelta(seconds=2))
    @circuit_breaker(_cb)
    @retry(max_attempts=2, delay=0.05, backoff=2.0, exceptions=(IOError,))
    async def get_account(self, account_id: str) → dict:
        resp = await self._http.get(f"/accounts/{account_id}")
        return resp.json()

```

Listing 13.13 — AccountGateway with full resilience stack

WIRING BACKEND IN A CLASS METHOD

Because Python evaluates class-body decorators before `__init__` runs, `self._cache` is not yet available there. The listing above passes `backend=None` as a placeholder to illustrate the stacking order. In practice, wrap `get_account` in `__init__` the same way as the handler examples: `self.get_account = cacheable(backend=cache, key=..., ttl=...)(self._do_get_account)`. Alternatively, use a module-level `InMemoryCache` instance for testing and swap it via the DI container in production.

How a call flows through the layers:

1. `@cacheable` checks the cache. On a hit, every inner layer is skipped entirely.
2. On a miss, `@fallback` becomes the outermost safety net.
3. `@rate_limiter` checks the token bucket; rejects the call if empty.
4. `@bulkhead` checks the permit counter; rejects if at capacity.

5. `@time_limiter` sets a two-second deadline for the layers below.
6. `@circuit_breaker` rejects immediately if the circuit is OPEN.
7. `@retry` attempts the HTTP call up to two times on `IOException`.
8. On success, `@cacheable` stores the response for 30 seconds.
9. If `IOException`, `OperationTimeoutException`, or `CircuitBreakerException` escapes, `@fallback` catches it and returns `DEGRADED`.

Note that `@cacheable` sits *above* `@fallback`. That means:

- A cached `DEGRADED` response from a previous failure cycle is returned as-is for up to 30 seconds without hitting the network.
- If you do not want to cache degraded responses, move `@cacheable` below `@fallback`, or use the `unless` predicate: `unless=lambda r: r.get("status") == "degraded"`.

Run it — make the downstream fail and watch the stack degrade

You do not need a live `AccountService` to verify the stack. Wire the resilience layers around a stub HTTP client that always raises, and assert that the caller still gets a usable `DEGRADED` response instead of an exception:


```
tests/resilience/test_gateway_stack.py
```

```
import pytest

from pyfly.resilience import fallback, retry

DEGRADED = {"status": "degraded", "balance_minor": None}

class _BrokenClient:
    async def get(self, path: str) → dict:
        raise IOError("AccountService unreachable")

@fallback(fallback_value=DEGRADED, on=(IOError,))
@retry(max_attempts=2, delay=0.0, exceptions=(IOError,))
async def get_account(client: _BrokenClient, account_id: str) → dict:
    resp = await client.get(f"/accounts/{account_id}")
    return resp

@pytest.mark.asyncio
async def test_degrades_instead_of_raising() → None:
    result = await get_account(_BrokenClient(), "acc-1")
    assert result == DEGRADED
```

Listing 13.13a — The stack degrades instead of throwing

Run it:

```
$ uv run --extra dev pytest tests/resilience/test_gateway_stack.py -q
.
[100%]
1 passed in 0.05s
```

`@retry` tried the call twice, both attempts raised `IOException`, and `@fallback` caught the final exception and returned `DEGRADED`. The caller never saw the error — exactly the behaviour you want when `AccountService` is having a bad day.

What just happened. You assembled a slice of the full stack — retry on the inside, fallback on the outside — and proved that a hard failure surfaces as a degraded-but-valid response. Add `@rate_limiter`, `@bulkhead`, `@time_limiter`, and `@circuit_breaker` between them in the order shown above and each one folds into the same flow: every exception they raise is caught by the outer `@fallback`, so the caller always receives a response it can use.

✓ What you built

This chapter closes Part IV. In Chapter 11 you split Lumen into independent services with typed HTTP clients. In Chapter 12 you added `DepositSaga` to coordinate multi-step operations with compensating transactions. Here you made the whole system fast and fault-tolerant.

Concretely, you learned:

- `@cacheable` short-circuits balance reads on a cache hit; the five-second TTL bounds staleness to an acceptable window. Applied to `GetBalanceHandler` by wrapping `_fetch` at construction time — `_fetch` calls `repository.find_by_id` and projects the resulting `WalletEntity` onto `BalanceDto` via `entity_to_balance_dto` (`Mapper.project` + the `@projection`-marked `BalanceView`).
- `@cache_put` refreshes the cache as a side-effect of each `DepositFunds` command. `_deposit` is decorated with `@transactional()`; it does `find_by_id → to_aggregate → mutate → upsert` as one committed unit of work, then updates the

cache with the returned balance. The key template must match the `@cacheable` key to hit the same slot.

- `@cache_evict` removes entries on wallet closure or administrative resets; `all_entries=True` flushes the entire cache in a single call.
- `CacheManager` mirrors writes to both Redis (primary) and `InMemoryCache` (fallback) and fails over transparently; it is the right default for any production deployment.
- `RateLimiter` + `@rate_limiter` cap inbound traffic with a token- bucket algorithm that allows controlled bursting.
- `Bulkhead` + `@bulkhead` isolate concurrency with a fail-fast semaphore that prevents one slow dependency from consuming all available resources.
- `@time_limiter` enforces deadlines using `asyncio.wait_for`, turning indefinitely hanging calls into bounded `OperationTimeoutException` errors.
- `@fallback` provides a degraded but functional response when every other layer has failed; the fallback method receives the original arguments and the caught exception via the `exc` keyword argument.
- `@retry` takes `max_attempts` as its only positional argument; all other parameters (`delay`, `backoff`, `max_delay`, `jitter`, `exceptions`) are keyword-only. It re-invokes operations a bounded number of times with exponential backoff.
- `@circuit_breaker` takes a `CircuitBreaker` **instance** — not keyword arguments — and opens the circuit after a failure threshold, short- circuiting subsequent calls during the recovery window so the downstream has time to recover.
- `ResilienceRegistry` (PyFly v26.6.110) materialises named `CircuitBreaker`, `RateLimiter`, `Bulkhead`, and time-limiter instances from `pyfly.resilience.*` config keys, so operations can tune thresholds in `pyfly.yaml` and inject the registry to look instances up by name — Spring Boot's `resilience4j.*.instances.<name>.*` parity.

- **Decorator order** matters: fallback outermost, then rate limiter, bulkhead, time limiter, circuit breaker, and retry innermost — with caching above the fallback to cache even degraded responses.

Lumen is now a multi-service, saga-coordinated, cached, and resilient system. Part V adds the final production concerns: observability — metrics, distributed tracing, and health endpoints — so you can see exactly what Lumen is doing in production.

Try it yourself

Exercise 1 — Conditional caching. The `GetBalance` handler is called far more often for active wallets than for test wallets. Add `condition=lambda query: not query.wallet_id.startswith("test-")` to the `cacheable(...)` call inside `GetBalanceHandler.__init__` and verify with a unit test using `InMemoryCache` that queries for test wallet ids always reach the repository.

Exercise 2 — Circuit breaker with rate-based threshold. Replace the consecutive-count circuit breaker in `AccountGateway` with a rate-based one: open the circuit when at least 60% of the last 20 calls fail. Construct `CircuitBreaker(failure_rate_threshold=0.6, window_size=20, recovery_timeout=60.0, expected=(IOError, TimeoutError))`. Two subtleties drive the test design. First, in rate mode the breaker stays not-tripped until the window is *full* — it requires a complete 20-call window before judging the rate — so a burst of failures alone never opens it. Second, the breaker only re-evaluates its trip condition on a *failure* (a success never opens it), so the call that crosses the threshold must itself be a failing call. Write a test that fires 8 succeeding calls followed by 12 failing ones (20 calls

total = one full window, ending on a failure). Assert that the circuit stays `CLOSED` through call 19 (the window is still partial), then `OPENS` on the 20th call, when the window fills and the failure rate reaches exactly $12 / 20 = 0.60$.

Exercise 3 — Evict by prefix. Lumen sometimes needs to invalidate all cache entries for a given wallet owner (GDPR deletion). Add a `purge_owner(owner_id: str)` method to a wallet admin service that calls

`backend.evict_by_prefix(f"wallet:balance:{owner_id}:")` directly (without a decorator), and write a test that pre-populates three wallet keys for one owner and one for another, calls `purge_owner`, and asserts that only the target owner's entries are gone.

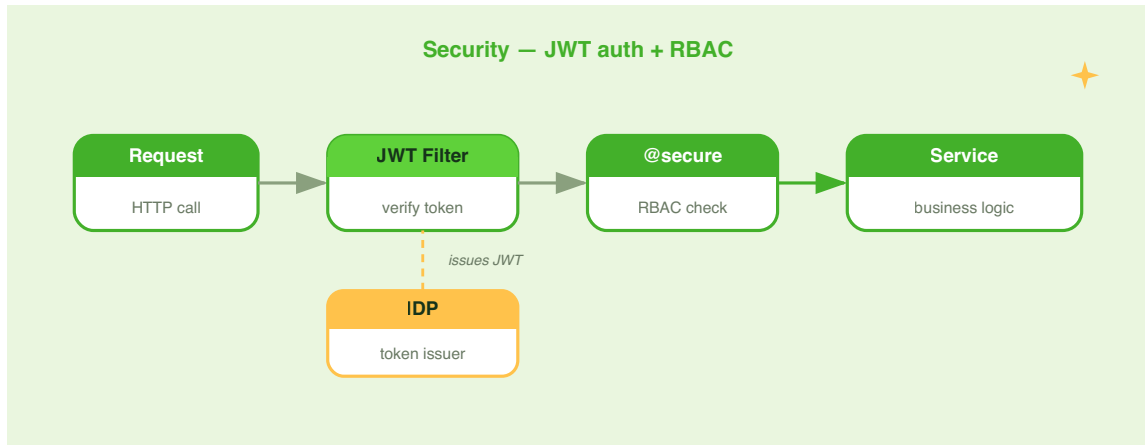
Exercise 4 — Config-driven resilience. Move `AccountGateway`'s hard-coded thresholds into `pyfly.yaml` under `pyfly.resilience.circuit-breaker.account-api`, `pyfly.resilience.rate-limiter.account-api`, and `pyfly.resilience.bulkhead.account-api`. Inject `ResilienceRegistry` into the gateway, look the three instances up by name, and write a test that asserts the materialised `CircuitBreaker.failure_threshold` matches the value you set in config. Note that `ResilienceRegistry.from_config(...)` expects a pyfly `Config`, not a plain dict — it calls `config.get_section("pyfly.resilience.circuit-breaker")` internally. Build one in the test from a nested dict, e.g. `Config({"pyfly": {"resilience": {"circuit-breaker": {"account-api": {"failure-threshold": 5}}}}})` (import `from pyfly.core.config import Config`), then pass that `Config` to `from_config`. Confirm that looking up a misspelled name raises `KeyError` with the list of available names.

PART V

Secure, Observe & Ship

CHAPTER 14

Security, Sessions & Identity



In Chapter 13 you made Lumen fast and fault-tolerant with caching and resilience decorators. Lumen's API now handles high concurrency without breaking under pressure — but it is wide open. Any caller can create wallets, read balances, or trigger deposits. Before you can ship Part V's remaining production concerns, you need to close that door.

This chapter locks Lumen down. You will:

- **Authenticate** every request with a signed JWT, using `JWTService` to issue and validate tokens and `SecurityMiddleware` to propagate the `SecurityContext` across the request scope.
- **Authorise** individual handlers and commands with the `@secure` decorator, specifying roles, permissions, or full security expressions.

- **Hash passwords** safely with `BcryptPasswordEncoder` so your user store is never a liability.
- **Manage server-side sessions** with `HttpSession`, a pluggable `SessionStore`, and a Redis backend for horizontal scaling.
- **Federate identity** to an external provider — Keycloak, AWS Cognito, or Azure AD — through the `IdpAdapter` port without changing a line of business logic.

If you have worked with Spring Security before, the shape will feel familiar: configure a filter chain, annotate individual methods, and swap the user-detail source for an IDP. PyFly calls those pieces `SecurityMiddleware + HttpSecurity`, `@secure`, and `IdpAdapter`, but the concepts map one-to-one.

This chapter is written as a guided tutorial. We build the security layer one piece at a time — issue a token, validate it, guard a handler, hash a password, store a session, federate to an IDP — and after each piece you will find a **Run it** checkpoint with the exact command to type and the output you should expect. If you have never wired authentication into a web service before, that is fine: each new term is glossed in plain language the first time it appears, and you can follow along by editing the Lumen sample as you read.

VERSION

The listings and config keys in this chapter target PyFly **v26.6.110**. If you are on an earlier release, a few property names differ — most notably the application port is now `pyfly.server.port` (Spring's `server.port`), and the actuator and admin dashboard live on a separate management port, `pyfly.management.server.port` (default `9090`).

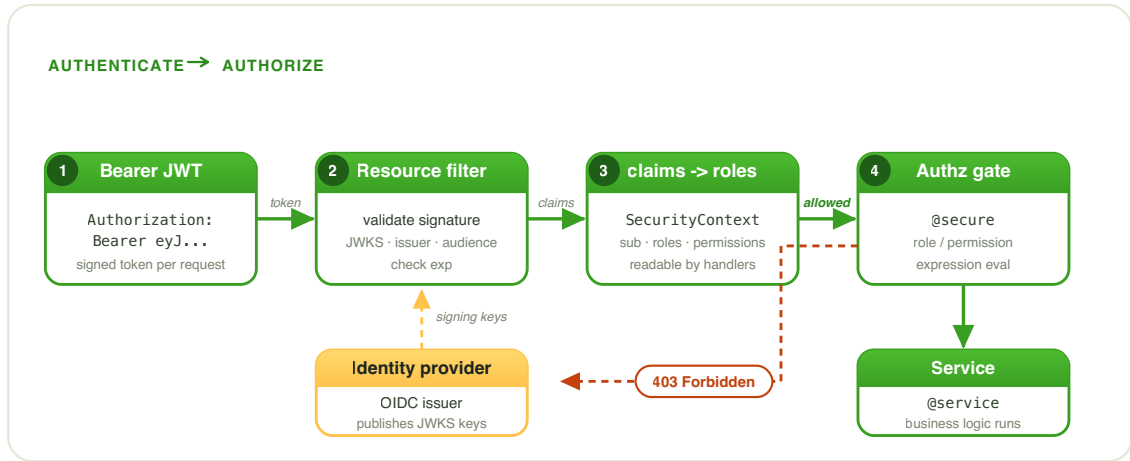


Figure 14.1 — Lumen's security layers. A JWT filter populates the `SecurityContext`; `HttpSecurity` enforces URL-level rules; `@secure` enforces handler-level rules; the IDP port delegates identity to an external provider.

Authentication with JWT

Why JSON Web Tokens?

Lumen is a stateless API. HTTP sessions would require sticky routing or a shared session store on every replica. JWT tokens let each service validate credentials independently — no shared state, no coordination, horizontal scaling by default.

A **JWT** (JSON Web Token, pronounced "jot") is a signed JSON payload. Think of it as a tamper-evident badge: the server stamps a small JSON document — *who you are*, *what roles you hold*, *when it expires* — and signs it with a secret key. The badge travels with every request. Because the signature can only be produced by someone holding the secret, the server can trust the badge without looking anything up in a database. Lumen's auth service issues a token on login; every subsequent request carries that token in the `Authorization` header; `SecurityMiddleware` validates the signature and unpacks the token into a `SecurityContext` that the rest of the request can read.

JWTService

`JWTService` wraps PyJWT with three focused operations:

Method	Description
<code>encode(payload)</code>	Sign a payload dict, adding <code>exp</code> if absent
<code>decode(token)</code>	Validate signature + <code>exp</code> ; raises <code>SecurityException</code> on failure
<code>to_security_context(token)</code>	Decode and extract <code>sub</code> , <code>roles</code> , <code>permissions</code> into a <code>SecurityContext</code>

The service always requires an `exp` claim — a token with no expiry is rejected at decode time. That invariant means every token in circulation has a bounded lifetime.

```
lumen/core/services/auth/auth_service.py
```

```
from pyfly.container import service
from pyfly.kernel.exceptions import UnauthorizedException
from pyfly.security import BcryptPasswordEncoder, JWTService, SecurityContext

@service
class AuthService:

    def __init__(
        self,
        jwt: JWTService,
        encoder: BcryptPasswordEncoder,
        user_repo,
    ) → None:
        self._jwt = jwt
        self._encoder = encoder
        self._users = user_repo

    async def login(self, username: str, password: str) → str:
        user = await self._users.find_by_username(username)
        if user is None or not self._encoder.verify(
            password, user.password_hash
        ):
            raise UnauthorizedException(
                "Invalid credentials", code="INVALID_CREDENTIALS"
            )
        # encode() auto-appends exp (default: 3 600 s from now)
        return self._jwt.encode({
            "sub": str(user.id),
            "roles": [user.role],
            "permissions": _permissions_for(user.role),
        })

    async def me(self, ctx: SecurityContext) → dict:
        user = await self._users.find_by_id(ctx.user_id)
        return {
            "id": str(user.id),
```

```

        "username": user.username,
        "role": user.role,
    }

def _permissions_for(role: str) → list[str]:
    MAP = {
        "USER": ["wallet:read", "wallet:deposit"],
        "ADMIN": [
            "wallet:read", "wallet:deposit",
            "wallet:create", "wallet:delete",
            "user:read", "user:write",
        ],
    }
    return MAP.get(role, [])

```

Listing 14.1 — Issuing a JWT on successful login

...

How it works. `login` fetches the user record and calls `BcryptPasswordEncoder.verify` to compare the supplied password against the stored hash. On success it calls `jwt.encode`, which auto-appends an `exp` claim `expiration_seconds` seconds from now (default `3600` — one hour). You never import `datetime`: the service computes the Unix-timestamp expiry as `int(time.time()) + expiration_seconds`. The caller receives a compact, self-contained token string.

Walking through the `login` method one step at a time:

Step 1 — Look up the user. `find_by_username` returns the stored user record, or `None` if no such username exists. We will treat both "no such user" and "wrong password" as the same failure so an attacker cannot tell which usernames are registered.

Step 2 — Verify the password. `self._encoder.verify(password, user.password_hash)` re-hashes the supplied password with the salt baked into the stored hash and compares the two in constant time. If the user was not found, or the passwords do not match, raise `UnauthorizedException` — PyFly renders this as a `401` with the machine-readable code `INVALID_CREDENTIALS`.

Step 3 — Build the claims. On success, assemble the JSON payload that will become the token's body: `sub` (the subject — the user's id), `roles`, and the `permissions` that role grants.

Step 4 — Sign and return. `self._jwt.encode({...})` signs the payload with the configured secret, stamps the mandatory `exp` claim, and returns the compact token string. That string is what the client stores and replays on every later request.

 JARGON: CLAIM

A *claim* is just one key/value statement inside the token's JSON body — `"sub": "42"` claims the subject is user 42. The token carries a bag of claims; the framework copies the security-relevant ones (`sub` , `roles` , `permissions`) into the `SecurityContext` .

💡 RUN IT — ISSUE A TOKEN IN THE REPL

You do not need a running server to see `JWTService` work. With Lumen's virtual environment active, open a Python REPL and sign a payload:

```
>>> from pyfly.security import JWTService
>>> jwt = JWTService(secret="dev-secret-change-me", algorithm="HS256")
>>> token = jwt.encode({"sub": "42", "roles": ["USER"]})
>>> token[:20]           # a compact "header.payload.signature" string
'eyJhbGciOiJIUzI1NiIs'
>>> ctx = jwt.to_security_context(token)
>>> ctx.user_id, ctx.roles
('42', ['USER'])
```

The token round-trips: `encode` added the `exp` claim for you, and `to_security_context` validated the signature and unpacked the claims into a `SecurityContext`. Try tampering with a character in the middle of the string and calling `jwt.decode(token)` again — you will get a `SecurityException`, because the signature no longer matches the payload.

The SecurityContext

`SecurityContext` is a frozen dataclass that carries authentication and authorisation data for a single request. The middleware creates it from the validated token; your handlers receive it as an injected parameter.

Field	Type	Description
<code>user_id</code>	<code>str</code> <code> </code> <code>None</code>	Authenticated user's id; <code>None</code> for anonymous
<code>roles</code>	<code>list[str]</code>	Roles granted in the token
<code>permissions</code>	<code>list[str]</code>	Fine-grained permissions
<code>attributes</code>	<code>dict[str, str]</code>	Extra claims (department, tenant, ...)

Key methods:

Method / Property	Returns	Description
<code>is_authenticated</code>	<code>bool</code>	<code>True</code> when <code>user_id</code> is not <code>None</code>
<code>has_role(role)</code>	<code>bool</code>	Exact match against <code>roles</code> list
<code>has_any_role(roles)</code>	<code>bool</code>	Set intersection — any of the listed roles
<code>has_permission(perm)</code>	<code>bool</code>	Exact match against <code>permissions</code> list
<code>SecurityContext.anonymous()</code>	<code>SecurityContext</code>	Create an unauthenticated context

The security filter

`SecurityMiddleware` (canonical location

`pyfly.web.adapters.starlette.security_middleware`, re-exported from

`pyfly.security`) sits at the Starlette middleware layer. For every request it:

1. Checks whether the path is in `exclude_paths`; if so, sets an anonymous context and continues.
2. Reads the `Authorization` header and strips the `Bearer` prefix.
3. Calls `jwt_service.to_security_context(token)`.
4. On success, stores the authenticated context in `request.state.security_context`.
5. On any `SecurityException` (expired, tampered, or missing `exp`), logs at DEBUG and sets an anonymous context instead.

The middleware **never rejects requests** — rejection is the job of `@secure` and `HttpSecurity`. An endpoint that requires the user can enforce it; a health-check endpoint can ignore the context entirely.

lumen/app.py

```
from pyfly.security import JWTService, SecurityMiddleware
from pyfly.web.adapters.starlette import create_app

def build_app(context):
    app = create_app(title="Lumen", context=context)

    jwt = context.get_bean(JWTService)
    app.add_middleware(
        SecurityMiddleware,
        jwt_service=jwt,
        exclude_paths=[
            "/docs",
            "/openapi.json",
            "/api/auth/login",
            "/api/auth/register",
        ],
    )
    return app
```

Listing 14.2 — Adding the security middleware

...

How it works. `exclude_paths` lists the paths where no token is expected. Login and register cannot require authentication because the token does not exist yet; docs paths are excluded so the API explorer works without credentials. Every other path goes through token validation.

i JARGON: MIDDLEWARE

Middleware is code that wraps every request, running before the route handler on the way in and after it on the way out. `SecurityMiddleware` uses the "in" pass to read the token and stash a `SecurityContext` on `request.state` so handlers downstream can read it without re-parsing the header.

What just happened. You now have the two halves of authentication wired up.

`AuthService.login` mints a token after checking the password; `SecurityMiddleware` reads that token on every later request and turns it into a `SecurityContext`. Crucially, the middleware is permissive — it never returns a `401`. A request with a bad or missing token simply arrives anonymous, and the decision to allow or reject it is deferred to the next two layers you will build: `HttpSecurity` (broad, URL-level) and `@secure` (precise, per-handler). Separating *who you are* (authentication) from *what you may do* (authorization) is what keeps each layer small and testable.

URL-level rules with `HttpSecurity`

`@secure` guards individual handler methods. `HttpSecurity` guards whole URL subtrees at the filter layer — before the route dispatcher runs. The two are complementary: `HttpSecurity` provides fast, broad policy at the edge; `@secure` adds fine-grained, per-handler enforcement behind it.

lumen/config/security_config.py

```

from pyfly.container import bean, configuration
from pyfly.security.http_security import HttpSecurity

@Configuration
class SecurityConfig:

    @bean
    def http_security(self) → HttpSecurity:
        hs = HttpSecurity()
        hs.authorize_requests() \
            .request_matchers("/idp/admin/**").has_role("ADMIN") \
            .request_matchers("/api/v1/wallets/**").authenticated() \
            .request_matchers(
                "/health", "/docs", "/openapi.json",
                "/idp/login", "/idp/refresh",
            ).permit_all() \
            .any_request().permit_all()
        return hs

```

Listing 14.3 — HttpSecurity DSL

...

How it works. Rules are evaluated in declaration order — first match wins. The `HttpSecurityFilter` runs at `HIGHEST_PRECEDENCE + 350`, after authentication filters have populated `request.state.security_context`, so every role and permission check has a fully hydrated context to inspect. The terminal methods — `has_role`, `has_any_role`, `has_permission`, `authenticated`, `permit_all`, and `deny_all` — cover every common policy; unsatisfied rules return RFC 7807 problem-detail JSON (`application/problem+json`) with the appropriate HTTP status.

Read the DSL chain top to bottom — that is exactly the order the filter evaluates it:

Step 1 — Open the rule list. `hs.authorize_requests()` starts the fluent builder. Every `.request_matchers(...)` call that follows registers one rule.

Step 2 — Lock down the admin subtree. `.request_matchers("/idp/admin/**").has_role("ADMIN")` requires the `ADMIN` role for anything under `/idp/admin`. The `**` glob matches any depth of path segments.

Step 3 — Require login for wallets. `.request_matchers("/api/v1/wallets/**").authenticated()` accepts any logged-in caller, regardless of role, for the wallet tree. Finer per-handler rules come later via `@secure`.

Step 4 — Allow the public paths. `.request_matchers("/health", "/docs", ...).permit_all()` lets health checks, the docs explorer, and the IDP login/refresh endpoints through with no token.

Step 5 — Set the default. `.any_request().permit_all()` is the catch-all for paths no earlier rule matched. Flip it to `.deny_all()` once every route is explicitly accounted for — "deny by default" is the safer production posture.

Step 6 — Return the builder. Return the configured `HttpSecurity` itself — do **not** call `hs.build()` here. Because `SecurityConfig` is a `@configuration` class, auto-configuration's `HttpSecurityFilterAutoConfiguration` (active whenever an `HttpSecurity` bean is present) picks up the `HttpSecurity` bean, calls `.build()` for you, and registers the resulting `HttpSecurityFilter` on the chain.

🔗 RUN IT — SEE THE GATE ACCEPT AND REJECT

Start the app with `uv run pyfly run` (it serves on `pyfly.server.port`, default `8080`). In another terminal, hit a guarded path without a token:

```
curl -i http://localhost:8080/api/v1/wallets
```

Expected — the gate rejects the anonymous request with an RFC 7807 body:

```
HTTP/1.1 401 Unauthorized
content-type: application/problem+json

{"type":"about:blank","title":"Unauthorized","status":401,
 "detail":"Authentication is required to access this resource.",
 "instance":"/api/v1/wallets"}
```

Now replay it with a token (use the `$TOKEN` you minted in the REPL above, or one from `/idp/login`):

```
curl -i -H "Authorization: Bearer $TOKEN" http://localhost:8080/api/v1/
wallets
```

Expected — `HTTP/1.1 200 OK` and a JSON page of wallets. The same URL, two outcomes, decided entirely by the token: that is the gate doing its job.

📖 TWO-LAYER DEFENSE

`HttpSecurity` provides fast URL-level policy before routes are even dispatched — good for blanket rules like "everything under `/api/v1/wallets` needs authentication." The `@secure` decorators on individual methods are the second, finer-grained layer. Use both together for defense in depth.

🍃 SPRING PARITY

`HttpSecurity` mirrors Spring Security's `HttpSecurity.authorizeHttpRequests()` chain. `request_matchers` corresponds to `requestMatchers`, `authenticated()` to `.authenticated()`, `has_role` to `hasRole`, and the `build()` that auto-configuration calls on your `HttpSecurity` bean finalises the underlying filter just as `build()` finalises the Spring filter chain. The fnmatch glob patterns (`/api/admin/**`) behave identically to Spring's Ant-style path matching.

Auto-configuration

You do not need to register `JWTService` or `BcryptPasswordEncoder` manually. Add the relevant properties to `pyfly.yaml` and auto-configuration wires everything:

`lumen/resources/pyfly.yaml`

```
pyfly:
  security:
    enabled: true
  jwt:
    secret: "${JWT_SECRET}"
    algorithm: HS256
  filter:
    enabled: true
    exclude-patterns: >-
      /docs,/openapi.json,
      /api/auth/login,/api/auth/register
  password:
    bcrypt-rounds: 12
```

Listing 14.4 — Security auto-configuration

...

Property	Default	Description
<code>pyfly.security.jwt.secret</code>	<code>change-me-in-production</code>	HMAC signing key — must be overridden
<code>pyfly.security.jwt.algorithm</code>	<code>HS256</code>	Signing algorithm
<code>pyfly.security.jwt.filter.enabled</code>	<i>(absent)</i>	Set <code>true</code> to register <code>SecurityFilter</code> bean automatically
<code>pyfly.security.jwt.exclude-patterns</code>	<i>(absent)</i>	Comma-separated paths to skip
<code>pyfly.security.password.bcrypt-rounds</code>	<code>12</code>	Bcrypt cost factor

Note where each key lives: `filter.enabled` is nested under `jwt.filter`, but `exclude-patterns` sits one level up, directly under `jwt` — that is the key the auto-configuration reads (`pyfly.security.jwt.exclude-patterns`). There is no `/actuator/health` in the exclude list any more: as of v26.6.110 the actuator and admin dashboard run on a **separate management port** (`pyfly.management.server.port`, default `9090`), so they never pass through the application's JWT filter at all.

⚠ PRODUCTION SECRET

Never commit the real JWT secret to source control. Use `${JWT_SECRET}` and inject the value from an environment variable or a secrets manager at deploy time.

⚠ THE MANAGEMENT PORT IS OPEN BY DEFAULT

For Spring parity, the management server (`/actuator/*` and the admin dashboard, default port `9090`) is **unauthenticated by default** — it does not share the application's security gate. In any non-local deployment, secure it explicitly:

```
pyfly:
  management:
    security:
      enabled: true      # apply the security gate to the management
port
```

Setting `pyfly.management.server.port: -1` disables the management endpoints entirely. The default actuator HTTP exposure is just `health,info` ; widen it with `pyfly.management.endpoints.web.exposure.include` .

💡 RUN IT — CONFIRM AUTO-WIRING ON STARTUP

With the keys above in `pyfly.yaml` , start the app and watch the boot log:

```
uv run pyfly run
```

Expected — you will see the app bind to `:8080` and the management server to `:9090` , and the security beans appear with no `@configuration` code of your own. A quick proof that the encoder bean exists:

```
curl -s http://localhost:8080/api/auth/login \
  -d '{"username":"alice","password":"hunter2"}' \
  -H 'content-type: application/json'
```

Expected — a JSON body containing an `access_token` (or a `401` with `INVALID_CREDENTIALS` if the credentials are wrong). Either way, the `JWTService` and `BcryptPasswordEncoder` beans were wired automatically.

Authorization with `@secure`

`@secure` is a function decorator that enforces authentication and authorisation on individual handlers and commands. It reads the `security_context` keyword argument that the middleware has already injected, then evaluates role, permission, and expression checks before the function body runs.

Signature

```
def secure(  
    roles: list[str] | None = None,  
    permissions: list[str] | None = None,  
    expression: str | None = None,  
    ) → Callable: ...
```

The decorated function must accept `security_context: SecurityContext` as a keyword argument — that is how `@secure` reaches the current user.

Role-based protection

Lumen's wallet endpoints (`/api/v1/wallets`) are the natural place to apply `@secure` .

The real `WalletController` injects the command and query buses; add

`security_context: SecurityContext` to each method that needs protection and stack `@secure` on top.

The controller exposes seven endpoints under `/api/v1/wallets` :

Method	Path	Handler	Guard
POST	<code>""</code>	<code>open_wallet</code>	USER or ADMIN
POST	<code>/ {wallet_id} / deposit</code>	<code>deposit</code>	USER/ADMIN + <code>wallet:deposit</code>
POST	<code>/ {wallet_id} / withdraw</code>	<code>withdraw</code>	USER/ADMIN + <code>wallet:deposit</code>
GET	<code>""</code>	<code>list_wallets</code>	authenticated
GET	<code>/ rich</code>	<code>list_rich_wallets</code>	authenticated
GET	<code>/ {wallet_id} / balance</code>	<code>wallet_balance</code>	USER or ADMIN
GET	<code>/ {wallet_id}</code>	<code>wallet_detail</code>	ADMIN only

The collection handlers (`list_wallets` , `list_rich_wallets`) are named with the `list_` prefix so the framework registers them before the `wallet_detail` / `wallet_balance` handlers — Starlette matches first-registered-wins, so the literal `/ rich` segment is found ahead of the `/ {wallet_id}` path variable.

lumen/web/controllers/wallet_controller.py

```

from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.core.services.wallets.get_balance_query import GetBalance
from lumen.core.services.wallets.get_wallet_query import GetWallet
from lumen.core.services.wallets.list_rich_wallets_query import
ListRichWallets
from lumen.core.services.wallets.list_wallets_query import ListWallets
from lumen.core.services.wallets.open_wallet_command import OpenWallet
from lumen.core.services.wallets.withdraw_funds_command import WithdrawFunds
from lumen.interfaces.dtos.v1.balance_dto import BalanceDto
from lumen.interfaces.dtos.v1.deposit_request import DepositRequest
from lumen.interfaces.dtos.v1.open_wallet_request import OpenWalletRequest
from lumen.interfaces.dtos.v1.page_dto import PageDto
from lumen.interfaces.dtos.v1.wallet_dto import WalletDto
from pyfly.container import rest_controller
from pyfly.cqrs import DefaultCommandBus, DefaultQueryBus
from pyfly.data import Pageable, Sort
from pyfly.kernel import ResourceNotFoundException
from pyfly.security import SecurityContext, secure
from pyfly.web import (
    Body,
    PathVar,
    QueryParam,
    Valid,
    get_mapping,
    post_mapping,
    request_mapping,
)

_NEWEST_FIRST = Sort.by("created_at").descending()

@rest_controller
@request_mapping("/api/v1/wallets")
class WalletController:
    """Digital-wallet REST API: open, deposit, withdraw, list, inspect."""

    def __init__(

```

```

        self,
        commands: DefaultCommandBus,
        queries: DefaultQueryBus,
    ) → None:
        self._commands = commands
        self._queries = queries

# Any authenticated user may open a wallet.
@secure(roles=["USER", "ADMIN"])
@post_mapping("", status_code=201)
async def open_wallet(
    self,
    request: Valid[Body[OpenWalletRequest]],
    security_context: SecurityContext,
) → dict[str, str]:
    wallet_id = await self._commands.send(
        OpenWallet(
            owner_id=request.owner_id,
            currency=request.currency,
        )
    )
    return {"wallet_id": wallet_id}

# Deposit: USER/ADMIN role + wallet:deposit permission.
@secure(roles=["USER", "ADMIN"], permissions=["wallet:deposit"])
@post_mapping("/{wallet_id}/deposit")
async def deposit(
    self,
    wallet_id: PathVar[str],
    request: Valid[Body[DepositRequest]],
    security_context: SecurityContext,
) → dict[str, int | str]:
    balance = await self._commands.send(
        DepositFunds(wallet_id=wallet_id, amount=request.amount)
    )
    return {"wallet_id": wallet_id, "balance_minor": balance}

# Withdraw: same guard as deposit.
@secure(roles=["USER", "ADMIN"], permissions=["wallet:deposit"])

```

```

@post_mapping("/{wallet_id}/withdraw")
async def withdraw(
    self,
    wallet_id: PathVar[str],
    request: Valid[Body[DepositRequest]],
    security_context: SecurityContext,
) → dict[str, int | str]:
    balance = await self._commands.send(
        WithdrawFunds(wallet_id=wallet_id, amount=request.amount)
    )
    return {"wallet_id": wallet_id, "balance_minor": balance}

# Paged list – collection routes registered before /{wallet_id}.
@secure(roles=["USER", "ADMIN"])
@get_mapping("")
async def list_wallets(
    self,
    page: QueryParam[int] = 1,
    size: QueryParam[int] = 20,
    security_context: SecurityContext = None,
) → PageDto[WalletDto]:
    result = await self._queries.query(
        ListWallets(
            pageable=Pageable.of(page, size, _NEWEST_FIRST)
        )
    )
    return PageDto.from_page(result)

# Rich list: wallets filtered by minimum balance.
@secure(roles=["USER", "ADMIN"])
@get_mapping("/rich")
async def list_rich_wallets(
    self,
    min_minor: QueryParam[int] = 0,
    page: QueryParam[int] = 1,
    size: QueryParam[int] = 20,
    security_context: SecurityContext = None,
) → PageDto[WalletDto]:
    result = await self._queries.query(

```

```

        ListRichWallets(
            min_minor=min_minor,
            pageable=Pageable.of(page, size, _NEWEST_FIRST),
        )
    )
    return PageDto.from_page(result)

# Single-wallet balance: any authenticated user.
@secure(roles=["USER", "ADMIN"])
@get_mapping("/{wallet_id}/balance")
async def wallet_balance(
    self,
    wallet_id: PathVar[str],
    security_context: SecurityContext,
) → BalanceDto:
    result = await self._queries.query(
        GetBalance(wallet_id=wallet_id)
    )
    if result is None:
        raise ResourceNotFoundException(
            f"Wallet {wallet_id!r} not found",
            code="WALLET_NOT_FOUND",
            context={"wallet_id": wallet_id},
        )
    return result

# Full wallet view: ADMIN only.
@secure(roles=["ADMIN"])
@get_mapping("/{wallet_id}")
async def wallet_detail(
    self,
    wallet_id: PathVar[str],
    security_context: SecurityContext,
) → WalletDto:
    result = await self._queries.query(
        GetWallet(wallet_id=wallet_id)
    )
    if result is None:
        raise ResourceNotFoundException(

```

```

        f"Wallet {wallet_id!r} not found",
        code="WALLET_NOT_FOUND",
        context={"wallet_id": wallet_id},
    )
    return result

```

Listing 14.5 — Role and permission guards on real Lumen endpoints

...

How it works. Stack `@secure` **above** `@post_mapping` / `@get_mapping` so authorisation runs before route binding. The framework injects `security_context` from `request.state.security_context`, which `SecurityMiddleware` already populated.

When you list multiple roles, the user needs **at least one** (OR semantics). When you list multiple permissions, the user needs **all** of them (AND semantics). When you supply both `roles` and `permissions`, both checks must pass independently.

The collection handlers (`list_wallets`, `list_rich_wallets`) sort alphabetically before `wallet_balance` / `wallet_detail`, so the framework registers the literal `""` and `/rich` routes first — ensuring Starlette resolves `/api/v1/wallets/rich` as a fixed segment rather than as a wallet id.

To add `@secure` to one handler, the recipe is always the same three steps:

Step 1 — Add the parameter. Give the method a `security_context: SecurityContext` keyword parameter. (For the `GET` list handlers that already have defaulted query params, default it to `None` so the parameter order stays valid: `security_context: SecurityContext = None`.)

Step 2 — Stack the decorator. Put `@secure(...)` **above** `@get_mapping` / `@post_mapping`. Order matters: the authorization check must run before the route binding does.

Step 3 — Choose the minimal guard. Pick the least-privilege rule that still lets the right users through — `roles=["USER", "ADMIN"]` for ordinary reads, an extra `permissions=["wallet:deposit"]` for money-moving writes, `roles=["ADMIN"]` for the full detail view.

🔗 RUN IT — WATCH A 403 FOR THE WRONG ROLE

`wallet_detail` is `ADMIN`-only. Call it with a `USER` token (the authentication succeeds, but authorization fails):

```
curl -i -H "Authorization: Bearer $USER_TOKEN" \
  http://localhost:8080/api/v1/wallets/abc-123
```

Expected — the request clears the gate (it *is* authenticated) but `@secure(roles=["ADMIN"])` rejects it. The exception handler renders a problem body that includes the machine-readable `code` :

```
HTTP/1.1 403 Forbidden
content-type: application/problem+json

{"type":"about:blank","title":"Forbidden","status":403,
 "detail":"Insufficient roles: requires one of ['ADMIN']",
 "code":"FORBIDDEN"}
```

Note the difference from the gate's `401` earlier: a missing or invalid token at the gate gives `401` ("I don't know who you are"), while a valid token without the required role gives `403 FORBIDDEN` ("I know who you are, and you may not do this"). The same `USER` token *would* succeed against `GET /api/v1/wallets/abc-123/balance`, which is guarded with `roles=["USER", "ADMIN"]`.

AMOUNTS IN MINOR UNITS

`DepositRequest.amount` is an `int` in **minor units** (cents). €10.50 is `1050`. This convention avoids floating-point rounding errors throughout the Money domain.

`WalletDto.balance_minor` carries the same integer; `WalletDto.balance` is a `float` rendered for display only.

Expression-based authorization

For policies that cannot be expressed with a flat role list, use the `expression` parameter. PyFly evaluates expressions via safe AST parsing — no `eval()` or `exec()` is involved anywhere in the chain.

lumen/web/controllers/wallet_controller.py

```

from pyfly.security import SecurityContext, secure

# ADMIN, or a MANAGER who also holds wallet:write.
@secure(
    expression=(
        "hasRole('ADMIN')"
        " or (hasRole('MANAGER') and hasPermission('wallet:write'))"
    )
)
async def approve_large_deposit(
    self,
    deposit_id: str,
    security_context: SecurityContext,
) → None:
    ...

# Any authenticated, non-guest user can see the wallet dashboard.
@secure(expression="isAuthenticated and not hasRole('GUEST')")
async def dashboard(
    self,
    security_context: SecurityContext,
) → dict:
    ...

```

Listing 14.6 — Security expressions on wallet endpoints

:::

Supported expression vocabulary (full set):

Token	Description
<code>hasRole('X')</code>	User has role <code>X</code>
<code>hasAnyRole('X', 'Y')</code>	User has at least one of the listed roles
<code>hasAuthority('X')</code>	User has role or permission <code>X</code>
<code>hasAnyAuthority('X', 'Y')</code>	At least one of the listed roles/permissions
<code>hasPermission('X')</code>	User has permission <code>X</code>
<code>isAuthenticated</code>	User is authenticated
<code>isAnonymous</code>	User is not authenticated
<code>permitAll</code>	Always allow
<code>denyAll</code>	Always deny
<code>principal / authentication</code>	The current <code>SecurityContext</code> object
<code>and / or / not</code>	Boolean operators
<code>(...)</code>	Grouping

EXPRESSIONS ARE SAFE

PyFly reduces each construct to `True` or `False`, then evaluates only a pure boolean AST. Any node that is not a constant, `BoolOp`, or `UnaryOp(Not)` raises `SecurityException(code="INVALID_EXPRESSION")`. This eliminates injection risks entirely.

Applying @secure to CQRS handlers

`@secure` is not limited to REST controllers. You can protect CQRS command handlers in exactly the same way — it fires before the handler body because the DI container injects `security_context` from `request.state.security_context` when it resolves the handler:

`lumen/core/services/wallets/deposit_funds_handler.py`

```
from pyfly.cQRS import command_handler
from pyfly.security import SecurityContext, secure

from lumen.core.services.wallets.deposit_funds_command import DepositFunds

@command_handler
class DepositFundsHandler:

    @secure(roles=["USER", "ADMIN"], permissions=["wallet:deposit"])
    async def handle(
        self,
        command: DepositFunds,
        security_context: SecurityContext,
    ) → int:
        # command.amount is in minor units (cents)
        ...
```

Listing 14.7 — @secure on a CQRS command handler

...

The check fires before any business logic runs.

Passwords

Why bcrypt?

MD5 and SHA-256 are engineered to be fast — ideal for data integrity, catastrophic for passwords. An attacker who steals your user table can test billions of SHA-256 guesses per second on commodity hardware. **Bcrypt** is deliberately slow and adjustably expensive: the cost factor (rounds) lets you tune the algorithm so that an attack requires orders of magnitude more time, without meaningfully affecting normal login latency.

BcryptPasswordEncoder

`BcryptPasswordEncoder` implements the `PasswordEncoder` protocol (a `runtime_checkable` Protocol with `hash` and `verify` methods):

`lumen/auth/password_service.py`

```
from pyfly.security import BcryptPasswordEncoder

encoder = BcryptPasswordEncoder(rounds=12)

# During registration — store only the hash, never the raw password
hashed = encoder.hash("correct-horse-battery-staple")

# During login — verify without storing the plaintext
is_match = encoder.verify("correct-horse-battery-staple", hashed)
# True

is_match = encoder.verify("wrong-password", hashed)
# False
```

Listing 14.8 — Hashing and verifying passwords

...

Parameter	Default	Notes
<code>rounds</code>	<code>12</code>	Each increment doubles hashing time. 12 is the recommended production default.

How it works. `hash` calls `bcrypt.gensalt(rounds=self._rounds)` to generate a fresh random salt, then `bcrypt.hashpw` to produce the hash. Both salt and hash are embedded in the returned string — the `$2b$12$...` prefix encodes the algorithm version and cost factor, making every stored hash self-describing. `verify` calls `bcrypt.checkpw`, which re-derives the hash from the raw password and the embedded salt and compares the result with a timing-safe equality check, preventing timing-oracle attacks.

The PasswordEncoder protocol

`PasswordEncoder` is a `runtime_checkable` Protocol. Any class that implements `hash(raw: str) → str` and `verify(raw: str, hashed: str) → bool` satisfies it — including `BcryptPasswordEncoder`. You can swap in `argon2` or `scrypt` at any time without touching service code:

```
from pyfly.security import PasswordEncoder

class Argon2PasswordEncoder:
    def hash(self, raw: str) → str: ...
    def verify(self, raw: str, hashed: str) → bool: ...

isinstance(Argon2PasswordEncoder(), PasswordEncoder) # True
```

 AUTO-CONFIGURATION

When `pyfly.security.enabled=true` and `bcrypt` is installed, PyFly auto-configures a `BcryptPasswordEncoder` bean with `rounds` read from `pyfly.security.password.bcrypt-rounds`. Declare the bean yourself in a `@configuration` class to override it without touching auto-configuration.

Sessions

Why server-side sessions?

JWT tokens are stateless — once issued, they cannot be revoked before they expire. If a user logs out, their token is still valid until `exp`. For many APIs that trade-off is acceptable. For browser-facing applications — Lumen's admin dashboard, for example — you need the server to be the authoritative source: log out must mean log out.

Server-side sessions give you that control. A `SessionStore` holds session data keyed by a random session ID. The browser receives only the session ID in a cookie. The server revokes a session instantly by deleting its store entry.

HttpSession

`HttpSession` wraps a session's data dictionary with typed accessors and tracks mutation state so the filter knows when to persist:

Property / Method	Description
<code>id</code>	UUID hex string session identifier
<code>is_new</code>	<code>True</code> if created during this request
<code>created_at</code>	Unix timestamp (<code>float</code>) of creation
<code>last_accessed</code>	Unix timestamp (<code>float</code>) of most recent access
<code>modified</code>	<code>True</code> if any attribute was written or session is new
<code>invalidated</code>	<code>True</code> if <code>invalidate()</code> was called
<code>previous_id</code>	Former id after <code>rotate_id()</code> (used by filter to clean up old entry)
<code>get_attribute(name)</code>	Return attribute value or <code>None</code>
<code>set_attribute(name, value)</code>	Write an attribute; marks session modified
<code>remove_attribute(name)</code>	Remove attribute if present
<code>get_attribute_names()</code>	List user-set attribute names (excludes internal <code>_*</code> keys)
<code>rotate_id()</code>	Assign a fresh session id, preserving all data
<code>invalidate()</code>	Mark for deletion on next filter pass
<code>get_data()</code>	Return the raw session dict (includes internal metadata)

```
lumen/core/services/auth/session_handler.py
```

```
from pyfly.session import HttpSession

async def post_login(
    username: str,
    password: str,
    session: HttpSession,
    auth_service,
) → dict:
    user = await auth_service.authenticate(username, password)

    # Rotate the id before writing auth state (session-fixation prevention)
    session.rotate_id()
    session.set_attribute("user_id", str(user.id))
    session.set_attribute("role", user.role)

    return {"message": "Logged in"}

async def logout(session: HttpSession) → dict:
    session.invalidate()
    return {"message": "Logged out"}
```

Listing 14.9 — Using the session after login

...

How it works. `rotate_id()` generates a fresh UUID session ID and records the old one in `session.previous_id`. When `SessionFilter` persists the session at the end of the request, it deletes the old store entry (the previous ID can no longer resolve to this session) and saves the new one. An attacker who obtained the pre-auth session ID cannot carry it into the authenticated session — the classic **session-fixation** mitigation.

The SessionStore protocol

All backends implement the `SessionStore` protocol:

```
class SessionStore(Protocol):
    async def get(
        self, session_id: str
    ) → dict[str, Any] | None: ...

    async def save(
        self, session_id: str,
        data: dict[str, Any],
        ttl: int,
    ) → None: ...

    async def delete(self, session_id: str) → None: ...

    async def exists(self, session_id: str) → bool: ...
```

Two adapters ship out of the box:

Adapter	Module	Notes
<code>InMemorySessionStore</code>	<code>pyfly.session.adapters.memory</code>	<code>asyncio.Lock</code> -guarded; single-process only; data lost on restart
<code>RedisSessionStore</code>	<code>pyfly.session.adapters.redis</code>	JSON-serialized; keys prefixed <code>pyfly:session:</code> ; TTL managed by Redis

SessionFilter

`SessionFilter` is an `OncePerRequestFilter` ordered at `HIGHEST_PRECEDENCE + 150`. It bookends every request:

1. Reads the session cookie (`PYFLY_SESSION` by default).

2. Loads the session from the store, or creates a new one.
3. Attaches it to `request.state.session`.
4. Calls `call_next(request)` — the rest of the filter chain and the handler run.
5. On the way back, persists a modified or new session, deletes an invalidated one, and re-issues the cookie with a rolling `max_age` (the TTL slides forward on every request).

Cookie attributes set by `SessionFilter` :

Attribute	Value	Reason
<code>httponly</code>	<code>True</code>	Prevents JavaScript access; XSS mitigation
<code>samesite</code>	<code>lax</code>	Blocks most cross-site request forgery flows
<code>secure</code>	configurable	Set <code>True</code> in production (HTTPS only)
<code>max_age</code>	<code>ttl</code>	Rolling expiry

Redis sessions in production

lumen/config/session_config.py

```
import redis.asyncio as aioredis

from pyfly.container import bean, configuration
from pyfly.session import SessionFilter
from pyfly.session.adapters.redis import RedisSessionStore

@Configuration
class SessionConfig:

    @bean
    def session_store(self) → RedisSessionStore:
        client = aioredis.from_url(
            "redis://localhost:6379/0"
        )
        return RedisSessionStore(client=client)

    @bean
    def session_filter(
        self,
        store: RedisSessionStore,
    ) → SessionFilter:
        return SessionFilter(
            store=store,
            cookie_name="LUMEN_SESSION",
            ttl=1800,
            secure=True,
        )
```

Listing 14.10 — Redis session store

...

How it works. `RedisSessionStore.save` JSON-serialises the session dictionary (including dataclass attributes such as `SecurityContext`, round-tripped via an allowlisted type-tag mechanism) and calls `client.set(key, raw, ex=ttl)`. The TTL is managed entirely by Redis, so expired sessions disappear server-side with zero cleanup overhead. Keys use the prefix `pyfly:session:` for namespace isolation. On read, only types on the allowlist are deserialised, eliminating arbitrary-object instantiation risks.

🔗 AUTO-CONFIGURATION

Add `pyfly.session.enabled: true` and `pyfly.session.store: redis` to `pyfly.yaml`. PyFly auto-configures `RedisSessionStore` (when `redis.asyncio` is installed) and `SessionFilter` for you. The `redis.url` defaults to `redis://localhost:6379/0`; override with `pyfly.session.redis.url`.

🌿 SPRING PARITY

`SessionFilter` mirrors Spring Session's `SessionRepositoryFilter`. `HttpSession` mirrors `javax.servlet.http.HttpSession` (or `jakarta.servlet.http.HttpSession` in Boot 3). `InMemorySessionStore` is equivalent to Spring Session's `MapSessionRepository`; `RedisSessionStore` is equivalent to `RedisSessionRepository`. The cookie attributes (`HttpOnly`, `SameSite=Lax`, rolling `Max-Age`) match Spring Session's defaults exactly.

Validating tokens from an external IdP (OAuth2 resource server)

So far Lumen has *minted its own tokens* with `JWTService` and a shared HMAC secret. That is perfect for a self-contained service. But in most real-world deployments, the tokens are issued by a dedicated identity provider — Keycloak, Microsoft Entra ID

(formerly Azure AD), or AWS Cognito — and your service's only job is to **validate** them. This is the **OAuth2 resource server** role, and as of v26.6.110 PyFly gives it to you with config alone: no code, and the same one configuration works across all three providers.

JARGON: RESOURCE SERVER

In OAuth2 terms, the IdP that issues tokens is the *authorization server*; your API, which holds the data callers want and checks their tokens, is the *resource server*. Lumen is a resource server. It never sees the user's password — it only verifies the badge the IdP already signed.

The difference from `JWTService` is the **signing scheme**. Your own tokens are signed with a symmetric secret (`HS256`) — the same key both signs and verifies, so only services that hold the secret can validate. IdP tokens are signed *asymmetrically* (`RS256`): the IdP keeps a private key and publishes the matching public keys at a **JWKS** endpoint (JSON Web Key Set). Your resource server downloads those public keys, caches them, and uses them to verify every token's signature — it never needs the IdP's secret at all.

Turning it on

`JWKSTokenValidator` does the work, and `OAuth2ResourceServerAutoConfiguration` wires it from `pyfly.security.oauth2.resource-server.*`. Here is the full Keycloak setup:

```
lumen/resources/pyfly.yaml
```

```
pyfly:
  security:
    oauth2:
      resource-server:
        enabled: true
        # Either point at the JWKS endpoint directly...
        jwks-uri: "https://keycloak.example.com/realms/lumen/protocol/openid-
connect/certs"
        # ...or give an issuer-uri and let OIDC discovery find jwks-uri +
issuer:
        # issuer-uri: "https://keycloak.example.com/realms/lumen"
        issuer: "https://keycloak.example.com/realms/lumen"
        audiences: "lumen-backend"
        validate-audience: true
        algorithms: "RS256"
        clock-skew-seconds: 60
        exclude-patterns: "/idp/login,/idp/refresh,/docs,/openapi.json"
```

Listing 14.11 — Multi-IdP OAuth2 resource server (Keycloak)

...

Step by step:

Step 1 — Enable it. `enabled: true` activates

`OAuth2ResourceServerAutoConfiguration` (PyJWT must be installed). That registers a `JWKSValidator` bean and the bearer-token filter on the application's chain.

Step 2 — Point at the keys. Set `jwks-uri` directly, or set `issuer-uri` and let PyFly fetch `<issuer-uri>/well-known/openid-configuration` to discover both the `jwks-uri` and the authoritative `issuer` — exactly like Spring's `issuer-uri`.

Step 3 — Pin the audience and issuer. `audiences` lists the values the token's `aud` claim must match (any one). `issuer` (when set) is checked against the token's `iss`. Together they ensure a token minted for a *different* application or realm cannot be replayed against Lumen.

Step 4 — Leave the defaults alone for claim mapping. You did not configure where roles live, yet it just works for Keycloak, Entra, and Cognito. That is the job of `ClaimMappings`, covered next.

How claims become a SecurityContext

Every IdP puts roles and scopes in different places. Keycloak nests realm roles under `realm_access.roles` and per-client roles under `resource_access.<client>.roles`; Entra uses a flat `roles` array (or `groups`); Cognito uses `cognito:groups`. `ClaimMappings` reconciles them with a small path language, so `has_role(...)` and `has_permission(...)` work the same regardless of provider.

Mapping field	Default search order	Becomes
<code>principal_claims</code>	<code>oid</code> , <code>sub</code>	<code>SecurityContext.user_id</code>
<code>authority_claims</code>	<code>roles</code> , <code>scopes</code> , <code>authorities</code> , <code>realm_access.roles</code> , <code>resource_access.*.roles</code> , <code>groups</code> , <code>cognito:groups</code>	<code>SecurityContext.roles</code>
<code>scope_claims</code>	<code>scp</code> , <code>scope</code> (space-split)	<code>SecurityContext.permissions</code>
<code>attribute_claims</code>	<i>(none)</i>	<code>SecurityContext.attributes</code>

Two rules make the defaults span every provider:

- **Dotted paths** descend into nested objects: `realm_access.roles` reads `payload["realm_access"]["roles"]`.

- A single-level ***** **wildcard** iterates every key at that level:

`resource_access.*.roles` collects the `roles` array from *every* client entry under `resource_access`. Path segments split on `.` only, so a colon-bearing claim name like `cognito:groups` is matched verbatim.

Roles are collected across *all* matching paths and de-duplicated, so a Keycloak token contributes both its realm roles and its client roles into one flat `roles` list.

JARGON: JWKS AND KID

A *JWKS* is the IdP's public-key directory. Each key carries a `kid` (key id); each token's header names the `kid` it was signed with. The validator looks up that exact key, caches the set (default 300s), and rotates automatically when the IdP publishes new keys — no redeploy needed.

If the built-in defaults do not match your IdP, override only what differs:


```
lumen/resources/pyfly.yaml
```

```
pyfly:
  security:
    oauth2:
      resource-server:
        enabled: true
        # Microsoft Entra ID (v2.0):
        issuer-uri: "https://login.microsoftonline.com/<tenant-id>/v2.0"
        audiences: "api://lumen-backend"
        principal-claim-names: "oid,sub"           # Entra's stable user id first
        authorities-claim-names: "roles,groups"
        scope-claim-names: "scp"                   # Entra delegated scopes
        attribute-claims: "tid,preferred_username"
        # AWS Cognito access tokens carry no 'aud' – disable audience checks:
        # validate-audience: false
        # authorities-claim-names: "cognito:groups"
```

Listing 14.12 — Tuning claim mapping for Entra ID and Cognito

...

Step 5 (when needed) — Cognito has no `aud`. Cognito access tokens carry `client_id` instead of `aud`, so set `validate-audience: false` for them; the signature, `iss`, and `exp` checks still apply.

 RUN IT — ACCEPT A REAL IDP TOKEN, REJECT A FORGED ONE

With the resource server enabled, replay a token your IdP issued against a guarded route:

```
curl -i -H "Authorization: Bearer $KEYCLOAK_TOKEN" \
  http://localhost:8080/api/v1/wallets
```

Expected — `HTTP/1.1 200 OK`. The filter fetched the JWKS, matched the token's `kid`, verified the `RS256` signature, the `iss`, the `aud`, and the `exp` (with 60s of clock-skew tolerance), then built a `SecurityContext` from the token's claims.

Now flip one character in the token and retry. Expected — `HTTP/1.1 401` (or an anonymous context that the gate then rejects), because the signature no longer matches any published key. You did not write or deploy a single line of validation code to get either outcome.

What just happened. You added a production-grade, multi-IdP token validator with configuration only. `JWSTokenValidator` verifies the signature against the IdP's published keys (so the secret never leaves the IdP), checks `iss` / `aud` / `exp`, and maps provider-specific claims into the same `SecurityContext` the rest of the chapter already uses. Your `@secure` decorators and `HttpSecurity` rules do not change at all — they read `roles` and `permissions` exactly as before, whether the token came from Lumen's own `JWTService` or from Keycloak.

 ERROR MODE: ANONYMOUS VS 401

By default (`authenticate-error-mode: anonymous`) a *present but invalid* token yields an anonymous context and the request continues to the `HttpSecurity` gate, which decides. Set `authenticate-error-mode: "401"` to reject an invalid token right at the filter with `WWW-Authenticate: Bearer error="invalid_token"` (RFC 6750). A *missing* token always falls through to the gate either way.

🍃 SPRING PARITY

This is PyFly's `spring-boot-starter-oauth2-resource-server`. The `resource-server.issuer-uri` / `jwt-uri` keys mirror `spring.security.oauth2.resourceserver.jwt.*`; OIDC discovery, the accepted audiences list, configurable algorithms, and the 60-second clock-skew default all match Spring Security's `JwtDecoder` and `JwtTimestampValidator`. `ClaimMappings` is the equivalent of a custom `JwtAuthenticationConverter` / `JwtGrantedAuthoritiesConverter`, but expressed declaratively in YAML.

External identity (IDP)

The problem with managing identity in-house

Lumen currently stores credentials in its own database, which means Lumen must implement password resets, MFA, email verification, account lockout, GDPR deletion, social login, and SSO — undifferentiated work that every service eventually needs. The industry answer is to delegate identity to a dedicated provider: Keycloak for on-premises, AWS Cognito for AWS-native stacks, Azure AD for Microsoft environments.

PyFly's `IdpAdapter` port makes that delegation pluggable behind a single interface. Swap the adapter and the business layer never knows.

IdpAdapter — the port

Every adapter must satisfy the `IdpAdapter` protocol:

```

class IdpAdapter(Protocol):
    name: str

    # User management
    async def create_user(
        self, user: IdpUser, password: str
    ) → IdpUser: ...
    async def get_user(self, user_id: str) → IdpUser | None: ...
    async def find_by_username(
        self, username: str
    ) → IdpUser | None: ...
    async def update_user(self, user: IdpUser) → IdpUser: ...
    async def delete_user(self, user_id: str) → bool: ...
    async def list_users(self, *, limit: int = 100) → list[IdpUser]: ...

    # Authentication
    async def login(
        self, request: LoginRequest
    ) → AuthResult: ...
    async def logout(self, access_token: str) → bool: ...
    async def refresh(
        self, refresh_token: str
    ) → AuthResult: ...
    async def introspect(
        self, access_token: str
    ) → SessionIntrospection: ...

    # Password / MFA
    async def change_password(
        self, request: PasswordChangeRequest
    ) → bool: ...
    async def reset_password(self, user_id: str) → str: ...

    # Roles
    async def assign_role(
        self, user_id: str, role: str
    ) → bool: ...
    async def revoke_role(

```

```
        self, user_id: str, role: str
    ) → bool: ...
    async def list_roles(self) → list[IdpRole]: ...
```

Key DTOs:

Class	Purpose
<code>IdpUser</code>	User record: <code>id</code> , <code>username</code> , <code>email</code> , <code>roles</code> , <code>attributes</code> , ...
<code>LoginRequest</code>	<code>username</code> , <code>password</code> , <code>mfa_code</code> (optional)
<code>AuthResult</code>	<code>user</code> , <code>access_token</code> , <code>refresh_token</code> , <code>expires_in</code> , <code>token_type</code>
<code>SessionIntrospection</code>	<code>active</code> , <code>user_id</code> , <code>username</code> , <code>scopes</code> , <code>expires_at</code>
<code>PasswordChangeRequest</code>	<code>user_id</code> , <code>old_password</code> , <code>new_password</code>
<code>IdpRole</code>	<code>name</code> , <code>description</code> , <code>scopes</code>

Keycloak adapter

lumen/config/idp_config.py

```
from pyfly.container import bean, configuration
from pyfly.idp import IdpAdapter, KeycloakIdpAdapter

@configuration
class IdpConfig:

    @bean
    def idp_adapter(self) → IdpAdapter:
        return KeycloakIdpAdapter(
            base_url="https://keycloak.example.com",
            realm="lumen",
            client_id="lumen-backend",
            client_secret="${KEYCLOAK_SECRET}",
            verify_ssl=True,
        )
```

Listing 14.13 — Wiring the Keycloak adapter

...

How it works. `KeycloakIdpAdapter` communicates with Keycloak's Admin REST API (`/admin/realms/{realm}/users`) and token endpoint (`/realms/{realm}/protocol/openid-connect/token`) via `httpx` . It caches the `client_credentials` admin token internally and re-fetches it within a ten-second safety margin of expiry — Keycloak's default client-credentials TTL is 60 s, so without this cache every admin call would require two network round trips.

Using the IDP in a service

lumen/core/services/auth/idp_auth_service.py

```

from pyfly.container import service
from pyfly.idp import IdpAdapter, IdpUser, LoginRequest
from pyfly.kernel.exceptions import UnauthorizedException

@service
class IdpAuthService:

    def __init__(self, idp: IdpAdapter) → None:
        self._idp = idp

    async def register(
        self,
        username: str,
        email: str,
        password: str,
        role: str = "USER",
    ) → str:
        user = IdpUser(
            username=username,
            email=email,
            roles=[role],
        )
        created = await self._idp.create_user(user, password)
        result = await self._idp.login(
            LoginRequest(
                username=username, password=password
            )
        )
        return result.access_token

    async def login(
        self, username: str, password: str
    ) → str:
        try:
            result = await self._idp.login(
                LoginRequest(

```



```

        username=username, password=password
    )
)
except PermissionError as exc:
    raise UnauthorizedException(
        "Invalid credentials",
        code="INVALID_CREDENTIALS",
    ) from exc
return result.access_token

async def introspect(self, token: str) → dict:
    info = await self._idp.introspect(token)
    return {
        "active": info.active,
        "user_id": info.user_id,
        "username": info.username,
        "scopes": info.scopes,
    }

```

Listing 14.14 — Using IdpAdapter in the auth service

...

How it works. `IdpAuthService` depends only on `IdpAdapter` — the DI container resolves the concrete `KeycloakIdpAdapter` at startup. The service layer never imports Keycloak, Cognito, or Azure-specific code. Switch provider by changing one line in `IdpConfig`; the service is untouched.

Auto-configuration and the built-in HTTP routes

Enable the IDP subsystem in `pyfly.yaml` and PyFly wires the adapter and a ready-made REST controller automatically:

```
lumen/resources/pyfly.yaml
```

```
pyfly:
  idp:
    enabled: true
    provider: keycloak
    keycloak:
      base-url: https://keycloak.example.com
      realm: lumen
      client-id: lumen-backend
      client-secret: "${KEYCLOAK_SECRET}"
```

Listing 14.15 — IDP auto-configuration

...

provider value	Adapter
internal-db	InternalDbIdpAdapter (bcrypt in-memory store)
keycloak	KeycloakIdpAdapter
cognito / aws-cognito	AwsCognitoIdpAdapter
azure-ad / azuread / entra	AzureAdIdpAdapter

When Starlette is present, `IdpAutoConfiguration` also registers an `IdpController` bean that exposes the full IDP API under `/idp`:

Route	Method	Description
<code>/idp/login</code>	POST	Authenticate (username + password + optional MFA)
<code>/idp/refresh</code>	POST	Refresh an access token
<code>/idp/logout</code>	POST	Revoke a token
<code>/idp/introspect</code>	POST	Inspect an active session
<code>/idp/admin/users</code>	POST	Create a user
<code>/idp/admin/users</code>	GET	List users
<code>/idp/admin/users/{user_id}</code>	GET / DELETE	Get or delete a user
<code>/idp/admin/users/{user_id}/roles/{role}</code>	POST / DELETE	Assign or revoke a role
<code>/idp/admin/roles</code>	GET	List all roles

💡 CUSTOM ADAPTERS

Any class that satisfies the `IdpAdapter` Protocol can be wired in as the `IdpAdapter` bean. Register it in a `@configuration` class and PyFly's `@conditional_on_missing_bean` skips auto-configuration entirely. This is the standard extension point for on-premises LDAP, in-house SSO, or test-double adapters.

 SPRING PARITY

`IdpAdapter` is PyFly's equivalent of Spring Security's `UserDetailsService` + `AuthenticationProvider` combination. `IdpUser` maps to `UserDetails`; `AuthResult` maps to the `Authentication` object returned by `AuthenticationManager.authenticate()`. `KeycloakIdpAdapter` plays the role of the Keycloak Spring Security adapter.

Logging users in with a form

Everything so far assumed a machine client: mint a JWT, replay it in the `Authorization` header, validate it on the way in. That is exactly right for Lumen's API. But Lumen also has an admin dashboard — a browser — and browsers do not carry Bearer headers. They carry cookies. For the dashboard you want the old, boring, correct thing: a username/password form that POSTs to a login URL, and a session cookie that rides along afterwards.

PyFly gives you that without writing a controller. `FormLoginFilter` intercepts a POST to the login URL, authenticates the credentials through a `ProviderManager`, and — crucially — **rotates the session id** before binding the authenticated `SecurityContext` to the session. The rotation is the session-fixation defense from the sessions section, applied automatically at the moment privileges change: whatever session id an attacker may have planted before login is thrown away and replaced. The context is stored in the session, and `OAuth2SessionSecurityFilter` restores it on every later request, so the dashboard stays logged in.

You declare the users and the URLs in `pyfly.yaml`. Passwords are stored **pre-hashed** — never plaintext — and the auto-configuration verifies them with `BcryptPasswordEncoder` through a `DaoAuthenticationProvider`:

```

pyfly:
  security:
    enabled: true
    form-login:
      enabled: true
      login-url: /login
      username-param: username
      password-param: password
      success-url: /dashboard
      failure-url: /login?error
      use-redirect: true
      users:
        alice:
          # bcrypt hash of "hunter2" — generate with
          BcryptPasswordEncoder.hash
          password-hash: "$2b$12$Q9.../iJ8wq6Yk5fJ0b0"
          roles: ADMIN
          permissions: wallet:read,wallet:create
          enabled: true
    logout:
      enabled: true
      logout-url: /logout
      success-url: /login?logout
      delete-cookies: PYFLY_SESSION
      use-redirect: true

```

`FormLoginFilter` runs at `HIGHEST_PRECEDENCE + 230`, just after the session-restoring filter, so a fresh login always overrides any prior anonymous context. `LogoutFilter` runs right behind it at `+235`: a POST to the logout URL invalidates the session, resets the request context to `SecurityContext.anonymous()`, and deletes the cookies you name in `delete-cookies`. Both filters speak two dialects — set `use-redirect: false` and they answer with JSON (`{"authenticated": true}` / `204`) instead of a `302`, which is what you want if the dashboard is a single-page app calling `fetch`.

The authentication itself is the `DaoAuthenticationProvider` from PyFly's authentication SPI. It is deliberately careful about one thing: an unknown username still incurs a password verification against a throw-away dummy hash, so the login endpoint cannot be used as a username oracle — a wrong password and a nonexistent user take the same time and raise the same `BadCredentialsException`. A disabled account (`enabled: false`) raises `DisabledException`.

🔗 RUN IT — LOG IN WITH A COOKIE, THEN LOG OUT

Start the app and POST the form (note `-d`, not JSON — this is a real HTML form submission):

```
curl -i -X POST http://localhost:8080/login \
  -d "username=alice&password=hunter2"
```

Expected — HTTP/1.1 302 Found, a `location: /dashboard` header, and a `set-cookie: PYFLY_SESSION=...` carrying the new (rotated) session id. Replay a wrong password and you get `302 → /login?error` instead. Now end the session:

```
curl -i -X POST http://localhost:8080/logout -b "PYFLY_SESSION=<id>"
```

Expected — `302 → /login?logout` and a `set-cookie` that expires `PYFLY_SESSION`. The session entry is gone from the store; the cookie no longer resolves to anything.

🍃 SPRING PARITY

This is Spring Security's `formLogin()` and `logout()` configurers. `FormLoginFilter` maps to `UsernamePasswordAuthenticationFilter`, `LogoutFilter` to Spring's `LogoutFilter`, and the config-driven `form-login.users` map is the equivalent of an in-memory `UserDetailsManager`. Session-id rotation on login matches Spring's default `changeSessionId` session-fixation strategy.

Stronger password hashing

bcrypt is a fine default, and it is the one Lumen has used so far. But two things are true at once: bcrypt is good enough today, and you will eventually want to move off it — to Argon2id, the algorithm OWASP now recommends — without a flag-day migration that forces every user to reset their password. The obstacle is that a stored hash is just a string; nothing about `$2b$12$...` tells your code *which* algorithm produced it once you have more than one in play.

`DelegatingPasswordEncoder` solves this the way Spring Security does: it prefixes every hash it produces with the encoder's id in braces — `{argon2}...`, `{bcrypt}...`, `{pbkdf2}...`, `{scrypt}...`. On verification it reads the prefix and dispatches to the matching encoder, so a single encoder bean can *read* every legacy hash while *writing* only the current default. A stored value with an unknown or missing prefix never matches — fail-closed by construction.

PyFly ships four `PasswordEncoder` adapters — `BcryptPasswordEncoder`, `Pbkdf2PasswordEncoder`, `ScryptPasswordEncoder`, and `Argon2PasswordEncoder` (Argon2id; install with `pip install pyfly[argon2]`) — plus the `create_delegating_password_encoder` factory, which builds a delegating encoder with **bcrypt** as the default id and the other three wired in for verification. To make Argon2id the *default* for new hashes while still reading Lumen's existing bcrypt column, construct the delegating encoder directly and pass `encoding_id="argon2"`:

```

from pyfly.container import bean, configuration
from pyfly.security import (
    Argon2PasswordEncoder,
    BcryptPasswordEncoder,
    DelegatingPasswordEncoder,
    Pbkdf2PasswordEncoder,
    ScryptPasswordEncoder,
)

@Configuration
class PasswordConfig:

    @bean
    def password_encoder(self) → DelegatingPasswordEncoder:
        return DelegatingPasswordEncoder(
            {
                "argon2": Argon2PasswordEncoder(), # OWASP-preferred default
                "bcrypt": BcryptPasswordEncoder(rounds=12),
                "pbkdf2": Pbkdf2PasswordEncoder(),
                "scrypt": ScryptPasswordEncoder(),
            },
            encoding_id="argon2",
        )

```

The win is **transparent on-login migration**. `DelegatingPasswordEncoder` exposes `upgrade_encoding(stored)`, which returns `True` when a stored hash was produced by anything other than the current default. Call it inside `login`, right after the password verifies, and re-hash in place. Each user is silently upgraded from bcrypt to Argon2id the next time they sign in — no reset email, no downtime, no second column:


```

async def login(self, username: str, password: str) → str:
    user = await self._users.find_by_username(username)
    if user is None or not self._encoder.verify(password, user.password_hash):
        raise UnauthorizedException(
            "Invalid credentials", code="INVALID_CREDENTIALS"
        )

    # Transparently re-hash legacy ({bcrypt}...) credentials to the current
    # default ({argon2}...) now that we hold the plaintext and it verified.
    if self._encoder.upgrade_encoding(user.password_hash):
        user.password_hash = self._encoder.hash(password)
        await self._users.save(user)

    return self._jwt.encode({"sub": str(user.id), "roles": [user.role]})

```

💡 RUN IT — SEE THE PREFIX AND THE UPGRADE IN THE REPL

```

>>> from pyfly.security import create_delegating_password_encoder
>>> enc = create_delegating_password_encoder() # bcrypt default
>>> h = enc.hash("hunter2")
>>> h[:8]
'{bcrypt}'
>>> enc.verify("hunter2", h)
True
>>> enc.upgrade_encoding(h) # already the default → no upgrade needed
False
>>> enc.upgrade_encoding("{pbkdf2}sha256$6000000$...") # legacy →
upgrade
True

```

The `{id}` prefix is what makes this work: verification routes by it, and `upgrade_encoding` compares it to the configured default.

JARGON: ARGON2ID

Argon2id is a memory-hard password hash — it forces an attacker to spend *memory*, not just CPU, per guess, which neutralises the GPU and ASIC farms that make bcrypt cracking cheap at scale. `Argon2PasswordEncoder` defaults to OWASP's interactive parameters (time cost 3, 64 MiB, 4 lanes).

Lumen as its own authorization server

Up to now Lumen has been a token *consumer* — minting simple JWTs for its own login, or validating tokens an external IdP issued. There is a third role Lumen can play: a real OAuth2 **authorization server** that issues tokens to *other* parties. You want this the moment Lumen has more than one moving part. A nightly ledger-reconciliation worker needs to call Lumen's API as itself, with no human in the loop. A first-party mobile app needs to sign a user in and act on their behalf. Those are two different OAuth2 grants, and `AuthorizationServer` speaks both.

Service-to-service: client_credentials

The reconciliation worker is a confidential client: it holds a secret and asks for a token in its own name. Register Lumen's clients in an `InMemoryClientRegistrationRepository` and enable the authorization server in config — the `AuthorizationServer` bean and an in-memory token store are auto-configured for you:

```

from pyfly.container import bean, configuration
from pyfly.security.oauth2 import (
    ClientRegistration,
    InMemoryClientRegistrationRepository,
)

@Configuration
class LumenClientsConfig:

    @bean
    def client_repository(self) → InMemoryClientRegistrationRepository:
        return InMemoryClientRegistrationRepository(
            # Machine client – service-to-service, no user.
            ClientRegistration(
                registration_id="lumen-reconciler",
                client_id="lumen-reconciler",
                client_secret="${RECONCILER_SECRET}",
                authorization_grant_type="client_credentials",
                scopes=["ledger:read"],
            ),
            # First-party mobile app – public client (no secret), PKCE only.
            ClientRegistration(
                registration_id="lumen-mobile",
                client_id="lumen-mobile",
                authorization_grant_type="authorization_code",
                redirect_uri="com.lumen.app://callback",
                scopes=["openid", "wallet:read", "wallet:deposit"],
            ),
        )

```

```

pyfly:
  security:
    oauth2:
      authorization-server:
        enabled: true
        secret: "${AS_SIGNING_SECRET}" # never the placeholder; HS256 ≥ 32
bytes
        issuer: "https://lumen.example.com"
        audience: "lumen-backend"
        access-token-ttl: 3600
        refresh-token-ttl: 86400
      token-store:
        provider: redis # memory | redis | postgres
        redis:
          url: "redis://localhost:6379/0"

```

The HTTP surface is `AuthorizationServerEndpoints`, which builds the standard OAuth2/OIDC routes off the `AuthorizationServer` bean. Mount them on the app:

```

from pyfly.security.oauth2 import AuthorizationServer,
AuthorizationServerEndpoints

def register_oauth2(app, context):
    server = context.get_bean(AuthorizationServer)
    app.router.routes.extend(AuthorizationServerEndpoints(server).routes())

```

That exposes the full set: `GET /oauth2/authorize`, `POST /oauth2/par`, `POST /oauth2/token`, `POST /oauth2/introspect`, `POST /oauth2/revoke`, `POST /oauth2/register`, `GET /oauth2/jwks`, and the two discovery documents `GET /.well-known/oauth-authorization-server` and `GET /.well-known/openid-configuration`.

 RUN IT — GET A MACHINE TOKEN

The worker authenticates with HTTP Basic and asks for the scope it was registered for:

```
curl -s -X POST http://localhost:8080/oauth2/token \
  -u lumen-reconciler:$RECONCILER_SECRET \
  -d grant_type=client_credentials \
  -d scope="ledger:read"
```

Expected — a JSON token response with `access_token`, `token_type: "Bearer"`, `expires_in: 3600`, a `refresh_token`, and `scope: "ledger:read"`. Now ask for a scope the client was **not** registered for:

```
curl -s -X POST http://localhost:8080/oauth2/token \
  -u lumen-reconciler:$RECONCILER_SECRET \
  -d grant_type=client_credentials -d scope="ledger:write admin"
```

Expected — `{"error": "invalid_scope", ...}` with HTTP `400`. A client can only ever obtain scopes it is registered for; unregistered scopes are rejected wholesale, never silently echoed. (Asking for `client_credentials` from a client registered only for `authorization_code` is rejected the same way, with `unauthorized_client`.)

User-facing: `authorization_code` + PKCE + an OIDC `id_token`

The mobile app is a public client — it ships no secret, because anything baked into an app binary is not a secret. Its proof of possession is **PKCE**: the app generates a random `code_verifier`, sends only its SHA-256 hash (`code_challenge`) when it asks for an authorization code, and reveals the verifier only when it redeems the code. An attacker who intercepts the code cannot redeem it without the verifier.

The flow has two legs. First the app sends the user to the authorize endpoint:

```
GET /oauth2/authorize?response_type=code
    &client_id=lumen-mobile
    &redirect_uri=com.lumen.app://callback
    &scope=openid%20wallet:read
    &code_challenge=E9MeIhoa20wvFrEMTJguCHaoeK1t8URWbuGJSstw-cM
    &code_challenge_method=S256
    &state=xyz
```

If the resource owner is not yet authenticated, the endpoint bounces them to the login URL and comes back. Once they are, Lumen issues a **single-use** authorization code and redirects to `redirect_uri?code=...&state=xyz&iss=...`. The `iss` parameter (RFC 9207) lets the app confirm which server answered — a mix-up defense. Then the app redeems the code, presenting the verifier:

```
curl -s -X POST http://localhost:8080/oauth2/token \
  -d grant_type=authorization_code \
  -d client_id=lumen-mobile \
  -d code="$CODE" \
  -d redirect_uri="com.lumen.app://callback" \
  -d code_verifier="$VERIFIER"
```

The response carries the `access_token`, a `refresh_token`, and — because the request included the `openid` scope — an OIDC `id_token` whose `aud` is the client id. The authorization server enforces the safety rails for you: the redirect URI must match the registration **exactly**, PKCE is **mandatory and must be S256**, and the code is **single-use**. Replay a code that was already redeemed and Lumen does not merely reject it — it treats the replay as an injection attack and **revokes the refresh token already issued from that code**.

Refresh rotation and reuse detection

Refresh tokens rotate. Every time the app refreshes, it gets a *new* refresh token and the old one is marked consumed:

```
curl -s -X POST http://localhost:8080/oauth2/token \  
  -d grant_type=refresh_token -d client_id=lumen-mobile \  
  -d refresh_token="$OLD_REFRESH"
```

The consumed token is not deleted — it is kept precisely so that a *replay* can be caught. If a stolen refresh token is presented after the legitimate client has already rotated it, the server detects the reuse and revokes the **entire token family** (the original token and every descendant), forcing both the attacker and the victim back through login. A legitimate client never replays a rotated token, so reuse is unambiguous evidence of theft. Both the reuse case and a presentation from the wrong client raise `INVALID_GRANT`.

Asymmetric signing and JWKS

For service-to-service tokens, HS256 with a shared secret is fine — Lumen signs and Lumen verifies. But once *other* services validate Lumen's tokens, you do not want to hand them your signing secret. Switch to asymmetric signing: Lumen holds a private key, signs with `RS256` (or `ES256` / `PS256`), and publishes only the matching **public** key at `/oauth2/jwks`. Provide an `AuthorizationServer` bean yourself to opt into asymmetric mode (a hand-declared bean suppresses the HS256 auto-configuration):

```

from pyfly.container import bean, configuration
from pyfly.security.oauth2 import (
    AuthorizationServer,
    InMemoryClientRegistrationRepository,
    InMemoryTokenStore,
)

@Configuration
class AsymmetricAuthServerConfig:

    @bean
    def authorization_server(
        self, clients: InMemoryClientRegistrationRepository
    ) → AuthorizationServer:
        return AuthorizationServer(
            secret="", # unused for asymmetric signing
            client_repository=clients,
            token_store=InMemoryTokenStore(),
            algorithm="RS256",
            private_key=open("lumen-signing-key.pem").read(),
            key_id="lumen-2026", # becomes the JWT header `kid`
            issuer="https://lumen.example.com",
        )

```

Now `GET /oauth2/jwks` returns the public JWK set (with the `kid` and `alg`), and any PyFly resource server can validate Lumen's tokens by pointing its `pyfly.security.oauth2.resource-server.jwks-uri` at it — exactly the Keycloak setup from earlier in this chapter, but with Lumen as the issuer. Lumen has come full circle: it issues the tokens *and* validates them, and the two halves share nothing but a public key.

 SPRING PARITY

`AuthorizationServer` + `AuthorizationServerEndpoints` is PyFly's `spring-authorization-server`. The endpoint paths, the discovery documents, mandatory-S256 PKCE, refresh-token rotation with reuse detection, and the `/oauth2/jwks` key set all match Spring Authorization Server's defaults.

Sender-constrained tokens

Every token in this chapter so far has been a *bearer* token: whoever holds it, uses it. That is the whole point — and the whole risk. A bearer token that leaks from a log, a proxy, or a compromised device is a fully-usable credential in an attacker's hands. For a wallet that moves money, "whoever holds it" is not a comfortable security model. Sender-constraining fixes it by binding the token to a key the legitimate client holds, so a stolen token is inert without that key.

PyFly supports both standard mechanisms, and the binding is carried in the token's `cnf` (confirmation) claim:

- **DPoP** (RFC 9449) — the client signs a fresh per-request *proof* JWT with its private key; the access token carries `cnf.jkt`, the SHA-256 thumbprint of that key (RFC 7638). The resource server verifies the proof and that its key's thumbprint matches `jkt`.
- **mTLS** (RFC 8705) — the access token carries `cnf["x5t#S256"]`, the thumbprint of the client's TLS certificate. The resource server compares it to the cert presented on the connection.

The authorization server binds the token automatically: if a client presents a `DPoP` header on the token request, Lumen stamps `cnf.jkt` into the issued access token. On the resource-server side you turn enforcement on:

```
pyfly:
  security:
    oauth2:
      resource-server:
        enabled: true
        jwks-uri: "https://lumen.example.com/oauth2/jwks"
        enforce-sender-constraints: true
        mtls-cert-header: "x-client-cert"    # set by your TLS-terminating
proxy
```

With `enforce-sender-constraints: true`, when a token carries a `cnf` claim the filter *requires* the matching proof: a `cnf.jkt` token with no valid `DPoP` proof header is rejected as `INVALID_TOKEN`; a `cnf["x5t#S256"]` token whose thumbprint does not match the presented client certificate is likewise rejected. A plain bearer token (no `cnf`) is unaffected — enforcement only kicks in for tokens that were issued sender-constrained, so you can roll it out gradually.

The verification primitives are public if you need them directly: `DPoPProofValidator`, `confirm_dpop_binding`, `confirm_mtls_binding`, `jwk_thumbprint`, `certificate_thumbprint`, and `access_token_hash`. For Lumen, the lesson is the proportional one: protect the money-moving scopes (`wallet:deposit`, `wallet:withdraw`) with sender-constrained tokens, and a leaked access token stops being a leaked withdrawal.

Secure by default

A theme runs through this whole chapter: the safe choice is the default, and you have to *opt out* of it, not opt in. That is deliberate. Security defaults that must be remembered are security defaults that get forgotten. Four of Lumen's hardening defaults are worth calling out together, because each one closes a mistake that is otherwise easy to ship.

The placeholder secret fails fast. PyFly's defaults ship a literal signing secret, `change-me-in-production`. If you leave it in place, the composition root **refuses to start** — both the symmetric `JWTService` (when its filter is enabled) and the authorization server raise `INSECURE_SIGNING_SECRET` rather than sign tokens anyone could forge. And because an HS256 key shorter than 32 bytes is weaker than the algorithm it feeds, a too-short HMAC secret fails the same way with `WEAK_SIGNING_SECRET`. There is no path to production with a guessable key.

CSRF protection is on. You did not enable it; it is enabled by default. The filter is *cookie-gated*: it only challenges unsafe requests that actually carry cookies, so Lumen's stateless, Bearer-token API traffic sails through untouched while the browser-facing dashboard is protected. Opt out per service with `pyfly.security.csrf.enabled: false`, or tighten to strict, always-on enforcement with `pyfly.security.csrf.cookie-gated: false`.

PKCE is on for the login flow. `ClientRegistration.use_pkce` defaults to `True`, public clients (those with no secret) are *forced* to use it regardless, and the authorization server's authorize step mandates the S256 method — there is no way to register an authorization-code client that skips proof-of-possession.

Resource-owner password grant is off. The legacy ROPC flow (`grant_type=password`), where an app collects the user's password and trades it for a token, is disabled on the Keycloak, Cognito, and Azure AD IdP adapters unless you

explicitly set `pyfly.idp.allow-password-grant: true`. The modern answer is the `authorization_code` + PKCE flow above; ROPC stays behind an opt-in for the rare legacy migration that still needs it.

```
# The opt-outs, gathered in one place — each one moves you OFF a safe default.
pyfly:
  security:
    csrf:
      enabled: false      # disable CSRF entirely (stateless-only
services)
      cookie-gated: false  # OR keep it on, but enforce strictly for
everyone
    idp:
      allow-password-grant: true # re-enable legacy ROPC (avoid unless forced)
```

The point is not that these defaults can never be relaxed — every one of them has a legitimate opt-out. The point is that relaxing them is a *visible, deliberate line in your config*, reviewed like any other change, instead of a safeguard you silently forgot to add.

Putting it together — Lumen's auth layer

The listing below shows the complete wiring: IDP adapter, JWT filter, URL-level rules, and a Redis session store for the admin dashboard.

lumen/config/security_full.py

```

from pyfly.container import bean, configuration
from pyfly.idp import IdpAdapter, KeycloakIdpAdapter
from pyfly.security.http_security import HttpSecurity

@configuration
class LumenSecurityConfig:

    @bean
    def idp_adapter(self) → IdpAdapter:
        return KeycloakIdpAdapter(
            base_url="https://keycloak.example.com",
            realm="lumen",
            client_id="lumen-backend",
            client_secret="${KEYCLOAK_SECRET}",
        )

    @bean
    def http_security(self) → HttpSecurity:
        hs = HttpSecurity()
        hs.authorize_requests() \
            .request_matchers(
                "/idp/login", "/idp/refresh",
                "/docs", "/openapi.json",
            ).permit_all() \
            .request_matchers(
                "/idp/admin/**"
            ).has_role("ADMIN") \
            .request_matchers(
                "/api/v1/wallets/**"
            ).authenticated() \
            .any_request().permit_all()
        return hs

```

Listing 14.16 — Full security configuration

...

With `pyfly.security.enabled=true`, `pyfly.session.enabled=true`, and `pyfly.session.store=redis` in `pyfly.yaml`, auto-configuration handles `JWTService`, `BcryptPasswordEncoder`, `SessionFilter`, and `RedisSessionStore`. The `@configuration` class above provides only what auto-configuration cannot infer: the Keycloak coordinates and the URL-level policy.

✓ What you built

This chapter opened Part V by closing Lumen's open front door. You:

- Used `JWTService` to issue signed tokens at login and to decode them back into a `SecurityContext` on every subsequent request. The `exp` claim is mandatory — `encode()` auto-adds it using a Unix timestamp so you never need to import `datetime`; tokens without `exp` are rejected at the boundary.
- Added `SecurityMiddleware` to the Starlette application so every request carries a populated `SecurityContext` by the time it reaches a handler.
- Declared URL-level policy with the `HttpSecurity` builder — a fluent DSL that produces an `HttpSecurityFilter` evaluated before the route dispatcher, covering Lumen's real `/api/v1/wallets/**` tree and the IDP admin routes.
- Protected Lumen's real wallet endpoints in `WalletController` with `@secure`, specifying roles, permissions, or full security expressions. The seven routes — `open_wallet`, `deposit`, `withdraw`, `list_wallets` (paged list), `list_rich_wallets` (filtered list), `wallet_balance`, and `wallet_detail` — each carry the minimal guard: `USER+ADMIN` for most, `wallet:deposit` permission for mutations, `ADMIN-only` for the full detail view. When `@secure` rejects a call it raises `SecurityException` (code `AUTH_REQUIRED`) for an unauthenticated caller or

`ForbiddenException` (code `FORBIDDEN`) for an authenticated caller who lacks the required role or permission — both render as HTTP `403` through the exception handler. The broader `HttpSecurity` gate, sitting in front of those handlers, is what returns the bare `401` for a missing or invalid token.

- Hashed passwords with `BcryptPasswordEncoder`, the default adapter for the `PasswordEncoder` protocol. The cost factor is tunable; the stored hash is self-describing; verification is timing-safe.
- Managed server-side sessions with `HttpSession` and the pluggable `SessionStore` protocol. In development, `InMemorySessionStore` requires no dependencies; in production, `RedisSessionStore` serialises to JSON and lets Redis manage TTL. The `SessionFilter` rolls the cookie TTL on every request and deletes it on invalidation.
- Validated tokens issued by an external IdP with the config-driven **OAuth2 resource server** (`JWSTokenValidator`). One block of `pyfly.security.oauth2.resource-server.*` config accepts Keycloak, Microsoft Entra ID, and AWS Cognito tokens out of the box: it verifies the `RS256` signature against the IdP's JWKS keys, checks `iss` / `aud` / `exp` with a 60-second clock-skew, and maps each provider's roles, scopes, and principal into the same `SecurityContext` via `ClaimMappings` — so `@secure` and `HttpSecurity` stay unchanged.
- Delegated identity to an external provider via the `IdpAdapter` port and the `KeycloakIdpAdapter` implementation. The auto-configured `IdpController` exposes login, refresh, logout, introspect, and admin user management under `/idp` with no extra code.
- Learned that the actuator and admin dashboard run on a **separate management port** (`pyfly.management.server.port`, default `9090`) that is **unauthenticated by default** for Spring parity, and that you secure it with `pyfly.management.security.enabled: true` (or disable it entirely with port `-1`).

Try it yourself

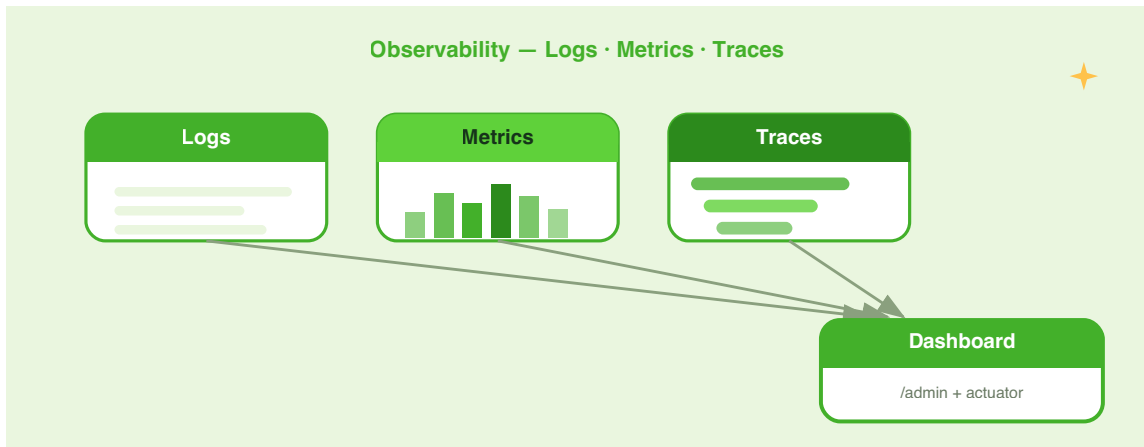
Exercise 1 — Role hierarchy. Lumen currently treats `ADMIN` and `USER` as independent roles. Add a "super-user" role `SUPER` that implicitly holds every `ADMIN` privilege. Implement a `RoleHierarchy` wrapper that pre-expands roles before storing them in `SecurityContext`, and update `_permissions_for` in Listing 14.1 so that a `SUPER`-role token passes every `@secure(roles=["ADMIN"])` check without explicitly carrying the `ADMIN` role. Write a unit test that creates a `SecurityContext(roles=["SUPER"])` after expansion and asserts `has_any_role(["ADMIN"])` returns `True`.

Exercise 2 — Session concurrency control. Lumen's admin users must not be logged in from more than two devices at the same time (a common financial compliance requirement). Enable `pyfly.session.concurrency.enabled: true` with `max-sessions: 2` and `strategy: evict-oldest` in `pyfly.yaml`. Write an integration test using `InMemorySessionStore` that creates three sessions for the same `user_id`, calls the `SessionConcurrencyController`, and asserts that the oldest session has been evicted while the two newest remain valid.

Exercise 3 — Custom IDP adapter. Lumen's staging environment uses a homegrown OAuth2 server. Implement a `StagingIdpAdapter` that satisfies `IdpAdapter`, backed by an in-memory user dictionary. The `login` method should issue a signed JWT using a `JWTService` injected through the constructor. Wire it as the `IdpAdapter` bean in a `@configuration` class tagged `@conditional_on_property("lumen.env", having_value="staging")` and confirm that the production `KeycloakIdpAdapter` bean is not created when that property is active.

CHAPTER 15

Observability & the Admin Dashboard



In Chapter 13 you surrounded Lumen's hot paths with caches and wrapped outbound calls in circuit breakers. In Chapter 14 you secured every endpoint with JWT authentication, role guards, and server-side sessions.

Lumen is now fast and safe — but it is still a black box. When a wallet deposit takes three seconds in production you need to know *where* those seconds went. When a downstream payment service degrades you need a dashboard that lights up red *before* your on-call engineer gets paged. When a compliance auditor asks why a particular transfer was rejected you need structured log records with enough context to reconstruct the decision.

This chapter adds eyes and ears to Lumen. The three pillars of observability answer three complementary questions about a running system:

Pillar	Question	PyFly module
Logging	"What happened, and in what context?"	<code>pyfly.logging</code>
Metrics	"How much? How fast? How often?"	<code>pyfly.observability.metrics</code>
Tracing	"What path did this request take?"	<code>pyfly.observability.tracing</code>

On top of those pillars sits the **Actuator** — production management endpoints that expose health, bean wiring, environment, and live logger levels — and the **Admin Dashboard**, an embedded browser UI that ties everything together in a single pane of glass.

 NEW IN V26.6.110

Two changes in this release shape how you reach everything in this chapter. First, the Actuator and Admin Dashboard now listen on a **separate management port** — `9090` by default — rather than the application port `8080`. Second, that management port is **open and unauthenticated by default** (Spring Boot's `management.server.port` model), so you can curl `http://localhost:9090/actuator/health` immediately, and you lock it down with `pyfly.management.security.enabled: true` when you ship. We will point out the right port at each step. If you have read older drafts that used `http://localhost:8080/actuator/...`, swap the port to `9090`.

Finally, you will see how **Aspect-Oriented Programming** (AOP) applies logging and metrics to every service method declaratively, without touching the methods themselves.

By the end of the chapter Lumen will produce structured JSON logs with correlation IDs and automatic PII masking, emit Prometheus metrics scraped by any standard collector, propagate OpenTelemetry trace spans across service boundaries, answer Kubernetes liveness and readiness probes, and display all of the above in a zero-configuration dashboard reachable at `/admin`.

HTML APPLICATIONS AND MODEL ADMINISTRATION

Appendix E follows the runnable catalog: `ModelAndView`, `Form[T]`, `reverse`, `static_url`, HTML errors, and CRUD over existing entities/documents. Data administration always requires authentication; a separate listener also requires `pyfly.management.security.enabled: true`.

Structured logging & PII redaction

Why structured logging?

Traditional log lines look like this:

```
[INFO] Order ord-123 created for customer acme corp (email: sales@acme.com)
```

Searching for `ord-123` in Elasticsearch works — until the format changes. And `sales@acme.com` landing in a log file may violate your data-protection policy without your team even noticing.

Structured logging replaces the interpolated string with an event name and explicit key-value pairs. The pairs render as JSON in production and as readable `key=value` in development. A log aggregation system ingests JSON natively; you query on `wallet_id` or `owner_id` as first-class fields, independent of message format.

get_logger

PyFly exposes a single factory function that returns a structured logger backed by `structlog` (when the `observability` extra is installed) or a zero-dependency stdlib shim otherwise. Both accept the same call signature.

JARGON: FACTORY FUNCTION

A *factory function* is just a function whose job is to build and hand back a configured object. `get_logger("lumen.wallet")` does the wiring for you — you never construct a logger by hand. The string you pass (`"lumen.wallet"`) is the *logger name*; it is conventionally the dotted module path, and it is what level overrides like `lumen.wallet:DEBUG` target.

Let us build a tiny example you can run, then graduate to the real wallet code. Follow these steps.

Step 1 — Create a throwaway demo module. Inside your Lumen project, create `src/lumen/logging_demo.py` with the contents of the listing below. (This is a scratch file for learning; delete it afterwards — the real logging lives inside the handlers later in the chapter.)

lumen/logging_demo.py

```

from pyfly.logging import get_logger

logger = get_logger("lumen.wallet")

logger.info("wallet_opened", wallet_id="wlt-001", owner_id="usr-42")
logger.warning("balance_low", wallet_id="wlt-001", remaining=300)
logger.error(
    "deposit_rejected",
    wallet_id="wlt-001",
    reason="insufficient_funds",
)

```

Listing 15.1 — Structured logger usage

Step 2 — Run it. Execute the module directly:

```
uv run python -m lumen.logging_demo
```

In development with `format: console` the output reads naturally:

```

10:30:00 [info    ] wallet_opened   wallet_id=wlt-001 owner_id=usr-42
10:30:01 [warning  ] balance_low    wallet_id=wlt-001 remaining=300
10:30:02 [error    ] deposit_rejected wallet_id=wlt-001
reason=insufficient_funds

```

Notice the shape: the *first* argument (`"wallet_opened"`) is the **event name**, not a sentence, and everything after it is a `key=value` pair. There is no string formatting, no f-string, no `%s`. That is the whole point of structured logging — the event name stays stable while the fields carry the variable data.

In production with `format: json` every line is a self-contained JSON object:

```
{
  "event": "wallet_opened",
  "wallet_id": "wlt-001",
  "owner_id": "usr-42",
  "timestamp": "2026-06-07T10:30:00Z",
  "level": "info",
  "logger": "lumen.wallet"
}
```

Configure logging in `pyfly.yaml`:

```
pyfly:
  logging:
    level:
      root: INFO
      lumen.wallet: DEBUG
      sqlalchemy.engine: WARNING
    format: console          # console | json | logfmt
```

`level.<name>` overrides the root level for any logger whose name begins with that prefix. `sqlalchemy.engine: WARNING` silences query logs without touching your code. An environment variable `PYFLY_LOGGING_LEVEL_ROOT=WARNING` overrides the config key, which is useful for staging builds.

What just happened. You called one function, `get_logger`, and got a logger that takes structured `key=value` fields. The `format` setting decides how those fields render: `console` for humans at your terminal, `json` for machines in production. You did not write any handler, formatter, or appender code — PyFly installed those on the root logger at startup. Flip `format: console` to `format: json` in `pyfly.yaml` and re-run the demo to see the same three events emitted as one JSON object per line, ready for a log aggregator to ingest.

💡 WHY NOT `STDLIB` `LOGGING` `DIRECTLY`?

`logging.getLogger("x").info("event", wallet_id="wlt-001")` raises `TypeError` — the `stdlib` rejects keyword arguments. `get_logger` guarantees the structured signature works regardless of which adapter is active.

Correlation IDs

Correlation IDs link every log line emitted during a single HTTP request. PyFly binds a `transaction_id` to the current async context automatically through `TransactionIdMiddleware`. Your handlers can bind additional fields — such as the authenticated user — so that those fields flow through all subsequent log calls without being passed explicitly:

`lumen/wallet/handler.py`

```
import structlog

from pyfly.logging import get_logger

logger = get_logger("lumen.wallet")

async def handle_deposit(wallet_id: str, amount: int, owner_id: str) → dict:
    structlog.contextvars.bind_contextvars(
        wallet_id=wallet_id,
        owner_id=owner_id,
    )

    logger.info("deposit_started", amount=amount)
    # ... business logic ...
    result = {"wallet_id": wallet_id, "new_balance": 1350}
    logger.info("deposit_completed", new_balance=result["new_balance"])

    structlog.contextvars.unbind_contextvars("wallet_id", "owner_id")
    return result
```

Listing 15.2 — Binding correlation context

Every `logger.*` call inside `handle_deposit` — including calls deep in downstream service methods — automatically carries `wallet_id` and `owner_id` without any further plumbing.

PII redaction

PII stands for *personally identifiable information* — emails, card numbers, national-ID numbers, and the like. **PII redaction** is enabled by default. Before any log record reaches an output handler, PyFly scans the rendered message for emails, credit-card numbers, IBANs, SSNs, JWTs, bearer tokens, and URL credentials. Detected patterns are replaced with `<EMAIL>`, `<CREDIT_CARD>`, and so on.

You can see this for yourself in a few seconds. Add one line to the demo from Step 1:

```
logger.info("contact_logged", email="alice@example.com")
```

Run `uv run python -m lumen.logging_demo` again. The output shows the value already masked — you never had to opt in:

```
10:30:03 [info    ] contact_logged email=<EMAIL>
```

That redaction pass runs for *every* logger in the process, including ones inside third-party libraries, which is why it catches leaks you did not write.

The regex engine ships with every install. The Presidio-backed NER engine — which also catches free-text names and addresses — is available via the `[pii]` extra and activates automatically when installed:

```
uv add "pyfly[observability,pii]"
python -m spacy download en_core_web_sm # lighter model; lg for higher
recall
```

Configure redaction in `pyfly.yaml`:


```

pyfly:
  logging:
    redaction:
      enabled: true           # default; set false to disable entirely
      engine: auto           # regex | presidio | auto (presidio if installed)
      mask: placeholder      # placeholder (<EMAIL>) | partial (****@acme.com)
      deny-fields:
        - password
        - token
        - secret
      presidio:
        score-threshold: 0.6
        languages: [en, es]

```

`deny-fields` lists structured log field *names* whose values are unconditionally replaced with `<REDACTED>`. Use it for fields like `password` where you know the value is sensitive without inspecting the content.

🌿 SPRING PARITY

Spring Boot does not ship built-in PII redaction; teams integrate Logback

`MaskingMessageConverter` or custom appenders manually. PyFly's redaction applies to *all* loggers — including third-party libraries — through a single `ProcessorFormatter` / `RedactionFilter` installed on the root handler. No per-library configuration is needed.

Rolling file appender

When logs go to a file rather than stdout, configure rotation in `pyfly.yaml`:

```
pyfly:
  logging:
    file:
      name: lumen.log
      path: ./logs
    rolling:
      max-size: 50MB
      max-history: 14
```

PyFly writes to `./logs/lumen.log` and rotates at 50 MB, keeping 14 rotated files before discarding the oldest. The same PII redaction pass applies to file output.

Metrics

The MetricsRegistry

A **metric** is a number you sample over time: a count of deposits, a latency in seconds, a memory figure. **Prometheus** is the de-facto open-source metrics database; it *scrapes* (periodically reads) an HTTP endpoint your app exposes and stores the numbers.

MetricsRegistry is PyFly's small front door to that world: a thin wrapper around the `prometheus_client` library that guarantees each metric name is registered only once. Duplicate calls to `counter()` or `histogram()` with the same name return the existing metric rather than raising a `ValueError`. Inject it from the DI container (auto-configured when `prometheus_client` is installed) or create it manually:

lumen/observability/metrics.py

```
from pyfly.observability import MetricsRegistry

registry = MetricsRegistry()

deposits_total = registry.counter(
    name="lumen.deposits.total",
    description="Deposit operations completed",
    labels=["status"],
)

deposit_duration = registry.histogram(
    name="lumen.deposits.duration",
    description="Deposit processing time in seconds",
    labels=["status"],
    buckets=(0.01, 0.05, 0.1, 0.25, 0.5, 1.0, 2.5, 5.0),
)
```

Listing 15.3 — Creating metrics

`counter()` and `histogram()` return native `prometheus_client.Counter` and `prometheus_client.Histogram` objects, so every Prometheus ecosystem tool — Grafana dashboards, alerting rules, recording rules — works without modification.

Method	Returns	Use for
<code>registry.counter(name, description, labels)</code>	<code>Counter</code>	Monotonically increasing counts
<code>registry.histogram(name, description, labels, buckets)</code>	<code>Histogram</code>	Durations, sizes, latency percentiles
<code>registry.counter(...)</code> called again	Same <code>Counter</code>	Safe deduplication

@timed — automatic duration histogram

`@timed` records how long an async or sync function takes to run, using a labeled histogram. It works on any callable and automatically adds `class`, `method`, and `exception` labels.

❗ JARGON: COUNTER VS HISTOGRAM

A **counter** only goes up — it answers "how many?" (deposits served, errors raised). A **histogram** buckets observed *values* — it answers "how long?" or "how big?" and lets Prometheus compute percentiles (p95 latency). Rule of thumb: count events with a counter; measure durations and sizes with a histogram.

Now let us wire the first real metric into Lumen. Here is the deposit handler you built in earlier chapters — the only change is the decorator on `do_handle`.

Step 1 — Import the metrics helpers. At the top of `src/lumen/core/services/wallets/deposit_funds_handler.py`, add `MetricsRegistry` and `timed` to the `pyfly.observability` import.

Step 2 — Create a module-level registry. Add `registry = MetricsRegistry()` above the class. Because the registry deduplicates by name, sharing one per module is safe.

Step 3 — Decorate `do_handle`. Put `@timed(...)` above `@transactional()` so the timer wraps the whole transactional unit of work.

lumen/core/services/wallets/deposit_funds_handler.py

```

from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker

from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.data.relational.sqlalchemy import transactional
from pyfly.domain import AggregateNotFound
from pyfly.eda import EventPublisher
from pyfly.observability import MetricsRegistry, timed

from lumen.core.mappers.wallet_mapper import to_aggregate, to_entity
from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.core.services.wallets.event_publishing import publish_domain_events
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository

registry = MetricsRegistry()

@command_handler
@service
class DepositFundsHandler(CommandHandler[DepositFunds, int]):
    """Credit funds to an existing wallet; returns the new balance."""

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events
        self._session_factory = session_factory

    @timed(registry, "lumen.deposit.duration", "Deposit handler latency")
    @transactional()
    async def do_handle(self, command: DepositFunds) → int:

```

```

entity = await self._repository.find_by_id(command.wallet_id)
if entity is None:
    raise AggregateNotFound("Wallet", command.wallet_id)
wallet = to_aggregate(entity)
wallet.deposit(Money(amount=command.amount, currency=wallet.currency))
await self._repository.upsert(to_entity(wallet))
await publish_domain_events(self._events, wallet.clear_events())
return wallet.balance.amount

```

Listing 15.4 — @timed on DepositFundsHandler

The decorator captures `start = time.perf_counter()`, calls the function, and observes the elapsed time in the `finally` block. The `exception` label is `"none"` on success and the exception type name on failure, so you can split latency by outcome in Grafana. The `class` and `method` labels are derived automatically from the function's qualified name.

Histogram names follow Micrometer dot.case convention: `"lumen.deposit.duration"` becomes `lumen_deposit_duration_seconds` in Prometheus, with a `_seconds` suffix added if absent.

Step 4 — Expose the Prometheus endpoint. By default the actuator web-exposes only `health` and `info` (see "The management port" below). Add `prometheus` to the exposure list in `pyfly.yaml` so the scrape endpoint appears:

```

pyfly:
  management:
    endpoints:
      web:
        exposure:
          include: "health,info,prometheus"

```

Step 5 — Run it and see the metric appear. Start Lumen, drive one deposit through the API, then scrape the management port.

```
# Terminal 1 – start the app (business API on 8080, management on 9090)
uv run pyfly run --server uvicorn

# Terminal 2 – open a wallet, then deposit into it
WALLET=$(curl -s -X POST localhost:8080/api/v1/wallets \
  -H 'Content-Type: application/json' \
  -d '{"owner_id":"usr-42","currency":"EUR"}' \
  | python -c "import sys,json;print(json.load(sys.stdin)['wallet_id'])")
curl -s -X POST localhost:8080/api/v1/wallets/$WALLET/deposit \
  -H 'Content-Type: application/json' -d '{"amount":1350}'

# Now scrape the metric – note the MANAGEMENT port 9090, not 8080
curl -s localhost:9090/actuator/prometheus | grep lumen_deposit_duration
```

You should see histogram lines like these (your numbers will differ):

```
lumen_deposit_duration_seconds_bucket{class="DepositFundsHandler",method="do_handle",exception="none"} 1.0
lumen_deposit_duration_seconds_count{class="DepositFundsHandler",method="do_handle",exception="none"} 1.0
lumen_deposit_duration_seconds_sum{class="DepositFundsHandler",method="do_handle",exception="none"} 0.013
```

The `_count` line confirms one observation; the `_sum` line is the total seconds spent. If `grep` finds nothing, you have not driven a deposit yet — the metric is created lazily on first call.

@counted — automatic invocation counter

`@counted` increments a counter on every function call. Lumen's `GetBalanceHandler` is a natural fit — every balance read increments the counter, tagged by outcome:

lumen/core/services/wallets/get_balance_handler.py

```

from pyfly.container import service
from pyfly.cqrs import QueryHandler, query_handler
from pyfly.observability import MetricsRegistry, counted

from lumen.core.mappers.wallet_mapper import entity_to_balance_dto
from lumen.core.services.wallets.get_balance_query import GetBalance
from lumen.interfaces.dtos.v1.balance_dto import BalanceDto
from lumen.models.repositories.wallet_repository import WalletRepository

registry = MetricsRegistry()

@query_handler
@service
class GetBalanceHandler(QueryHandler[GetBalance, BalanceDto | None]):

    def __init__(self, repository: WalletRepository) → None:
        super().__init__()
        self._repository = repository

    @counted(registry, "lumen.balance.reads", "Balance queries served")
    async def do_handle(self, query: GetBalance) → BalanceDto | None:
        entity = await self._repository.find_by_id(query.wallet_id)
        return entity_to_balance_dto(entity) if entity is not None else None

```

Listing 15.5 — @counted on GetBalanceHandler

On success the counter is incremented with labels `class="GetBalanceHandler"`, `method="do_handle"`, `result="success"`, and `exception="none"`. On failure it uses `result="failure"` and `exception=<TypeName>`, then re-raises the original exception. The counter name receives a `_total` suffix in Prometheus automatically, per the naming convention.

You can stack both decorators on the same method. The following listing shows Lumen's `WithdrawFundsHandler` timed and counted simultaneously:

```
lumen/core/services/wallets/withdraw_funds_handler.py
```

```
from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker

from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.data.relational.sqlalchemy import transactional
from pyfly.domain import AggregateNotFound
from pyfly.eda import EventPublisher
from pyfly.observability import MetricsRegistry, counted, timed

from lumen.core.mappers.wallet_mapper import to_aggregate, to_entity
from lumen.core.services.wallets.event_publishing import publish_domain_events
from lumen.core.services.wallets.withdraw_funds_command import WithdrawFunds
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository

registry = MetricsRegistry()

@command_handler
@service
class WithdrawFundsHandler(CommandHandler[WithdrawFunds, int]):
    """Debit funds from a wallet; returns the new balance in minor units."""

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
    ) → None:
        super().__init__()
        self._repository = repository
        self._events = events
        self._session_factory = session_factory

    @timed(registry, "lumen.withdrawal.duration", "Withdrawal latency")
    @counted(registry, "lumen.withdrawals", "Withdrawal attempts")
    @transactional()
```

```

async def do_handle(self, command: WithdrawFunds) → int:
    entity = await self._repository.find_by_id(command.wallet_id)
    if entity is None:
        raise AggregateNotFound("Wallet", command.wallet_id)
    wallet = to_aggregate(entity)
    wallet.withdraw(Money(amount=command.amount,
currency=wallet.currency))
    await self._repository.upsert(to_entity(wallet))
    await publish_domain_events(self._events, wallet.clear_events())
    return wallet.balance.amount

```

Listing 15.6 — Stacking `@timed` and `@counted`

`amount` is an `int` in minor units (e.g. 1050 = €10.50) — the `Money` value object enforces the type. Each invocation produces both a histogram observation and a counter increment.

What just happened. You added cross-cutting measurement to three handlers by changing nothing but the decorator stack. `@timed` answers "how long did it take?", `@counted` answers "how often, and did it succeed?", and stacking them gives you both for free. The handler bodies — the actual business logic — never mention metrics. After running a few deposits and withdrawals you can scrape `localhost:9090/actuator/prometheus` again and watch `lumen_withdrawals_total` and `lumen_balance_reads_total` climb.

Prometheus scrape endpoint

The actuator (covered in the next section) exposes the metrics registry for scraping. When the actuator is enabled and `prometheus_client` is installed, two endpoints are mounted automatically with no additional code:

- `GET /actuator/metrics` — Micrometer-compatible JSON listing all metric names
- `GET /actuator/prometheus` — standard text-exposition format for scraping

Point your Prometheus `scrape_configs` at `/actuator/prometheus` and all `MetricsRegistry` metrics appear alongside built-in process metrics (CPU, memory, threads, GC). By default the actuator listens on the **management port** (`9090`), not the application port (`8080`) — so the scrape target is `http://<host>:9090/actuator/prometheus`. See "The management port" below.

🌿 SPRING PARITY

`MetricsRegistry` mirrors Spring's `MeterRegistry` from Micrometer. `@timed` corresponds to Spring's `@Timed` and `@counted` to `@Counted`. Dot.case names (`lumen.deposit.duration`) match the Micrometer convention that Spring Boot Actuator's `/actuator/metrics` also exposes.

Server-layer observability

Everything you metered so far lives *inside* your application: `@timed` wraps a handler, `@counted` increments on a method call, `MetricsFilter` records `http_server_requests_seconds` as a request passes through the filter chain, and `process_metrics` samples CPU and memory. All of that describes the work Lumen does — but none of it describes the **server** running Lumen. How many connections is uvicorn holding open right now? How many requests are in flight *at the server layer*, before they ever reach a handler? How long has this worker been bound, as distinct from the Python process uptime? Until now those numbers were invisible.

i NEW IN V26.6.113

Server-layer observability adds a family of `server_*` metrics about the ASGI server itself — uvicorn, granian, or hypercorn — alongside the existing application-layer meters. They are on by default (enabled by the web and core starters, mirroring `pyfly.observability.metrics.enabled`), appear at `/actuator/prometheus` with no extra wiring, and feed a new live **Observability** view in the admin dashboard.

How it works — three cooperating sources

You might expect PyFly to simply read the server's own statistics. On the in-process `serve_async` path it does, but that path is not how you run in production.

`pyfly run --server uvicorn --workers 4` calls `uvicorn.run(workers=N)`, which **forks worker subprocesses** — each builds its *own* server object in its own process. The adapter bean your worker holds is not the object actually serving traffic, so its `server_state` is unreachable across the process boundary. PyFly therefore layers three mechanisms, each writing to the same Prometheus registry that `/actuator/prometheus` already exposes:

1. A pure-ASGI middleware (`ServerMetricsASGIMiddleware`) wraps the app at the **outermost** layer. This is the **primary** source: it runs in every worker, for every server type and worker count, and it sees every connection and request before any handler does. It emits `server_active_connections`, `server_in_flight_requests`, and `server_requests_total`.
2. A `ServerMetricsBinder`, started from the in-worker ASGI lifespan beside `register_process_metrics` and the `ManagementServer`, emits `server_workers` (read from the `_PYFLY_WORKERS` env var that `pyfly run` sets), `server_uptime_seconds` (since this worker bound), `server_started_total`, `server_stopped_total`, and optionally `server_native_connections`.
3. A best-effort `ServerStatsPort`, implemented per adapter. On the in-process `serve_async` path the uvicorn adapter surfaces its *true* socket connection count

and total request count from `uvicorn.Server.server_state`. The granian and hypercorn adapters report workers and uptime only — granian's Rust runtime and hypercorn's lack of a server handle leave the connection fields `None` — so native stats are enrichment, never the foundation.

The middleware is the uniform floor; native stats are best-effort icing on top. That split is why the numbers are always present, regardless of how many workers you fork or which server you chose.

The metric catalog

Every meter below carries two labels: `server` (the server type) and `worker_pid` (the process that emitted it). Prometheus names follow the usual exposition conventions.

Metric	Type	Meaning
<code>server_active_connections</code>	gauge	Open ASGI connections (HTTP + WebSocket). Approximate — not true sockets; idle keep-alive sockets the server holds are invisible to the ASGI layer.
<code>server_in_flight_requests</code>	gauge	HTTP requests currently being handled.
<code>server_requests_total</code>	counter	Completed HTTP requests at the server layer.
<code>server_workers</code>	gauge	Configured worker processes.
<code>server_uptime_seconds</code>	gauge	Seconds since this worker bound — distinct from <code>process_uptime_seconds</code> .
<code>server_started_total</code>	counter	Worker-start lifecycle events.
<code>server_stopped_total</code>	counter	Worker-stop lifecycle events.
<code>server_native_connections</code>	gauge	uvicorn's <i>true</i> socket connection count, including idle keep-alive. Present only on the <code>serve_async</code> path; absent for granian and hypercorn.

❗ TWO KINDS OF CONNECTION COUNT, TWO KINDS OF UPTIME

`server_active_connections` counts what the ASGI layer can see; an idle HTTP keep-alive socket holding no live connection is invisible to it, so this gauge under-counts what the kernel sees. `server_native_connections`, when available, is the server's own socket count and *does* include those idle keep-alives — which is exactly why it usually reads higher. Likewise, `server_uptime_seconds` measures how long *this worker* has been bound, while `process_uptime_seconds` measures the Python process; after a worker respawn they diverge.

Enabling it

Server-layer observability is on by default, so you do not have to do anything to get the `server_*` meters. The knobs let you tune the sample cadence, opt into native access logging, or turn the whole subsystem off:

```
pyfly:
  server:
    observability:
      enabled: true           # default; mirrors
                             pyfly.observability.metrics.enabled
      sample-interval-seconds: 5.0 # how often the binder samples gauges
      access-log: false         # opt-in native server access logging
```

Like the rest of the metrics stack, it requires the `observability` extra (`prometheus_client`); without it the subsystem degrades to a silent no-op rather than failing startup. Remember that `server_*` is exposed on the **management port** (`9090`) alongside the other actuator metrics — and that `prometheus` must be in your `pyfly.management.endpoints.web.exposure.include` list for the scrape endpoint to appear.

Run it — scrape the server metrics. Start Lumen and scrape the management port, filtering for the new family:

```
uv run pyfly run --server uvicorn
curl -s localhost:9090/actuator/prometheus | grep '^server_'
```

You should see exposition lines like these (your PIDs and numbers will differ):


```

server_workers{server="uvicorn",worker_pid="48211"} 1.0
server_uptime_seconds{server="uvicorn",worker_pid="48211"} 37.4
server_active_connections{server="uvicorn",worker_pid="48211"} 2.0
server_in_flight_requests{server="uvicorn",worker_pid="48211"} 1.0
server_requests_total{server="uvicorn",worker_pid="48211"} 128.0
server_started_total{server="uvicorn",worker_pid="48211"} 1.0
server_native_connections{server="uvicorn",worker_pid="48211"} 5.0

```

Notice `server_native_connections` reads higher than `server_active_connections` here: the extra sockets are idle keep-alive connections the kernel holds that the ASGI layer never sees. Drive a deposit on `8080` and re-scrape to watch `server_requests_total` climb. Switch to `--server granian` and re-scrape: the gauges and counters are still there, but `server_native_connections` is gone — granian's Rust runtime offers no handle to read it.

Multi-worker aggregation

With `--workers 1` a scrape is trivially complete: one worker, one set of mmapmed values. With `workers > 1` each forked worker would otherwise expose only *its own* numbers on *its own* listener, and a single scrape would see just one worker. PyFly closes that gap by auto-enabling `prometheus_client`'s multiprocess mode. Before forking, `pyfly run` sets `PROMETHEUS_MULTIPROC_DIR` (`pyfly/observability/multiprocess.py`); each worker writes its metrics to mmap files in that directory; and `/actuator/prometheus` aggregates across every worker through a `MultiProcessCollector`. The upshot: **one scrape reflects the whole fleet of workers**, and the `worker_pid` label lets you still break the numbers down per process.

⚠ CUSTOM COLLECTORS ARE NOT AGGREGATED

Multiprocess mode aggregates only `Counter`, `Gauge`, `Histogram`, and `Summary` values. Custom Python collectors — the `process_*` and `system_*` metrics — are *not* merged across workers, because the registry cannot serialize arbitrary collector state through the mmap files. The `server_*` and `http_server_requests_*` meters are ordinary counters and gauges, so they aggregate correctly; treat the per-process CPU and memory figures as single-worker samples when reading a multi-worker scrape.

The admin Observability view

Server metrics are also surfaced in the dashboard, under **Monitoring**, in a new live **Observability** view. It opens with stat cards for workers, uptime, active connections, in-flight requests, and requests-per-second; below them sit rolling charts of the same series, and a per-worker breakdown table keyed on `worker_pid` so you can spot one hot or stalled worker in a fleet. The view links across to the existing Metrics and Traces views for drill-down. It is backed by `GET /admin/api/observability` for the initial snapshot and the SSE stream `/admin/api/sse/observability` for live updates — the same push-not-poll model the rest of the dashboard uses.

📄 SCOPE: ASYNC-ONLY, GUNICORN-READY

This release keeps the server stack async-only ASGI — granian, then uvicorn, then hypercorn — and does **not** add gunicorn. The `ServerStatsPort` and the multiprocess design are deliberately gunicorn-ready, though, so a future sync-worker adapter can plug in without reworking the metric plumbing. For local development, `docker-compose.yml` now ships Prometheus and Grafana services that scrape `/actuator/prometheus` (config in `ops/prometheus/prometheus.yml`), so you can watch the `server_*` series on a dashboard end to end.

Distributed tracing

@span — OpenTelemetry span decorator

JARGON: TRACE, SPAN, OPENTELEMETRY

A **span** is one timed, named step of work — "fetch wallet", "persist deposit". A **trace** is the whole tree of spans for a single request, root to leaf. **OpenTelemetry** (often "OTel") is the vendor-neutral standard for producing traces; once your spans speak OTel, any compatible viewer — Jaeger, Tempo, Honeycomb — can display them. PyFly emits OTel spans so you are never locked to one tool.

`@span` wraps an async or sync function in an OpenTelemetry span. Each span is a timed, named unit of work. Spans nest automatically through OpenTelemetry's context propagation, so a `@span`-decorated function called from inside another

`@span`-decorated function produces a parent-child relationship in your trace viewer:

lumen/wallet/service.py

```

from pyfly.observability import span

class DepositFundsHandler:

    @span("deposit-funds")
    async def do_handle(self, command):
        balance = await self._fetch_wallet(command.wallet_id)
        await self._persist_deposit(command.wallet_id, command.amount)
        return balance + command.amount

    @span("fetch-wallet")
    async def _fetch_wallet(self, wallet_id: str) → int:
        # ... repository.find(wallet_id) ...
        return 1000

    @span("persist-deposit")
    async def _persist_deposit(self, wallet_id: str, amount: int) → None:
        # ... repository.add(wallet) ...
        pass

```

Listing 15.7 — @span on CQRS handler methods

In a trace viewer this appears as:

```

deposit-funds [120 ms]
  +-- fetch-wallet [15 ms]
  +-- persist-deposit [90 ms]

```

`@span` creates a tracer named `"pyfly"` via `trace.get_tracer("pyfly")`. When the decorated function raises, the span automatically records the error: it sets status to `ERROR`, calls `current_span.record_exception(exc)`, then re-raises so callers see the original exception unmodified. Sync functions are supported identically — no `await` on the decorated side.

OpenTelemetry auto-configuration

PyFly wires up a `TracerProvider` with a `BatchSpanProcessor` automatically when `opentelemetry-api` and `opentelemetry-sdk` are installed:

```
uv add opentelemetry-api opentelemetry-sdk opentelemetry-exporter-otlp
```

Configure the exporter in `pyfly.yaml`:

```
pyfly:
  observability:
    tracing:
      enabled: true
      service-name: "${pyfly.app.name}"
      exporter: otlp
      otlp:
        endpoint: "http://localhost:4318"
```

Run it — see spans without a collector. Standing up Jaeger or Tempo is overkill while you are learning. Set the **console exporter** instead, which prints each span to stdout. Lumen ships with `tracing.enabled: false`; flip it on and choose `console`:

```
pyfly:
  observability:
    tracing:
      enabled: true
      exporter: console
```

Restart with `uv run pyfly run --server uvicorn`, drive one deposit through the API as in the metrics step, and watch the terminal. Each `@span` prints a JSON block showing its name, the parent span id, and the elapsed time — the same parent-child structure the diagram above sketches, just in text form. Switch back to `exporter: otlp` (or remove the override) once you have a real collector.

Exporter selection rules:

- `exporter: otlp` — uses OTLP/HTTP; requires `opentelemetry-exporter-otlp`
- `exporter: console` — prints spans to stdout; useful for local debugging
- `exporter: none` — records spans but discards them (useful in tests)
- Unset — auto-selects OTLP if `pyfly.observability.tracing.otlp.endpoint` or the `OTEL_EXPORTER_OTLP_ENDPOINT` environment variable is configured; otherwise logs a single info line and discards spans silently

💡 CUSTOM TRACERPROVIDER

If you need gRPC export or a non-standard SDK configuration, register your own `TracerProvider` before PyFly starts — `trace.set_tracer_provider(...)`. PyFly detects an existing global provider and skips auto-configuration.

Context propagation — inbound and outbound

Spans stay correlated *within* a process automatically. Keeping the same trace *across* multiple services requires extracting the upstream trace context from inbound HTTP headers and injecting the current context into every outbound call.

PyFly handles both ends without any per-handler code.

Inbound — `TracingFilter`:

`TracingFilter` is wired into Lumen's filter chain immediately after `CorrelationFilter` by `create_app()`. For each request it reads the `W3C traceparent` header, opens a `SERVER` span as a child of the upstream context, and keeps that span active for the lifetime of the request. Every `@span` created during the request — and every log line — belongs to the caller's distributed trace:

```
# Simplified view of what TracingFilter does per-request:
parent = extract_context(request.headers) # parse W3C traceparent
with tracer.start_as_current_span(
    f"{request.method} {request.url.path}",
    context=parent,
    kind=trace.SpanKind.SERVER,
) as span:
    response = await call_next(request)
    span.set_attribute("http.response.status_code", response.status_code)
```

When OpenTelemetry is not installed, the filter is a transparent pass-through.

Outbound — HttpxClientAdapter:

`HttpxClientAdapter` calls `inject_headers()` on every outbound request so downstream services can continue the same trace:

`lumen/client/inventory_client.py`

```
from pyfly.client.adapters.httpx_adapter import HttpxClientAdapter

class InventoryClient:

    def __init__(self) → None:
        self._http = HttpxClientAdapter(
            base_url="http://inventory-service:8080"
        )

    async def check_stock(self, sku: str) → dict:
        # The active traceparent is injected automatically into
        # the outbound request headers – no manual plumbing required.
        resp = await self._http.request("GET", f"/skus/{sku}")
        return resp.json()
```

Listing 15.8 — Trace propagation

Logs carry `trace_id` and `span_id`:

`StructlogAdapter` registers a processor that stamps the active span's IDs on every log record. No code change required — any `get_logger(...)` call inside an active span gains `trace_id` and `span_id` fields automatically:

```
{
  "event": "deposit_completed",
  "wallet_id": "wlt-001",
  "new_balance": 1350,
  "trace_id": "1a4b3145ed8f2dd11172ee3584123f4a",
  "span_id": "d2a62aaa81b0ad66",
  "timestamp": "2026-06-07T10:30:00Z",
  "level": "info",
  "logger": "lumen.wallet"
}
```

With `trace_id` in every log record you can jump from a Grafana Loki search for `wallet_id=wlt-001` directly to the correlated Tempo trace view, and from there to the Prometheus latency charts for that time window — all three pillars joined on a single identifier.

The low-level propagation helpers are available if you ever need them directly:

```
from pyfly.observability.propagation import (
    extract_context, # parse traceparent from inbound headers
    inject_headers,  # write traceparent into outbound headers
    current_trace_ids, # → (trace_id, span_id) hex, or None
    has_otel,         # True if opentelemetry is importable
)
```


Health checks & the Actuator

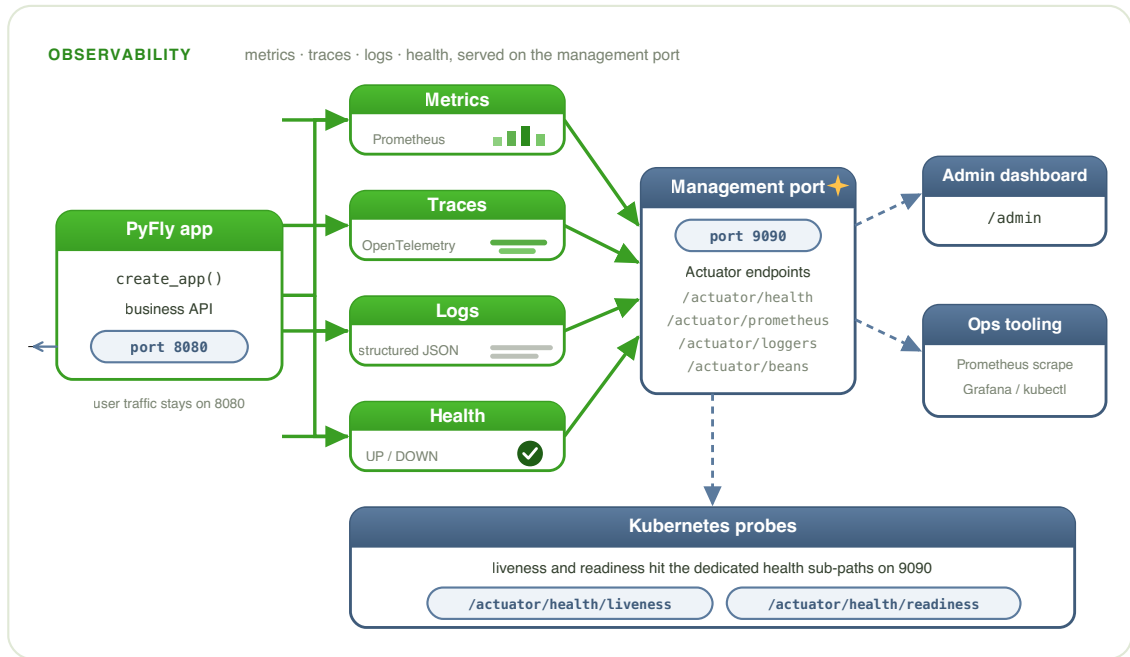


Figure 15.1 — The Actuator exposes health, beans, loggers, and Prometheus metrics over HTTP. Kubernetes liveness and readiness probes hit the dedicated sub-paths.

The **Actuator** gives Kubernetes and your ops tooling a stable contract: a set of management endpoints that expose health, bean wiring, environment state, and metric scrapers. You configure it once and every tool from `kubectl` to Grafana can consume it without custom code.

Enabling the Actuator

The Actuator is **on by default** when a PyFly context is present — you do not have to enable it at all to get `/actuator/health` and `/actuator/info`. You only touch the flag to turn it *off*, or to be explicit. Pass `actuator_enabled=True` to `create_app()`, or set the flag in `pyfly.yaml`:

```

pyfly:
  management:
    enabled: true          # default; the actuator is on unless you set
false
  app:
    name: lumen
    version: 1.0.0
    description: Lumen wallet service

```

❗ CONFIG KEY CHANGED

The enable flag is now `pyfly.management.enabled`. The older `pyfly.web.actuator.enabled` still works as a legacy alias, but new code should use the `pyfly.management.*` namespace, which is where the port, security, and endpoint-exposure settings also live.

When enabled, `create_app()` automatically scans the DI container for `HealthIndicator` beans, creates a `HealthAggregator`, instantiates all built-in endpoints, and mounts them at `/actuator/*`.

Run it — your first health check. Lumen already enables the actuator, so just start the app and curl the health endpoint on the **management port**:

```

uv run pyfly run --server uvicorn
curl -s localhost:9090/actuator/health

```

A healthy app returns HTTP 200 with:

```

{"status": "UP"}

```

If you get connection-refused, confirm you used `9090` (management), not `8080` (business). The same `/actuator/health` on `8080` returns 404 — the business port deliberately does not carry management endpoints.

The management port

By default these `/actuator/*` endpoints — and the `/admin` dashboard from the next section — are served on a **separate management port** (`9090`), not the application port (`8080`). This is Spring Boot's `management.server.port` model: keep health checks, Prometheus scraping, and the admin console off the public port, exposing only the business API to the internet while ops tooling reaches `9090` from inside the cluster.

The management port is a second in-process listener (not extra workers), so it adds no deployment complexity. Configure it via `pyfly.management.server.port` (env `PYFLY_MANAGEMENT_SERVER_PORT`): set it **equal** to `pyfly.server.port` to serve everything on one port, or to `-1` to disable the management web endpoints. A Kubernetes deployment therefore points liveness/readiness probes and the Prometheus `ServiceMonitor` at port `9090`, and the `Service / Ingress` for user traffic at `8080`.

⚠ THE MANAGEMENT PORT IS OPEN BY DEFAULT

As of v26.6.110 the management port is **unauthenticated by default** — the `/actuator/*` and `/admin` routes answer any caller that can reach `9090`. This is intentional (Spring Boot's model): the port is meant to be reachable only from inside your cluster, behind network isolation, never exposed on the public internet. If you cannot guarantee that isolation, turn on the app's security filters for the management port too:

```
pyfly:
  management:
    security:
      enabled: true
```

With that flag set, the same authentication, role guards, and CSRF rules that protect your business API also gate the management port.

By default the actuator web-exposes only `health` and `info` — again matching Spring Boot, which keeps potentially sensitive endpoints (`beans` , `env` , `threaddump` , `prometheus`) off the wire until you opt in. Widen the set with `pyfly.management.endpoints.web.exposure.include` :

```
pyfly:
  management:
    endpoints:
      web:
        exposure:
          include: "health,info,metrics,prometheus,loggers"  # or "*" for all
```

So the Prometheus scrape and runtime-logger steps later in this chapter assume you have added `prometheus` and `loggers` to this list. The `*` shortcut exposes everything and is convenient in development.

Built-in endpoints

Endpoint	Method	Description
<code>/actuator</code>	GET	HAL-style index of all enabled endpoints
<code>/actuator/health</code>	GET	Aggregated status: <code>UP</code> (200) or <code>DOWN</code> (503)
<code>/actuator/health/liveness</code>	GET	Kubernetes liveness probe sub-path
<code>/actuator/health/readiness</code>	GET	Kubernetes readiness probe sub-path
<code>/actuator/beans</code>	GET	All registered DI beans with stereotype and scope
<code>/actuator/env</code>	GET	Active configuration profiles
<code>/actuator/info</code>	GET	Application name, version, description
<code>/actuator/loggers</code>	GET, POST	List loggers; change levels at runtime
<code>/actuator/metrics</code>	GET	Micrometer-compatible JSON metric names
<code>/actuator/prometheus</code>	GET	Prometheus text-exposition scrape target
<code>/actuator/threaddump</code>	GET	Snapshot of all live threads and stack traces
<code>/actuator/refresh</code>	POST	Evict refresh-scoped beans; re-bind config

Custom HealthIndicator

Any `@component` bean with an `async def health(self) → HealthStatus` method is automatically discovered and registered as a health indicator. Lumen's `WalletRepository` is a good candidate — `count()` issues a lightweight `SELECT COUNT(*)` against the live database session without mutating any data:

lumen/health/indicators.py

```

from pyfly.actuator import HealthStatus
from pyfly.container import component

from lumen.models.repositories.wallet_repository import WalletRepository

@component
class WalletRepositoryHealthIndicator:
    """Checks the wallet store is reachable with a lightweight probe."""

    def __init__(self, repository: WalletRepository) → None:
        self._repository = repository

    async def health(self) → HealthStatus:
        try:
            # count() issues SELECT COUNT(*) – fast, read-only probe.
            total = await self._repository.count()
            return HealthStatus(
                status="UP",
                details={"store": "wallet-repository", "rows": total},
            )
        except Exception as exc:
            return HealthStatus(
                status="DOWN",
                details={"error": str(exc)},
            )

@component
class DatabaseHealthIndicator:
    """Checks database connectivity via a lightweight SELECT 1."""

    def __init__(self, session_factory) → None:
        self._factory = session_factory

    async def health(self) → HealthStatus:
        try:

```

```

    async with self._factory() as session:
        await session.execute("SELECT 1")
    return HealthStatus(
        status="UP",
        details={"type": "postgresql", "pool_active": 3},
    )
except Exception as exc:
    return HealthStatus(
        status="DOWN",
        details={"error": str(exc)},
    )

```

Listing 15.9 — HealthIndicator beans

`HealthStatus.status` accepts four values: `"UP"`, `"DOWN"`, `"OUT_OF_SERVICE"`, or `"UNKNOWN"`. The aggregator applies a severity ordering (`DOWN > OUT_OF_SERVICE > UP > UNKNOWN`) and returns the worst-case status across all indicators. If any indicator's `health()` method raises, that indicator is treated as `"DOWN"` with `details={"error": "check failed"}`; the exception is logged but does not crash the health endpoint.

Run it — watch a custom indicator surface. Drop the listing above into `src/lumen/health/indicators.py`, restart Lumen, and ask for the detailed health report (the `?show-details` form, or simply scrape it from the management port):

```
curl -s localhost:9090/actuator/health
```

A healthy response now includes your component by name:

```
{
  "status": "UP",
  "components": {
    "WalletRepositoryHealthIndicator": {
      "status": "UP",
      "details": {"store": "wallet-repository", "rows": 42}
    },
    "DatabaseHealthIndicator": {
      "status": "UP",
      "details": {"type": "postgresql", "pool_active": 3}
    }
  }
}
```

You wrote no registration code: the `@component` stereotype plus the `async def health()` method is the entire contract. At startup the actuator scanned the container, found anything that looks like a health indicator, and folded it into the aggregator.

BUILDING PROBES BY HAND — `APP.STATE.PYFLY_HEALTH_AGGREGATOR`

New in v26.6.110, the live `HealthAggregator` is reachable at `app.state.pyfly_health_aggregator` on the Starlette app object. This is the *same* aggregator the `/actuator/health` route uses, whether the actuator runs on the main app or on the separate management port. If you ever need a bespoke readiness gate — say, a custom ASGI route that returns 503 until a one-time warm-up completes — you can read this aggregator directly or register extra indicators on it after `create_app()`, instead of going through HTTP. It is exposed only by the Starlette adapter.

Changing log levels at runtime

The loggers endpoint lets you inspect and change log levels without restarting Lumen — invaluable when a production incident needs DEBUG output for exactly one package. First add `loggers` to the exposure list (`pyfly.management.endpoints.web.exposure.include: "health,info,loggers"`), then drive it from the **management port** `9090`:

```
# List all loggers with configured and effective levels
curl http://localhost:9090/actuator/loggers

# Enable DEBUG for the wallet module — takes effect immediately
curl -X POST http://localhost:9090/actuator/loggers/lumen.wallet \
  -H "Content-Type: application/json" \
  -d '{"configuredLevel": "DEBUG"}'

# Reset to inherit from parent
curl -X POST http://localhost:9090/actuator/loggers/lumen.wallet \
  -H "Content-Type: application/json" \
  -d '{"configuredLevel": null}'
```

What just happened. You changed a logger's level on a *running* process. The POST took effect immediately — no restart, no redeploy. In a real incident you would flip exactly one package to DEBUG, capture the noisy output you need, then POST `null` to put it back the way it was, all without disturbing the rest of the service.

The endpoint uses Spring Boot's level vocabulary (`OFF` , `ERROR` , `WARN` , `INFO` , `DEBUG` , `TRACE`) and is drop-in compatible with Spring Boot Actuator tooling.

Custom actuator endpoint

To expose a custom endpoint, implement the `ActuatorEndpoint` protocol and annotate the class with `@component`. PyFly discovers it during context startup and mounts it at `/actuator/{endpoint_id}` automatically:

lumen/actuator/git_info.py

```

from pyfly.container import component

@component
class GitInfoEndpoint:
    """Exposes build metadata at /actuator/git."""

    @property
    def endpoint_id(self) → str:
        return "git"

    @property
    def enabled(self) → bool:
        return True

    async def handle(self, context=None) → dict:
        return {
            "branch": "main",
            "commit": {
                "id": "5c6f83b",
                "time": "2026-06-07T08:30:00Z",
            },
            "build": {
                "version": "1.0.0",
            },
        }

```

Listing 15.10 — Custom actuator endpoint

Kubernetes probe configuration

Point your pod spec at the dedicated liveness and readiness sub-paths so Kubernetes can make independent restart and traffic decisions. Because the actuator lives on the management port, the probes target **9090**, while your **Service** routes user traffic to **8080**:

```

livenessProbe:
  httpGet:
    path: /actuator/health/liveness
    port: 9090
  initialDelaySeconds: 10
  periodSeconds: 30
readinessProbe:
  httpGet:
    path: /actuator/health/readiness
    port: 9090
  initialDelaySeconds: 5
  periodSeconds: 10

```

The separate sub-paths let you group indicators independently — an in-flight migration that degrades readiness need not trigger a liveness restart and container recreation.

SPRING PARITY

PyFly's Actuator mirrors Spring Boot Actuator. `HealthIndicator`, `HealthStatus`, `HealthAggregator`, `ActuatorEndpoint`, and `ActuatorRegistry` correspond directly to their Spring counterparts. The loggers endpoint uses the same Spring Boot level vocabulary and `configuredLevel` / `effectiveLevel` response shape, making it compatible with Spring Boot Admin and Actuator-aware tooling out of the box. `MetricsAutoConfiguration` and `MetricsActuatorAutoConfiguration` mirror Spring Boot's Micrometer auto-configuration: when `prometheus_client` is installed, `/actuator/prometheus` appears without any manual wiring.

The Admin Dashboard

The **Admin Dashboard** is a zero-build, zero-dependency browser UI served directly from the `pyfly.admin` package. One configuration line enables it; navigate to `/admin` — no separate server, no `npm` build step.

Enabling the dashboard

```
pyfly:
  admin:
    enabled: true
    title: "Lumen Admin"
    theme: auto          # auto | light | dark
    refresh_interval: 5000
```

The dashboard auto-discovers beans, health indicators, loggers, scheduled tasks, HTTP mappings, caches, CQRS handlers, sagas, and metrics from the running `ApplicationContext`. It presents them in **15 built-in views** with real-time Server-Sent Event (SSE) updates — no WebSocket, no polling loop in your code.

WHERE TO FIND IT: THE MANAGEMENT PORT

Like the actuator, the dashboard is served on the **management port**. With Lumen's defaults that is `http://localhost:9090/admin`, not `8080/admin`. Set `pyfly.management.server.port` equal to `pyfly.server.port` if you would rather serve the dashboard on the same port as your API. And remember: that port is open by default — gate it with `pyfly.management.security.enabled` or the dashboard's own `require_auth` (shown under "Security") before exposing it anywhere untrusted.

Run it — open the dashboard. Enable it in `pyfly.yaml`, start Lumen, and open the URL in a browser:

```
pyfly:
  admin:
    enabled: true
    title: "Lumen Admin"
```

```
uv run pyfly run --server uvicorn
# then visit http://localhost:9090/admin
```

You should see the Overview view populate within a second or two: app name and uptime, a green health badge, and bean counts grouped by stereotype. Drive a few deposits through `localhost:8080` and watch the Health and Metrics panels update live — the page never reloads, because the data arrives over SSE.

JARGON: SSE (SERVER-SENT EVENTS)

SSE is a one-way streaming channel: the browser opens a single long-lived HTTP connection and the server *pushes* events down it as they happen. It is simpler than a WebSocket (which is two-way) and is exactly the right tool for a dashboard that only needs to *receive* updates. You write no polling loop; the framework handles the stream.

Built-in views

Dashboard section:

View	Description
Overview	App info, uptime, health badge, bean counts by stereotype
Health	Component status with color-coded UP / DOWN / UNKNOWN badges; live SSE

Application section:

View	Description
Beans	All DI beans with stereotype, scope, and dependency graph
Environment	Active profiles and masked environment variables
Configuration	Resolved config tree for all namespaces with source tracking
Loggers	Logger levels with runtime level-change UI; TRACE and OFF supported

Monitoring section:

View	Description
Metrics	CPU, memory, threads, GC, uptime; optional Prometheus metrics; live trend chart
Scheduled Tasks	All <code>@scheduled</code> tasks with cron expressions and status
HTTP Traces	Request/response traces with p50/p90/p95/p99 latency, error-rate bar
Log Viewer	Live-tail with level filters, search, and pause/resume

Infrastructure section:

View	Description
Mappings	All HTTP routes with handler, parameters, return type, and docstring
Caches	Adapter type, entry count, per-key eviction, bulk evict-all
CQRS	Command and query handlers with bus pipeline introspection
Transactions	Saga step DAGs and TCC participant coverage; in-flight count

Fleet section (server mode):

View	Description
Instances	All registered remote application instances with health status

Real-time SSE streams

The dashboard never polls the backend with `setInterval`. It opens a single `EventSource` connection and the server pushes events as they occur:

SSE endpoint	Event name	What it streams
<code>/admin/api/sse/health</code>	<code>health</code>	Status change whenever aggregate health changes
<code>/admin/api/sse/metrics</code>	<code>metrics</code>	Full metric-name list at each refresh interval
<code>/admin/api/sse/traces</code>	<code>trace</code>	Individual HTTP traces as they arrive
<code>/admin/api/sse/logfile</code>	<code>log</code>	New log records from the in-memory ring buffer
<code>/admin/api/sse/beans</code>	<code>beans</code>	Bean registry snapshot at each refresh interval

The log viewer ring buffer holds 2,000 records; the HTTP traces ring buffer holds 500. Admin and actuator paths (`/admin/*` , `/actuator/*`) are excluded from trace capture automatically so they do not pollute the trace panel.

Runtime logger management

The Loggers view uses the same `/admin/api/loggers/{name}` endpoint as the actuator. Click a logger row to open an inline level selector and submit — the new level takes effect immediately, and the UI re-fetches to confirm the change. The Reset button sends `null` to return the logger to `NOTSET` (inherit from parent).

Custom view extension

To add your own sidebar view, implement `AdminViewExtension` and annotate with `@component`. The dashboard discovers it at startup:

lumen/admin/deployment_view.py

```

from pyfly.container import component

@component
class DeploymentView:
    """Shows deployment metadata in the admin sidebar."""

    @property
    def view_id(self) → str:
        return "deployments"

    @property
    def display_name(self) → str:
        return "Deployments"

    @property
    def icon(self) → str:
        return "upload-cloud"

    async def get_data(self, context=None) → dict:
        return {
            "last_deploy": "2026-06-07T08:00:00Z",
            "version": "1.0.0",
            "environment": "production",
        }

```

Listing 15.11 — Custom admin view

`view_id` sets the sidebar URL fragment (`#deployments`), `display_name` appears in the sidebar menu, and `icon` maps to a Feather icon. `get_data()` is called by `GET /admin/api/views` and can query the DI container, a database, or any external source.

Security

Restrict dashboard access to operators in production:


```
pyfly:
  admin:
    enabled: true
    require_auth: true
    allowed_roles:
      - ADMIN
      - OPS
```

When `require_auth: true`, every `/admin/api/*` route — data, mutation, SSE, and instance-registry endpoints — requires an authenticated principal whose roles overlap with `allowed_roles`. Unauthenticated requests receive `401`; authenticated users who lack every listed role receive `403`. The static SPA shell remains public so the dashboard can boot and display the error message.

Fleet monitoring — server mode

For a fleet of Lumen instances, run one dedicated admin-server and point every application instance at it:

```
# Admin server instance
pyfly:
  admin:
    enabled: true
  server:
    enabled: true
    poll_interval: 10000
    instances:
      - name: lumen-1
        url: http://lumen-1:8080
      - name: lumen-2
        url: http://lumen-2:8080
```

```
# Each application instance
pyfly:
  admin:
    enabled: true
    client:
      url: http://admin-server:8080
    auto_register: true
```

`StaticDiscovery` seeds the registry from the YAML list. `AdminClientRegistration` registers the instance on startup and deregisters on shutdown. HTTP calls use `httpx` when available and fall back to `urllib.request`; registration errors are silently swallowed so an unreachable admin server never aborts application startup.

🍃 SPRING PARITY

PyFly Admin maps directly to Spring Boot Admin. `server.enabled: true` replaces `@EnableAdminServer`. `client.url` replaces `spring.boot.admin.client.url`. The Vaadin/React frontend is replaced with a vanilla-JS SPA that requires no build tooling. SSE streams replace Spring Boot Admin's WebSocket notifications. The built-in Log Viewer replaces the Spring Boot Admin logfile viewer backed by `/actuator/logfile`; PyFly's ring buffer approach avoids the file-path configuration that Spring Boot Admin requires.

AOP for cross-cutting concerns

What is AOP?

Aspect-Oriented Programming separates cross-cutting concerns — logging, metrics, security, auditing — from business logic. Without AOP, every service method begins with `logger.info(...)` and ends with `metrics.increment(...)`. With AOP, you write that logic once in an `@aspect` class and apply it to every matching method via a pointcut expression — the methods themselves stay clean.

PyFly's AOP module ships five advice types:

Advice	Decorator	Runs
Before	<code>@before</code>	Before the target method
After returning	<code>@after_returning</code>	After the method succeeds
After throwing	<code>@after_throwing</code>	After the method raises
After (finally)	<code>@after</code>	Always, success or failure
Around	<code>@around</code>	Wraps the entire call; must call <code>jp.proceed()</code>

@aspect — declaring an aspect

📌 JARGON: ASPECT, ADVICE, POINTCUT, WEAVING

Four words travel together in AOP. An **aspect** is the class that bundles a cross-cutting concern. **Advice** is a single piece of behaviour inside it (the body of `@before`, `@around`, and so on). A **pointcut** is the pattern string — like `"service.*.*"` — that decides *which* methods the advice applies to. **Weaving** is the act of stitching the advice into those methods at startup. You write aspects; PyFly does the weaving.

`@aspect` marks a class as a PyFly aspect. The class is automatically registered in the DI container as a singleton and receives injected dependencies via `__init__`. No explicit base class is required.

Build the logging aspect in three moves.

Step 1 — Declare the class. Create `src/lumen/aspects/logging_aspect.py`, mark the class `@aspect`, and give it a module-level `logger`.

Step 2 — Add advice methods. Each method is decorated with one advice type (`@before`, `@after_returning`, `@after_throwing`) and a pointcut string. The `jp:JoinPoint` argument carries the intercepted call's details.

Step 3 — Set ordering. `@order(-50)` makes this aspect run earlier than higher-numbered ones — useful when you want logging to bracket the metrics aspect you write next.

lumen/aspects/logging_aspect.py

```

from pyfly.aop import aspect, before, after_returning, after_throwing,
JoinPoint
from pyfly.container.ordering import order
from pyfly.logging import get_logger

logger = get_logger("lumen.audit")

@aspect
@order(-50)
class AuditLoggingAspect:
    """Logs entry, exit, and failure for every service method."""

    @before("service.*.*")
    def log_entry(self, jp: JoinPoint) → None:
        logger.info(
            "method_called",
            cls=type(jp.target).__name__,
            method=jp.method_name,
        )

    @after_returning("service.*.*")
    def log_return(self, jp: JoinPoint) → None:
        logger.info(
            "method_returned",
            cls=type(jp.target).__name__,
            method=jp.method_name,
        )

    @after_throwing("service.*.*")
    def log_error(self, jp: JoinPoint) → None:
        logger.error(
            "method_raised",
            cls=type(jp.target).__name__,
            method=jp.method_name,

```

```
        exc=type(jp.exception).__name__,
    )
```

Listing 15.12 — A logging aspect

The pointcut `"service.*.*"` matches every public method on every `@service`-stereotype bean. `*` matches exactly one dot-separated segment; `**` matches one or more. Partial globs are supported within a segment: `"service.*.do_handle"` matches all `do_handle` methods on all service-stereotype handlers.

Qualified names follow the pattern `"{stereotype}.{ClassName}.{method_name}"`, so `service.DepositFundsHandler.do_handle` uniquely identifies the `do_handle` method on `DepositFundsHandler`.

💡 @BEFORE HANDLERS MUST BE SYNCHRONOUS

`@before`, `@after_returning`, `@after_throwing`, and `@after` handlers are always called synchronously by the weaver. Only `@around` handlers can be async (and must `await jp.proceed()` when advising an async method).

@around — metrics without decorators

`@around` is the most powerful advice type. It wraps the entire method execution; call `await jp.proceed()` to invoke the original method (or the next advice in the chain) and add behaviour on either side:

lumen/aspects/metrics_aspect.py

```

import time

from pyfly.aop import JoinPoint, around, aspect
from pyfly.container.ordering import order
from pyfly.observability import MetricsRegistry

registry = MetricsRegistry()

@aspect
@order(50)
class MetricsAspect:
    """Records duration and call counts for every service method."""

    @around("service.*.*")
    async def record_metrics(self, jp: JoinPoint):
        start = time.perf_counter()
        exc_name = "none"
        try:
            result = await jp.proceed()
            return result
        except Exception as exc:
            exc_name = type(exc).__name__
            raise
        finally:
            elapsed = time.perf_counter() - start
            histogram = registry.histogram(
                f"service.{jp.method_name}.duration",
                f"Duration of {jp.method_name}",
                labels=["exception"],
            )
            histogram.labels(exception=exc_name).observe(elapsed)

```

Listing 15.13 — @around metrics aspect

`@order(-50)` on `AuditLoggingAspect` and `@order(50)` on `MetricsAspect` ensure the logging aspect fires first in the advice chain. `HIGHEST_PRECEDENCE = -(2^31)` runs earliest; `LOWEST_PRECEDENCE = 2^31 - 1` runs last.

Automatic weaving — AspectBeanPostProcessor

In production you never call `weave_bean()` manually. `AopAutoConfiguration` registers `AspectBeanPostProcessor` unconditionally. During context startup the post-processor:

1. Collects every bean whose class has `__pyfly__aspect__ = True` into an `AspectRegistry`.
2. For every non-aspect bean, checks whether any registered pointcut matches any public method.
3. Wraps matching methods in-place with the full advice chain via `weave_bean()`.

The result is zero-configuration AOP: define aspects, define services, start the application — the weaver wires them together.

What just happened. You wrote two aspects — one for logging, one for metrics — and never edited a single handler. At startup, `AspectBeanPostProcessor` matched their pointcuts against your service beans and wove the advice into the matching methods in place. From now on, every `@service` method emits entry/exit logs and a duration histogram automatically. Add a new handler tomorrow and it inherits the same observability the moment its pointcut matches — nothing to remember, nothing to copy-paste. That is the payoff of AOP: the cross-cutting behaviour lives in one place, and the business code stays clean.

JoinPoint reference

Every advice handler receives a `JoinPoint` dataclass:

Attribute	Available in	Description
<code>target</code>	All	The bean instance being intercepted
<code>method_name</code>	All	Name of the intercepted method
<code>args</code>	All	Positional arguments passed to the method
<code>kwargs</code>	All	Keyword arguments passed to the method
<code>return_value</code>	<code>@after_returning</code> , <code>@after</code>	Return value (after success)
<code>exception</code>	<code>@after_throwing</code> , <code>@after</code>	The raised exception (or <code>None</code>)
<code>proceed</code>	<code>@around</code> only	Callable; <code>await</code> -able for async methods

Putting it together — full observability on `DepositFundsHandler`

Lumen's deposit handler with all three observability pillars applied — zero observability code inside the business logic:

lumen/core/services/wallets/deposit_funds_handler.py

```

from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker

from pyfly.container import service
from pyfly.cqrs import CommandHandler, command_handler
from pyfly.data.relational.sqlalchemy import transactional
from pyfly.domain import AggregateNotFound
from pyfly.eda import EventPublisher
from pyfly.logging import get_logger
from pyfly.observability import MetricsRegistry, counted, span, timed

from lumen.core.mappers.wallet_mapper import to_aggregate, to_entity
from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.core.services.wallets.event_publishing import publish_domain_events
from lumen.models.entities.v1.money import Money
from lumen.models.repositories.wallet_repository import WalletRepository

logger = get_logger("lumen.wallet")
registry = MetricsRegistry()

@command_handler
@service
class DepositFundsHandler(CommandHandler[DepositFunds, int]):
    """
    Credits funds to an existing wallet and returns the new balance
    (in minor units, e.g. 1350 = €13.50).

    Logging, metrics, and tracing are applied by decorators and aspects;
    the business logic here stays free of cross-cutting concerns.
    """

    def __init__(
        self,
        repository: WalletRepository,
        events: EventPublisher,
        session_factory: async_sessionmaker[AsyncSession],
    ) → None:

```

```

    super().__init__()
    self._repository = repository
    self._events = events
    self._session_factory = session_factory

    @timed(registry, "lumen.wallet.deposit.duration", "Deposit latency")
    @counted(registry, "lumen.wallet.deposits", "Total deposit attempts")
    @span("wallet-deposit")
    @transactional()
    async def do_handle(self, command: DepositFunds) → int:
        logger.info("deposit_started", wallet_id=command.wallet_id,
                    amount=command.amount)
        entity = await self._repository.find_by_id(command.wallet_id)
        if entity is None:
            raise AggregateNotFound("Wallet", command.wallet_id)
        wallet = to_aggregate(entity)
        wallet.deposit(Money(amount=command.amount, currency=wallet.currency))
        await self._repository.upsert(to_entity(wallet))
        await publish_domain_events(self._events, wallet.clear_events())
        logger.info("deposit_completed", wallet_id=command.wallet_id,
                    new_balance=wallet.balance.amount)
        return wallet.balance.amount

```

Listing 15.14 — *DepositFundsHandler with full observability*

How it works. `@span` opens an OpenTelemetry span. `@timed` records the `do_handle` duration. `@counted` increments the call counter. `AuditLoggingAspect` fires `@before` and `@after_returning` on every service method. `MetricsAspect` adds its own `@around` histogram. PII redaction strips sensitive values from log output automatically. The actuator exposes `/actuator/health`, `/actuator/prometheus`, and `/actuator/loggers`. The Admin Dashboard shows live traces and log records.

`command.amount` is an `int` in minor units — enforced by the `DepositFunds` command's validator (`amount > 0`). The `Money` value object wraps it with the wallet's `Currency`, preventing cross-currency arithmetic at the domain boundary.

Seven lines of decorators and one `get_logger` call — and Lumen's deposit path is fully observable.

Run it — confirm nothing broke. Observability decorators wrap behaviour around your handlers; they must not change the result those handlers return. Run the existing Lumen suite to prove the deposit path still behaves:

```
uv run --extra dev pytest -q
```

Every test should stay green:

```
..... [100%]
41 passed in 0.3s
```

Then do one full manual lap with the app running: drive a deposit on `8080`, confirm `/actuator/health` is `UP` on `9090`, scrape `/actuator/prometheus` and see `lumen_wallet_deposits_total` increment, and open `http://localhost:9090/admin` to watch the same event flow through the live Metrics and Log Viewer panels. All three pillars — logs, metrics, traces — now describe a single deposit, joined by one `trace_id`.

✓ What you built

You started with a production-ready but opaque service. By the end of this chapter Lumen:

- Emits **structured JSON logs** with correlation IDs, async-context-bound fields, and automatic PII redaction via regex and optional Presidio NER.

- Exports **Prometheus metrics** from `MetricsRegistry`, tagged with `@timed` duration histograms and `@counted` invocation counters, scraped at `/actuator/prometheus`.
- Propagates **OpenTelemetry traces** end-to-end: `TracingFilter` opens a SERVER span from the inbound `traceparent` header, `@span` creates child spans for handler calls, `HttpxClientAdapter` injects context into outbound requests, and `StructlogAdapter` stamps every log record with `trace_id` and `span_id`.
- Answers **Kubernetes health probes** at `/actuator/health/liveness` and `/actuator/health/readiness` via auto-discovered `HealthIndicator` beans, including a `WalletRepositoryHealthIndicator` that probes the wallet store.
- Surfaces all of the above in the **embedded Admin Dashboard** at `/admin`, with 15 built-in views, real-time SSE streams, a live log tail, an HTTP trace analytics panel, and runtime logger-level management.
- Applies logging and metrics **cross-cuttingly** via `@aspect`, `@before`, `@after_returning`, `@after_throwing`, and `@around`, woven automatically by `AspectBeanPostProcessor` without touching handler code.

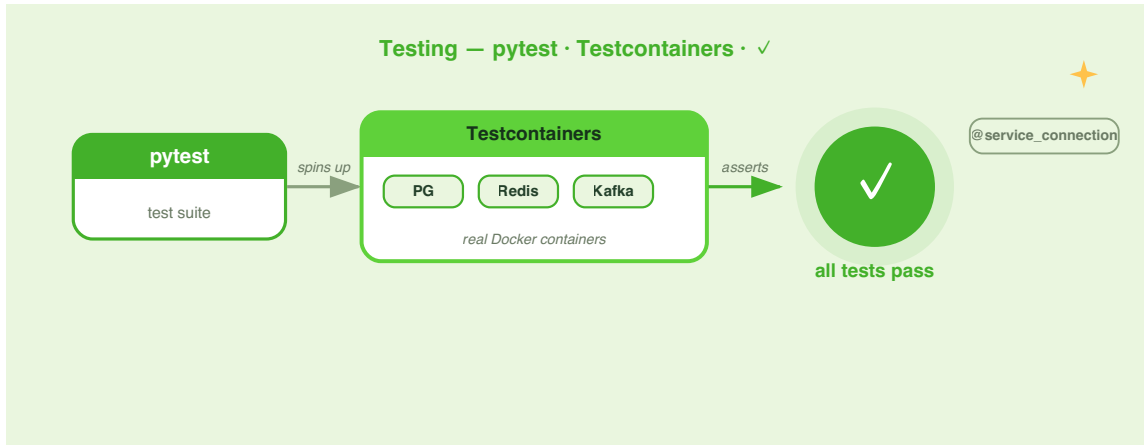
Try it yourself

1. **PII redaction audit.** Add a log statement to `DepositFundsHandler.do_handle` that includes a fake email address as a field value (`customer_email="alice@example.com"`) and a `token` field with an arbitrary string. Run Lumen locally with `format: console`, observe that the email is replaced with `<EMAIL>` and the token field value is replaced with `<REDACTED>`. Then switch `engine: presidio` (after installing `pyfly[pii]` and `en_core_web_sm`) and compare the output.

2. **Custom HealthIndicator.** Write a `StripeHealthIndicator` that calls `https://status.stripe.com/api/v2/status.json` with `httpx`, parses `indicator.status`, and returns `UP` if the value is `"none"` or `DOWN` otherwise. Register it as a `@component` and verify that `/actuator/health` includes a `StripeHealthIndicator` component. Test with a mocked `httpx` call that raises `httpx.ConnectError` and verify the aggregated status becomes `DOWN`.
3. **Metrics aspect with per-method thresholds.** Extend `MetricsAspect` from Listing 15.13 with a configurable `slow_threshold` (default 0.5 s) injected from `pyfly.yaml` via `@config_properties`. When a service method exceeds the threshold, emit a `logger.warning("slow_method", ...)` with `method_name` and `elapsed` fields. Write a pytest test that uses a `FakeClock` or `unittest.mock.patch("time.perf_counter")` to simulate a slow call and asserts the warning is logged.

CHAPTER 16

Testing PyFly Applications



The wallet works. Deposits land, balances update, events propagate through the bus, and the saga coordinator rolls back cleanly on failure. What you do not yet have is **confidence** that it will keep working after the next refactor. That confidence comes from tests — tests that run in milliseconds, prove the domain model enforces its invariants, verify the CQRS pipeline dispatches correctly, exercise the repository's derived queries and Specification predicates against a real SQLite database, and boot the full application context in an integration test that proves the entire DI + persistence composition.

PyFly treats testing as a first-class concern. The `pyfly.testing` module ships higher-level helpers — `PyFlyTestCase`, `create_test_container`, `assert_event_published`, Testcontainers wiring — that you can reach for when you

need them. Lumen's own test suite does not use them: it wires real components directly from `conftest.py`, uses standard pytest fixtures, and covers every layer of the pyramid with no boilerplate. That is the approach this chapter teaches.

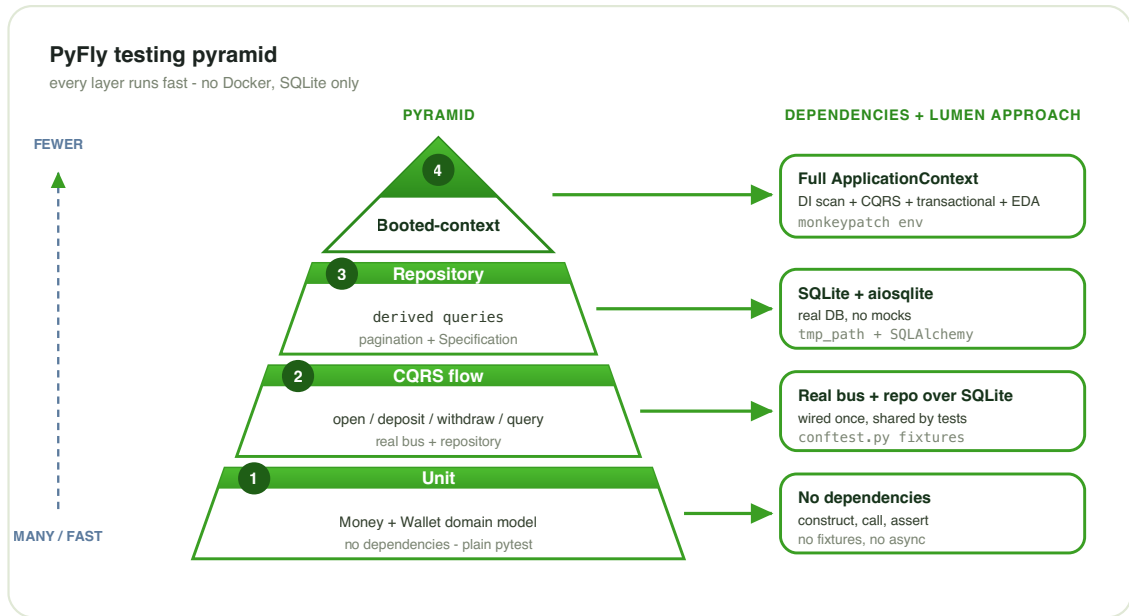


Figure 16.1 — PyFly's testing pyramid. Fast unit tests form the wide base; integration tests sit in the middle; real-DB adapter tests and a booted-context integration test crown the top.

The pyramid has four levels. **Unit tests** sit at the base — many of them, running in milliseconds, exercising the domain model with no dependencies. **CQRS flow tests** occupy the next tier — the full open/deposit/withdraw/query cycle routed through the real bus and the real repository, all wired in `conftest.py`. **Repository tests** exercise derived queries, pagination, and Specification predicates against SQLite. At the peak, a **booted-context integration test** starts the real `ApplicationContext` — DI scan, CQRS auto-config, `RepositoryBeanPostProcessor`, `@transactional` seam, EDA — and drives the full lifecycle.

Level	Dependencies	Speed	Lumen approach
Unit	None	Fast	plain pytest, no fixtures
CQRS flow	Real bus + repository over SQLite	Fast	conftest.py fixtures
Repository	SQLite + aiosqlite	Fast	<code>tmp_path</code> + SQLAlchemy
Booted-context	Full ApplicationContext + SQLite	Fast	<code>monkeypatch</code> env override

The project uses pytest with `pytest-asyncio` in **auto mode**. Enable it once in `pyproject.toml` and every `async def test_*` function is collected automatically:

```
[tool.pytest.ini_options]
asyncio_mode = "auto"
testpaths = ["tests"]
pythonpath = ["src"]
```

A quick gloss before we start, since three terms recur in this chapter. A **fixture** is a reusable piece of test setup — pytest builds it once, hands it to your test, and tears it down afterward. A **conftest.py** is a special file pytest discovers automatically; any fixtures you declare there become available to every test in the package without an import. And `pytest-asyncio` **auto mode** simply means you can write `async def test_...` and pytest will `await` it for you — no per-test decorator required. Each of these appears again with a concrete example below; this is just the map.

Install dev dependencies and run the suite:

```
uv run --extra dev pytest -q
```

The bare `uv sync` (without `--extra dev`) omits the dev group, so pytest is not installed. Always include `--extra dev` when running tests.

Run it. From the `samples/lumen` directory, run the whole suite once now so you have a baseline before changing anything:

```
uv run --extra dev pytest -q
```

You should see a row of dots — one per test — followed by a summary line:

```
.....
[100%]
41 passed in 0.28s
```

Forty-one passing tests, under a third of a second, no Docker and no external process. That speed is the whole point of the pyramid: the fast base catches most regressions before the slower integration layers ever run. If you instead see `No module named pytest`, you forgot `--extra dev` — re-run with it.

Unit-testing the domain

The domain model — `Money` and `Wallet` — has no framework dependencies. It never touches a database, a message bus, or an HTTP client. That purity makes it the ideal target for fast unit tests: construct objects, call methods, assert outcomes. No mocks, no fixtures, no `async`.

Testing Money

`Money` is a frozen dataclass. Every operation either succeeds and returns a new `Money` instance, or raises `BusinessRuleViolation`. Each violation carries a `.rule` string that names the violated invariant — useful for asserting the exact rule in tests.

A quick gloss on two terms. A **value object** is an object defined entirely by its values — two `Money(1050, EUR)` instances are equal because their fields are equal, not because they are the same object in memory. A **frozen dataclass** is Python's way of making such an object immutable: once constructed, you cannot reassign its fields. Together they make `Money` safe to pass around freely — no caller can mutate it behind your back, so it never needs defensive copying.

Let us build the test file one assertion-group at a time. Each step below maps to one `def test_...` function in the listing that follows.

Step 1 — equality is structural. `test_value_equality_is_structural` asserts that two `Money` values with the same amount and currency are equal, and that differing in either field makes them unequal. This is the value-object contract.

Step 2 — immutability is enforced. `test_money_is_immutable` tries to assign to `money.amount` and expects an exception. The frozen dataclass raises `FrozenInstanceError`, proving you cannot mutate a value after construction.

Step 3 — arithmetic returns new values. `test_add_and_subtract_same_currency` checks that `add` and `subtract` produce the expected `Money`, never mutating the operands.

Step 4 — the convenience surface. `test_zero_factory_and_major_units` covers the `Money.zero(currency)` factory, the `major_units` property, and the `__str__` formatting.

Step 5 — invariants reject bad input. The last two tests assert that mixing currencies and passing a non-integer amount each raise `BusinessRuleViolation` with a specific `.rule` string — `"money-currency-mismatch"` and `"money-amount-integer"`.

tests/test_money.py

```

from __future__ import annotations

import pytest
from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.money import Money

from pyfly.domain import BusinessRuleViolation


def test_value_equality_is_structural() → None:
    assert Money(1050, Currency.EUR) == Money(1050, Currency.EUR)
    assert Money(1050, Currency.EUR) != Money(1050, Currency.USD)
    assert Money(1050, Currency.EUR) != Money(999, Currency.EUR)


def test_money_is_immutable() → None:
    money = Money(1050, Currency.EUR)
    with pytest.raises(Exception): # frozen dataclass → FrozenInstanceError
        money.amount = 0 # type: ignore[misc]


def test_add_and_subtract_same_currency() → None:
    a = Money(1050, Currency.EUR)
    b = Money(450, Currency.EUR)
    assert a.add(b) == Money(1500, Currency.EUR)
    assert a.subtract(b) == Money(600, Currency.EUR)


def test_zero_factory_and_major_units() → None:
    assert Money.zero(Currency.USD) == Money(0, Currency.USD)
    assert Money(1050, Currency.EUR).major_units == 10.5
    assert str(Money(1050, Currency.EUR)) == "10.50 EUR"


def test_currency_mismatch_is_rejected() → None:
    with pytest.raises(BusinessRuleViolation) as exc:
        Money(100, Currency.EUR).add(Money(100, Currency.USD))

```

```

assert exc.value.rule == "money-currency-mismatch"

def test_non_integer_amount_is_rejected() → None:
    with pytest.raises(BusinessRuleViolation) as exc:
        Money(10.5, Currency.EUR) # type: ignore[arg-type]
    assert exc.value.rule == "money-amount-integer"

```

Listing 16.1 — Pure unit tests for the Money value object

Every test is synchronous — no `async`, no `await`, no fixtures. Pytest collects the module-level functions automatically. `Currency.EUR` is an enum value, not a plain string, matching the domain model's type contract exactly.

Run it. Run just this file to see the unit base of the pyramid in action:

```
uv run --extra dev pytest tests/test_money.py -q
```

Expected output:

```

.....
[100%]
6 passed in 0.02s

```

Six tests, twenty milliseconds. No database connected, no event bus started — these tests construct a plain object and assert on its behaviour. That is what makes the base of the pyramid so wide and so fast.

What just happened. You proved the entire `Money` contract — equality, immutability, arithmetic, and every invariant — without touching a single piece of framework infrastructure. Pure-domain code stays pure-domain testable. When one of these fails, you know the bug is in the value object itself, not in wiring, a session, or the bus.

 MINOR-UNIT ARITHMETIC

`Money` stores amounts in **minor units** (integer cents). `Money(1050, Currency.EUR)` represents €10.50 — verified by `major_units == 10.5` and `str(...) == "10.50 EUR"`. The `Money.zero(currency)` factory returns a `Money(0, currency)`, useful for initialising wallet balances.

Testing the Wallet aggregate

`Wallet` enforces several invariants: the owner must be a non-blank string, deposits must be positive, withdrawals must not overdraw, and amounts must match the wallet's currency. Each violation carries a `.rule` attribute for precise assertion.

One term first. An **aggregate** (or **aggregate root**) is a cluster of domain objects treated as a single unit for consistency — here, the `Wallet` and its balance. Every change goes through the aggregate's methods, so the aggregate is the one place that guarantees its invariants hold. That makes it the natural unit to test: drive it through its public methods and assert the rules never break.

The pattern every aggregate test follows is **arrange, act, assert**. Arrange: construct the wallet into a known state. Act: call one method. Assert: check the balance, the emitted event, or the raised violation. Watch for it in each test:

Step 1 — opening emits an event. `test_open_creates_empty_wallet` opens a wallet and asserts the balance is zero and exactly one `WalletOpened` event is queued.

Step 2 — opening validates its arguments. `test_open_requires_owner` passes a blank owner and expects `BusinessRuleViolation` with rule `"wallet-owner-required"`.

Step 3 — the happy path of deposit then withdraw.

`test_deposit_then_withdraw_happy_path` deposits, asserts the balance and the `FundsDeposited` event, then withdraws and asserts the balance and the `FundsWithdrawn` event. Note the `clear_events()` call in the arrange step — more on that just below the listing.

Step 4 — invariants reject bad operations. The final three tests prove a withdrawal cannot overdraw, a deposit must be positive, and an amount must match the wallet's currency. Each asserts the exact `.rule` string, and the overdraw test also asserts the balance was left unchanged and no event was raised — proof the invariant fired *before* any state changed.

tests/test_wallet_aggregate.py

```

from __future__ import annotations

import pytest
from lumen.interfaces.enums.v1.currency import Currency
from lumen.models.entities.v1.money import Money
from lumen.models.entities.v1.wallet_entity import (
    FundsDeposited,
    FundsWithdrawn,
    Wallet,
    WalletOpened,
)

from pyfly.domain import BusinessRuleViolation


def test_open_creates_empty_wallet() → None:
    wallet = Wallet.open("wlt-1", "owner-1", Currency.EUR)
    assert wallet.owner_id == "owner-1"
    assert wallet.currency is Currency.EUR
    assert wallet.balance == Money.zero(Currency.EUR)
    [event] = wallet.pending_events()
    assert isinstance(event, WalletOpened)
    assert event.wallet_id == "wlt-1"
    assert event.currency == "EUR"


def test_open_requires_owner() → None:
    with pytest.raises(BusinessRuleViolation) as exc:
        Wallet.open("wlt-x", " ", Currency.EUR)
    assert exc.value.rule == "wallet-owner-required"


def test_deposit_then_withdraw_happy_path() → None:
    wallet = Wallet.open("wlt-2", "owner-2", Currency.EUR)
    wallet.clear_events()

    wallet.deposit(Money(1000, Currency.EUR))

```



```

assert wallet.balance == Money(1000, Currency.EUR)
[event] = wallet.clear_events()
assert isinstance(event, FundsDeposited)
assert event.amount == 1000
assert event.balance == 1000

wallet.withdraw(Money(400, Currency.EUR))
assert wallet.balance == Money(600, Currency.EUR)
[event] = wallet.clear_events()
assert isinstance(event, FundsWithdrawn)
assert event.amount == 400
assert event.balance == 600

def test_withdraw_cannot_overdraw() → None:
    wallet = Wallet.open("wlt-3", "owner-3", Currency.EUR)
    wallet.deposit(Money(500, Currency.EUR))
    wallet.clear_events()
    with pytest.raises(BusinessRuleViolation) as exc:
        wallet.withdraw(Money(501, Currency.EUR))
    assert exc.value.rule == "wallet-insufficient-funds"
    # invariant held: balance unchanged, no event raised
    assert wallet.balance == Money(500, Currency.EUR)
    assert wallet.pending_events() == []

def test_deposit_must_be_positive() → None:
    wallet = Wallet.open("wlt-4", "owner-4", Currency.EUR)
    with pytest.raises(BusinessRuleViolation) as exc:
        wallet.deposit(Money(0, Currency.EUR))
    assert exc.value.rule == "wallet-deposit-positive"

def test_currency_mismatch_is_rejected() → None:
    wallet = Wallet.open("wlt-5", "owner-5", Currency.EUR)
    with pytest.raises(BusinessRuleViolation) as exc:
        wallet.deposit(Money(100, Currency.USD))
    assert exc.value.rule == "wallet-currency-mismatch"

```

Listing 16.2 — Unit tests for the Wallet aggregate root

Three details deserve attention. First, `Wallet.open` takes three positional arguments: a pre-generated `wallet_id`, an `owner_id`, and a `Currency` enum value — the aggregate does not generate its own ID. Second, `pending_events()` returns buffered events without draining; `clear_events()` returns and drains. The `test_deposit_then_withdraw_happy_path` test calls `clear_events()` after opening so each assertion sees exactly one event. Third, `FundsDeposited` and `FundsWithdrawn` carry `amount` (the operation amount in minor units) and `balance` (the post-operation running balance) — not `new_balance`. Always verify real event dataclass fields before asserting them.

Run it.

```
uv run --extra dev pytest tests/test_wallet_aggregate.py -q
```

Expected output:

```
.....
[100%]
6 passed in 0.02s
```

What just happened. The trickiest line to read is the assignment `[event] = wallet.clear_events()`. That is a list unpacking: it asserts the returned list has **exactly one** element and binds it to `event` in a single step. If the aggregate had raised zero or two events, the unpacking itself would raise a `ValueError` and the test would fail — so the shape of the event stream is checked for free. This is why the happy-path test calls `clear_events()` right after opening: it drains the `WalletOpened` event so the next unpacking sees only the `FundsDeposited` event you are actually asserting on.

 SPRING PARITY

Testing a DDD aggregate in isolation is the same discipline in any stack. In Spring / jMolecules you would call the aggregate's methods directly and check `aggregate.domainEvents()` (provided by `AbstractAggregateRoot`) before calling `afterDomainEventPublication()` to drain the buffer. PyFly's `clear_events()` plays the same role — drain, assert, move on.

Wiring the test stack with `confest.py`

The CQRS and event-listener tests need real infrastructure: a SQLite-backed `WalletRepository`, an event bus, command and query handlers, and a running bus. Rather than recreating this in every test module, Lumen declares the wiring once in `tests/confest.py`. Pytest discovers the file automatically and makes the fixtures available to every test in the package.

The key difference from a hand-rolled adapter test is that the `repository` fixture uses the **real framework** `WalletRepository` — the same Spring-Data-style class the application boots — and runs it through the **real** `RepositoryBeanPostProcessor`, which compiles derived-query stubs from method names at startup.

This file is the heart of the chapter, so we will read it top to bottom as a series of small, layered steps. Each fixture builds on the one above it. The thing to keep in mind: pytest links fixtures by **name**. When a fixture function declares a parameter, pytest looks for a fixture with that name, builds it, and injects it. That is how a small graph of independent fixtures composes into a full test stack.

Step 1 — make the sample importable. The first few lines push the sample's `src/` onto `sys.path` so `import lumen...` resolves. The `pythonpath = ["src"]` setting in `pyproject.toml` does the same thing for pytest's own collection; this line covers direct imports inside `conftest.py` before pytest's path tweaks apply.

Step 2 — `session_factory` : a database engine. This async fixture creates an in-memory SQLite engine, runs `Base.metadata.create_all` to build the schema, and yields an `async_sessionmaker`. A **session factory** is a callable that hands out fresh database sessions; the framework's relational auto-configuration creates one exactly like this at startup. The single shared engine keeps the in-memory database alive for the whole test — close it and the data vanishes.

Step 3 — `repository` : the real Spring-Data-style repository. It constructs the framework `WalletRepository`, then calls `RepositoryBeanPostProcessor().after_init(repo, "walletRepository")`. A **post-processor** is a hook that runs against a freshly created bean — here it reads method names like `find_by_owner_id` and compiles them into real queries. Skip this call and those methods stay stubs that raise `NotImplementedError`.

Step 4 — `event_bus` : the in-memory event bus. A one-line fixture that yields an `InMemoryEventBus` — the same publisher the application uses in non-Kafka deployments.

Step 5 — `audit_listener` : a subscriber on that bus. It creates the `WalletAuditListener` and subscribes its handler to the bus, reading the event patterns straight off the decorated method's `__pyfly_event_patterns__` attribute — exactly what the `ApplicationContext` does when it auto-wires listeners at startup.

Step 6 — `command_bus` and `query_bus` : the CQRS dispatchers. Each builds a `HandlerRegistry`, registers the real handlers (passing them the `repository`, `event_bus`, and `session_factory` they need), and yields a bus. **CQRS** — Command/

Query Responsibility Segregation — simply means writes go through a command bus and reads through a query bus; a **handler** is the function that actually processes one command or query.

tests/conftest.py

```

from __future__ import annotations

import sys
from collections.abc import AsyncIterator
from pathlib import Path

import pytest_asyncio
from sqlalchemy.ext.asyncio import (
    AsyncSession,
    async_sessionmaker,
    create_async_engine,
)

# Make the sample's `src/` importable
_HERE = Path(__file__).resolve().parent
_SRC = _HERE.parent / "src"
sys.path.insert(0, str(_SRC))

from lumen.core.services.listeners import WalletAuditListener
from lumen.core.services.wallets import (
    DepositFundsHandler,
    GetBalanceHandler,
    GetWalletHandler,
    ListRichWalletsHandler,
    ListWalletsHandler,
    OpenWalletHandler,
    WithdrawFundsHandler,
)
from lumen.models.entities.v1.wallet_orm import WalletEntity
from lumen.models.repositories import WalletRepository

from pyfly.cqrs import DefaultCommandBus, DefaultQueryBus, HandlerRegistry
from pyfly.data.relational.sqlalchemy import Base
from pyfly.data.relational.sqlalchemy.post_processor import (
    RepositoryBeanPostProcessor,
)
from pyfly.eda.adapters.memory import InMemoryEventBus

```

```

@pytest_asyncio.fixture
async def session_factory() →
    AsyncIterator[async_sessionmaker[AsyncSession]]:
        """An in-memory SQLite engine + session factory, schema created.

        Mirrors the framework's relational auto-configuration: build the
        async engine and run Base.metadata.create_all.
        """
        engine = create_async_engine("sqlite+aiosqlite:///memory:")
        async with engine.begin() as conn:
            await conn.run_sync(Base.metadata.create_all)
        factory = async_sessionmaker(engine, expire_on_commit=False)
        try:
            yield factory
        finally:
            await engine.dispose()

@pytest_asyncio.fixture
async def repository(
    session_factory: async_sessionmaker[AsyncSession],
) → AsyncIterator[WalletRepository]:
    """The framework WalletRepository, post-processed.

    RepositoryBeanPostProcessor compiles derived-query stubs
    (e.g. find_by_owner_id) onto the bean – the same step the
    ApplicationContext runs at startup.
    """
    session = session_factory()
    repo = WalletRepository(WalletEntity, session)
    RepositoryBeanPostProcessor().after_init(repo, "walletRepository")
    try:
        yield repo
    finally:
        await session.close()

```

```

@pytest_asyncio.fixture
async def event_bus() → AsyncIterator[InMemoryEventBus]:
    """A real in-memory EDA bus – the same EventPublisher used in
    production."""
    yield InMemoryEventBus()

@pytest_asyncio.fixture
async def audit_listener(
    event_bus: InMemoryEventBus,
) → AsyncIterator[WalletAuditListener]:
    """The wallet audit projection, subscribed to the bus exactly as the
    ApplicationContext auto-wires it at startup."""
    listener = WalletAuditListener()
    method = listener.on_wallet_event
    for pattern in method.__pyfly_event_patterns__:
        event_bus.subscribe(pattern, method)
    yield listener

@pytest_asyncio.fixture
async def command_bus(
    repository: WalletRepository,
    event_bus: InMemoryEventBus,
    session_factory: async_sessionmaker[AsyncSession],
) → AsyncIterator[DefaultCommandBus]:
    registry = HandlerRegistry()
    registry.register_command_handler(
        OpenWalletHandler(
            repository=repository,
            events=event_bus,
            session_factory=session_factory,
        )
    )
    registry.register_command_handler(
        DepositFundsHandler(
            repository=repository,
            events=event_bus,
            session_factory=session_factory,

```



```

    )
)
registry.register_command_handler(
    WithdrawFundsHandler(
        repository=repository,
        events=event_bus,
        session_factory=session_factory,
    )
)
yield DefaultCommandBus(registry=registry)

@pytest_asyncio.fixture
async def query_bus(
    repository: WalletRepository,
) → AsyncIterator[DefaultQueryBus]:
    registry = HandlerRegistry()
    registry.register_query_handler(GetWalletHandler(repository=repository))
    registry.register_query_handler(GetBalanceHandler(repository=repository))
    registry.register_query_handler(
        ListWalletsHandler(repository=repository)
    )
    registry.register_query_handler(
        ListRichWalletsHandler(repository=repository)
    )
    yield DefaultQueryBus(registry=registry)

```

Listing 16.3 — *conftest.py*: real framework components wired with no mocks

Each fixture is declared with `@pytest_asyncio.fixture` (not the bare `@pytest.fixture`) so `pytest-asyncio` manages the async iterator lifecycle. `asyncio_mode = "auto"` in `pyproject.toml` makes async fixtures and tests work without per-function decorators — but the fixture decorator itself must still be `pytest_asyncio.fixture`.

The `session_factory` fixture is shared. `repository` and `command_bus` both receive it, so the same in-memory SQLite engine backs reads, writes, and the `@transactional` boundary the handlers open. The `audit_listener` and `command_bus` fixtures both receive `event_bus`; pytest instantiates that once per test and shares it between them, so events published by the command handlers are visible to the listener.

What just happened. You wired a complete, production-shaped test stack — engine, repository, event bus, listener, and both CQRS buses — entirely from fixtures, with no mocks. The two facts that make it work are worth holding onto. First, **fixtures compose by name**: `command_bus` asks for `repository`, `event_bus`, and `session_factory` simply by naming them as parameters, and pytest threads the graph together. Second, **a fixture requested by two others is built once per test**: both `repository` and `command_bus` name `session_factory`, so they share one engine — the write a command makes is the read a query sees. Get this file right and every test in the next four sections is a two-line call.

🍃 SPRING PARITY

`conftest.py` is PyFly's `@TestConfiguration` plus the shared `application-test.properties` of a Spring Boot project — a single place that declares the beans every test reuses. A `@pytest_asyncio.fixture` is the rough equivalent of a `@Bean` method in that config: pytest builds it lazily, injects it where its name is requested, and tears it down afterward, just as the Spring test context manages bean lifecycle and injection.

 NO MOCKS ANYWHERE

Every component in `conftest.py` is the real production implementation.

`WalletRepository` is the same class the application boots.

`RepositoryBeanPostProcessor` is the same post-processor the `ApplicationContext` runs at startup to compile derived-query stubs. `InMemoryEventBus` is the same bus used in non-Kafka deployments. The goal is to test the real code paths, not the wiring.

Testing the CQRS flow end to end

With the fixtures from `conftest.py`, exercising the full command/query cycle is a matter of calling `command_bus.send(...)` and `query_bus.query(...)`. No handler is instantiated in the test body — the bus dispatches to the handler already registered in the fixture.

Notice how short each test body becomes now that the wiring lives in `conftest.py`. A test declares the fixtures it needs as parameters, then reads as plain narrative:

Step 1 — request the buses. Each test signature lists `command_bus` and/or `query_bus`. pytest sees those names, builds the fixture graph from `conftest.py`, and injects the ready buses.

Step 2 — send commands. `await command_bus.send(OpenWallet(...))` returns the new wallet id; subsequent `DepositFunds` and `WithdrawFunds` commands return the running balance. You assert on each return value as you go.

Step 3 — query the read side. `await query_bus.query(GetWallet(...))` and `GetBalance(...)` reload the persisted state and return DTOs (data-transfer objects — plain read models). You assert their fields match what the commands wrote.

Step 4 — prove the error paths. The remaining tests send a command that must fail — an overdraw, a non-positive deposit, an unknown wallet — wrapped in `pytest.raises(CommandProcessingException)`. That context manager asserts the block raises the named exception; if it does not, the test fails.

tests/test_cqrs_flow.py

```

from __future__ import annotations

import pytest
from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.core.services.wallets.get_balance_query import GetBalance
from lumen.core.services.wallets.get_wallet_query import GetWallet
from lumen.core.services.wallets.open_wallet_command import OpenWallet
from lumen.core.services.wallets.withdraw_funds_command import WithdrawFunds
from lumen.interfaces.enums.v1.currency import Currency

from pyfly.cqrs import DefaultCommandBus, DefaultQueryBus

@pytest.mark.asyncio
async def test_full_wallet_lifecycle(
    command_bus: DefaultCommandBus,
    query_bus: DefaultQueryBus,
) → None:
    wallet_id = await command_bus.send(
        OpenWallet(owner_id="u-1", currency=Currency.EUR)
    )
    assert isinstance(wallet_id, str) and wallet_id.startswith("wlt-")

    balance = await command_bus.send(
        DepositFunds(wallet_id=wallet_id, amount=1500)
    )
    assert balance == 1500

    balance = await command_bus.send(
        WithdrawFunds(wallet_id=wallet_id, amount=500)
    )
    assert balance == 1000

    wallet = await query_bus.query(GetWallet(wallet_id=wallet_id))
    assert wallet is not None
    assert wallet.id == wallet_id
    assert wallet.owner_id == "u-1"

```

```

assert wallet.currency is Currency.EUR
assert wallet.balance_minor == 1000
assert wallet.balance == 10.0

balance_dto = await query_bus.query(GetBalance(wallet_id=wallet_id))
assert balance_dto is not None
assert balance_dto.balance_minor == 1000
assert balance_dto.balance == 10.0

@pytest.mark.asyncio
async def test_get_wallet_returns_none_for_unknown_id(
    query_bus: DefaultQueryBus,
) → None:
    assert await query_bus.query(
        GetWallet(wallet_id="wlt-does-not-exist")
    ) is None

@pytest.mark.asyncio
async def test_overdraw_is_rejected_through_the_bus(
    command_bus: DefaultCommandBus,
) → None:
    from pyfly.cqrs.exceptions import CommandProcessingException

    wallet_id = await command_bus.send(
        OpenWallet(owner_id="u-2", currency=Currency.EUR)
    )
    await command_bus.send(DepositFunds(wallet_id=wallet_id, amount=100))

    with pytest.raises(CommandProcessingException):
        await command_bus.send(
            WithdrawFunds(wallet_id=wallet_id, amount=999)
        )

@pytest.mark.asyncio
async def test_validation_rejects_non_positive_deposit(
    command_bus: DefaultCommandBus,

```

```

) → None:
    from pyfly.cqrs.exceptions import CommandProcessingException

    wallet_id = await command_bus.send(
        OpenWallet(owner_id="u-3", currency=Currency.EUR)
    )
    with pytest.raises(CommandProcessingException):
        await command_bus.send(DepositFunds(wallet_id=wallet_id, amount=0))

@pytest.mark.asyncio
async def test_deposit_to_unknown_wallet_is_rejected(
    command_bus: DefaultCommandBus,
) → None:
    from pyfly.cqrs.exceptions import CommandProcessingException

    with pytest.raises(CommandProcessingException):
        await command_bus.send(
            DepositFunds(wallet_id="wlt-nope", amount=100)
        )

```

Listing 16.4 — End-to-end CQRS tests through the real bus

`test_full_wallet_lifecycle` is the primary smoke test: it sends every command in the natural order and then queries both the full wallet DTO and the balance DTO. The wallet DTO exposes `balance_minor` (integer minor units) and `balance` (major units as a float); both derive from the same `WalletEntity` row stored through the repository.

The error-path tests verify that the bus surfaces domain violations correctly.

`CommandProcessingException` is the bus's wrapper for any exception raised inside a handler — including `BusinessRuleViolation` from the aggregate. Calling code never sees the raw domain exception; it always sees the bus wrapper.

Run it.

```
uv run --extra dev pytest tests/test_cqrs_flow.py -q
```

Expected output:

```
.....
[100%]
5 passed in 0.05s
```

What just happened. This is the first layer that touches real infrastructure, and it still runs in milliseconds. The command went through the real bus, the real handler opened a real `@transactional` unit of work on a real SQLite session, committed it, and the query read it back — the exact path that runs in production, minus the HTTP layer. Because the handler is registered in the fixture rather than constructed in the test, you are testing the *dispatch* as well as the logic: if the command-to-handler routing broke, these tests would catch it.

ASYNCIO_MODE = "AUTO" AND @PYTEST.MARK.ASYNCIO

With `asyncio_mode = "auto"` every async test is collected and run automatically. The `@pytest.mark.asyncio` decorator is **not required** but is harmless and makes the async intent explicit at a glance. Lumen keeps it for clarity.

Testing the repository adapter

The CQRS flow tests prove the full open/deposit/withdraw/query pipeline. The repository adapter tests go one level deeper: they directly exercise the `WalletRepository` API — CRUD, **derived queries** compiled from method names, `Pageable / Page` pagination, and `Specification` predicates — against a temporary

SQLite file database. No Docker, no external process, no network. pytest's built-in `tmp_path` fixture provides a temporary directory cleaned up automatically after each test.

The local fixture `_make_repo` mirrors what the `ApplicationContext` does at startup: construct the repository and run `RepositoryBeanPostProcessor.after_init` to compile the derived-query stubs. Without that call, methods like `find_by_owner_id` would raise `NotImplementedError`.

Two terms before the listing. A **derived query** is a repository method whose body is generated from its *name*: `find_by_owner_id` becomes `WHERE owner_id = :value`, no SQL written by hand. A **Specification** is a reusable, composable predicate object you pass to a query — `balance_at_least(1000)` is one — for filters too dynamic to bake into a method name. **Pagination** wraps both: `Pageable.of(page, size, sort)` describes which slice you want, and the query returns a `Page` carrying the items plus `total`, `total_pages`, and `has_next`.

This section uses a **file-based** SQLite database (via `tmp_path`) rather than the in-memory one, so it can prove a stronger property — durability across a reconnect. Here is the shape of each test:

Step 1 — `sqlite_factory` : a temp-file engine. The fixture builds a SQLite engine backed by a real file under pytest's `tmp_path` (a fresh temp directory per test, auto-deleted afterward), creates the schema, and yields both the factory and the URL. Yielding the URL is what lets one test reconnect with a brand-new engine.

Step 2 — CRUD and persistence.

`test_upsert_inserts_then_updates_and_persists` upserts a row, upserts it again with a new balance to prove update-in-place, commits, reads it back — then disposes the engine and reconnects with a *fresh* one to prove the data truly hit disk.

Step 3 — the unknown-id path. `test_find_by_id_unknown_returns_none` confirms a miss returns `None`, not an error.

Step 4 — derived query. `test_derived_find_by_owner_id` inserts wallets for two owners and asserts `find_by_owner_id("alice")` returns only Alice's — proof the post-processor compiled the method-name convention correctly.

Step 5 — Specification + pagination.

`test_specification_find_rich_paged_and_sorted` and `test_find_all_pageable_counts_and_pages` exercise the `find_rich` / `find_all_by_spec` predicate path and the `find_all(pageable)` paging path, asserting `total`, `total_pages`, `has_next`, and the exact ordering of `page.items`.

tests/test_sql_wallet_repository.py

```

from __future__ import annotations

from collections.abc import AsyncIterator
from datetime import UTC, datetime, timedelta
from pathlib import Path

import pytest
import pytest_asyncio
from sqlalchemy.ext.asyncio import (
    AsyncSession,
    async_sessionmaker,
    create_async_engine,
)

from lumen.models.entities.v1.wallet_orm import WalletEntity
from lumen.models.repositories.wallet_repository import (
    WalletRepository,
    balance_at_least,
)

from pyfly.data import Pageable, Sort
from pyfly.data.relational.sqlalchemy import Base
from pyfly.data.relational.sqlalchemy.post_processor import (
    RepositoryBeanPostProcessor,
)

def _entity(
    wid: str,
    owner: str,
    minor: int,
    *,
    currency: str = "EUR",
    age_days: int = 0,
) → WalletEntity:
    created = datetime.now(UTC) - timedelta(days=age_days)
    return WalletEntity(
        id=wid,

```

```

        owner_id=owner,
        currency=currency,
        balance_minor=minor,
        created_at=created,
    )

def _make_repo(session: AsyncSession) → WalletRepository:
    repo = WalletRepository(WalletEntity, session)
    # Mirror the ApplicationContext: compile derived-query stubs.
    RepositoryBeanPostProcessor().after_init(repo, "walletRepository")
    return repo

@pytest_asyncio.fixture
async def sqlite_factory(
    tmp_path: Path,
) → AsyncIterator[tuple[async_sessionmaker[AsyncSession], str]]:
    """A temp-file SQLite engine + session factory, schema created.

    Yields the session factory and the database URL so the test can
    reconnect with a fresh engine to verify true persistence.
    """
    db_url = f"sqlite+aiosqlite:/// {tmp_path / 'wallets.db'}"
    engine = create_async_engine(db_url)
    async with engine.begin() as conn:
        await conn.run_sync(Base.metadata.create_all)
    factory = async_sessionmaker(engine, expire_on_commit=False)
    try:
        yield factory, db_url
    finally:
        await engine.dispose()

@pytest.mark.asyncio
async def test_upsert_inserts_then_updates_and_persists(
    sqlite_factory: tuple[async_sessionmaker[AsyncSession], str],
) → None:
    factory, db_url = sqlite_factory

```

```

async with factory() as session:
    repo = _make_repo(session)
    await repo.upsert(_entity("wlt-1", "owner-42", 0, currency="USD"))
    # update: same PK, new balance
    await repo.upsert(_entity("wlt-1", "owner-42", 2500, currency="USD"))
    await session.commit()

    got = await repo.find_by_id("wlt-1")
    assert got is not None
    assert got.owner_id == "owner-42"
    assert got.currency == "USD"
    assert got.balance_minor == 2500
    assert await repo.count() == 1

# prove persistence: reconnect with a brand-new engine/session
fresh_engine = create_async_engine(db_url)
fresh_factory = async_sessionmaker(fresh_engine, expire_on_commit=False)
try:
    async with fresh_factory() as fresh_session:
        fresh_repo = _make_repo(fresh_session)
        persisted = await fresh_repo.find_by_id("wlt-1")
        assert persisted is not None, "wallet should survive a reconnect"
        assert persisted.balance_minor == 2500
finally:
    await fresh_engine.dispose()

@pytest.mark.asyncio
async def test_find_by_id_unknown_returns_none(
    sqlite_factory: tuple[async_sessionmaker[AsyncSession], str],
) → None:
    factory, _ = sqlite_factory
    async with factory() as session:
        repo = _make_repo(session)
        assert await repo.find_by_id("wlt-nope") is None

@pytest.mark.asyncio

```

```

async def test_derived_find_by_owner_id(
    sqlite_factory: tuple[async_sessionmaker[AsyncSession], str],
) → None:
    factory, _ = sqlite_factory
    async with factory() as session:
        repo = _make_repo(session)
        await repo.upsert(_entity("wlt-1", "alice", 100))
        await repo.upsert(_entity("wlt-2", "alice", 200))
        await repo.upsert(_entity("wlt-3", "bob", 300))
        await session.commit()

        owned = await repo.find_by_owner_id("alice")
        assert sorted(w.id for w in owned) == ["wlt-1", "wlt-2"]
        assert await repo.find_by_owner_id("nobody") == []

@pytest.mark.asyncio
async def test_specification_find_rich_paged_and_sorted(
    sqlite_factory: tuple[async_sessionmaker[AsyncSession], str],
) → None:
    factory, _ = sqlite_factory
    async with factory() as session:
        repo = _make_repo(session)
        # age_days drives created_at so we can assert newest-first ordering.
        await repo.upsert(_entity("wlt-poor", "a", 50, age_days=3))
        await repo.upsert(_entity("wlt-mid", "b", 1000, age_days=2))
        await repo.upsert(_entity("wlt-rich", "c", 5000, age_days=1))
        await session.commit()

        # Specification: balance_minor ≥ 1000, newest first, page size 1.
        newest_first = Sort.by("created_at").descending()
        page = await repo.find_rich(1000, Pageable.of(1, 1, newest_first))
        assert page.total == 2 # mid + rich
        assert page.total_pages == 2
        assert page.has_next is True
        assert [w.id for w in page.items] == ["wlt-rich"]

        page2 = await repo.find_rich(1000, Pageable.of(2, 1, newest_first))
        assert [w.id for w in page2.items] == ["wlt-mid"]

```

```

# The bare predicate also works through find_all_by_spec.
rich = await repo.find_all_by_spec(balance_at_least(5000))
assert [w.id for w in rich] == ["wlt-rich"]

@pytest.mark.asyncio
async def test_find_all_pageable_counts_and_pages(
    sqlite_factory: tuple[async_sessionmaker[AsyncSession], str],
) → None:
    factory, _ = sqlite_factory
    async with factory() as session:
        repo = _make_repo(session)
        for i in range(5):
            await repo.upsert(
                _entity(f"wlt-{i}", "owner", i * 100, age_days=5 - i)
            )
        await session.commit()

    page = await repo.find_all(
        Pageable.of(1, 2, Sort.by("created_at").descending())
    )
    assert page.total == 5
    assert page.total_pages == 3
    assert len(page.items) == 2
    # newest first → wlt-4 (age 1 day), then wlt-3
    assert [w.id for w in page.items] == ["wlt-4", "wlt-3"]

```

Listing 16.5 — Repository tests: CRUD, derived query, pagination, Specification

Four things to note. First, `_make_repo` calls

`RepositoryBeanPostProcessor().after_init(repo, ...)` — without this, `find_by_owner_id` is still a stub and raises `NotImplementedError`. The post-processor compiles the method name into a SQLAlchemy `WHERE owner_id = :owner_id` clause. Second, `upsert` is the repository's insert-or-update; after each batch of upserts, `await session.commit()` flushes to SQLite. Third, `find_rich` takes

a minimum balance and a `Pageable` ; it delegates to `find_all_by_spec_paged(balance_at_least(min), pageable)` . Fourth, the two-engine pattern in `test_upsert_inserts_then_updates_and_persists` proves true durability: data committed through one engine is readable by a completely fresh engine and session.

Run it.

```
uv run --extra dev pytest tests/test_sql_wallet_repository.py -q
```

Expected output:

```
.....
[100%]
5 passed in 0.06s
```

What just happened. The standout is the two-engine reconnect in the first test. Many "persistence" tests pass even when nothing was written to disk, because the same session caches the object in memory and hands it back on read. By disposing the engine entirely and opening a *second* one against the same file URL, this test forces a real round-trip to storage — if `upsert` or `commit` were silently not persisting, the reconnect would return `None` and the test would fail. That is the difference between testing your code and testing your cache.

SPRING PARITY

This test layer is the Python equivalent of `@DataJpaTest` with an embedded H2 database in Spring Boot. `@DataJpaTest` loads only the JPA layer (entities, repositories, Flyway) and wires a fresh in-memory H2 for every test class. The `sqlite_factory` fixture does the same: create the schema, run the tests, dispose the engine. No Docker, no external process.

💡 DERIVED QUERIES ARE METHOD-NAME CONVENTIONS

`WalletRepository.find_by_owner_id` is declared as a stub (`raise NotImplementedError`). `RepositoryBeanPostProcessor` inspects the method name at startup — `find_by_owner_id` → `WHERE owner_id = :value` — and replaces the stub with a real coroutine. Testing this method therefore also tests that the post-processor convention is working correctly.

Testing the event listener

`WalletAuditListener` listens for domain events published by the command handlers. Testing it end to end — command runs on the bus, handler publishes events, listener receives them — requires all three components to share the same `InMemoryEventBus`. The `conftest.py` fixtures already arrange this: both `command_bus` and `audit_listener` accept an `event_bus` argument, and pytest injects the same instance into both.

An **event listener** is just a method that the framework subscribes to the bus so it runs whenever a matching event is published. Lumen's `WalletAuditListener` keeps an in-memory audit log and a running balance per wallet — a tiny **projection** (a read model built by folding events). Testing it is the clearest demonstration of the shared-fixture trick: because `command_bus` and `audit_listener` name the same `event_bus`, an event a command publishes is an event the listener observes, with no glue in the test body.

The tests follow one rhythm:

Step 1 — drive commands. Open a wallet, deposit, withdraw — all through `command_bus`.

Step 2 — read the projection. Call `audit_listener.entries_for(wallet_id)` and assert the recorded event types, in order, plus the `running_total`.

Step 3 — assert the negative. One test deliberately overdraws — a command that must fail — and asserts the audit log records nothing from it. A failed operation leaves no trace.

tests/test_event_listener.py

```

from __future__ import annotations

import pytest
from lumen.core.services.listeners import WalletAuditListener
from lumen.core.services.wallets.deposit_funds_command import DepositFunds
from lumen.core.services.wallets.open_wallet_command import OpenWallet
from lumen.core.services.wallets.withdraw_funds_command import WithdrawFunds
from lumen.interfaces.enums.v1.currency import Currency

from pyfly.cqrs import DefaultCommandBus

@pytest.mark.asyncio
async def test_listener_observes_wallet_events(
    command_bus: DefaultCommandBus,
    audit_listener: WalletAuditListener,
) → None:
    wallet_id = await command_bus.send(
        OpenWallet(owner_id="u-1", currency=Currency.EUR)
    )
    await command_bus.send(DepositFunds(wallet_id=wallet_id, amount=1500))
    await command_bus.send(WithdrawFunds(wallet_id=wallet_id, amount=400))

    entries = audit_listener.entries_for(wallet_id)
    assert [e.event_type for e in entries] == [
        "WalletOpened",
        "FundsDeposited",
        "FundsWithdrawn",
    ]

    # The payload carried the real domain-event fields.
    deposited = entries[1]
    assert deposited.payload["amount"] == 1500
    assert deposited.payload["currency"] == "EUR"
    assert deposited.payload["balance"] == 1500
    assert deposited.event_id # the aggregate's DomainEvent.event_id

```

```

    # The running-total projection reflects deposit - withdrawal.
    assert audit_listener.running_total(wallet_id) == 1100

@pytest.mark.asyncio
async def test_listener_records_nothing_before_any_command(
    audit_listener: WalletAuditListener,
) → None:
    assert audit_listener.entries == []
    assert audit_listener.running_total("anything") == 0

@pytest.mark.asyncio
async def test_event_type_matches_domain_event_class_names(
    command_bus: DefaultCommandBus,
    audit_listener: WalletAuditListener,
) → None:
    # A rejected withdrawal raises no event – it must not appear in the log.
    wallet_id = await command_bus.send(
        OpenWallet(owner_id="u-2", currency=Currency.USD)
    )
    await command_bus.send(DepositFunds(wallet_id=wallet_id, amount=100))

    from pyfly.cqrs.exceptions import CommandProcessingException

    with pytest.raises(CommandProcessingException):
        await command_bus.send(
            WithdrawFunds(wallet_id=wallet_id, amount=9999)
        )

    types = [e.event_type for e in audit_listener.entries_for(wallet_id)]
    assert types == ["WalletOpened", "FundsDeposited"]
    assert audit_listener.running_total(wallet_id) == 100

```

Listing 16.6 — Event listener tests: command publishes, listener observes

`test_listener_observes_wallet_events` is the core integration proof: three commands produce three events, the listener records all three in order, the payload fields match the aggregate's event dataclass fields, and the `running_total` projection equals the arithmetic result. No bus mock, no event capture list — the production listener runs on the production bus.

`test_event_type_matches_domain_event_class_names` proves a domain invariant: a rejected command (overdraw) raises no event. The audit log must never record a side effect from a failed operation.

Run it.

```
uv run --extra dev pytest tests/test_event_listener.py -q
```

Expected output:

```
...
[100%]
3 passed in 0.04s
```

What just happened. No part of the test connected the listener to the bus — the `audit_listener` fixture did that in `conftest.py`, subscribing the handler to the same `event_bus` the `command_bus` publishes through. So sending a command and then reading `entries_for(...)` exercises the real publish/subscribe path end to end. The negative test is the subtle one: it proves the audit log is driven by *events*, not by *attempts* — a rejected withdrawal raises a `BusinessRuleViolation` before any event is emitted, so nothing reaches the listener.

 EVENT_TYPE IS THE CLASS NAME

PyFly's event publisher sets `event_type` to the domain event class name:

```
"WalletOpened", "FundsDeposited", "FundsWithdrawn". The
@event_listener(event_types=["WalletOpened", "FundsDeposited",
"FundsWithdrawn"]) decorator on WalletAuditListener.on_wallet_event names
those three types explicitly; the framework stores them on the method as
__pyfly_event_patterns__, which the audit_listener fixture reads to subscribe. The
test asserts the string class names directly.
```

Booted-context integration test

The unit tests, CQRS flow tests, and repository tests each wire one layer of the stack.

The booted-context integration test wires everything at once: it starts the real

`ApplicationContext` — DI component scan, CQRS auto-configuration, relational auto-configuration, `RepositoryBeanPostProcessor`, `@transactional` seam, EDA event bus — then resolves the `DefaultCommandBus` and `DefaultQueryBus` from the context and drives the full wallet lifecycle.

The **ApplicationContext** is PyFly's runtime container — the object that scans for components, builds beans, runs post-processors, and holds the wired application together. Booting it is the most faithful test you can write short of starting an HTTP server: every piece of wiring the framework does at startup actually happens.

The database URL is overridden via an environment variable so the test never touches the developer's `lumen.db`. Here is the plan:

Step 1 — isolate the database. The `booted_context` fixture uses pytest's `monkeypatch` fixture to set `PYFLY_DATA_RELATIONAL_URL` to a temp-file SQLite path under `tmp_path`, *before* the app boots. `monkeypatch` is pytest's safe way to set an environment variable for the duration of one test and automatically restore it afterward — so this test can never clobber your real `lumen.db`.

Step 2 — boot the real application. It constructs `PyFlyApplication(LumenApplication, config_path=...)` and `await app.startup()`. That single call runs the entire startup sequence: component scan, all auto-configurations, the `RepositoryBeanPostProcessor`, and the event bus. The fixture yields `app.context` and, in its `finally` block, closes the shared session and calls `app.shutdown()`.

Step 3 — resolve beans and drive the lifecycle. The test calls `ctx.get_bean(DefaultCommandBus)` and `ctx.get_bean(DefaultQueryBus)` — pulling the *same* buses the application would use — then runs `open` → `deposit` → `withdraw` → `list` → `rich` → `balance`, asserting each result.

tests/test_app_context_integration.py

```

from __future__ import annotations

import logging
import sys
from collections.abc import AsyncIterator
from pathlib import Path

import pytest
import pytest_asyncio
from sqlalchemy.ext.asyncio import AsyncSession

_HERE = Path(__file__).resolve().parent
_SRC = _HERE.parent / "src"
sys.path.insert(0, str(_SRC))

@pytest_asyncio.fixture
async def booted_context(
    tmp_path: Path,
    monkeypatch: pytest.MonkeyPatch,
) → AsyncIterator[object]:
    """Boot the full LumenApplication against an isolated SQLite file."""
    db_path = tmp_path / "lumen-it.db"
    monkeypatch.setenv(
        "PYFLY_DATA_RELATIONAL_URL",
        f"sqlite+aiosqlite:/// {db_path}",
    )
    # Silence the pool-GC warning that aiosqlite emits at teardown.
    logging.getLogger(
        "sqlalchemy.pool.impl.AsyncAdaptedQueuePool"
    ).setLevel(logging.CRITICAL)

    from lumen.app import LumenApplication
    from pyfly.core import PyFlyApplication

    app = PyFlyApplication(
        LumenApplication,

```



```

        config_path=str(_HERE.parent / "pyfly.yaml"),
    )
    await app.startup()
    try:
        yield app.context
    finally:
        await app.context.get_bean(AsyncSession).close()
        await app.shutdown()

@pytest.mark.asyncio
async def test_full_lifecycle_through_booted_context(
    booted_context: object,
) → None:
    from lumen.core.services.wallets.deposit_funds_command import DepositFunds
    from lumen.core.services.wallets.get_balance_query import GetBalance
    from lumen.core.services.wallets.get_wallet_query import GetWallet
    from lumen.core.services.wallets.list_rich_wallets_query import (
        ListRichWallets,
    )
    from lumen.core.services.wallets.list_wallets_query import ListWallets
    from lumen.core.services.wallets.open_wallet_command import OpenWallet
    from lumen.core.services.wallets.withdraw_funds_command import
WithdrawFunds
    from lumen.interfaces.enums.v1.currency import Currency
    from pyfly.cqrs import DefaultCommandBus, DefaultQueryBus
    from pyfly.data import Pageable

    ctx = booted_context
    commands = ctx.get_bean(DefaultCommandBus)
    queries = ctx.get_bean(DefaultQueryBus)

    # --- open → deposit → withdraw, each a committed unit of work -----
    w1 = await commands.send(OpenWallet(owner_id="u-1",
currency=Currency.EUR))
    w2 = await commands.send(OpenWallet(owner_id="u-2",
currency=Currency.EUR))
    assert w1.startswith("wlt-") and w2.startswith("wlt-")

```

```

    assert await commands.send(DepositFunds(wallet_id=w1, amount=5000)) ==
5000
    assert await commands.send(WithdrawFunds(wallet_id=w1, amount=1500)) ==
3500
    assert await commands.send(DepositFunds(wallet_id=w2, amount=100)) == 100

# --- persistence survived: reload the aggregate via the query side ---
reloaded = await queries.query(GetWallet(wallet_id=w1))
assert reloaded is not None
assert reloaded.owner_id == "u-1"
assert reloaded.balance_minor == 3500

# --- paged list (find_all(pageable) + Page.map) -----
page = await queries.query(ListWallets(pageable=Pageable.of(1, 10)))
assert page.total == 2
assert {w.id for w in page.items} == {w1, w2}

# --- Specification: only wallets with balance ≥ 1000 -----
rich = await queries.query(
    ListRichWallets(min_minor=1000, pageable=Pageable.of(1, 10))
)
assert rich.total == 1
assert [w.id for w in rich.items] == [w1]

everyone = await queries.query(
    ListRichWallets(min_minor=0, pageable=Pageable.of(1, 10))
)
assert everyone.total == 2

# --- projection-backed balance -----
balance = await queries.query(GetBalance(wallet_id=w1))
assert balance is not None
assert balance.balance_minor == 3500
assert balance.balance == 35.0

```

Listing 16.7 — Booted-context integration: full DI + persistence composition

`booted_context` uses pytest's built-in `monkeypatch` fixture to set `PYFLY_DATA_RELATIONAL_URL` before the application boots. The framework reads this environment variable during relational auto-configuration, so the context uses the isolated temp-file SQLite database for the life of the test, then disposes of it when the fixture tears down.

`test_full_lifecycle_through_booted_context` exercises every query type the application exposes: `GetWallet` (aggregate reload), `ListWallets` (paginated list using `find_all(pageable)`), `ListRichWallets` (Specification predicate using `find_all_by_spec_paged`), and `GetBalance` (projection-backed balance). It proves that the `RepositoryBeanPostProcessor`, the `@transactional` boundary around each command handler, and the DI wiring all compose correctly in a single boot.

Run it.

```
uv run --extra dev pytest tests/test_app_context_integration.py -q
```

Expected output:

```
.
[100%]
1 passed in 0.15s
```

One test, but the heaviest one in the suite: it actually started the framework. If the DI scan missed a bean, an auto-configuration mis-wired, or the `@transactional` boundary failed to commit, this is the test that catches it — which is exactly why it sits at the peak of the pyramid and why there is only one of it.

What just happened. The environment-variable override is the load-bearing trick. The framework reads `pyfly.data.relational.url` from config during relational auto-configuration, and PyFly maps any config key to a `PYFLY_*` environment variable (dots and dashes become underscores, uppercased), so `PYFLY_DATA_RELATIONAL_URL`

overrides the `url` in `pyfly.yaml`. Setting it with `monkeypatch` *before* `app.startup()` is what redirects the whole booted application to a throwaway database — and `monkeypatch` undoes the change when the fixture tears down, so no other test is affected.

💡 A DEDICATED TEST PROFILE (V26.6.110)

Setting one variable is fine for a single override. When a project needs a whole block of test-only settings, PyFly's **profile** mechanism (Spring parity) is cleaner: drop a `pyfly-test.yaml` next to `pyfly.yaml` with your test overrides, then activate it by setting `PYFLY_PROFILES_ACTIVE=test` (or `pyfly.profiles.active: test` in config). On boot, PyFly overlays `pyfly-test.yaml` on top of the base `pyfly.yaml`, so values like the database URL or `ddl-auto` apply only under that profile. Lumen does not need a profile — one env override covers its single test-only setting — but reach for one as the test configuration grows.

🍃 SPRING PARITY

This test is the Python equivalent of `@SpringBootTest` with an embedded H2 database. `@SpringBootTest` loads the full application context, including all auto-configurations and the JPA layer; you set `spring.datasource.url` in `application-test.properties` to redirect to H2. PyFly's environment variable override (`monkeypatch.setenv`) plays the same role. Both approaches prove that the composed application works end to end without any external infrastructure.

Framework testing helpers

The tests above cover Lumen's full pyramid with standard pytest primitives and PyFly's real production components. For larger applications or teams that prefer more structure, `pyfly.testing` ships higher-level helpers that mirror Spring Boot's testing annotations.

`PyFlyTestCase` + `mock_bean(T)` work like `@MockBean` in `@SpringBootTest`. Declare `repo = mock_bean(WalletDomainRepository)` on the class body; `setup()` installs an `AsyncMock(spec=T)` into the application context and wires it into any collaborator that depends on it.

`create_test_container(overrides={Interface: Implementation})` builds a DI container with fakes registered for specific interfaces. Resolve the class under test from it and its dependencies are already injected.

`assert_event_published(events, event_type, payload_contains=...)` scans a captured `EventEnvelope` list for the first envelope with the given type, optionally checks payload keys, and returns the envelope for further assertions.

`assert_no_events_published(events)` fails if the list is non-empty.

Testcontainers integration (`postgres_container()`, `redis_container()`, `pyfly_config(container, base={...})`) is PyFly's equivalent of Spring Boot's `@Testcontainers` + `@ServiceConnection`. Start a real Postgres container; `pyfly_config` rewrites the sync `psycpg2://` URL to `postgresql+asyncpg://` and merges it into a `Config` ready to boot an `ApplicationContext`. Install support with:

```
pip install 'pyfly[testcontainers]'
```

Guard every Testcontainers test with `@requires_docker` so it skips cleanly on machines without Docker and runs automatically on CI runners that have it:

```

from pyfly.testing import postgres_container, pyfly_config, requires_docker

@requires_docker
async def test_wallet_round_trip_against_real_postgres():
    with postgres_container() as pg:
        config = pyfly_config(pg, base={"pyfly.data.enabled": True})
        assert config.get("pyfly.data.relational.url").startswith(
            "postgresql+asyncpg://"
        )
    ...

```

Lumen does not use these helpers — SQLite covers the persistence layer without Docker, and the in-memory bus covers event routing. Reach for them when your project has infrastructure that cannot be reproduced without a real daemon.

Run it. You have now walked every layer file by file. Run the whole suite one more time to confirm the full pyramid is green together:

```
uv run --extra dev pytest -q
```

Expected output — the same **41 passed** you started with, now with a mental model of exactly what each dot proves:

```

.....
[100%]
41 passed in 0.28s

```

✓ What you built

The six test files this chapter built add up to 26 passing tests, exercising every layer of the pyramid. Together with the saga tests from Chapter 12 and the event-sourcing tests from Chapter 9, Lumen's full suite is **41 passing tests** — the count you saw when you ran `uv run --extra dev pytest -q` at the start.

At the base, `test_money.py` and `test_wallet_aggregate.py` prove the domain model's arithmetic, immutability, and invariant rules. All tests are synchronous, pure Python functions — no fixtures, no DI, no `async`. The `BusinessRuleViolation.rule` attribute makes each assertion specific to the exact violated invariant.

In the middle, `conftest.py` wires the real components — the framework `WalletRepository` over an in-memory SQLite engine, `InMemoryEventBus`, all five command and query handlers (including `ListWalletsHandler` and `ListRichWalletsHandler`), and `WalletAuditListener` — into reusable async fixtures that pytest shares automatically across modules. The `RepositoryBeanPostProcessor` is applied to the repository fixture exactly as the `ApplicationContext` applies it at startup. `test_cqrs_flow.py` dispatches commands and queries through the real bus and checks every field of the query DTOs. `test_event_listener.py` proves that the audit listener observes exactly the events produced by successful commands and nothing from rejected ones.

`test_sql_wallet_repository.py` exercises the `WalletRepository` directly against a temporary SQLite file, covering the full CRUD surface, the derived query `find_by_owner_id` (compiled from the method name by `RepositoryBeanPostProcessor`), the `find_all(pageable)` API that returns a `Page`

with total count and page metadata, and the `Specification` predicate path via `find_rich` / `find_all_by_spec`. The two-engine reconnect pattern proves true durability.

At the peak, `test_app_context_integration.py` boots the real `LumenApplication` with the database URL overridden to an isolated SQLite file, then drives the full open → deposit → withdraw → list → rich → balance lifecycle through the context-resolved buses. This single test proves that the DI scan, CQRS auto-configuration, `RepositoryBeanPostProcessor`, and `@transactional` boundary all compose correctly.

Concretely, you learned:

- `asyncio_mode = "auto"` + `pythonpath = ["src"]` in `pyproject.toml` — all async tests run without decorators; the `src/` layout is importable.
- `uv run --extra dev pytest -q` — the bare `uv sync` drops the dev group; always include `--extra dev` to get pytest.
- `@pytest_asyncio.fixture` — async fixture lifecycle managed by pytest-asyncio; plain `@pytest.fixture` does not handle async generators.
- **Shared fixture instances** — when two fixtures request the same fixture name (e.g., `event_bus`, `session_factory`), pytest resolves it once per test and shares the instance.
- `RepositoryBeanPostProcessor().after_init(repo, name)` — must be called in tests that exercise derived queries; without it, the method stubs raise `NotImplementedError`.
- **Derived queries** (`find_by_owner_id`) — declared as stubs; the post-processor compiles them to `WHERE owner_id = :value` at startup.

- **Pageable.of(page, size, sort) + Page** — the `find_all(pageable)` API returns a `Page` with `total`, `total_pages`, `has_next`, and `items`; assert each field for pagination correctness.
 - **Specification predicates** — `balance_at_least(n)` is passed to `find_rich` / `find_all_by_spec` to filter by an arbitrary predicate without adding a new derived-query method.
 - **pending_events() vs clear_events()** — `pending_events()` reads without draining; `clear_events()` drains. Always call `clear_events()` in arrange steps so assertions only see events from the act step.
 - **BusinessRuleViolation.rule** — assert the exact rule string, not just the exception class, to prove that the right invariant fired.
 - **monkeypatch.setenv** — override configuration before booting the context in integration tests; the framework reads environment variables during auto-configuration.
 - **PYFLY_* config overrides** — every config key maps to an environment variable (`pyfly.data.relational.url` → `PYFLY_DATA_RELATIONAL_URL`); set it with `monkeypatch.setenv` before booting to redirect the whole application.
 - **Test profile (v26.6.110)** — for a block of test-only settings, add a `pyfly-test.yaml` overlay and activate it with `PYFLY_PROFILES_ACTIVE=test` (Spring `application-test.yaml` parity).
 - **Framework helpers** (`PyFlyTestCase`, `mock_bean`, `create_test_container`, `Testcontainers`) — available in `pyfly.testing` for projects that need them; Lumen keeps things simple with real components.
-

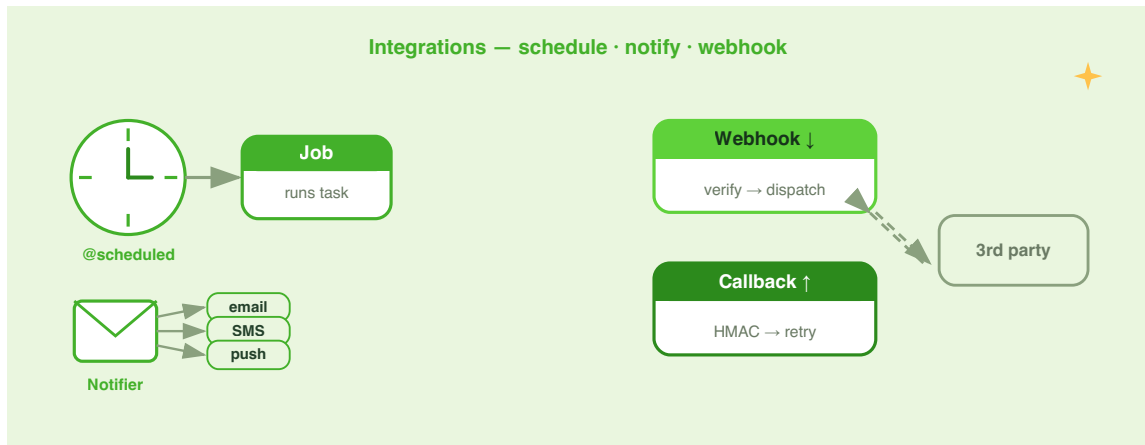
Try it yourself

- 1. Add a test for zero-amount withdrawal.** In `test_wallet_aggregate.py`, add `test_withdraw_zero_is_rejected`. Open a wallet, deposit 500 EUR, then attempt `wallet.withdraw(Money(0, Currency.EUR))`. Assert that a `BusinessRuleViolation` is raised and check its `.rule` attribute. Compare the rule name with the deposit equivalent — are the rules symmetric?
- 2. Test an unknown wallet via the CQRS bus.** In `test_cqrs_flow.py`, add `test_withdraw_from_unknown_wallet_is_rejected`. Send only a `WithdrawFunds(wallet_id="wlt-ghost", amount=50)` command without opening a wallet first. Assert that `CommandProcessingException` is raised. Confirm that the repository still holds no wallets by querying `GetWallet(wallet_id="wlt-ghost")` and asserting `None`.
- 3. Extend the listener test with a second wallet.** In `test_event_listener.py`, add a test that opens two wallets (`u-A` and `u-B`), deposits different amounts into each, and then calls `audit_listener.entries_for(wallet_id_A)` and `audit_listener.entries_for(wallet_id_B)` separately. Assert that each returns exactly two entries (`WalletOpened` + `FundsDeposited`) and that the payload `amount` values differ. This proves that `entries_for` filters by wallet ID, not by event type.
- 4. Add a multi-owner pagination test.** In `test_sql_wallet_repository.py`, add a test that inserts ten wallets with two different owners, calls `find_by_owner_id` for each owner, and then calls `find_all` with `Pageable.of(1, 3, Sort.by("balance_minor").descending())`. Assert that `page.total == 10`,

`page.total_pages == 4`, and that the first item in `page.items` has the highest `balance_minor`. This proves that pagination is independent of the derived-query filter.

CHAPTER 17

Scheduling, Notifications, Webhooks & Callbacks



In Chapter 16 you gave Lumen a full test harness — unit tests for the domain, CQRS flow tests through the real bus, and a SQLite adapter test that proves true persistence. Lumen is now well-tested and resilient. But it still lives entirely inside its own process: it reacts to requests, but it never reaches out unprompted.

Real financial platforms are different. They send nightly account statements, fire an SMS the moment funds land, receive payment-status webhooks from a payment provider at midnight, and call back partner systems to confirm that a disbursement was booked. That is four distinct integration patterns — scheduling, notifications, inbound webhooks, and outbound callbacks — and this chapter covers all of them.

By the end of the chapter Lumen will:

- run a **nightly scheduled job** that tallies daily wallet balances using `@scheduled` with a cron expression;
- send a **"funds received" email and push notification** via PyFly's pluggable `EmailService` and `PushService` ports, triggered by the real `FundsDeposited` domain event;
- accept **inbound webhooks** from an illustrative payment provider, verifying the HMAC-SHA256 signature, deduplicating replays, and dispatching to a typed listener;
- dispatch **outbound callbacks** to partner systems, signing each payload, retrying on transient failures, and recording every delivery attempt.

Install the two optional extras before you start:

```
uv add "pyfly[scheduling,notifications]"
```

NEW TERM: OPTIONAL EXTRAS

An *extra* is an opt-in slice of a package's dependencies. `pyfly` keeps its core lean and ships heavier capabilities — scheduling, notifications, and the rest — behind named extras so you only install what you use. `pyfly[scheduling,notifications]` pulls in both. The square-bracket syntax is standard Python packaging; `uv add` records it in your `pyproject.toml` so the next `uv sync` reinstalls them. If you later see `ModuleNotFoundError` for `croniter` or a notifications provider, you skipped this line — re-run it.

This chapter targets PyFly **v26.6.110**. Every code listing below matches the real Lumen source under `samples/lumen/src/lumen`, and every framework API was checked against `pyfly` itself, so what you build here runs unchanged.

Scheduled tasks

Why schedule instead of trigger?

Many operations in a financial platform cannot wait for an HTTP request to arrive. Nightly reconciliation must run at 02:00 regardless of whether any user is active. A cache-warming pass should fire 30 seconds after startup — before real traffic arrives — not when the first slow request triggers a miss. A heartbeat metric should emit every 10 seconds so the ops dashboard shows a live signal, not a stale reading.

PyFly's scheduling module provides a declarative, decorator-driven way to define all three patterns without manually managing threads, event loops, or timer wheels.

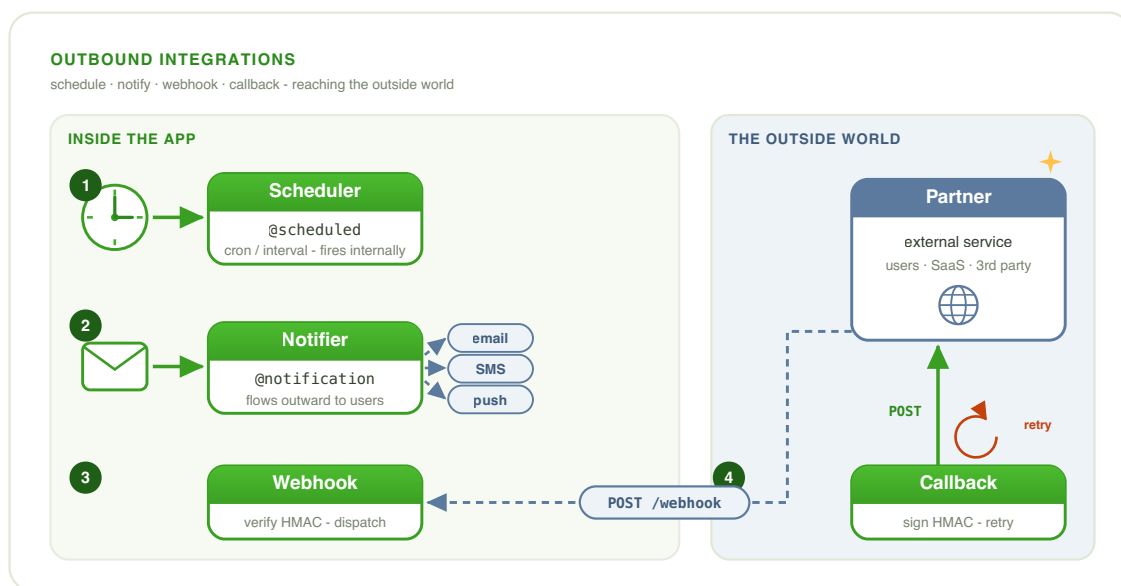


Figure 17.1 — The four\

integration layers added in this chapter. Scheduled tasks fire\ internally; notifications flow outward to users; inbound webhooks\ arrive from partners; outbound callbacks close the feedback loop.

The @scheduled decorator

`@scheduled` marks any `async` method on a `@service` bean for periodic execution. It accepts exactly one *trigger*: `fixed_rate`, `fixed_delay`, or `cron`. Providing zero or more than one trigger raises a `ValueError` at decoration time, so mistakes surface at startup rather than silently at three in the morning.

NEW TERM: TRIGGER

A *trigger* is the rule that decides *when* a scheduled method runs. You pick exactly one: `cron` (a calendar expression like "every day at 02:00"), `fixed_rate` (a steady interval such as "every 10 seconds"), or `fixed_delay` (a gap measured after each run finishes). One method, one trigger.

Let us build the nightly rollup one decision at a time.

Step 1 — Create the service file. In the Lumen tree, add `daily_rollup.py` under a `ledger` package. The class is an ordinary `@service` — a plain Python class that PyFly registers in its dependency-injection container and constructs for you. Because it is a managed bean, you can ask for the `WalletRepository` in the constructor and the framework hands it over; you never call `new` yourself.

Step 2 — Pick the trigger. This job must run once a night, on the clock, so the trigger is `cron`. The expression `"0 2 * * *"` reads field-by-field as *minute 0, hour 2, every day-of-month, every month, every day-of-week* — i.e. 02:00 every day.

Step 3 — Write the work. Inside the method, load every wallet, sum the persisted `balance_minor` integers, and (for now) log the total. In production you would write the snapshot to a reporting table or ship it downstream; logging keeps the example focused on the *scheduling*, not the bookkeeping.

lumen/ledger/daily_rollup.py

```

from datetime import timedelta

from lumen.models.repositories.wallet_repository import WalletRepository
from pyfly.container import service
from pyfly.scheduling import scheduled

@service
class DailyRollupService:
    """Tallies all wallet balances once per night at 02:00 UTC."""

    def __init__(self, wallet_repo: WalletRepository) → None:
        self._wallets = wallet_repo

    @scheduled(cron="0 2 * * *")
    async def run(self) → None:
        wallets = await self._wallets.find_all()
        # find_all() returns WalletEntity rows; balance_minor is the
        # persisted integer (cents).
        total_minor_units = sum(w.balance_minor for w in wallets)
        # Persist or ship the nightly snapshot; here we log it.
        print(
            f"[rollup] {len(wallets)} wallets, "
            f"total {total_minor_units / 100:.2f} "
            f"(minor units: {total_minor_units})"
        )

```

Listing 17.1 — Nightly wallet balance rollup with @scheduled

How it works. `@scheduled(cron="0 2 * * *")` fires every day at 02:00 UTC. The scheduler calculates `seconds_until_next()` via `CronExpression`, sleeps exactly that long, then submits the method to the executor.

That is all the wiring Lumen needs. With `pyfly[scheduling]` installed, `SchedulingAutoConfiguration` automatically:

1. registers a `TaskScheduler` bean;
2. scans every `@service` bean for `@scheduled` methods;
3. starts the scheduler during `ApplicationContext` startup;
4. stops it gracefully on shutdown.

No `SchedulerManager` required.

NEW TERM: AUTO-CONFIGURATION

Auto-configuration is PyFly noticing what is on your classpath and wiring the matching machinery for you. `SchedulingAutoConfiguration` only activates when `croniter` (pulled in by the `scheduling` extra) is importable — so the scheduler appears the moment you install the extra and stays absent otherwise. This is the same "convention over configuration" idea Spring Boot made famous; you can always override a bean by declaring your own.

Run it. Waiting until 02:00 to see your first tick is no fun, so temporarily change the trigger to fire every minute — `@scheduled(cron="* * * * *")` — and start the app from the `samples/lumen` directory:

```
uv run pyfly run --server uvicorn
```

At the top of the next minute you should see your rollout line in the logs (an empty database simply reports zero wallets):

```
[rollup] 0 wallets, total 0.00 (minor units: 0)
```

Open a wallet and deposit into it (see the curl recipes in Chapter 7), wait for the next minute, and the totals move:

```
[rollup] 1 wallets, total 15.00 (minor units: 1500)
```

Stop the app with Ctrl-C and **change the trigger back to** `"0 2 * * *"` before committing — the every-minute cron was only a probe.

WHAT JUST HAPPENED

You did not start a thread, open an event loop, or register a timer. You wrote one `@service` with one `@scheduled` method, and the framework discovered it at startup, computed the next fire time from the cron expression, slept until then, and ran your coroutine — repeating forever. The scheduler is *declarative*: you state *when*, PyFly handles *how*.

fixed_rate vs. fixed_delay

`fixed_rate` measures from the **start** of one execution to the start of the next.

`fixed_delay` measures from the **end** of one execution to the start of the next. Use

`fixed_rate` for heartbeats and metrics where you need a steady cadence regardless of execution time. Use `fixed_delay` when you need a guaranteed breathing gap — for example, when polling an upstream API that rate-limits on request frequency.

lumen/health/monitor.py

```

from datetime import timedelta

from pyfly.container import service
from pyfly.scheduling import scheduled

@service
class HealthMonitor:

    @scheduled(
        fixed_rate=timedelta(seconds=10),
        initial_delay=timedelta(seconds=5),
    )
    async def heartbeat(self) → None:
        """Emit a liveness metric every 10 s, starting 5 s after startup."""
        # metrics.gauge("lumen.up", 1)
        pass

@service
class ExchangeRatePoller:

    def __init__(self, fx_repo) → None:
        self._repo = fx_repo

    @scheduled(fixed_delay=timedelta(minutes=5))
    async def poll(self) → None:
        """Fetch the latest exchange rates, then wait 5 min before
        repeating."""
        rates = await self._repo.fetch_latest()
        await self._repo.store(rates)

```

Listing 17.2 — fixed_rate heartbeat and fixed_delay poll

`initial_delay` postpones the first run; it is available for both `fixed_rate` and `fixed_delay` (ignored for `cron` triggers, which always wait for the next matching calendar instant).

CronExpression

`CronExpression` is also usable directly — convenient when you need to display upcoming schedule times in a UI or validate a user-supplied expression before storing it.

`lumen/ledger/schedule_preview.py`

```
from pyfly.scheduling import CronExpression

def preview_rollup_schedule(expression: str, n: int = 5) → list[str]:
    """Return the next N fire times for a given cron expression."""
    cron = CronExpression(expression)
    return [str(t) for t in cron.next_n_fire_times(n)]
```

Listing 17.3 — Using CronExpression standalone

Run it. `CronExpression` needs no running app, so the fastest way to build intuition is the REPL. From `samples/lumen`:

```
uv run python -c "from pyfly.scheduling import CronExpression; \
print(*CronExpression('0 2 * * *').next_n_fire_times(3), sep='\n')"
```

You should see the next three 02:00 instants, one per line (your dates will differ):

```
2026-06-16 02:00:00+00:00
2026-06-17 02:00:00+00:00
2026-06-18 02:00:00+00:00
```

Notice the `+00:00` — fire times are UTC unless you pass a `zone` (covered next). This is also the cleanest way to sanity-check a user-supplied expression before you store it: an invalid string raises `ValueError` immediately.

`CronExpression` accepts both the standard 5-field format (`min hour dom month dow`) and the Spring-style 6-field format with seconds first (`sec min hour dom month dow`). The Spring `?` wildcard is normalised to `*` transparently.

Expression	Fires
<code>* * * * *</code>	Every minute
<code>0 * * * *</code>	Every hour, on the hour
<code>0 2 * * *</code>	Every day at 02:00
<code>0 9 * * 1-5</code>	Weekdays at 09:00
<code>30 2 1 * *</code>	1st of each month at 02:30
<code>*/15 * * * *</code>	Every 15 minutes
<code>0 0 12 * * *</code>	Noon every day (6-field, seconds-first)

Time-zone-aware cron

Cron expressions are evaluated in **UTC** by default. Pass `zone` with an IANA time-zone name to evaluate fire times in a specific zone instead:

```
@scheduled(cron="0 2 * * *", zone="America/New_York")
async def close_books(self) → None:
    """02:00 New York time — follows DST automatically."""
    ...
```

The same `zone` parameter is available on `CronExpression`:

```
cron = CronExpression("0 9 * * *", zone="Europe/Madrid")
next_run = cron.next_fire_time() # zone-aware datetime
```

DST transitions are handled by the `zoneinfo` database; no manual offset adjustment is required.

Distributed locking

When multiple Lumen instances run behind a load balancer, every instance schedules the same `@scheduled` methods. Without coordination, the nightly rollup fires once per instance and writes duplicate records. The `lock` parameter solves this the same way Spring's `@SchedulerLock` (ShedLock) does: before each tick the scheduler tries to acquire a named lock and **skips the run** if the lock is already held elsewhere.

```
@scheduled(cron="0 2 * * *", lock=True, lock_ttl=timedelta(minutes=5))
async def run(self) → None:
    """lock=True auto-names the lock 'DailyRollupService.run'."""
    ...
```

- `lock=True` — derives the lock name from `"ClassName.method_name"`.
- `lock="shared-name"` — explicit name; useful when two methods must be mutually exclusive.
- `lock_ttl` — safety-valve TTL; set it comfortably longer than the job's worst-case runtime.

By default `TaskScheduler` uses `LocalLock`, which always acquires — single-instance behaviour is unchanged. For cross-process coordination, implement `DistributedLock` and register it as a bean:

lumen/infra/redis_lock.py

```

from pyfly.container import bean, configuration
from pyfly.scheduling import DistributedLock

class RedisLock:
    """Best-effort named lock backed by Redis SET NX PX."""

    def __init__(self, redis) → None:
        self._redis = redis

    async def try_acquire(self, name: str, ttl: float) → bool:
        ok = await self._redis.set(
            f"pyfly:lock:{name}", "1",
            nx=True, px=int(ttl * 1000),
        )
        return ok is True

    async def release(self, name: str) → None:
        await self._redis.delete(f"pyfly:lock:{name}")

@configuration
class LockConfig:

    @bean
    def distributed_lock(self) → DistributedLock:
        import redis.asyncio as aioredis
        client = aioredis.from_url("redis://localhost:6379/1")
        return RedisLock(client)

```

Listing 17.4 — Redis-backed DistributedLock

`SchedulingAutoConfiguration` detects the `DistributedLock` bean in the container automatically and passes it to the `TaskScheduler`. Any object with conforming `try_acquire` and `release` coroutines satisfies the protocol.

@async_method

`@async_method` marks a method for fire-and-forget execution via the `TaskExecutorPort`. The caller returns immediately; the framework routes the coroutine through the configured executor in the background:

```
from pyfly.scheduling import async_method

@service
class AlertService:

    @async_method
    async def send_alert(self, msg: str) → None:
        """Caller does not await – AlertService dispatches asynchronously."""
        ...
```

Under the hood `@async_method` sets `__pyfly_async__ = True` on the function; the framework detects this flag and submits the coroutine to the `TaskExecutorPort`.

🍃 SPRING PARITY

`@scheduled(fixed_rate=...)` mirrors Spring's `@Scheduled(fixedRate=...)`.
`@scheduled(fixed_delay=...)` mirrors `@Scheduled(fixedDelay=...)`.
`@scheduled(cron=...)` mirrors `@Scheduled(cron=...)`. `zone=` mirrors Spring's `zone` attribute. `lock=True` mirrors ShedLock's `@SchedulerLock`. `@async_method` mirrors Spring's `@Async`.

Configuration reference

```
pyfly:
  scheduling:
    enabled: true           # set false to disable all loops
    executor:
      type: asyncio        # 'asyncio' (default, in-loop) or 'thread'
      max-workers: 4       # worker threads when type is 'thread'
    lock:
      provider: none       # none | memory | redis | postgres
```

When `enabled` is `false`, `TaskScheduler` starts no loops and all `@scheduled` methods are silently skipped.

The `executor.type` chooses how each tick runs. The default `asyncio` runs the coroutine on the application event loop — ideal for short, I/O-bound jobs like the rollup. Switch to `thread` (a pool of `executor.max-workers` threads) when a job does heavy CPU work or calls a blocking library, so it cannot stall the loop.

🔗 CHOOSING A LOCK PROVIDER

`lock.provider` selects the backend behind `@scheduled(lock=...)`, described next: `none` (the default — no coordination), `memory` (mutual exclusion within one process), `redis`, or `postgres` (true cross-instance coordination with no code change). On `redis` / `postgres` PyFly builds the `DistributedLock` bean for you from `pyfly.scheduling.lock.redis.url` or the app's existing `AsyncEngine`; the hand-rolled `@bean` in Listing 17.4 is the do-it-yourself alternative when you need custom semantics.

Notifications

Lumen needs to tell customers that their money has arrived — email for the balance confirmation, and optionally an SMS or mobile push for the real-time alert.

PyFly's notifications module defines three **port protocols** and three **default services**. Your business logic depends on the protocols; the concrete provider adapters — SMTP, SendGrid, Twilio, Firebase — live behind the port boundary and can be swapped without touching a single line of domain code.

NEW TERM: PORT AND ADAPTER

A *port* is an interface your code talks to — here, "something that can send an email". An *adapter* is a concrete implementation of that port — `SmtplibEmailProvider`, `SendGridEmailProvider`, and so on. The pattern (also called *hexagonal architecture*) means your deposit logic depends only on the `EmailService` port, never on a specific vendor. Swapping SMTP for SendGrid is a one-line change in a configuration class; the domain code never notices.

The port hierarchy

Protocol	Service class	Method
<code>EmailProvider</code>	<code>DefaultEmailService</code>	<code>send(EmailMessage) → NotificationResult</code>
<code>SmsProvider</code>	<code>DefaultSmsService</code>	<code>send(SmsMessage) → NotificationResult</code>
<code>PushProvider</code>	<code>DefaultPushService</code>	<code>send(PushMessage) → NotificationResult</code>

`DefaultEmailService`, `DefaultSmsService`, and `DefaultPushService` are thin wrappers: each delegates to a provider, catches any provider exception, and returns a structured `NotificationResult` with `status=FAILED` and the error string rather than propagating the exception. A transient SendGrid outage does not take down the deposit handler.

Messages and results

`FundsDeposited` carries `amount` and `balance` as **integer minor units** (cents). The `Money.major_units` property converts them for display — `Money(25000, Currency.EUR).major_units` is `250.0`. Keep that in mind when formatting notification bodies.

lumen/notifications/models_overview.py

```

from pyfly.notifications import (
    EmailMessage,
    NotificationResult,
    PushMessage,
    SmsMessage,
)

# Email – full field set
email = EmailMessage(
    to=["alice@example.com"],
    sender="no-reply@lumenbank.com",
    subject="Funds received",
    body_text="EUR 250.00 has been credited to your wallet.",
    body_html=(
        "<p><strong>EUR 250.00</strong> has been credited "
        "to your wallet.</p>"
    ),
)

# SMS – compact
sms = SmsMessage(
    to="+346000000001",
    body="Lumen: EUR 250.00 received. New balance: EUR 750.00.",
    sender="LUMEN",
)

# Push – structured payload
push = PushMessage(
    device_tokens=["FCM_TOKEN_GOES_HERE"],
    title="Funds received",
    body="EUR 250.00 credited",
    data={"wallet_id": "w-001", "amount_minor": 25000},
)

```

Listing 17.5 — The core DTOs

`NotificationResult` carries `id`, `provider`, `status` (`EmailStatus.SENT` | `DELIVERED` | `FAILED` | ...), an optional `provider_id` (e.g. the SendGrid message ID), and an optional `error`.

Wiring the SMTP provider

For development and self-hosted deployments, `SmtplibEmailProvider` runs `smtplib` from a thread pool so the async event loop is never blocked.

Step 1 — Build the provider as a `@bean`. A `@configuration` class is PyFly's place to assemble objects the container cannot construct on its own — here, an SMTP client that needs a host, credentials, and TLS settings. Each `@bean` method returns one ready-to-use object; the framework caches it and injects it wherever the return type is requested.

Step 2 — Wrap it in `DefaultEmailService`. The second `@bean` takes the provider and returns it as an `EmailService` — the *port* your domain code depends on. The declared return type matters: by returning `EmailService`, every class that asks for an `EmailService` receives this wrapper, and none of them learn that SMTP is behind it.

lumen/notifications/config.py

```

from pyfly.container import bean, configuration
from pyfly.notifications import DefaultEmailService, EmailService
from pyfly.notifications.providers.smtp import SmtplibEmailProvider

@configuration
class NotificationConfig:

    @bean
    def email_provider(self) → SmtplibEmailProvider:
        return SmtplibEmailProvider(
            "smtp.lumenbank.internal",
            port=587,
            username="notifications",
            password="s3cr3t",
            use_tls=True,
        )

    @bean
    def email_service(
        self, provider: SmtplibEmailProvider,
    ) → EmailService:
        return DefaultEmailService(provider=provider)

```

Listing 17.6 — SMTP provider wired as a @bean

`SmtplibEmailProvider` accepts `host`, `port` (default `587`), `username`, `password`, and `use_tls` (default `True`). Swap it for `SendGridEmailProvider` or `ResendEmailProvider` by changing one `@bean` method — `DefaultEmailService` is indifferent to the provider behind it.

🔔 AVAILABLE PROVIDERS

`pyfly.notifications` ships eight built-in adapters: `DummyEmailProvider` / `DummySmsProvider` / `DummyPushProvider` (log-only, for dev/tests), `SmtplibEmailProvider`, `SendGridEmailProvider`, `ResendEmailProvider` (email), `TwilioSmsProvider` (SMS), and `FirebasePushProvider` (push). All satisfy their respective `*Provider` protocol.

Sending a "funds received" notification

Lumen publishes a `FundsDeposited` domain event every time the `deposit()` command succeeds (see Chapter 8). The right place to trigger the notification is an EDA listener subscribed to that event — not the command handler itself, which keeps the deposit path free of notification concerns.

`FundsDeposited` carries `wallet_id: str`, `amount: int` (minor units), `currency: str`, and `balance: int` (new balance, minor units). The listener converts `amount` to a display string via `amount / 100`.

Step 1 — Subscribe to the event. Stack

`@event_listener(event_types=["FundsDeposited"])` on an `async` method of a `@service`. At startup PyFly discovers the stamped method and auto-subscribes it to the `EventPublisher` bus — exactly the same mechanism `WalletAuditListener` used back in Chapter 8. You never wire a bus by hand.

Step 2 — Read the payload. The handler receives an `EventEnvelope`. Its `payload` is a plain dict of the event's fields, so you pull `wallet_id`, `amount`, `currency`, and `balance` out with `.get(...)` and coerce the amounts to ints. Because amounts are minor units, dividing by 100 gives the display value: `25000` becomes `250.00`.

Step 3 — Send through the ports. Inject `EmailService` and `PushService` in the constructor and call `.send(...)` on each. Both return a `NotificationResult` rather than raising — a flaky provider degrades gracefully instead of crashing the listener.

lumen/wallet/deposit_notification_listener.py

```

from pyfly.container import service
from pyfly.eda import EventEnvelope, event_listener
from pyfly.notifications import (
    EmailMessage,
    EmailService,
    PushMessage,
    PushService,
)

@service
class DepositNotificationListener:
    """Sends email + push when a FundsDeposited event is observed."""

    def __init__(
        self,
        email_service: EmailService,
        push_service: PushService,
    ) → None:
        self._email = email_service
        self._push = push_service

    @event_listener(event_types=["FundsDeposited"])
    async def on_funds_deposited(
        self, envelope: EventEnvelope,
    ) → None:
        payload = envelope.payload
        wallet_id = str(payload.get("wallet_id", ""))
        amount_minor = int(payload.get("amount", 0))
        currency = str(payload.get("currency", "EUR"))
        balance_minor = int(payload.get("balance", 0))
        amount_str = f"{amount_minor / 100:.2f} {currency}"
        balance_str = f"{balance_minor / 100:.2f} {currency}"

        # Fetch contact details from a wallet profile service in prod;
        # hardcoded here for illustration.
        email = "customer@example.com"

```



```

device_token = "FCM_TOKEN_GOES_HERE"

await self._email.send(EmailMessage(
    to=[email],
    sender="no-reply@lumenbank.com",
    subject=f"Funds received: {amount_str}",
    body_text=(
        f"{amount_str} has been credited to wallet "
        f"{wallet_id}. New balance: {balance_str}."
    ),
))

await self._push.send(PushMessage(
    device_tokens=[device_token],
    title="Funds received",
    body=f"{amount_str} credited",
    data={
        "wallet_id": wallet_id,
        "amount_minor": amount_minor,
        "currency": currency,
    },
))

```

Listing 17.7 — Notifying on FundsDeposited

Both calls return a `NotificationResult` ; inspect the `status` field to log failures or schedule retries.

Run it. You do not want a real SMTP server while developing, so swap the provider for the log-only `DummyEmailProvider` . In your `@configuration` class, return a `DummyEmailProvider` (and `DummyPushProvider`) instead of the SMTP one, then start the app and trigger a deposit:

```

uv run pyfly run --server uvicorn
# in a second terminal, open a wallet and deposit (see Chapter 7), e.g.:
curl -s -X POST localhost:8080/api/v1/wallets/<wallet-id>/deposit \
  -H 'content-type: application/json' -d '{"amount":25000}'

```

The deposit publishes `FundsDeposited`, the listener fires, and the dummy providers log the messages they "sent":

```
[dummy email] to=['customer@example.com'] subject=Funds received: 250.00 EUR
[dummy push] tokens=1 title=Funds received
```

The `DummyEmailProvider` also keeps every message it received in a `.sent` list — which is exactly what the test in Exercise 2 asserts against, no SMTP server required.

WHAT JUST HAPPENED

The deposit command knew nothing about email. It simply did its job and raised a `FundsDeposited` domain event. The notification logic lives in a separate listener that *reacts* to that event, so the deposit path stays clean and you can add, remove, or change notifications without touching the command handler. That separation — publish a fact, let interested parties react — is the whole point of event-driven architecture.

SPRING PARITY

`EmailService` / `SmsService` / `PushService` are the Python equivalents of Spring's `JavaMailSender` (email) and third-party Spring integrations for SMS and push. The hexagonal port/adaptor split is identical: your domain code depends on the protocol; concrete provider adapters live in the infrastructure layer.

Inbound webhooks

An illustrative payment provider POSTs a `payment_intent.succeeded` event to Lumen whenever a customer tops up their wallet by card. Lumen must:

1. **verify the HMAC-SHA256 signature** to reject forged payloads;

2. **deduplicate** replays using the idempotency key so a retry does not credit a wallet twice;
3. **dispatch** the verified event to a typed listener.

PyFly's `pyfly.webhooks` module handles all three steps.

NEW TERMS: WEBHOOK, HMAC, IDEMPOTENCY

A *webhook* is an HTTP POST that an external system sends *to you* when something happens — the inbound mirror of the outbound callbacks later in this chapter. Because anyone can POST to a public URL, the provider signs each request with a shared secret using *HMAC* (a keyed hash); recomputing the hash over the exact bytes received and comparing proves the payload is genuine and untampered. *Idempotency* means "safe to receive more than once": providers retry on network blips, so you store an idempotency key and ignore a repeat — otherwise one card top-up could credit a wallet twice.

WebhookEvent and AbstractWebhookEventListener

Every inbound event is modelled as a `WebhookEvent` dataclass:

```
@dataclass
class WebhookEvent:
    id: str                # auto-generated UUID
    source: str            # e.g. "payment-provider"
    event_type: str        # from body["type"]
    headers: dict[str, str]
    body: dict[str, Any]
    raw_body: bytes
    received_at: datetime
    idempotency_key: str | None
```

Subclass `AbstractWebhookEventListener` and set `source` to the name you will pass to `WebhookProcessor.process()`:

lumen/webhooks/payment_listener.py

```

from pyfly.container import service
from pyfly.webhooks import AbstractWebhookEventListener, WebhookEvent

@service
class PaymentWebhookListener(AbstractWebhookEventListener):
    source = "payment-provider"

    def __init__(self, deposit_handler) → None:
        self._handler = deposit_handler

    async def handle(self, event: WebhookEvent) → None:
        if event.event_type == "payment_intent.succeeded":
            pi = event.body.get("data", {}).get("object", {})
            wallet_id = pi.get("metadata", {}).get("wallet_id")
            amount_minor = int(pi.get("amount_received", 0))
            currency_code = pi.get("currency", "EUR").upper()
            if wallet_id and amount_minor > 0:
                # Delegate to the CQRS command handler so the aggregate
                # enforces the balance invariant and raises FundsDeposited.
                from lumen.core.services.wallets.deposit_funds_command import
(
                    DepositFunds,
                )
                await self._handler.handle(
                    DepositFunds(
                        wallet_id=wallet_id,
                        amount=amount_minor,
                    )
                )

    async def on_error(
        self, event: WebhookEvent, error: BaseException,
    ) → None:
        # Override to DLQ or page on-call.
        pass

```

Listing 17.8 — Payment-provider webhook listener

`on_error` is called when `handle` raises; the default is a no-op. Override it to publish to a dead-letter queue or emit a metric.

WebhookProcessor — verify, dedupe, dispatch

`WebhookProcessor` wires together a signature validator, an idempotency store, and a list of listeners. Assemble it in a `@configuration` class so it is a single shared bean:

- `listeners` is the list of `AbstractWebhookEventListener` subclasses to fan events out to (just `PaymentWebhookListener` for now);
- `signature_validators` maps each `source` name to the validator that proves its requests are genuine — here an `HmacSignatureValidator` keyed by the webhook secret your provider gave you;
- an `event_store` (omitted here, so the default in-memory one is used) remembers idempotency keys.

lumen/webhooks/processor_config.py

```

from pyfly.container import bean, configuration
from pyfly.webhooks import (
    HmacSignatureValidator,
    WebhookProcessor,
)
from lumen.webhooks.payment_listener import PaymentWebhookListener

@configuration
class WebhookConfig:

    @bean
    def webhook_processor(
        self, payment_listener: PaymentWebhookListener,
    ) → WebhookProcessor:
        return WebhookProcessor(
            listeners=[payment_listener],
            signature_validators={
                "payment-provider": HmacSignatureValidator(
                    secret="whsec_REPLACE_ME",
                ),
            },
        )

```

Listing 17.9 — Assembling WebhookProcessor

`HmacSignatureValidator` expects the `sha256=<hex>` header format and uses `hmac.compare_digest` for a constant-time comparison. Change the `header_prefix` parameter if your provider uses a different scheme.

Handling a webhook in an HTTP handler

Call `processor.process()` from your inbound HTTP handler. Pass the raw request body (unmodified bytes) — the validator computes the HMAC over the exact bytes received.

⚠ READ THE BODY AS RAW BYTES, NOT PARSED JSON

The signature is computed over the *exact bytes* the provider sent. If you parse the JSON and re-serialize it, key order or whitespace can shift and the HMAC will no longer match — every legitimate request would be rejected. Always pass `await request.body()` (the untouched bytes) to `process()`, as the handler below does.

lumen/webhooks/payment_handler.py

```

from pyfly.container import rest_controller
from pyfly.web import post_mapping, request_mapping
from pyfly.webhooks import WebhookProcessor
from starlette.requests import Request
from starlette.responses import Response

@rest_controller
@request_mapping("/webhooks")
class PaymentWebhookHandler:

    def __init__(self, processor: WebhookProcessor) → None:
        self._processor = processor

    @post_mapping("/payment")
    async def receive(self, request: Request) → Response:
        # Read the untouched bytes – the HMAC is computed over exactly
        # what the provider sent (see the warning above).
        raw_body = await request.body()
        headers = {
            "X-Signature": request.headers.get(
                "X-Webhook-Signature", ""
            ),
            "X-Idempotency-Key": request.headers.get(
                "X-Idempotency-Key", ""
            ),
        }
        try:
            await self._processor.process(
                source="payment-provider",
                raw_body=raw_body,
                headers=headers,
            )
        except ValueError:
            return Response(content=b"invalid signature", status_code=400)
        return Response(content=b"ok", status_code=200)

```


Listing 17.10 — Inbound payment-provider webhook endpoint

The `process()` signature accepts `signature_header` and `idempotency_header` keyword arguments to override the default header names (`X-Signature` and `X-Idempotency-Key`).

What happens inside `process()` :

1. The validator is looked up by `source` ; if none is registered, a `NoOpSignatureValidator` is used — always passes, safe for development but not production.
2. If signature validation fails, `ValueError` is raised immediately and no listeners are called.
3. The raw body is decoded as JSON; on failure the raw bytes are stored under `body["_raw"]` .
4. If `idempotency_key` is present and already seen, the event is returned but listeners are **not** called.
5. Each listener for the source is called in registration order; if one raises, the error is logged and `on_error` is called before continuing to the next listener.

Run it. Test the endpoint the way a real provider would — by signing the exact body. Pick the same secret you put in `HmacSignatureValidator` (`whsec_REPLACE_ME` in Listing 17.9), compute the HMAC with `openssl` , and POST it. Start the app (`uv run pyfly run --server uvicorn`), then in a second terminal:

terminal

```
BODY='{ "type": "payment_intent.succeeded", "data": { "object":
{"amount_received": 25000, "currency": "eur", "metadata": { "wallet_id": "<wallet-
id>" } } } }'
SIG=$(printf '%s' "$BODY" | openssl dgst -sha256 -hmac "whsec_REPLACE_ME" |
sed 's/^.* //')

curl -s -X POST localhost:8080/webhooks/payment \
-H "X-Webhook-Signature: sha256=$SIG" \
-H "X-Idempotency-Key: evt-001" \
-H 'content-type: application/json' \
-d "$BODY"
```

Listing 17.10a — Sign and POST a webhook

A correctly signed request returns `ok` and credits the wallet (you will see the `FundsDeposited` notification logs from earlier fire too):

ok

Now prove the two guarantees. POST the **same** command again with the same `X-Idempotency-Key` — it still returns `ok`, but the wallet is *not* credited a second time (the duplicate is dropped before any listener runs). Then tamper with one byte of `$BODY` *without* recomputing `$SIG` and POST again — the signature no longer matches, so the handler returns:

invalid signature

That is the verify-dedupe-dispatch pipeline working end to end, and it mirrors Exercise 3 almost exactly.

❗ WHAT JUST HAPPENED

A stranger on the internet POSTed JSON to your service, and three guards ran before a single line of your business logic: the signature check rejected forgeries, the idempotency store rejected replays, and only then did the typed listener translate the event into a CQRS `DepositFunds` command — so the wallet aggregate still enforced its own invariants. You wrote the `handle()` body; PyFly supplied the gauntlet around it.

❗ IN-MEMORY IDEMPOTENCY STORE

The default `InMemoryWebhookEventStore` holds seen keys in a Python `set`. For production clusters, implement the `WebhookEventStore` protocol backed by Redis or a database:

```
class RedisEventStore:
    async def already_processed(
        self, key: str,
    ) → bool: ...
    async def remember(self, key: str) → None: ...
```

Pass it as `event_store=RedisEventStore(...)` to `WebhookProcessor`.

Outbound callbacks

When Lumen books a disbursement to a partner bank, that partner expects a `DisbursementSettled` POST to their webhook URL — signed, retried on failure, and auditable. PyFly's `pyfly.callbacks` module handles the outbound side.

i NEW TERM: OUTBOUND CALLBACK

A *callback* here is the reverse of the inbound webhook you just built: *Lumen* is now the sender, POSTing an event to a partner's URL. The module gives you the same trust and reliability machinery on the way out — it signs each payload (so the partner can verify it), retries transient failures with backoff, and records every attempt so you have an audit trail when a partner asks "did you ever tell us about transaction X?".

Subscriptions and config

Each partner is modelled as a `CallbackConfig` — a tenant-scoped record that holds the webhook secret, event subscriptions, and retry policy.

i NEW TERM: TENANT

A *tenant* is one isolated customer or organisation inside a shared application — here, "*lumen*". Callbacks are tenant-scoped so a multi-tenant deployment can hold each tenant's partner URLs, secrets, and retry policy separately and never cross the wires. With a single tenant you simply pass the same `tenant_id` everywhere.

Step 1 — Describe each subscription. A `CallbackSubscription` pairs an `event_type` with the `target_url` to POST it to. Use the exact event name ("`DisbursementSettled`") to route one event, or "`*`" as a catch-all that matches every event for the tenant — handy for an audit endpoint that wants the full firehose.

Step 2 — Wrap them in a `CallbackConfig` and save it. The config carries the shared `secret` (used to sign every payload), the retry policy (`max_attempts` , `backoff_ms`), and the list of subscriptions. Persist it through a `CallbackConfigRepository` — `InMemoryCallbackConfigRepository` for now, a database-backed one in production.

lumen/callbacks/register_partner.py

```

from pyfly.callbacks import (
    CallbackConfig,
    CallbackSubscription,
    InMemoryCallbackConfigRepository,
    InMemoryCallbackExecutionRepository,
)

async def register_clearance_bank(configs) → None:
    await configs.save(CallbackConfig(
        tenant_id="lumen",
        name="clearance-bank",
        secret="cb-secret-xyz",
        max_attempts=5,
        backoff_ms=2_000,
        subscriptions=[
            CallbackSubscription(
                event_type="DisbursementSettled",
                target_url=(
                    "https://api.clearancebank.example.com"
                    "/hooks/lumen"
                ),
            ),
            CallbackSubscription(
                event_type="*",
                target_url=(
                    "https://audit.clearancebank.example.com"
                    "/all-events"
                ),
            ),
        ],
    ))

```

Listing 17.11 — Registering a partner callback

`event_type="*"` is a catch-all: every event dispatched for the tenant matches. Named types match only the exact event type string.

Dispatching an event

`CallbackDispatcher.dispatch()` fans the event out to every matching subscription:

lumen/callbacks/dispatcher_config.py

```

from pyfly.callbacks import (
    CallbackDispatcher,
    InMemoryCallbackConfigRepository,
    InMemoryCallbackExecutionRepository,
)
from pyfly.container import bean, configuration

@configuration
class CallbackConfig_:

    @bean
    def callback_configs(self) → InMemoryCallbackConfigRepository:
        return InMemoryCallbackConfigRepository()

    @bean
    def callback_executions(
        self,
    ) → InMemoryCallbackExecutionRepository:
        return InMemoryCallbackExecutionRepository()

    @bean
    def callback_dispatcher(
        self,
        configs: InMemoryCallbackConfigRepository,
        executions: InMemoryCallbackExecutionRepository,
    ) → CallbackDispatcher:
        return CallbackDispatcher(
            configs=configs,
            executions=executions,
        )

```

Listing 17.12 — Wiring and calling CallbackDispatcher

Then in the domain service — note the payload uses `amount` in minor units (cents), so `50_000` is EUR 500.00:

```

results = await dispatcher.dispatch(
    "lumen",          # tenant_id
    "DisbursementSettled",
    {"id": "txn-009", "amount": 50_000, "currency": "EUR"},
)

```

`dispatch()` returns one `CallbackExecution` record per matching subscription, each with `status`, `attempts`, `response_status`, and `delivered_at`.

Run it. The dispatcher's default HTTP sender does not actually call the network — it logs the request it *would* make and returns `200` — which is perfect for seeing the wiring without standing up a partner server. Drop this into a script (or `uv run python`) from `samples/lumen`:

lumen/callbacks/try_dispatch.py

```

import asyncio

from pyfly.callbacks import (
    CallbackConfig,
    CallbackDispatcher,
    CallbackSubscription,
    InMemoryCallbackConfigRepository,
    InMemoryCallbackExecutionRepository,
)

async def main() → None:
    configs = InMemoryCallbackConfigRepository()
    await configs.save(CallbackConfig(
        tenant_id="lumen",
        name="clearance-bank",
        secret="cb-secret-xyz",
        subscriptions=[CallbackSubscription(
            event_type="DisbursementSettled",
            target_url="https://api.clearancebank.example.com/hooks/lumen",
        )],
    ))
    dispatcher = CallbackDispatcher(
        configs=configs,
        executions=InMemoryCallbackExecutionRepository(),
    )
    results = await dispatcher.dispatch(
        "lumen",
        "DisbursementSettled",
        {"id": "txn-009", "amount": 50_000, "currency": "EUR"},
    )
    for r in results:
        print(r.status, r.attempts, r.response_status, r.target_url)

asyncio.run(main())

```

Listing 17.12a — Dispatch against the default (log-only) sender

You should see one delivered execution, with the signed request logged just above it:

```
would POST https://api.clearancebank.example.com/hooks/lumen headers={'X-
Pyfly-Signature': 'sha256=...', 'Content-Type': 'application/json'}
body={'id': 'txn-009', 'amount': 50000, 'currency': 'EUR'}
DELIVERED 1 200 https://api.clearancebank.example.com/hooks/lumen
```

DELIVERED 1 200 reads as: status DELIVERED, succeeded on attempt 1, HTTP 200.

To send for real, pass your own `http=` sender (an `httpx / aiohttp` POST) to `CallbackDispatcher`; the signing, retry, and audit logic stay identical.

WHAT JUST HAPPENED

One `dispatch()` call looked up every subscription the tenant has for that event type, signed the payload, POSTed it, and wrote a `CallbackExecution` record for each — all without your domain service knowing how many partners are listening or how retries work. Adding a partner later is just another saved `CallbackConfig`; the disbursement code never changes.

HMAC signing and retry logic

When `CallbackConfig.secret` is set, `CallbackDispatcher` signs the canonical JSON payload before every POST using HMAC-SHA256:

```
# canonical body — compact, keys sorted
canonical = json.dumps(payload, separators=(",", ":"), sort_keys=True)
sig = hmac.new(secret, canonical.encode(), hashlib.sha256).hexdigest()
headers["X-Pyfly-Signature"] = f"sha256={sig}"
```

The recipient can verify the signature using PyFly's own `HmacSignatureValidator` — the same class used for inbound webhooks.

Retry policy:

- The dispatcher retries up to `max_attempts` times (default `5`).
- Between retries it applies exponential backoff: `delay = min(backoff_ms * 2^(attempt-1), 300_000) ms`.
- Only *transient* HTTP status codes trigger a retry: `408`, `429`, `500`, `502`, `503`, `504`, or any `≥ 500`.
- Permanent client errors (`4xx` except `408 / 429`) mark the execution as `FAILED` immediately without retrying.
- On success (`2xx`) the execution is marked `DELIVERED` and `delivered_at` is stamped.

`lumen/callbacks/models_overview.py`

```
from pyfly.callbacks import CallbackStatus

# After a successful delivery:
assert execution.status == CallbackStatus.DELIVERED
assert execution.delivered_at is not None
assert execution.response_status == 200

# After all retries are exhausted:
assert execution.status == CallbackStatus.FAILED
assert execution.attempts == 5
assert execution.last_error is not None
```

Listing 17.13 — *CallbackExecution status lifecycle*

SSRF protection — authorized domains

The `authorized_domains` field on `CallbackConfig` acts as an allowlist. When set, `CallbackDispatcher` checks that the target URL's hostname matches one of the allowed domains before making any outbound request. A URL that fails the check is immediately marked `FAILED` with `last_error="Domain not authorized"` — no HTTP request is made.

```
from pyfly.callbacks import AuthorizedDomain, CallbackConfig

config = CallbackConfig(
    tenant_id="lumen",
    name="safe-config",
    secret="s3cr3t",
    authorized_domains=[
        AuthorizedDomain(domain="clearancebank.example.com"),
    ],
    subscriptions=[...],
)
```

Subdomains of allowed domains are also accepted (e.g.

`api.clearancebank.example.com`).

NEW TERM: SSRF

Server-Side Request Forgery is an attack where a malicious value tricks your server into making an HTTP request it should not — for example, a partner URL pointing at `http://169.254.169.254/` (a cloud metadata endpoint) to steal credentials. Because callback URLs can come from partner-supplied config, the `authorized_domains` allowlist closes that door: a host that is not on the list is marked `FAILED` *before* any request leaves the process. To verify, add an `AuthorizedDomain` for one host, then dispatch a subscription whose `target_url` points elsewhere — the returned `CallbackExecution` will read `status=FAILED`, `attempts=0`, and `last_error="Domain not authorized"`, and nothing is sent.

 SPRING PARITY

PyFly's `@scheduled` / `CronExpression` / `TaskScheduler` trio mirrors Spring's `@Scheduled` / `CronExpression` / `ThreadPoolTaskScheduler`. Notification ports map to Spring's `MailSender` / `JavaMailSender`. `WebhookProcessor` corresponds to a Spring `@RestController` + Spring Security's HMAC `HmacRequestMatcher`. `CallbackDispatcher` with HMAC signing and retry mirrors Spring's `WebhookPublisher` pattern from Spring Modulith.

Run it — confirm the suite is still green. You added four new integration patterns; make sure nothing else regressed. From the `samples/lumen` directory:

```
uv run --extra dev pytest -q
```

You should see every existing test still pass:

```
.....
[100%]
41 passed in 0.28s
```

The three exercises below add scheduling, notification, and webhook tests of their own — re-run this command after each to watch the count climb. Remember the `--extra dev` flag; the bare `uv sync` omits `pytest`.

✓ What you built

You extended Lumen into a connected system that operates independently of incoming requests:

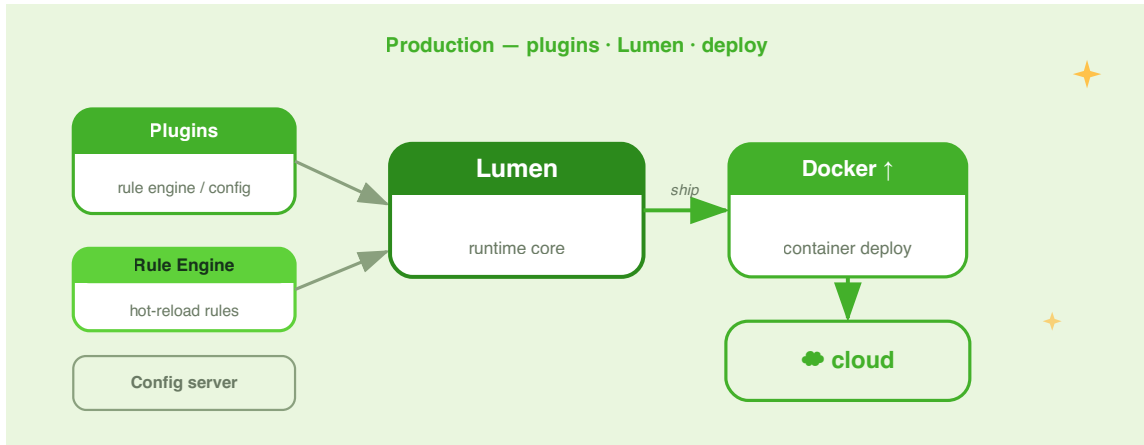
- **Scheduled tasks** — `@scheduled` with `cron`, `fixed_rate`, and `fixed_delay` triggers runs work on a calendar or timer. `CronExpression` drives fire-time calculations, including time-zone-aware scheduling with DST handled automatically. `lock=True` serialises cluster-wide execution via the `DistributedLock` port. `@async_method` offloads fire-and-forget work to the executor.
 - **Notifications** — `EmailService`, `SmsService`, and `PushService` port protocols decouple business logic from provider adapters (SMTP, SendGrid, Resend, Twilio, Firebase). `DefaultEmailService` and its siblings catch provider errors and return structured `NotificationResult` values rather than propagating exceptions. `DepositNotificationListener` subscribes to the real `FundsDeposited` EDA event and converts minor-unit amounts to display strings before sending.
 - **Inbound webhooks** — `AbstractWebhookEventListener` defines typed consumers. `WebhookProcessor` gates every event with `HmacSignatureValidator` and `WebhookEventStore` before dispatching to listeners. `on_error()` hooks allow DLQ integration without breaking the dispatch loop.
 - **Outbound callbacks** — `CallbackDispatcher` fans events out to `CallbackSubscription` targets, signs payloads with HMAC-SHA256 under the `X-Pyfly-Signature` header, and retries on transient failures with exponential backoff. `CallbackExecution` records provide a full delivery audit trail. `authorized_domains` prevents SSRF.
-

Try it yourself

1. **Cluster lock.** Add a second instance of `DailyRollupService` to a test that spins up two `ApplicationContext` instances sharing a `FakeDistributedLock` (one that only returns `True` for the first acquirer). Assert that `run()` is called exactly once across both instances for a single cron tick.
2. **Provider swap.** Replace `SmtplibEmailProvider` with a `DummyEmailProvider` in the test suite. Write a test that deposits EUR 100 (10 000 minor units) into wallet `w-001` via the deposit handler, triggering a `FundsDeposited` event. The provider records every message it received in its `.sent` list, so assert that `provider.sent[-1].body_text` contains the wallet ID and the formatted amount (`100.00 EUR`).
3. **Signature replay attack.** Write a test that calls `PaymentWebhookHandler.receive()` twice with the same raw body, headers, and idempotency key. Assert that the first call returns 200 and credits the wallet (triggering `FundsDeposited`), and the second call also returns 200 (the duplicate is silently ignored by `InMemoryWebhookEventStore`) but does **not** credit the wallet a second time.

CHAPTER 18

Extending PyFly & Going to Production



Lumen is no longer a toy. Over the past seventeen chapters you built a wallet service from a single annotated class to a full event-driven microservice — CQRS, sagas, EDA, HTTP clients, caching, resilience, observability, security, and a test suite backed by real containers. You know how to run it, debug it, and deploy it.

This final chapter is about the distance between "it works on my laptop" and "it runs reliably for real users." That distance has three dimensions. First: **extensibility** — the ability to add behaviour without forking the framework. Second: the features your domain may need right now — a rules engine, centralised config, multiple languages, real-time push, a CLI. Third: the operational habits that separate a weekend project from a production service.

You will move quickly. Every section picks one topic, shows the minimal working API, and connects it to Lumen. By the end you will have a complete picture of the PyFly ecosystem and a production checklist worth keeping close.

CONVENTIONS IN THIS CHAPTER

The listings and config keys here target PyFly **v26.6.110**. Two deployment facts from that release run through the whole "Going to production" section, so it helps to know them up front:

- The application listens on `pyfly.server.port` (default `8080`), the Spring `server.port` parity key. The old `pyfly.web.port` / `PYFLY_WEB_PORT` keys were removed in v26.6.102.
- The Actuator and Admin Dashboard run on a **separate management port** (`pyfly.management.server.port`, default `9090`), open and unauthenticated by default. We cover this in detail when we wire up health probes.

Each feature below is built the same way: a short "why", the code, then a **Run it** checkpoint that shows the exact command and the output you should see. Type the commands as you go — that is how the ideas stick.

Plugins and extension points

Why a plugin system?

The core framework is intentionally small. Optional features — formatters, audit sinks, notification channels — should be pluggable so teams can compose only what they need and ship additions without touching framework internals.

PyFly's `pyfly.plugins` module mirrors Spring's plugin registry: you declare an **extension point** (a named slot), **extensions** (concrete contributions), and bundle them into a **plugin** with a lifecycle.

New jargon, in plain terms: an **extension point** is a labelled hole in your application — "anything that wants to be an audit sink plugs in here." An **extension** is one thing that fills the hole. A **plugin** is the package that ships one or more extensions together and can be started and stopped as a unit. If you have ever defined a Java interface and let callers register implementations of it, you already know the shape — the decorators below just make the wiring declarative.

We will build the smallest useful example: a console audit sink that prints every event it is handed.

Step 1 — Declare the extension point. This is the contract. Every audit sink must promise a `record(event)` method. The `id="audit-sinks"` string is the name other code uses to find every sink later.

Step 2 — Write a plugin that contributes one sink. The `@plugin` decorator wraps a class with a mandatory `id` and `version`; the inner `@extension` class is the actual contribution. It inherits the extension-point interface (`AuditSink`) so the registry can confirm it honours the contract.

Step 3 — Give the plugin a lifecycle. The `start` and `stop` methods are where you open and close resources (a file handle, a network connection). The plugin manager calls them for you.

Here are all three steps in one file.

lumen/plugins/audit.py

```

from pyfly.plugins import (
    PluginManager,
    extension,
    extension_point,
    plugin,
)

@extension_point(id="audit-sinks")
class AuditSink:
    """Interface that all audit sinks must implement."""

    def record(self, event: dict) → None: ...

@plugin(id="console-audit", version="1.0.0")
class ConsoleAuditPlugin:

    @extension(point="audit-sinks", priority=10)
    class ConsoleSink(AuditSink):
        name = "console"

        def record(self, event: dict) → None:
            print(f"[AUDIT] {event}")

        async def start(self) → None:
            print("ConsoleAuditPlugin started")

        async def stop(self) → None:
            print("ConsoleAuditPlugin stopped")

```

Listing 18.1 — Declaring an audit-sink plugin

How it works. `@extension_point(id="audit-sinks")` registers a named slot and declares the interface every contribution must implement. `@plugin(id="console-audit", version="1.0.0")` declares a plugin class with a mandatory `id` and `version`. `@extension(point="audit-sinks", priority=10)` marks an inner class as a contribution to that slot; the inner class must inherit the extension-point interface so the registry can validate it. Higher priority wins first position when iterating results.

Loading and running the plugin:

`lumen/plugins/runner.py`

```
import asyncio

from pyfly.plugins import PluginManager

from lumen.plugins.audit import ConsoleAuditPlugin

async def main() → None:
    manager = PluginManager()
    await manager.add(ConsoleAuditPlugin)
    await manager.start_all()

    sinks = await manager.registry.get("audit-sinks")
    for sink in sinks:
        sink.record({"action": "deposit", "amount_minor": 100})

    await manager.stop_all()

asyncio.run(main())
```

Listing 18.2 — Driving the plugin lifecycle

`PluginManager.add()` inspects the class for nested `@extension_point` declarations, then registers each `@extension` contribution. `start_all()` invokes each plugin's `init` then `start` hooks in dependency order; `stop_all()` reverses the sequence, calling `stop` then `unload`. Circular dependencies raise `PluginResolutionError` before any code runs.

💡 RUN IT

Save both listings under `src/lumen/plugins/` and run the runner module directly:

```
uv run python -m lumen.plugins.runner
```

You should see the lifecycle hooks fire, the single event get recorded, and the clean shutdown:

```
ConsoleAuditPlugin started
[AUDIT] {'action': 'deposit', 'amount_minor': 100}
ConsoleAuditPlugin stopped
```

The order is the whole point: `start` runs before any sink is used, your code iterates the registered sinks, and `stop` runs last. If you add a second plugin later, `start_all()` boots both in dependency order and `stop_all()` shuts them down in reverse — you never manage that ordering by hand.

What just happened. You added a capability to the application — audit recording — without editing a single line of framework code. The framework discovered your plugin, validated that its extension honours the `AuditSink` contract, ran its lifecycle, and handed you the registered sinks to call. That is the core promise of a plugin system: open for extension, closed for modification.

Method	Description
<code>await manager.add(cls)</code>	Scan and register a plugin class
<code>await manager.start_all()</code>	<code>init</code> → <code>start</code> in dependency order
<code>await manager.stop_all()</code>	<code>stop</code> → <code>unload</code> in reverse order
<code>await manager.remove(plugin_id)</code>	Unload one plugin; returns <code>False</code> if unknown
<code>await manager.registry.get(point_id)</code>	Extensions for a slot, priority-sorted

SPRING PARITY

`@plugin` / `@extension_point` / `@extension` mirror `@Plugin` / `ExtensionPoint` / `@Extension` from Spring's plugin API. `PluginManager.start_all()` plays the role of the Spring plugin container's lifecycle management. The dependency-order boot and reverse-order shutdown are identical in semantics.

Business rules with the Rule Engine

Most real-world services carry logic that belongs to the business, not the code: "flag orders over 500,000 cents", "block shipments to sanctioned regions", "apply a surcharge after hours." Hard-coding those thresholds in Python means a rebuild every time the business changes its mind.

PyFly's `pyfly.rule_engine` gives product owners a YAML dial they can turn without touching source code.

A **rule engine**, in plain terms, is a small interpreter for "if this, then that" statements that live in data rather than code. You feed it a bag of facts (the *context*) and a set of rules; it checks each rule's condition against the facts and, for the ones that match, performs the listed actions — usually writing a flag back into the context. The win is that the *rules* are editable by non-programmers, while the *engine* that runs them is fixed and tested.

Defining rules in YAML

We will build a fraud-and-limit check in two steps: first write the rules as data, then wire a service that runs them.

Step 1 — Write the rules as YAML. Each rule names a **when** condition and a list of **then** actions. Because the rules are plain data, a product owner can review them in a pull request and a config server can hot-swap them at runtime.

Rules live in a separate YAML file that any team member can review. The evaluator parses this file once at startup — or on each fetch if you hot-reload from a Config Server (see the next section).

`lumen/rules/transaction_rules.yaml`

```
id: transaction-rules
name: Lumen transaction rules

rules:
  - id: daily-limit
    priority: 20
    when:
      op: ge
      field: transaction.daily_total
      value: 500000
    then:
      - type: set
        target: flags.limit_exceeded
        value: true
      - type: log
        value: "daily limit exceeded"

  - id: fraud-country
    priority: 10
    when:
      op: in
      field: transaction.country
      value: ["XX", "YY", "ZZ"]
    then:
      - type: set
        target: flags.fraud_risk
        value: true
      - type: log
        value: "high-risk country detected"

  - id: high-value
    priority: 5
    when:
      op: ge
      field: transaction.amount
      value: 100000
    then:
```



```
- type: set
  target: flags.high_value
  value: true
```

Listing 18.3 — Fraud and daily-limit rules (amounts in minor units)

Each rule has a `when` condition and a list of `then` actions. Amounts are always **integer minor units** (cents), so `1000000` is €1,000.00. Conditions use these operators:

Comparison	Logical
<code>eq</code> , <code>ne</code> , <code>gt</code> , <code>ge</code> , <code>lt</code> , <code>le</code>	<code>and</code> , <code>or</code> , <code>not</code>
<code>in</code> , <code>not_in</code> , <code>regex</code>	(with <code>conditions: [...]</code>)

Actions are `set` (write to a context path), `increment`, or `log`. Subclass `RuleEvaluator` and override `_execute_action` to add `call`, `calculate`, or any custom verb.

Step 2 — Wire a service that loads and runs the rules. The service parses the YAML once at construction, then exposes an `assess()` method that builds a context, runs the evaluator, and returns the flags the rules set.

lumen/rules/risk_service.py

```

from pathlib import Path

from pyfly.container import service
from pyfly.rule_engine import RuleSetEvaluator, RuleSetLoader

@service
class RiskService:
    """Evaluate transaction-level rules and return risk flags."""

    def __init__(self) → None:
        yaml_text = (
            Path(__file__).parent / "transaction_rules.yaml"
        ).read_text()
        self._ruleset = RuleSetLoader.from_yaml(yaml_text)
        self._evaluator = RuleSetEvaluator()

    def assess(
        self,
        amount: int,
        daily_total: int,
        country: str,
    ) → dict:
        ctx = {
            "transaction": {
                "amount": amount,
                "daily_total": daily_total,
                "country": country,
            },
            "flags": {},
        }
        self._evaluator.evaluate(self._ruleset, ctx)
        return ctx["flags"]

```

Listing 18.4 — Evaluating rules against a transaction

How it works. `RuleSetLoader.from_yaml(text)` parses the YAML into an AST. `RuleSetEvaluator.evaluate(ruleset, ctx)` walks every rule in priority order, evaluates the `when` clause, and applies matching `then` actions — mutating `ctx` in place and returning a `List[EvaluationResult]`. The `flags` dict is the authoritative output: a downstream handler rejects, queues, or flags the transaction based on whatever keys are set. `amount` and `daily_total` are in minor units (cents) to match the Lumen domain.

RUN IT

Exercise the service from a Python REPL to see the rules fire. A €6,000.00 transfer (600000 minor units) trips both the high-value and daily-limit thresholds:

```
uv run python -c "
from lumen.rules.risk_service import RiskService
print(RiskService().assess(amount=600000, daily_total=600000,
country='US'))
"
```

Expected output — the flags the matching rules set, and nothing else:

```
{'high_value': True, 'limit_exceeded': True}
```

Now drop the amount to 50000 (€500.00) with a clean country and you get back an empty `{}` — no rule matched, so nothing was flagged. You changed the *outcome* without touching any Python: that is the dial the YAML gives your product owners.

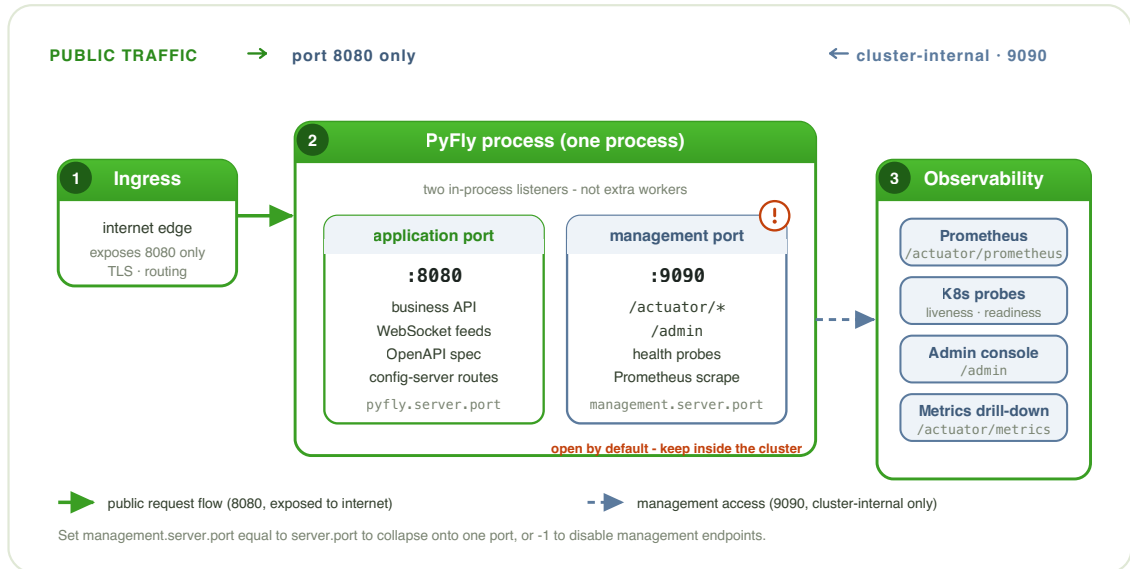


Figure 18.1 — Rule evaluation at the service boundary. YAML rules are parsed once at startup; each transaction passes through the evaluator as a mutable context dict.

💡 HOT-RELOAD RULES WITHOUT REDEPLOYMENT

Store `transaction_rules.yaml` in the Config Server (see the next section) and re-parse on every fetch. Your rules engine becomes a live dial the business controls.

Centralised config (Config Server)

As Lumen grows to multiple services, each carries its own copy of database URLs, timeouts, and feature flags. The Config Server module removes that duplication: one service owns the truth; every other service fetches on startup.

A **config server**, in plain terms, is a small HTTP service whose entire job is to hand out configuration. Instead of baking timeouts and URLs into each service's own `pyfly.yaml`, you store them in one place and every service asks for its bundle at boot. Change the value once, restart (or refresh) the consumers, and the whole fleet moves together.

Running the server

Enable the server in `pyfly.yaml`:

```
pyfly:
  config-server:
    enabled: true
    backend:
      root: /etc/lumen/config
```

That is all. PyFly auto-configures a `ConfigServer` backed by a `FilesystemConfigBackend` and mounts the HTTP routes automatically:

Method	Path	Purpose
GET	<code>/app/{profile}</code>	Fetch merged config bundle
GET	<code>/app/{profile}/{label}</code>	Fetch for a specific label
POST	<code>/app/{profile}</code>	Save a config bundle
GET	<code>/_list</code>	List all stored bundles

The response shape is Spring Cloud Config-compatible, so an existing Spring service can consume the same endpoint without changes.

Saving and fetching config programmatically

`lumen/config/seed.py`

```
import asyncio

from pyfly.config_server import (
    ConfigClient,
    ConfigServer,
    FilesystemConfigBackend,
)

async def seed() → None:
    server = ConfigServer(FilesystemConfigBackend("/etc/lumen/config"))
    await server.save(
        "wallet",
        "prod",
        {
            "db.url": "postgres://db:5432/lumen",
            "cache.ttl": 30,
        },
    )
    bundle = await server.fetch("wallet", "prod")
    print(bundle)

asyncio.run(seed())
```

Listing 18.5 — Seeding and reading a config bundle

 RUN IT

With `pyfly.config-server.enabled: true` in `pyfly.yaml`, start the app and curl the bundle you seeded. Remember: the config-server routes are served on the **application** port (`8080`), not the management port.

```
uv run pyfly run
# in another terminal:
curl http://localhost:8080/wallet/prod
```

The response is Spring Cloud Config-shaped — your seeded keys arrive inside a `propertySources` array:

```
{
  "name": "wallet",
  "profiles": ["prod"],
  "propertySources": [
    {"name": "wallet-prod", "source": {"db.url": "postgres://db:5432/
lumen", "cache.ttl": 30}}
  ]
}
```

Because the shape matches Spring Cloud Config, an existing Spring Boot service can point its `spring.config.import=configserver:` at this same URL and consume it unchanged.

Client services fetch on startup with:

lumen/config/bootstrap.py

```

from pyfly.config_server import ConfigClient

async def load_remote() → dict:
    client = ConfigClient(
        url="http://config:8888",
        application="wallet",
        profile="prod",
        label="main",
    )
    return await client.fetch()

```

Listing 18.6 — Fetching remote config at startup

`ConfigClient.fetch()` GETs `{url}/{application}/{profile}/{label}`, merges the `propertySources` array (highest priority first), and returns a flat `{dotted_key: value}` dict. In normal operation you never call `ConfigClient` directly — set `pyfly.cloud.config.uri` in `pyfly.yaml` and `PyFlyApplication` calls it automatically during bootstrap, merging the result into the application `Config` as a high-precedence source.

FALLBACK PRIORITY

The server assembles up to four overlay layers: `{app}/{profile}`, `{app}/default`, `application/{profile}`, `application/default`. A client merges them with the first source winning, so environment-specific overrides always beat shared defaults.

Internationalisation (i18n)

Lumen's error messages and notifications currently live as Python string literals. When users speak different languages, that approach does not scale.

i18n is shorthand for *internationalisation* (the 18 letters between the "i" and the "n"). In practice it means pulling every user-facing string out of your code into per-language **resource bundles**, then choosing the right bundle at runtime based on the caller's preferred language. Your code refers to a string by a stable key like `wallet.deposit_ok`; the framework looks up the translation.

Enable the i18n subsystem with a single flag:

```
pyfly:
  i18n:
    enabled: true
    base-path: i18n/
    default-locale: en
```

Writing resource bundles

Step 1 — Write one bundle per language. Files are named

`messages_<locale>.yaml`. Keys are shared across languages; only the values change.

The `{0}`, `{1}` markers are positional placeholders the framework fills in at render time.

```
# i18n/messages_en.yaml
wallet:
  deposit_ok: "Deposited {0} minor units to wallet {1}."
  limit_exceeded: "Daily limit exceeded. Maximum is {0} minor units."

# i18n/messages_es.yaml
wallet:
  deposit_ok: "Se depositaron {0} unidades menores en la billetera {1}."
  limit_exceeded: "Se superó el límite diario. El máximo es {0} unidades menor
es."
```

`ResourceBundleMessageSource` resolves keys with dot notation and substitutes `{n}` placeholders (zero-based) following `MessageFormat` semantics. Missing codes fall back to `default-locale`; if they are absent there too, `getMessage` raises `KeyError`.

Using MessageSource in a service

Step 2 — Inject `MessageSource` and resolve the locale from the request. The service below reads the caller's `Accept-Language` header, picks the matching bundle, and renders the message with the runtime arguments.

lumen/i18n/notification_service.py

```

from pyfly.container import service
from pyfly.i18n import AcceptHeaderLocaleResolver, MessageSource

@service
class NotificationService:
    """Renders user-facing messages in the caller's preferred locale."""

    def __init__(
        self,
        messages: MessageSource,
        locale_resolver: AcceptHeaderLocaleResolver,
    ) → None:
        self._messages = messages
        self._resolver = locale_resolver

    def deposit_confirmation(
        self,
        request,
        amount_minor: int,
        wallet_id: str,
    ) → str:
        locale = self._resolver.resolve_locale(request)
        return self._messages.get_message_or_default(
            "wallet.deposit_ok",
            default="Deposit successful.",
            args=(amount_minor, wallet_id),
            locale=locale,
        )

```

Listing 18.7 — Locale-aware notification service

`AcceptHeaderLocaleResolver` parses the `Accept-Language` header and returns the highest-`q` primary subtag. Use `FixedLocaleResolver` for single-language deployments or tests. Auto-configuration registers both when `pyfly.i18n.enabled: true`; inject either `MessageSource` (the port protocol) or the concrete `ResourceBundleMessageSource` — both work.

RUN IT

The fastest way to prove the lookup works is a test that resolves the bundle directly — no HTTP server required. Save Listing 18.7a under `tests/`, then run just this test:

```
uv run --extra dev pytest tests/test_messages.py -q
```

Expected:

```
1 passed in 0.05s
```

Swap `locale="es"` for `locale="en"` and the assertion would need the English string instead — same key, different bundle. That is the whole point of `i18n`: your code never changes, only the resolved locale does.

tests/test_messages.py

```

from pyfly.i18n.adapters.resource_bundle import ResourceBundleMessageSource

def test_deposit_message_in_spanish() → None:
    messages = ResourceBundleMessageSource(base_path="i18n/",
    default_locale="en")
    text = messages.get_message(
        "wallet.deposit_ok", args=(100, "w-001"), locale="es"
    )
    assert text == "Se depositaron 100 unidades menores en la billetera
    w-001."

```

Listing 18.7a — Asserting a translated message

🍃 SPRING PARITY

`MessageSource`, `ResourceBundleMessageSource`, `AcceptHeaderLocaleResolver`, and `FixedLocaleResolver` are direct name equivalents of the Spring MVC i18n stack. The API differs only in the use of positional `{n}` placeholders instead of SpEL inside message strings.

Real-time updates with WebSocket

The Lumen admin dashboard currently polls for balance changes. A WebSocket endpoint eliminates the poll — the server pushes an update the instant a deposit commits.

A **WebSocket** is a two-way connection that stays open. A normal HTTP request is one-shot: the client asks, the server answers, the line closes. With a WebSocket the line stays up, so the *server* can send data whenever it likes — perfect for live updates. The URL scheme is `ws://` (or `wss://` over TLS) instead of `http://`.

lumen/web/balance_ws_controller.py

```

import asyncio

from pyfly.container import rest_controller
from pyfly.web import request_mapping
from pyfly.websocket import WebSocketSession, websocket_mapping

@rest_controller
@request_mapping("/ws")
class BalanceFeedController:
    """Streams balance updates to connected clients."""

    def __init__(self, wallet_service) → None:
        self._wallet = wallet_service
        self._clients: set[WebSocketSession] = set()

    @websocket_mapping("/balance/{wallet_id}")
    async def balance_feed(self, session: WebSocketSession) → None:
        wallet_id = session.path_params["wallet_id"]
        await session.accept()
        self._clients.add(session)
        try:
            while True:
                balance = await self._wallet.get_balance(wallet_id)
                await session.send_json(
                    {"wallet_id": wallet_id, "balance_minor": balance}
                )
                await asyncio.sleep(1)
        finally:
            self._clients.discard(session)

    async def on_disconnect(self, session: WebSocketSession) → None:
        self._clients.discard(session)

```

Listing 18.8 — Live balance feed via WebSocket

How it works. `@websocket_mapping("/balance/{wallet_id}")` mounts the endpoint at `ws://<host>/ws/balance/{wallet_id}`. The full path is the controller's `@request_mapping` base (`/ws`) concatenated with the decorator's path.

🔗 RUN IT

Start the app, then open the live feed with any WebSocket client. Using `websocat` (`brew install websocat`):

```
uv run pyfly run
# in another terminal:
websocat ws://localhost:8080/ws/balance/w-001
```

You should see a JSON frame arrive roughly once a second, pushed by the server without you asking again:

```
{"wallet_id": "w-001", "balance_minor": 5000}
{"wallet_id": "w-001", "balance_minor": 5000}
```

Deposit into `w-001` from another terminal and watch the `balance_minor` value jump on the next frame — no polling, no refresh. Press `Ctrl+C` to close; the controller's `on_disconnect` runs and the session is dropped from the broadcast set.

`WebSocketSession` exposes the connection lifecycle:

Method	Description
<code>await accept(subprotocol=None)</code>	Complete the handshake
<code>await send_json(data)</code>	Serialise and send a JSON message
<code>await send_text(data)</code>	Send a plain string
<code>await receive_text()</code>	Block until a text message arrives
<code>await receive_json()</code>	Block until a JSON message arrives
<code>await close(code=1000, reason=None)</code>	Close the connection cleanly

`session.path_params`, `session.query_params`, and `session.headers` expose connection metadata. WebSocket routes are auto-discovered alongside HTTP routes — no extra configuration required.

The optional `on_disconnect` method is invoked automatically by the registrar after the `@websocket_mapping` handler returns or the client closes the connection (only if the connection was accepted first), giving controllers a safe place to release resources.

💡 BROADCASTING

Keep a `set[WebSocketSession]` on the controller and fan out with `for client in list(self._clients): await client.send_json(payload)`. Because controller beans are singletons the set lives for the lifetime of the application.

Shell commands and startup runners

Not every feature lives behind an HTTP endpoint. Database seed scripts, one-time data migrations, and scheduled batch jobs are better expressed as CLI commands that run inside the same DI container — sharing services, configuration, and repositories with the main application.

The key idea here: a **shell command** is just a method that runs from the terminal but still has full access to your application's wired services. You do not write a separate script that re-creates the database connection by hand — the command receives the same `WalletService` your HTTP controllers use, because it lives inside the same DI container.

@shell_component and @shell_method

`lumen/cli/wallet_commands.py`

```

from pyfly.shell import (
    shell_argument,
    shell_component,
    shell_method,
    shell_option,
)

@shell_component
class WalletCommands:
    """Operational commands for the Lumen wallet service."""

    def __init__(self, wallet_service) → None:
        self._wallet = wallet_service

    @shell_method(group="wallet", help="Deposit funds into a wallet")
    @shell_argument("wallet_id", help="Target wallet identifier")
    @shell_option("--amount", help="Amount in minor units (integer cents)")
    async def deposit(
        self, wallet_id: str, amount: int = 100
    ) → str:
        result = await self._wallet.deposit(wallet_id, amount)
        return f"New balance: {result['balance_minor']} minor units"

    @shell_method(group="wallet", help="Show current balance")
    @shell_argument("wallet_id", help="Wallet to inspect")
    async def balance(self, wallet_id: str) → str:
        data = await self._wallet.get_balance(wallet_id)
        return f"{wallet_id}: {data['balance_minor']} minor units"

```

Listing 18.9 — DI-powered shell commands

Enable the shell in `pyfly.yaml`:

```
pyfly:
  shell:
    enabled: true
```

PyFly auto-configures a `ClickShellAdapter` and wires every `@shell_method` at startup. The group name becomes a sub-command:

```
python -m lumen wallet deposit w-001 --amount 500
python -m lumen wallet balance w-001
python -m lumen          # no args → drops into REPL mode
```

💡 RUN IT

Deposit 500 minor units into a wallet straight from the command line — the command runs your real `WalletService` against the real database:

```
uv run python -m lumen wallet deposit w-001 --amount 500
```

Expected output (the return value of your `deposit` method, printed by the shell adapter):

```
New balance: 500 minor units
```

Run `uv run python -m lumen wallet balance w-001` and you will see the same 500 reflected back — proof that the command and the HTTP API are talking to the same persistent state, not a throwaway in-memory copy.

CommandLineRunner — one-shot post-startup tasks

For tasks that run once at startup — seeding, warm-up, connection checks — implement `CommandLineRunner`:

lumen/runners/seed_runner.py

```

from pyfly.container import service
from pyfly.shell import CommandLineRunner

@service
class SeedRunner(CommandLineRunner):
    """Seed the database with a default admin wallet on first boot."""

    def __init__(self, wallet_service) → None:
        self._wallet = wallet_service

    async def run(self, args: list[str]) → None:
        if "--seed" in args:
            await self._wallet.ensure_default_wallet()
            print("Default wallet ensured.")

```

Listing 18.10 — Post-startup database seeder

Any bean whose class implements `async def run(self, args: list[str]) → None` structurally satisfies the `CommandLineRunner` protocol. The framework detects it via `isinstance()` (the protocol is `@runtime_checkable`) after `ApplicationReadyEvent` fires, then invokes it with the raw CLI arguments. Use `@order(n)` to control execution order when multiple runners coexist.

 RUN IT

The runner fires during application startup and receives the process's `sys.argv[1:]`, so launch the app through its CLI entry point with the flag appended:

```
uv run python -m lumen --seed
```

After the banner and the route table, you will see the runner's confirmation line:

```
Default wallet ensured.
```

Boot without `--seed` and the line is absent — the `if "--seed" in args` guard skips the work. That is the difference between a *runner* (runs every boot, you gate it yourself) and a one-off shell command (runs only when you invoke its name).

 SPRING PARITY

`@shell_component`, `@shell_method`, `@shell_option`, `@shell_argument`, and `CommandLineRunner` are direct equivalents of Spring Shell's `@ShellComponent`, `@ShellMethod`, `@ShellOption`, `@ShellArgument`, and Spring Boot's `CommandLineRunner` interface. Click replaces JLine as the terminal library, but the programming model is identical.

Generating an SDK from the OpenAPI spec

When Lumen exposes an HTTP API, downstream services should call it via a generated client — not hand-written `httpx` calls that drift out of sync. PyFly builds and serves an OpenAPI 3.1 spec automatically at `/openapi.json`.

An **OpenAPI spec** is a machine-readable description of your HTTP API: every path, every parameter, every request and response shape. An **SDK** (software development kit) is the client library a tool generates *from* that spec — typed methods that wrap the HTTP calls for you. The chain is: your controllers → the spec → a generated client. Because each link is mechanical, the client can never silently drift out of sync with the server.

`OpenAPIGenerator` assembles the spec from route metadata collected by `ControllerRegistrar`:

- **Info** — populated from `title`, `version`, and `description` passed to `create_app()`.
- **Paths** — one operation per `@get_mapping` / `@post_mapping` etc., with parameters inferred from `PathVar[T]`, `QueryParam[T]`, `Header[T]`, and `Body[BaseModel]` type hints.
- **Schemas** — Pydantic models registered in `components.schemas` via `model_json_schema()` and referenced with `$ref`.

With the spec available, generate a Python client in two steps — download the spec, then run the generator.

Step 1 — Download the spec from a running instance. The spec lives on the application port (`8080`), alongside your business routes.

Step 2 — Run the OpenAPI generator to turn that JSON into an installable Python package.

```
# Download the spec from a running instance
curl http://localhost:8080/openapi.json -o lumen-spec.json

# Generate a Python client package
openapi-generator-cli generate \
  -i lumen-spec.json \
  -g python \
  -o lumen-client \
  --package-name lumen_client
```

The generated `lumen_client` package contains typed models and a `DefaultApi` with one method per operation. Consumer services add it as a dependency and call it without knowing anything about HTTP:

`payment/services/wallet_client.py`

```
from lumen_client import ApiClient, Configuration, DefaultApi

class WalletGateway:
    """Typed façade over the generated Lumen client SDK."""

    def __init__(self, base_url: str) → None:
        cfg = Configuration(host=base_url)
        self._api = DefaultApi(ApiClient(cfg))

    def get_balance(self, wallet_id: str) → int:
        result = self._api.get_wallet_balance(wallet_id)
        return result.balance_minor
```

Listing 18.11 — Consuming the generated Lumen SDK

 KEEP THE SPEC VERSIONED

Check `lumen-spec.json` into the Lumen repository and regenerate client packages in CI whenever the spec changes. Downstream teams pin to a specific spec version via their dependency manager — the same discipline Java teams use with Maven artifact versions.

Going to production

Packaging with Docker

Containerising, in plain terms, means freezing your app and everything it needs to run — Python, dependencies, your code, your config — into a single image that runs identically on your laptop, in CI, and in production. A `Dockerfile` is the recipe for building that image.

`pyfly new` generates a `Dockerfile` for every archetype. For a web service it looks like this after production hardening. Read it top to bottom — each line is one step in the build:


```

FROM python:3.12-slim

WORKDIR /app
COPY pyproject.toml uv.lock ./
RUN pip install uv && \
    uv sync --no-dev --extra web --extra data-relational \
        --extra security --extra observability

COPY src/ src/
COPY pyfly.yaml .

# 8080 = application traffic; 9090 = the management port (actuator + admin).
EXPOSE 8080 9090
CMD ["pyfly", "run", "--host", "0.0.0.0", \
    "--port", "8080", "--server", "granian", "--workers", "2"]

```

Walking the recipe: `FROM` picks a small Python base image; `COPY` + `uv sync` install exactly the dependency extras you name (and nothing else); the second `COPY` adds your source and config; `EXPOSE` documents the two ports the container listens on; `CMD` is the command that runs when the container starts. The `--port 8080` flag here sets `pyfly.server.port` for this process — the management port stays at its `9090` default unless you override `pyfly.management.server.port`.

Install only the extras your service actually uses — the `full` meta-extra pulls in Kafka, RabbitMQ, and MongoDB drivers even when you need none of them.

 RUN IT

Build the image and run it, mapping both ports out of the container:

```
docker build -t lumen:local .
docker run --rm -p 8080:8080 -p 9090:9090 lumen:local
```

Then confirm both listeners answer — the business port on **8080** and the management port on **9090** :

```
curl http://localhost:8080/openapi.json # app: returns the spec
curl http://localhost:9090/actuator/health # management: {"status":"UP"}
```

Two ports, one process. That separation is what the rest of this section builds on.

Environment variables and secrets

Never bake secrets into `pyfly.yaml`. PyFly resolves `${ENV_VAR}` placeholders anywhere in configuration:

```
pyfly:
  data:
    relational:
      url: ${DATABASE_URL}

  security:
    jwt:
      secret: ${JWT_SECRET}
```

PyFly reads the actual values from the container environment at startup. In Kubernetes, back those variables with a `Secret`; in Docker Compose, use an `.env` file that is never committed. The `pyfly doctor` command checks that required tools are present but does not validate secrets — that responsibility remains yours.

Graceful shutdown

PyFly honours graceful shutdown by default. Set `pyfly.server.graceful-timeout` (seconds) to control how long the server waits for in-flight requests to complete before forcing exit:

```
pyfly:
  server:
    graceful-timeout: 30
```

SIGTERM triggers the shutdown sequence: the server stops accepting new connections, `ApplicationContext.stop()` runs `@pre_destroy` hooks and `stop_all()` for plugins, and the process exits cleanly. In Kubernetes set `terminationGracePeriodSeconds` to at least five seconds more than `graceful-timeout`.

Server selection

The ASGI server is selected by priority at runtime:

Priority	Server	Install extra
1	Granian (Rust/tokio)	<code>granian</code>
2	Uvicorn	<code>web</code> (default)
3	Hypercorn	<code>hypercorn</code>

For production, prefer Granian — it delivers roughly 3× the throughput of Uvicorn with native HTTP/2. Pair it with uvloop on Linux for an additional 2–4× event-loop speedup:

```
uv add "pyfly[web-fast]" # granian + uvloop in one shot
```

Pin the choice in YAML to avoid surprises on machines where multiple servers happen to be installed:

```
pyfly:
  server:
    type: granian
    event-loop: uvloop
    workers: 4
    graceful-timeout: 30
```

The two-port deploy

This is the deployment fact you most need to internalise for v26.6.110. PyFly runs on **two ports**, both inside one process:

- the **application port** — `pyfly.server.port`, default `8080` — serves your business API, your WebSocket feeds, the OpenAPI spec, and the config-server routes;
- the **management port** — `pyfly.management.server.port`, default `9090` — serves the Actuator (`/actuator/*`) and the Admin Dashboard (`/admin`).

The management port is a second in-process listener, not extra worker processes, so it costs almost nothing. Two tuning options matter at deploy time: set

`pyfly.management.server.port` **equal** to `pyfly.server.port` to collapse everything onto one port, or set it to `-1` to disable the management web endpoints entirely. The env override is `PYFLY_MANAGEMENT_SERVER_PORT`.

Why split them? Because it lets you expose only `8080` to the internet while keeping health checks, Prometheus scrapes, and the admin console on `9090`, reachable only from inside your cluster.

⚠ THE MANAGEMENT PORT IS OPEN BY DEFAULT

As of v26.6.110 the management port is **unauthenticated by default** (Spring Boot's `management.server.port` model): anything that can reach `9090` can read `/actuator/*` and `/admin`. That is intentional — the port is meant to sit behind network isolation, never on the public internet. If you cannot guarantee that isolation, apply the app's security filters to the management port too:

```
pyfly:
  management:
    security:
      enabled: true
```

With that flag set, the same authentication, role guards, and CSRF rules that protect your business API also gate `9090`.

Health endpoints

PyFly exposes Spring Boot-style actuator endpoints out of the box. They live on the **management port** (`9090`):

Endpoint	Purpose
<code>GET /actuator/health</code>	Aggregate health (UP / DOWN)
<code>GET /actuator/health/liveness</code>	Kubernetes liveness probe
<code>GET /actuator/health/readiness</code>	Kubernetes readiness probe
<code>GET /actuator/metrics</code>	Per-metric drill-down (e.g. <code>/actuator/metrics/http.server.requests</code>)
<code>GET /actuator/prometheus</code>	Prometheus scrape endpoint

Only `health` and `info` are exposed over HTTP by default (Spring Boot's secure default). To publish metrics and the Prometheus scrape endpoint, add them to the exposure list in `pyfly.yaml`:

```
pyfly:
  management:
    endpoints:
      web:
        exposure:
          include: health,info,metrics,prometheus
    endpoint:
      health:
        show-details: when-authorized
```

RUN IT

Hit the probes on the management port — they answer immediately because that port is open by default:

```
curl http://localhost:9090/actuator/health
curl http://localhost:9090/actuator/health/readiness
```

Expected — a JSON status object the orchestrator can parse:

```
{"status": "UP"}
```

A readiness check that fails (a database still warming up, say) returns `{"status": "DOWN"}` with HTTP `503`, which is exactly what Kubernetes needs to hold traffic back until the pod is ready.

Then wire your Kubernetes deployment — note every probe and scrape points at the **management port** `9090`, not `8080`:

```

livenessProbe:
  httpGet:
    path: /actuator/health/liveness
    port: 9090
  initialDelaySeconds: 10
  periodSeconds: 15
readinessProbe:
  httpGet:
    path: /actuator/health/readiness
    port: 9090
  initialDelaySeconds: 5
  periodSeconds: 10

```

🍃 SPRING PARITY

The two-port split is a direct port of Spring Boot's `management.server.port` : app traffic on `server.port` , actuator and management UI on a dedicated `management.server.port` . The open-by-default posture, the `-1` -to-disable convention, and `management.security.enabled` to lock it down all mirror Spring Boot's behaviour.

📄 CUSTOM PROBES FROM THE LIVE HEALTHAGGREGATOR

New in v26.6.110, the live `HealthAggregator` is reachable at `app.state.pyfly_health_aggregator` (Starlette adapter only). Register an extra readiness indicator after `create_app()` — for example a check that pings a downstream service — and it shows up on `/actuator/health` regardless of whether the actuator runs on the shared or the separate management port.

What just happened. You now have the full shape of a PyFly deployment: one container, two ports (`8080` for users, `9090` for ops), secrets injected as environment variables, a Granian server tuned with explicit workers, a graceful-shutdown window

that drains in-flight requests, and Kubernetes probes wired to the management port. The checklist below is the same picture as a list you can paste into a pull-request template.

The production checklist

- [] All secrets are environment variables — none are in `pyfly.yaml` or source control.
- [] Docker image installs only the extras the service uses.
- [] Server is pinned to Granian + uvloop; `workers` is set explicitly (not left at `1` for multi-core machines).
- [] `graceful-timeout` is at least 15 s; Kubernetes `terminationGracePeriodSeconds` is at least 5 s more.
- [] Liveness and readiness probes point at the **management port** (`9090`), not the application port, and are tested.
- [] `/actuator/health` (on `9090`) returns UP before traffic is sent.
- [] The management port is network-isolated, or `pyfly.management.security.enabled: true` is set — it is open by default.
- [] The Prometheus scrape endpoint (`/actuator/prometheus` on `9090`) is added to the exposure list and scraped by the monitoring stack.
- [] Structured logging is enabled (`pyfly[observability]`) and log level is `INFO` in production, not `DEBUG` .
- [] OpenTelemetry exporter is pointed at the production collector.
- [] Database migrations (`pyfly db upgrade`) run in a pre-deploy step, not at application startup.
- [] The generated OpenAPI spec is versioned in CI and downstream SDK packages pin to a specific spec revision.
- [] `pyfly doctor` passes on every developer machine and CI runner.

✓ What you built

Seventeen chapters ago you typed `@pyfly_application` and watched the DI container wire your first `@service`. Today that same container powers a production wallet platform. Here is what you constructed along the way.

Chapters 1–3 gave you the foundation: a first-class DI container, flexible configuration (YAML, env, profiles, SpEL expressions), and a complete HTTP layer with request mapping, filters, content negotiation, and a JSON serialisation layer you can replace without touching a single controller.

Chapters 4–6 introduced persistence. You mapped entities with SQLAlchemy, managed schema evolution with Alembic, and structured the domain with DDD tactical patterns — aggregates, value objects, and repositories that keep business logic out of infrastructure.

Chapters 7–9 brought the architecture alive. CQRS split reads from writes at the handler level. EDA let services react to events without polling. Event sourcing made every state change a first-class fact — replayable, auditable, and the foundation for projections.

Chapters 10–12 taught Lumen to leave its own process. Resilient HTTP clients called downstream services without cascading failures. Sagas orchestrated multi-step transactions across service boundaries with automatic compensation when any step went wrong.

Chapters 13–15 hardened the platform. Caching cut database pressure. Rate limiters, bulkheads, timeouts, and circuit breakers turned every dependency into a controlled blast radius. Distributed tracing, structured logging, and a live admin dashboard gave you eyes inside the system at every layer.

Chapters 16–17 closed the feedback loop. A structured test suite — unit, integration, and Testcontainers-backed persistence tests — made the platform safe to change. Scheduled tasks, push notifications, webhooks, and callbacks let Lumen reach out to the world on its own schedule.

Chapter 18 showed you what lies beyond the core: a plugin system for open extension, a YAML rule engine for business-owned logic, a Config Server for fleet-wide configuration, i18n for global audiences, WebSocket for real-time UX, a Shell module for operational tooling, an OpenAPI spec that generates typed client SDKs automatically, and the production habits that keep all of it running — packaged as one container that listens on two ports, `8080` for users and `9090` for operations.

PyFly is not magic. Every abstraction in this book has a cost, and you now understand what that cost is: a DI container that starts in milliseconds but requires you to think about bean scopes; an async HTTP server that handles thousands of concurrent connections but requires you to avoid blocking calls; a saga engine that survives partial failures but requires you to write compensating transactions.

Understanding the cost is what separates a practitioner from a reader who merely copies patterns. You are now a practitioner.

Try it yourself

1. **Add a custom plugin.** Implement an `AuditPlugin` that contributes an extension to an `"audit-sinks"` extension point. The extension class must implement an `AuditSink` interface and write each event to a file. In a test, call `PluginManager.registry.get("audit-sinks")` and assert that your extension is returned with the expected `name`.

2. **Ship a rules change without redeployment.** Store `transaction_rules.yaml` in the Config Server (`pyfly.config-server.enabled: true`). Write a `RiskService` that fetches the YAML via `ConfigClient` on each call to `assess()` (or caches it with a short TTL). Update the `value` threshold through the `POST /{app}/{profile}` route and verify that `assess()` picks up the new value without restarting the service.
 3. **Localise a rejection message.** Add `wallet.limit_exceeded` to `i18n/messages_en.yaml` and `i18n/messages_es.yaml` . Wire a `NotificationService` that reads the locale from the `Accept-Language` header and returns the correct string. Write two tests — one with `Accept-Language: en` , one with `Accept-Language: es` — and assert each returns the right message.
-

Lumen is ready for production. For what comes next — new modules, community plugins, and release notes — visit the framework documentation at github.com/fireflyframework/fireflyframework-pyfly. Every concept in this book lives in that repository; the source is the ultimate reference.

Appendices

APPENDIX A

Spring Boot → PyFly Cheat-Sheet

If you have shipped production services with Spring Boot, the concepts in PyFly will feel immediately familiar: stereotypes, constructor injection, `@configuration` + `@bean` factories, typed config binding, derived queries, saga orchestration — they are all here. What changes is the syntax (Python decorators instead of Java annotations), the runtime model (native `async/await` instead of Project Reactor or servlet threads), and a handful of deliberate design choices made to fit idiomatic Python. This cheat-sheet maps every Spring Boot concept you already know to its PyFly equivalent, so you can start reading and writing PyFly code without relearning the architecture from scratch.

Application Bootstrap & Stereotypes

Spring Boot	PyFly	Notes
<code>@SpringBootApplication</code>	<code>@pyfly_application</code>	Combines <code>@EnableAutoConfiguration</code> + <code>@ComponentScan</code> . <code>scan_packages</code> replaces classpath scanning — list packages explicitly.
<code>SpringApplication.run(...)</code>	<code>PyFlyApplication(App); await app.startup()</code>	Entry point is async.
<code>@Component</code>	<code>@component</code>	Generic singleton managed bean.
<code>@Service</code>	<code>@service</code>	Business-logic layer.
<code>@Repository</code>	<code>@repository</code>	Data-access layer.
<code>@RestController</code>	<code>@rest_controller</code>	API endpoint class. No <code>@Controller</code> (no view rendering).
<code>@Configuration</code>	<code>@configuration</code>	Bean factory class.
<code>@Bean</code>	<code>@bean</code>	Factory method inside a <code>@configuration</code> class. Return type hint is the bean's registered type.
<code>@Primary</code>	<code>@primary</code> or <code>@bean(primary=True)</code>	Default when multiple candidates exist.
<code>@Order(N)</code>	<code>@order(N)</code>	Lifecycle and injection ordering.
<code>@Lazy</code>	<code>@lazy</code>	Bean is not created until first resolution.

Dependency Injection

Spring Boot	PyFly	Notes
Constructor <code>@Autowired</code> (implicit in modern Spring)	Plain constructor with type-hinted params	Container reads <code>__init__</code> type hints automatically.
Field <code>@Autowired</code>	<code>field: T = Autowired()</code>	<code>Autowired(required=False)</code> for optional.
<code>@Qualifier("name")</code>	<code>Annotated[T, Qualifier("name")]</code>	Python's <code>Annotated</code> carries the qualifier without losing the base type.
<code>Optional<T></code> injection	<code>Optional[T]</code> param	Resolves to <code>None</code> when no bean is registered.
<code>List<T></code> injection	<code>list[T]</code> param	Collects all registered implementations of <code>T</code> .
<code>Map<String, T></code> injection	<code>dict[str, T]</code> param	<code>{bean-name: bean}</code> for every named bean of type <code>T</code> .
<code>Repository<User></code> generic injection	<code>Repository[User, int]</code> param	Container matches on generic type arguments, honours <code>@primary</code> for ties.
<code>ObjectFactory<T></code> / <code>Provider<T></code>	<code>Provider[T]</code>	Deferred resolution; each <code>.get()</code> re-resolves — safe for <code>TRANSIENT</code> beans.

 PREFER CONSTRUCTOR INJECTION

Constructor injection keeps dependencies visible in the class signature, prevents missing-dependency bugs at startup rather than at runtime, and lets you write plain-Python unit

tests without a container: `svc = WalletService(repo=MockRepo(),
events=MockEvents())` .

Conditions & Auto-Configuration

Spring Boot	PyFly	Notes
<code>@ConditionalOnProperty</code>	<code>@conditional_on_property</code>	Register when a config key equals a specific value.
<code>@ConditionalOnClass</code>	<code>@conditional_on_class("module")</code>	Register when a Python module is importable.
<code>@ConditionalOnMissingBean</code>	<code>@conditional_on_missing_bean(T)</code>	Register when no bean of type <code>T</code> exists yet.
<code>@ConditionalOnBean</code>	<code>@conditional_on_bean(T)</code>	Register only if a bean of type <code>T</code> is present.
<code>@ConditionalOnSingleCandidate</code>	<code>@conditional_on_single_candidate(T)</code>	Exactly one candidate, or one marked <code>@primary</code> .
<code>@ConditionalOnWebApplication</code>	<code>@conditional_on_web_application()</code>	Web stack (Starlette/ FastAPI) present.
<code>@ConditionalOnResource</code>	<code>@conditional_on_resource(path)</code>	Filesystem path exists.

Spring Boot	PyFly	Notes
@ConditionalOnExpression	@conditional_on_expression("#{...}")	SpEL-lite expression — supports \$ {key:default} + arithmetic, comparison, boolean. AST-parsed, no eval .

Lifecycle Hooks & Scopes

Spring Boot	PyFly	Notes
<code>@PostConstruct</code>	<code>@post_construct</code>	Called after DI; can be <code>async def</code> .
<code>@PreDestroy</code>	<code>@pre_destroy</code>	Called on graceful shutdown; can be <code>async def</code> .
Default (singleton) scope	Default (singleton)	One instance per application.
<code>@SessionScope</code>	<code>@component(scope=Scope.SESSION)</code>	One instance per <code>HttpSession</code> .
Custom <code>Scope</code> SPI	<code>Container.register_scope(name, handler)</code>	Implement the <code>ScopeHandler</code> protocol.
<code>@RefreshScope</code> (Spring Cloud)	<code>@refresh_scope</code>	Evicted and rebuilt on <code>POST /actuator/refresh</code> .
<code>ContextRefresher.refresh()</code>	<code>ContextRefresher.refresh()</code> (injectable)	Evicts refresh-scoped beans, resets <code>@config_properties</code> , returns the changed keys.
<code>ApplicationEventPublisher</code>	<code>ApplicationEventPublisher</code> (injectable)	<code>await publisher.publish(event)</code> .
<code>@EventListener</code>	<code>@app_event_listener</code>	Dispatch by <code>isinstance</code> ; sync listeners are allowed.

Configuration & Profiles

Spring Boot	PyFly	Notes
<code>application.yml</code>	<code>pyfly.yaml</code>	Same hierarchical structure.
<code>application-{profile}.yml</code>	<code>pyfly-{profile}.yaml</code>	Profile overlays.
<code>spring.profiles.active=dev</code>	<code>PYFLY_PROFILES_ACTIVE=dev</code> (env var) or <code>pyfly.profiles.active: dev</code> in <code>pyfly.yaml</code>	Activation is identical in priority order.
<code>@ConfigurationProperties(prefix=...)</code>	<code>@config_properties(prefix=...)</code> on a <code>@dataclass</code> (from <code>pyfly.core import config_properties</code>)	Pydantic-backed: validated and frozen at startup.
<code>@Value("\${key}")</code>	<code>field: str = Value("\${key}")</code> (from <code>pyfly.core import Value</code>)	Raises on missing key.
<code>@Value("\${key:default}")</code>	<code>field: str = Value("\${key:default}")</code>	Default after the colon.
<code>@Value("#{expr}")</code> SpEL	<code>Value("#{...}")</code> SpEL-lite	Arithmetic, comparison, boolean, <code>\${...}</code> substitution, <code>env</code> mapping. Constructor injection: <code>Annotated[bool, Value("#{...}")]</code> .
<code>@Bean @Profile("dev")</code>	<code>@bean(profile="dev")</code>	Profile expression (<code>& \ ! ()</code>) on any <code>@bean</code> .

Spring Boot	PyFly	Notes
Boolean <code>@Profile("prod & cloud")</code>	<code>profile="prod & cloud"</code>	Spring Boot 2.4+ operators; legacy comma-OR form still works.
<code>spring.application.name</code>	<code>pyfly.app.name</code>	Application name key in <code>pyfly.yaml</code> .

Property-source priority (lowest → highest):

1. `pyfly-defaults.yaml` (framework built-ins)
2. `pyfly.yaml` (application defaults)
3. `pyfly-{profile}.yaml` (profile overlays)
4. Environment variables (runtime, highest priority)

This is identical to Spring Boot's property-source ordering.

Web Layer

Imports: `from pyfly.web import (Body, PathVar, QueryParam, Valid, get_mapping, post_mapping, put_mapping, delete_mapping, patch_mapping, request_mapping)`. Stereotypes: `from pyfly.container import rest_controller, service, repository, component, configuration`. 404: `from pyfly.kernel import ResourceNotFoundException`.

Spring Boot	PyFly	Notes
<code>@RestController</code>	<code>@rest_controller</code>	
<code>@RequestMapping("/path")</code>	<code>@request_mapping("/path")</code>	Class-level prefix.
<code>@GetMapping</code>	<code>@get_mapping</code>	All handler methods are <code>async def</code> .
<code>@PostMapping</code>	<code>@post_mapping</code>	
<code>@PutMapping</code>	<code>@put_mapping</code>	
<code>@DeleteMapping</code>	<code>@delete_mapping</code>	
<code>@PatchMapping</code>	<code>@patch_mapping</code>	
<code>@PathVariable Long id</code>	<code>id: PathVar[int]</code> <code>param</code>	Type annotation required; matched by name.
<code>@RequestParam(defaultValue="0") int page</code>	<code>page:</code> <code>QueryParam[int] = 0</code>	Python default replaces <code>defaultValue</code> .
<code>@RequestBody @Valid T body</code>	<code>body:</code> <code>Valid[Body[T]]</code>	Pydantic deserialization + validation combined.
<code>@RequestHeader("X-Token") String t</code>	<code>t: Header[str]</code>	
<code>@ResponseStatus(HttpStatus.CREATED)</code>	<code>@post_mapping("/", status_code=201)</code>	Status code on the mapping decorator.
<code>@ControllerAdvice + @ExceptionHandler</code>	<code>@exception_handler</code> or built-in exception hierarchy	<code>ResourceNotFoundException</code> → 404, <code>ValidationException</code> → 422, etc.

Spring Boot	PyFly	Notes
<code>spring.mvc.problemdetails.enabled</code>	<code>pyfly.web.problem-details.enabled: true</code>	RFC 7807 <code>application/problem+json</code> responses.

JSON & Content Negotiation

Spring Boot	PyFly	Notes
<code>spring.jackson.property-naming-strategy</code>	<code>pyfly.web.json.property-naming-strategy</code>	<code>as-is</code> (default) or <code>camelCase</code> .
<code>spring.jackson.default-property-inclusion: non_null</code>	<code>pyfly.web.json.exclude-none: true</code>	
Jackson <code>ObjectMapper</code>	<code>PyFlyJsonSerializer</code>	Central serialization boundary.
Jackson <code>Module</code> / custom serializer	<code>JsonSerializers.register(Type, encode=fn)</code>	Non-Pydantic type encoders.
<code>@JsonNaming(CamelCase...)</code>	<code>CamelModel</code> base class	Opt-in camelCase model — <code>order_id</code> serializes as <code>orderId</code> .
<code>HttpMessageConverter</code>	<code>MessageConverter</code> / <code>MessageConverterRegistry</code>	Built-in: JSON (first) then XML; add custom converters via <code>registry.add(...)</code> .

Data Access

Imports: `from pyfly.data.relational.sqlalchemy import (Repository, BaseEntity, Base, Specification, transactional, Propagation, Isolation)`
`from pyfly.data import Page, Pageable, Sort` `from pyfly.data.query import query` `from pyfly.container import repository`

Repositories

Spring Boot	PyFly	Notes
<code>JpaRepository<E, ID> / CrudRepository</code>	<code>Repository[E, ID]</code>	<code>from pyfly.data.relational.sqla import Repository ; decorate @repository . Subclass with type params; AsyncSession injected automatically at startup</code>
<code>@Repository</code> <code>interface WalletRepo</code> <code>extends JpaRepository<...></code>	<code>@repository class</code> <code>WalletRepository(Repository[WalletEntity, str])</code>	Class, not interface; body holds derived-query stubs and custom methods only.
<code>findByOwnerId(String id)</code>	<code>async def find_by_owner_id(self, owner_id: str) → list[WalletEntity]: ...</code>	Stub body <code>...</code> triggers compile by <code>RepositoryBeanPostProcessor</code> at startup. Prefixes: <code>find_by_</code> , <code>count_by_</code> , <code>exists_by_</code> , <code>delete_by_</code> .
<code>@Query("SELECT u FROM User u WHERE ...")</code>	<code>@query("SELECT u FROM User u WHERE ...")</code> (JPQL-like) or <code>@query("SELECT ...", native=True)</code> (raw SQL)	<code>from pyfly.data.query import query . Params use :name syntax</code>
<code>Specification<T></code>	<code>Specification(lambda root, q: q.where(root.status == "ACTIVE"))</code>	Compose with <code>& / \ / ~ ;</code> ; <code>repo.find_all_by_spec(spec)</code> <code>repo.find_all_by_spec_page(pageable)</code> .

Entities

Spring Boot	PyFly	Notes
<code>@Entity</code> <code>class Order</code>	<code>class Order(Base)</code>	from pyfly.data.relational.sqlalchemy import Base . Registers the table in Base.metadata .
<code>@Entity</code> + surrogate UUID PK + audit columns	<code>class Order(BaseEntity)</code>	from pyfly.data.relational.sqlalchemy import BaseEntity . Inherits id: UUID , created_at , updated_at , created_by , updated_by — all mapped as SQLAlchemy 2.0 Mapped / mapped_column .
<code>@Column</code> / <code>@Id</code>	<code>id: Mapped[str] = mapped_column(String(64), primary_key=True)</code>	SQLAlchemy 2.0 typed columns. Base (no audit) lets the entity own its own PK type, as Lumen's WalletEntity does with a str id.
<code>@SoftDelete</code> (Hibernate 6)	<code>SoftDeleteMixin + SoftDeleteRepository</code>	from pyfly.data.relational.sqlalchemy import SoftDeleteMixin . Adds deleted_at ; repository filters it automatically.
Optimistic locking <code>@Version</code>	<code>VersionedMixin</code>	Adds a version: int column; SQLAlchemy raises StaleDataError on conflict.

Pagination & Sorting

Spring Boot	PyFly	Notes
<pre>Pageable / PageRequest.of(page, size, Sort.by(...).descending())</pre>	<pre>Pageable.of(page, size, Sort.by("created_at").descending())</pre>	<pre>from pyfly.data import Pageable, Sort Sort.by(*fields) returns ascending; .descending() flips all.</pre>
<pre>Page<T></pre>	<pre>Page[T]</pre>	<pre>from pyfly.data import Page Attributes: .items, .total, .p transforms items, preserving metadata.</pre>
<pre>page.getContent() / page.getTotalElements()</pre>	<pre>page.items / page.total</pre>	<pre>Python naming; .total_pages derived as ceil(total / size).</pre>
<pre>repo.findAll(pageable)</pre>	<pre>await repo.find_all(pageable)</pre>	<pre>Returns Page[T].</pre>
<pre>repo.findAll(spec, pageable)</pre>	<pre>await repo.find_all_by_spec_paged(spec, pageable)</pre>	<pre>Applies WHERE, ORDER BY, and LIMIT/OFFSET in one call.</pre>

Transactions & **save**

Spring Boot	PyFly	Notes
<code>@Transactional</code>	<code>@transactional()</code>	from pyfly import t _session injected success,
<code>@Transactional(propagation = REQUIRES_NEW)</code>	<code>@transactional(propagation=Propagation.REQUIRES_NEW)</code>	Full Prop REQUIRE NEVER ,
<code>@Transactional(isolation = READ_COMMITTED)</code>	<code>@transactional(isolation=Isolation.READ_COMMITTED)</code>	Full Isol
<code>@Transactional(readOnly = true)</code>	<code>@transactional(read_only=True)</code>	Routes to Routing read_on
<code>repo.save(entity)</code>	<code>await repo.save(entity)</code>	Calls se flushes b @transa
<code>session.merge(entity)</code> (upsert)	<code>await repo.upsert(entity)</code> (<i>extend</i>) or <code>session.merge(entity)</code> + flush	No built-i your repo WalletR handles l the PK.
<code>AbstractRoutingDataSource</code>	<code>RoutingSessionFactory</code>	factory force a si
Multiple <code>DataSource</code> beans	<code>NamedDataSources</code>	Config: pyfly.d inject Na .get("<

Projections & Mapper

Spring Boot	PyFly	Notes
Interface projection <code>interface OrderSummary { ... }</code>	<code>@projection class OrderSummary: id: str; status: str</code>	<code>from pyfly.data.projection import projection</code> . A concrete dataclass (not a <code>Protocol</code>), not a JDK proxy; registered with <code>mapper.register_projection(src, proj)</code> .
<code>repo.findAll(cls, Projection.class)</code>	<code>mapper.project(entity, OrderSummary)</code>	<code>from pyfly.data.mapper import Mapper</code> .
MapStruct <code>@Mapper</code>	<code>Mapper</code> + <code>@mapping</code> decorator	Runtime reflection; no codegen. <code>mapper.map(obj, TargetDTO)</code> , <code>mapper.map_list(...)</code> .

Migrations

Spring Boot	PyFly	Notes
Flyway migrations	<code>pyfly.data.migrations.*</code>	Same concept: versioned SQL scripts run at startup.

Caching

Spring Boot	PyFly	Notes
<code>@Cacheable(value="cache", key="#id")</code>	<code>@cacheable(backend=cache, key="item:{id}")</code>	<code>{param}</code> template syntax in the key.
<code>@Cacheable(condition=..., unless=...)</code>	<code>@cacheable(condition=..., unless=...)</code>	<code>condition</code> bypasses on args; <code>unless</code> skips storing based on result.
<code>@CacheEvict</code>	<code>@cache_evict(backend=cache, key="item:{id}")</code>	
<code>@CachePut</code>	<code>@cache_put(backend=cache, key="item:{id}")</code>	Update cache after mutation.
<code>CacheManager</code> (auto-configured)	<code>InMemoryCache</code> / <code>RedisCacheAdapter</code> (auto-configured)	Redis selected when the <code>redis</code> extra is installed; falls back to in-memory.

Messaging & Events

Spring Boot	PyFly	Notes
<code>@KafkaListener(topics=..., groupId=...)</code>	<code>@message_listener(topic=..., group_id=...)</code>	Handler is <code>async def</code>
<code>KafkaTemplate.send(topic, event)</code>	<code>await publisher.publish(dest, event_type, payload)</code>	<code>EventPublisher</code> port; (from <code>pyfly.eda</code> import <code>EventPublisher</code>); swap adapters via config.
<code>@RetryableTopic</code> / DLT	<code>@message_listener(retries=3, retry_delay=1.0, dead_letter_topic="...")</code>	Linear-backoff retry; exhausted messages routed to DLQ with <code>x-original-topic</code> / <code>x-exception</code> headers.
<code>ApplicationEvent</code>	<code>EventEnvelope</code>	Domain event container
<code>@EventListener</code>	<code>@event_listener(event_types=["TypeName"])</code>	In-process EDA handler; <code>event_type</code> is the class name string. No <code>@domain_event_listener</code>
<code>ApplicationEventPublisher</code>	<code>ApplicationEventPublisher</code> (injectable)	<code>await publisher.publish(e)</code> for Spring-style app events

💡 MESSAGING VS EDA

PyFly separates **broker messaging** (`pyfly.messaging` — Kafka/RabbitMQ transport) from **domain events** (`pyfly.eda` — `EventEnvelope` + `EventBus`). Start with `InMemoryEventBus` inside a monolith; switch to a Kafka adapter later by changing one configuration key, not your handlers.

Security

Spring Boot	PyFly	Notes
<code>SecurityAutoConfiguration</code>	<code>JwtAutoConfiguration</code> + <code>PasswordEncoderAutoConfiguration</code>	Split by optional dep
JWT filter chain	<code>JwtAutoConfiguration</code> auto-wired	Enable with <code>pyfly.security.jw</code>
<code>@PreAuthorize("hasRole('ADMIN')")</code>	<code>@pre_authorize("hasRole('ADMIN')")</code>	Same SpEL subset: <code>hasAnyRole</code> , <code>hasA</code> , <code>isAuthenticated</code> , <code>denyAll</code> , <code>#param</code> , AST-walked, no <code>eva</code>
<code>@PostAuthorize("returnObject.owner == principal")</code>	<code>@post_authorize("returnObject.owner == principal")</code>	<code>returnObject</code> is bo method's return valu
<code>RoleHierarchy</code> bean	<code>RoleHierarchy.from_string("ADMIN > USER")</code> + <code>set_role_hierarchy(...)</code>	<code>expand(roles)</code> con <code>hasRole</code> / <code>hasAnyRo</code>
<code>ClientRegistration</code> (OAuth2 auth-code)	<code>ClientRegistration(...)</code>	PKCE: add <code>use_pkce</code> generates <code>code_ver</code> <code>code_challenge</code> (S
<code>maximumSessions</code> / <code>SessionRegistry</code>	<code>SessionConcurrencyController</code> + <code>SessionRegistry</code>	Cap per-principal co (<code>evict-oldest</code> or Enable: <code>pyfly.session.con</code> <code>true</code> .

Scheduling

Spring Boot	PyFly	Notes
<code>@Scheduled(fixedRate = 5000)</code>	<code>@scheduled(fixed_rate=5.0)</code>	Spring uses milliseconds; PyFly uses seconds .
<code>@Scheduled(fixedDelay = 1000)</code>	<code>@scheduled(fixed_delay=1.0)</code>	
<code>@Scheduled(cron = "0 0 2 * * ?")</code>	<code>@scheduled(cron="0 2 * * *")</code>	Spring: 6-field (seconds-first). PyFly: standard 5-field; also accepts the 6-field Spring form and <code>?</code> .
<code>@Scheduled(cron=..., zone=...)</code>	<code>@scheduled(cron=..., zone="America/New_York")</code>	IANA time zone; ignored for <code>fixed_rate</code> / <code>fixed_delay</code> .
ShedLock / <code>@SchedulerLock</code>	<code>@scheduled(lock=True, lock_ttl=30)</code> + <code>DistributedLock</code> bean	Skips a tick when the lock is held elsewhere. Defaults to in-process <code>LocalLock</code> ; register a Redis <code>DistributedLock</code> for cross-process single-firing.
<code>@EnableScheduling</code>	<code>SchedulingAutoConfiguration</code>	Auto-enabled when <code>croniter</code> is installed. No explicit <code>@Enable...</code> needed.

Resilience

```
from pyfly.resilience import retry, CircuitBreaker, circuit_breaker,
RateLimiter, rate_limiter, Bulkhead, bulkhead, time_limiter, fallback
```

Spring (Resilience4j)	PyFly	Notes
<code>@Retry</code>	<code>@retry(max_attempts=3, *, delay=0.1, backoff=2.0, jitter=0.1, exceptions=(IOError,))</code>	<code>delay / backoff / jitter</code> are keyword-only. <code>jitter</code> is a float fraction in <code>[0,1]</code> .
<code>@CircuitBreaker</code>	<code>breaker = CircuitBreaker(...); @circuit_breaker(breaker)</code>	Pass a <code>CircuitBreaker</code> instance. Count-based (<code>failure_threshold</code>) or rate-based (<code>failure_rate_threshold</code> + <code>window_size</code>).
<code>@RateLimiter</code>	<code>limiter = RateLimiter(max_tokens=100, refill_rate=100/60); @rate_limiter(limiter)</code>	Token-bucket; pass an instance.
<code>@Bulkhead</code>	<code>bh = Bulkhead(max_concurrent=10); @bulkhead(bh)</code>	Concurrency cap; pass an instance.
<code>@TimeLimiter</code>	<code>@time_limiter(timeout=timedelta(seconds=2))</code>	Raises <code>asyncio.TimeoutError</code> on breach.
<code>fallbackMethod</code>	<code>@fallback(fallback_method=fn)</code> or <code>@fallback(fallback_value=v)</code>	Static or callable fallback.

AOP

Spring Boot	PyFly	Notes
<code>@Aspect + @Component</code>	<code>@aspect + @component</code>	
<code>@Before("execution(...)")</code>	<code>@before("execution(...)")</code>	
<code>@After</code>	<code>@after</code>	Always runs (like <code>finally</code>).
<code>@Around</code>	<code>@around</code>	Call <code>await</code> <code>join_point.proceed()</code> to continue.
<code>@AfterReturning</code>	<code>@after_returning</code>	
<code>@AfterThrowing</code>	<code>@after_throwing</code>	
<code>@EnableAspectJAutoProxy</code>	<code>AopAutoConfiguration</code>	Always active; no opt-in needed.

Pointcut DSL: `execution(* pkg.services.*.*(..))` for method patterns;
`annotation(timed)` for decorator-targeted matching.

Observability & Actuator

Spring Boot	PyFly
<code>management.endpoints.web.exposure.include</code>	<code>pyfly.management.endpoints.web.exposure.include</code>
<code>/actuator/health</code>	<code>/actuator/health</code>
<code>/actuator/info</code>	<code>/actuator/info</code>
<code>/actuator/beans</code>	<code>/actuator/beans</code>
<code>/actuator/env</code>	<code>/actuator/env</code>
<code>POST /actuator/refresh</code> (Spring Cloud)	<code>POST /actuator/refresh</code>
Micrometer <code>@Timed</code>	<code>@timed("metric_name")</code>
Micrometer <code>@Counted</code>	<code>@counted("metric_name")</code>
Prometheus <code>MetricsRegistry</code>	<code>MetricsRegistry</code> (Prometheus backend)
Sleuth / Micrometer Tracing (W3C)	<code>TracingFilter</code> (inbound) + <code>HttpxClientAdapter</code> (outbound)

CQRS & Sagas

```
from pyfly.cqrs import (Command, CommandHandler, DefaultCommandBus, Query,
QueryHandler, DefaultQueryBus, command_handler, query_handler) from
pyfly.transactional.saga.annotations import (saga, saga_step, Input,
FromStep)
```

Spring Boot / Axon	PyFly	Notes
<code>@CommandHandler</code>	<code>@command_handler</code> + <code>@service</code> stacked	Both decorators required; <code>@service</code> registers the bean. Override <code>do_handle(self, cmd)</code> .
<code>@QueryHandler</code>	<code>@query_handler</code> + <code>@service</code> stacked	Same rule. Override <code>do_handle(self, qry)</code> . Bus method: <code>.query(...)</code> .
Controller bus injection	<code>commands: DefaultCommandBus,</code> <code>queries: DefaultQueryBus</code>	Inject the concrete classes, not the protocol. <code>commands.send(cmd)</code> , <code>queries.query(qry)</code> .
<code>@Repository</code> + port	<code>@repository</code> on a class that inherits the <code>@runtime_checkable</code> <code>Protocol</code> port	Port is a <code>Protocol</code> ; adapter class inherits it.
<code>@EventHandler</code> (event-sourcing)	<code>@event_handler</code>	Sourced from <code>EventStore</code> .
<code>@Saga</code>	<code>@saga(name="...",</code> <code>layer_concurrency=N)</code> + <code>@service</code>	Both decorators required for DI + engine registration.
<code>@SagaStep(id=...,</code> <code>compensate=...)</code>	<code>@saga_step(id="...",</code> <code>compensate="method_name")</code>	
<code>@Input</code>	<code>Annotated[T, Input()]</code>	Marker is an instance : <code>Input()</code> . Injects the saga's initial payload.
<code>@FromStep("id")</code>	<code>Annotated[T, FromStep("id")]</code>	Marker is an instance : <code>FromStep("step-id")</code> . Injects a prior step's result.

Spring Boot / Axon	PyFly	Notes
<code>@Tcc</code>	<code>@tcc(name="...")</code>	TCC (Try-Confirm-Cancel) transaction class.
<code>@TccParticipant</code>	<code>@tcc_participant(id="...", order=N)</code>	
<code>@TryMethod</code> / <code>@ConfirmMethod</code> / <code>@CancelMethod</code>	<code>@try_method</code> / <code>@confirm_method</code> / <code>@cancel_method</code>	TCC three-phase methods.
<code>@FromTry</code>	<code>Annotated[T, FromTry]</code>	Inject the try-phase result into confirm/cancel.

Event sourcing: `from pyfly.eventsourcing import AggregateRoot, EventSourcedRepository`. Aggregate uses `self.when(EventType, handler_fn)` to register apply-handlers. Data: `from pyfly.data.relational.sqlalchemy import Base` (requires `pyfly[data-relational]`).

Integration Testing

Spring Boot	PyFly	Notes
<code>@SpringBootTest</code>	<code>service_slice(*beans) / slice_context(...)</code>	Minimal started context; <code>overrides</code> accept a class or pre-built instance.
<code>@WebMvcTest</code>	<code>web_slice(*controllers, overrides=...) → (context, client)</code>	Starts minimal context + <code>PyFlyTestClient</code> .
<code>@DataJpaTest</code>	<code>data_slice(*beans) → context</code>	Data-layer slice.
<code>@Testcontainers + @Container</code>	<code>with postgres_container() as pg:</code>	Python context manager handles lifecycle.
<code>@ServiceConnection</code>	<code>pyfly_config(pg) / pyfly_config_for(pg)</code>	Maps container connection details into PyFly config keys.
<code>@DynamicPropertySource</code>	<code>pyfly_config(*containers, base=...)</code>	One-call <code>Config</code> for several containers.
<code>PostgreSQLContainer</code>	<code>postgres_container()</code>	URL auto-rewritten to <code>asyncpg</code> .
<code>MySQLContainer</code>	<code>mysql_container()</code>	URL rewritten to <code>aiomysql</code> .
<code>GenericContainer (Redis)</code>	<code>redis_container()</code>	Cache + session URLs wired.
<code>KafkaContainer</code>	<code>kafka_container()</code>	
<code>@requires_docker</code>	<code>@requires_docker</code>	Skips test cleanly when Docker daemon is absent.

Install with: `pip install 'pyfly[testcontainers]'`

Embedded Server

Spring Boot	PyFly	Notes
<code>server.port</code>	<code>pyfly.server.port</code>	Application HTTP listen port (default 8080).
<code>server.address</code>	<code>pyfly.server.host</code>	Application bind address.
<code>management.server.port</code>	<code>pyfly.management.server.port</code>	Separate management (actuator + admin) port; default 9090.
Tomcat (default)	Granian (default)	Rust/tokio HTTP runtime; highest priority.
Jetty (fallback)	Uvicorn (fallback)	Ecosystem-standard ASGI fallback.
Undertow (alternative)	Hypercorn (alternative)	Advanced protocol support (HTTP/3).
<code>server.tomcat.*</code>	<code>pyfly.server.granian.*</code>	Server-specific tuning.
<code>WebServer</code> interface	<code>ApplicationServerPort</code> protocol	Contract for the embedded ASGI server.
<code>EventLoopGroup</code> (Netty)	<code>EventLoopPort</code> protocol	Contract for the I/O runtime.
<code>server.type: auto</code>	<code>pyfly.server.type: auto</code> (default)	Granian → Uvicorn → Hypercorn cascade.

💡 **AUTO-CONFIGURATION CASCADE**

PyFly's auto-configuration uses the same conditional-bean pattern as Spring Boot: if Granian is installed, `GranianServerAdapter` wins; if not, Uvicorn is tried next; then Hypercorn. Override at any point by providing your own `ApplicationServerPort` bean.

Native HTML and static resources

Spring / Django	PyFly
Spring <code>ModelAndView</code>	<code>ModelAndView(template, model)</code>
Django <code>reverse</code> / template URL tag	<code>reverse(request, name, **params)</code> / <code>reverse(name, **params)</code>
Django static helper / template tag	<code>static_url(request, path)</code> / <code>static_url(path)</code>
Django <code>ModelAdmin</code>	Explicit <code>ModelAdmin</code> over existing SQLAlchemy/Beanie models

Appendix E explains configuration, security, and persistence boundaries.

APPENDIX B

MongoDB & Document Data

PyFly's document data layer wraps MongoDB through **Beanie ODM** and **Motor** (the async MongoDB driver). The API mirrors the relational adapter deliberately — the same `MongoRepository[T, ID]` base class, the same derived query naming convention, the same `Page / Pageable / Sort` vocabulary — so switching between relational and document storage touches only the document class definition and the repository base class, not the service layer.

All concrete types live in `pyfly.data.document.mongodb`. Shared types (`Page` , `Pageable` , `Sort`) come from `pyfly.data`.

Installation and configuration

Install the extra:

```
terminal
```

```
uv add "pyfly[data-document]"
```

Listing B.1 — Install the data-document extra

Enable the adapter in `pyfly.yaml`:

pyfly.yaml

```
pyfly:
  data:
    document:
      enabled: true
      uri: "mongodb://localhost:27017"
      database: "myapp"
      min_pool_size: 5
      max_pool_size: 50
```

Listing B.2 — Minimal MongoDB configuration

Configuration reference

pyfly.yaml key	Type	Default	Description
pyfly.data.document.enabled	bool	false	Enable the MongoDB adapter
pyfly.data.document.uri	str	mongodb://localhost:27017	Connection URI
pyfly.data.document.database	str	pyfly	Database name
pyfly.data.document.min_pool_size	int	0	Motor pool minimum
pyfly.data.document.max_pool_size	int	100	Motor pool maximum

Every key has a matching environment variable: replace dots with underscores and uppercase — e.g. `PYFLY_DATA_DOCUMENT_URI`. For MongoDB Atlas or a replica set:

pyfly.yaml

```

# Atlas
pyfly:
  data:
    document:
      enabled: true
      uri: >-
        mongodb+srv://user:secret@cluster.mongodb.net/
        ?retryWrites=true&w=majority
      database: production_db

# Replica set (required for transactions)
# pyfly:
#   data:
#     document:
#       uri: "mongodb://m1:27017,m2:27017,m3:27017/?replicaSet=rs0"

```

Listing B.3 — Atlas and replica-set URIs

BaseDocument

`BaseDocument` extends `beanie.Document` with an audit trail. Every document class in a PyFly application should inherit from it.

Field	Type	Default	Description
<code>id</code>	<code>PydanticObjectId</code>	Auto-generated	Document primary key (ObjectId)
<code>created_at</code>	<code>datetime</code>	<code>datetime.now(UTC)</code>	Insert timestamp
<code>updated_at</code>	<code>datetime</code>	<code>datetime.now(UTC)</code>	Last-update timestamp
<code>created_by</code>	<code>str</code> <code>None</code>	<code>None</code>	Creator identifier
<code>updated_by</code>	<code>str</code> <code>None</code>	<code>None</code>	Last-updater identifier

`use_state_management = True` is set on the base `Settings` class, enabling Beanie's change-tracking so `save_changes()` produces efficient partial updates.

A typical document class:

catalog/product_document.py

```

from pydantic import BaseModel, Field
from beanie import Indexed, PydanticObjectId
from pyfly.data.document.mongodb import BaseDocument

class Dimensions(BaseModel):
    width_cm: float
    height_cm: float
    depth_cm: float

class ProductDocument(BaseDocument):
    name: str
    sku: Indexed(str, unique=True)
    description: str = ""
    price: float = Field(gt=0)
    category: Indexed(str)
    tags: list[str] = Field(default_factory=list)
    dimensions: Dimensions | None = None
    active: bool = True

class Settings:
    name = "products"

```

Listing B.4 — ProductDocument with index and nested model

The `Settings.name` attribute sets the MongoDB collection name. Omitting it causes Beanie to derive the name from the class name, which is rarely what you want.

For compound or descending indexes use `Settings.indexes`:

catalog/product_document.py

```

from pymongo import IndexModel, ASCENDING, DESCENDING
from pyfly.data.document.mongodb import BaseDocument

class OrderDocument(BaseDocument):
    customer_id: str
    status: str
    total: float
    region: str

    class Settings:
        name = "orders"
        indexes = [
            IndexModel(
                [("customer_id", ASCENDING), ("status", ASCENDING)],
                name="idx_customer_status",
            ),
            IndexModel(
                [("region", ASCENDING), ("total", DESCENDING)],
                name="idx_region_total",
            ),
        ]

```

Listing B.5 — Compound index via Settings.indexes

Spring Data ↔ MongoRepository mapping

The table below shows how Spring Data MongoDB concepts map to PyFly's document layer. The surface is intentionally identical to `Repository[T, ID]` in the relational adapter, so the same service-layer patterns apply to both.

Spring Data MongoDB	PyFly	Notes
<code>MongoRepository<E, ID></code>	<code>MongoRepository[E, ID]</code>	<code>from pyfly.data.document.mongodb import MongoRepository</code> ; decorate with <code>@repository</code> .
<code>@Document class Product + @Id</code>	<code>class ProductDocument(BaseDocument)</code>	<code>from pyfly.data.document.mongodb import BaseDocument</code> . Inherits <code>id</code> (<code>Beanie PydanticObjectId</code>), <code>created_at</code> , <code>updated_at</code> , <code>created_by</code> , <code>updated_by</code> . Collection name set in <code>class Settings: name = "products"</code> .
<code>findByCategory(String c)</code>	<code>async def find_by_category(self, category: str) → List[ProductDocument]: ...</code>	Stub body <code>...</code> compiled by <code>MongoRepositoryBeanPostProcessor</code> at startup. Same prefixes as relational: <code>find_by_</code> , <code>count_by_</code> , <code>exists_by_</code> , <code>delete_by_</code> .
<code>@Query("{ 'status': ? @ }")</code>	<code>@query('{"status": ":status"}')</code>	<code>from pyfly.data.query import query</code> . JSON filter or aggregation pipeline; <code>:param</code> substitution.
<code>MongoSpecification</code>	<code>MongoSpecification(lambda root, q: {"active": True})</code>	<code>from pyfly.data.document.mongodb import MongoSpecification</code> . Compose with <code>&</code> / <code> </code> / <code>~</code> ; run via <code>find_all_by_spec(spec)</code> or <code>find_all_by_spec_paged(spec, pageable)</code> .
<code>PageRequest.of(page, size, Sort.by(...))</code>	<code>Pageable.of(page, size, Sort.by("name").descending())</code>	<code>from pyfly.data import Pageable, Sort</code> — identical to the relational adapter.

Spring Data MongoDB	PyFly	Notes
<code>Page<T></code>	<code>Page[T]</code>	<code>.items</code> , <code>.total</code> , <code>.page</code> , <code>.size</code> , <code>.total</code> — same as relational.

MongoRepository[T, ID]

Subclass `MongoRepository[T, ID]` and annotate with `@repository`. The framework extracts the document type and ID type from the generic parameters at class-definition time via `__init_subclass__`. No `__init__` is required.

catalog/product_repository.py

```

from beanie import PydanticObjectId
from pyfly.container import repository
from pyfly.data.document.mongodb import MongoRepository

from catalog.product_document import ProductDocument

@repository
class ProductRepository(MongoRepository[ProductDocument, PydanticObjectId]):

    # --- derived query method stubs (compiled at startup) ---

    async def find_by_category(
        self, category: str
    ) → list[ProductDocument]: ...

    async def find_by_active_and_category(
        self, active: bool, category: str
    ) → list[ProductDocument]: ...

    async def find_by_price_greater_than_order_by_price_desc(
        self, min_price: float
    ) → list[ProductDocument]: ...

    async def find_by_name_containing(
        self, fragment: str
    ) → list[ProductDocument]: ...

    async def count_by_category(self, category: str) → int: ...

    async def exists_by_sku(self, sku: str) → bool: ...

    async def delete_by_active(self, active: bool) → int: ...

```

Listing B.6 — ProductRepository: CRUD + derived queries

Built-in CRUD methods

Method	Return type	Description
<code>save(entity)</code>	<code>T</code>	Insert or update via Beanie <code>entity.save()</code>
<code>find_by_id(id)</code>	<code>T None</code>	Find by primary key
<code>find_all(**filters)</code>	<code>list[T]</code>	Find all; keyword args become equality filters
<code>find_all(sort)</code>	<code>list[T]</code>	Fetch all documents, ordered by a <code>Sort</code>
<code>find_all(pageable)</code>	<code>Page[T]</code>	Paged query: counts the total, applies the Pageable's sort, slices with skip/limit, returns <code>Page[T]</code>
<code>stream_all(sort)</code>	<code>AsyncIterator[T]</code>	Stream all documents (the <code>Flux<T></code> analogue); optional <code>Sort</code> and equality filters
<code>delete(entity)</code>	<code>None</code>	Delete a loaded document instance
<code>delete_by_id(id)</code>	<code>None</code>	Delete by primary key; no-op if not found
<code>count()</code>	<code>int</code>	Count all documents in the collection
<code>exists_by_id(id)</code>	<code>bool</code>	True if a document with this ID exists
<code>save_all(entities)</code>	<code>list[T]</code>	Bulk insert via <code>insert_many</code>
<code>find_all_by_id(ids)</code>	<code>list[T]</code>	Find all with IDs in a list

Method	Return type	Description
<code>delete_all_by_id(ids)</code>	<code>None</code>	Delete all with IDs in a list
<code>delete_all(entities=None)</code>	<code>None</code>	Delete the given documents; with no args, truncate the entire collection
<code>find_all_by_spec(spec)</code>	<code>list[T]</code>	Find matching a <code>MongoSpecification</code>
<code>find_all_by_spec_paged(spec, pageable)</code>	<code>Page[T]</code>	Find matching a <code>MongoSpecification</code> with pagination and sort

`find_all(**filters)` translates keyword arguments into MongoDB equality filters:

```
# {"status": "PENDING", "customer_id": "abc"}
orders = await repo.find_all(status="PENDING", customer_id="abc")
```

Derived query methods

PyFly compiles stub methods on `MongoRepository` subclasses into real MongoDB queries at startup. The naming convention is identical to the relational adapter and to Spring Data: `{prefix}_by_{predicates}[_order_by_{fields}]`.

Prefixes: `find_by`, `count_by`, `exists_by`, `delete_by`

Connectors: `_and_`, `_or_`

Operator mapping

Method suffix	MongoDB filter	Args consumed
<i>(none, default)</i>	<code>{field: value}</code>	1
<code>_not</code>	<code>{field: {"\$ne": value}}</code>	1
<code>_greater_than</code>	<code>{field: {"\$gt": value}}</code>	1
<code>_greater_than_equal</code>	<code>{field: {"\$gte": value}}</code>	1
<code>_less_than</code>	<code>{field: {"\$lt": value}}</code>	1
<code>_less_than_equal</code>	<code>{field: {"\$lte": value}}</code>	1
<code>_between</code>	<code>{field: {"\$gte": low, "\$lte": high}}</code>	2
<code>_like</code>	<code>{field: {"\$regex": pattern}}</code> (SQL % → .*)	1
<code>_containing</code>	<code>{field: {"\$regex": ".*val.*", "\$options": "i"}}</code>	1
<code>_in</code>	<code>{field: {"\$in": values}}</code>	1 (list)
<code>_is_null</code>	<code>{field: None}</code>	0
<code>_is_not_null</code>	<code>{field: {"\$ne": None}}</code>	0

Ordering: append `_order_by_{field}_{asc|desc}`. Multiple sort fields are chained:

```
# sort=[("name", ASC), ("created_at", DESC)]
async def find_by_active_order_by_name_asc_created_at_desc(
    self, active: bool
) → list[ProductDocument]: ...
```

The `MongoRepositoryBeanPostProcessor` detects stubs (bodies containing only `...` or `pass`) and replaces them with compiled callables. Source: `src/pyfly/data/document/mongodb/post_processor.py` and `src/pyfly/data/document/mongodb/query_compiler.py`.

Custom queries with `@query`

For queries that cannot be expressed through naming conventions, `@query` accepts a MongoDB filter document (`{...}`) or aggregation pipeline (`[...]`) as a JSON string. Named parameters use `:param_name` syntax.

catalog/order_repository.py

```

from pyfly.container import repository
from pyfly.data.document.mongodb import MongoRepository
from pyfly.data.query import query

from catalog.order_document import OrderDocument

@repository
class OrderRepository(MongoRepository[OrderDocument, str]):

    @query('{"status": ":status", "total": {"$gte": ":min_total"}}')
    async def find_by_status_min_total(
        self, status: str, min_total: float
    ) → list[OrderDocument]: ...

    @query(
        ' [{"$match": {"customer_id": ":cid"}}, '
        ' {"$group": {"_id": "$category", '
        ' "total": {"$sum": "$amount"}}}] '
    )
    async def totals_by_category(
        self, cid: str
    ) → list[dict]: ...

```

Listing B.7 — @query filter and aggregation examples

Substitution rules:

- A JSON string value that is *exactly* `:param_name` is replaced by the Python value, preserving its type (`int`, `bool`, `list`, etc.).
- `:param_name` embedded inside a larger string is replaced via `str(value)`.
- Dicts and lists are recursed. Non-string JSON values pass through unchanged.

`MongoQueryExecutor` parses the query template once at startup, detects whether it is a filter or pipeline, and substitutes parameters at call time.

Pagination

catalog/product_service.py

```
from pyfly.data import Page, Pageable, Sort
from pyfly.data.document.mongodb import MongoRepository

from catalog.product_document import ProductDocument

async def list_products(
    repo: MongoRepository[ProductDocument, str],
    page: int = 1,
    size: int = 20,
) → Page[ProductDocument]:
    pageable = Pageable.of(
        page=page,
        size=size,
        sort=Sort.by("name"),
    )
    return await repo.find_all(pageable)
```

Listing B.8 — Paginated product listing

`find_all(pageable)` counts the total, applies the `Pageable`'s sort, slices with `.skip((page-1)*size)` and `.limit(size)` on the Beanie query, and returns `Page[T]`. `Pageable` is 1-based, so page `1` is the first page.

Transaction management

Multi-document transactions require a **replica set** deployment. Standalone MongoDB does not support them.

pyfly.yaml

```
# Start MongoDB: mongod --replSet rs0 --bind_ip localhost
# Init (once, in mongosh): rs.initiate()
pyfly:
  data:
    document:
      enabled: true
      uri: "mongodb://localhost:27017/?replicaSet=rs0"
      database: myapp
```

Listing B.9 — Single-node replica set for local development

The `@mongo_transactional` decorator wraps an async function in a Motor session and transaction. The function's Beanie operations participate automatically:

billing/transfer.py

```

from motor.motor_asyncio import AsyncIOMotorClient
from pyfly.data.document.mongodb import mongo_transactional

from billing.account_document import AccountDocument

def make_transfer_fn(client: AsyncIOMotorClient):
    @mongo_transactional(client)
    async def transfer(
        from_id: str, to_id: str, amount: float
    ) → None:
        src = await AccountDocument.get(from_id)
        dst = await AccountDocument.get(to_id)
        if src is None or dst is None or src.balance < amount:
            raise ValueError("Invalid transfer")
        src.balance -= amount
        dst.balance += amount
        await src.save()
        await dst.save()
    return transfer

```

Listing B.10 — Atomic fund transfer with `@mongo_transactional`

On success the transaction commits; on any exception it aborts and re-raises. Unlike the relational `@reactive_transactional`, the Motor session is not injected as an argument — Beanie picks it up through the Motor context.

The `motor_client` bean is registered automatically by `DocumentAutoConfiguration` when the adapter is enabled. Inject it into your service via the DI container.

⚠ REPLICA SET REQUIRED

`@mongo_transactional` will raise an error against a standalone MongoDB instance. Use the `?replicaSet=rs0` URI fragment (see Listing B.9) even for local dev.

Auto-configuration

`DocumentAutoConfiguration` activates when:

1. `beanie` is importable (`@conditional_on_class("beanie")`), and
2. `pyfly.data.document.enabled` is `"true"` in config.

It registers three beans automatically:

Bean	Type	Role
<code>motor_client</code>	<code>AsyncIOMotorClient</code>	Async MongoDB connection pool
<code>mongo_post_processor</code>	<code>MongoRepositoryBeanPostProcessor</code>	Compiles derived query stubs
<code>odm_initializer</code>	<code>BeanieInitializer</code>	Calls <code>init_beanie()</code> at startup

`BeanieInitializer.start()` discovers `BaseDocument` subclasses in two passes: first from every registered `MongoRepository._entity_type` (set by `__init_subclass__`), then directly registered `BaseDocument` subclasses. This means defining a repository is sufficient — you do not need to register document models separately.

Source files: `src/pyfly/data/document/auto_configuration.py` , `src/pyfly/data/document/mongodb/initializer.py` .

Testing

For unit tests, use [mongomock-motor](#) or point at a dedicated test database:

pyfly-test.yaml

```
pyfly:
  data:
    document:
      enabled: true
      database: "myapp_test"
```

Listing B.11 — Test database configuration

For integration tests, PyFly's Testcontainers support spins up a real MongoDB container automatically — see the testing chapter and

`@ServiceConnection(MongoDBContainer)`.

APPENDIX C

ECM: Content & E-Signature

`pyfly.ecm` provides hexagonal abstractions for enterprise document management: blob storage, document metadata, folder hierarchies, and e-signature workflows. The module ships fully wired adapters for local storage, AWS S3, and Azure Blob Storage, plus e-signature adapters for DocuSign, Adobe Sign, and Logalty — all selectable without changing a line of application code, purely via YAML.

NO EXTRA DEPENDENCY FOR THE MODULE ITSELF

The core ECM module (`pyfly.ecm`) has no third-party dependencies beyond the standard library. Cloud storage adapters (`boto3` , `azure-storage-blob`) and e-signature SDK calls are made through lightweight async wrappers in each adapter class; install only what you need.

Domain model

All ECM types are plain Python dataclasses defined in `src/pyfly/ecm/models.py` .

Class	Key fields	Description
<code>Document</code>	<code>id</code> , <code>name</code> , <code>folder_id</code> , <code>content_type</code> , <code>size_bytes</code> , <code>metadata</code> , <code>versions</code>	Logical document record
<code>DocumentVersion</code>	<code>version</code> , <code>content_hash</code> , <code>size_bytes</code> , <code>storage_uri</code>	Immutable version snapshot
<code>Folder</code>	<code>id</code> , <code>name</code> , <code>parent_id</code> , <code>path</code>	Folder node in the hierarchy
<code>Recipient</code>	<code>name</code> , <code>email</code> , <code>role</code>	E-signature recipient
<code>SignatureRequest</code>	<code>document_id</code> , <code>recipients</code> , <code>subject</code> , <code>message</code>	Signature request payload
<code>ESignatureEnvelope</code>	<code>id</code> , <code>provider</code> , <code>status</code> , <code>provider_envelope_id</code>	Envelope tracking record

`ESignatureStatus` is a `StrEnum` with values `DRAFT`, `SENT`, `SIGNED`, `DECLINED`, and `EXPIRED`.

Ports (contracts)

Four `Protocol` classes defined in `src/pyfly/ecm/ports.py` form the hexagonal boundary. Your application services depend on these contracts; adapters implement them.

DocumentStoragePort

```
class DocumentStoragePort(Protocol):
    name: str
    async def upload(
        self, document: Document, content: bytes
    ) → DocumentVersion: ...
    async def download(
        self, document: Document, version: int | None = None
    ) → bytes: ...
    async def delete(
        self, document: Document, version: int | None = None
    ) → bool: ...
```

MetadataStoragePort

```
class MetadataStoragePort(Protocol):
    async def save(self, document: Document) → Document: ...
    async def get(self, document_id: str) → Document | None: ...
    async def list(
        self, folder_id: str | None = None, *, limit: int = 100
    ) → list[Document]: ...
    async def delete(self, document_id: str) → bool: ...
```

FolderRepositoryPort

```
class FolderRepositoryPort(Protocol):
    async def save(self, folder: Folder) → Folder: ...
    async def get(self, folder_id: str) → Folder | None: ...
    async def list(self, parent_id: str | None = None) → list[Folder]: ...
    async def delete(self, folder_id: str) → bool: ...
```

ESignatureAdapter

```
class ESignatureAdapter(Protocol):
    name: str
    async def send(
        self, request: SignatureRequest
    ) → ESignatureEnvelope: ...
    async def get(self, envelope_id: str) → ESignatureEnvelope | None: ...
    async def cancel(self, envelope_id: str) → bool: ...
```

DocumentService

`DocumentService` orchestrates the storage and metadata ports. It is the primary entry point for document upload, download, listing, and deletion.

contracts/ecm_usage.py

```

from pyfly.ecm import (
    DocumentService,
    LocalFilesystemStorageAdapter,
)
from pyfly.ecm.in_memory import (
    InMemoryFolderRepository,
    InMemoryMetadataStorage,
)

service = DocumentService(
    storage=LocalFilesystemStorageAdapter("/var/ecm/docs"),
    metadata=InMemoryMetadataStorage(),
    folders=InMemoryFolderRepository(),
)

async def demo() → None:
    # Upload
    doc = await service.upload(
        name="contract.pdf",
        content=b"%PDF-1.4 ...",
        content_type="application/pdf",
        metadata={"department": "legal", "year": "2026"},
    )

    # Download (latest version)
    content = await service.download(doc.id)

    # Download a specific version
    v1_content = await service.download(doc.id, version=1)

    # List documents in a folder
    docs = await service.list(folder_id=doc.folder_id)

```

```
# Delete (returns False if blob delete partially fails)
ok = await service.delete(doc.id)
```

Listing C.1 — *DocumentService: upload, download, delete*

delete semantics

`DocumentService.delete(document_id)` removes both the stored blob and the metadata record and returns the **logical AND** of the two delete results. If the document ID is unknown it returns `False` immediately. If the blob delete fails the metadata record is still removed (so the logical document is gone), but the method returns `False` to signal a partial delete. This contract is verified in the source at `src/pyfly/ecm/services.py`.

Folder management

`contracts/ecm_folders.py`

```
from pyfly.ecm import DocumentService, Folder

async def setup_folders(service: DocumentService) → None:
    root = await service.create_folder(
        Folder(name="contracts", path="/contracts")
    )
    legal = await service.create_folder(
        Folder(name="legal", parent_id=root.id, path="/contracts/legal")
    )
    doc = await service.upload(
        name="nda.pdf",
        content=b"...",
        folder_id=legal.id,
        content_type="application/pdf",
    )
```

Listing C.2 — *Creating folder hierarchies*

❗ FOLDER PORT IS OPTIONAL

Pass `folders=None` to `DocumentService` if you do not need folder support. Calling `create_folder` on such an instance raises `RuntimeError`.

ESignatureService

`ESignatureService` wraps an `ESignatureAdapter` and exposes three operations: `request`, `get`, and `cancel`.

contracts/esignature_usage.py

```

from pyfly.ecm import (
    ESignatureService,
    NoOpESignatureAdapter,
    Recipient,
    SignatureRequest,
)

esig = ESignatureService(adapter=NoOpESignatureAdapter())

async def send_for_signature(document_id: str) → str:
    envelope = await esig.request(
        SignatureRequest(
            document_id=document_id,
            recipients=[
                Recipient(
                    name="Alice Johnson",
                    email="alice@example.com",
                    role="signer",
                ),
            ],
            subject="Please sign your employment contract",
            message="Review and sign at your earliest convenience.",
        )
    )
    return envelope.id

```

Listing C.3 — Requesting an e-signature

Storage adapters

Adapter class	pyfly.ecm.storage.provider value	Notes
<code>LocalFilesystemStorageAdapter</code>	<code>local</code> (default)	Stores blobs as <code><base_dir>/<id>/v<n></code>
<code>AwsS3StorageAdapter</code>	<code>s3</code> or <code>aws</code>	Requires <code>boto3</code> ; async via <code>run_in_executor</code>
<code>AzureBlobStorageAdapter</code>	<code>azure</code>	Requires <code>azure-storage-blob</code>

All three implement `DocumentStoragePort` . `LocalFilesystemStorageAdapter` creates the base directory on construction and stores each version as a separate file, making it suitable for development and testing without external services.

E-signature adapters

Adapter class	<code>pyfly.ecm.esignature.provider</code> value	Notes
<code>NoOpESignatureAdapter</code>	<code>noop</code> (default)	Returns stub envelopes; safe for dev/test
<code>DocuSignESignatureAdapter</code>	<code>docusign</code>	REST API calls via <code>httpx</code>
<code>AdobeSignESignatureAdapter</code>	<code>adobe</code>	REST API calls via <code>httpx</code>
<code>LogaltyESignatureAdapter</code>	<code>logalty</code>	REST API calls via <code>httpx</code>

Implement `ESignatureAdapter` to add a custom provider (e.g. an in-house signing service). Register your implementation as a bean and inject it into `ESignatureService`.

Auto-configuration

`EcmAutoConfiguration` activates when `pyfly.ecm.enabled=true` is set in config. It registers five beans:

Bean	Default implementation
<code>metadata_storage</code>	<code>InMemoryMetadataStorage</code>
<code>folder_repository</code>	<code>InMemoryFolderRepository</code>
<code>document_storage</code>	<code>LocalFilesystemStorageAdapter</code> (temp dir)
<code>document_service</code>	<code>DocumentService</code> wiring all three above
<code>esignature_adapter</code>	<code>NoOpESignatureAdapter</code>
<code>esignature_service</code>	<code>ESignatureService</code> wrapping the adapter

The storage and e-signature adapters are selected from config so you switch providers purely via YAML — no code change required:

pyfly.yaml

```

pyfly:
  ecm:
    enabled: true
    storage:
      provider: s3
      local:
        base-dir: /var/ecm
      s3:
        bucket: my-docs-bucket
        region: eu-west-1
        key-prefix: documents/
      azure:
        container: my-container
        connection-string: "DefaultEndpointsProtocol=https;..."
        account-url: "https://acct.blob.core.windows.net"
        key-prefix: documents/
    esignature:
      provider: docusign
      docusign:
        base-url: "https://demo.docusign.net/restapi"
        account-id: "your-account-id"
        access-token: "your-access-token"
      adobe:
        api-base: "https://api.na4.adobesign.com"
        access-token: "your-token"
      logalty:
        api-base: "https://api.logalty.com"
        api-key: "your-api-key"

```

Listing C.4 — Full ECM YAML with S3 storage and DocuSign

💡 START WITH LOCAL + NOOP

In development, omit the `provider` keys and let the defaults take effect: `local` storage uses a fresh temp directory and `noop` e-signature returns deterministic stub data. No external services needed.

Implementing a custom storage adapter

Implement `DocumentStoragePort` and register it as a `@bean` :

infra/gcs_storage.py

```

from pyfly.ecm.models import Document, DocumentVersion
from pyfly.ecm.ports import DocumentStoragePort

class GcsStorageAdapter:
    """Google Cloud Storage adapter (skeleton)."""

    name = "gcs"

    def __init__(self, bucket: str, key_prefix: str = "") → None:
        self._bucket = bucket
        self._prefix = key_prefix

    async def upload(
        self, document: Document, content: bytes
    ) → DocumentVersion:
        version_no = (
            (document.versions[-1].version + 1) if document.versions else 1
        )
        key = f"{self._prefix}{document.id}/v{version_no}"
        # ... upload content to GCS ...
        return DocumentVersion(
            version=version_no,
            content_hash="sha256-placeholder",
            size_bytes=len(content),
            storage_uri=f"gs://{self._bucket}/{key}",
        )

    async def download(
        self, document: Document, version: int | None = None
    ) → bytes:
        target = version or document.versions[-1].version
        key = f"{self._prefix}{document.id}/v{target}"
        # ... download from GCS ...
        return b""

    async def delete(

```

```

        self, document: Document, version: int | None = None
    ) → bool:
        # ... delete object(s) from GCS ...
        return True

```

Listing C.5 — Custom storage adapter skeleton

Source files: - `src/pyfly/ecm/models.py` — domain dataclasses - `src/pyfly/ecm/ports.py` — Protocol contracts - `src/pyfly/ecm/services.py` — `DocumentService`, `ESignatureService` - `src/pyfly/ecm/adapters/` — bundled storage and e-signature adapters - `src/pyfly/ecm/in_memory.py` — `InMemoryMetadataStorage`, `InMemoryFolderRepository` - `src/pyfly/ecm/auto_configuration.py` — `EcmAutoConfiguration`

APPENDIX D

CLI & Troubleshooting

The `pyfly` CLI provides project scaffolding, application startup, database migrations, and environment diagnostics. It is built with **Click** for command parsing and **Rich** for coloured terminal output.

Install: `uv add "pyfly[cli]"` (or `uv sync --extra cli`). Requires Click, Rich, Jinja2, and questionnaire.

Entry point: `pyfly` — registered as a console script in `pyproject.toml`.

Command reference

Command	Description
<code>pyfly new [NAME]</code>	Scaffold a new project (interactive or direct)
<code>pyfly run</code>	Start the ASGI application server
<code>pyfly info</code>	Show Python version and installed extras
<code>pyfly doctor</code>	Diagnose environment (Python, tools, PyFly)
<code>pyfly db init</code>	Initialise an Alembic migration environment
<code>pyfly db migrate [-m MSG]</code>	Auto-generate a migration revision
<code>pyfly db upgrade [REVISION]</code>	Apply pending migrations (default: <code>head</code>)
<code>pyfly db downgrade REVISION</code>	Revert to a previous revision
<code>pyfly license</code>	Display the Apache 2.0 license text
<code>pyfly sbom [--json]</code>	Software Bill of Materials table
<code>pyfly --version</code>	Print the installed PyFly version
<code>pyfly --help</code>	Print the banner and all commands

pyfly new

Creates a complete project directory from one of seven archetypes.

```
pyfly new <name> [--archetype ARCHETYPE] [--features FEAT,...] [--directory DIR]
pyfly new                                     # interactive mode (questionary TUI)
```

Archetypes

Archetype	Default features	What is generated
<code>core</code>	<i>(none)</i>	DI container, config, Dockerfile
<code>web-api</code>	<code>web</code>	REST controllers, services, repositories (Todo CRUD)
<code>fastapi-api</code>	<code>fastapi</code>	FastAPI stack, native OpenAPI
<code>web</code>	<code>web</code>	Server-rendered HTML with Jinja2 templates
<code>hexagonal</code>	<code>web</code>	Ports & adapters, domain/application/infra/api layers
<code>library</code>	<i>(none)</i>	PEP 561 <code>py.typed</code> package, no PyFly runtime
<code>cli</code>	<code>shell</code>	Shell commands with DI, no ASGI entry point

Available `--features` values

Feature	What it adds
<code>web</code>	Starlette HTTP server, REST controllers, OpenAPI docs
<code>fastapi</code>	FastAPI HTTP server, native OpenAPI
<code>granian</code>	Granian ASGI server (Rust/tokio)
<code>hypercorn</code>	Hypercorn ASGI server (HTTP/2, HTTP/3)
<code>data-relational</code>	SQLAlchemy ORM (async), Alembic migrations
<code>data-document</code>	Beanie ODM, Motor (MongoDB)
<code>eda</code>	In-memory event bus, Kafka + RabbitMQ support
<code>cache</code>	Caching layer (in-memory default; Redis if installed)
<code>client</code>	Resilient HTTP client (httpx + retry/circuit-breaker)
<code>security</code>	JWT authentication, bcrypt password hashing
<code>scheduling</code>	Cron-based task scheduling
<code>observability</code>	Prometheus metrics, OpenTelemetry tracing
<code>cqrs</code>	Command/Query Responsibility Segregation
<code>shell</code>	DI-powered interactive shell commands

Examples

terminal

```
# Minimal core service
pyfly new wallet-service

# REST API with SQL data layer
pyfly new lumen --archetype web-api --features web,data-relational

# Hexagonal service with cache and security
pyfly new payment-svc --archetype hexagonal \
    --features web,data-relational,cache,security

# MongoDB-backed microservice
pyfly new catalog-svc --archetype web-api \
    --features web,data-document

# CLI application with interactive shell
pyfly new admin-tool --archetype cli

# Interactive mode (arrow keys + checkboxes)
pyfly new
```

Listing D.1 — *pyfly new examples*

In interactive mode the wizard prompts for name, package name (pre-filled from the project name), archetype (single-select with arrows), and features (multi-select with space bar). A confirmation summary is shown before creation. Press Ctrl+C at any prompt to exit cleanly.

Project names containing hyphens are automatically converted to valid Python package names (`my-service` → `my_service`). Use the **Package name** prompt in interactive mode to override this derived value.

pyfly run

Starts the ASGI application using the auto-selected server.

```
pyfly run [--host HOST] [--port PORT] [--server SERVER]
          [--workers N] [--reload] [--app MODULE:VAR]
```

Option	Default	Description
<code>--host</code>	<code>0.0.0.0</code>	Bind address
<code>--port</code>	Config or <code>8080</code>	App port (CLI → <code>pyfly.server.port</code> → 8080)
<code>--server</code>	Auto-detect	<code>granian</code> , <code>uvicorn</code> , or <code>hypercorn</code>
<code>--workers</code>	Config or <code>0</code>	Worker processes (<code>0</code> = cpu count)
<code>--reload</code>	off	Auto-reload on code changes
<code>--app</code>	Auto-discovered	Import path, e.g. <code>myapp.main:app</code>

Server priority (when `--server` is omitted): Granian › Uvicorn › Hypercorn. The first importable server wins.

Application discovery (when `--app` is omitted):

- `pyfly.app.module` in `pyfly.yaml`
- `app.module` in `pyfly.yaml` (flat layout)
- `src/<package>/main.py` auto-scan

`pyfly run` adds `src/` to `sys.path` automatically for src-layout projects.

terminal

```
# Development with auto-reload
pyfly run --reload

# Production: Granian, bind all interfaces, all CPU cores
pyfly run --host 0.0.0.0 --port 80 --server granian --workers 0

# Explicit application path
pyfly run --app my_service.main:app
```

Listing D.2 — pyfly run examples

pyfly info

Displays two Rich tables: Python version, platform, and architecture; plus installation status of every PyFly optional module. Useful after a selective install to confirm which adapters are available.

```
pyfly info
```

pyfly doctor

Runs a comprehensive environment health check.

```
pyfly doctor
```

Check	Pass condition	Failure impact
Python version	<code>>= 3.12</code>	Fatal — doctor reports fail
Virtual environment	<code>sys.prefix != sys.base_prefix</code>	Warning only
<code>git</code> on PATH	<code>shutil.which("git")</code> returns a path	Fatal
<code>uv</code> on PATH	<code>shutil.which("uv")</code> returns a path	Fatal
<code>uvicorn</code> on PATH	present	Advisory (-)
<code>alembic</code> on PATH	present	Advisory (-)
<code>ruff</code> on PATH	present	Advisory (-)
<code>mypy</code> on PATH	present	Advisory (-)
PyFly importable	<code>import pyfly</code> succeeds	Fatal

Exits with **"All checks passed!"** or **"Some issues found."**.

pyfly db

Database migration commands backed by [Alembic](#).

REQUIRES DATA-RELATIONAL EXTRA

All `pyfly db` subcommands require Alembic (`pyfly[data-relational]`).

pyfly db init

```
pyfly db init
```

Creates `alembic/` and `alembic.ini` in the current directory. Overwrites `alembic/env.py` with a PyFly template that wires `async_engine_from_config` and `Base.metadata` from `pyfly.data.relational.sqlalchemy`. Exits with an error if `alembic/` already exists.

pyfly db migrate

```
pyfly db migrate [-m "description"]
```

Runs `alembic revision --autogenerate`. Compares current `Base.metadata` against the live database and writes a new version file to `alembic/versions/`. Requires `alembic.ini` (run `pyfly db init` first).

pyfly db upgrade / downgrade

```
pyfly db upgrade [REVISION]      # default: head
pyfly db downgrade REVISION      # e.g. -1, base, abc123
```

Typical migration workflow

terminal

```
# 1. Initialise Alembic (once per project)
pyfly db init

# 2. Generate initial migration from your entity models
pyfly db migrate -m "initial schema"

# 3. Apply migrations
pyfly db upgrade

# 4. After modifying entities, generate a new revision
pyfly db migrate -m "add order status column"
pyfly db upgrade

# 5. Roll back one step
pyfly db downgrade -1

# 6. Revert all migrations
pyfly db downgrade base
```

Listing D.3 — Database migration lifecycle

pyfly sbom

Prints a Rich table of every PyFly dependency with required and installed versions. The `--json` flag outputs machine-readable JSON for compliance pipelines.

```
pyfly sbom
pyfly sbom --json
```

Typical development workflow

terminal

```
# Verify environment
pyfly doctor

# Scaffold the project
pyfly new lumen --archetype web-api \
  --features web,data-relational,security

cd lumen

# Check what was installed
pyfly info

# Initialise migrations
pyfly db init
pyfly db migrate -m "initial schema"
pyfly db upgrade

# Develop with auto-reload
pyfly run --reload

# Add a new column, migrate, upgrade
pyfly db migrate -m "add shipment_id to orders"
pyfly db upgrade
```

Listing D.4 — End-to-end workflow from setup to development

Troubleshooting

Common problems and fixes

Symptom	Likely cause	Fix
<code>command not found: pyfly</code>	PATH not updated after install	<code>source ~/.zshrc</code> (or <code>~/.bashrc</code>); or <code>export PATH="\$HOME/.pyfly/bin:\$PATH"</code>
Python 3.11 found (3.12 required)	Old Python on PATH	<code>brew install python@3.12</code> (macOS) or <code>sudo apt install python3.12</code> (Ubuntu)
<code>ModuleNotFoundError: No module named 'starlette'</code>	<code>web</code> extra not installed	<code>uv add "pyfly[web]"</code> or <code>uv sync --extra web</code>
<code>ModuleNotFoundError: No module named 'beanie'</code>	<code>data-document</code> extra missing	<code>uv add "pyfly[data-document]"</code>
<code>ModuleNotFoundError: No module named 'sqlalchemy'</code>	<code>data-relational</code> extra missing	<code>uv add "pyfly[data-relational]"</code>
<code>alembic is not installed</code>	Missing Alembic	<code>uv add "pyfly[data-relational]"</code>
<code>No application found</code>	No <code>pyfly.yaml</code> or <code>--app</code> flag	Add <code>pyfly.app.module: myapp.main:app</code> to <code>pyfly.yaml</code> , or pass <code>--app myapp.main:app</code>
<code>No ASGI server found</code>	No server extra installed	<code>uv add "pyfly[web]"</code> (adds Uvicorn)
<code>venv module not found</code>	Split-package Python (Debian/Ubuntu)	<code>sudo apt install python3.12-venv</code>
Directory ' <code>alembic</code> ' already exists	<code>db init</code> run twice	<code>rm -rf alembic alembic.ini</code> then re-run <code>pyfly db init</code>
<code>uv cache clean</code> fixes slow installs	Corrupt uv cache	Run <code>uv cache clean</code> then retry

Symptom	Likely cause	Fix
Ctrl+C in <code>pyfly new</code> leaves no files	Interactive mode cancelled cleanly	Re-run <code>pyfly new</code> ; no cleanup needed

Uninstalling

For an installer-based installation (`bash install.sh`):

terminal

```
# Remove the installation directory
rm -rf ~/.pyfly

# Remove the PATH line from your shell profile
# Delete the line: export PATH="$HOME/.pyfly/bin:$PATH" # PyFly Framework
```

Listing D.5 — Uninstall PyFly

For a manual `uv` / `pip` installation: `uv remove pyfly` or `pip uninstall pyfly` .

Extending the CLI

The CLI is built on Click. Add custom commands by importing the `cli` group and calling `cli.add_command` :

myapp/generate.py

```
import click
from pyfly.cli.main import cli

@click.command()
@click.argument("entity_name")
def generate(entity_name: str) → None:
    """Generate boilerplate for a new entity."""
    click.echo(f"Generating {entity_name} entity ...")

cli.add_command(generate, name="generate")
```

Listing D.6 — Adding a custom CLI command

Register the command module in your `pyproject.toml` entry points or import it in your application's `__init__.py` before the CLI is invoked.

APPENDIX E

Native Web Applications & Model Administration

PyFly can serve HTML pages and administer the same data used by its REST APIs. This appendix follows the runnable `samples/webapp` catalog. Its `Product` is an ordinary PyFly `BaseEntity`; HTML forms and the dashboard read and write the same rows. Existing applications can register their own SQLAlchemy entities or Beanie documents without defining a second persistence model.

Lumen remains the book's wallet service. The catalog is a separate, smaller companion for browser interactions. Do not expose Lumen's balance or ledger as generic writable records: money movement must continue through its domain commands. A custom administration provider can delegate to those commands.

E.1. Run the complete catalog

From the framework checkout, enter `samples/webapp`. The sample's project file resolves PyFly from the surrounding checkout, so it exercises this implementation.

```

cd samples/webapp
uv sync --group dev
export WEBAPP_ADMIN_PASSWORD=\
    "$(python -c 'import secrets; print(secrets.token_urlsafe(24))')"
export WEBAPP_EDIT_TOKEN_KEY=\
    "$(python -c 'import secrets; print(secrets.token_urlsafe(48))')"
export WEBAPP_DATABASE_URL="sqlite+aiosqlite:///catalog.db"
export PYFLY_SECURITY_CSRF_COOKIE_SECURE=false
uv run python -m catalog.init_db
uv run uvicorn catalog.app:create_webapp --factory \
    --host 127.0.0.1 --port 8080

```

Open <http://127.0.0.1:8080/>, sign in as `admin` using the generated password, and create a product. Open `/admin`, select **Datasources** → **Products**, and find the same record. Edit its price there, then reload the catalog. Stop and restart the server: the SQLite file preserves the product.

The initialization command creates the sample table only if absent. The server does not create or migrate schema. The sample deliberately places management on port 8080 to share authentication with the catalog; PyFly's default separate management port is 9090. Use migrations for deployed databases and HTTPS with secure cookies. The cookie override above is for local HTTP development.

E.2. One model, two browser interfaces

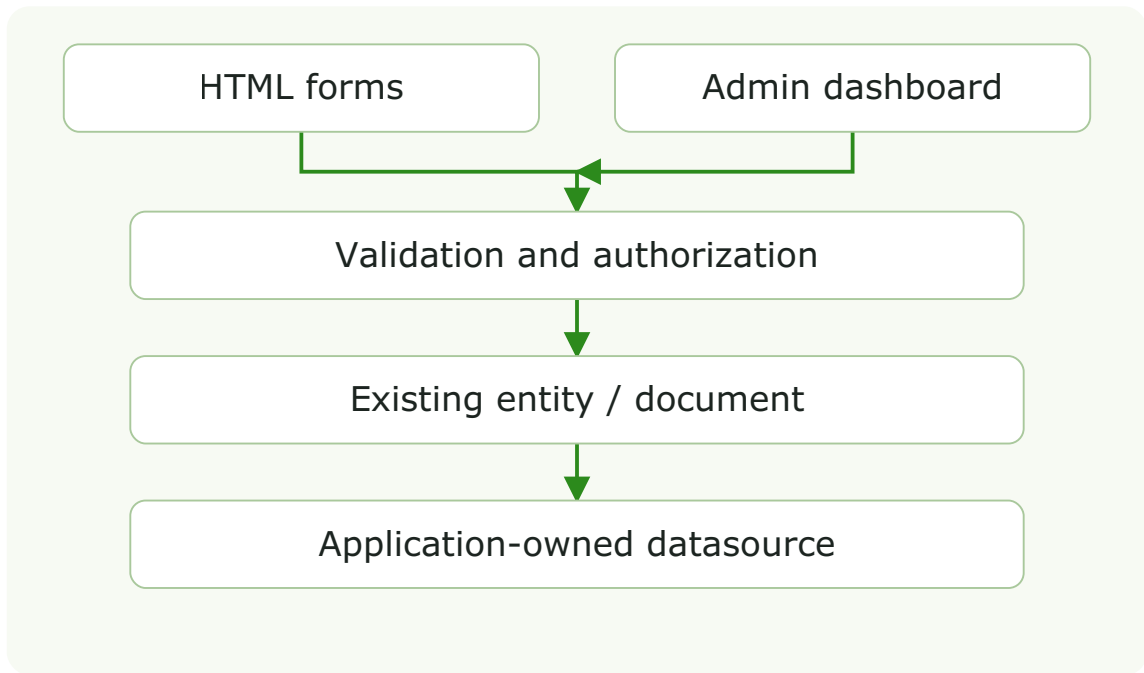


Figure E.1 — Both interfaces use the same models and datasource.

The catalog has three layers of metadata with different jobs:

Type	Responsibility
<code>Product(BaseEntity)</code>	Persisted columns, keys, constraints, and audit fields.
<code>ProductWrite(BaseModel)</code>	Accepted input and validation shared by both interfaces.
<code>ModelAdmin</code>	Visible fields, writable fields, operations, and source selection.

`ProductForm` extends the input schema with an edit token. Input schemas do not create tables and `ModelAdmin` does not replace the entity. The catalog owns its session factory; its HTML controller and dashboard use an administration service backed by that same factory. Applications may instead keep their usual services for public pages and expose a separate, restricted administration registration.

catalog/models.py

```
from decimal import Decimal

from pydantic import BaseModel, Field
from sqlalchemy import Numeric, String
from sqlalchemy.orm import Mapped, mapped_column

from pyfly.data.relational.sqlalchemy.entity import BaseEntity


class Product(BaseEntity):
    __tablename__ = "webapp_products"
    name: Mapped[str] = mapped_column(String(80), unique=True)
    price: Mapped[Decimal] = mapped_column(Numeric(10, 2))


class ProductWrite(BaseModel):
    name: str = Field(min_length=1, max_length=80)
    price: Decimal = Field(ge=0, max_digits=10, decimal_places=2)


class ProductForm(ProductWrite):
    edit_version: str = ""
```

Listing E.1 — The existing entity and shared input schema

E.3. Render pages and resolve URLs

Install `pyfly[webapp]` for Starlette, Jinja, and browser security; add `data-relational` for SQL administration or `data-document` for MongoDB. The same view contracts work with the `fastapi` extra. Enable rendering explicitly:

pyfly.yaml

```
pyfly:
  web:
    templates:
      enabled: true
      directories: []
      packages: ["catalog:templates"]
    static:
      enabled: true
      directories: []
      packages: ["catalog:static"]
    errors:
      html-enabled: true
```

Listing E.2 — Package resources used by the catalog

Filesystem directories are relative to the process working directory. Package roots travel with an installed application. Set `directories: []` when relying only on packages, otherwise PyFly also checks the default local directories. Templates support inheritance, includes, async helpers, and HTML autoescaping. Undefined variables fail by default, making misspelled context keys visible.

Use `@controller` and return `ModelAndView("catalog.html", context)` for a page. `ModelAndView` also accepts `status_code` and response `headers`. The existing `@rest_controller` continues to provide JSON behavior. Assign unique route names with `@get_mapping("/", name="catalog")` to avoid hardcoded paths.

In templates, `reverse('catalog')` generates a local URL and `static_url('catalog.css')` resolves a static asset. In Python, import `reverse` and `static_url` from `pyfly.web` and pass `request` as the first argument. `Redirect(reverse(request, "catalog"))` produces a local 303 redirect after a successful POST. Both helpers include mounted prefixes and the configured static path. The existing `url_for('catalog')` helper provides an absolute URL.

Pass decoded names and parameters: the helpers encode spaces, Unicode, `#`, and `?`. Asset paths are relative to the static roots, for example `images/logo.svg` or `downloads/manual.pdf`, without a leading `/` or `..` segments. Missing route names raise a resolution error; URL generation does not check file existence. Jinja `{% include 'partials/menu.html' %}` loads a fragment from the template roots; it neither generates a URL nor exposes server files.

Explicitly opt in to external HTTP(S) redirects with `allow_external=True`. Register a `TemplateEngine` bean to replace Jinja or `TemplateContextProcessor` beans to add request-specific context. View values override processor values; URL and CSRF helpers remain framework-owned.

E.4. Bind forms and redisplay validation

The sample's create action accepts `product: Form[ProductForm]`. PyFly parses the form, converts its fields, and validates the Pydantic model before calling the controller. The action passes `product.model_dump(exclude={"edit_version"})` to `AdminDataService.create` and then redirects. The edit action instead passes the record ID and `product.edit_version` to `AdminDataService.update`.

Each POST form must send a CSRF token:

catalog/templates/form.html

```

<input type="hidden" name="{{ csrf_field }}" value="{{ csrf_token }}">
<input type="hidden" name="edit_version"
      value="{{ values.get('edit_version', '') }}">
<label for="name">Name</label>
<input id="name" name="name" value="{{ values.get('name', '') }}"
      required maxlength="80">
<label for="price">Price</label>
<input id="price" name="price" inputmode="decimal"
      value="{{ values.get('price', '') }}" required>

```

Listing E.3 — Fields inside the sample form

Catch `FormValidationException` using `@exception_handler` and return the form template with `error.values`, `error.errors`, and status 422. The complete handler is in `catalog/controllers.py`. Secret-like fields and secret Pydantic types are removed from redisplay values; validation messages omit submitted input. Keep escaping enabled. A normal GET does not invalidate the token in another open tab; session rotation or invalidation rotates it.

Aliases and repeated list fields are supported. Repeated scalar fields are rejected. An unchecked checkbox is absent, so use a Boolean default of `False` and a checked value of `"true"`. Multipart text and `File[UploadedFile]` share bounded parsing. Defaults are 1,000 fields, 20 files, 1 MiB per part, and 16 MiB per request, configured under `pyfly.web.forms`.

E.5. Own the HTML error experience

Add `errors/404.html` and `errors/500.html` to the configured template roots. To map a status explicitly, set `pyfly.web.errors.templates` to a mapping such as `{"404": "errors/missing.html"}`. Resolution tries that mapping, the exact status,

the status family such as `errors/4xx.html`, and the configured `default-template` (`errors/error.html`). A missing template advances to the next candidate; a broken error template falls back to a built-in escaped page.

Error templates receive `status`, `title`, `message`, `path`, and `transaction_id`, plus `reverse`, `static_url`, and `url_for` for navigation and shared assets. An error layout can use `static_url('catalog.css')`; it should not depend on application context processors or CSRF/form helpers. Server errors hide exception details by default. Browser errors negotiate HTML from `Accept`; REST and data-admin API errors remain JSON. Authentication and CSRF failures preserve their status and security headers. Error responses use `no-store` caching and `Vary: Accept`.

Branded pages and the welcome screen

The framework includes `pyfly/base.html`, `pyfly/welcome.html`, and `pyfly/error.html`, with the PyFly logo, green palette, responsive layouts, and automatic light/dark colors. Only the `web` scaffold enables browser features automatically. API, microservice, hexagonal, library, and CLI projects keep them disabled; installing an extra alone does not activate pages. An existing API may deliberately opt in through configuration.

When templates are enabled and no application controller handles `/`, the welcome page appears automatically. Startup-discovered home controllers take precedence, as the catalog's controller does. Set `pyfly.web.welcome.enabled: false` to remove the fallback, or `welcome.template` to select another template. A minimal webapp using only bundled templates sets `templates.enabled: true` and `templates.directories: []`.

Configure `pyfly.web.branding` once for normal views and errors. `name` defaults to `PyFly`; `tagline` to `Build something that matters.`; `primary-color` to `#4cbb2f`; and `accent-color` to `#c2e85f`. Colors accept six-digit hex values. `logo-url` and

`favicon-url` accept HTTP(S) URLs or application-relative paths such as `/static/catalog-logo.svg`; the ASGI mount prefix is added to relative paths. An empty logo uses the bundled PyFly logo, and an empty favicon omits the icon. Customize `documentation-url`, `support-url`, and `footer`; empty link URLs hide those links.

Templates receive `branding`, `home-url`, and `web_asset_url`. The framework serves its logo and styles under `branding.assets-path`, default `/_pyfly/web`, only when templates or HTML errors are enabled. Permit this reserved prefix through security rules if pre-login pages need it. External styles work with `style-src 'self'`. Application assets still use `static_url`, while `web_asset_url('web.css')`, `web_asset_url('theme.css')`, and `web_asset_url('logo.png')` address bundled resources.

Extend a bundled template and override `title`, `head`, `navigation`, `content`, or `footer`. For example, the sample's error pages extend `pyfly/error.html` and replace the navigation block with a catalog link. Application template roots precede bundled roots. Keep an extending template at a different path from its parent to avoid recursive inheritance. A custom template engine must supply the chosen welcome template or disable the fallback.

Development stack traces

`pyfly.web.errors.include-stacktrace` accepts `never`, `on-debug` (the default), and `always`. Enable `pyfly.web.debug: true` in a local development profile, or use `debug=True` in the adapter factory, to activate the default policy. Environment overrides work too: `PYFLY_WEB_DEBUG=true`. Production defaults hide the diagnostic panel, and URL query parameters cannot enable it. PyFly owns negotiated debug pages when HTML errors are enabled; REST controllers still return JSON.

Custom error templates receive `show_stacktrace`, `exception_type`, `exception_message`, `stack_trace`, and `stack_frames`. Each frame has `filename`, `lineno`, `function`, and `source`. The trace refers to the original exception even when an exception converter changes the HTTP status. Disabled diagnostics return empty strings and an empty frame list. Early security failures may have no exception.

`max-stack-frames` defaults to 30 (1–200 allowed); `max-trace-length` defaults to 20,000 characters (256–100,000 allowed). Traces include the most recent frames of the current exception, not exception chains or local variables. Request headers, cookies, query parameters, and bodies are not collected. Exception text and source lines may still contain secrets, so public production deployments should keep traces off. Jinja escapes trace text; never mark it `safe`.

`pyfly/error.html` includes the diagnostic panel. In a standalone custom layout use `{% if show_stacktrace %}<pre>{{ stack_trace }}</pre>{% endif %}`. A broken application error template falls back to an independent bundled renderer, retaining the original trace. HTML errors also have an escaped branded fallback when Jinja is absent.

E.6. Register the existing entity for administration

In a normal scanned application, expose a `ModelAdmin` bean. The following is an alternative to the sample's explicit registry and provider wiring:

catalog/administration.py

```

from catalog.models import Product, ProductWrite
from pyfly.admin import ModelAdmin
from pyfly.container import bean, configuration

@Configuration
class Administration:
    @bean
    def products(self) → ModelAdmin:
        return ModelAdmin(
            "products", Product, label="Products",
            datasource="primary",
            fields=("id", "name", "price", "updated_at"),
            editable_fields=("name", "price"),
            search_fields=("name",), filter_fields=("name",),
            ordering=("name",),
            operations=("list", "read", "create", "update", "delete"),
            create_schema=ProductWrite, update_schema=ProductWrite,
        )

```

Listing E.4 — Register the existing Product

The primary source borrows the existing `async_session_factory` bean. A named source borrows `NamedDataSources.get(name)`. For an already initialized Beanie document use `datasource="document"`. An explicit `provider` takes precedence. Admin does not create clients, migrate schema, or close application-owned pools.

Enable `pyfly.admin.data.enabled`, configure `allowed-roles`, and provide `edit-token-key` from an environment secret of at least 32 characters when writes are enabled. Share the key across workers; rotating it invalidates open edits. Registrations default to list/read only. No table becomes exposed merely because it exists in a datasource. Primary keys, audit/version fields, and binary fields are read-only; writable fields must be explicitly selected.

E.7. Permissions, conflicts, and business rules

Administration always requires an authenticated existing `SecurityContext` and an allowed data-admin role, even when monitoring uses `require-auth: false`. A separate management listener must enable `pyfly.management.security.enabled`; startup fails otherwise. Configure authentication and path rules on that listener. Every API mutation also needs the CSRF cookie and matching `X-XSRF-TOKEN` header, including requests authenticated by a bearer token.

Allowed operations are the intersection of global configuration, the registration, and `has_permission(operation, context, record=None)`. A hook can narrow permission, never expand either allowlist. Use `scope(context)` for server-owned equality restrictions such as tenant IDs. Scope limits counts, lists, reads, and mutations; creates inherit scope and updates cannot move records outside it. Object-level permission checks alone do not hide list rows.

The dashboard groups sources and resources, pages records, and supports configured search/filter/sort fields. Create/edit forms expose typed controls, null values, and validation feedback. `relations={"category_id": "categories"}` obtains choices from the authorized target resource and validates selections before writing. Nested relationship writes require application-specific behavior.

An edit token represents the full persisted snapshot, including hidden fields. PATCH sends it as `editToken`; DELETE sends it in `If-Match`. PostgreSQL locks the row before comparison; SQLite holds an immediate transaction; MongoDB makes one conditional replace/delete against the original BSON document. A stale edit returns 409 and the dashboard preserves input until the user deliberately reloads. Constraint conflicts also return 409; invalid input returns 422. Verify equivalent locking before enabling concurrent writes on other SQL dialects.

Decimals and large scalar integers preserve precision as strings at the browser boundary. `AdminFieldAdapter(type, serialize, parse)` adapts custom field types. For service-level invariants or Beanie action hooks, implement `AdminDataProvider` and pass it as `provider`. The built-in Mongo provider validates the resulting document but does not execute arbitrary Beanie action hooks. A custom provider must enforce scope, validation, and atomic stale-edit checks while delegating to the application's business operations. Successful audit events record the actor, resource, identifier, operation, field names, and correlation ID, without values.

E.8. Configure, test, and troubleshoot

All new properties use the existing YAML/TOML, profile, and environment binding. For example `PYFLY_WEB_TEMPLATES_AUTO_RELOAD=true` enables local template reload, and `PYFLY_ADMIN_DATA_OPERATIONS=list,read` makes administration globally read-only. Use profile files for development differences and environment secrets for credentials. Templates, static resources, HTML errors, and model administration are opt-in.

Run `uv run pytest` inside `samples/webapp`. Its temporary database tests exercise rendering, validation, CSRF, create/edit persistence, and shared admin IDs. Framework tests additionally cover both web adapters, real PostgreSQL/MongoDB, browser CRUD, and installed-wheel variants. See `docs/modules/webapps.md` for exact commands and the complete property/API reference.

Symptom	Check
Template root fails at startup	Root exists; package is installed; unused filesystem roots are disabled.
403 on a local form POST	Secure cookie on HTTP, missing hidden field, or a rotated session token.
Datasources is absent or empty	Data admin enabled, registration discovered, role and list permission granted.
Management startup rejects configuration	Enable management security and wire real authentication.
409 when saving	Reload the latest record; inspect uniqueness/constraint failures.
Wrong static links under a prefix	Use <code>reverse / static_url</code> and configure the ASGI root path.

Try it yourself

1. Create a product in the catalog, edit it in the dashboard, and verify its ID and price through a fresh database session.
2. Open an edit in two tabs. Save the first, then save the second. Confirm a 409 prevents silently overwriting the newer value.
3. Restrict global operations to `list,read`. Confirm direct mutation requests fail as well as the write controls disappearing from the dashboard.
4. Change the 404 template, request a missing page with `Accept: text/html`, then request it with `Accept: application/json` and compare the responses.

APPENDIX

Glossary

Adapter — A concrete class that implements a port by delegating to a specific library or infrastructure technology (PostgreSQL, Redis, Kafka, etc.). Adapters live at the edge of the hexagonal architecture and can be swapped without touching the domain or application layers. In PyFly, the async SQLAlchemy session factory,

`RedisCacheAdapter`, and `KafkaMessageBroker` are all adapters (Chapters 5, 10, 13).

Aggregate root — The single entry point to a cluster of domain objects that must stay consistent together; all state changes are routed through its methods, which enforce invariants and emit domain events. External code loads and saves only the root, never its inner objects. In PyFly, `AggregateRoot[ID]` is the state-stored base class; `Wallet` is Lumen's state-stored aggregate root and `LedgerAccount` is its event-sourced counterpart (Chapters 6, 9).

AOP (Aspect-Oriented Programming) — A technique for adding cross-cutting concerns — logging, metrics, security checks — declaratively to methods without modifying their source code. PyFly's `@aspect` and `@before` / `@after` / `@around` advice decorators implement AOP; Chapter 15 uses it to attach observability to every service method.

ApplicationContext — The central DI container that discovers beans, resolves dependencies, and manages their lifecycle from startup to shutdown. It fires `ContextRefreshedEvent` once all beans are wired and `ContextClosedEvent` on graceful shutdown. `ApplicationContext` is the PyFly equivalent of Spring's `ApplicationContext` (Chapter 2).

Async/await — Python's native coroutine syntax for non-blocking I/O. PyFly is built async-native: every HTTP handler, repository method, bus call, and service client is declared `async def` and scheduled on the event loop. Blocking calls inside an `async def` function freeze the loop and must be avoided (Chapter 1).

Autowiring — The mechanism by which the DI container resolves a bean's constructor arguments from type hints alone, with no factory code required. PyFly inspects `__init__` signatures at startup and injects matching beans automatically; the `@primary` annotation selects the preferred implementation when multiple candidates exist (Chapter 2).

Bean — Any Python object that the DI container creates, wires, and manages. You declare a class as a bean by applying a stereotype decorator (`@service` , `@repository` , `@component` , `@configuration` , or `@rest_controller`), or by annotating a factory method with `@bean` inside a `@configuration` class (Chapter 2).

BFF (Backend for Frontend) — An API gateway layer that sits in front of multiple microservices and composes their capabilities into a user-journey-focused API, tailored to a specific frontend client. In Chapter 11, Lumen's BFF tier aggregates wallet and payment data so the mobile app makes one request instead of two.

Bounded context — A DDD-level scope within which a domain model has a single, unambiguous meaning. Services in a microservices architecture often map one-to-one to bounded contexts: `WalletService` owns the wallet model; `PaymentsService` owns the payment model. Shared kernel or anti-corruption layers are needed when two contexts must exchange data (Chapter 6).

Bulkhead — A resilience pattern that limits the number of concurrent calls to a particular resource or downstream service, preventing a slow dependency from exhausting the thread or coroutine pool and degrading unrelated paths. PyFly's `@bulkhead(max_concurrent=N)` decorator implements the semaphore bulkhead (Chapter 13).

Circuit breaker — A resilience pattern that counts consecutive failures to a remote dependency; once the failure threshold is reached, the breaker trips to the open state and short-circuits further calls with a fast error, giving the dependency time to recover before calls resume. PyFly's `@circuit_breaker` decorator and `@service_client` inject this behaviour automatically (Chapters 11, 13).

Command (CQRS) — An immutable, frozen dataclass that expresses a single write intent — "open a wallet", "deposit funds" — and carries the exact data needed to fulfil it. Commands inherit from `Command[R]` where `R` is the handler's return type, and flow through the `CommandBus`. They may optionally implement `validate()` and `authorize()` (Chapter 7).

CommandBus — The pipeline that receives a `Command`, runs validation and authorization, dispatches it to the matching `CommandHandler`, and then publishes any domain events buffered by the aggregate. One handler is registered per command type; the bus enforces this constraint at startup (Chapter 7).

Compensation — A forward operation that semantically reverses the effect of a previously completed saga step when a later step fails. Unlike a database rollback, compensation is a new write that explicitly undoes the earlier change (for example, "refund payment" compensates "capture payment"). Each `@saga_step` names its compensation method (Chapter 12).

Component — A generic stereotype for a managed bean that does not fit the more specific roles of `@service`, `@repository`, or `@rest_controller`. `@component` registers the class with the container and enables injection, but carries no additional semantic meaning (Chapter 2).

Configuration class — A class decorated with `@configuration` that groups `@bean` factory methods. The container treats it as a source of bean definitions, calling each factory method and registering the return value as a named, typed bean. Profile guards and conditional annotations can be applied to the class or to individual factory methods (Chapter 3).

Container (DI) — See *ApplicationContext*.

Convention over configuration — The principle that sensible defaults eliminate boilerplate: a class annotated `@service` is automatically singleton-scoped, discovered by component scan, and injected by type without any XML or explicit registration. PyFly adopts this as a core design value, requiring explicit overrides only when the default is wrong (Chapter 1).

CQRS (Command Query Responsibility Segregation) — An architectural pattern that separates write operations (commands) from read operations (queries) into distinct code paths, buses, and potentially distinct data models. Reads can be cached independently of writes; handlers can be tested in isolation; cross-cutting concerns are applied uniformly by the respective bus (Chapter 7).

Dead-letter queue (DLQ) — A dedicated message queue or topic that receives messages which could not be processed after the configured number of retries. PyFly's `@message_listener` routes poison messages to the DLQ automatically, preventing them from blocking healthy messages (Chapter 10).

Dependency injection (DI) — A design pattern in which an object declares its collaborators as constructor parameters and an external container provides the concrete instances. DI decouples construction from use, making it straightforward to swap implementations and to test classes with fakes or mocks (Chapter 2).

Domain event — An immutable record of a business fact that has already occurred — "wallet opened", "funds deposited". Domain events are produced by aggregate roots via `raise_event()`, published through the `EventPublisher` port, and consumed by independent listeners. In event sourcing, they are also the source of truth for aggregate state (Chapters 6, 8, 9).

DTO (Data Transfer Object) — A plain, serializable object used to carry data across a layer boundary — between the HTTP controller and the service, or between a service and its API client — without exposing internal domain types. In PyFly, Pydantic `BaseModel` subclasses serve as request/response DTOs (Chapter 4).

EDA (Event-Driven Architecture) — An architectural style in which services communicate by publishing and subscribing to events rather than by direct synchronous calls. Producers and consumers are decoupled: a producer does not know which consumers exist. PyFly's `EventPublisher` and `@event_listener` are the intra-process EDA primitives; `MessageBrokerPort` extends the pattern across process boundaries (Chapters 8, 10).

Entity — A domain object with a stable identity that persists over time and through state changes. Two entities are equal if and only if they share the same non-null `id`. In PyFly, `Entity[TID]` tracks identity; `BaseEntity` adds audit columns (`created_at` , `updated_at`) for the persistence layer (Chapters 5, 6).

Event sourcing — A persistence strategy in which every state change is stored as an immutable domain event in an append-only stream. The current state of an aggregate is computed by replaying all events from the stream. PyFly's `pyfly.eventsourcing` module provides `AggregateRoot`, `EventStore`, `EventSourcedRepository`, snapshot support, and a `ProjectionRunner` (Chapter 9).

EventEnvelope — The metadata wrapper that packages a domain event payload for delivery: event ID, event type, aggregate stream ID, sequence number, timestamp, correlation ID, and causation ID. Every event reaching a listener arrives in an `EventEnvelope`; you never construct one manually (Chapters 8, 9).

EventStore — The append-only persistence layer for an event-sourced system. It records domain events indexed by stream ID and sequence number, enforces optimistic concurrency via the `version` token, and supports range queries for replay. PyFly ships `SQLAlchemyEventStore` and `InMemoryEventStore` (Chapter 9).

Hexagonal architecture — An architectural style that places the domain and application logic at the centre, surrounded by ports (interfaces), with adapters at the edges. Business code depends only on ports; adapters implement those ports using specific technologies. This is the organising principle of every PyFly module (Chapters 1, 2, 5).

Idempotency — The property that performing the same operation multiple times produces the same result as performing it once. Idempotent handlers are essential in messaging and saga compensation: network retries or at-least-once delivery guarantees mean a message may arrive more than once. Deduplication keys or idempotency tokens are used to detect and discard duplicate executions (Chapters 10, 12).

Migration — A versioned, ordered script that evolves a relational database schema without destroying data. PyFly integrates Alembic for migration management; `pyfly db migrate` auto-generates a migration from entity changes and `pyfly db upgrade` applies pending migrations (Chapter 5).

Outbox pattern — A technique for publishing domain events reliably alongside a database write: both the state change and the event records are written in the same local transaction; a background relay process reads unsent events from the outbox table and forwards them to the broker. This eliminates the two-phase commit between the database and the message broker (Chapters 9, 12).

Port — A Python `Protocol` class that defines the interface a piece of business logic depends on, without specifying any implementation. The DI container wires the concrete adapter that satisfies the protocol at startup. Ports enable the hexagonal architecture and make adapters swappable with zero business-logic changes (Chapters 1, 2, 5).

Primary bean — When multiple beans satisfy the same type, the one annotated `@primary` is preferred for injection. Without a `@primary` annotation the container raises an ambiguity error. `@primary` is how you designate the production adapter among several alternatives (Chapter 2).

Profile — A named activation tag that selects which beans and configuration values are active at runtime. PyFly loads `pyfly-{profile}.yaml` over the base `pyfly.yaml` and activates beans annotated `@profile("prod")` only when `prod` is in the active profile list. The active profile is set via `PYFLY_PROFILES_ACTIVE` or `pyfly.yaml` (Chapter 3).

Projection — A read model derived by consuming a stream of events. A projection subscribes to specific event types and incrementally builds a queryable view — a balance cache, an audit table, a dashboard aggregate — without touching the write model. In event sourcing, `ProjectionRunner` replays the `EventStore` to rebuild projections from scratch (Chapters 8, 9).

Query (CQRS) — An immutable dataclass that expresses a read intent — "get wallet balance", "list transactions". Queries inherit from `Query[R]` and flow through the `QueryBus`, which can cache results transparently. Queries never mutate state (Chapter 7).

QueryBus — The pipeline that receives a `Query`, optionally returns a cached result, dispatches to the matching `QueryHandler`, and optionally stores the result in cache. Separating the query bus from the command bus allows different cross-cutting behaviour — caching, read-replica routing — for read paths (Chapter 7).

Rate limiter — A resilience component that caps the number of requests an endpoint or service client can accept within a time window, preventing overload from bursty traffic. PyFly's `RateLimiter(max_tokens=N, refill_rate=r)` + `@rate_limiter(limiter)` implements a token-bucket limiter (Chapter 13).

Repository — A collection-like abstraction over the persistence layer that allows the application to load and save aggregates or entities without any SQL in the business code. PyFly's `CrudRepository[E, ID]` provides typed `find_by_id`, `save`, `delete`, and derived-query helpers; custom implementations annotated `@repository` replace the in-memory defaults (Chapters 2, 5).

Retry — A resilience pattern that re-executes a failed operation after a delay, up to a configured maximum attempt count. Retries handle transient failures — brief network glitches, momentary 503s — without operator intervention. PyFly's

`@retry(max_attempts=N, backoff=...)` implements exponential back-off with jitter; `@service_client` includes retry by default (Chapters 11, 13).

Saga — A sequence of local transactions coordinated by a central orchestrator. Each step commits to its own service's database; if a later step fails, the orchestrator calls the compensating transaction for each already-committed step in reverse order. PyFly's `@saga` and `@saga_step` decorators implement the orchestrated saga pattern with a parallel-execution DAG (Chapter 12).

Serialization — The process of converting an in-memory object to a wire format (JSON, Protobuf, Avro) for HTTP responses or message publishing, and deserializing the reverse. PyFly's central `PyFlyJsonSerializer` bean (`from pyfly.web import PyFlyJsonSerializer`) configures Pydantic-backed JSON serialization once and applies it to every HTTP response and message envelope (Chapters 4, 10).

Service — A managed bean decorated with `@service` that houses business logic and orchestrates calls to repositories, event publishers, and other services. Services are the application layer in hexagonal architecture: they translate incoming commands into domain operations and persist results (Chapter 2).

Snapshot — A point-in-time serialized copy of an event-sourced aggregate's state, stored alongside the event stream to accelerate replay. On load, the repository restores the snapshot and replays only the events that occurred after the snapshot version, capping replay time regardless of stream length (Chapter 9).

Stereotype — A decorator that registers a class as a bean and signals its architectural role: `@service`, `@repository`, `@component`, `@configuration`, or `@rest_controller`. All stereotypes are technically equivalent in the container; the semantic difference is for human readers and tooling (Chapter 2).

TCC (Try-Confirm-Cancel) — A distributed-transaction pattern in which each participant first *reserves* a resource (Try), then the coordinator either *confirms* all reservations or *cancels* them based on whether all Tries succeeded. TCC is useful when exact, immediate reservation semantics are required — for example, holding funds before capturing a payment. PyFly's `@tcc(name="...")` + `@tcc_participant(id="...", order=N)` decorators implement the TCC protocol alongside the saga engine (Chapter 12).

Testcontainers — A library that starts real Docker containers (PostgreSQL, Redis, Kafka) for integration tests and tears them down when the suite finishes. PyFly's `pyfly.testing` module provides `postgres_container` and `redis_container` fixtures that wire Testcontainers into the DI container via `@ServiceConnection`-style configuration (Chapter 16).

Value object — An immutable domain object identified by its value rather than by an identity field. Two value objects are equal if all their fields are equal. In PyFly, `ValueObject` is the base class; apply `@dataclass(frozen=True)` to enforce immutability. `Money` — an integer amount in **minor units** (cents) and a `Currency` `StrEnum` — is Lumen's canonical value object; `Wallet` is the aggregate root that owns it (Chapter 6).

Webhook — An inbound HTTP callback that an external provider (Stripe, Twilio, etc.) calls to notify your service of an asynchronous event, such as a payment-status change. PyFly's `@webhook_listener` decorator verifies the HMAC-SHA256 signature, deduplicates replays via a nonce cache, and routes the payload to a typed handler (Chapter 17).

Workflow — A long-running variant of the saga pattern where steps may pause for minutes, hours, or human approval before resuming. Workflows persist their state between steps so they survive process restarts; PyFly's `@workflow` and `@workflow_step` decorators provide this capability on top of the same saga engine (Chapter 12).